

BOUNDARY LAWS, INTRINSIC PERIODIC RIGIDITY, AND GLOBAL PHYSICAL RECONSTRUCTION IN DISPERSING BILLIARDS

QIAN QI

ABSTRACT. We determine the two labelled contact jets of a smooth dispersing billiard channel from its gap and the signed endpoint law of each contact type at one fixed positive offset. The laws arise from long alternating collision bridges. On a collar independent of the flight number, the normalized mixed endpoint derivative has a nonlinear relative limit with every fixed mixed geometric and offset derivative. A four-density identity cancels the unknown flux amplitudes and recovers the unsymmetrized half-line actions. A weighted nonlinear half-line inverse, finite-truncation envelope differentiation with smooth finite remainders, and homogeneous degree isolation then recover the contact jets through explicit two-by-two blocks of determinant one. No reflection symmetry, equal contacts, or supplied curvatures are assumed.

For connected real-analytic boundaries, intrinsic channel data determine complete obstacle images in their separate frames. Signature-rigid incidence matching and a rank-two anchoring pair based at one channel frame recover the Euclidean period lattice, including its Gram form and scale. A rooted signature-rigid spanning tree places every remaining obstacle orbit. On compact analytic classes we obtain stable unique finite-signature matching and a uniform finite-data inverse modulus. The common-orientation orbit of the entire marked signed datum yields the corresponding classification up to one global Euclidean isometry; this orbit is distinct from pointwise sign erasure.

The local information depends on the retained record. Endpoint-residual-time observations are equivalent to a moving-ceiling Poisson experiment, with explicit kernels controlling layer, bulk, and corner errors. Removing residual time gives a linearly vanishing endpoint density and a nonregular Gaussian limit; waiting counts supply a separate faster normal information channel. In the anchored finite-dimensional local models, the endpoint and count-endpoint Gaussian approximations hold in Le Cam distance on every fixed compact local parameter set. These scalar and finite-strip Poisson families have common dominating measures, although a designated reference member need not dominate them. For anchored homothetic channels, the information loss from deleting normal endpoint coordinates is maximal when the observed contact varies, whereas observing only the fixed contact gives equivalent matched endpoint experiments.

Finally, a charged near-onset pilot and single-offset transverse-law tests give uniform global reconstruction from common physical position records, using only even bridges whose minimum flight number diverges. We also prove a direct graph-interpolation benchmark in the same position experiment. This separates its noiseless geometric sampling from the intrinsic inverse based only on transverse laws. All preparation failures are charged, and no finite-dimensional parametrization of the compact analytic class is used.

CONTENTS

1.	Introduction	3
2.	Geometric setting and the principal relative theorem	9

Date: September 14, 2026.

2020 Mathematics Subject Classification. 37D50, 37C30, 70H09, 53A04, 62G05.

Key words and phrases. Dispersing billiards, inverse problems, boundary laws, single-offset density inversion, contact rigidity, analytic rigidity, lattice holonomy, Poisson boundary experiments, moving support, Le Cam equivalence, observable calibration, global statistical reconstruction.

Part 1. Relative laws, signed inversion, and intrinsic rigidity	13
3. Uniform thresholds and conditional contact metrics	13
4. Shortest channels in arbitrary periodic configurations	14
5. A parameter-dependent boundary-value and relative-flux lemma	16
6. Full-phase integration and the threshold law	18
7. Two nonlinear boundary layers	20
8. From Jacobi identities to a relative physical law	26
9. Hyperbolic coordinates and the intrinsic boundary measure	29
10. Boundary energy profiles and nonlinear parity compatibility	37
11. Determination of two independent even contacts	42
12. Signed endpoint rigidity and the all-order determinant-one inverse	45
13. An amplitude-free inverse from one endpoint law	52
14. Intrinsic lifted-channel graphs and analytic table rigidity	59
15. Rank-two holonomy and recovery of the periodic lattice	67
16. Finite analytic signatures and stability of intrinsic gluing	69
17. Physical coordinates and finite-window observation	72
18. Two-flight determination and the finite observation design	76
19. The signed one-flight benchmark and the restriction to long bridges	80
20. Direct geometric sampling in the same long-even experiment	81
Part 2. Boundary information and physical local experiments	83
21. Relative boundary laws for physical experiments	83
22. Adaptive observation and stopped experiments	85
23. Boundary information after residual-time integration	88
24. Vector boundary information for a moving-support local family	91
25. Physical realization of anchored finite contact-jet families	100
26. A physical endpoint experiment on fixed laboratory windows	101
27. Compact local experiments	107
28. The complete endpoint–time acquisition experiment at its fastest local scale	108
29. Two-sided deficiency for the moving-ceiling experiment	111
30. Common domination and the information in exact positions	114
31. Waiting counts and endpoint information after residual-time coarsening	117
32. Intrinsic normal–tangent geometry of the count–endpoint limit	120
33. The endpoint critical experiment	122
34. The Abel geometry of the boundary invariant	125
35. The inverse autoconvolution problem	128
36. Regularized observation of the complete invariant	128
Part 3. Observable calibration and global physical reconstruction	133
37. From noisy finite jets to global analytic table loss	133
38. Common physical observations and charged channel calibration	135
39. Single-offset statistical separation and global physical reconstruction	138
40. Compatible analytic variations and uniform finite positive designs	142
Part 4. Complete auxiliary proofs and applications	144
Appendix A.1. Information beyond the leading hierarchy	144
Appendix A.2. Statistical determination of an analytic contact	146
Appendix A.3. Finite-flight contact recovery and coefficient sensitivity	150
Appendix A.4. Reconstruction of boundary profiles from physical counts	153

Appendix A.5.	Acquisition from integrated physical flux	157
Appendix A.6.	Calibration within the smooth physical experiment	159
Appendix A.7.	Critical boundary experiments and support separation	163
Appendix A.8.	Curvatures invisible to all threshold locations	169
Appendix A.9.	A sharp pairwise inverse through a triple collision	172
Appendix A.10.	Three amplitudes in the constrained physical family	175
Appendix A.11.	Exact inversion from four fixed physical windows	177
Appendix A.12.	Joint curvature and onset accuracy	180
Appendix A.13.	Reconstruction with unknown smooth remainder functions	183
Appendix A.14.	The sharp cost of unknown smooth remainders	185
Appendix A.15.	Relation to action, spectral, and statistical inverse problems	187
Appendix A.16.	Conditional endpoint metrics and curvature observability	190
Appendix A.17.	Leading-data factorization and quantitative acquisition	193
Appendix A.18.	Local calibration from the onset probabilities	195
Appendix A.19.	A sharp stability scale at the circular configuration	198
Appendix A.20.	Measurable reconstruction on compact parameter sets	200
Appendix A.21.	Locally calibrated reconstruction from unlabelled counts	201
Appendix A.22.	Physical indistinguishability in local count designs	204
Appendix A.23.	Count-only reconstruction from a fixed local bracket	206
Appendix A.24.	Marked onsets and source domains	208
Appendix A.25.	Record topology and differentiated moving tests	208
Appendix A.26.	The circular collision hierarchy	210
Appendix A.27.	Stationary flux and the complete marked event	212
Appendix A.28.	Localization to actual alternating itineraries	213
Appendix A.29.	A boundary-value calculus uniform in the itinerary length	214
Appendix A.30.	Uniform threshold integration and its response	216
Appendix A.31.	The joint impact record and physical sampling cuts	218
Appendix A.32.	Mechanical consequences of the threshold coefficients	219
References		221

1. INTRODUCTION

The geometry of a dispersing billiard can be observed through trajectories, lengths, collision positions, or probability laws of selected collision words. These data have different inverse content. An exact planar collision position lies on the unknown boundary and supplies a geometric sample. Its signed transverse projection does not supply the missing normal coordinate. We study the geometry encoded by the distribution of the latter observations for long alternating bridges near a closest pair of strictly convex scatterers.

Our main local result determines both labelled contact jets from the gap and one signed conditional endpoint law of each contact type at a single positive offset. The bridge probability decays exponentially with the number of flights, but its conditional law retains a nonlinear profile formed by repeated returns to both contacts. We construct this profile as a relative limit on a collar independent of the flight number and invert it on the geometric image of the billiard observation map. Neither reflection symmetry, equal contacts, nor supplied curvatures are required. The inverse combines cancellation of the unknown flux amplitude with a finite-remainder analysis of the nonlinear half-line action. This gives a triangular reconstruction of the labelled contact jets at every prescribed finite order. For connected analytic boundaries, intrinsic

channel matching then recovers complete obstacles and the Euclidean period lattice under the marked incidence hypotheses stated below.

We also construct globally consistent estimators in an unregistered physical position-sensor experiment, with a charged calibration pilot and only long even bridges. We give a direct graph-sampling proof available in that same richer experiment. Thus the distinction between geometric sampling and the inverse from transverse laws is part of the theorem statements, rather than an inference from the flight restriction. The local information results separately identify the effects of deleting residual times and waiting counts.

1.1. Relative boundary laws. For a selected closest-pair channel let $g > 0$ be its gap, let $\kappa_0, \kappa_1 > 0$ be its labelled contact curvatures, and put

$$c_b = 1 + g\kappa_b, \quad \gamma = \operatorname{arcosh} \sqrt{c_0 c_1}.$$

If $W_j(u, v)$ is the stationary length of a j -flight bridge with signed transverse endpoints, define

$$E_j = W_j - jg, \quad b_j = \frac{-W_{j,uv}}{-W_{j,uv}(0, 0)}.$$

There are strictly convex actions S_b and positive amplitudes B_b , normalized by $S_b(0) = S'_b(0) = 0$ and $B_b(0) = 1$, such that

$$\|E_j - S_0 \oplus S_p\|_{C^k} + \|b_j - B_0 \otimes B_p\|_{C^k} \leq C_k \tau^j, \quad p = j \bmod 2, \quad 0 < \tau < 1. \quad (1.1)$$

The order k is fixed but arbitrary, including the mixed geometric and offset derivatives stated in Theorem 2.1. The endpoint and offset collars do not shrink with j . The local forward construction requires neither finite horizon nor global reflection symmetry. Its weighted inverse, differentiated operator comparison, and uniform hyperbolicity margins are proved in Part I.

At excess time $d > 0$ the successful endpoint-residual-time limit has density proportional to

$$B_0(u)B_p(v)\mathbf{1}_{\{0 < r < d - S_0(u) - S_p(v)\}}.$$

For same-type even bridges, integrating out r gives a density $f_{b,d}(u, v)$ proportional to $(d - S_b(u) - S_b(v))_+ B_b(u)B_b(v)$. The family of supports as d varies determines the unsymmetrized action by a threshold slice. The density itself carries more information: only one positive d is needed. Proposition 13.8 gives a support-preserving action change in the functional density class, showing why one support alone does not supply the same invariant.

1.2. A single-offset inverse and intrinsic periodic determination. On a small interior square, write

$$R_b(u, v) = \frac{f_{b,d}(u, v)f_{b,d}(0, 0)}{f_{b,d}(u, 0)f_{b,d}(0, v)}.$$

All unknown amplitude and normalization factors cancel, leaving

$$1 - R_b(u, v) = t_b(u)t_b(v), \quad t_b(u) = \frac{S_b(u)}{d - S_b(u)}. \quad (1.2)$$

At a fixed nonzero anchor a , strict convexity gives $t_b(a) > 0$. Consequently

$$t_b(u) = \frac{1 - R_b(u, a)}{\sqrt{1 - R_b(a, a)}}, \quad S_b(u) = \frac{dt_b(u)}{1 + t_b(u)}.$$

These are identities of continuous densities determined by the law; point values of a density are not assumed to be individual observations. The fixed-anchor formula has a locally Lipschitz inverse in every fixed interior C^M norm, uniformly on compact nondegenerate classes. It avoids taking derivatives of a square root at the minimum of S_b .

The gap is a separate datum, obtained from physical onset gg . If $a_b = S_b''(0)$, then

$$\gamma = \operatorname{arsinh}(g\sqrt{a_0a_1}), \quad c_b = \cosh \gamma \sqrt{a_b/a_{1-b}}, \quad \kappa_b = (c_b - 1)/g.$$

Write $q_n = (\psi_0^{(n)}(0), \psi_1^{(n)}(0))^t$ for the next graph jets and $s_n = (S_0^{(n)}(0), S_1^{(n)}(0))^t$. For $n \geq 3$,

$$s_n = M_n q_n + R_n, \quad M_n = \begin{pmatrix} \coth(n\gamma) & \mathfrak{r}_0^n \operatorname{csch}(n\gamma) \\ \mathfrak{r}_1^n \operatorname{csch}(n\gamma) & \coth(n\gamma) \end{pmatrix}, \quad \mathfrak{r}_b = \sqrt{c_b/c_{1-b}}, \quad \det M_n = 1, \quad (1.3)$$

where R_n is determined by lower graph jets. Establishing this filtration requires more than the density cancellation. We invert the nonlinear half-line stationarity operator on a weighted sequence space, differentiate finite truncations of its action, control the endpoint tails, and isolate the first homogeneous occurrence of each new graph jet. A finite Taylor-remainder estimate proves that equal graph jets through order M give equal action jets through order M , even for smooth graphs differing by flat terms. Smooth boundary images are not identified by their jets; the boundary-image conclusion uses analyticity. The inverse is fixed-order stable; determinant one alone is not a bound uniform in the infinite jet order.

For a periodic table the deck group is a marked abstract $\mathbb{Z}^2$, with no supplied Euclidean Gram form. A channel label (a, b, ℓ) specifies its obstacle orbits and abstract deck displacement. Each channel has its own oriented frame; relative placements between channels are not supplied. A signature-rigid transition has a unique framed point with the prescribed complete oriented curvature signature. In the presence of analytic symmetries the general gluing theorem gives a gluing space, not an arbitrarily selected phase.

Theorem 1.1 (Determination from single-offset transverse laws). *For a smooth strictly convex channel, its gap and two same-type signed limiting endpoint laws at one known sufficiently small positive offset determine every finite labelled contact jet. No longitudinal position, residual time, waiting count, flux amplitude, or contact curvature is supplied as part of these conditional-law data.*

For connected real-analytic incident boundaries the data determine their complete boundary images in the separate channel frames. Suppose the marked data are realizable by a periodic table and the measured channel graph contains a rank-two anchoring pair based at one channel frame, in the sense of Definition 15.1, and a rooted signature-rigid spanning tree reaching every obstacle orbit. Then the data determine the entire marked periodic table, including its Euclidean lattice, up to one simultaneous element of $\operatorname{SE}(2)$. On a compact class with uniform collars the same positive offset can be used for every channel and both types.

Proof. Theorem 13.1 supplies the action germs. The all-order contact inverse is Theorem 12.1. Analytic continuation and the intrinsic gluing classification are proved in Theorem 14.3; rank-two metric recovery is Theorem 15.3. Propagation along the rooted spanning tree is Theorem 15.4. To use the same frame for these last two steps, apply Lemma 14.11 to root the supplied tree at an obstacle met by the anchoring pair. A signature-rigid incidence on that pair supplies the unique framed signature required by the lemma. Thus the two cycle holonomies and the tree placements are reconstructed in one common frame, without an additional registration hypothesis. Their composition is Theorem 13.4. $\square$

The last lattice step is $L = VM^{-1}$, where V consists of recovered translation holonomies and M of their independent marked deck vectors. The difficult input is the recovery and unique matching of complete framed curve images, not this matrix identity. Likewise, marked deck labels and signed transverse conventions are part of the intrinsic data map, not quantities inferred from an unlabelled trajectory.

1.3. Physical acquisition and the same-experiment comparison. For global physical acquisition each channel detector reports the first and last planar collision positions in a common oriented sensor frame for that channel's two types, together with residual preparation time or a failure symbol. Frames of distinct channels are not registered. Designs use only the marked channel, type, even flight number, and physical observation time. All preparations and stopping decisions are retained and charged. These common Borel spaces are defined in (38.1). The anchored scalar local experiment is a different, explicitly coarser model; no unknown contact chart is silently treated as observed.

Let $\mathfrak{K}$ be the compact marked analytic class of Section 16, with uniform positive geometry, holomorphic curvature collars, persistent channels, the preceding uniqueness structure, and nondegenerate rank-two holonomies. Let d_q be the labelled C^q distance in a root-frame gauge.

Theorem 1.2 (Uniform physical reconstruction at one post-pilot offset). *There are finite deterministically capped policies π_s and measurable table estimators $\hat{T}_s$ in the common physical position experiment such that every flight number used by π_s is even and at least s , and for every fixed $q \geq 0$ and $\epsilon > 0$,*

$$\sup_{T \in \mathfrak{K}} \Pr_T^{\pi_s} \{d_q(\hat{T}_s, T) > \epsilon\} \longrightarrow 0.$$

No gap, contact center, tangent, inter-channel placement, or lattice metric is supplied. After the pilot all channels and both types use one fixed positive offset from the estimated onset, independent of s . The estimator uses the estimated gaps and bounded tests of the successful transverse endpoint pairs; it does not require their normal coordinates or residual times in its post-pilot criterion.

Proof. Theorem 38.2 gives a charged near-onset pilot at the already selected final even flight number. The single-offset inverse makes gaps and the two endpoint laws at a fixed $d_0 > 0$ jointly separating. Lemma 39.1 selects finitely many bounded Lipschitz tests, and Theorem 39.2 gives a finite-template estimator with a deterministic preparation cap. The compact inverse modulus and Theorem 39.3 give the conclusion. $\square$

The theorem does not claim that half-line inversion is necessary in this position experiment. Proposition 20.1 gives a second route using exactly the same long even designs, charged pilot, and position records. Once the frame error is at most ϵ , interpolation of $M + 1$ separated graph samples at scale a gives

$$|\hat{\psi}_b^{(m)}(0) - \psi_b^{(m)}(0)| \leq C_M \{a^{M+1-m} + \epsilon a^{-m}\}, \quad m \leq M.$$

Each required bin has positive preparation probability and a finite charged cap. Choosing $\epsilon \leq a^{M+1}$ gives fixed-order consistency without estimating an action. Arbitrarily large even flight numbers do not exclude this comparison.

The intrinsic theorem has different data: no graph ordinate is available. Its single-offset distributional inverse and its all-order alternating contact recursion give a reconstruction from genuinely coarsened data. Exact law injectivity is not statistical equivalence to noiseless position sampling. Proposition 30.3 identifies the precise dependence on the observed contact type in a registered homothetic family. For any finite number of successful endpoint pairs, the scalar-to-position deficiency is one when the even design starts on the varying obstacle, while the reverse deficiency is zero. When both observed endpoints instead lie on the fixed facing obstacle, the matched endpoint experiments are equivalent. Residual time is deleted from both records in this comparison; retaining it in both gives the same dichotomy. Intermediate collisions are not silently added to either record. These comparisons do not alter the two-type law inverse above.

The finite separator library and compact inverse moduli may depend non-effectively on $\mathfrak{K}$. Neither global estimator asserts an analytic minimax rate or polynomial-time design construction.

Separately, Theorem 16.1 supplies a uniform finite signature with local embedding and global separation bounds, and Lemma 16.2 gives unique matching under controlled C^2 perturbations. The compact inverse modulus does not replace that matching proof. Budget-indexed implementations run the selected stage afresh; a cumulative archive of earlier stages is not claimed to have a diverging minimum flight number.

1.4. Local information under coarsening. In the anchored finite-dimensional family, residual-time retention leaves positive density at a moving ceiling. At shape scale k_n^{-1} , the successful endpoint–time experiment is equivalent to a Poisson boundary-shift experiment; the corresponding gap scale is $(j_n k_n)^{-1}$. Theorem 29.2 gives parameter-independent kernels in both directions, controlling the layer, reconstructed bulk, and the intersection with the fixed residual-time face. Its bulk estimate uses the explicit bounded $O(k_n^{-1})$ relative-density perturbation, not bare trace convergence.

The finite-strip Poisson family is jointly dominated by a common Poisson envelope. It need not be absolutely continuous with respect to its zero-shift member. The scalar endpoint family is likewise jointly dominated by Lebesgue measure with the required cemetery atoms. Its common-collar censoring produces the stronger property of domination by the censored reference member. Proposition 30.1 gives both reference measures and the Poisson Radon–Nikodym formula explicitly.

Removing residual time produces linear vanishing at the endpoint support boundary and the scale $k_n \delta_n^2 \log(1/\delta_n) \rightarrow 1$. The reference-dominated representative has a uniform LAN expansion. Lemma 24.6 identifies its finite Gaussian limits; the uniform moving-boundary modulus and finite-net argument then prove convergence of the original endpoint family in two-sided Le Cam distance on every fixed compact local set, in Theorem 27.2 and Corollary 27.3. Waiting counts identify the normal quotient to a level set of the hyperbolic exponent at scale $(j_n \sqrt{k_n})^{-1}$, while endpoint information acts on the tangent directions at their slower scale. Theorem 31.3 gives the compact two-sided Gaussian product approximation for this count–endpoint record. Deleting those counts is a further coarsening. These assertions concern their declared scalar record spaces, not the support-singular planar position experiment or the complete growing collision array.

1.5. Comparison with related inverse problems. Marked lengths, dynamical conjugacy, and boundary-law determination pose different inverse problems. De Simoi, Kaloshin and Leguil [5] determine analytic open dispersing billiards from marked lengths under their symmetry and genericity hypotheses. Florio and Leguil [11] obtain smooth conjugacies from periodic length data for the hyperbolic systems covered by their hypotheses, including the stated open-billiard application. This is a dynamical conclusion, distinct from identifying obstacle images by a Euclidean isometry.¹ Finamore and Leguil [12] use an enriched marked length spectrum and a CAT(0) construction to recover finite-horizon Sinai billiards up to isometry.

The data in Theorem 1.1 are instead onsets and signed transverse laws of selected channels, with marked deck labels and signature-rigid incidences. The local observation deletes the normal endpoint coordinates while retaining the two labelled contact types. The proof first identifies its nonlinear limiting density from physical bridges, and then recovers the unsymmetrized actions and the actual contact jets. Only the subsequent boundary-image step uses analyticity. This composition does not pass through a supplied normal-form chart or a dynamical conjugacy. Neither the channel laws and incidence data nor the ordinary or enriched marked length spectrum are derived from the other observation map here; the respective hypotheses and conclusions therefore remain distinct.

¹We use the revised version v5 of [11]. Its revision notice removes the earlier spectral-rigidity assertion affected by the error in Proposition 3.1; the smooth-conjugacy result is the comparison used here.

The cofactor identity and inverse-decay estimates used in the forward argument are classical tools, as detailed in Section 8. Their role here is to control a nonlinear Hessian perturbation in trace norm along the whole physical bridge, before passing through a normalized moving-sublevel integral. The inverse then requires finite-truncation envelope differentiation and a smooth finite-remainder theorem before homogeneous degree isolation yields the contact recursion. These are the analytic and geometric steps connecting the relative law to the determination theorem; the determinant-one identity and the final lattice equation record their explicit inversion, rather than replacing those steps.

Within the present geometry, the mixed-type one-flight support inverse of Proposition 19.1 and the two-flight calculations remain useful benchmarks. The direct-position result above is stronger as a comparison to our physical protocol because it uses the same long even record restriction. The additional inverse content lies in recovering geometry from the single-offset transverse laws, not in prohibiting one short flight while supplying noiseless geometric samples.

Nonregular support-dependent estimation and exceptional normal rates have an established history; see Smith [33]. Meister and Reiss [36] prove equivalence of a nonregular regression experiment to Poisson experiments whose intensity supports encode the regression boundary. A Gaussian/Poisson contrast or Poissonization alone is therefore not asserted as a new principle. The results here identify the billiard-specific nonlinear density, its amplitude-free and all-order contact inverses, and the kernels linking explicitly retained observation levels, with the rare-event preparation cost included.

1.6. Organization. Part I proves the relative law, the all-order signed contact inverse, the single-offset density inversion, intrinsic analytic gluing, lattice recovery, and finite-signature stability. It retains the short-flight comparisons and includes the same-experiment direct-position benchmark. Part II proves the boundary information results, distinguishes common domination from reference absolute continuity, proves the observed-type information dichotomy, and compares the residual-time and count coarsenings. Part III constructs observable calibration and the single-offset global estimator, and separately gives compatible analytic variation bundles. The appendices retain the complete auxiliary derivations and applications.

1.7. Proof architecture and the role of the observation map. The principal analytic assertion is the relative nonlinear boundary law; the principal inverse assertion is recovery of the two labelled contact jets from signed transverse laws. Their composition precedes the global and statistical applications. We describe the dependencies to distinguish the nonlinear arguments from the algebraic and compactness steps that complete those applications.

Relative asymptotics. The finite Jacobi calculation identifies both the endpoint Hessian and the normalization of the mixed derivative. The perturbation of the interior Hessian is summable in trace norm, uniformly in the number of flights. Lemma 5.2 therefore controls a relative determinant, rather than an absolute error divided by an exponentially small reference twist. The half-line construction then separates the two endpoint layers. Gluing the two half-line solutions and estimating the discarded middle block gives the factorization in Theorem 7.2. The uniform sublevel integral and its fixed-domain differentiation yield Theorem 2.1. Every derivative order is finite and fixed; the constants may depend on that order and on the corresponding boundary norms. The offset collar is independent of the flight number. Geometric differentiation is channel-centered, with physical time written as $t = jg(\xi) + d$.

Signed inversion. The four-density identity in Theorem 13.1 removes the unknown amplitudes at one positive offset. This algebra recovers actions, not graph jets directly. To pass to the contact geometry one must differentiate the half-line action through finite truncations, cancel the interior variations by stationarity, and control the terminal term before taking the limit. The smooth-remainder argument proves finite-jet factorization for actual smooth graphs, including

graphs differing by flat terms. Homogeneous degree isolation then gives the triangular recursion of Theorem 12.1, with the explicit two-by-two blocks in (1.3). Invertibility is established at each fixed order; no uniform inverse on an unrestricted infinite jet space is inferred from the determinant identity.

Intrinsic global reconstruction. Only after the local inverse is proved does analyticity extend the recovered germs to connected boundary images. The gluing problem retains the full oriented signatures and the marked deck labels. A common-frame rank-two anchoring pair recovers the period metric. Lemma 14.11 transports the root of the supplied signature-rigid tree to an obstacle met by that pair. The tree then places every remaining obstacle orbit in the same frame. No new incidence, deck displacement, or observation is needed. Realizability and unique signature matching are hypotheses of this global conclusion, not consequences of the matrix identity for the lattice. The finite-signature embedding and noisy matching theorem supply the local separation and uniqueness needed before the compact inverse-modulus argument. Neither compactness nor analytic continuation alone gives a numerical reconstruction rate.

Statistical and physical consequences. The transfer estimates are used only after the boundary law is available. Finite likelihood-vector convergence and a separate uniform-modulus argument supply the compact-local Gaussian comparisons. The likelihood-tilting and original-alternative moment estimates justify the quadratic-loss statements without assuming that total variation controls unbounded losses. The moving-ceiling comparison uses two actual Markov kernels, together with separate layer, bulk, and corner estimates. Finally, the global physical protocol includes a charged position-based pilot. Its post-pilot transverse statistic is coarser than the total position record, and the direct-position benchmark belongs to that same richer acquisition experiment. The smooth-nuisance envelope in Section A.14 is a distinct comparison model; its lower bound does not assert realizability of every splice by a billiard table. These distinctions are part of the mathematical statements and remain in force throughout the applications.

The complete auxiliary proofs are included in the appendices. No statistical equivalence between different observation maps is used in the deterministic contact or periodic rigidity arguments.

2. GEOMETRIC SETTING AND THE PRINCIPAL RELATIVE THEOREM

Fix a full-rank lattice $\Lambda \subset \mathbb{R}^2$ and finitely many bounded C^∞ strictly convex obstacles $D_1, \dots, D_s$ with positive curvature. Distinct lattice translates have disjoint closures. Let $A > 0$ be the free area per lattice cell and prepare the unit-speed billiard with normalized phase volume $dq d\theta / (2\pi A)$ on the torus complement. We observe the collision count and, when specified, the first and last collision positions. Finite horizon, congruence and symmetry are not assumed for the general forward or boundary-law results.

Consider a fixed configuration and a smooth finite-dimensional family $D_{a,\xi}$ through it, with ξ in a compact set K_0 contained in a sufficiently small parameter neighborhood. At every fixed derivative order, constants may depend on the corresponding higher norms of this family, on positive separation and curvature bounds, and on the lattice, but not on the collision order.

Let $\mathcal{E}$ be the finite set, modulo common lattice translation, of oriented closest pairs attaining the minimum gap at the reference configuration. An index $e = (a, b, \ell)$ represents a segment from $D_{a,\xi}$ to $D_{b,\xi} + \ell$; reversing it gives $\bar{e} = (b, a, -\ell)$. Write

$$\begin{aligned} g_* &= \min_{e \in \mathcal{E}} g_e, & c_{e,b} &= 1 + g_e \kappa_{e,b} \quad (b = 0, 1), \\ c_e &= \sqrt{c_{e,0} c_{e,1}}, & \gamma_e &= \operatorname{arcosh} c_e, & d_{j,e} &= t - j g_e. \end{aligned} \tag{2.1}$$

Here $\kappa_{e,0}$ and $\kappa_{e,1}$ are the positive curvatures at the two contact points and need not be equal. A channel event $E_{j,e}(d)$ consists of the $j+1$ impacts alternating between the specified facing patches

in the time window $t = jg_e + d$. These patches are fixed around the reference contacts before observation and define a measurable part of the physical record. The localization theorem below proves that the relevant pairs continue smoothly and that a common onset collar is available.

Parameter derivatives are multi-index derivatives in a fixed Euclidean parameter chart. Unless a statement explicitly says otherwise, every geometric derivative is taken in the *channel-centered* coordinates (ξ, d) , with

$$t = jg_e(\xi) + d, \quad (2.2)$$

so the offset d is held fixed while the physical time moves with the channel onset. In particular the mixed geometric derivatives in Theorem 2.1 are not derivatives at fixed physical time. This convention also applies to the parameter derivatives of $S_b, B_b, \mathcal{F}_p$ and to the uniform constants below. Matrix norms are Euclidean operator norms unless a trace norm is specified.

2.1. Local multiplier convention. Suppress the channel index for the local hyperbolic calculations and set

$$\begin{aligned} \varrho &= e^{-\gamma}, & \lambda &= \varrho^2 = e^{-2\gamma}, \\ \mathfrak{r}_b &= \sqrt{c_b/c_{1-b}}, & r_b &= \mathfrak{r}_b \varrho \quad (b = 0, 1). \end{aligned} \quad (2.3)$$

The number $\mathfrak{r}_b$ is a curvature ratio, whereas r_b is the derivative at zero of the physical first-hit map φ_b . Thus $\mathfrak{r}_0 \mathfrak{r}_1 = 1$, $r_0 r_1 = \lambda$, and $R_b = \varphi_{1-b} \circ \varphi_b$ has multiplier λ . The integer j counts flights; n iterations of R_b mean $j = 2n$. The symbol ρ in exponential estimates denotes a freely chosen strict decay margin, not the exact factor ϱ .

2.2. The nonlinear relative law. Fix an oriented channel e and abbreviate $g = g_e$, $\gamma = \gamma_e$. Use transverse endpoint coordinates u, v at its two contacts. Let $W_j(u, v)$ be the stationary length of its j alternating flights, $E_j = W_j - jg$, $d_j^0 = -W_{j,uv}(0, 0)$, and $b_j = -W_{j,uv}/d_j^0$. For $p = j \bmod 2$ put

$$a_b = \frac{\sqrt{c_0 c_1} \sinh \gamma}{g c_{1-b}}, \quad q_p = \sqrt{a_0 a_p}, \quad d_j^0 = \frac{q_p}{\sinh(j\gamma)}.$$

Theorem 2.1 (Relative boundary law). *There are positive smooth functions B_b and strictly convex smooth functions S_b , $b = 0, 1$, with*

$$S_b(0) = S'_b(0) = 0, \quad S''_b(0) = a_b, \quad B_b(0) = 1,$$

a common endpoint box, a number $d_ > 0$, and $0 < \tau < 1$ such that, for every fixed derivative order k and every $j \geq 1$,*

$$\|E_j - S_0 \oplus S_p\|_{C^k} + \|b_j - B_0 \otimes B_p\|_{C^k} \leq C_k \tau^j, \quad p = j \bmod 2. \quad (2.4)$$

The action difference vanishes to second order at the origin. Define

$$\mathcal{F}_p(d) = \frac{q_p}{\pi d^2} \int (d - S_0(u) - S_p(v))_+ B_0(u) B_p(v) du dv. \quad (2.5)$$

Then $\mathcal{F}_p$ extends smoothly to $[0, d_]$, is positive there, and has value one at zero. The exact selected-channel probabilities obey*

$$\left\| \frac{2A \sinh(j\gamma)}{d^2} \Pr(E_{j,e}(d)) - \mathcal{F}_p(d) \right\|_{C^k([0, d_*])} \leq C_k \tau^j. \quad (2.6)$$

The norm includes every fixed mixed geometric and offset derivative in the centered convention (2.2). The limiting conditional endpoint-residual law at $d > 0$ has density proportional to

$$B_0(u) B_p(v) \mathbf{1}_{\{r > 0, S_0(u) + S_p(v) + r < d\}}.$$

All assertions hold uniformly on the compact local families specified above. For analytic families the endpoint constructions are analytic.

Proof and dependencies. Lemma 5.2 constructs the finite bridge and its relative determinant. Lemma 7.1 constructs S_b, B_b with the normalizations above. Theorem 7.2 proves (2.4) by comparing two separated perturbation blocks in trace norm, including mixed derivatives. Theorem 7.3 proves (2.6) using the common Morse domain of Lemma 6.1 and identifies the conditional density. The full proofs occur in Part I. $\square$

2.3. From relative laws to physical observation protocols. The relative comparison retains the probability of obtaining a selected record. In Theorem 21.1, the finite and boundary subprobability laws are compared at each table in common endpoint and residual-time coordinates, with their missing-event atom included. In unscaled physical positions the same boundary embeddings are used for the two laws at that table. This proves a bound between statistical families; it does not supply a sampling kernel that knows the unknown table's contact center or tangent.

For predetermined physical windows t_i , the excess $d_i(T) = t_i - j_i g_e(T)$ is used only in the proof. The acquisition need not observe the gap. For adaptive designs and stopped procedures, Theorem 22.3 compares the complete declared transcript, with a bound of the form

$$\|\mathbf{P}_T^\pi - \mathbf{Q}_T^\pi\|_{\text{TV}} \leq C \mathbb{E}_{\mathbf{P}_T^\pi} \sum_{i=1}^{\mathcal{T}} Y_i \tau^{j(a_i)}.$$

Here $\mathcal{T}$ is the stopping time, a_i the chosen design and Y_i its success indicator. A deterministically capped batch stopping no later than its k -th success has error at most $Ck\tau^J$ when all its flight numbers are at least J . No failed preparation is removed from the cost.

There are two different realizations of endpoint coordinates in this article. The local information theorems in Section 26 concern an anchored registered family: the reported scalar transverse coordinates are literal laboratory coordinates fixed throughout that family, and the programmed windows are $t_{j,\ell} = jg_0 + d_\ell$. The local gap parameter is part of the likelihood, not supplied to the acquisition. The following information hierarchy concerns this declared scalar local record and its coarsenings. It does not assert that the local Poisson or Gaussian limit exhausts all information in a two-dimensional position sensor.

The global unknown-contact experiment is specified separately in Section 38. Each channel sensor records both components of its first and last endpoint positions before any transverse projection. A single sensor frame is shared by the two types of that channel, while different channel frames are not registered. The charged pilot of Theorem 38.2 estimates the contact positions, tangent and onset. The subsequent transverse normalization is then a measurable function of those estimates and a fresh physical record. Proposition 38.3 bounds its error for the finite bounded Lipschitz tests used by the global estimator. The parameter-dependent ideal normalization appears only in this error bound, not among the observations.

Thus the same-table physical-to-boundary comparison, the anchored local information experiment, the observable within-channel calibration, and the intrinsic inter-channel gluing have different roles. Their hypotheses are not exchanged. The earlier centered-window and fixed-table coarsening results remain in the auxiliary compendium with their stated preparation costs and calibration assumptions.

2.4. Observation hierarchy and the information retained by each coarsening. The same physical preparation generates several statistically distinct experiments. To compare their sigma-fields, fix a parameter-independent policy π with a deterministic preparation cap N , as in Section 22, and start with its complete stopped physical history. This history retains every chosen design, all collision points, labels and flight times within every preparation, the success/failure sequence, and the stopping time $T \leq N$. Independent policy randomness may also be recorded when it belongs to the specified experiment. Full collision histories are encoded in the common

physical coordinates as finite marked arrays. Their countable disjoint union, followed by the finite products and disjoint unions imposed by $T \leq N$, gives one standard Borel stopped-history space. Geometry changes its law, not its underlying measurable space. No total-variation replacement theorem for this full growing collision array is asserted here.

The per-preparation endpoint–time map is a building block, not a separate stopped experiment. On the prescribed bridge event it retains the first and last signed endpoints and the residual preparation-to-first-impact time; on failure it retains a failure atom. This is the one-record map for which Theorem 21.1 proves the finite-to-boundary comparison. Applying it separately to the preparations in the stopped history defines the following hierarchy on a single sample space.

- (1) *Full stopped collision history.* The complete history just described retains all within-preparation collision data, designs, failures and the stopping time.
- (2) *Stopped endpoint–time acquisition transcript.* Replace each within-preparation history by its endpoint–time record, retaining every design, every failure, and T . Recorded policy randomness is retained consistently. Theorem 22.3 compares the laws of this transcript, with every failed preparation charged.
- (3) *Count–endpoint transcript.* Delete residual times from item 2, but retain the complete design list, the success/failure sequence, the successful endpoint pairs, and the stopping time. The failure sequence may equivalently be encoded by waiting counts when the final censored failure run and the stopping information are also retained.
- (4) *Endpoint–output experiment.* Delete waiting counts and residual times, retaining the successful endpoint pairs required by the specified output rule and its cap cemetery symbol. Any output labels are precisely those specified by that rule, rather than an unannounced extra transcript.

Let H^π , T_{ET}^π , T_{CE}^π , and T_{out}^π denote these four records. Their deterministic forgetting maps give, after pullback to the full stopped-history space,

$$\sigma(T_{\text{out}}^\pi) \subseteq \sigma(T_{\text{CE}}^\pi) \subseteq \sigma(T_{\text{ET}}^\pi) \subseteq \sigma(H^\pi). \quad (2.7)$$

Common randomized output maps are Markov coarsenings as well; their independent seeds can be adjoined when a sigma-field formulation is needed. The coarsened record is always the output of the same acquisition policy. It is not the history of a different policy obtained by retrospectively withholding observations from the design rule. In particular, retained adaptive designs can encode part of a deleted record. The asymptotic identifications below are those of the protocols and hypotheses in the cited theorems, not a claim that deleting times from every adaptive policy produces the same limiting experiment.

These common Markov coarsenings produce genuinely different asymptotic experiments. At the fastest local geometric scale, the endpoint–time protocol of Theorem 28.3 converges to its Poisson boundary-shift experiment; the positive density at the moving residual-time ceiling yields shape resolution of order k^{-1} . At that scale the waiting counts are asymptotically ancillary under $j = o(\sqrt{k})$, with the other hypotheses of that theorem in force.

After residual time is removed, the ceiling density is integrated into a linearly vanishing endpoint density. The count channel then matters. In the multirate coordinate of Theorem 31.3, the waiting counts yield a Gaussian shift in the hyperbolic-exponent direction at shape scale $(j\sqrt{k})^{-1}$, while the iso-hyperbolic endpoint directions are visible at the slower boundary scale $k\delta^2 \log(1/\delta) \asymp 1$. Removing the waiting counts in that protocol produces the endpoint-output experiment and its endpoint Gaussian limit.

This hierarchy is consistent with the earlier fixed-table comparison. If $q_j = e^{-j\gamma}$, the endpoint–residual record and the endpoint-only record have different critical scales: the former reacts at order kq_j , whereas the latter reacts at order $kq_j^2 \log(1/q_j)$. Thus a weak one-record limit

does not determine the information scale, and integrating out residual time is not an innocuous presentation choice.

The distinction between statistical information and physical cost is also maintained. For a selected bridge, success has probability of order $d^2 e^{-j\gamma}$, so obtaining k successful records requires an exponentially larger number of preparations. Failed preparations remain in the stopped transcript whenever that transcript is the experiment under study. Conditioning on success is used only when a theorem explicitly studies the successful mark law.

Finally, the record hierarchy should not be confused with geometric registration. The local statistical realization in Section 26 uses one fixed laboratory chart so that all nearby parameters live on a common observation space. The global rigidity theorem of Section 14, by contrast, does not supply relative Euclidean placements between distinct measured channels; those placements are reconstructed through analytic gluing.

A common choice of orientation is a third, separate issue. The unoriented classification uses the orbit of the entire marked signed law collection under one simultaneous reversal, as specified in Definition 14.12. The reversal transports the lattice orientation convention and all channel signs together; it does not alter deck labels or forget the relative sign choices. This law-valued quotient is not another item in the above hierarchy of Markov coarsenings. In particular, replacing each endpoint pair by its absolute values, averaging a law with its reflection, or independently discarding the orientation of each channel gives different data. The signed inverse is not invoked as an inverse of any of those coarsenings. The precise common-orientation classification and its proof are given in Theorem 14.14.

Part 1. Relative laws, signed inversion, and intrinsic rigidity

3. UNIFORM THRESHOLDS AND CONDITIONAL CONTACT METRICS

Theorem 3.1 (Uniform thresholds for periodic dispersing configurations). *There is $\varepsilon_0 > 0$, independent of every integer $j \geq 1$, such that $\varepsilon_0 < \min_{K_0} g_*/4$ and, for $0 \leq t - jg_* < \varepsilon_0$,*

$$\Pr_{\xi}(N_t \geq j+1) = \Pr_{\xi}(N_t = j+1) = \sum_{e \in \mathcal{E}} \frac{d_{j,e,+}^2}{2A(\xi) \sinh(j\gamma_e(\xi))} [1 + d_{j,e,+} \mathcal{R}_{j,e}(\xi, d_{j,e,+})]. \quad (3.1)$$

Every fixed mixed derivative of $\mathcal{R}_{j,e}$ is bounded uniformly in j ; it has a smooth right extension at offset zero. For $t \leq jg_$ the probability is zero. At the selected channel's own window $t = jg_e + d$, $0 < d < \varepsilon_0$, the same individual summand is exactly $\Pr_{\xi}(E_{j,e}(d))$, whether or not e minimizes the gap.*

The exponent two is independent of j , but its coefficient is exponentially small. The essential estimate is relative to that coefficient. The proof uses a weighted Dirichlet construction and a trace-norm estimate for its relative determinant, rather than a long initial-value iteration. The local hypotheses for this construction are verified at every reference configuration above. In the triangular one-obstacle near-circle class, $\mathcal{E}$ consists of the six unit lattice directions, so (3.1) is its specialization, with all competing onsets.

For the next theorem, lift the first and last collision positions to the fixed contact patches in $\mathbb{R}^2$, and denote them by Q_0, Q_j . Define the two scalar conditional variances

$$V_{j,b}(d) = \frac{1}{d} \left\{ \mathbb{E}(|Q_b|^2 \mid E_{j,e}(d)) - |\mathbb{E}(Q_b \mid E_{j,e}(d))|^2 \right\}, \quad b = 0, j. \quad (3.2)$$

These are translation- and rotation-invariant within the chosen lifts. Their definition requires neither the unknown contact point nor its tangent or curvature. The gap g_e is identified by its channel onset; quantitative timing calibration is a separate experiment below.

Theorem 3.2 (Uniform two-variance curvature recovery). *For every odd $j \geq 1$ the right limits $v_{j,0}, v_{j,1}$ of $V_{j,0}(d), V_{j,j}(d)$ exist, and*

$$(v_{j,0}, v_{j,1}) = \frac{g_e \coth(j\gamma_e)}{3 \sinh \gamma_e} \left(\sqrt{\frac{c_{e,1}}{c_{e,0}}}, \sqrt{\frac{c_{e,0}}{c_{e,1}}} \right). \quad (3.3)$$

Each difference $V_{j,b}(d) - v_{j,b}$ is d times a smooth function with uniform fixed-order derivative bounds. At known g_e , these two limits determine the ordered pair $(\kappa_{e,0}, \kappa_{e,1})$ without knowing A . The inverse is given by

$$\begin{aligned} \frac{\coth(j\gamma)}{\sinh \gamma} &= \frac{3}{g} \sqrt{v_0 v_1}, & r &= \sqrt{v_1/v_0}, \\ \kappa_0 &= \frac{r \cosh \gamma - 1}{g}, & \kappa_1 &= \frac{r^{-1} \cosh \gamma - 1}{g}. \end{aligned} \quad (3.4)$$

On every compact positive gap–curvature box this observation map and its inverse are Lipschitz, with constants independent of odd j . They remain uniformly conditioned at $\kappa_0 = \kappa_1$. If both measured variances have error at most δ , a reconstruction constrained to such a box has curvature error at most $C(d + \delta)$.

Thus the full conditional metric retains information lost by the scalar flux coefficient. Theorem A.16.3 realizes the scalar ambiguity by actual smooth periodic obstacles with the same gap, area and every leading count amplitude, while their endpoint curvatures vary. The two variances distinguish that family. The statement is local contact-curvature recovery, not global shape rigidity. Conditioning on an exponentially rare event is not free: Proposition A.16.2 records the number of independent physical preparations required.

4. SHORTEST CHANNELS IN ARBITRARY PERIODIC CONFIGURATIONS

The relevant localization is a consequence of separated strict convexity, not proximity to a particular model shape.

Lemma 4.1 (A geometric criterion for the complete event). *At any reference configuration of Section 2, the minimum distance between distinct lifted obstacles is positive and is attained by finitely many oriented pairs modulo common translation. Every such pair has a unique closest segment, positive-definite distance Hessian and positive clearance from the remaining obstacles. Its contact points, length and curvatures continue smoothly under small smooth deformations. There are fixed contact and direction neighborhoods and $\varepsilon_0 > 0$ such that every physical sequence with j complete flights and total length at most $jg_* + \varepsilon_0$ alternates in one of these neighborhoods. The constants are independent of j .*

Proof. Choose bounded representatives of the finitely many obstacles. The distance between D_a and $D_b + \lambda$ tends to infinity with $|\lambda|$, uniformly in a, b . Only finitely many lattice translates can have distance below any fixed bound. Each of those finitely many positive distances is attained; disjoint closures make their minimum positive. Among pairs not attaining this minimum, the next distance is separated from it by a positive amount. Pairs outside the finite list already exceed a larger fixed bound. These statements persist under a small Hausdorff perturbation, with a strict gap to all pairs outside $\mathcal{E}$.

The closest displacement vector between two disjoint convex bodies is unique: if two different displacement vectors both minimized its norm, their midpoint would have smaller norm and would still be a displacement. For that vector, each endpoint is a support point in its normal direction. Strict convexity makes those support points unique. Rotate the segment to $[0, g]$ and

use a common transverse coordinate. The facing graphs are $(-\psi_0(u), u)$ and $(g + \psi_1(v), v)$, with $\psi_b(0) = \psi'_b(0) = 0$ and $\psi''_b(0) = \kappa_b > 0$. The distance Hessian at its minimum is

$$\frac{1}{g} \begin{pmatrix} 1 + g\kappa_0 & -1 \\ -1 & 1 + g\kappa_1 \end{pmatrix}. \quad (4.1)$$

Its determinant and its diagonal entries are positive. The implicit function theorem gives smooth contact points and distance. Compactness keeps both eigenvalues uniformly positive and all chosen charts uniform.

A reference minimum segment cannot meet a third obstacle in its open interior: the initial part would give a smaller distance from its first obstacle to that third obstacle. The closed segment also stays a positive distance from every other obstacle, since its endpoints belong only to its own two disjoint bodies and only finitely many other bodies are nearby. Small contact patches and small geometric perturbations preserve this clearance. Strict convexity makes each connecting chord leave the first obstacle and enter the second transversely, with no earlier hit. Thus these local chords are physical first hits.

At a fixed outgoing obstacle, distinct minimizing channels give distinct normal outgoing collision states. If two shared a contact and outgoing normal, their other contacts at the same distance would coincide, contradicting disjointness. The finite collection of states is therefore separated. Choose their neighborhoods so small that normal reflection of a flight arriving along e can belong to a short outgoing channel only in the neighborhood of $\bar{e}$. This property is open and uniform.

Every physical free flight joins distinct lifted obstacles: a straight ray leaving a convex body cannot return to it before another hit. Hence its length is at least g_* . If a sum of j such lengths is at most $jg_* + \varepsilon_0$, every summand is at most $g_* + \varepsilon_0$. The gap to the other pairs restricts that summand to $\mathcal{E}$. Away from the unique contact patches the distance to that pair's minimum is bounded below; decrease ε_0 to exclude this complement. Each short flight is then nearly normal. The separated outgoing states and specular reflection force every following short flight to be the reverse channel. Induction uses the same neighborhoods at each impact and introduces no length-dependent constant. Grazing states and their regular preimages have phase measure zero. $\square$

The lemma also applies to a specified nonminimal closest chord with positive clearance, for its selected local itinerary. It need not then exhaust the whole count event. All original six near-circle channels satisfy the criterion. For example, the criterion also applies to arbitrarily eccentric ellipses and to noncongruent obstacles on any fixed lattice, provided the indicated disjointness and positive curvature conditions hold; there is no small eccentricity assumption.

For one channel the exact one-flight actions are

$$\ell_b(u, v) = \sqrt{(g + \psi_b(u) + \psi_{1-b}(v))^2 + (v - u)^2}, \quad b = 0, 1. \quad (4.2)$$

Uniformly on a small fixed box,

$$\ell_b(u, v) = g + \frac{c_b u^2 + c_{1-b} v^2 - 2uv}{2g} + O((|u| + |v|)^3), \quad (4.3)$$

$$\ell_b(u, v) - g \geq \psi_b(u) + \psi_{1-b}(v) \geq c_{\text{loc}}(u^2 + v^2). \quad (4.4)$$

The full broken-action Hessian has strictly positive diagonal-dominance margins: κ_b in endpoint rows and $2\kappa_b$ in interior rows at zero. Every row perturbation involves only neighboring coordinates. The same small box therefore gives strict convexity at every length.

5. A PARAMETER-DEPENDENT BOUNDARY-VALUE AND RELATIVE-FLUX LEMMA

Let $W_{j,e}(\xi, u, v)$ be the stationary value of the sum of (4.2), with $y_0 = u, y_j = v$. We construct that value on one common neighborhood. Endpoint Hessians are denoted by H ; scalar relative remainders are denoted by $\mathcal{R}$.

Lemma 5.1 (Alternating Jacobi reduction). *Put $c = \sqrt{c_0 c_1}$, $\gamma = \operatorname{arcosh} c$, $\sigma_i = \sqrt{c_{1-(i \bmod 2)}}$, and $D_j = \operatorname{diag}(\sigma_0, \sigma_j)$. The quadratic endpoint Hessian is*

$$H_{j,e} = \frac{c \sinh \gamma}{g} D_j^{-1} \begin{pmatrix} \coth(j\gamma) & -\operatorname{csch}(j\gamma) \\ -\operatorname{csch}(j\gamma) & \coth(j\gamma) \end{pmatrix} D_j^{-1}. \quad (5.1)$$

Consequently

$$d_{j,e}^0 := -W_{uv}(\xi, 0, 0) = \frac{c \sinh \gamma}{g \sigma_0 \sigma_j \sinh(j\gamma)}, \quad \sqrt{\det H_{j,e}} = \frac{c \sinh \gamma}{g \sigma_0 \sigma_j}, \quad \frac{d_{j,e}^0}{\sqrt{\det H_{j,e}}} = \operatorname{csch}(j\gamma). \quad (5.2)$$

The endpoint eigenvalues have positive uniform upper and lower bounds. The period-two return multiplier is $e^{2\gamma}$.

Proof. The interior equation is $y_{i+1} - 2c_{i \bmod 2} y_i + y_{i-1} = 0$. Set $y_i = \sigma_i z_i$. Since $c_{i \bmod 2} \sigma_i / \sigma_{i+1} = c$, the recurrence becomes $z_{i+1} - 2c z_i + z_{i-1} = 0$ and the whole quadratic action becomes

$$\frac{c}{2g} \sum_{i=0}^{j-1} [c(z_i^2 + z_{i+1}^2) - 2z_i z_{i+1}].$$

Its endpoint solution is $z_i = \sinh((j-i)\gamma) z_0 / \sinh(j\gamma) + \sinh(i\gamma) z_j / \sinh(j\gamma)$. The first endpoint derivative is $c(c z_0 - z_1)/g$. The identity $c \sinh(j\gamma) - \sinh((j-1)\gamma) = \sinh \gamma \cosh(j\gamma)$, and its reversed version, give (5.1). Now use $\coth^2 - \operatorname{csch}^2 = 1$. The middle matrix has eigenvalues $\tanh(j\gamma/2)$ and $\coth(j\gamma/2)$. Since $\gamma \geq \gamma_* > 0$ and the σ_i have uniform positive bounds, this proves the eigenvalue bounds. The two-periodic scaling conjugates the two-step recurrence to the square of $\begin{pmatrix} 2c & -1 \\ 1 & 0 \end{pmatrix}$. The nonzero one-flight twist $1/g$ identifies position-pair and collision position-momentum coordinates, so its unstable eigenvalue is the stated physical return multiplier. In particular the formula at $j = 1$ is exactly (4.1). $\square$

Lemma 5.2 (Uniform nonlinear bridge and relative determinant). *Suppose g and the two graph functions in (4.2) range over a compact smooth parameter family, with g, κ_0, κ_1 bounded above and bounded away from zero. There is a common endpoint neighborhood on which the stationary segment exists uniquely and is smooth. For some fixed $\rho < 1$, with $a = |u| + |v|$ and $w_i = \rho^i + \rho^{j-i}$,*

$$y_i^{\text{lin}} = \sigma_i \left[\frac{\sinh((j-i)\gamma)}{\sinh(j\gamma)} \frac{u}{\sigma_0} + \frac{\sinh(i\gamma)}{\sinh(j\gamma)} \frac{v}{\sigma_j} \right], \quad (5.3)$$

$$|y_i| \leq C a w_i, \quad |y_i - y_i^{\text{lin}}| \leq C a^2 w_i,$$

$$W_{j,e} - jg = \frac{1}{2}(u, v) H_{j,e}(u, v)^t + O(a^3),$$

$$-W_{uv}(\xi, u, v) = d_{j,e}^0(\xi) b_{j,e}(\xi, u, v), \quad b_{j,e}(\xi, 0, 0) = 1, \quad b_{j,e} = 1 + O(a) > 0. \quad (5.4)$$

Every fixed mixed derivative of y_i and $y_i - y_i^{\text{lin}}$ is bounded by $C_k w_i$ times respectively $a^{\max(1-|\beta|, 0)}$ and $a^{\max(2-|\beta|, 0)}$, where β counts endpoint derivatives. The action remainder and $b_{j,e}$ have uniform fixed-order bounds. If both graphs are even, the bridge correction is $O(a^3 w_i)$, the action remainder is $O(a^4)$, and $b_{j,e} - 1 = O(a^2)$ is even. In an analytic family these bounds hold on a common complex neighborhood and are analytic bounds.

Proof. Let H_{int}^0 be the quadratic interior Hessian. Its inverse, for $1 \leq i \leq k < j$, is

$$G_{ik} = \frac{g\sigma_i\sigma_k}{c} \frac{\sinh(i\gamma) \sinh((j-k)\gamma)}{\sinh \gamma \sinh(j\gamma)}, \quad G_{ki} = G_{ik}. \quad (5.5)$$

Substitution verifies the homogeneous recurrence away from $i = k$, zero endpoint values and the unit jump at $i = k$. In particular $|G_{ik}| \leq Ce^{-\gamma^*|i-k|}$. Choose $e^{-\gamma^*} < \rho_0 < \rho < 1$. A geometric-series sum yields

$$\sum_k \rho_0^{|i-k|} (\rho^k + \rho^{j-k}) \leq C_\rho w_i. \quad (5.6)$$

For its ρ^k part, split at $k = i$ and sum the ratios ρ_0/ρ and $\rho_0\rho$; the other part follows by reversal. Thus G is bounded independently of j on the norm $\|z\|_w = \max_i |z_i|/w_i$.

Parameter differentiation of (5.5) has the same property. To see this without a hidden growing dimension, write the hyperbolic quotient as a bounded factor times $e^{-\gamma(k-i)}$, using factors $1 - e^{-2r\gamma}$ for its distances to the endpoints. Derivatives of those factors produce $r^m e^{-2r\gamma}$, uniformly bounded for each fixed m ; derivatives of the first exponential produce $|i - k|^m e^{-\gamma|i-k|} \leq C_m \rho_0^{|i-k|}$. All derivatives of the positive σ_i are bounded. The same argument for the endpoint solution gives $|\partial_\xi^\alpha y_i^{\text{lin}}| \leq C_\alpha a w_i$. Consequently every fixed parameter derivative of G is bounded in the weighted norm, with no j factor.

The interior gradient is $H_{\text{int}}^0(y - y^{\text{lin}}) + \mathcal{N}_\xi(y)$, including the fixed endpoint coordinates in the local remainder. Its i th component depends on only three neighboring coordinates and vanishes quadratically there. Adjacent weights have bounded ratios. On $\|y\|_w \leq C_0 a$, its weighted norm is $O(a^2)$ and its derivative has norm $O(a)$. The map $y \mapsto y^{\text{lin}} - G\mathcal{N}_\xi(y)$ preserves this ball and has Lipschitz constant at most $1/2$ when a is uniformly small. This proves existence and the first estimates. The diagonal-dominance bound after (4.4) gives uniqueness throughout the contact box, not just inside the weighted ball.

For clarity, the derivative induction takes place in these same weighted spaces. One derivative of the fixed-point equation gives

$$(I + GD_y \mathcal{N}_\xi) Dy = Dy^{\text{lin}} - (DG)\mathcal{N}_\xi - GD_{\text{explicit}} \mathcal{N}_\xi.$$

Its left inverse has norm at most two. At order k , the only term containing an order- k derivative of y is the same left operator; every other term contains derivatives of G , explicit parameter or endpoint derivatives of the local remainder, and lower derivatives of y . The local chain rule and bounded neighboring-weight ratios bound their products in $\|\cdot\|_w$. Induction proves the estimates at order k . Parameter derivatives preserve vanishing at zero; Taylor's integral formula in (u, v) therefore gives precisely the powers of a stated in the lemma, including the correction of order two.

For every $p > 0$, $\sum_i w_i^p \leq C_p$. Sum the cubic action remainders using this inequality. The linear bridge is stationary for the quadratic action with endpoints fixed, so its change on substituting the nonlinear bridge is $O(a^4)$. This proves the action estimate. The same summability and the derivative induction prove all its fixed-order bounds.

For the relative estimate put $b_i = -\ell_{i \bmod 2, uv}(y_i, y_{i+1})$ and $H_{\text{int}} = H_{\text{int}}^0 + E$. Schur complementation gives

$$-W_{uv} = \frac{\prod_{i=0}^{j-1} b_i}{\det H_{\text{int}}}. \quad (5.7)$$

Indeed, the endpoint-to-interior entries are $-b_0, -b_{j-1}$ and the corner entry of the inverse tridiagonal interior matrix is $b_1 \cdots b_{j-2} / \det H_{\text{int}}$. At $j = 1$ the determinant is empty and equals

one. The local expansions and endpoint weights give

$$\sum_i |\log(gb_i)| \leq Ca, \quad \|E\|_1 \leq \sum_{i,k} |E_{ik}| \leq Ca, \quad \|(H_{\text{int}}^0)^{-1}\| \leq C.$$

The trace-norm bound uses tridiagonality; each matrix unit has trace norm one. Take the endpoint neighborhood so small that $\|(H_{\text{int}}^0)^{-1}E\| < 1/2$. Then

$$\log \frac{\det H_{\text{int}}}{\det H_{\text{int}}^0} = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} \text{tr}[(H_{\text{int}}^0)^{-1}E]^m = O(a). \quad (5.8)$$

The absolute trace is bounded by one trace norm times $m-1$ operator norms. Summing therefore costs $C\|E\|_1$, not the matrix dimension. Divide (5.7) by its value at zero and subtract logarithms to obtain (5.4).

Every differentiated E has uniformly bounded trace norm by the already proved coordinate bounds. The quadratic Hessian and each of its parameter derivatives are uniformly banded with bounded entries; hence derivatives of its inverse have bounded operator norm, by $\partial H^{-1} = -H^{-1}(\partial H)H^{-1}$ and induction. Termwise differentiation of (5.8) is justified by geometric convergence even after its fixed polynomial factors in m . Every term has at least one trace-class factor. The differentiated sum of $\log(gb_i)$ is controlled by the same endpoint weights. This proves all derivative bounds for the logarithmic ratio and its exponential. Positive twists and a positive interior determinant give positivity.

If the graphs are even, the action expansion is even, its nonlinear gradient is cubic, and its Hessian and twist errors are quadratic. The preceding estimates then give the improved orders. Uniqueness implies simultaneous reversal of every coordinate, proving evenness. For a jointly analytic family all local functions extend to a common small complex neighborhood of the fixed compact real parameter set. Shrink it to retain the strict exponential and inverse-norm margins. The same contraction is uniform there. The determinant series remains convergent, and Cauchy estimates on a smaller neighborhood prove the analytic assertion. No assertion about analytic extension is used for a merely smooth family. $\square$

6. FULL-PHASE INTEGRATION AND THE THRESHOLD LAW

Let $P(\xi)$ be the sum of obstacle perimeters in one cell. The collision section measure is $ds dp/(2P)$, the mean roof $\pi A/P$, and the stationary collision intensity $P/(\pi A)$. These identities follow from a flight-tube change of variables: irrational torus lines hit an open obstacle, rational directions form a null angular set, and the positive minimal gap excludes finite-time accumulation. In particular, in first-impact endpoint coordinates the full phase measure is

$$\frac{-W_{uv}}{2\pi A(\xi)} du dv dr. \quad (6.1)$$

Indeed, first variation gives $p = -W_u/(ds/du)$; the arclength factor cancels in $|ds dp| = |W_{uv}| du dv$. The positive relative twist makes the endpoint map injective at fixed u . The localized stationary path is physical by Lemma 4.1. Thus (6.1) is not a substituted transverse ensemble. The total angle-boundary flux is $2P$, and its flight-tube integral is $2\pi A$; this also proves the normalization without an upper roof bound.

Lemma 6.1 (Uniform smooth radial integration). *There are smooth Morse inverses $\Phi_{j,e}(\xi, w)$ on a common ball, with all fixed derivatives bounded uniformly in j , such that*

$$W_{j,e}(\xi, \Phi_{j,e}(\xi, w)) - jg_e = |w|^2/2, \quad D_w \Phi_{j,e}(\xi, 0) = H_{j,e}^{-1/2}.$$

The phase density (6.1) in (w, r) is

$$\frac{a_{j,e}(\xi, w)}{2\pi A(\xi) \sinh(j\gamma_e)} dw dr, \quad a_{j,e}(\xi, 0) = 1, \quad a_{j,e} = 1 + O(|w|) > 0, \quad (6.2)$$

with the same derivative bounds. If a is any such uniformly smooth amplitude, then

$$\frac{2}{\pi} \int_{|z|<1} (1 - |z|^2) a(\xi, \sqrt{2d} z) dz = a(\xi, 0) + dH(\xi, d),$$

where H is smooth for $d \geq 0$ and has uniform fixed-order bounds. No evenness of a or of the scatterer is required.

Proof. For $E = W - jg$, Taylor's integral formula writes $E(y) = \frac{1}{2}y^t M(y)y$, where $M(y) = 2 \int_0^1 (1-s) D^2 E(sy) ds$ is uniformly positive near zero. The map $y \mapsto M(y)^{1/2}y$ has uniformly invertible differential at zero and small nonlinear differential on a common ball. Its inverse is obtained by contraction, then differentiated to each finite order. This construction uses the uniform endpoint Hessian, not the exponentially small twist as an inverse bound. Inserting the Jacobian and (5.2) into (6.1) gives (6.2).

For the last assertion replace $a(w)$ by $(a(w) + a(-w))/2$ in the integral. Every odd Taylor term vanishes. Taylor expansion to any prescribed order, with a sufficiently high differentiated remainder, then yields an expansion in integer powers of d . One can also set $r = \sqrt{d}$ first: the integral is a smooth even function of r with uniform bounds. Successive division of its even Taylor remainder by r^2 shows that it is a smooth function of r^2 on the right. Using two more endpoint derivatives for each division proves the uniform bounds for every fixed derivative of H . $\square$

Proof of Theorem 3.1. In a channel's window $jg_e + d$, each of its j complete flights is at least g_e . Hence the allowed first residual time satisfies $0 < r < d - (W - jg_e) < d < g_*$. The preceding roof cannot truncate this interval, and the next roof cannot fit after the last hit. The first-impact phase formula (6.1) therefore applies to this selected channel at its own threshold: integrating the residual variable gives exactly that interval length. The coercivity (4.4) implies $|y_0|^2 + |y_j|^2 + 2 \sum_{0 < i < j} |y_i|^2 \leq Cd$ on its sublevel. It places the entire active set inside the common Morse coordinates, so no chart-boundary term occurs. Its exact probability is

$$\frac{1}{2\pi A \sinh(j\gamma_e)} \int_{|w|^2 < 2d} (d - |w|^2/2) a_{j,e}(\xi, w) dw.$$

Since the quadratic integral is πd^2 , Lemma 6.1 proves the channel assertion, including its derivative bounds.

For the complete event let $t = jg_* + \varepsilon$ with $0 < \varepsilon < \varepsilon_0$. No $j+2$ hits are possible, since $t < (j+1)g_*$. Lemma 4.1 partitions every contributing j -flight segment into exactly one of the finitely many oriented channels. Its available excess there is $d_{j,e} = \varepsilon - j(g_e - g_*) \leq \varepsilon$. Channels with nonpositive excess contribute zero. Summing the exact channel formulas gives (3.1). The minimum-roof bound proves the negative-side assertion. $\square$

Proof of Theorem A.24.1. Write the mark on a stationary bridge as $w_{j,e} + V_{j,e}$, subtracting the normal value separately at each impact. Chord directions and normal reflections are smooth maps of neighboring transverse coordinates, with uniformly positive chord denominators. Lemma 5.2 therefore gives $|V_{j,e}| \leq C(|u| + |v|)$ and uniform fixed-order bounds for all its derivatives. The normal states now depend on ξ , but each summand's deviation vanishes at $(u, v) = 0$ for every ξ ; parameter differentiation preserves its endpoint localization. The finite chain rule and $\sum_i (\rho^i + \rho^{j-i}) \leq C$ prove the bounds. Multiplying the Morse amplitude by $e^{qV_{j,e}}$ and applying Lemma 6.1 proves (A.24.1).

For the convergence assertion, exact parity counting gives

$$w_{j,e} = (j+1)\bar{f}_e + \mathbf{1}_{\{j \text{ even}\}}(a_{e,0} - a_{e,1})/2. \quad (6.3)$$

For fixed real q the parity factor is bounded above and below by positive constants, as is the normalized integral of $a_{j,e}e^{qV}$ on a sufficiently small collar. Also $\text{csch}(j\gamma_e) \asymp 2e^{-j\gamma_e}$, uniformly in j . Thus a channel's normalized term is comparable to $e^{j(q\bar{f}_e - \gamma_e)}$. Positivity rules out cancellation: equality in (A.24.2) gives a nondecaying term, and a positive exponent gives a growing term. This proves necessity as well as sufficiency, including the right limit at zero. On a compact subset of the strict domain there is a common margin $\delta > 0$. Parameter and source derivatives of the leading factor cost only a polynomial in j ; offset derivatives act on the uniform smooth remainder. Every fixed derivative is bounded by $C(1+j)^m e^{-\delta j}$ and is summable. The vector-source assertion is the same argument with scalar products. $\square$

Corollary 6.2 (Moving threshold interfaces). *Within the collar of (3.1), the probability is a C^1 function of the smooth physical variables (ξ, t) , with smooth pieces separated by $t = jg_e(\xi)$. For an orientation e , along a smooth scalar path $s \mapsto (\xi(s), t(s))$ entering $d_{j,e} > 0$ transversely, its contribution has second-derivative jump from the inactive to the active side*

$$\frac{(t'(s) - j \partial_s g_e)^2}{A(\xi) \sinh(j\gamma_e)}. \quad (6.4)$$

Coincident contributions, including the reversed orientation, add with their respective activation signs. The marked formula has the additional factor $e^{qw_{j,e}}$. One-sided and distributional higher derivatives follow by differentiating the positive-part expression, with polynomial-in- j coefficient bounds.

Proof. Each summand is $d_+^2[C(\xi) + d_+C(\xi)H(\xi, d_+)]$. It and its first derivatives match at zero. Only $2C(\partial_s d)^2$ survives in the second-derivative jump. All other terms contain a factor d . Finite addition proves the claim at intersections. For higher derivatives the ordinary product and chain rules on each side, and the derivatives of d_+^2 in the distributional sense, give the stated formulas; no differentiation of g_* is used. $\square$

In the triangular one-obstacle specialization, fix $R_* \in (0, 1/2)$. For a concrete change of minimizing mechanism, take $h_s(\theta) = R_* + s \cos(2\theta)$. Differentiating the closest distance at the circular normal chord gives $g'_e(0) = -2 \cos(2 \arg e)$: the endpoint variation is the normal support variation, since the unperturbed endpoint gradient is zero. The horizontal pair has derivative -2 and the other two pairs derivative 1 . Reflection symmetry keeps those two lengths equal. Consequently the horizontal pair minimizes for small positive s , while the other two minimize for small negative s . Formula (3.1) continues through $s = 0$. A nonminimal pair contributes exactly while $j(g_e - g_*) < \varepsilon$; the parameter transition scale is therefore ε/j at a transverse length crossing. This assertion is about activation of physical channels, not a new family of symbolic itineraries.

For the conditional record, put $w = \sqrt{2d}z$ and $r = d\eta$. The fixed latent domain and its probability measure are

$$\mathcal{U} = \{(z, \eta) : |z| < 1, 0 < \eta < 1 - |z|^2\}, \quad \text{dm} = \frac{2}{\pi} dz d\eta. \quad (6.5)$$

This notation will also be used for the general record response.

7. TWO NONLINEAR BOUNDARY LAYERS

Fix one oriented channel and suppress its index and the smooth geometric parameter. All bounds below include each fixed number of parameter derivatives on the compact families

specified in Section 2. Shrinking the common contact box is allowed once, independently of the flight number. The functions $E_j = W_j - jg$ and $b_j = -W_{j,uv}/d_j^0$ refer to the actual nonlinear stationary action and its normalized twist, not their quadratic models.

7.1. The half-line construction. For a starting type $b \in \{0, 1\}$ put $\sigma_i^{(b)} = \sqrt{c_{1-((b+i) \bmod 2)}}$. The quadratic half-line interior Hessian acts on sequences indexed by $i \geq 1$, with fixed zero boundary at $i = 0$. Its diagonal entries are $2c_{(b+i) \bmod 2}/g$ and its adjacent entries are $-1/g$. Its inverse is

$$G_{ik}^{(b)} = \frac{g\sigma_i^{(b)}\sigma_k^{(b)}}{2c\sinh\gamma} \left(e^{-\gamma|i-k|} - e^{-\gamma(i+k)} \right), \quad i, k \geq 1. \quad (7.1)$$

The formula follows either from the recurrence and its unit jump or from (5.5) by sending the right endpoint to infinity. It is a bounded positive operator on $\ell^2(\mathbb{N})$, and is bounded on $\sup_i \rho^{-i}|x_i|$ whenever $e^{-\gamma} < \rho < 1$. Every fixed parameter derivative obeys the same assertion after leaving a strict exponential margin.

Lemma 7.1 (Semi-infinite stationary segments). *For $|u| < u_*$ there is a unique small stationary half-line segment $x^{(b)}(u) = (x_i^{(b)}(u))_{i \geq 0}$ with $x_0^{(b)} = u$ and $x_i^{(b)} \rightarrow 0$. It is smooth, and on a common neighborhood*

$$\begin{aligned} x_i^{(b)}(u) &= \frac{\sigma_i^{(b)}}{\sigma_0^{(b)}} e^{-i\gamma} u + O(u^2 \rho^i), \\ |\partial_\xi^\alpha \partial_u^r x_i^{(b)}(u)| &\leq C_{\alpha,r} \rho^i |u|^{\max(1-r,0)}. \end{aligned} \quad (7.2)$$

The boundary action

$$S_b(u) = \sum_{i \geq 0} \left[\ell_{(b+i) \bmod 2}(x_i^{(b)}(u), x_{i+1}^{(b)}(u)) - g \right] \quad (7.3)$$

is smooth and strictly convex near zero. It satisfies

$$S_b(0) = S'_b(0) = 0, \quad a_b := S''_b(0) = \frac{c \sinh \gamma}{g c_{1-b}} > 0. \quad (7.4)$$

These assertions are analytic on a common smaller neighborhood for jointly analytic graph families.

Proof. Use the half-line Green operator in the weighted contraction from Lemma 5.2. The linear solution is the first term of (7.2). The nonlinear gradient is local, vanishes quadratically, and its derivative has norm $O(|u|)$ on the weighted ball of radius $C|u|$. Thus the map has contraction constant at most one half on a common ball. Differentiating its fixed-point equation leaves the same invertible operator on the highest derivative; every other term is a product of a local derivative and weighted lower ones. The convolution estimate used in (5.6) proves (7.2) by induction.

Uniqueness also holds among all sufficiently small stationary sequences tending to zero, not just among elements of that weighted ball. Their difference solves a homogeneous Jacobi equation whose averaged Hessian is strictly diagonally dominant. Its inverse on bounded sequences is bounded by the reciprocal dominance margin: write the Hessian as its positive diagonal times $I - T$, with $\|T\|_\infty < 1$. With zero left boundary the difference is zero. This argument applies to every fixed small contact box.

Every summand of (7.3) is $O(u^2 \rho^{2i})$. Its differentiated sums converge by the same weighted bounds. Interior stationarity cancels the first-variation terms, leaving only the left endpoint derivative; the boundary term at a truncation tends to zero. At quadratic order this derivative is $(c_b u - x_1^{(b), \text{lin}})/g$. Since $c_b \sigma_0^{(b)}/\sigma_1^{(b)} = c$, it equals $a_b u$. This proves (7.4), hence strict convexity.

The uniform complex contraction and Cauchy estimates, as in Lemma 5.2, prove the analytic assertion. $\square$

Let $H_b(u)$ be the half-line interior Hessian evaluated on this segment, and write $\Delta H_b(u) = H_b(u) - H_b(0)$. It is tridiagonal. Locality and (7.2) imply

$$\|\Delta H_b(u)\|_1 \leq C|u|, \quad \|\partial_\xi^\alpha \partial_u^r \Delta H_b(u)\|_1 \leq C_{\alpha,r}. \quad (7.5)$$

Here and below $\|\cdot\|_1$ on operators is the trace norm. Indeed, the sum of absolute entries bounds that norm, and the entries are bounded by sums of neighboring endpoint weights. Define

$$U_b(u) = \sum_{i \geq 0} \log \left(g[-\ell_{(b+i) \bmod 2, uv}(x_i^{(b)}, x_{i+1}^{(b)})] \right), \quad (7.6)$$

$$B_b(u) = \exp \left\{ U_b(u) - \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} \operatorname{tr}[(G^{(b)} \Delta H_b(u))^m] \right\}. \quad (7.7)$$

This also defines the Fredholm determinant in the denominator of B_b : no additional determinant normalization is implicit. The series converges absolutely after requiring $\|G^{(b)} \Delta H_b\| < 1/2$. Each trace is bounded by one trace norm times $m-1$ operator norms. The edge sum converges by endpoint localization. Consequently $B_b > 0$, $B_b(0) = 1$, and every fixed mixed derivative is uniformly bounded. Differentiation of the trace series retains a trace-class factor, so its bound does not involve the number of sites of a truncation.

7.2. Relative factorization.

Theorem 7.2 (Nonlinear two-boundary factorization). *There are a common endpoint box and $\tau \in (0, 1)$ such that for every fixed k ,*

$$\|E_j - S_0 \oplus S_p\|_{C^k} + \|b_j - B_0 \otimes B_p\|_{C^k} \leq C_k \tau^j, \quad p = j \bmod 2. \quad (7.8)$$

The norms include mixed geometric-parameter and endpoint derivatives; $S_0 \oplus S_p$ means $S_0(u) + S_p(v)$ and $B_0 \otimes B_p$ means $B_0(u)B_p(v)$. The action difference has zero value and first endpoint differential at zero, and therefore is $O(\tau^j(|u| + |v|)^2)$. The amplitude difference vanishes at zero. The fixed-order constants, but not j , may depend on higher boundary norms. Both limits are analytic for analytic families.

Proof. We give the gluing and determinant estimates separately. Reverse a half-line segment starting at type p at the right endpoint. Its i th coordinate in a length- j indexing is $x_{j-i}^{(p)}(v)$; the reversal is consistent because the flight action is symmetric under reversal together with interchange of endpoint types. For $0 < i < j$ put

$$z_i = x_i^{(0)}(u) + x_{j-i}^{(p)}(v), \quad z_0 = u, \quad z_j = v.$$

The endpoint overwrites cost $O(\rho^j)$ in every fixed differentiated norm. In each interior stationarity equation the linear terms cancel. The remaining interaction is a sum of products of left and right coordinates or their neighboring values. Its size is bounded by $C\rho^i \rho^{j-i} = C\rho^j$. Thus, for the interior residual $r(z)$,

$$\|\partial^\alpha r(z)\|_{\ell^1} \leq C_\alpha(1+j)\rho^j. \quad (7.9)$$

The assertion for derivatives follows from the local chain rule; parameter derivatives of the separated linear exponentials introduce only fixed powers of indices, absorbed by a strict exponential margin.

Let y be the finite stationary segment already constructed in Lemma 5.2. The identity

$$r(z) - r(y) = \left[\int_0^1 H_{\text{int}}(y + t(z - y)) \, dt \right] (z - y)$$

has a symmetric strictly diagonally dominant matrix with a uniform positive margin. Its inverse has bounded ℓ^∞ norm by diagonal splitting and a geometric series; symmetry gives the same ℓ^1 bound. It follows that $\|y - z\|_{\ell^1} \leq C(1 + j)\rho^j$. Differentiating this identity leaves the same uniformly invertible operator on the highest difference derivative. All other terms contain lower difference derivatives and uniformly bounded local coefficients. Induction proves the corresponding ℓ^1 estimate at every fixed order, with a polynomial in j multiplying an exponential. Fixing a slower exponential absorbs every such fixed polynomial.

For the action, subtract the two half-line sums. The missing tails are exponentially small. On overlapping edges the difference between the value at the sum and the sum of the two values minus g is a local interaction bounded by $C\rho^j$. The replacement of z by y is controlled by its ℓ^1 error and bounded local derivatives. The same argument after differentiation proves the first term of (7.8). Both finite and limiting actions have zero value and gradient at zero; Taylor's integral remainder yields the additional quadratic vanishing.

For the normalized twist, the finite cofactor formula gives

$$\log b_j = \sum_{i=0}^{j-1} \log(g[-\ell_{i \bmod 2, uv}(y_i, y_{i+1})]) - \log \det(I + G_j \Delta H_j), \quad (7.10)$$

where $G_j = (H_{\text{int}}^0)^{-1}$ and $\Delta H_j = H_{\text{int}} - H_{\text{int}}^0$. The edge sum converges exponentially to $U_0(u) + U_p(v)$ by exactly the preceding local interaction and tail estimates. It remains to control the determinant relatively.

Here are the needed finite-dimensional-to-half-line details. For $j \geq 6$ choose $r = \lfloor j/3 \rfloor$ and retain the first and last r interior sites. Set to zero all perturbation entries outside their two diagonal blocks, including crossing entries. The trace-norm error is bounded by

$$C_k(1 + j)^{M_k} \rho^r \quad (7.11)$$

after k fixed derivatives. This uses tridiagonality and the decay of the two endpoint layers, not a dimension times operator-norm estimate. Within each retained block, replacing the finite segment by its own half-line segment has an exponentially small entrywise sum, by (7.9) and the opposite tail.

For a perturbation supported in these blocks, the identity $\det(I + AB) = \det(I + BA)$ reduces the determinant to that support. The left compression of G_j differs from the compression of $G^{(0)}$ by an exponentially small operator norm. The reversed right compression differs in the same way from $G^{(p)}$. To check the claim, use (5.5): after expanding hyperbolic sines into exponentials, the reflection from the remote boundary has a factor $e^{-\gamma(2j-i-k)}$ for $i, k \leq r$. The two off-diagonal block kernels have a factor $e^{-\gamma(j-2r)}$. Summing at most r^2 entries and differentiating any fixed number of times only adds a polynomial factor. Consequently the compressed Green matrix differs exponentially from the direct sum of the two compressed half-line operators. The same estimate holds after multiplication by the bounded trace-norm perturbation.

For completeness, if T, T' are the resulting trace-class operators with norms at most $q < 1$, telescoping powers gives

$$|\text{tr}(T^m - (T')^m)| \leq mq^{m-1} \|T - T'\|_1.$$

Summation with coefficient $1/m$ proves local Lipschitz continuity of the logarithmic determinant in trace norm. Its differentiated series has the same property with a fixed polynomial factor in m , still summable. This justifies each truncation and comparison above, for all fixed mixed derivatives. Sending the block size to infinity at either end costs $O(\rho^r)$ again. The direct-sum logarithm is the sum of the half-line logarithms in (7.7).

We have proved exponential convergence of (7.10) to $\log B_0(u) + \log B_p(v)$. Exponentiation preserves the bounds, since all logarithms range over one bounded interval. Choose $\tau < 1$ slower

than the finitely many exponential rates in the gluing, truncation, and off-diagonal estimates. Polynomial factors at each fixed derivative order are absorbed in C_k . The finitely many $j < 6$ are absorbed by enlarging those constants. At zero the finite and limiting normalized twists are one, which proves the last vanishing assertion. Uniform complex versions of the same estimates prove analyticity. In particular the argument never divides an absolute action error by the exponentially small reference twist. $\square$

7.3. Physical limiting laws. For $p \in \{0, 1\}$ let $q_p = \sqrt{a_0 a_p}$ and define, for $d > 0$,

$$\mathcal{F}_p(d) = \frac{q_p}{\pi d^2} \int_{\mathbb{R}^2} (d - S_0(u) - S_p(v))_+ B_0(u) B_p(v) \, du \, dv. \quad (7.12)$$

The integral is restricted to the common contact box; for the collars below its support lies strictly inside that box, so the displayed extension by zero is unambiguous.

Theorem 7.3 (Fixed-offset long-bridge law). *There is $d_* > 0$ independent of j such that $\mathcal{F}_p$ has a smooth right extension to $[0, d_*]$, with $\mathcal{F}_p(0) = 1$ and $\mathcal{F}_p > 0$. On this interval,*

$$\left\| \frac{2A \sinh(j\gamma)}{d^2} \Pr(E_{j,e}(d)) - \mathcal{F}_p(d) \right\|_{C^k} \leq C_k \tau^j, \quad p = j \bmod 2, \quad (7.13)$$

for every fixed mixed parameter and offset derivative order. The conditional endpoint and first-residual-time law converges to

$$\frac{B_0(u) B_p(v)}{I_p(d)} \mathbf{1}_{\{r > 0, S_0(u) + S_p(v) + r < d\}} \, du \, dv \, dr, \quad I_p(d) = \int (d - S_0 - S_p)_+ B_0 B_p \, du \, dv. \quad (7.14)$$

For the scaled variables $(u/\sqrt{d}, v/\sqrt{d}, r/d)$ the convergence has bounded-Lipschitz error at most $C\tau^j$, uniformly for $0 < d \leq d_*$. Any fixed number of collisions at the two ends may be included, with their limiting coordinates given by the corresponding half-line segments. The two ends are coupled by the common residual energy constraint; (7.14) is not a product probability.

Proof. At parity p , (5.2) and (7.4) give $d_j^0 = q_p / \sinh(j\gamma)$. Full-phase integration therefore gives exactly

$$F_j(d) = \frac{q_p}{\pi d^2} \int (d - E_j)_+ b_j \, du \, dv.$$

To compare this integral down to $d = 0$, use the constructive Morse maps of Lemma 6.1 for both E_j and $S_0 \oplus S_p$. Their quadratic matrices are uniformly positive. The matrix $2 \int_0^1 (1-s) D^2 E(sy) \, ds$, its positive square root and the inverse coordinate map depend locally Lipschitz-continuously in each fixed C^k norm on E , at the cost of finitely many additional derivatives. This follows by differentiating the defining inverse equation and using the uniform inverse differential; the positive-square-root map has bounded derivatives on the common compact positive cone. Thus their inverse maps and transformed amplitudes differ by $C_k \tau^j$. The inverse maps both vanish at zero, so their difference at w is also $O(\tau^j |w|)$.

Put $w = \sqrt{2d} z$. The normalized integrals are over the same unit disk with weight $1 - |z|^2$. Their odd Taylor terms vanish. Taylor's integral remainder, to two additional endpoint orders for each requested offset derivative, bounds the difference of the smooth functions of d by $C_k \tau^j$. This proves (7.13) including its derivatives at zero; it is not differentiation of a moving sharp boundary without justification. The quadratic integral at zero is $\pi d^2 / q_p$, proving $\mathcal{F}_p(0) = 1$. Positivity and a smaller common collar give a uniform positive lower bound.

For conditional laws retain $r = d\eta$ and the fixed domain $\mathcal{U}$ in (6.5). The normalized densities there differ in L^1 by $C\tau^j$, since the denominators are uniformly positive. The two endpoint maps, after division by $\sqrt{d}$, differ uniformly by $C\tau^j$. Couple the common part of the latent densities and use these maps; the remaining mass is $O(\tau^j)$ on a bounded scaled domain. This

proves the bounded-Lipschitz estimate. A fixed number of end coordinates follows from the differentiated gluing estimate and its vanishing at zero. First variations give directions from neighboring coordinates when desired. The explicit density (7.14) follows by returning to (u, v, r) coordinates. $\square$

Corollary 7.4 (Nonlinear weights at a tied onset). *Suppose a fixed finite set of selected ground channels has the same gap. Let $\gamma_{\min}$ be the least of their exponents. At a fixed $0 < d \leq d_*$, along a parity subsequence their conditional weights in the complete maximal-count event converge to*

$$\frac{\mathbf{1}_{\{\gamma_e = \gamma_{\min}\}} \mathcal{F}_{e,p}(d)}{\sum_{e': \gamma_{e'} = \gamma_{\min}} \mathcal{F}_{e',p}(d)}.$$

The convergence is exponential. At $d = 0$ the minimizing leading weights are equal, whereas at positive offset the boundary laws supply their corrections.

Proof. Divide the finite sum in (3.1) by $d^2 e^{-j\gamma_{\min}}/A$. Theorem 7.3 and $\operatorname{csch}(j\gamma) = 2e^{-j\gamma}/(1 - e^{-2j\gamma})$ give the limit. All limiting denominators are positive. A finite positive gap between distinct exponents and the factorization rate give an exponential error; if all exponents tie only the latter rate is needed. $\square$

Corollary 7.5 (The first poles of the onset series). *For a fixed geometry put $\mathcal{G}(z, d) = \sum_{j \geq 1} z^j \operatorname{Pr}(E_{j,e}(d))/d^2$. There is $\nu > 0$ such that, with $x = ze^{-\gamma}$,*

$$\mathcal{G}(z, d) = \frac{\mathcal{F}_1(d)x + \mathcal{F}_0(d)x^2}{A(1 - x^2)} + \mathcal{H}(z, d), \quad (7.15)$$

where $\mathcal{H}$ is holomorphic for $|z| < e^{\gamma+\nu}$ and smooth in $d \in [0, d_]$. Every fixed geometric and offset derivative has the same assertion on strictly smaller common domains. The positive singularity is a simple pole at $z = e^\gamma$, with residue $-e^\gamma[\mathcal{F}_0(d) + \mathcal{F}_1(d)]/(2A)$. The possible pole at $-e^\gamma$ is removable exactly when $\mathcal{F}_0(d) = \mathcal{F}_1(d)$.*

Proof. The coefficient in question is

$$\frac{e^{-j\gamma}}{A(1 - e^{-2j\gamma})} F_j(d) = \frac{e^{-j\gamma}}{A} \mathcal{F}_{j \bmod 2}(d) + O(e^{-(\gamma+\nu)j}),$$

for any positive ν strictly smaller than $-\log \tau$ and 2γ . Fixed parameter derivatives add only fixed powers of j , absorbed by a smaller strict margin. The error series is normally convergent, as are its stated derivatives. Summing its even and odd leading terms gives (7.15). Positivity gives the nonzero positive residue. At the negative point the numerator is $\mathcal{F}_0 - \mathcal{F}_1$, proving the removability criterion. $\square$

The series in (7.15) uses a different physical window $jd + d$ at each coefficient. It is not the generating function of all itineraries in one observation window. Its continuation is a consequence of a nonlinear relative-flux limit for these specified events.

7.4. Differentiated operator bounds and moving poles. For use in differentiated forms of the relative factorization, the operator estimates can be recorded in a single common norm convention. Let G denote either a finite reference Green matrix or a half-line Green operator, and let ΔH be the corresponding tridiagonal Hessian perturbation on its stationary segment. For every fixed k ,

$$\max_{|\alpha| \leq k} \|\partial_\xi^\alpha G\|_{\text{op}} \leq C_k, \quad \max_{|\alpha| + |\beta| \leq k} \|\partial_\xi^\alpha \partial_Y^\beta \Delta H\|_1 \leq C_k, \quad \|G \Delta H\|_{\text{op}} \leq q < 1. \quad (7.16)$$

Here Y is the finite endpoint vector or the single half-line endpoint. There is no assertion that derivatives of $G \Delta H$ are themselves small. They are bounded in trace norm, whereas its zeroth-order operator norm is small on the fixed contact box.

For completeness, the first bound follows directly from the Green kernels: their parameter derivatives introduce fixed powers of the indices into exponentially summable kernels, absorbed by the strict exponential margin. The Schur row and column estimates give the operator bound independently of matrix size. For the second, each tridiagonal entry is a local smooth function of neighboring stationary coordinates. The endpoint localization and its differentiated bounds make the sum of its absolute entries bounded. This sum bounds the trace norm. The last inequality follows by choosing the common box once, using $\Delta H(0) = 0$. The product rule consequently gives trace-norm bounds for every fixed derivative of $T = G\Delta H$.

These estimates also explain the differentiated determinant comparison. Differentiating $\text{tr}(T^n)$ a fixed k times produces at most $C_k n^k$ products with at most k differentiated factors. One factor is bounded in trace norm and all others in operator norm. The undifferentiated factors contribute the geometric decay q^{n-k} ; the finitely many $n < k$ are harmless. In a difference of two such expressions, telescope one factor at a time. The differentiated trace-norm differences from the block comparison in the proof of Theorem 7.2 supply $C_k \tau^j$, and summation in n is uniform. Thus differentiated logarithmic determinants retain a trace-class factor and never acquire a factor equal to the number of interior sites.

In Corollary 7.5, simple poles refer to the undifferentiated series at a fixed geometry. If a parameter moves γ , differentiation of its displayed rational principal part can increase the pole order. After k parameter derivatives it is at most $k + 1$. For example, $\partial_\xi(1 - ze^{-\gamma})^{-1} = -ze^{-\gamma}(\partial_\xi \gamma)/(1 - ze^{-\gamma})^2$. The differentiated remainder remains holomorphic on the stated strictly smaller common domains. This separates the regularity of the remainder from the order of the moving principal poles.

8. FROM JACOBI IDENTITIES TO A RELATIVE PHYSICAL LAW

The proof of the relative law uses three different types of information. The first is finite-dimensional algebra, the second is localization of nonlinear perturbations, and the third is stability of a physical sublevel integral. We separate these statements before deriving a nonlinear compatibility law from their combination.

8.1. The classical operator input. For a periodic discrete Lagrangian orbit, the discrete Hill formula of Bolotin and Treschev [2, Theorem 2.1] relates $\det(P - I)$ to the periodic action Hessian divided by the product of mixed edge determinants. The Hessian in our endpoint problem instead has Dirichlet boundary conditions. Its relevant scalar identity is the following elementary Schur-complement formula. We include its proof to specify the boundary convention, not as a new determinant identity.

Proposition 8.1 (Dirichlet cofactor and normalization). *Let $L(x_0, \dots, x_j) = \sum_{i=0}^{j-1} \ell_i(x_i, x_{i+1})$ have a smooth stationary interior at fixed endpoints $x_0 = u, x_j = v$. Suppose its interior Hessian H is positive definite and tridiagonal, with mixed edge derivatives $\ell_{i,12} = -J_i < 0$. For the stationary value $W(u, v)$,*

$$-W_{uv} = \frac{\prod_{i=0}^{j-1} J_i}{\det H}. \quad (8.1)$$

For a reference interior Hessian H^0 and reference edges J_i^0 , put $G^0 = (H^0)^{-1}$ and $\Delta H = H - H^0$. Then

$$\frac{-W_{uv}}{(-W_{uv})^0} = \frac{\prod_i (J_i/J_i^0)}{\det(I + G^0 \Delta H)}. \quad (8.2)$$

If $H = D(I - T)$, D is diagonal with $D_{ii} \geq d_0 > 0$, T has only its two adjacent diagonals nonzero, and $\|T\|_{\ell^\infty \rightarrow \ell^\infty} \leq q < 1$, then

$$|(H^{-1})_{ik}| \leq \frac{q^{|i-k|}}{d_0(1-q)}. \quad (8.3)$$

These estimates have constants independent of the number of sites.

Proof. For $j \geq 2$, differentiating the stationary equations eliminates the interior and gives $W_{uv} = -J_0 J_{j-1} (H^{-1})_{1,j-1}$. The corner cofactor of a tridiagonal matrix is $J_1 \cdots J_{j-2}$, so Cramer's rule proves (8.1). At $j = 1$ the interior determinant is the empty determinant, equal to one, and the same assertion is immediate. Taking the ratio at the reference point proves (8.2). Finally, $H^{-1} = \sum_{n \geq 0} T^n D^{-1}$. The (i, k) entry of T^n vanishes for $n < |i - k|$, while every entry has absolute value at most q^n . Summing the remaining series proves (8.3). $\square$

The separated inverse structure is classical: Meurant's Theorem 2.3 [15] gives inverse entries by two scalar recurrences, and his Theorem 2.4 describes their monotone decrease under strict diagonal dominance. Nabben [16] studies quantitative inverse decay for nonsymmetric tridiagonal and band matrices. Our use of a weighted Green operator belongs to this operator framework. In particular neither the corner cofactor nor exponential inverse decay is claimed here as a new mechanism.

What must additionally be proved for the present probability problem is summability of the *nonlinear perturbation evaluated on the entire stationary bridge*. A bound on each matrix entry, without its location, would not give the required determinant limit. The relevant continuity statement is recorded next.

Proposition 8.2 (A dimension-independent relative determinant bound). *For trace-class operators T, T' on a common Hilbert space with $\|T\|, \|T'\| \leq q < 1$, define*

$$\mathcal{L}(T) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \operatorname{tr}(T^n).$$

Then

$$|\mathcal{L}(T) - \mathcal{L}(T')| \leq (1 - q)^{-1} \|T - T'\|_1. \quad (8.4)$$

Suppose T, T' depend smoothly on finitely many parameters, their trace-class derivatives through order k are uniformly bounded, and the corresponding derivatives of $T - T'$ have trace norm at most ϵ . Then

$$\|\mathcal{L}(T) - \mathcal{L}(T')\|_{C^k} \leq C_k \epsilon.$$

The constants depend on q , the parameter dimension and the stated bounds, not on the dimension of a finite-dimensional truncation.

Proof. The algebraic telescoping formula for $T^n - (T')^n$ and the trace ideal inequality give $|\operatorname{tr}(T^n - (T')^n)| \leq nq^{n-1} \|T - T'\|_1$. Division by n and summation prove the first assertion. After k parameter derivatives, each product has at most k differentiated factors. In the difference of the two differentiated products, telescope the factors once more and place the difference factor in trace norm. Every other differentiated factor has a bounded operator norm. All but at most $k + 1$ of the factors still have norm at most q . There are at most $C_k n^{k+1}$ terms. The finitely many products with $n \leq k + 1$ are bounded separately; the other terms are bounded by $C_k \epsilon n^{k+1} q^{n-k-1}$. Their sum, including $1/n$, converges. The same estimates justify differentiating the original series by uniform convergence. No smallness of a differentiated operator has been used. $\square$

In the billiard application, Lemma 5.2 gives $|x_i| \leq C(|u|\rho^i + |v|\rho^{j-i})$, with the corresponding localized derivative bounds. Since a Hessian entry depends on only neighboring coordinates, its

perturbation has a uniformly summable entrywise absolute value. Matrix units have trace norm one; hence

$$\|\Delta H_j\|_1 \leq C(|u| + |v|), \quad \|\partial^\alpha \Delta H_j\|_1 \leq C_\alpha.$$

The two-block construction in Theorem 7.2 then compares the normalized finite determinant with the direct sum of the two half-line determinants. Its trace-norm error tends to zero exponentially. Proposition 8.2 converts that specific geometric estimate into convergence of the logarithmic amplitude. Inverse decay supplies access to the blocks; the localized perturbation estimate supplies their summable determinant error.

8.2. The passage through physical integration. A second, logically separate requirement is uniformity as the offset vanishes. The following statement isolates this passage. All integrals are over a fixed neighborhood containing the indicated sublevel sets. The parameter set is compact and finite dimensional.

Proposition 8.3 (Stability of normalized sublevel integration). *Let $E, \tilde{E}$ be smooth functions of $y \in \mathbb{R}^2$ and of parameters, with value and y -gradient zero at $y = 0$. Assume their Hessians at zero are uniformly positive and their higher derivatives are uniformly bounded on a common neighborhood. Let $a, \tilde{a}$ be smooth amplitudes with uniform derivative bounds. There is a common $d_* > 0$ such that*

$$\mathcal{J}(E, a)(d) = d^{-2} \int (d - E(y))_+ a(y) \, dy$$

extends smoothly from $d > 0$ to $[0, d_]$. For each fixed k , bounds through a sufficiently high finite order $r(k)$ give*

$$\|\mathcal{J}(E, a) - \mathcal{J}(\tilde{E}, \tilde{a})\|_{C^k} \leq C_k(\|E - \tilde{E}\|_{C^{r(k)}} + \|a - \tilde{a}\|_{C^{r(k)}}). \quad (8.5)$$

The norms on the left include mixed parameter and offset derivatives.

Proof. Taylor's integral formula gives $E(y) = y^t M_E(y) y / 2$, with $M_E(y) = 2 \int_0^1 (1-s) D^2 E(sy) \, ds$. After decreasing the common neighborhood, the maps $y \mapsto M_E(y)^{1/2} y$ and their counterparts for $\tilde{E}$ have uniformly invertible differentials. Their inverses Φ_E are obtained by the contraction used in Lemma 6.1. Subtracting the inverse equations gives their Lipschitz dependence in C^0 on E ; differentiating and isolating the highest derivative on the same uniformly invertible differential proves the corresponding finite-order estimates by induction. Smoothness of the positive matrix square root is used only on a compact positive-definite set.

Set $A_E(w) = a(\Phi_E(w)) |\det D\Phi_E(w)|$. The change of variables $w = \sqrt{2d} z$ gives the fixed-domain formula

$$\mathcal{J}(E, a)(d) = 2 \int_{|z| < 1} (1 - |z|^2) A_E(\sqrt{2d} z) \, dz. \quad (8.6)$$

The inverse-map estimates bound $A_E - A_{\tilde{E}}$ in each fixed C^s norm by a sufficiently high finite norm of the two input differences. The integral in (8.6), as a function of $t = \sqrt{2d}$, is smooth and even. A smooth even function $h(t)$ has a smooth right representation in t^2 with its first k derivatives controlled by derivatives of h through order $2k$. For example $h'(t)/t = \int_0^1 h''(st) \, ds$; iteration, applied also to the even function h'' , proves the assertion and its difference estimate. Parameter derivatives commute with these integrals. Applying it to (8.6) proves the smooth extension and (8.5). All sublevel sets lie inside the common chart by the uniform quadratic lower bound, so there is no additional chart-boundary term. $\square$

Apply this proposition to (E_j, b_j) and $(S_0 \oplus S_p, B_0 \otimes B_p)$. The exact physical multiplier is $q_p / (2\pi A \sinh(j\gamma))$, by (6.1). Thus the exponentially small flux is retained *outside* a uniformly stable normalized integral. In this form the proof yields all fixed-order derivatives on a closed

collar and the rare-event-preserving experiment transfer, rather than only an identity at the normal orbit.

The operator identities above, the trace estimate and the smooth integration principle are reusable elementary ingredients. The billiard-specific work is to verify their simultaneous hypotheses for physical stationary records with arbitrary collision order. The resulting law has, in addition to the experiment comparison in Part II, the nonlinear parity consequence established below. This consequence concerns the complete boundary functions at positive offset and does not follow from a finite list of leading amplitudes.

9. HYPERBOLIC COORDINATES AND THE INTRINSIC BOUNDARY MEASURE

The relative product law has a classical analytic model. We first calculate that model in physical endpoint coordinates, and then identify the boundary amplitudes intrinsically for the smooth billiards of Theorem 2.1. These are complementary statements: the first assumes normalizing charts; the second constructs a one-dimensional scalar density with the amplitude of the already proved half-line law. The scalar coordinate itself is classical, as recalled in Lemma 9.2. No smooth two-dimensional normal-form theorem is used in this identification.

9.1. The analytic mixed-boundary calculation. De Simoi–Kaloshin–Leguil [5], in the normal-form discussion on printed page 10 of arXiv:1905.00890v4, recall the analytic symplectic normal form near a hyperbolic period-two orbit. The local existence statement is distinct from the symmetry assumptions in their geometric inverse theorem; see also their discussion on printed page 13. The following physical endpoint calculation was supplied in the independent memorandum [24]. We include its proof and normalizations. The proposition is conditional on the indicated charts, and is not attributed as a probability theorem to [5].

Proposition 9.1 (Relative flux in analytic hyperbolic coordinates). *Let*

$$N(s, p) = (\Delta(sp)s, \Delta(sp)^{-1}p), \quad \Delta(0) = \lambda \in (0, 1), \quad (9.1)$$

where Δ is positive and analytic near zero. Let

$$\mathcal{C}_L(s, p) = (U(s, p), \mathcal{P}(s, p)), \quad \mathcal{C}_R(q, t) = (V(q, t), \mathcal{Q}(q, t))$$

be analytic canonical charts sending zero to zero, with $U_s(0, 0)V_t(0, 0) \neq 0$. For the local branch of $\mathcal{C}_R \circ N^n \circ \mathcal{C}_L^{-1}$, prescribe the initial physical position u and terminal physical position v . There are a fixed box in (u, v) , an integer n_0 , and $\sigma \in (0, 1)$ such that this boundary problem has a unique local solution for all $n \geq n_0$. Its physical mixed flux $J_n = |\partial P / \partial v|_u$ satisfies

$$\left\| \frac{J_n(u, v)}{J_n(0, 0)} - \mathcal{B}_L(u)\mathcal{B}_R(v) \right\|_{C^k} \leq C_k \sigma^n \quad (9.2)$$

for every fixed k , where, writing $U(\alpha(u), 0) = u$ and $V(0, \beta(v)) = v$,

$$\mathcal{B}_L(u) = \frac{U_s(0, 0)}{U_s(\alpha(u), 0)}, \quad \mathcal{B}_R(v) = \frac{V_t(0, 0)}{V_t(0, \beta(v))}. \quad (9.3)$$

These functions are positive and equal one at zero. If the physical branch has a generating action W_n , normalized by $P = -W_{n,u}$ and $Q = W_{n,v}$, then there are functions $\mathcal{S}_L, \mathcal{S}_R$, vanishing with their first derivatives at zero, for which

$$\|W_n - W_n(0, 0) - \mathcal{S}_L \oplus \mathcal{S}_R\|_{C^k} \leq C_k \sigma^n. \quad (9.4)$$

All assertions hold with mixed parameter derivatives for a supplied jointly smooth family of these functions and charts on common domains, provided their fixed-order norms are uniformly bounded, their projection derivatives are uniformly nonzero, and $0 < \lambda_- \leq \lambda \leq \lambda_+ < 1$.

Proof. Temporarily prescribe $s = s_0$ and $t = p_n$ rather than u, v . The invariant $I = sp$ and the orbit endpoints obey

$$I = st\Delta(I)^n, \quad p_0 = t\Delta(I)^n, \quad q = s_n = s\Delta(I)^n. \quad (9.5)$$

For a single map put $\lambda_+ = \lambda$. Choose $\lambda_+ < \mu < \sigma < 1$ and decrease the I interval so that $\Delta \leq \mu$. On a fixed sufficiently small (s, t) box, $|st|n\mu^{n-1}\|\Delta'\|_\infty < 1/2$ for every $n \geq 1$, since $\sup_n n\mu^{n-1} < \infty$. The right side of the first identity in (9.5) maps a common small closed I interval into itself. It is a contraction there. Its solution satisfies $|I| \leq C\mu^n$. The intermediate coordinates are $(s\Delta(I)^i, t\Delta(I)^{n-i})$, so the whole orbit remains inside the chosen local chart.

Set

$$a_n = \Delta(I)^n, \quad D_n = 1 - nI \frac{\Delta'(I)}{\Delta(I)}.$$

The contraction estimate gives $|D_n| \geq 1/2$. Differentiating (9.5), without dividing by either endpoint, gives

$$T_n := \left. \frac{\partial p_0}{\partial t} \right|_s = \left. \frac{\partial q}{\partial s} \right|_t = \frac{\Delta(I)^n}{D_n}. \quad (9.6)$$

For example, $I_t = sa_n/D_n$ and $(p_0)_t = a_n + tn\Delta(I)^{n-1}\Delta'(I)I_t = a_n/D_n$. This also proves the formula on both axes.

Here are the differentiated relative estimates. At every fixed order, derivatives of the function $\Delta(I)^n$ with I treated as an independent variable are bounded by $C_k(1+n)^k\mu^n$; a finite number of small n are absorbed in C_k . Implicit differentiation of the first equation in (9.5) leaves D_n on the highest derivative of I . Its remaining terms contain the preceding derivatives of I , derivatives of st , and derivatives of Δ^n . Induction therefore gives

$$\|I\|_{C^k} \leq C_k(1+n)^{m_k}\mu^n.$$

These norms can include the supplied parameters. Since $\log(\Delta(I)/\lambda) = Ih(I)$ with a jointly smooth bounded function h , the same induction gives

$$\left\| n \log \frac{\Delta(I)}{\lambda} \right\|_{C^k} + \|D_n - 1\|_{C^k} \leq C_k(1+n)^{r_k}\mu^n.$$

The identity at $I = 0$ holds throughout the parameter family; thus parameter derivatives of λ do not leave an unsuppressed term in this normalized expression. Exponentiating and using $|D_n| \geq 1/2$ proves

$$\|\lambda^{-n}T_n - 1\|_{C^k} \leq C_k\sigma^n. \quad (9.7)$$

One fixed strict margin $\mu < \sigma$ absorbs the polynomial at each fixed order; the constants, not the margin, depend on that order.

The physical boundary map in the mixed coordinates is

$$\Phi_n(s, t) = (U(s, ta_n), V(sa_n, t)), \quad \Phi_\infty(s, t) = (U(s, 0), V(0, t)). \quad (9.8)$$

It converges to Φ_∞ in every fixed derivative norm at an exponential rate. Choose fixed inner and outer boxes on which the two limiting scalar projections are diffeomorphisms, with the closure of the inner box strictly inside the image of the outer one. Conjugating Φ_n by Φ_∞^{-1} gives the identity plus a uniformly small C^1 perturbation. The equation for its inverse is a contraction on a small ball about each point of the inner box, uniformly in that point. This gives a unique inverse on a common nonshrinking physical box for all $n \geq n_0$. Subtracting the inverse equations gives its C^0 convergence. Differentiation isolates the same uniformly invertible differential and proves the C^k convergence by induction. In particular $(s, t) = \Phi_n^{-1}(u, v)$ tends to $(\alpha(u), \beta(v))$ at the stated rate.

Put $\mathcal{D}_n = \det D\Phi_n$. Canonicity of the initial chart and (9.6) give, on the boundary surface,

$$du \wedge dP = ds \wedge dp_0 = T_n ds \wedge dt, \quad du \wedge dv = \mathcal{D}_n ds \wedge dt.$$

Consequently the exact physical, rather than canonical, flux is

$$J_n(u, v) = \left| \frac{T_n(s, t)}{\mathcal{D}_n(s, t)} \right|, \quad (s, t) = \Phi_n^{-1}(u, v). \quad (9.9)$$

At zero this denominator is exactly

$$\mathcal{D}_n(0, 0) = U_s(0, 0)V_t(0, 0) - \lambda^{2n}U_p(0, 0)V_q(0, 0). \quad (9.10)$$

On the physical box its limit is $U_s(\alpha(u), 0)V_t(0, \beta(v))$, uniformly bounded away from zero. Equations (9.7)–(9.10) therefore prove (9.2) and (9.3), including all fixed mixed derivatives. This step is necessary: canonical boundary coordinates are not physical configuration observations.

Finally define

$$\mathcal{S}_L(u) = - \int_0^u \mathcal{P}(\alpha(x), 0) \, dx, \quad \mathcal{S}_R(v) = \int_0^v \mathcal{Q}(0, \beta(y)) \, dy.$$

The physical first variations of W_n converge to the two displayed integrands, with all fixed derivatives, by (9.8). Integrate the difference of the gradients along the two coordinate segments from $(0, 0)$ to (u, v) . The values at the origin fix the additive constants and prove (9.4). $\square$

For the two-step billiard return the contracting eigenvalue in (9.1) is $\lambda = e^{-2\gamma}$, and n returns mean $j = 2n$ flights. At contact b its stable and unstable linear directions in physical position–momentum coordinates are $(1, -a_b)$ and $(1, a_b)$, by (5.1). They are nonvertical, regardless of equality or inequality of the curvatures. Use the same canonical chart at both ends. Write its first derivative as

$$DC(0) = \begin{pmatrix} A & B \\ -a_b A & a_b B \end{pmatrix}.$$

Canonicity gives $AB = (2a_b)^{-1}$. Thus (9.10) and (9.9) recover the exact physical normalization

$$J_n(0, 0) = \frac{2a_b \lambda^n}{1 - \lambda^{2n}} = \frac{a_b}{\sinh(2n\gamma)}. \quad (9.11)$$

This is an exact agreement with the Jacobi formula, not merely an agreement of leading exponential orders. If both constructions apply to the same analytic billiard, uniqueness of the normalized limit and evaluation at the other endpoint zero give $B_b = \mathcal{B}_L = \mathcal{B}_R$. Their action limits agree as well, with the stated zero normalizations. The normal-form function Δ alone does not specify the physical amplitudes: the projection charts enter (9.3) essentially.

9.2. The classical scalar comparison. Smooth linearization on a one-dimensional contracting branch is classical. It requires neither a symplectic chart on a neighborhood nor an analytic hypothesis. This is the one-dimensional case of Sternberg’s theorem; see [25] and the precise regularity statement in the introduction of Eynard–Bontemps–Navas [26]. We record the elementary product construction in the normalization needed here. The scalar comparison and its application to the width were pointed out in the memorandum [27]. Its role is to separate a classical coordinate from the identification of the physical determinant amplitude.

Lemma 9.2 (Normalized scalar linearization). *Let R_ξ be a jointly smooth family on a fixed interval about zero, with parameter ξ in a compact set, such that*

$$R_\xi(0) = 0, \quad R'_\xi(0) = \lambda_\xi, \quad 0 < \ell \leq R'_\xi(u) \leq q < 1.$$

The family is understood to be defined on an open neighborhood of the compact parameter set. On a common smaller interval the series

$$\begin{aligned} h_\xi(u) &= \log \frac{R'_\xi(u)}{\lambda_\xi}, \\ L_\xi(u) &= \sum_{i=0}^{\infty} h_\xi(R_\xi^i(u)), \quad \mathfrak{B}_\xi(u) = \exp L_\xi(u), \quad \mathfrak{z}_\xi(u) = \int_0^u \mathfrak{B}_\xi(x) \, dx \end{aligned} \quad (9.12)$$

converges with every fixed mixed parameter and endpoint derivative. The coordinate $\mathfrak{z}_\xi$ is increasing, normalized by $\mathfrak{z}_\xi(0) = 0$ and $\mathfrak{z}'_\xi(0) = 1$, and satisfies

$$\mathfrak{B}_\xi(R_\xi(u))R'_\xi(u) = \lambda_\xi \mathfrak{B}_\xi(u), \quad \mathfrak{z}_\xi \circ R_\xi = \lambda_\xi \mathfrak{z}_\xi. \quad (9.13)$$

It is the unique normalized differentiable local linearizer. For every fixed k and $\sigma \in (\lambda_+, 1)$, where $\lambda_+ = \max_\xi \lambda_\xi$, one has

$$\|\lambda_\xi^{-n} R_\xi^n - \mathfrak{z}_\xi\|_{C^k} + \|\lambda_\xi^{-n} (R_\xi^n)' - \mathfrak{B}_\xi\|_{C^k} \leq C_{k,\sigma} \sigma^n. \quad (9.14)$$

Here and below such norms include any fixed collection of mixed parameter and endpoint derivatives on the common domain.

Proof. Suppress ξ in compositions, but not in differentiation. Put $x_n = R_\xi^n(u)$. Since $R_\xi(0) = 0$ and $|R'_\xi| \leq q$, $|x_n| \leq q^n |u|$. At every fixed positive mixed order m , differentiation of $x_{n+1} = R_\xi(x_n)$ leaves $R'_\xi(x_n)$ multiplying the highest derivative of x_n . Every other term either contains a lower derivative of x_n , or is a pure parameter derivative of R_ξ evaluated at x_n . The latter vanishes at $x_n = 0$, because the fixed point is common to the family. Induction and the mean value theorem therefore give

$$\sup_{|\alpha|+r=m} |\partial_\xi^\alpha \partial_u^r x_n| \leq C_m (1+n)^{M_m} q^n.$$

Indeed, at the inductive step the right-hand forcing in the highest order recurrence is bounded by a polynomial in n times q^n ; iteration of that recurrence only increases the polynomial degree. Uniform bounds on the finitely many derivatives of R_ξ involved follow from compactness. This argument covers endpoint derivatives, parameter derivatives and their mixtures.

The function h_ξ is smooth, $h_\xi(0) = 0$, and all its pure parameter derivatives vanish at zero. The same chain-rule argument bounds each fixed mixed derivative of $h_\xi(x_n)$ by $C_m (1+n)^{N_m} q^n$. Summability proves convergence of (9.12) with all the asserted derivatives. In particular $L_\xi(0) = 0$ and $\mathfrak{B}_\xi(0) = 1$. Shifting the summation index gives $L_\xi(u) - L_\xi(R_\xi(u)) = h_\xi(u)$. Exponentiation proves the first identity in (9.13). Integration from zero proves the second, since both sides vanish there.

If $\tilde{\mathfrak{z}}$ is another normalized differentiable linearizer, set $H = \tilde{\mathfrak{z}} \circ \mathfrak{z}^{-1}$. Then $H(0) = 0$, $H'(0) = 1$, and $H(\lambda z) = \lambda H(z)$. For every z in a sufficiently small interval, $H(z) = \lambda^{-n} H(\lambda^n z) \rightarrow z$. This proves uniqueness.

The exact relation $R_\xi^n = \mathfrak{z}_\xi^{-1} \circ (\lambda_\xi^n \mathfrak{z}_\xi)$ gives the sharper rate based on λ_+ rather than the preliminary contraction bound q . The inverse function theorem on a common smaller interval makes $\mathfrak{z}_\xi^{-1}$ jointly smooth. Its value and first derivative at zero are 0 and 1; write it as $x + x^2 A_\xi(x)$. Thus

$$\lambda_\xi^{-n} R_\xi^n(u) - \mathfrak{z}_\xi(u) = \lambda_\xi^n \mathfrak{z}_\xi(u)^2 A_\xi(\lambda_\xi^n \mathfrak{z}_\xi(u)).$$

Every fixed mixed derivative is bounded by $C_k (1+n)^{N_k} \lambda_+^n$; the positive lower bound on λ_ξ controls derivatives of its powers. Apply the same argument with one additional endpoint

derivative for the density estimate. A fixed strict margin $\lambda_+ < \sigma$ absorbs the polynomial factors. Finally, the finite product identity

$$\exp\left\{\sum_{i=0}^{n-1} h_\xi(R_\xi^i(u))\right\} = \lambda_\xi^{-n}(R_\xi^n)'(u) \quad (9.15)$$

follows directly from the chain rule and agrees with the density limit. $\square$

The alternating scalar identity is equally classical. Suppose that $\varphi_b(0) = 0$, $\varphi_b'(0) = r_b > 0$ and $R_b = \varphi_{1-b} \circ \varphi_b$, with $r_0 r_1 = \lambda < 1$. Let $\mathfrak{z}_b$ be the normalized scalar linearizer of R_b . The identity $\varphi_b \circ R_b = R_{1-b} \circ \varphi_b$ shows that $r_b^{-1} \mathfrak{z}_{1-b} \circ \varphi_b$ is another normalized linearizer of R_b . Uniqueness gives

$$\mathfrak{z}_{1-b} \circ \varphi_b = r_b \mathfrak{z}_b, \quad \mathfrak{B}_{1-b}(\varphi_b(u)) \varphi_b'(u) = r_b \mathfrak{B}_b(u).$$

These scalar facts by themselves do not identify the amplitude in a normalized finite physical bridge. That identification is the content of the following theorem and its exact Schur-concatenation proof.

9.3. A smooth intrinsic characterization. We now return to the full smooth setting. Write $W_{j,b}$ for the stationary action starting at type b , and let

$$\varphi_b(u) = x_1^{(b)}(u), \quad \mathfrak{r}_b = \sqrt{\frac{c_b}{c_{1-b}}}, \quad r_b = \mathfrak{r}_b \varrho, \quad R_b = \varphi_{1-b} \circ \varphi_b, \quad \varrho = e^{-\gamma}, \quad \lambda = \varrho^2.$$

Here $x^{(b)}$ is the physical half-line in Lemma 7.1. Thus φ_b is the first-hit position along the stable branch, not a freely chosen change of variables. Notice that $r_0 r_1 = \lambda$ and $0 < r_b < 1$. Indeed $\sqrt{c_b}/c_{1-b} e^{-\gamma} = c_b/(c_0 c_1 + \sqrt{c_0 c_1}(c_0 c_1 - 1)) < 1$, since $c_{1-b} > 1$.

Theorem 9.3 (The boundary amplitude as a linearizing density). *On a fixed smaller collar, uniformly for the smooth compact families of Theorem 2.1, the half-line amplitudes satisfy*

$$B_{1-b}(\varphi_b(u)) \varphi_b'(u) = r_b B_b(u), \quad b = 0, 1. \quad (9.16)$$

The increasing smooth coordinates

$$\zeta_b(u) = \int_0^u B_b(x) \, dx \quad (9.17)$$

are normalized by $\zeta_b(0) = 0$, $\zeta_b'(0) = 1$ and obey

$$\zeta_{1-b}(\varphi_b(u)) = r_b \zeta_b(u), \quad \zeta_b(R_b(u)) = \lambda \zeta_b(u). \quad (9.18)$$

For each b , ζ_b is the unique differentiable local coordinate with that normalization which linearizes R_b . Moreover, for a fixed $\sigma \in (\lambda_+, 1)$, where λ_+ is the maximum of λ on the compact family,

$$\|\lambda^{-n} R_b^n - \zeta_b\|_{C^k} + \|\lambda^{-n} (R_b^n)' - B_b\|_{C^k} \leq C_k \sigma^n. \quad (9.19)$$

The norms include every fixed mixed parameter and endpoint derivative. No normalizing chart for the two-dimensional return map is a hypothesis.

Proof. Put $o = 1 - b$. Uniqueness of the half-line tail gives the exact relations

$$S_b(u) = \ell_b(u, \varphi_b(u)) - g + S_o(\varphi_b(u)), \quad \ell_{b,2}(u, \varphi_b(u)) + S_o'(\varphi_b(u)) = 0. \quad (9.20)$$

The second relation also follows from first variation of the middle endpoint. Its w derivative is $\ell_{b,22}(u, w) + S_o''(w)$, uniformly positive on the small box. The implicit function theorem and the already constructed half-line therefore give

$$\varphi_b'(u) = \frac{-\ell_{b,12}(u, \varphi_b(u))}{\ell_{b,22}(u, \varphi_b(u)) + S_o''(\varphi_b(u))} > 0. \quad (9.21)$$

At zero this is $(c_o + ga_o)^{-1} = \sqrt{c_b/c_o}e^{-\gamma} = r_b$. After decreasing the common interval, both first-hit maps are contractions into it. All their mixed derivatives are controlled by the same positive denominator. The graph of stable initial physical states is $P = -S'_b(u)$: differentiating the first identity in (9.20) gives $P = -\ell_{b,1}$, while the terminal momentum is $\ell_{b,2} = -S'_o$. The reversed unstable graph is $P = S'_b(u)$. This verifies the dynamical interpretation directly from the physical generating action.

We prove the amplitude identity from exact finite concatenation. For a tail of j flights starting at o , let $w_j = w_j(u, v)$ solve

$$\ell_{b,2}(u, w_j) + W_{j,o,1}(w_j, v) = 0.$$

Strict convexity identifies the resulting stationary value with $W_{j+1,b}(u, v)$. Set $H_j = \ell_{b,22}(u, w_j) + W_{j,o,11}(w_j, v)$. Eliminating the middle endpoint gives the exact Schur identity

$$-W_{j+1,b,12}(u, v) = \frac{-\ell_{b,12}(u, w_j)}{H_j}[-W_{j,o,12}(w_j, v)]. \quad (9.22)$$

The denominator is uniformly positive. Theorem 7.2 and its differentiated action limit imply $w_j \rightarrow \varphi_b(u)$ and $H_j \rightarrow \ell_{b,22}(u, \varphi_b(u)) + S''_o(\varphi_b(u))$ on the fixed box. The inverse equations, with the same positive denominator, give this convergence in every fixed mixed norm.

The terminal type of both sides of (9.22) is $c = (o + j) \bmod 2$. The exact reference constants are

$$d_{j,o}^0 = \frac{\sqrt{a_o a_c}}{\sinh(j\gamma)}, \quad d_{j+1,b}^0 = \frac{\sqrt{a_b a_c}}{\sinh((j+1)\gamma)}.$$

Their ratio tends to $\sqrt{a_o/a_b}e^\gamma = r_b^{-1}$, with all fixed parameter derivatives. Divide (9.22) by $d_{j+1,b}^0$ and let j tend to infinity along either parity. Relative factorization and (9.21) give

$$B_b(u)B_c(v) = r_b^{-1}\varphi'_b(u)B_o(\varphi_b(u))B_c(v).$$

Since $B_c > 0$, cancellation proves (9.16). Integration from zero proves (9.18). This argument uses the previously proved relative theorem to identify its amplitude; it does not claim that a scalar linearization alone proves the finite-bridge theorem.

For uniqueness, conjugate any other normalized differentiable linearizer by ζ_b^{-1} . The resulting function h has $h(0) = 0$, $h'(0) = 1$ and $h(\lambda z) = \lambda h(z)$. Hence $h(z) = \lambda^{-n}h(\lambda^n z) \rightarrow z$, so $h(z) = z$. For the quantitative assertion, the exact identity is

$$R_b^n(u) = \zeta_b^{-1}(\lambda^n \zeta_b(u)).$$

Write $\zeta_b^{-1}(x) = x + x^2 h_b(x)$, where h_b is jointly smooth on a common interval. Division by λ^n leaves $\zeta_b(u) + \lambda^n \zeta_b(u)^2 h_b(\lambda^n \zeta_b(u))$. At each fixed derivative order this last term, including parameter derivatives of λ , is bounded by $C_k(1+n)^{m_k} \lambda_+^n$. Its endpoint derivative proves the second estimate in (9.19); the fixed strict margin gives the displayed rate. This also proves all claimed uniformities without extending the scalar coordinate to a symplectic chart off the invariant branch. $\square$

Corollary 9.4 (Physical determinant and scalar product). *For the physical return $R_b = \varphi_{1-b} \circ \varphi_b$ in Theorem 9.3, put $h_b(u) = \log(R'_b(u)/\lambda)$. The physical edge product U_b , Green operator $G^{(b)}$ and Hessian perturbation ΔH_b of (7.6)–(7.7) satisfy*

$$\frac{\exp U_b(u)}{\det(I + G^{(b)} \Delta H_b(u))} = B_b(u) = \exp \left\{ \sum_{i=0}^{\infty} h_b(R_b^i(u)) \right\}. \quad (9.23)$$

All expressions have the same value 1 at zero, and the series converges in every fixed mixed norm on the common collar. More precisely, the finite scalar product has the exact remainder

$$\log B_b(u) - \sum_{i=0}^{n-1} h_b(R_b^i(u)) = \log B_b(R_b^n(u)). \quad (9.24)$$

For each fixed k and each $\sigma \in (\lambda_+, 1)$ the right side has C^k norm at most $C_{k,\sigma}\sigma^n$.

Proof. The first equality is the original relative Fredholm determinant construction, with its convergent trace series. Theorem 9.3 identifies its positive density by exact physical concatenation. Composing its two first-flight cocycles gives $B_b(R_b(u))R_b'(u) = \lambda B_b(u)$. Let $\mathfrak{B}_b$ be the scalar density in Lemma 9.2. The positive quotient $B_b/\mathfrak{B}_b$ is invariant under R_b and equals one at zero. Iteration to the contracting fixed point makes the quotient identically one. This proves (9.23). Telescoping the logarithmic cocycle proves (9.24) for every finite n . For the mixed norm bound use $R_b^n = \zeta_b^{-1}(\lambda^n \zeta_b)$ and write $\log B_b(\zeta_b^{-1}(x)) = x C_b(x)$ with C_b jointly smooth. The remainder is then $\lambda^n \zeta_b(u) C_b(\lambda^n \zeta_b(u))$. The same differentiated estimate as in (9.14) proves the stated bound. $\square$

In particular, the infinite scalar product is not introduced as a new linearization theorem. Equation (9.23) is an identification of two normalized constructions for the same physical branch. The finite-bridge trace-norm comparison, the physical reference twist, the first-hit selection and the residual-time integration still have their separate proofs. The scalar product does not replace them.

Corollary 9.5 (Stable-action width determined by the even laws). *Put $\hat{S}_b = S_b \circ \zeta_b^{-1}$. Then the limiting laws can be written*

$$\mathcal{F}_{bc}(d) = \frac{\sqrt{a_b a_c}}{\pi d^2} \int (d - \hat{S}_b(x) - \hat{S}_c(y))_+ dx dy. \quad (9.25)$$

For small $E > 0$, let $u_{b,-}(E) < 0 < u_{b,+}(E)$ solve $S_b(u) = E$. The two even laws determine the normalized widths

$$\begin{aligned} \mathcal{W}_b(E) &:= \sqrt{\frac{a_b}{2}} [\zeta_b(u_{b,+}(E)) - \zeta_b(u_{b,-}(E))] \\ &= \int_0^E x^{-1/2} \mathcal{V}_b(x) dx. \end{aligned} \quad (9.26)$$

Thus the smooth invariant is the complete normalized sublevel width of the half-line action in its intrinsic linearizing coordinate.

Proof. The change of variables $x = \zeta_b(u)$ has differential $dx = B_b(u) du$. This proves (9.25). The pushforward calculation in (10.1) gives

$$\int_{S_b(u) < E} B_b(u) du = \sqrt{\frac{2}{a_b}} \int_0^E x^{-1/2} \mathcal{V}_b(x) dx.$$

It is also exactly $\zeta_b(u_{b,+}(E)) - \zeta_b(u_{b,-}(E))$. Theorem 10.1 determines $\mathcal{V}_b$ from the even law, proving the assertion. That theorem's proof uses only the half-line formula and Volterra uniqueness, not this corollary. $\square$

Conversely, on $E > 0$,

$$\mathcal{V}_b(E) = \sqrt{E} \mathcal{W}_b'(E), \quad \mathcal{W}_b(0) = 0, \quad \mathcal{W}_b(E) = 2\sqrt{E} + O(E^{3/2}). \quad (9.27)$$

The first expression extends smoothly to $E = 0$ with value one. Thus the normalized width and the energy profile are equivalent coordinates on the same invariant, not two independent

invariants. The pushforward theorem determines a sublevel measure; the width formula expresses that measure in the classical stable coordinate.

The normalization in (9.26) is useful when the separate curvatures are not supplied. The even laws determine $\mathcal{W}_b$ without knowing a_b separately. With a_b supplied, they also determine the unnormalized coordinate width. When both participating contact graphs are even, S_b and B_b are even and ζ_b is odd, so this width specifies both inverse branches of $\tilde{S}_b$. For a general contact it specifies their difference. The physical contact-graph inverse in Section 11 is a further geometric argument, not an inference that a symmetrized width determines an arbitrary asymmetric boundary.

9.4. The comparison at the level of the physical theorem. Proposition 9.1 explains the analytic relative product mechanism, including its physical projection and exact reference normalization. If the limiting endpoint Hessians are positive, Proposition 8.3 also transports its estimates through the normalized residual integral. At an isolated analytic billiard channel the positivity and first-hit property are verified in Sections 4–5. An odd-flight analytic comparison can likewise compose the terminal chart with one physical flight. Thus neither the parity distinction nor the absence of global finite horizon is an obstruction to that local analytic mechanism. They are requirements handled simultaneously in our physical theorem, not separate claims of a new normal-form effect.

Theorem 2.1 does not supply normalizing charts as data or hypotheses. It starts with smooth obstacle graphs and positive geometric margins. Its half-line and finite-bridge contractions, localized trace-ideal estimates, and common sublevel charts prove the mixed parameter and offset bounds on one closed collar, with the exact phase normalization and the physical event selection. A pointwise analytic normal form supplies none of these uniform chart estimates for an arbitrary smooth family. The conditional family clause in Proposition 9.1 states exactly what would be needed to use that route instead. No assertion that such a smooth normal-form route is impossible, or absent from all prior literature, is required here.

Theorem 9.3 makes the geometric meaning of the determinant amplitude explicit in the full smooth class: it is the derivative of the normalized scalar linearizer along the physical stable branch, with an exact first-flight cocycle linking the two contact types. Corollary 9.5 identifies what the complete invariant measures in that coordinate. These conclusions connect the analytic and classical scalar interpretations to the constructed smooth amplitude. Equation (9.23) gives their exact relation; (9.27) records that the width is a coordinate on the previously determined invariant. Neither interpretation is counted as an additional rigidity theorem.

The remaining steps concern information that the forward benchmark does not reconstruct: the Volterra inverse for the complete widths; the explicit independent-contact geometric inverse, its analytic physical image and its two-flight coordinates; and the regularized binary experiment with its charged smooth-class calibration. The preparation theorem uses certified physical smoothness bounds, not access to Δ or to a canonical chart. Its sufficient rate is not claimed to be optimal. The central result is the joint passage from smooth physical records to an intrinsic boundary measure, to its complete observable invariant, and to a finite-preparation inverse. The classical analytic product mechanism is part of that explanation, not the novelty claimed for the package.

9.5. An exact mixed-parameter benchmark. The strict exponential margin in Lemma 9.2 is part of its mixed-parameter assertion. The following exact family, recorded in the independent memorandum [29], explains why the multiplier itself cannot in general replace that margin.

Proposition 9.6 (A strict-margin example). *On a common interval about zero put*

$$R_{\lambda,a}(u) = \frac{\lambda u}{1 + a(1 - \lambda)u}, \quad z_a(u) = \frac{u}{1 + au}, \quad B_a(u) = (1 + au)^{-2}.$$

Let $0 < \lambda_- \leq \lambda \leq \lambda_+ < 1$ and let a range over a compact interval containing zero in its interior. For a sufficiently small common endpoint interval these maps are smooth contractions and

$$R_{\lambda,a}^n(u) = \frac{\lambda^n u}{1 + a(1 - \lambda^n)u}, \quad (9.28)$$

$$\lambda^{-n}(R_{\lambda,a}^n)'(u) = [1 + a(1 - \lambda^n)u]^{-2}, \quad (9.29)$$

$$\lambda^{-n}R_{\lambda,a}^n(u) - z_a(u) = \frac{a\lambda^n u^2}{[1 + a(1 - \lambda^n)u](1 + au)}. \quad (9.30)$$

The normalized errors converge with every fixed mixed derivative at rate $C_{k,\sigma}\sigma^n$ for each $\lambda_+ < \sigma < 1$. There is no such bound with $C\lambda_+^n$ for the mixed derivative $\partial_\lambda \partial_a$ of (9.30) on a nontrivial endpoint interval.

Proof. The denominators are uniformly positive after restricting the endpoint interval. Also $R_{\lambda,a}'(u) = \lambda/[1 + a(1 - \lambda)u]^2$ is then positive and uniformly less than one. Direct substitution gives $z_a \circ R_{\lambda,a} = \lambda z_a$ and $z_a^{-1}(x) = x/(1 - ax)$. Iteration and one endpoint differentiation prove (9.28)–(9.29). Subtraction proves (9.30). At each fixed order, differentiation gives a polynomial in n times λ_+^n , multiplied by uniformly bounded rational functions; the lower bound on λ controls the finitely many inverse powers introduced. A strict margin absorbs that polynomial. The density estimate follows either directly or by one additional endpoint derivative.

At $a = 0$ the mixed derivative of the error is exactly

$$\partial_\lambda \partial_a (\lambda^{-n} R_{\lambda,a}^n(u) - z_a(u)) \Big|_{a=0} = n\lambda^{n-1}u^2.$$

At $\lambda = \lambda_+$ and any fixed $u \neq 0$, its ratio to λ_+^n is nu^2/λ_+ , which is unbounded. Finally the exact identity

$$\lambda^{-n}(R_{\lambda,a}^n)'(u)B_a(R_{\lambda,a}^n(u)) = B_a(u)$$

checks the scalar product remainder with no asymptotic expansion. $\square$

This is a benchmark for mixed estimates, not a billiard realization or an additional linearization theorem. The physical theorem retains its already strict exponential margin and its independent relative finite-bridge proof.

10. BOUNDARY ENERGY PROFILES AND NONLINEAR PARITY COMPATIBILITY

Fix one of the general channels of Theorem 2.1 and retain observations from both orientations. For an even number of flights, the first and last impacts have the same contact type; for an odd number they have different types. Write $\mathcal{F}_{00}, \mathcal{F}_{11}$ for the two even limiting functions and $\mathcal{F}_{01} = \mathcal{F}_{10}$ for the odd one. Their normalization is that of (2.6), with the same free area A and multiplier γ . Thus they are scalar functions derived from selected physical event probabilities, not postulated independent boundary densities.

10.1. An intrinsic positive factorization. For each contact type put $a_b = S_b''(0) > 0$ and introduce the signed Morse coordinate and its increasing inverse,

$$t_b(u) = \operatorname{sgn}(u)\sqrt{2S_b(u)}, \quad u = \chi_b(t).$$

The value at $u = 0$ is defined by smooth extension. Set

$$\begin{aligned} f_b(t) &= B_b(\chi_b(t))\chi_b'(t), \\ \mathcal{V}_b(x) &= \frac{\sqrt{a_b}}{2} \{f_b(\sqrt{2x}) + f_b(-\sqrt{2x})\}, \quad x \geq 0. \end{aligned} \quad (10.1)$$

We call $\mathcal{V}_b$ the normalized boundary energy profile. It is a symmetrized weighted energy datum: no assertion that it determines a nonsymmetric contact graph is implicit in this terminology.

Theorem 10.1 (Nonlinear boundary compatibility). *On a sufficiently small common closed collar the functions $\mathcal{V}_b$ are smooth, positive, and satisfy $\mathcal{V}_b(0) = 1$. For $b, c \in \{0, 1\}$ the limiting physical laws have the representation*

$$\mathcal{F}_{bc}(d) = \frac{2}{\pi d^2} \int_{\substack{x, y > 0 \\ x+y < d}} (d-x-y) \frac{\mathcal{V}_b(x)\mathcal{V}_c(y)}{\sqrt{xy}} dx dy. \quad (10.2)$$

Let $*$ denote Volterra convolution on the positive half-line, used only at arguments in this collar, and define

$$k_b(x) = x^{-1/2}\mathcal{V}_b(x), \quad K_{bc}(d) = \frac{\pi}{2} \frac{d^2}{dd^2} \{d^2 \mathcal{F}_{bc}(d)\}. \quad (10.3)$$

Then $K_{bc} = k_b * k_c$, $K_{bc}(0) = \pi$, and

$$K_{01} * K_{01} = K_{00} * K_{11}. \quad (10.4)$$

Within the class of smooth normalized profiles, $\mathcal{F}_{00}$ and $\mathcal{F}_{11}$ determine $\mathcal{V}_0, \mathcal{V}_1$ uniquely and therefore determine the complete odd law $\mathcal{F}_{01}$ on the same collar. The assertions hold without equal-curvature, even-graph or symmetry assumptions. All the forward identities have the mixed geometric regularity of the relative law.

Proof. Strict convexity makes the signed Morse coordinate smooth and increasing near zero, with derivative $\sqrt{a_b}$ at zero. Consequently f_b is smooth, positive and has value $a_b^{-1/2}$. Its even part is a smooth function of t^2 , by the even-function argument in Proposition 8.3. This proves the asserted properties of $\mathcal{V}_b$, including its parameter regularity. Reversing the choice of transverse direction merely interchanges the two terms in (10.1).

Under $x = S_b(u) = t^2/2$, the pushforward of $B_b(u) du$ has, for $x > 0$, density

$$\frac{f_b(\sqrt{2x}) + f_b(-\sqrt{2x})}{\sqrt{2x}} = \sqrt{\frac{2}{a_b}} x^{-1/2} \mathcal{V}_b(x).$$

Insert this expression twice into (2.5), whose prefactor for types b, c is $\sqrt{a_b a_c}/(\pi d^2)$. The Hessian factors cancel and give (10.2). This calculation also proves equality of the two odd orientations. It is valid for each even orientation by applying the relative law after interchanging the contact types.

The kernels k_b are locally integrable. Fubini's theorem rewrites (10.2) as

$$\frac{\pi}{2} d^2 \mathcal{F}_{bc}(d) = \int_0^d (d-s) (k_b * k_c)(s) ds.$$

The convolution is continuous down to zero: substitution $x = ds$ gives

$$(k_b * k_c)(d) = \int_0^1 \frac{\mathcal{V}_b(ds)\mathcal{V}_c(d(1-s))}{\sqrt{s(1-s)}} ds,$$

whose value at zero is π . Differentiation proves (10.3). Associativity and commutativity of these locally integrable convolutions, again by Fubini on a bounded simplex, give (10.4). No extension of the physical law beyond the collar is used.

It remains to prove uniqueness, since a formal square root of Taylor coefficients alone would not suffice for smooth functions. Suppose $k = x^{-1/2}V(x)$ and $\ell = x^{-1/2}\tilde{V}(x)$, with smooth $V(0) = \tilde{V}(0) = 1$, have $k * k = \ell * \ell$ on the collar. Put $f = k - \ell$, $a = k + \ell$ and $k_*(x) = x^{-1/2}$. Then $a * f = 0$ and

$$a(x) = 2x^{-1/2} + x^{1/2}w(x)$$

for a smooth w . The identity $k_* * k_* = \pi$ gives

$$(k_* * a)(x) = 2\pi + R(x), \quad R(x) = x \int_0^1 \frac{s^{1/2}}{(1-s)^{1/2}} w(xs) \, ds.$$

Here $R(0) = 0$ and R' is bounded. Convoluting $a * f = 0$ with k_* and differentiating its absolutely continuous integral yields

$$2\pi f + R' * f = 0$$

almost everywhere. Hence $|f(x)| \leq C \int_0^x |f(s)| \, ds$. The absolutely continuous function $\int_0^x |f(s)| \, ds$ has zero initial value, and the elementary Gronwall inequality makes it zero throughout the collar. Thus $k = \ell$. Applying this argument to each of K_{00} and K_{11} proves uniqueness of both profiles; (10.2) then determines the odd law. $\square$

Theorem 10.2 (Stability of the complete boundary energy profile). *Fix $D > 0$ in the common collar. Let $V, \tilde{V}$ be two smooth normalized profiles, with $V(0) = \tilde{V}(0) = 1$ and $\|V'\|_\infty, \|\tilde{V}'\|_\infty \leq B$ on $[0, D]$. Let $F, \tilde{F}$ be their same-type laws defined by (10.2), and put $K = (\pi/2)(d^2 F)''$, $\tilde{K} = (\pi/2)(d^2 \tilde{F})''$. Then*

$$\|V - \tilde{V}\|_{C^0([0, D])} \leq \frac{D}{\pi} e^{BD/2} \|K' - \tilde{K}'\|_{C^0([0, D])} \leq C_{B, D} \|F - \tilde{F}\|_{C^3([0, D])}. \quad (10.5)$$

Thus the two complete boundary profiles are locally Lipschitz functions of the two same-type laws on their physical image, when the latter are given this specified differentiated norm. The resulting odd law is Lipschitz in C^0 on uniformly bounded profile sets.

Proof. Use $k = x^{-1/2}V$, $\ell = x^{-1/2}\tilde{V}$, $f = k - \ell$, $a = k + \ell$, $k_* = x^{-1/2}$, and put $Q = K - \tilde{K} = a * f$. The convolution in the preceding proof has the more direct expression

$$(k_* * a)(x) = 2\pi + R(x), \quad R(x) = \int_0^1 \frac{V(xs) + \tilde{V}(xs) - 2}{\sqrt{s(1-s)}} \, ds.$$

It follows that $R(0) = 0$ and

$$\|R'\|_\infty \leq 2B \int_0^1 \frac{s}{\sqrt{s(1-s)}} \, ds = \pi B.$$

Since $Q(0) = 0$, differentiation under a convolution, or integration by parts before differentiation, gives

$$2\pi f + R' * f = (k_* * Q)' = k_* * Q'.$$

The right side is bounded by $2\sqrt{x} \|Q'\|_\infty$. The normalized initial values imply that f extends continuously with $f(0) = 0$. Therefore

$$|f(x)| \leq \frac{\sqrt{D}}{\pi} \|Q'\|_\infty + \frac{B}{2} \int_0^x |f(s)| \, ds.$$

Gronwall's inequality gives $\|f\|_\infty \leq \sqrt{D} e^{BD/2} \|Q'\|_\infty / \pi$. Multiplication by $\sqrt{x}$ proves the first estimate. For the second,

$$Q'(d) = \frac{\pi}{2} \{6(F - \tilde{F})'(d) + 6d(F - \tilde{F})''(d) + d^2(F - \tilde{F})'''(d)\},$$

which is bounded by the asserted C^3 norm. Finally the integral (10.2) is a bilinear map of the profiles with a positive kernel of normalized total mass one. Subtracting its two products proves the C^0 Lipschitz assertion for the odd law. $\square$

This is stability of normalized energy profiles, not of an arbitrary nonsymmetric contact graph. It is an inverse on the image of the forward profile map; no claim that every positive scalar function is a convolution square is needed. The derivative norm in (10.5) is part of the observation statement. The unregularized theorem does not replace it by uniform error of raw probabilities. The finite-preparation theorem in Section A.4 adds regularization and supplied smoothness bounds to address that different data problem. For finite jets, an explicit polynomial inverse is available without solving a function-valued equation.

10.2. Finite jets and observable identities. For a smooth right germ h , write $[d^n]h = h^{(n)}(0)/n!$. Fix $M \geq 0$ and work in the truncated polynomial ring $\mathbb{R}[z]/(z^{M+1})$. Introduce

$$\begin{aligned}\mathcal{M}_{bc}^{[M]}(z) &= \sum_{n=0}^M \frac{(n+2)!}{2} [d^n] \mathcal{F}_{bc} z^n, \\ \mathcal{A}_b^{[M]}(z) &= \sum_{n=0}^M \left(\frac{1}{2}\right)_n [x^n] \mathcal{V}_b z^n, \quad \left(\frac{1}{2}\right)_0 = 1.\end{aligned}\tag{10.6}$$

The notation $(1/2)_n$ denotes the rising factorial. Truncation is important: even for analytic input, no convergence radius of a factorially reweighted infinite series is being asserted.

Theorem 10.3 (Rank-one jets and stable finite-order inversion). *The transformed jets satisfy*

$$\mathcal{M}_{bc}^{[M]} = \mathcal{A}_b^{[M]} \mathcal{A}_c^{[M]}, \quad (\mathcal{M}_{01}^{[M]})^2 = \mathcal{M}_{00}^{[M]} \mathcal{M}_{11}^{[M]} \pmod{z^{M+1}}.\tag{10.7}$$

Both profiles' order- M jets are recovered from the two same-type laws by the following triangular recursion. If $m_{bb,n} = [z^n] \mathcal{M}_{bb}^{[M]}$ and $\alpha_{b,n} = [z^n] \mathcal{A}_b^{[M]}$, then

$$\alpha_{b,0} = 1, \quad \alpha_{b,n} = \frac{1}{2} \left(m_{bb,n} - \sum_{r=1}^{n-1} \alpha_{b,r} \alpha_{b,n-r} \right), \quad [x^n] \mathcal{V}_b = \frac{\alpha_{b,n}}{(1/2)_n}.\tag{10.8}$$

For fixed M these reconstruction maps, and the resulting prediction of the odd jet, are Lipschitz on bounded sets of input coefficients. They are smooth with respect to geometric parameters on the compact families of the relative law.

Proof. Write $v_{b,r} = [x^r] \mathcal{V}_b$. The beta integral on the simplex gives, for $r, s \geq 0$,

$$\int_{x,y>0, x+y<d} (d-x-y) x^{r-1/2} y^{s-1/2} dx dy = \frac{\pi(1/2)_r(1/2)_s}{(r+s+2)!} d^{r+s+2}.$$

This follows by first setting $x = dX, y = dY$ and integrating in Y , then in X ; both integrals are the ordinary beta integral. Taylor's integral remainders, bounded on the scaled simplex, justify coefficient extraction for smooth rather than only analytic profiles. Thus

$$[d^n] \mathcal{F}_{bc} = \frac{2}{(n+2)!} \sum_{r+s=n} (1/2)_r (1/2)_s v_{b,r} v_{c,s}.\tag{10.9}$$

Multiplying by $(n+2)!/2$ gives the first identity in (10.7); its determinant identity is an immediate polynomial consequence. All constant coefficients are one. Comparing the degree- n coefficient of $\mathcal{M}_{bb}^{[M]} = (\mathcal{A}_b^{[M]})^2$ leaves $2\alpha_{b,n}$ as the only unknown term, proving (10.8). Induction makes every output coefficient a polynomial in finitely many input coefficients, divided only by fixed positive numbers. Its derivative is bounded on each bounded coefficient box, proving the Lipschitz assertion. Finite parameter differentiation of these polynomial identities proves the last assertion without any infinite-order uniformity claim. $\square$

Corollary 10.4 (The first two nonlinear compatibility identities). *With primes denoting right offset derivatives at zero,*

$$2\mathcal{F}'_{01} = \mathcal{F}'_{00} + \mathcal{F}'_{11}, \quad (10.10)$$

$$\mathcal{F}''_{01} = \frac{\mathcal{F}''_{00} + \mathcal{F}''_{11}}{2} - \frac{3}{16}(\mathcal{F}'_{00} - \mathcal{F}'_{11})^2. \quad (10.11)$$

In particular the odd second derivative is at most the mean of the two even second derivatives, with the exact deficit displayed above.

Proof. The first two nonconstant coefficients of $\mathcal{M}_{bc}$ are $3\mathcal{F}'_{bc}$ and $6\mathcal{F}''_{bc}$. Equate degrees one and two in $\mathcal{M}_{01}^2 = \mathcal{M}_{00}\mathcal{M}_{11}$ and substitute the degree-one equality in the degree-two equality. This gives the two formulas, including the negative square coefficient. $\square$

These identities concern nonlinear offset coefficients. The zeroth-order condition merely says that every normalized law has value one. Neither the zeroth-order condition nor the multiplier alone determines the first two coefficients. The geometric variation in Proposition A.1.1 and the realization in Theorem A.1.2 exhibit nonconstant boundary data at fixed leading selected-channel invariants. The compatibility theorem organizes such data without imposing the identical-even hypothesis of the separate graph-jet inverse in the appendix.

10.3. Finite physical records and a quantitative defect. The preceding identities are exact statements about the limiting functions. The relative law also controls how they can be read from long but finite records. For $n \geq 1$ define

$$\begin{aligned} F_{bb,n}(d) &= \frac{2A \sinh(2n\gamma)}{d^2} \Pr(E_{2n,e_b}(d)), \quad e_0 = e, \quad e_1 = \bar{e}, \\ F_{01,n}(d) &= \frac{2A \sinh((2n+1)\gamma)}{d^2} \Pr(E_{2n+1,e}(d)). \end{aligned} \quad (10.12)$$

They are extended to $d = 0$ by their exact smooth right limits. These definitions use the selected patch labels and the normalization A, γ ; they are not unlabelled binary data with unknown normalizer. Estimation of these auxiliary quantities is a separate observation problem.

Corollary 10.5 (Finite-record compatibility and profile-jet recovery). *Let M and the number of geometric derivatives be fixed. Apply the transform (10.6) to the three finite laws in (10.12), obtaining $\mathcal{M}_{bc,n}^{[M]}$. With the maximum coefficient norm on truncated polynomials,*

$$\|(\mathcal{M}_{01,n}^{[M]})^2 - \mathcal{M}_{00,n}^{[M]}\mathcal{M}_{11,n}^{[M]}\| \leq C_M \tau^{2n}. \quad (10.13)$$

The same estimate holds for each fixed mixed geometric derivative. Suppose the measured order- M jets of the two same-type finite laws have coefficient error at most ϵ , with their known constant coefficients fixed to one. The recursion (10.8) reconstructs both boundary profile jets, and predicts the odd limiting jet, with error

$$C_M(\epsilon + \tau^{2n}). \quad (10.14)$$

For every fixed subcollar $[0, D] \subset [0, d_]$, the finite kernels $K_{bc,n} = (\pi/2)(d^2 F_{bc,n})''$ also satisfy*

$$\|K_{01,n} * K_{01,n} - K_{00,n} * K_{11,n}\|_{C^0([0,D])} \leq C_D \tau^{2n}.$$

Proof. Theorem 2.1, applied to each orientation and parity, bounds $F_{bc,n} - \mathcal{F}_{bc}$ in every fixed C^k norm by $C_k \tau^{2n}$. Coefficient extraction and the finite transform preserve this order. The determinant in (10.7) is a polynomial in bounded coefficients; subtract its value at the limiting jets to prove (10.13). The product rule proves the parameter derivative assertion. Apply the finite polynomial inverse of Theorem 10.3 to the two measured jets to get (10.14); this is also defined for noisy jets outside the physical image. Finally the relative C^2 estimate bounds $K_{bc,n} - K_{bc}$

uniformly on $[0, D]$. For bounded functions there, $\|f * g\|_\infty \leq D\|f\|_\infty\|g\|_\infty$. Telescoping the four products and using (10.4) proves the last claim. $\square$

The two oriented even laws suffice to reconstruct both normalized boundary energy profiles and to predict the odd law. Odd records are needed for the independent compatibility test, not as a third datum for this reconstruction. Neither conclusion recovers full collision arrays or an arbitrary billiard table. Unlike the fixed-flight parametric inverse below, the compatibility defect and its vanishing rate use the long-bridge relative theorem directly. Section A.4 supplies a regularized finite-preparation version of the full-profile inverse, with the required normalization and smoothness bounds stated explicitly.

11. DETERMINATION OF TWO INDEPENDENT EVEN CONTACTS

The boundary invariant has a geometric inverse on a class substantially larger than the identical-contact class. In this section each facing graph is even in the common transverse coordinate, but the two graphs are otherwise unrelated. Their curvatures may be different or equal. This is an additional hypothesis for geometric determination, not for Theorems 2.1 and 10.1. The gap and the two labelled contact curvatures are fixed. They are not inferred separately from the single multiplier.

Write, for $b = 0, 1$,

$$\psi_b(y) = \frac{\kappa_b}{2}y^2 + \sum_{m \geq 2} \frac{q_{b,2m}}{(2m)!}y^{2m}, \quad c_b = 1 + g\kappa_b, \quad c = \sqrt{c_0c_1}, \quad \gamma = \operatorname{arcosh} c, \quad \varrho = e^{-\gamma}. \quad (11.1)$$

For smooth graphs the series specifies jets only. Put

$$a_b = \frac{c \sinh \gamma}{gc_{1-b}}, \quad f_{b,k} = [d^k] \mathcal{F}_{bb}(d), \quad q_m = (q_{0,2m}, q_{1,2m})^t, \quad f_{m-1} = (f_{0,m-1}, f_{1,m-1})^t.$$

The two even laws are obtained by observing the two orientations of one selected physical channel. In particular, their indices retain contact labels even when the curvatures coincide.

Theorem 11.1 (Block-triangular two-contact inverse). *For every $m \geq 2$, the vector f_{m-1} depends only on $g, \kappa_0, \kappa_1, q_2, \dots, q_m$, and is affine in q_m . Its derivative with respect to that last vector is*

$$\frac{\partial f_{m-1}}{\partial q_m} = - \begin{pmatrix} P_m & Q_m \\ Q_m & P_m \end{pmatrix} \operatorname{diag}(K_{m,0}, K_{m,1}), \quad K_{m,b} = \frac{4}{2^m(m+1)(m!)^2 a_b^m}, \quad (11.2)$$

where

$$\begin{aligned} E_m &= \frac{1}{e^{4(m-1)\gamma} - 1} - \frac{1}{e^{4m\gamma} - 1}, \\ O_m &= \frac{1}{2 \sinh(2(m-1)\gamma)} - \frac{1}{2 \sinh(2m\gamma)}, \\ P_m &= \coth(2m\gamma) + 2mE_m, \quad Q_m = \operatorname{csch}(2m\gamma) + 2mO_m. \end{aligned} \quad (11.3)$$

Both eigenvalues of the symmetric factor are positive. More explicitly,

$$P_m + Q_m = \coth(m\gamma) + 2m \left\{ \frac{1}{e^{2(m-1)\gamma} - 1} - \frac{1}{e^{2m\gamma} - 1} \right\}, \quad (11.4)$$

$$P_m - Q_m = m \tanh((m-1)\gamma) - (m-1) \tanh(m\gamma) > 0. \quad (11.5)$$

Consequently, for every finite $M \geq 2$, the map

$$(q_2, \dots, q_M) \mapsto (f_1, \dots, f_{M-1}) \quad (11.6)$$

has a block-triangular analytic inverse on its image. The inverse is locally Lipschitz on compact finite-jet sets and positive compact g, κ_0, κ_1 boxes. These bounds require no separation of κ_0 and κ_1 , and no separation of the higher jets.

Equality of the two normalized even-law germs, for two pairs of real-analytic even graphs at the same labelled g, κ_0, κ_1 , implies equality of both graph germs. For smooth compact parameter families, replacing the limiting coefficients by their even finite-flight coefficients recovers the jet vector through order $2M$ with error at most $C_M \tau^j$. No bound uniform in M is asserted.

11.1. The two boundary contributions.

Proof. We first establish the last-jet derivative; the resulting block invertibility will then give the inverse. Set $\mathbf{r}_b = \sqrt{c_b/c_{1-b}}$. The linear half-line orbit starting at type b is

$$x_i^{(b)}(u) = t_i^{(b)} u + O(u^3 \rho^i), \quad t_i^{(b)} = \begin{cases} \varrho^i, & i \text{ even}, \\ \mathbf{r}_b \varrho^i, & i \text{ odd}. \end{cases} \quad (11.7)$$

The cubic remainder follows from evenness and the half-line contraction of Lemma 7.1. Notice that $a_b/a_{1-b} = \mathbf{r}_b^2$.

For clarity, finite-jet dependence can be proved before making any analyticity assumption on the graphs. Differentiate the stationary half-line equations at zero in their weighted sequence space. At each order, the highest orbit derivative is multiplied by the same positive quadratic Jacobi operator. Its inverse is (7.1); all remaining terms contain lower orbit jets and finite graph jets. This gives an induction in the endpoint degree. The weighted derivative estimates justify the differentiated action sums and the differentiated trace-class determinant series. The coefficient of d^{m-1} in the residual-time integral uses the action through degree $2m$ and the amplitude through degree $2m-2$. Indeed, scaling the two endpoints by $\sqrt{d}$ gives degrees $d^{n/2-1}$ for an action term of endpoint degree n and $d^{n/2}$ for an amplitude term. The common Morse coordinates justify this calculation with smooth remainders down to $d=0$. Thus only graph jets through order $2m$ can enter.

Keep all lower jets fixed and vary $q_{r,2m}$. The first varying homogeneous term of a flight length is the sum of the corresponding pure endpoint powers divided by $(2m)!$. Its mixed derivative is zero. In the stationary action, a boundary site is counted once and an interior site twice. The envelope identity therefore gives, with $s_{b,2m} = S_b^{(2m)}(0)$,

$$\begin{aligned} \partial_{q_{b,2m}} s_{b,2m} &= 1 + 2 \sum_{k \geq 1} \varrho^{4mk} = \coth(2m\gamma), \\ \partial_{q_{1-b,2m}} s_{b,2m} &= 2\mathbf{r}_b^{2m} \sum_{k \geq 0} \varrho^{2m(2k+1)} = \mathbf{r}_b^{2m} \operatorname{csch}(2m\gamma). \end{aligned} \quad (11.8)$$

This can equally be computed on finite stationary segments and then passed to the half-line. Every differentiated sum is absolutely convergent. Orbit corrections of degree at least three cannot change the degree- $2m$ terms displayed here.

Write $\beta_{b,2m-2} = [u^{2m-2}]B_b(u)$, using ordinary coefficients. At this degree the edge-product variation vanishes, since the varying length term is pure. The varying interior Hessian is diagonal, with its i th diagonal entry, at a site of varied type, equal to

$$\frac{2(t_i^{(b)})^{2m-2} u^{2m-2}}{(2m-2)!}.$$

The variation of the stationary orbit starts in degree $2m-1$. An unvaried Hessian perturbation starts in degree two. Hence orbit variation does not affect the degree under consideration, and only the linear term of the logarithmic determinant contributes there. The diagonal half-line Green entry is

$$G_{ii}^{(b)} = \frac{1 - \varrho^{2i}}{2a_{(b+i) \bmod 2}}.$$

It follows that

$$\begin{aligned}\partial_{q_{b,2m}}\beta_{b,2m-2} &= -\frac{E_m}{a_b(2m-2)!}, \\ \partial_{q_{1-b,2m}}\beta_{b,2m-2} &= -\frac{\mathfrak{r}_b^{2m}O_m}{a_b(2m-2)!}.\end{aligned}\tag{11.9}$$

For the first formula sum $(1 - \varrho^{4k})\varrho^{4(m-1)k}$ over $k \geq 1$. For the second sum $(1 - \varrho^{2(2k+1)})\varrho^{(2m-2)(2k+1)}$ over $k \geq 0$ and use $a_{1-b}^{-1} = \mathfrak{r}_b^2 a_b^{-1}$. The sums are respectively E_m and O_m . Passing from $\log B_b$ to B_b does not change the first varying coefficient: multiplication by a positive-degree unvaried term would raise its degree.

Residual time and the invertible block. For the same-type law both endpoint quadratic actions have coefficient a_b . Under the density proportional to $(d - a_b(u^2 + v^2)/2)_+$ one has

$$\begin{aligned}\int (d - a_b(u^2 + v^2)/2)_+ du dv &= \frac{\pi d^2}{a_b}, \\ \mathbb{E}(u^{2k}) &= \frac{2\binom{2k}{k}}{2^k(k+1)(k+2)a_b^k} d^k.\end{aligned}\tag{11.10}$$

These identities follow from the angular average $\binom{2k}{k}/4^k$ and a one-dimensional radial integration. First variation of the action has no moving-boundary term, because the residual-time weight is zero at the boundary. Its contribution from $s_{b,2m}$ to $f_{b,m-1}$ is $-K_{m,b}s_{b,2m}$. The amplitude contribution from $\beta_{b,2m-2}$ has coefficient

$$\frac{4\binom{2m-2}{m-1}}{2^{m-1}m(m+1)a_b^{m-1}}.$$

Combining these coefficients with (11.8)–(11.9) gives

$$\partial_{q_{b,2m}}f_{b,m-1} = -K_{m,b}P_m, \quad \partial_{q_{1-b,2m}}f_{b,m-1} = -K_{m,b}\mathfrak{r}_b^{2m}Q_m.$$

But $K_{m,b}\mathfrak{r}_b^{2m} = K_{m,1-b}$. This proves (11.2), including the unequal-curvature factors. Lower nonlinear action and amplitude terms can multiply a first variation only at higher endpoint degree. Thus the computation remains valid at arbitrary fixed lower even jets. The last-jet derivatives are constant in that jet, proving the asserted affinity.

Both E_m and O_m are positive. The identities

$$\coth(2x) + \operatorname{csch}(2x) = \coth x, \quad \coth(2x) - \operatorname{csch}(2x) = \tanh x$$

and the geometric sums give (11.4). For the difference,

$$E_m - O_m = \frac{1}{2}\{\tanh((m-1)\gamma) - \tanh(m\gamma)\},$$

which proves (11.5). The function $\tanh x$ is strictly concave for $x > 0$ and has value zero at zero. Applying strict concavity at $(m-1)\gamma = ((m-1)/m)m\gamma$ proves the strict inequality in (11.5). Thus

$$\det \frac{\partial f_{m-1}}{\partial q_m} = K_{m,0}K_{m,1}(P_m + Q_m)(P_m - Q_m) > 0.$$

There is no denominator involving $\kappa_0 - \kappa_1$.

To write the inverse explicitly, suppose all lower graph jets are already known and let R_m denote their contribution to f_{m-1} . For $z_m = R_m - f_{m-1}$, solve

$$q_m = \operatorname{diag}(K_{m,0}^{-1}, K_{m,1}^{-1}) \frac{1}{P_m^2 - Q_m^2} \begin{pmatrix} P_m & -Q_m \\ -Q_m & P_m \end{pmatrix} z_m.\tag{11.11}$$

The recursion proves uniqueness at every finite order. Its coefficients are analytic in the positive geometric parameters and in lower finite jets: the operations are local length expansions, inversion of a positive quadratic operator, and convergent geometric sums. On any fixed compact set and finite collection of orders, the positive denominators have positive minima. The recursion and its derivatives therefore give the stated locally Lipschitz inverse.

For analytic graphs, equality of all recovered even jets implies equality of the two convergent graph series; their odd jets vanish by assumption. For smooth families, the differentiated relative theorem gives $|f_{j,b,k} - f_{b,k}| \leq C_M \tau^j$ through $k = M - 1$. Apply the finite-order Lipschitz inverse. For large j its inputs stay in the chosen compact neighborhood. The algebraic recursion is defined for every finite coefficient vector at fixed positive geometric data; the finitely many remaining even j are covered by increasing the constant. This completes the proof. $\square$

Corollary 11.2 (Labelled analytic boundary determination). *In the two-independent-even contact class, the gap, the two labelled contact curvatures, and the two normalized even-law germs determine both real-analytic contact graphs. If their participating obstacles have connected real-analytic closed boundaries, each boundary is determined in its contact frame. The conclusion allows equal contact curvatures and does not require congruent obstacles or identical graphs. It makes no determination of unobserved obstacles or of the lattice.*

Proof. The theorem determines each analytic graph germ. To compare two candidate closed boundaries at one contact, parametrize both by oriented arclength from that contact and its fixed tangent. Their curvatures are real-analytic functions on the real line, continued around the closed curves, and agree on an interval. The identity theorem makes them agree everywhere. Uniqueness for the planar Frenet equations with the common initial frame makes the two boundary images identical. Apply the argument at each participating contact. This is exact analytic continuation; no stability of distant boundary points from noisy local data is asserted. $\square$

At equal curvatures, restricting (11.2) to the diagonal $q_{0,2m} = q_{1,2m}$ recovers exactly Theorem A.2.1. The new direction is the antisymmetric one, whose nonvanishing coefficient is (11.5). A single identical-contact calculation does not see that coefficient. Without the evenness assumption, odd graph jets also enter the symmetrized energy datum. The theorem does not identify their fibres; the general relative law and full energy-profile inverse remain valid without that assumption.

12. SIGNED ENDPOINT RIGIDITY AND THE ALL-ORDER DETERMINANT-ONE INVERSE

The scalar onset laws integrate out the signs of the endpoint coordinates and therefore retain only a symmetrized profile. The physical endpoint record is strictly richer. If the signed endpoint positions are retained, the support recovers the two half-line actions before symmetrization. We prove here that those two actions determine the complete labelled contact jets. The proof is written at the level needed by the inverse theorem: the nonlinear half-line stationarity operator is inverted quantitatively in a weighted Banach space, the infinite action is differentiated through finite truncations, and the first occurrence of each new graph jet is isolated by homogeneous degree.

Fix oriented transverse coordinates at the two labelled contacts. For $b \in \{0, 1\}$ use an even number of flights and the orientation for which the first and last impacts have type b . Let $\Pi_{bb,d}^{\text{end},\infty}$ be the limiting conditional endpoint law after residual time has been integrated out. By Theorem 2.1, on a common endpoint box its interior density is proportional to

$$(d - S_b(u) - S_b(v))_+ B_b(u) B_b(v), \quad (12.1)$$

where $B_b > 0$.

Theorem 12.1 (All-order signed-endpoint rigidity). *Let the two labelled facing contacts be C^∞ and strictly convex. No reflection symmetry, equality of the two contacts, evenness, or supplied contact curvature is assumed. Suppose the selected channel and its signed endpoint coordinates are retained.*

There is $d_ > 0$ such that the two families*

$$\{\Pi_{00,d}^{\text{end},\infty} : 0 < d < d_*\}, \quad \{\Pi_{11,d}^{\text{end},\infty} : 0 < d < d_*\}$$

determine the half-line actions S_0, S_1 on a fixed contact neighborhood. If physical time is retained, the onset determines the gap g . Hence the gap, both contact curvatures, and every finite labelled graph jet are determined by the signed endpoint laws.

In the fixed oriented contact frames set $q_{b,n} = \psi_b^{(n)}(0)$, $q_{b,2} = \kappa_b$, and $q_n = (q_{0,n}, q_{1,n})^t$. For each finite $M \geq 2$ Taylor's formula gives

$$\psi_b(y) = \sum_{n=2}^M \frac{q_{b,n}}{n!} y^n + R_{b,M}(y), \quad |\partial_y^r R_{b,M}(y)| \leq C_M |y|^{M+1-r}, \quad 0 \leq r \leq M+1. \quad (12.2)$$

Here the bound uses a C^{M+1} bound on a common contact interval. For smooth contacts no convergence of an infinite Taylor series is asserted. Equal finite graph jets, including arbitrary smooth remainders, are treated in Lemma 12.4 below. Put

$$a_b = S_b''(0), \quad s_n = (S_0^{(n)}(0), S_1^{(n)}(0))^t.$$

Then

$$\gamma = \text{arsinh}(g\sqrt{a_0 a_1}), \quad c = \cosh \gamma, \quad c_0 = c\sqrt{a_0/a_1}, \quad c_1 = c\sqrt{a_1/a_0}, \quad \kappa_b = \frac{c_b - 1}{g}. \quad (12.3)$$

For every $n \geq 3$, s_n depends only on the leading geometry and $q_3, \dots, q_n$, is affine in q_n , and

$$D_{q_n} s_n = M_n := \begin{pmatrix} \coth(n\gamma) & \mathbf{r}_0^n \text{csch}(n\gamma) \\ \mathbf{r}_1^n \text{csch}(n\gamma) & \coth(n\gamma) \end{pmatrix}, \quad \mathbf{r}_b = \sqrt{c_b/c_{1-b}}. \quad (12.4)$$

The determinant of M_n is exactly one. For every fixed M , the finite signed-jet forward map through order M is C^1 with an everywhere invertible differential; on compact positive geometric and finite-jet sets its least singular value is bounded away from zero. Its inverse is locally Lipschitz there.

For real-analytic contacts, equality of the two signed endpoint-law germs as $d \downarrow 0$ and of the physical onset determines the two labelled analytic contact germs. If the participating obstacle boundaries are connected and real analytic, their boundary images are determined in the fixed contact frames. For a smooth compact family and every fixed M , the corresponding finite even-flight support families recover both graph jets through order M with error $O_M(\tau^j)$.

12.1. Support thresholds and the leading geometry.

Recovery of the actions and the quadratic geometry. Positivity of B_b makes the interior support in (12.1) exactly

$$\{(u, v) : S_b(u) + S_b(v) < d\}.$$

Define

$$T_b(u, v) = \inf\{d > 0 : (u, v) \in \text{int supp } \Pi_{bb,d}^{\text{end},\infty}\}.$$

Then

$$T_b(u, v) = S_b(u) + S_b(v), \quad S_b(u) = T_b(u, 0), \quad (12.5)$$

because $S_b(0) = 0$. Thus the signed support recovers the unsymmetrized action.

For a selected j -flight itinerary the infimum of physical times at which the near-onset event has positive probability is fg , hence the time axis gives g . The half-line Hessian identity

$$a_b = \frac{c \sinh \gamma}{g c_{1-b}}, \quad c = \sqrt{c_0 c_1} = \cosh \gamma$$

gives

$$a_0 a_1 = \frac{\sinh^2 \gamma}{g^2}, \quad \frac{a_0}{a_1} = \frac{c_0}{c_1},$$

which is equivalent to (12.3). No curvature is supplied as an external parameter. $\square$

12.2. A quantitative weighted inverse for the nonlinear half-line. Let $\ell_b(y, z)$ be the local flight length from a point of type b with transverse coordinate y to the facing point of type $1 - b$ with coordinate z . For a half-line beginning at type b , set $x_0 = u$ and, for $i \geq 1$, write

$$\mathcal{F}_{b,i}(x, u; q) := \partial_2 \ell_{b+i-1}(x_{i-1}, x_i) + \partial_1 \ell_{b+i}(x_i, x_{i+1}) = 0, \quad (12.6)$$

where contact types are read modulo two. Fix a compact positive geometric set. Its hyperbolic exponent has a positive minimum $\gamma_- > 0$. Choose

$$e^{-\gamma_-} < \rho < 1, \quad (12.7)$$

and define

$$X_\rho = \left\{ x = (x_i)_{i \geq 1} : \|x\|_\rho := \sup_{i \geq 1} \rho^{-i} |x_i| < \infty \right\}.$$

Lemma 12.2 (Weighted Green bound and nonlinear inverse). *Let $H_b^0(q) = D_x \mathcal{F}_b(0, 0; q)$ be the quadratic half-line Jacobi operator. Its inverse has kernel*

$$G_{ik}^{(b)} = \frac{g \sigma_i^{(b)} \sigma_k^{(b)}}{2c \sinh \gamma} \left(e^{-\gamma|i-k|} - e^{-\gamma(i+k)} \right), \quad i, k \geq 1. \quad (12.8)$$

There is $C_0 < \infty$, uniform on the compact geometric set, such that

$$\|G^{(b)} y\|_\rho \leq C_0 \|y\|_\rho. \quad (12.9)$$

After shrinking a common endpoint box, the unique stationary half-line $x^{(b)}(u; q)$ satisfies

$$\|x^{(b)}(u; q)\|_\rho \leq C_1 |u|, \quad (12.10)$$

and, with $H_b(u, q) = D_x \mathcal{F}_b(x^{(b)}(u; q), u; q)$,

$$\|H_b(u, q) - H_b^0(q)\|_{X_\rho \rightarrow X_\rho} \leq C_2 |u|. \quad (12.11)$$

Consequently, after choosing u_* so that $C_0 C_2 u_* \leq 1/2$,

$$\|H_b(u, q)^{-1}\|_{X_\rho \rightarrow X_\rho} \leq 2C_0, \quad |u| \leq u_*. \quad (12.12)$$

For every fixed finite collection of endpoint and geometric derivatives, the corresponding derivatives of $x^{(b)}$ are uniformly bounded in X_ρ .

Proof. The factors $\sigma_i^{(b)}$ take only two positive values on the compact geometric set, so (12.8) gives $|G_{ik}^{(b)}| \leq C e^{-\gamma_-|i-k|}$. If $|y_k| \leq \|y\|_\rho \rho^k$, then for $k \leq i$

$$e^{-\gamma_-(i-k)} \rho^k = \rho^i (e^{-\gamma_-}/\rho)^{i-k},$$

while for $k \geq i$

$$e^{-\gamma_-(k-i)} \rho^k = \rho^i (\rho e^{-\gamma_-})^{k-i}.$$

Both geometric series are uniformly summable because of (12.7); the reflected term in (12.8) is smaller. This proves (12.9).

The nonlinear part of (12.6) is local and vanishes quadratically at the zero orbit. On a weighted ball $\|x\|_\rho \leq R|u|$, its derivative differs from H_b^0 by at most $C(|u| + \|x\|_\rho)$ in X_ρ operator norm. The Green bound therefore makes the standard fixed-point map a contraction on a common ball and gives (12.10). Substitution of that bound gives (12.11). Factor

$$H_b(u, q) = H_b^0(q)[I + G^{(b)}(H_b(u, q) - H_b^0(q))].$$

The bracket differs from the identity by an operator of norm at most $1/2$, so its Neumann series converges and proves (12.12).

Differentiate the fixed-point or stationarity equation. At every order the highest orbit derivative is multiplied by the same operator $H_b(u, q)$; all remaining terms are finite sums of local coefficient derivatives and lower orbit derivatives. The uniform inverse bound, the weighted product estimate, and induction give the last assertion. This is the quantitative version of the half-line construction in Lemma 7.1. $\square$

12.3. Derivative bookkeeping, the infinite action, and the envelope limit. The stationarity equations contain first derivatives of the local flight lengths. Hence differentiating them k times can expose graph derivatives of order $k+1$. We do not infer the jet order of the action from that stationarity count. Instead the action filtration is proved directly from its first variation.

Set

$$\mathcal{A}_b(u, q) = \sum_{i \geq 0} [\ell_{b+i}(x_i^{(b)}(u; q), x_{i+1}^{(b)}(u; q)) - g]. \quad (12.13)$$

The weighted orbit bounds imply absolute and locally uniform convergence of this series and of every fixed finite collection of its mixed derivatives. Its value is $S_b(u)$.

Lemma 12.3 (Finite-truncation envelope identity). *Fix a finite jet coordinate $q_{r,m}$, realized by a smooth local graph variation which equals $y^m/m!$ in a neighborhood containing the statistical contact patch. Then, uniformly for $|u| \leq u_*$,*

$$\partial_{q_{r,m}} S_b(u) = \sum_{i \geq 0} \partial_{q_{r,m}} \ell_{b+i}(x_i^{(b)}(u), x_{i+1}^{(b)}(u)). \quad (12.14)$$

The series and every fixed finite number of u -derivatives converge absolutely. In particular no derivative of the stationary orbit appears in the right-hand side.

Proof. Let

$$A_{b,N}(u, q) = \sum_{i=0}^{N-1} [\ell_{b+i}(x_i^{(b)}, x_{i+1}^{(b)}) - g].$$

Differentiate with respect to $q_{r,m}$. The coefficients of $\partial_{q_{r,m}} x_i$ telescope into the stationarity equations for $1 \leq i \leq N-1$. They vanish. The only orbit-variation term left is the right boundary term

$$R_N = \partial_2 \ell_{b+N-1}(x_{N-1}^{(b)}, x_N^{(b)}) \partial_{q_{r,m}} x_N^{(b)}.$$

The first factor is $O(|u|\rho^N)$ by stationarity and the local Taylor expansion at the zero orbit; the second is $O(\rho^N)$ after any prescribed finite collection of u -derivatives, by Lemma 12.2. Hence

$$|\partial_u^k R_N| \leq C_k \rho^{2N} \quad (12.15)$$

for every fixed k (after harmless adjustment at $u=0$). Thus the boundary term tends to zero uniformly with the derivatives used below.

The direct parameter derivatives of the i th flight are bounded by a fixed multiple of a summable power of ρ^i , because both endpoints of that flight are $O(|u|\rho^i)$ up to a shift of one index. The Weierstrass test therefore permits passage to the limit $N \rightarrow \infty$ and termwise finite-order differentiation. This proves (12.14) and the asserted tail control. $\square$

12.4. Finite smooth remainders and factorization through jets. The coordinate variation in Lemma 12.3 alone does not justify independence from all smooth remainders. The next lemma treats the remainder as a function, before choosing any finite jet coordinates.

Lemma 12.4 (Smooth finite-jet factorization). *Fix $M \geq 2$. Let $\psi_0^{[0]}, \psi_1^{[0]}$ and $\psi_0^{[1]}, \psi_1^{[1]}$ be two smooth strictly convex anchored graph pairs with the same gap and the same graph jets through order M . On a common sufficiently small interval their half-line actions satisfy*

$$|S_b^{[1]}(u) - S_b^{[0]}(u)| \leq C_M |u|^{M+1} \sum_{i \geq 0} \rho^{(M+1)i} = \frac{C_M}{1 - \rho^{M+1}} |u|^{M+1}, \quad b \in \{0, 1\}. \quad (12.16)$$

Consequently $j_0^M S_b^{[1]} = j_0^M S_b^{[0]}$. The constants and contact interval can be chosen uniformly with positive gap and curvature margins and bounded C^{M+2} graph norms on a common initial interval. For additional differentiated estimates, the corresponding higher smooth bounds are required. Bounds on a finite list of jet coefficients alone do not supply these functional remainder bounds.

Proof. Put $\Delta\psi_r = \psi_r^{[1]} - \psi_r^{[0]}$ and $\psi_{r,t} = \psi_r^{[0]} + t\Delta\psi_r$, $0 \leq t \leq 1$. Strict convexity and the anchor conditions persist along this interpolation on a common contact interval. Equal quadratic jets and the common gap fix the quadratic half-line operator. The contraction proof of Lemma 12.2, uniformly in t , gives stationary orbits $x^{(b,t)}$ with

$$|x_i^{(b,t)}(u)| \leq C|u|\rho^i, \quad i \geq 0, \quad x_0^{(b,t)} = u. \quad (12.17)$$

The same implicit-function argument gives C^M dependence on u and smooth dependence on t at the finite orders used here. In particular $\|\partial_t x^{(b,t)}\|_\rho$ is uniformly bounded. These assertions are local statements about graph pairs; no globally realized interpolation of periodic tables is required.

For a flight of starting type r , write

$$h_{r,t}(y, z) = g + \psi_{r,t}(y) + \psi_{1-r,t}(z), \quad \ell_{r,t}(y, z) = \sqrt{h_{r,t}(y, z)^2 + (z - y)^2}.$$

Taylor's remainder bound for the equal M -jets gives $|\Delta\psi_r(y)| \leq C_M |y|^{M+1}$. At fixed endpoints,

$$\partial_t \ell_{r,t}(y, z) = \frac{h_{r,t}(y, z)}{\ell_{r,t}(y, z)} \{\Delta\psi_r(y) + \Delta\psi_{1-r}(z)\}, \quad |\partial_t \ell_{r,t}(y, z)| \leq C_M (|y|^{M+1} + |z|^{M+1}). \quad (12.18)$$

The ratio is between zero and one since the separation is positive.

Apply finite-truncation differentiation to

$$A_{b,N,t}(u) = \sum_{i=0}^{N-1} [\ell_{b+i,t}(x_i^{(b,t)}, x_{i+1}^{(b,t)}) - g].$$

Interior orbit derivatives cancel by stationarity, and $\partial_t x_0 = 0$. Thus its derivative is the sum of the direct variations in (12.18), plus exactly the boundary term

$$B_{N,t} = \partial_2 \ell_{b+N-1,t}(x_{N-1}^{(b,t)}, x_N^{(b,t)}) \partial_t x_N^{(b,t)}.$$

The first factor is $O(|u|\rho^N)$ and the second is $O(\rho^N)$; hence $B_{N,t} \rightarrow 0$ uniformly in t and on the endpoint interval. By (12.17), the direct variations are bounded by $C_M |u|^{M+1} \rho^{(M+1)i}$, a summable majorant. Instead of assuming an interchange of infinite differentiation, integrate the exact finite identity in t and then let $N \rightarrow \infty$. Uniform action convergence, the vanishing boundary term and dominated convergence give

$$S_b^{[1]}(u) - S_b^{[0]}(u) = \int_0^1 \sum_{i \geq 0} \partial_t \ell_{b+i,t}(x_i^{(b,t)}, x_{i+1}^{(b,t)}) dt.$$

This proves (12.16). The actions are C^M by the weighted construction. Taylor's formula then implies equality of their derivatives at zero through order M : a nonzero Taylor coefficient of degree at most M would contradict the displayed $O(|u|^{M+1})$ bound. Every constant used is uniform under the stated functional bounds and positive margins. $\square$

In particular, the finite map from graph jets to action jets is independent of the chosen smooth representatives. One may compute it on polynomial graph representatives with smooth cutoffs on a common small interval. On a compact positive finite-jet set these representatives and intervals can be chosen with uniform finite smooth bounds. This is the finite-dimensional map used below, not an identification of a smooth function with its formal Taylor series.

A flat variation illustrates the distinction. For sufficiently small ϵ and a smooth cutoff equal to one near zero, the two graphs

$$\psi^{[0]}(y) = \frac{\kappa}{2}y^2, \quad \psi^{[1]}(y) = \frac{\kappa}{2}y^2 + \begin{cases} \epsilon e^{-1/y^2}, & y \neq 0, \\ 0, & y = 0 \end{cases}$$

are strictly convex near zero and have identical jets of every finite order, without being equal as functions. Applied for every M , the lemma shows equality of every finite action jet. It does not assert equality of the action functions or of the smooth boundary images. Analytic boundary determination below uses analyticity only after the finite-jet inverse has been established.

Lemma 12.5 (Jet filtration and homogeneous isolation). *Let $m \geq 3$. Holding the lower graph jets fixed, one has*

$$\partial_{q_r, m} S_b(u) = O(u^m). \quad (12.19)$$

Together with Lemma 12.4, this shows that $S_b^{(n)}(0)$ depends only on the graph jets through order n , including independence from smooth remainders and flat variations. For $m = n$, the coefficient of u^n in $\partial_{q_r, n} S_b(u)$ is obtained by substituting only the linear half-line orbit into the pure endpoint monomials. The error is $O(u^{n+1})$, uniformly on compact smooth classes with the required finite derivative bounds. The resulting finite-jet coefficient is uniformly controlled on compact positive finite-jet sets.

Proof. For a flight joining transverse coordinates y, z , let $h(y, z)$ denote its positive horizontal separation. In the anchored local graph representation its length is

$$\ell(y, z) = \sqrt{h(y, z)^2 + (z - y)^2}.$$

If the varied m th graph jet occurs at the y endpoint, then, on the region where the cutoff equals one,

$$\partial_{q_r, m} \ell(y, z) = \frac{h(y, z)}{\ell(y, z)} \frac{y^m}{m!} = \frac{y^m}{m!} + O((|y| + |z|)^{m+2}); \quad (12.20)$$

for a variation at the z endpoint the same formula holds with z^m . Indeed $h/\ell = 1 + O((|y| + |z|)^2)$ at the normal contact. The signs are positive for both facing graphs because increasing either inward graph height increases the horizontal separation.

Lemma 7.1 and Lemma 12.2 give

$$x_i^{(b)}(u) = t_i^{(b)}u + R_i^{(b)}(u), \quad |R_i^{(b)}(u)| \leq C|u|^2\rho^i, \quad (12.21)$$

with

$$t_i^{(b)} = \begin{cases} e^{-i\gamma}, & i \text{ even}, \\ \mathbf{r}_b e^{-i\gamma}, & i \text{ odd}. \end{cases} \quad (12.22)$$

Hence $(x_i^{(b)}(u))^m = (t_i^{(b)})^m u^m + O(u^{m+1}\rho^{mi})$. Insert this and (12.20) into the absolutely convergent envelope sum (12.14). Every summand is $O(u^m\rho^{mi})$, proving (12.19). At degree m only

the first term of (12.20) and the linear orbit in (12.22) survive; every nonlinear orbit correction raises the endpoint degree by at least one. Summability is uniform, so the total remainder is $O(u^{m+1})$. $\square$

This lemma is the derivative-order bookkeeping needed by the arbitrary-order inverse. The orbit derivative of order k may depend on a $(k+1)$ st graph derivative, but the degree- n action jet cannot: a graph variation of order $m > n$ starts at endpoint degree m in the exact envelope formula.

12.5. The determinant-one block at every degree.

Proposition 12.6 (Last-jet block). *For every $n \geq 3$, after the lower graph jets are fixed, s_n is affine in q_n and its derivative is the matrix M_n in (12.4). In particular*

$$\det M_n = 1. \quad (12.23)$$

Proof. By Lemma 12.5, the degree- n first variation is the pure endpoint-power sum on the linear orbit. A boundary site is counted once and every interior site twice. For a half-line starting at type b , varying the n th jet of the starting contact gives

$$\partial_{q_b,n} S_b^{(n)}(0) = 1 + 2 \sum_{k \geq 1} e^{-2n\gamma k} = \frac{1 + e^{-2n\gamma}}{1 - e^{-2n\gamma}} = \coth(n\gamma), \quad (12.24)$$

whereas varying the opposite contact gives

$$\partial_{q_{1-b,n}} S_b^{(n)}(0) = 2\mathfrak{r}_b^n \sum_{k \geq 0} e^{-n\gamma(2k+1)} = \mathfrak{r}_b^n \operatorname{csch}(n\gamma). \quad (12.25)$$

All multiplicities are explicit and the sums converge absolutely. Interchanging the labels gives the second row of (12.4). Since $\mathfrak{r}_0 \mathfrak{r}_1 = 1$,

$$\det M_n = \coth^2(n\gamma) - (\mathfrak{r}_0 \mathfrak{r}_1)^n \operatorname{csch}^2(n\gamma) = 1.$$

The derivative of s_n with respect to q_n is independent of q_n : the linear orbit depends only on the quadratic geometry and the degree- n homogeneous coefficient above contains no higher occurrence of q_n . Therefore the dependence is affine. $\square$

The inverse block is

$$M_n^{-1} = \begin{pmatrix} \coth(n\gamma) & -\mathfrak{r}_0^n \operatorname{csch}(n\gamma) \\ -\mathfrak{r}_1^n \operatorname{csch}(n\gamma) & \coth(n\gamma) \end{pmatrix}. \quad (12.26)$$

If R_n is the contribution of the already reconstructed lower jets, then

$$q_n = M_n^{-1}(s_n - R_n). \quad (12.27)$$

For fixed maximal order M , all coefficients and derivatives in this finite recursion are uniformly bounded on compact positive geometric and finite-jet sets.

Proposition 12.7 (Quantitative tangent isomorphism). *Fix $g > 0$ and $M \geq 2$, and define*

$$\mathcal{A}_M : (\kappa_0, \kappa_1, q_3, \dots, q_M) \mapsto (s_2, s_3, \dots, s_M), \quad s_2 = (a_0, a_1)^t.$$

Then $\mathcal{A}_M$ is C^1 and $D\mathcal{A}_M$ is an isomorphism at every strictly convex labelled contact jet. On every compact positive geometric and finite-jet set there is $c_M > 0$ such that

$$\|D\mathcal{A}_M[h]\| \geq c_M \|h\|. \quad (12.28)$$

Proof. Lemma 12.4 makes $\mathcal{A}_M$ well defined independently of smooth representatives. Choose the local polynomial representatives just described. The weighted implicit-function construction and differentiated action bounds give C^1 dependence on their finite jet coordinates; the result is independent of this choice. Formula (12.3) is a smooth inverse of the quadratic map $(\kappa_0, \kappa_1) \mapsto (a_0, a_1)$ at fixed g . This gives an invertible leading 2×2 diagonal block. For $n \geq 3$ the new pair q_n enters s_n through the diagonal block M_n , while all dependence on lower jets is strictly below the block diagonal. Hence $D\mathcal{A}_M$ is block lower triangular with invertible diagonal blocks. The least singular value is a positive continuous function on a compact set, which proves (12.28). $\square$

Completion of Theorem 12.1. The support threshold and quadratic recovery were proved first. Lemma 12.4 proves independence from arbitrary smooth remainders at each finite order. Lemmas 12.2–12.5 and Proposition 12.6 give the all-order recursion (12.27); Proposition 12.7 gives the quantitative tangent inverse.

For analytic contacts, equality of all recovered labelled jets gives equality of the convergent graph germs. Parametrize two candidate connected analytic boundaries by oriented arclength from the recovered contact frame. Their analytic curvature functions agree on an interval and hence everywhere by the identity theorem; uniqueness for the planar Frenet equations gives equal boundary images. This is exact analytic continuation, not a stability claim for distant boundary points.

For the actual finite even-flight law, positivity of the physical flux makes its endpoint support threshold the exact stationary excess action $E_{j,b}(u, v)$. The differentiated relative estimate gives

$$\|E_{j,b} - (S_b \oplus S_b)\|_{C^M} \leq C_M \tau^j.$$

Taking the slice $v = 0$ and applying the finite-order locally Lipschitz inverse proves the $O_M(\tau^j)$ reconstruction statement. $\square$

Remark 12.8 (Oriented intrinsic meaning). Odd graph jets change sign when the chosen transverse orientation is reversed. Equivalently, the recovered data determine the full oriented curvature jet with respect to arclength at each labelled contact. Thus the inverse is intrinsic after the labelled contact point, tangent line, and an orientation of that line have been fixed; no reflection convention is hidden in the theorem.

13. AN AMPLITUDE-FREE INVERSE FROM ONE ENDPOINT LAW

An endpoint position in the plane supplies a point of the unknown boundary. A signed transverse endpoint supplies only one coordinate of that point. The result below concerns the latter datum. It recovers the two half-line actions from one positive-offset endpoint law of each type, without retaining longitudinal positions, residual times, waiting counts, or the unknown flux amplitudes. The gap remains a separate onset datum. This strengthens the support-family inverse of Theorem 12.1: a continuum of offsets is unnecessary when the density, rather than only its support, is retained.

13.1. A four-density identity. Fix a known $d > 0$. Let $I = [-r, r]$, and suppose on a neighborhood of I that $S(0) = S'(0) = 0$, $S'' > 0$, and $B > 0$. Shrink r until $\max_I S < d/4$. The endpoint law has on I^2 the positive smooth density

$$f(u, v) = Z^{-1} B(u) B(v) \{d - S(u) - S(v)\}, \quad Z > 0. \quad (13.1)$$

Its continuation outside this interior square may vanish at the moving boundary. A law in this class determines the displayed continuous density uniquely: two continuous versions equal almost everywhere on an open set are equal everywhere there. Thus values on the slices below are determined by the probability law, although a finite sample does not supply them without estimation.

Choose a fixed $a \in I \setminus \{0\}$. For a positive density f define

$$R_f(u, v) = \frac{f(u, v)f(0, 0)}{f(u, 0)f(0, v)}, \quad (13.2)$$

$$q_f = \sqrt{1 - R_f(a, a)}, \quad T_f(u) = \frac{1 - R_f(u, a)}{q_f}, \quad (13.3)$$

$$\mathcal{S}_d(f)(u) = \frac{dT_f(u)}{1 + T_f(u)}, \quad \mathcal{B}(f)(u) = \frac{f(u, 0)}{f(0, 0)}(1 + T_f(u)). \quad (13.4)$$

The square root in (13.3) is the positive scalar root at the fixed nonzero anchor, not a pointwise square root differentiated at the degenerate minimum $u = 0$.

Theorem 13.1 (Single-offset law inversion). *For (13.1), all quantities in (13.3)–(13.4) are defined and*

$$\mathcal{S}_d(f) = S, \quad \mathcal{B}(f) = B/B(0) \quad \text{on } I. \quad (13.5)$$

Consequently one same-type signed endpoint law at one known positive offset determines the complete action germ and the normalized amplitude germ. Neither an amplitude model nor its normalizing constant is supplied.

Proof. Set $t(u) = S(u)/(d - S(u))$. Direct cancellation in (13.2) gives

$$R_f(u, v) = \frac{d\{d - S(u) - S(v)\}}{(d - S(u))(d - S(v))} = 1 - t(u)t(v). \quad (13.6)$$

Strict convexity and the normalization of S imply $S(a) > 0$, hence $q_f = t(a) > 0$. Therefore $T_f(u) = t(u)$, and solving this relation for $S(u)$ gives the first identity in (13.5). Also

$$\frac{f(u, 0)}{f(0, 0)} = \frac{B(u)}{B(0)} \frac{d - S(u)}{d} = \frac{B(u)/B(0)}{1 + t(u)},$$

which proves the second. No symmetry of S or B was used. In particular values at negative and positive u remain separate and all odd derivatives of S are retained. $\square$

13.2. Fixed-order stability and finite flights.

Proposition 13.2 (Interior stability of the law inverse). *Fix an integer $M \geq 0$, the interval I , the anchor a , and $d > 0$. On a compact family of densities (13.1) with uniformly positive f , uniformly bounded C^M norms, and $q_f \geq q_* > 0$, there are $\varepsilon_*, C_M > 0$ such that*

$$\|\tilde{f} - f\|_{C^M(I^2)} < \varepsilon_* \implies \|\mathcal{S}_d(\tilde{f}) - S\|_{C^M(I)} + \|\mathcal{B}(\tilde{f}) - B/B(0)\|_{C^M(I)} \leq C_M \|\tilde{f} - f\|_{C^M(I^2)}.$$

The perturbed function need not factorize or have total mass one. It must be defined on a neighborhood of the closed square. The constants are fixed-order constants; they are not uniform as $M \rightarrow \infty$, $a \rightarrow 0$, or $d \downarrow 0$.

Proof. Evaluation on each fixed slice is a bounded linear map on C^M . All density denominators in (13.2) have a common positive lower bound. Products and reciprocal functions are locally Lipschitz in C^M on sets with such bounds, as follows by differentiating $hh^{-1} = 1$ inductively. Thus the ratios are locally Lipschitz in C^M . The scalar $1 - R_f(a, a)$ is at least q_*^2 , so its positive square root has a bounded derivative on a common neighborhood. The denominator $q_{\tilde{f}}$ is then at least $q_*/2$. For the exact density, $1 + T_f \geq 1$; after a further uniform reduction of ε_* , $1 + T_{\tilde{f}} \geq 1/2$. Applying the same product and reciprocal bounds to (13.4) proves the claim. This proof avoids differentiation of $\sqrt{t(u)^2}$ at zero. $\square$

The hypotheses can be arranged uniformly on any compact smooth contact class with positive curvature and gap bounds. Choose one d inside its common positive-offset collar, then a common I on which $S_b(u) < d/4$. A fixed nonzero $a \in I$ has $S_b(a) \geq ca^2$, hence a common positive anchor bound. The amplitude and normalization bounds give a positive lower bound for f on I^2 .

Corollary 13.3 (Finite-flight law reconstruction). *Fix $M \geq 2$ on such a compact class. Let $f_{j,b,d}$ be the successful signed endpoint density for an even j -flight bridge, in the exact contact transverse chart, at the fixed offset d . On the selected interior square, for all sufficiently large even j ,*

$$\|\mathcal{S}_d(f_{j,b,d}) - S_b\|_{C^M(I)} \leq C_M \tau^j. \quad (13.7)$$

More generally an estimated density $\hat{f}_{j,b,d}$ with interior C^M error at most ϵ , together with a gap estimate of error at most ϵ_g , recovers both labelled graph jets through order M with error at most

$$C_M(\tau^j + \epsilon + \epsilon_g), \quad (13.8)$$

provided the errors are sufficiently small. This is a stability statement in a density norm, not an assertion that total variation alone controls arbitrarily high derivatives.

Proof. In the transverse chart the finite endpoint density is a positive normalizing constant times $b_j(u, v)(d - E_j(u, v))_+$. The differentiated relative theorem gives $b_j = B_b \otimes B_b + O_{C^M}(\tau^j)$ and $E_j = S_b \oplus S_b + O_{C^M}(\tau^j)$. On the chosen square the cutoff argument is bounded away from zero. Over the whole common endpoint box the inequality $|x_+ - y_+| \leq |x - y|$, the uniform amplitude bounds, and its finite volume give an $O(\tau^j)$ error for the normalizing integral. That integral has a positive uniform lower bound at fixed d . Division therefore proves $\|f_{j,b,d} - \hat{f}_{j,b,d}\|_{C^M(I^2)} \leq C_M \tau^j$. Apply Proposition 13.2.

Before applying the contact-jet recursion, subtract the zeroth and first Taylor terms of the estimated action. This bounded projection fixes the true anchored action, preserves the error bound, and does not change any coefficient of order at least two. Thus a small finite-flight or density error is not interpreted as an unknown change of contact origin or tangent. For completeness, if $a_b = S_b''(0)$, the leading inversion is

$$\gamma = \operatorname{arsinh}(g\sqrt{a_0 a_1}), \quad c_b = \cosh \gamma \sqrt{a_b/a_{1-b}}, \quad \kappa_b = (c_b - 1)/g.$$

At each $3 \leq n \leq M$, use $q_n = M_n^{-1}(s_n - R_n)$, where R_n is already determined by lower-order graph jets and M_n is the determinant-one block of (12.4). Each of these finitely many operations is smooth on a neighborhood of the compact positive geometric set. Induction in n gives (13.8). No infinite-order condition bound is inferred from the determinant identity. $\square$

13.3. The intrinsic data map.

Theorem 13.4 (Periodic determination from one endpoint law of each type). *For every measured channel e , retain its gap g_e , its signed transverse coordinate conventions, and the two conditional endpoint laws $\Pi_{e,00,d_e}^{\text{end},\infty}$ and $\Pi_{e,11,d_e}^{\text{end},\infty}$ at one known sufficiently small positive offset d_e . No longitudinal coordinate, residual time, waiting count, or amplitude is part of these conditional-law data. Then:*

- (1) *For smooth strictly convex contacts the data determine every finite labelled contact jet, without reflection symmetry or supplied curvatures.*
- (2) *For connected real-analytic incident boundaries they determine the complete boundary images in their separate channel frames.*
- (3) *Suppose the data are realized by a labelled periodic dispersing billiard with connected real-analytic strictly convex obstacle boundaries and the marked lifted-channel graph is connected. Assume that this graph contains a rank-two anchoring pair of signature-rigid*

cycles based at one channel frame and a rooted signature-rigid spanning tree reaching every obstacle orbit, as in Theorem 15.4. Then the data determine the complete marked periodic table and the Euclidean realization of its rank-two lattice, including its Gram form and scale, up to one simultaneous element of $\text{SE}(2)$.

On a compact class all d_e can be the same fixed $d_0 > 0$.

Proof. Theorem 13.1 recovers each action on a neighborhood of zero. The weighted half-line construction, finite-truncation envelope identity and homogeneous isolation in Theorem 12.1 then recover the labelled graph jets. The density cancellation is not a replacement for that inverse: it supplies its unsymmetrized action input. Analytic continuation gives the complete incident boundary images. The intrinsic gluing theorem matches their framed signatures.

For part (3), fix the common starting channel frame of the anchoring pair, and choose an obstacle orbit r incident to that frame. A signature-rigid incidence on an anchoring cycle gives a framed point of this obstacle whose complete oriented curvature signature occurs uniquely. Lemma 14.11 therefore permits us to root the supplied spanning tree at r . This reorients the existing tree; it introduces no channel, incidence, deck displacement, or registration datum. In particular, the analytic image at the new root is already placed in the chosen anchoring frame.

The two signature-rigid cycles determine translation holonomies $V = (v_1 \ v_2)$ in this same frame. If M has their two independent marked deck displacements as columns, then Theorem 15.3 gives

$$L = VM^{-1}, \quad L^T L = M^{-T} V^T V M^{-1}.$$

This determines the lattice, including its scale, but does not by itself place obstacle orbits outside the anchoring cycles. Those placements follow by induction along the rerooted tree. At each edge, the already placed parent boundary has a unique framed point with the prescribed oriented signature. Matching that frame fixes the adjacent channel; its recovered analytic image places the child obstacle, with the marked deck displacement interpreted through the same matrix L . The tree reaches every obstacle orbit, as required by Theorem 15.4.

To verify uniqueness in one gauge, consider two admissible realizations of the data and align their starting channel frames by one element of $\text{SE}(2)$. The local inverse and analytic continuation give the same incident boundary images. Unique signature matching along the two cycles gives the same V , hence the same L . At each successive tree edge, unique matching at the already common parent frame gives the same child placement in both realizations. Induction identifies every obstacle orbit; the common lattice then identifies all its translates. The realizability hypothesis supplies existence and admissibility of this reconstruction, including compatibility with the remaining measured channels. Changing the starting frame applies one simultaneous Euclidean motion to the entire reconstruction, not independent motions to its obstacles. None of these steps uses ambient position samples or an externally supplied lattice metric. Uniform collar bounds give the last assertion. $\square$

This theorem concerns an intrinsic, law-valued observation map with specified signed transverse origins and axes, as does the preceding support-family inverse. It does not treat an unknown centering transformation as a common statistical kernel. Section 38 obtains such origins and axes from a charged physical pilot, and Section 39 implements the law inverse by bounded tests after that pilot. Exact longitudinal positions are available in that richer acquisition protocol and permit the independent direct sampling route of Section 20. Neither route is claimed to be statistically equivalent to the other.

13.4. Equivariant reconstruction of perturbed densities. The inverse on the exact factorized family is independent of its anchor. An extension to nearby densities is not. We now specify an extension which commutes with a change of the common transverse orientation, including

away from the factorized family. The magnitude of the anchor is fixed once; its sign is part of the coordinate convention, not an additional observation.

Write $\varsigma(u) = -u$ and $f^r(u, v) = f(-u, -v)$. For each nonzero $\alpha \in I$, make the anchor in (13.3)–(13.4) explicit:

$$q_\alpha(f) = \sqrt{1 - R_f(\alpha, \alpha)}, \quad T_\alpha(f)(u) = \frac{1 - R_f(u, \alpha)}{q_\alpha(f)}, \quad (13.9)$$

$$S_{d,\alpha}(f) = \frac{dT_\alpha(f)}{1 + T_\alpha(f)}, \quad B_\alpha(f)(u) = \frac{f(u, 0)}{f(0, 0)} \{1 + T_\alpha(f)(u)\}. \quad (13.10)$$

For $M \geq 0$, let $\mathcal{U}_\alpha^M$ be the set of real C^M functions on a neighborhood of I^2 satisfying

$$\inf_{I^2} f > 0, \quad 1 - R_f(\alpha, \alpha) > 0, \quad \inf_I \{1 + T_\alpha(f)\} > 0. \quad (13.11)$$

All these conditions are open in the restricted C^M norm. Neither factorization nor a mass normalization is required. Fix $0 < a < r$. In orientation sector $\epsilon \in \{+1, -1\}$ use $\alpha = \epsilon a$ and set

$$\mathcal{F}_{d,a}(\epsilon, f) = (\epsilon, S_{d,\epsilon a}(f), B_{\epsilon a}(f)).$$

The common reversal acts by $(\epsilon, f) \mapsto (-\epsilon, f^r)$, and on the output by changing the tag and precomposing both functions with ς . For a channel collection the same tag is used for every contact and edge; fixed positive anchor magnitudes may differ between edges.

Proposition 13.5 (Transported-anchor equivariance and stability). *Reflection is an isometric bijection $\mathcal{U}_\alpha^M \rightarrow \mathcal{U}_{-\alpha}^M$, and on this entire domain*

$$\begin{aligned} T_{-\alpha}(f^r) &= T_\alpha(f) \circ \varsigma, \\ S_{d,-\alpha}(f^r) &= S_{d,\alpha}(f) \circ \varsigma, \quad B_{-\alpha}(f^r) = B_\alpha(f) \circ \varsigma. \end{aligned} \quad (13.12)$$

In particular $\mathcal{F}_{d,a}$ is equivariant, also for nonfactorized perturbations. On sets on which

$$\inf f \geq m > 0, \quad \|f\|_{C^M} \leq K, \quad q_{\epsilon a}(f) \geq q_* > 0, \quad \inf(1 + T_{\epsilon a}(f)) \geq \nu > 0, \quad (13.13)$$

it is Lipschitz in each sector with one constant $C_M = C(M, I, a, d, m, K, q_, \nu)$, the same in the two sectors. The estimate holds for every pair of functions satisfying these bounds, not only for a perturbation compared with an exact model member.*

Every compact exact-model family satisfying the hypotheses of Proposition 13.2, together with its reflected copy, has a neighborhood of this form when the anchor is transported. On the exact family the output is $(\epsilon, S, B/B(0))$, independently of the admissible anchor magnitude.

Proof. Substitution in the four-density ratio gives

$$R_{f^r}(u, v) = R_f(-u, -v), \quad q_{-\alpha}(f^r) = q_\alpha(f).$$

It follows that $T_{-\alpha}(f^r)(u) = T_\alpha(f)(-u)$. Substituting in (13.10), including the density ratio on the axis, proves all of (13.12). Each strict inequality in (13.11) is preserved. Reflection is an involution and preserves the supremum of every partial derivative of order at most M , proving the domain and isometry assertions.

For completeness, all uniform estimates can be proved between the two endpoints, without requiring a segment to remain in the formula domain. Restriction to the axis or to a fixed slice has operator norm at most one for the usual maximum-derivative norm. Leibniz's rule bounds products in C^M . If $h \geq m$, differentiating $hh^{-1} = 1$ bounds the C^M norm of h^{-1} in terms of m^{-1} and $\|h\|_{C^M}$. For two such functions, the identity $h^{-1} - k^{-1} = (k - h)h^{-1}k^{-1}$ gives the corresponding difference bound. Applying these estimates to R_f gives

$$\|R_f - R_g\|_{C^M(I^2)} \leq C\|f - g\|_{C^M(I^2)}.$$

At the chosen anchor,

$$|q_\alpha(f) - q_\alpha(g)| = \frac{|R_f(\alpha, \alpha) - R_g(\alpha, \alpha)|}{q_\alpha(f) + q_\alpha(g)} \leq \frac{C}{2q_*} \|f - g\|_{C^M(I^2)}.$$

The lower bound on q_α now gives a difference bound for T_α . Its C^M norm is uniformly bounded by the same estimates. The reciprocal bound using ν , followed by the product bound, proves the claims for $S_{d,\alpha}$ and B_α . Reflection preserves every bound used, so the constants agree in the two sectors.

For the last assertion, on an exact model $1 + T_\alpha \geq 1$ and $q_\alpha = S(\alpha)/(d - S(\alpha)) > 0$. The compact lower bounds give a common neighborhood with positive density denominators, anchor radical and $1 + T_\alpha$. Its reflected copy has exactly the same bounds. Exact recovery and anchor independence follow from (13.6). No differentiated square root at the action minimum is taken. $\square$

Corollary 13.6 (Orbit stability and finite-order contact reconstruction). *The transported-anchor action and amplitude reconstruction descends to the common-orientation quotient with the same fixed-order Lipschitz constant. On a compact positive contact class, the finite-jet reconstruction of Corollary 13.3, with this anchor convention, is also equivariant. Its quotient error bound through any fixed order $M \geq 2$ remains $C_M(\tau^j + \varepsilon + \varepsilon_g)$. Here ε is the interior C^M density error and ε_g is the gap error, as in that corollary.*

Proof. For tagged representatives (ϵ, f) , (δ, g) , let $\eta \in \{0, 1\}$ be the unique value with $\epsilon = (-1)^\eta \delta$. Define their orbit distance by

$$\bar{d}_M([\epsilon, f], [\delta, g]) = \|f - g^{r^\eta}\|_{C^M(I^2)}.$$

It is independent of the chosen representatives because reflection is an isometry. Define the output distance by the sum of the two corresponding function norms, using the same tag alignment. Equivariance and the same-sector estimate in Proposition 13.5 prove the quotient estimate with the same constant. For finitely many contact laws use the corresponding sum or maximum product norm throughout.

The anchored projection $Ph(u) = h(u) - h(0) - uh'(0)$ is bounded on $C^M(I)$ and satisfies $P(h \circ \varsigma) = (Ph) \circ \varsigma$. Reflection fixes the gap and both quadratic action coefficients. It therefore fixes the leading curvatures and γ in (12.3). At degree n , it multiplies both graph and action coefficients by $(-1)^n$. The finite smooth-jet factorization and reflection of the half-line action show that the lower-jet remainder obeys

$$R_n(((-1)^m q_m)_{2 \leq m < n}) = (-1)^n R_n((q_m)_{2 \leq m < n}).$$

One may verify this identity on smooth polynomial representatives; finite smooth-jet factorization makes it independent of their cutoffs and remainders. The block M_n depends only on the reflection-invariant quadratic geometry. Thus the recursion $q_n = M_n^{-1}(s_n - R_n)$ commutes with reflection, by induction in n . Its finitely many operations are smooth on the compact positive set and on a sufficiently small neighborhood of it. Combining their Lipschitz bounds with the density estimate and the finite-flight estimate proves the stated error bound. The alignment argument above then gives the same bound on orientation orbits. $\square$

The extension just constructed is an action, amplitude and finite-jet reconstruction on its specified neighborhood. An arbitrary perturbed density is not asserted to be the exact law of a periodic billiard, and its output is not asserted to admit analytic continuation or global incidence gluing. The exact global classification still uses the exact law image. All constants remain fixed-order constants.

Remark 13.7 (Why a fixed signed anchor cannot be used in both sectors). For a fixed signed anchor the identity is instead $S_{d,a}(f^r) = S_{d,-a}(f) \circ \varsigma$. This cannot in general be replaced by $S_{d,a}(f) \circ \varsigma$. For example, take $d = 1$ and

$$f_\zeta(u, v) = \frac{2}{\pi}(1 - u^2 - v^2)_+ \{1 + \zeta uv(u + v)\}, \quad 0 < \zeta \leq 1/100.$$

This is a probability density on the unit disk: its perturbation integrates to zero under simultaneous reversal, and its multiplier is positive. On $[-1/3, 1/3]^2$ it is a positive smooth perturbation of the exact density with $S(u) = u^2$. Set $a = 1/4$, $D = 1/225$, and $h = 2\zeta a^3(224/225)$. Direct substitution yields

$$T_a(f_\zeta^r)(a)^2 = D + h, \quad T_a(f_\zeta)(-a)^2 = \frac{D^2}{D - h}, \quad (D + h) - \frac{D^2}{D - h} = -\frac{h^2}{D - h} \neq 0.$$

Both unsquared quantities are positive. Hence the action outputs differ as well. At $\zeta = 1/100$ the last difference is $-49/2092500$. This is a witness concerning an off-model formula, not physical-table nonidentifiability. Transporting the anchor restores the identity exactly, without removing the perturbation neighborhood.

13.5. What one density adds to one support. The family of support thresholds determines the action, but the support at one offset need not do so. The following functional comparison isolates the additional information used by the four-density identity.

Proposition 13.8 (A support-preserving action change). *On $[-1, 1]$ let $d = 1$, $S(u) = u^2$, and, for $0 < \epsilon \leq 1/10$, set*

$$\phi_\epsilon(s) = s + \epsilon s(1 - s)(s - 1/2), \quad \tilde{S}(u) = \phi_\epsilon(S(u)).$$

Both actions vanish to second order at zero, are strictly convex, and are different. Nevertheless

$$S(u) + S(v) < 1 \iff \tilde{S}(u) + \tilde{S}(v) < 1. \quad (13.14)$$

Their endpoint densities with any positive separable amplitudes cannot have the same four-density invariant on an interior neighborhood of zero.

Proof. For $0 \leq s \leq 1$,

$$\phi'_\epsilon(s) = 1 + \epsilon(-3s^2 + 3s - 1/2) \geq 1 - \epsilon/2 > 0, \quad \phi_\epsilon(1 - s) = 1 - \phi_\epsilon(s).$$

Consequently, for $s, t \in [0, 1]$, $s + t < 1$ is equivalent to $\phi_\epsilon(t) < \phi_\epsilon(1 - s) = 1 - \phi_\epsilon(s)$. This proves (13.14). Also

$$\tilde{S}''(u) = 2\{\phi'_\epsilon(s) + 2s\phi''_\epsilon(s)\}_{s=u^2}, \quad \phi'_\epsilon(s) + 2s\phi''_\epsilon(s) = 1 + \epsilon(-15s^2 + 9s - 1/2) \geq 1 - 13\epsilon/2 > 0.$$

The zero value and derivative at the origin follow by composition. For $0 < S(u) < 1/2$ the correction is nonzero, so the actions differ on every neighborhood of zero. On a sufficiently small interior square, (13.6) and the fixed-anchor inverse recover S from its four-density invariant, irrespective of the positive amplitude. Equality of the invariants would therefore make the two actions equal, a contradiction. $\square$

This example belongs to the functional action-density class. It is not an assertion that both actions have been realized by physical billiard tables satisfying the global hypotheses. The physical theorem uses the relative-law construction for that realization. The comparison shows precisely why retaining density, rather than just one support, enters Theorem 13.1.

13.6. Analytic continuation of a registered boundary germ.

Lemma 13.9 (A registered analytic germ determines the connected boundary). *Let $C, C' \subset \mathbb{R}^2$ be connected embedded real-analytic strictly convex closed curves. Suppose there are points $p \in C$, $p' \in C'$ and an orientation-preserving Euclidean motion A carrying the oriented point–tangent frame at p to that at p' , such that $A(C)$ and C' agree on a nonempty boundary arc containing p' . Then*

$$A(C) = C'.$$

Equivalently, equality of a nonempty registered analytic boundary germ implies equality of the complete connected boundary image.

Proof. Parametrize C and C' by positively oriented arclength from p and p' , with lengths L and L' . Their positive signed curvature functions κ and κ' are real analytic. Equality of the registered arcs gives equality of the curvature germs at the origins. Analytic continuation on the universal arclength covers therefore gives one real-analytic positive function $\tilde{\kappa} : \mathbb{R} \rightarrow (0, \infty)$ which is both L -periodic and L' -periodic and restricts to the two curvature functions.

Strict convexity gives total turning

$$\int_0^L \tilde{\kappa}(s) ds = 2\pi = \int_0^{L'} \tilde{\kappa}(s) ds.$$

If, say, $L' < L$, positivity would give $\int_{L'}^L \tilde{\kappa} > 0$, contradicting these two equalities; the opposite inequality is identical. Hence $L = L'$.

Finally the planar Frenet system

$$X' = T, \quad T' = \tilde{\kappa}N, \quad N' = -\tilde{\kappa}T$$

has unique solutions for a prescribed oriented point–tangent frame. The motion A matches those initial data, so the two arclength parametrizations agree on $[0, L]$. Their complete boundary images are therefore equal. $\square$

14. INTRINSIC LIFTED-CHANNEL GRAPHS AND ANALYTIC TABLE RIGIDITY

The signed one-channel inverse is formulated in a channel-adapted oriented frame. In a periodic billiard a measured channel also carries a deck translation. A global inverse theorem must therefore reconstruct both the relative placement of distinct channel frames and their placement relative to the periodic lattice.

Let $\Lambda_{\text{abs}} \simeq \mathbb{Z}^2$ be a marked abstract lattice with its fixed Euclidean Gram form and the orientation induced by its marked basis. An oriented Euclidean realization is an orientation-preserving linear isometry

$$\iota : \Lambda_{\text{abs}} \otimes \mathbb{R} \longrightarrow \mathbb{R}^2.$$

The quotient obstacle orbits are labelled $1, \dots, N$. An oriented measured channel is indexed exactly as in Section 2:

$$e = (a_e, b_e, \ell_e), \quad \ell_e \in \Lambda_{\text{abs}},$$

and joins a lift of obstacle a_e to the lift of obstacle b_e translated by ℓ_e . Its two incidences are denoted $(e, -)$ and $(e, +)$, with

$$r(e, -) = a_e, \quad \nu(e, -) = 0, \quad r(e, +) = b_e, \quad \nu(e, +) = \ell_e. \quad (14.1)$$

Reversing the channel replaces (a, b, ℓ) by $(b, a, -\ell)$.

Definition 14.1 (Intrinsic signed lifted-channel datum). For an oriented channel $e = (a, b, \ell)$, its intrinsic datum consists of

- (1) the lifted channel label (a, b, ℓ) in the marked abstract lattice;

- (2) the physical onset g_e ;
- (3) the two same-type signed endpoint-law germs

$$\{\Pi_{00,d}^{e,\text{end},\infty} : 0 < d < d_e\}, \quad \{\Pi_{11,d}^{e,\text{end},\infty} : 0 < d < d_e\};$$

- (4) the sign convention for the transverse endpoint coordinate.

The law is recorded in the channel-adapted frame in which the first contact is $(0, 0)$, the second is $(g_e, 0)$, and the oriented transverse axes are those used by the signed endpoint coordinates. No Euclidean placement of this frame relative to the lattice or to any other measured channel is supplied.

Theorem 12.1 and Lemma 13.9 recover two complete oriented analytic boundary images in the edge frame,

$$C_{e,-}, \quad C_{e,+},$$

representing the lifted boundaries at deck offsets 0 and ℓ_e . The remaining inverse problem is a periodic Euclidean gluing problem.

14.1. Analytic signatures and local matching. Let C be a connected oriented real-analytic strictly convex closed curve, parametrized by oriented arclength $s \in \mathbb{R}/L\mathbb{Z}$, with signed curvature $\kappa_C(s) > 0$. At $p = C(s_0)$ define

$$\mathbf{j}_C(p) = (\kappa_C(s_0), \kappa'_C(s_0), \kappa''_C(s_0), \dots).$$

Let $\text{Sym}^+(C) \subset \text{SE}(2)$ be the orientation-preserving Euclidean symmetry group of the boundary image.

Lemma 14.2 (Coincident analytic signatures are exactly symmetry orbits). *For framed points $p, q \in C$,*

$$\mathbf{j}_C(p) = \mathbf{j}_C(q)$$

if and only if an element of $\text{Sym}^+(C)$ carries the oriented framed point p to the oriented framed point q .

Proof. The reverse implication is immediate. For the forward implication choose arclength so that $p = C(0)$ and $q = C(a)$. Equality of the complete curvature jets gives equality of the real-analytic germs $\kappa_C(s)$ and $\kappa_C(s + a)$, hence equality for every s by the identity theorem. Let $A \in \text{SE}(2)$ send the oriented point–tangent frame at 0 to that at a . Uniqueness for the planar Frenet system gives $AC(s) = C(s + a)$ for all s , so $A \in \text{Sym}^+(C)$. $\square$

Thus a circle has a genuine continuous phase ambiguity and a cyclically symmetric analytic obstacle may have a finite one. These are actual inverse ambiguities, not registration conventions.

14.2. Admissible periodic gluings. Let $\tau_v(x) = x + v$. An *algebraic periodic gluing* of the intrinsic data is a pair

$$(\iota, (A_e)_{e \in E}), \quad A_e \in \text{SE}(2), \tag{14.2}$$

where ι is an oriented Euclidean realization of Λ_{abs} , such that whenever two incidences (e, σ) , (f, σ') carry the same quotient obstacle label,

$$\tau_{-\iota(\nu(e, \sigma))} A_e C_{e, \sigma} = \tau_{-\iota(\nu(f, \sigma'))} A_f C_{f, \sigma'} \tag{14.3}$$

as oriented labelled boundary images. Thus every recovered lift is translated back by its deck vector before copies of the same quotient obstacle are identified.

The common values in (14.3) define quotient obstacle images $C_1, \dots, C_N$. Extend them by all deck translations $\iota(\Lambda_{\text{abs}})$. We call the gluing *admissible* if this periodic family is a dispersing billiard table: distinct translated obstacle closures are disjoint and every measured edge has the

prescribed facing contacts, onset and selected closest-pair channel geometry. These are conditions on the reconstructed curves and their candidate placements.

The group $\text{SE}(2)$ acts by

$$B = (R, t) : \quad \iota \mapsto R\iota, \quad A_e \mapsto B \circ A_e.$$

Both (14.3) and admissibility are invariant. Let $\mathfrak{G}_{\text{per}}(\mathcal{D})$ be the quotient of the admissible periodic gluings by this action.

Theorem 14.3 (Intrinsic periodic reconstruction and holonomy classification). *Let two labelled periodic dispersing billiard tables have the same marked abstract Euclidean lattice, the same obstacle-orbit labels, and the same connected measured lifted-channel graph. Assume every obstacle boundary is connected, real analytic and strictly convex, and that the two tables have identical intrinsic signed lifted-channel data for every measured edge. Then:*

- (1) *every incident lifted obstacle boundary is recovered as a complete analytic image in its own channel frame;*
- (2) *the observations determine the admissible periodic gluing space $\mathfrak{G}_{\text{per}}(\mathcal{D})$ without a common laboratory registration;*
- (3) *labelled periodic dispersing tables realizing the observations, modulo one simultaneous orientation-preserving Euclidean motion of the lattice and all obstacle images, are in bijection with $\mathfrak{G}_{\text{per}}(\mathcal{D})$.*

In particular, if $\mathfrak{G}_{\text{per}}(\mathcal{D})$ is a singleton, the complete labelled periodic table is intrinsically determined up to one global $\text{SE}(2)$ motion.

For the common-orientation orbit of the entire marked datum, as defined in Definition 14.12, the corresponding classification is modulo $\text{E}(2)$, by Theorem 14.14. This assertion concerns one common orientation choice, not pointwise deletion of endpoint signs.

Proof. The first assertion follows from Theorem 12.1 and Lemma 13.9. Hence every curve entering (14.3) is determined by the observations.

A table realization supplies its lattice realization ι and the actual placement A_e of every channel frame. The incidence (e, σ) lies on the lift $D_{r(e, \sigma)} + \iota(\nu(e, \sigma))$, so translating it back by the deck vector gives the same quotient obstacle image at every incidence with that label. The resulting gluing is admissible, and simultaneous global Euclidean motion changes only its quotient representative.

Conversely an admissible gluing defines

$$C_r = \tau_{-\iota(\nu(e, \sigma))} A_e C_{e, \sigma}$$

for any incidence with $r(e, \sigma) = r$. The gluing constraint makes this independent of the choice. The periodic lifts are $C_r + \iota(\lambda)$, $\lambda \in \Lambda_{\text{abs}}$. For a measured edge $e = (a, b, \ell)$, its placed endpoint curves are exactly C_a and $C_b + \iota(\ell)$, with the observed onset and oriented signed contact convention. Admissibility makes the full periodic family a dispersing billiard table with the declared measured channels. These constructions are inverse modulo the common $\text{SE}(2)$ action. $\square$

14.3. Cycle holonomy and an intrinsic uniqueness criterion. If two incidences of the same obstacle orbit have contact signatures occurring at unique framed points, Lemma 14.2 gives a unique orientation-preserving Euclidean congruence between their recovered curve copies. We call such an incidence transition *signature-rigid*.

Consider a closed oriented channel cycle

$$c = e_1^{\epsilon_1} \cdots e_m^{\epsilon_m}, \quad \epsilon_i \in \{\pm 1\},$$

for which every required incidence transition is signature-rigid. Its marked deck displacement is

$$\eta_c = \sum_{i=1}^m \epsilon_i \ell_{e_i} \in \Lambda_{\text{abs}}. \quad (14.4)$$

Compose the unique data-determined incidence congruences around the cycle in the starting channel frame and denote the resulting Euclidean motion by H_c .

Lemma 14.4 (Periodic cycle-holonomy identity). *For realizable intrinsic data, H_c has identity rotational part. Hence $H_c = \tau_{v_c}$ for a data-determined vector v_c . For every admissible periodic gluing and every placement $A_{e_0}(x) = R_{e_0}x + t_{e_0}$ of the starting channel frame,*

$$R_{e_0}v_c = \iota(\eta_c). \quad (14.5)$$

Equivalently, after using the global gauge to set the starting frame rotation to the identity, $\iota(\eta_c) = v_c$.

Proof. For two consecutive incidences of the same quotient obstacle, the gluing equation identifies their placed curve copies after the corresponding deck translations. The unique signature-rigid congruence between the two recovered edge-frame copies is therefore conjugate, by their placements, to translation by the difference of their deck offsets. Compose these identities around the cycle. All intermediate edge placements telescope. The signed deck differences sum to η_c , giving

$$A_{e_0}H_cA_{e_0}^{-1} = \tau_{\iota(\eta_c)}.$$

Conjugation by A_{e_0} preserves the rotational part, so H_c is a translation. Comparing translation vectors gives (14.5). $\square$

Definition 14.5 (Lattice-anchoring cycle). A signature-rigid cycle is *lattice-anchoring* if $\eta_c \neq 0$. Because the marked basis fixes the orientation and Gram form of the abstract lattice, the image of one nonzero vector determines the orientation-preserving realization ι relative to the starting channel frame through Lemma 14.4.

Definition 14.6 (Rooted signature-rigid spanning tree). Fix a quotient obstacle orbit r_* . A spanning tree rooted at r_* is *signature-rigid* if its edges can be ordered so that each edge joins an already reached obstacle orbit to one new orbit, and its contact on the already reached orbit has a curvature signature occurring at a unique framed point of that obstacle. For a one-vertex quotient graph the empty tree is allowed.

Corollary 14.7 (Fully intrinsic periodic table rigidity). *Under the hypotheses of Theorem 14.3, assume the measured network contains a lattice-anchoring cycle and, rooted at an obstacle orbit met by that cycle, a signature-rigid spanning tree. Then $\mathfrak{S}_{\text{per}}(\mathcal{D})$ is a singleton. Hence the intrinsic signed lifted-channel data determine the complete labelled periodic billiard table up to one simultaneous global element of $\text{SE}(2)$. No absolute contact point, absolute tangent frame, inter-channel placement, or lattice orientation relative to a channel frame is supplied in advance.*

Proof. Use the global Euclidean gauge to place the starting frame of the anchoring cycle. The cycle places at least one recovered obstacle lift and therefore a root obstacle image. Since $\eta_c \neq 0$, Lemma 14.4 fixes the orientation-preserving lattice realization ι relative to that frame. Traverse the rooted spanning tree. At each edge, the unique signature on the already placed obstacle fixes the edge-frame transform, and the known deck correction $\iota(\ell_e)$ fixes the new quotient obstacle image. Induction places all obstacle orbits. Thus there is at most one algebraic gluing modulo the initial global gauge. Since the hypotheses concern data realized by a dispersing table, that unique gluing is admissible. Hence the admissible gluing space is a singleton. $\square$

Corollary 14.8 (One-root registration is sufficient once the lattice is known). *Suppose an oriented Euclidean realization of the periodic lattice and the Euclidean placement of one root channel frame relative to that lattice are supplied as ambient calibration. Then a rooted signature-rigid spanning tree determines every obstacle image and every remaining measured channel frame. Thus one root registration suffices; a common laboratory registration of all channels is not required.*

Remark 14.9 (Symmetry, connectivity, holonomy and admissibility). The sufficient conditions above are not necessary. Finite obstacle symmetries produce finitely many possible incidence matches, and several cycles can eliminate those choices. The exact object controlling uniqueness is the admissible periodic gluing space $\mathfrak{G}_{\text{per}}(\mathcal{D})$. It separates four genuine issues: obstacle symmetry, channel-network connectivity, periodic lattice holonomy and global geometric admissibility.

Corollary 14.10 (Common laboratory registration as a special case). *If the Euclidean placement of the lattice and of every measured channel frame is supplied externally, then the periodic gluing is fixed a priori. A family of measured channels meeting every obstacle orbit therefore determines the whole analytic table in that registration, recovering the registered v22 statement as a special case.*

14.4. Transporting the root to an anchoring cycle. The root of a signature-rigid spanning tree need not be prescribed at the starting frame of an anchoring cycle. The next lemma shows that these two choices are compatible under the existing signature hypotheses. Thus the propagation of obstacle placements requires no additional registration or root-incidence assumption.

Lemma 14.11 (Rerooting a signature-rigid spanning tree). *Let a rooted signature-rigid spanning tree join a finite collection of connected real-analytic strictly convex obstacle boundaries. Suppose one vertex r has a framed boundary point at which the complete oriented curvature signature occurs uniquely. Then the same undirected tree, rooted instead at r , is signature-rigid. In particular, an obstacle orbit met by a signature-rigid anchoring cycle may be chosen as the root.*

Proof. First, uniqueness of the complete signature at one framed point of a boundary C implies

$$\text{Sym}^+(C) = \{\text{id}\}.$$

Indeed, every symmetry carries that framed point to one with the same signature, by Lemma 14.2. Uniqueness makes it fix the point and its oriented unit tangent. An orientation-preserving Euclidean motion fixing such a frame is the identity. The same lemma then shows that the complete signature occurs uniquely at every framed point of C . Thus uniqueness is a property of this boundary, rather than of a specially chosen incidence on it.

Let r_0 be the original root. Every vertex which has a child in the original rooted tree has a unique-signature contact on an outgoing edge, by Definition 14.6. The preceding argument makes all its framed boundary signatures unique. Write the unique tree path from r_0 to r as

$$r_0 = v_0, v_1, \dots, v_k = r.$$

Rerooting reverses precisely the edges of this path. On the reversed edge from v_i to v_{i-1} , with $1 \leq i < k$, the new parent v_i was the old parent of v_{i+1} and therefore has unique signatures at every framed point. For $i = k$, the same conclusion follows from the hypothesis on r . Every reversed edge consequently has a unique-signature contact on its new parent. Off the path, the parent-child relations are unchanged and retain their original property. This proves the assertion; when $k = 0$, including the one-vertex empty tree, nothing is changed.

Every required incidence transition on a signature-rigid cycle has a unique framed signature by its definition. A vertex met by that cycle therefore satisfies the hypothesis on r , proving the last assertion. $\square$

For the rank-two recovery of Theorem 15.4, fix the common starting channel frame of the anchoring pair and choose a root obstacle met there. Lemma 14.11 transports any supplied rooted signature-rigid spanning tree to that obstacle. The two cycle holonomies determine the lattice in this frame, and induction along the rerooted tree then places every obstacle orbit. The same argument applies to the single-offset reconstruction in Theorem 13.4. The tree is only reoriented; no edge, obstacle label, deck displacement, or observation is added or removed.

14.5. The common-orientation quotient of the marked datum. There are two different operations on signed observations: changing one orientation convention for the entire experiment, and discarding signs from individual observations. Only the former is involved in passing from oriented to unoriented Euclidean classification. We define it at the level of the complete law-valued datum.

Let $J(x, y) = (x, -y)$ and $r(u, v) = (-u, -v)$. For a signed channel-law collection $\mathcal{D}$, write $\mathcal{R}_0\mathcal{D}$ for the collection obtained by replacing, for every measured edge, both contact types and every retained offset,

$$\Pi_{bb,d}^{e,\text{end},\infty} \quad \text{by} \quad r_*\Pi_{bb,d}^{e,\text{end},\infty}. \quad (14.6)$$

The same involution is used throughout the collection. The obstacle labels, the directed edge labels, the marked deck elements and the onset gaps are unchanged. In particular this operation does not reverse the directed channel (a, b, ℓ) . For a density it reads $f_{e,b,d}^r(u, v) = f_{e,b,d}(-u, -v)$. When residual time is present its coordinate is left fixed, and when a full transcript is present its designs, counts, failure symbols and stopping rule are transported together with the same change of endpoint coordinates.

The orientation of a marked lattice requires some care. In the preceding signed formulation ι is required to preserve the orientation of the marked basis. A reflection of the physical table does not preserve this sector. To describe the common reflection action, let $\epsilon \in \{+1, -1\}$ denote the orientation character of ι . Write $\mathcal{D}_\epsilon = (\epsilon, \mathcal{D})$, and let $\mathfrak{G}_{\text{per}}^\epsilon(\mathcal{D})$ be the gluing space defined by (14.2)–(14.3), except that the isometry ι has orientation character ϵ . The placements A_e are still proper Euclidean motions of the consistently oriented channel frames. The original signed statement is the $\epsilon = +1$ sector. The proof of its classification applies in either sector: it uses the incidence equations and Euclidean isometry of ι , not the sign of its determinant.

The character ϵ records the common orientation convention relative to the marked lattice; it is not a supplied lattice angle or a laboratory registration. In an unoriented representative it must be transported with the transverse convention, rather than erased independently.

Definition 14.12 (Common-orientation datum). On the complete marked signed data define

$$\mathcal{R}(\epsilon, \mathcal{D}) = (-\epsilon, \mathcal{R}_0\mathcal{D}), \quad \mathcal{R}^2 = \text{id}. \quad (14.7)$$

The common-orientation datum is the two-element orbit

$$\overline{\mathcal{D}} = \{(\epsilon, \mathcal{D}), (-\epsilon, \mathcal{R}_0\mathcal{D})\}.$$

Thus it retains all relative sign choices, including their relation to the marked lattice convention, but not a choice of the one common orientation. It is an orbit of the entire collection of probability laws, not a mixture of those laws and not a pushforward under pointwise absolute values.

Lemma 14.13 (Equivariance of recovery and gluing). *Reflection induces a well-defined involution between the two signed gluing spaces belonging to $\overline{\mathcal{D}}$. On representatives it is*

$$(\iota, (A_e), (C_{e,\sigma})) \longmapsto (J\iota, (JA_eJ), (JC_{e,\sigma})). \quad (14.8)$$

Reflected boundary images are given their canonical positive boundary orientation. The recovered quotient obstacle images are JC_r , and all labelled deck translations are transported by the same J .

Proof. A reflected contact graph is $\psi_b^r(y) = \psi_b(-y)$. Reflection preserves flight lengths, the reflection law, arclength and the collision flux measure. Sending every endpoint coordinate to its negative therefore gives (14.6); the gap and residual-time coordinate do not change. Consequently the signed inverse and analytic globalization recover exactly the reflected boundary images. Equivalently the half-line actions and their graph jets satisfy

$$S_b^r(u) = S_b(-u), \quad (q_{b,n})^r = (-1)^n q_{b,n}. \quad (14.9)$$

These identities also follow directly by reflecting each finite stationary bridge and then taking the half-line limit. They do not require convergence of an infinite Taylor series for a smooth graph.

Since $\det J = -1$, the realization $J\iota$ lies in the opposite lattice sector while $JA_e J \in \text{SE}(2)$. The identity $J\tau_v = \tau_{Jv}J$ gives, incidence by incidence,

$$\tau_{-J\iota(\nu)}(JA_e J)(JC_{e,\sigma}) = J(\tau_{-\iota(\nu)}A_e C_{e,\sigma}). \quad (14.10)$$

Thus every gluing equation is preserved. Reflection also preserves disjointness, strict convexity, facing contacts, flight lengths and the selected channel geometry; hence it preserves admissibility. Applying the construction twice gives the original gluing.

Finally a different proper representative $B \circ A_e$, with lattice realization $R_B \iota$, changes the reflected representative by the single proper motion $JB J$. Formula (14.8) is therefore well defined after the proper Euclidean quotient. $\square$

Theorem 14.14 (Unoriented marked periodic classification). *Under the analytic and measured-network hypotheses of Theorem 14.3, let $\overline{\mathcal{D}}$ be the common-orientation datum of Definition 14.12. Define*

$$\overline{\mathfrak{G}}_{\text{per}}(\overline{\mathcal{D}}) = \left(\mathfrak{G}_{\text{per}}^\epsilon(\mathcal{D}) \sqcup \mathfrak{G}_{\text{per}}^{-\epsilon}(\mathcal{R}_0 \mathcal{D}) \right) / \langle \mathcal{R} \rangle. \quad (14.11)$$

The labelled periodic tables realizing this datum, modulo one common element of $\text{E}(2)$ acting on the lattice and every obstacle image, are in bijection with $\overline{\mathfrak{G}}_{\text{per}}(\overline{\mathcal{D}})$. In particular, a singleton signed gluing space gives a singleton unoriented gluing space and determines the table up to one common Euclidean isometry. The conclusion applies to a single positive offset at each contact type whenever the signed reconstruction is made using the single-offset inverse.

Proof. Choose either representative of $\overline{\mathcal{D}}$. The signed classification, in its indicated lattice sector, identifies proper-motion classes of realizing tables with the corresponding signed gluing space. Lemma 14.13 identifies the classifications for the two representatives by the same reflection of the entire table.

More explicitly, if two realizing tables are related by a proper Euclidean motion, the signed classification identifies their signed gluing classes. If they are related by an improper motion, write that motion as BJ with $B \in \text{SE}(2)$. Reflection first interchanges the two signed data representatives by $\mathcal{R}$, and B then identifies their proper classes. Conversely, equality in (14.11) is represented either by equality in one signed gluing space or by equality after its common reflected identification. The inverse signed classification yields respectively a proper or an improper global Euclidean motion of the realizing tables. This proves both directions of the bijection. Reflection maps the two signed gluing spaces bijectively, so the singleton assertion follows. The argument uses only the signed inverse, not the number of offsets used to recover its actions. $\square$

Corollary 14.15 (Holonomy and fixed-order stability under the quotient). *The signature-rigid gluing criteria are invariant under the common reflection. If two independent marked cycle columns form M , their recovered holonomy columns form V , and $L = VM^{-1}$, then*

$$V^r = JV, \quad L^r = JL, \quad (L^r)^\top L^r = L^\top L. \quad (14.12)$$

Thus the common-orientation quotient retains the recovered Gram form and scale. Any fixed-order stability estimate for the equivariant signed inverse on its exact law image, valid for the

representative pairs used in reflection-invariant orbit metrics, descends with the same constant to those metrics. On arbitrary perturbed densities the corresponding action, amplitude and finite-jet assertion uses the transported-anchor extension of Proposition 13.5 and Corollary 13.6. It does not use a fixed signed anchor in both orientation sectors.

Proof. Canonical positive arclength reverses under reflection. Hence the curvature signature transforms as $(\kappa, \kappa', \kappa'', \dots) \mapsto (\kappa, -\kappa', \kappa'', \dots)$. Equality and uniqueness of signature matches are preserved. Every incidence congruence is conjugated by J , as is its product around a cycle. Since $J\tau_v J = \tau_{Jv}$, the holonomy vector becomes Jv ; marked cycle columns are unchanged. Formula (14.12) follows from $J^T J = I$. A rooted rigid spanning tree is carried to the same rooted rigid combinatorial tree with all its matches reflected.

For the stability assertion on exact laws, let F be the equivariant signed inverse between the relevant fixed-order data and reconstruction spaces, with reflection-invariant metrics d and d' , and suppose $d'(F(D), F(E)) \leq C_M d(D, E)$ for every representative pair entering the orbit comparison. If the estimate is formulated sectorwise, use only same-sector representative comparisons; their reflected estimates have the same constant. Define the orbit distances by

$$\bar{d}([D], [E]) = \min_{\eta \in \{0,1\}} d(D, \mathcal{R}^\eta E), \quad \bar{d}'([X], [Y]) = \min_{\eta \in \{0,1\}} d'(X, \mathcal{R}^\eta Y).$$

For a sectorwise metric the minima are restricted to the nonempty set of same-sector comparisons. Choose a minimizing η and use equivariance of F . Then

$$\bar{d}'([F(D)], [F(E)]) \leq d'(F(D), F(\mathcal{R}^\eta E)) \leq C_M \bar{d}([D], [E]).$$

For contact jets and densities on symmetric compact boxes, reflection invariance follows from (14.9) and the usual fixed-order supremum norms. This statement retains the order-dependent constants and hypotheses of the signed estimates; it asserts no order-independent stability for analytic continuation. For perturbed densities the same argument applies to the explicitly equivariant extension in (13.12), on its stated domain. The fixed-anchor formula alone has no such off-model equivariance; see Remark 13.7. No arbitrary perturbed density is thereby declared globally realizable. $\square$

Remark 14.16 (Law-valued orientation orbits are not folded records). For a continuously distributed endpoint pair, the pushforward under $(u, v) \mapsto (|u|, |v|)$ has density, on the open positive quadrant,

$$f_{\text{abs}}(a, b) = f(a, b) + f(-a, b) + f(a, -b) + f(-a, -b). \quad (14.13)$$

The coordinate axes have zero mass in this setting. This sum is not the law-valued orbit $\{f, f \circ r\}$. Even pushing each observed pair to its orbit under simultaneous reversal is a sample-space coarsening, not an orbit of the complete probability law. Randomly symmetrizing a law is a further operation, and independently reversing different channel collections gives an action of $\{\pm 1\}^E$, rather than the single diagonal group used above.

For example, on $[-1, 1]^2$, the positive normalized symmetric densities

$$f_1(u, v) = \frac{1}{4}(1 + \alpha(u + v)), \quad f_2(u, v) = \frac{1}{4}(1 + 2\alpha(u + v)), \quad 0 < \alpha < \frac{1}{4},$$

have the same folded density, identically one on $(0, 1)^2$, but their law-valued common-reversal orbits are different: the coefficient of $u + v$ can only change sign under r , not double. This example distinguishes the observation maps in a class of densities. It is not a claim that these two densities are realized by physical billiard tables, nor a physical nonidentifiability theorem for folded billiard observations. All rigidity statements above use the specified law-valued common-orientation datum.

15. RANK-TWO HOLONOMY AND RECOVERY OF THE PERIODIC LATTICE

The preceding intrinsic reconstruction theorem was stated relative to a marked abstract lattice carrying a prescribed Euclidean Gram form. The holonomy identity in fact contains more information. With two independent deck cycles based in one channel frame, the Euclidean realization of the marked lattice, including its metric and absolute scale, is determined by the channel data. We record that stronger form here and use it in the global statistical theorem below.

Let $\Lambda_{\text{top}} \simeq \mathbb{Z}^2$ be only an oriented marked free abelian group of rank two. No inner product on $\Lambda_{\text{top}} \otimes \mathbb{R}$ is supplied. A periodic Euclidean realization is an orientation-preserving linear isomorphism

$$L : \Lambda_{\text{top}} \otimes \mathbb{R} \longrightarrow \mathbb{R}^2.$$

It induces the Gram form

$$G_L(\eta, \zeta) = \langle L\eta, L\zeta \rangle_{\mathbb{R}^2}. \quad (15.1)$$

The lifted channel label $e = (a, b, \ell)$, with $\ell \in \Lambda_{\text{top}}$, remains part of the marked acquisition protocol. Thus the deck class is known combinatorially, but neither its Euclidean length nor its direction is supplied.

An algebraic gluing is now defined exactly as in Section 14, except that the unknown lattice variable is $L \in \text{GL}^+(2, \mathbb{R})$, rather than an isometry from a pre-metrized lattice. The incidence relation is

$$\tau_{-L\nu(e, \sigma)} A_e C_{e, \sigma} = \tau_{-L\nu(f, \sigma')} A_f C_{f, \sigma'} \quad (15.2)$$

whenever the two incidences carry the same quotient-obstacle label. Admissibility again means that the periodic translates form a dispersing billiard and that the declared measured edges are the prescribed facing channels. Simultaneous Euclidean motion acts on the whole realization in the usual way.

Definition 15.1 (Rank-two anchoring pair). Two signature-rigid closed channel cycles c_1, c_2 , based at the same oriented channel frame, form a *rank-two anchoring pair* if their marked deck displacements

$$\eta_i = \sum_{e \in c_i} \epsilon_e \ell_e \in \Lambda_{\text{top}}, \quad i = 1, 2,$$

are linearly independent in $\Lambda_{\text{top}} \otimes \mathbb{R}$.

The incidence congruences are determined entirely by the recovered analytic curve copies. Hence, as in Lemma 14.4, their composition around c_i is a data-determined Euclidean motion $H_i = \tau_{v_i}$. The next proposition is the metric-free version of the cycle-holonomy identity.

Proposition 15.2 (Metric-free cycle holonomy). *For realizable intrinsic data and every signature-rigid cycle c based at an edge frame whose placement has rotational part R_0 ,*

$$R_0 v_c = L \eta_c. \quad (15.3)$$

In particular, no Euclidean structure on the abstract deck group is used in deriving the identity.

Proof. The proof of Lemma 14.4 uses only the deck-addition law and Euclidean translations of the realized table. Replacing $\iota(\nu)$ there by $L\nu$ leaves the telescoping argument unchanged:

$$A_{e_0} H_c A_{e_0}^{-1} = \tau_{L\eta_c}.$$

The rotational part of H_c is therefore the identity, and comparison of translation vectors gives (15.3). $\square$

Theorem 15.3 (Recovery of lattice shape, scale and Gram form). *Assume the intrinsic signed channel data contain a rank-two anchoring pair c_1, c_2 . Then the data determine the full Euclidean realization of the marked lattice relative to the common starting channel frame, up to the one global rotation already contained in the global $\text{SE}(2)$ gauge. More precisely, after fixing that gauge so that the starting-frame rotation is the identity, there is a unique linear map*

$$L : \Lambda_{\text{top}} \otimes \mathbb{R} \longrightarrow \mathbb{R}^2, \quad L\eta_i = v_i \quad (i = 1, 2). \quad (15.4)$$

Consequently the full marked Gram matrix, including absolute scale, is the data-determined matrix

$$[G_L]_{\mathcal{B}} = M_{\mathcal{B}}^{-T} V^T V M_{\mathcal{B}}^{-1}, \quad (15.5)$$

where $M_{\mathcal{B}}$ has as columns the coordinates of η_1, η_2 in the marked basis $\mathcal{B}$, and $V = (v_1 \ v_2)$.

Proof. The vectors η_1, η_2 form a basis of $\Lambda_{\text{top}} \otimes \mathbb{R}$. Equation (15.3) therefore prescribes a unique linear map L once the common starting-frame rotation has been fixed. Realizability implies that L is invertible and orientation preserving, so v_1, v_2 are linearly independent with the compatible orientation. Formula (15.5) is simply $G_L = L^T L$ written in the marked basis. A change of the global Euclidean gauge left-multiplies L by a rotation and hence leaves G_L unchanged. $\square$

Theorem 15.4 (Uncalibrated intrinsic periodic rigidity). *Let two labelled periodic dispersing billiard tables have the same oriented marked deck group $\Lambda_{\text{top}} \simeq \mathbb{Z}^2$, the same obstacle-orbit labels and the same connected measured lifted-channel graph. No lattice Gram form, lattice scale, lattice direction, common channel registration or absolute contact frame is supplied. Assume every obstacle boundary is connected, real analytic and strictly convex, the intrinsic signed channel data agree, the measured graph contains a rank-two anchoring pair, and a rooted signature-rigid spanning tree reaches every obstacle orbit. Then the complete marked periodic table and the Euclidean Gram form of its lattice are determined by the observations up to one simultaneous global element of $\text{SE}(2)$.*

Proof. The signed one-channel inverse and analytic continuation recover every incident obstacle curve in its own channel frame. Fix the common starting frame of the rank-two anchoring pair, and choose an incident obstacle orbit r . A signature-rigid incidence on an anchoring cycle gives a framed point of this obstacle with a unique complete oriented curvature signature. Lemma 14.11 therefore roots the supplied spanning tree at r , without changing an edge, a deck label or the observation map. The recovered image at this root is already placed in the chosen anchoring frame.

Theorem 15.3 fixes the complete lattice realization in this same frame. The rerooted signature-rigid spanning tree then propagates the placements of all quotient obstacle images exactly as in Corollary 14.7, with every deck correction interpreted through that one lattice realization. Thus there is at most one algebraic gluing. Existence is not hidden in the word “admissible”: it is provided by the standing hypothesis that the data are generated by an actual periodic dispersing table. Hence that algebraic gluing is admissible and is the unique realization modulo the initial global Euclidean gauge. $\square$

Remark 15.5 (What is marked and what is recovered). The deck label ℓ is a combinatorial lift label in the marked covering and is part of the acquisition design; its Euclidean vector is not. Unknown or corrupted deck-class labels constitute a different inverse problem and are not silently inferred here. The classification of all admissible gluings is a bookkeeping theorem; the rigidity content is the rank-two holonomy recovery plus signature-rigid propagation. Theorem 15.4 therefore removes the metric calibration singled out in the preceding formulation while keeping the measurement protocol explicit.

16. FINITE ANALYTIC SIGNATURES AND STABILITY OF INTRINSIC GLUING

Exact analytic uniqueness, finite-signature matching, and a compact inverse modulus are three different assertions. We prove them in that order. An outside-arc gap alone does not prove uniqueness inside the correct arc. The local assertion will instead follow from a finite jet embedding, whose order is selected uniformly from the complete analytic signatures.

Fix the marked measured graph, two signature-rigid anchoring cycles with independent marked deck displacements, and a rooted signature-rigid spanning tree. Let $\mathfrak{K}$ be a compact class of their realizations, modulo one simultaneous global element of $\mathrm{SE}(2)$. The designation *signature-rigid* has its meaning in Section 14: each transition used by these cycles and the tree has a unique matching oriented framed point, not merely a unique point after arbitrarily choosing a symmetry phase. Obstacles not used for such transitions may have exact symmetries; the classification theorem for general gluing spaces remains unchanged.

Assume uniform positive curvature, gap, separation, diameter, free-area and persistent-channel collar bounds. For some $\rho, B > 0$, the oriented arclength curvature functions extend holomorphically to strips of width ρ , with absolute value at most B . The class is closed in a fixed root-frame gauge and compact in every fixed labelled C^q topology. A channel margin $\chi > 0$ excludes switching to a competing closest pair. The data-determined holonomy vectors satisfy

$$\sigma_{\min}(v_1(T) \ v_2(T)) \geq \sigma_{\mathrm{hol}} > 0. \quad (16.1)$$

The compactness assumption includes the lattice and marked placements; it is not inferred from a bound on the curvature of a single obstacle alone. These are the uniform geometric assumptions used below.

A transition obstacle has trivial orientation-preserving symmetry group. Indeed any such nonidentity symmetry carries each boundary point to a different point with the same complete oriented signature, contradicting the designated unique framed match. A nonidentity rigid motion preserving a compact strictly convex body is a rotation about an interior point and has no boundary fixed point. In particular, transition obstacles are not circles. This observation uses the exact signature-rigidity assumption already required for unique global gluing; it is not a new reflection or genericity assumption.

16.1. Uniform finite embeddings. Parametrize each transition obstacle by normalized oriented arclength $x \in \mathbb{R}/\mathbb{Z}$, with its origin at a designated incidence. The physical arclength derivative is denoted by ∂_s . Define

$$J_{T,i}^{[m]}(x) = (\kappa_{T,i}(x), \partial_s \kappa_{T,i}(x), \dots, \partial_s^m \kappa_{T,i}(x)).$$

All derivatives with respect to x below are of this vector-valued map. Uniform upper and lower perimeter bounds make normalized and physical arclength distances equivalent. Fixed positive weights could be placed on the coordinates without changing any assertion.

Theorem 16.1 (Uniformly selected finite signature embedding). *There are an integer m , numbers $v, r, \sigma > 0$, and $U, B_2 < \infty$, uniform on $\mathfrak{K}$ and its finite list of transition obstacles, such that, writing $J = J_{T,i}^{[m]}$,*

$$|J'(x)| \geq v, \quad |J'(x)| \leq U, \quad |J''(x)| \leq B_2, \quad (16.2)$$

$$|J(x) - J(y)| \geq (v/2) \operatorname{dist}(x, y), \quad \operatorname{dist}(x, y) \leq r, \quad (16.3)$$

$$|J(x) - J(y)| \geq \sigma, \quad \operatorname{dist}(x, y) \geq r. \quad (16.4)$$

Thus a sufficiently rich finite signature, not necessarily an initially chosen scalar curvature signature, is a uniform embedding of every transition boundary.

Proof. If every derivative of positive order of an analytic curvature function vanished at one point, it would be constant by the identity theorem. Since transition obstacles are not circles, at each (T, i, x) some component of the derivative of a finite signature is nonzero. The parameter space of these triples is compact: use normalized arclength and the finite obstacle list. Each nonvanishing condition persists on a neighborhood. A finite subcover gives an order m_1 for which $|(J^{[m_1]})'|$ is bounded below by a positive constant v . Holomorphic strip bounds give a uniform second derivative bound for this finite vector. Taylor's integral formula then gives

$$|J^{[m_1]}(y) - J^{[m_1]}(x)| \geq v|y - x| - \frac{1}{2}\|(J^{[m_1]})''\|_\infty|y - x|^2 \geq (v/2)|y - x|$$

on a uniform short arc. Choose its radius $r < 1/4$.

For pairs at normalized arclength distance at least r , equality of all signature coordinates would, by Lemma 14.2, give a nonidentity orientation-preserving symmetry. No such symmetry exists on a transition obstacle. Each pair is therefore separated by some finite coordinate. The set of (T, i, x, y) with distance at least r is compact. A second finite subcover supplies an order $m \geq m_1$ and a positive minimum separation σ . Adding coordinates preserves the local lower bound already proved for the m_1 -block. The strip bounds supply finite uniform U, B_2 for the enlarged vector. This proves (16.2)–(16.4). $\square$

The argument also separates a prescribed target arc from its complement: shrink the radius below r , using (16.3) on the intermediate annulus and (16.4) outside. Therefore an existing outside-arc margin may be retained, but it is not used to infer (16.3). The latter is an independently proved consequence of analytic nonconstancy and compactness.

Lemma 16.2 (Unique matching for noisy finite signatures). *There are $\epsilon_0, C > 0$, uniform on the class, with the following property. Fix a true target $J(x_0)$. Suppose a recovered signature curve $\tilde{J}$, expressed in the same normalized arclength parameter, and a target vector y satisfy*

$$\|\tilde{J} - J\|_{C^2} \leq \epsilon, \quad |y - J(x_0)| \leq \epsilon, \quad 0 < \epsilon < \epsilon_0.$$

Then the minimum of $|\tilde{J}(x) - y|^2$ on the parameter circle is attained at one point $\hat{x}$, and

$$\text{dist}(\hat{x}, x_0) \leq C\epsilon. \tag{16.5}$$

Locally this match is Lipschitz in the target vector and in the recovered curve in C^1 . A mere C^0 perturbation of an arbitrary signature curve is not asserted to have a unique nearest point.

Proof. Choose a radius $a < r/4$ so small that $2B_2Ua < v^2/16$, decreasing it if necessary to handle bounded perturbations of the derivatives. Theorem 16.1 gives a positive uniform separation between $J(x_0)$ and every $J(x)$ outside this arc, namely at least $\min(va/2, \sigma)$. For sufficiently small ϵ , no minimizer can lie outside the arc: the value of the residual norm at x_0 is at most 2ϵ , while outside it is larger.

On the arc put $f(x) = \frac{1}{2}|\tilde{J}(x) - y|^2$. Its second derivative is

$$f''(x) = |\tilde{J}'(x)|^2 + (\tilde{J}(x) - y) \cdot \tilde{J}''(x).$$

The derivative lower bound, the choice of a , and then a sufficiently small ϵ_0 give $f'' \geq v^2/8$ throughout the arc. Hence the minimum is unique and interior. At the minimum the residual norm is at most 2ϵ , so

$$|J(\hat{x}) - J(x_0)| \leq 4\epsilon.$$

The local embedding bound proves (16.5), for example with $C = 8/v$. Finally compare the equations $f'(\hat{x}) = 0$ for two nearby data sets. The derivative of f' is at least $v^2/8$, and its change under a perturbation of y and $\tilde{J}$ is bounded by a constant times $|\Delta y| + \|\Delta \tilde{J}\|_{C^1}$. The mean value theorem gives the stated local Lipschitz dependence. $\square$

For example, a scalar curvature with a strict minimum can take each nearby larger value twice inside the correct arc. Adding a finite number of its oriented derivatives as above resolves this failure; declaring the original scalar match unique would not. Exact finite symmetries can instead be handled on a fixed symmetry quotient for a classification problem, but they must not be assigned an arbitrary phase and called a unique Euclidean gluing. The unique-reconstruction theorem uses the signature-rigid transitions specified at the start of the section.

16.2. Propagation through the marked graph.

Lemma 16.3 (Persistence of the prescribed gluing). *On $\mathfrak{R}$, the prescribed incidence matches are unique and continuous. More precisely, if their recovered finite signature curves and targets satisfy the hypotheses of Lemma 16.2, and the recovered point–tangent frames are close in C^1 , then the incidence congruences and their finite graph compositions depend locally Lipschitz-continuously on these quantities. The lattice map obeys*

$$L = VM^{-1}, \quad \|\tilde{L} - L\| \leq \|\tilde{V} - V\| \|M^{-1}\|, \quad (16.6)$$

where the fixed invertible matrix M has the two marked deck vectors as columns and $V = (v_1 \ v_2)$. Rank two persists for $\|\tilde{V} - V\| < \sigma_{\text{hol}}/2$.

Proof. The matching lemma supplies the unique in-arc point; its global separation excludes other arcs. Point and oriented tangent determine the required orientation-preserving rigid motion between the two recovered curve frames. The elementary formula for that motion is smooth in the point and unit tangent data on a fixed orientation chart. Products of a fixed finite number of such motions are locally Lipschitz on bounded point data. Consequently the tree placements and the two cycle holonomies inherit a uniform local bound, with a constant depending on the finite graph. The channel margin prevents a different closest pair from replacing the selected one. Formula (16.6) is direct matrix multiplication by the fixed M^{-1} , not inversion of a noisy matrix V . The singular-value perturbation inequality proves rank persistence. Also

$$\|\tilde{L}^t \tilde{L} - L^t L\| \leq 2\|L\| \|\tilde{L} - L\| + \|\tilde{L} - L\|^2,$$

so the Gram form and subsequent deck corrections are stable as well. $\square$

This lemma concerns already recovered signature curves in the stated norms. It does not turn noisy contact Taylor coefficients into a well-conditioned unrestricted analytic continuation algorithm. The next argument supplies the distinct, non-effective continuity route needed for global statistical consistency.

16.3. The compact inverse modulus. Normalize the action coordinates by

$$x_{e,b,m}(T) = \rho^m S_{e,b}^{(m)}(0; T)/m!, \quad m \geq 2,$$

using any fixed positive scale ρ , and the gaps by g_* . For a fixed enumeration $|e|$ of the finite channel set define

$$\begin{aligned} d_{\mathcal{D}}(T, T') &= \sum_{e,b} \sum_{m \geq 2} 2^{-(m+|e|+3)} \min\{1, |x_{e,b,m}(T) - x_{e,b,m}(T')|\} \\ &\quad + \sum_e 2^{-(|e|+2)} \min\{1, |g_e(T) - g_e(T')|/g_*\}. \end{aligned} \quad (16.7)$$

The truncation by one and the summable weights make this metric compatible with coordinatewise convergence of the intrinsic data. This conclusion requires no assertion that the action holomorphic radius equals the curvature holomorphic radius.

Theorem 16.4 (Uniform inverse modulus). *For every fixed $q \geq 0$,*

$$\omega_q(t) = \sup\{d_q(T, T') : T, T' \in \mathfrak{K}, d_{\mathcal{D}}(T, T') \leq t\} \quad (16.8)$$

is finite, nondecreasing, and tends to zero as $t \downarrow 0$. If $\mathcal{D}_M$ consists of the gaps and action jets through order M , then

$$d_q(T, T') \leq \omega_q(C\|\mathcal{D}_M(T) - \mathcal{D}_M(T')\|_M + C2^{-M}). \quad (16.9)$$

The finite norm is the maximum norm of the normalized retained coordinates; C depends only on the finite graph and the chosen weights.

Proof. Equality of the complete action data and gaps gives the full analytic contact germs by Theorem 12.1. Analytic continuation, exact signature-rigid gluing, and Theorem 15.3 determine the marked periodic table in the root-frame gauge. This exact injectivity uses the complete signatures; it does not depend on a possibly noisy finite matching rule. The complete data map is continuous coordinate by coordinate, by the half-line inverse estimates, and hence continuous in (16.7). A continuous injection from compact $\mathfrak{K}$ into this metric data space has a uniformly continuous inverse on its image. This proves the assertion about ω_q . The terms with $m > M$ in the data metric sum to at most $C2^{-M}$; the retained terms are bounded by the normalized finite maximum norm times C . Substitution gives (16.9). $\square$

Remark 16.5 (Two distinct stability statements). The local finite-signature estimates are proved by immersion, separation and strict convexity of the matching criterion. The global modulus is proved by exact injectivity and compactness. Neither proof substitutes for the other. The tail $C2^{-M}$ is a product-metric truncation bound, not an exponential analytic-continuation error rate. No effective formula for ω_q or a uniform-in-order jet condition number is asserted.

17. PHYSICAL COORDINATES AND FINITE-WINDOW OBSERVATION

The block inverse is a statement about actual boundary laws. We now show that its two sets of coordinates vary independently in physical tables while the complete selected leading hierarchy remains fixed. This also gives physical alternatives for a finite-dimensional observation theorem. The dimension is fixed throughout this section; none of its statistical bounds is a lower bound for the full infinite-dimensional profile experiment.

17.1. Independent realization at fixed leading data.

Theorem 17.1 (A physical open image of two-contact jets). *Fix $M \geq 2$. For any two curvature radii R_0, R_1 sufficiently close to one, there is a real-analytic family of positively curved periodic billiards on $3\mathbb{Z} \times 4\mathbb{Z}$, with $2(M-1)$ parameters, for which the selected horizontal channel has*

$$g = 1, \quad A = 12 - \pi, \quad \kappa_b = R_b^{-1}.$$

Both contact graphs are even, but their higher jets are independent. The parameters can be taken to be either

$$(q_{b,2m})_{b=0,1; 2 \leq m \leq M} \quad \text{or} \quad (f_{b,k})_{b=0,1; 1 \leq k \leq M-1}.$$

Each of these maps is a local analytic diffeomorphism, with inverse derivative bounded on a sufficiently small compact neighborhood. Every member has the same leading selected count and endpoint-metric coefficients at every collision order, in both orientations. The statement includes $R_0 = R_1$ and requires no equality of higher jets.

Proof. Use one obstacle per cell and its support function. Put

$$w_0(\theta) = \frac{1 + \cos \theta}{2}, \quad w_1(\theta) = \frac{1 - \cos \theta}{2},$$

and consider the finite-dimensional analytic family

$$h(\theta) = 1 + \alpha \sin^2 \theta + \beta \cos \theta \sin^2 \theta + \sum_{m=2}^M \{s_{0m} w_0(\theta) + s_{1m} w_1(\theta)\} \sin^{2m} \theta + z \sin^{2M+2} \theta. \quad (17.1)$$

At the two horizontal normals the support values equal one, the support derivatives vanish, and the curvature radii are

$$h(0) + h''(0) = 1 + 2\alpha + 2\beta, \quad h(\pi) + h''(\pi) = 1 + 2\alpha - 2\beta.$$

Thus set $\alpha = (R_0 + R_1 - 2)/4$ and $\beta = (R_0 - R_1)/4$. No s or z changes these data. The obstacle area is

$$\mathcal{S}(h) = \frac{1}{2} \int_0^{2\pi} (h^2 - (h')^2) d\theta.$$

At the unit disk, $\partial_z \mathcal{S} = \int_0^{2\pi} \sin^{2M+2} \theta d\theta > 0$. The analytic implicit-function theorem therefore gives a unique small $z = z(\alpha, \beta, s)$ for which $\mathcal{S} = \pi$. This function is analytic jointly in all the displayed parameters.

For small parameters, $h + h''$ is positive, so the support function defines a strictly convex analytic closed curve. Its right and left horizontal support points remain $(1, 0)$ and $(-1, 0)$. The selected chord from this obstacle to its translate by $(3, 0)$ has length one and normal incidence. The unit-disk configuration has positive clearance and a strict gap between its horizontal nearest neighbors and all other translates; these properties persist on a small neighborhood. For the infinitely many distant translates this follows from a uniform bound on the obstacle diameter, leaving only finitely many translates to check. Thus the constructed chords are physical first hits and form the selected ground channel with its reverse. The free cell area is $12 - \pi$. The function h is even in θ , so the curve is invariant under reflection across the horizontal axis. Each horizontal contact graph is consequently even in its transverse coordinate. Reflection across the vertical axis is not imposed.

At normal zero the s_{0m} perturbation starts in order $2m$, whereas its s_{1m} counterpart starts in order $2m + 2$. At normal π these roles are interchanged. The area compensator starts in order $2M + 2$ at both normals. For completeness, the support-envelope calculation in a contact frame gives

$$\frac{\partial q_{b,2m}}{\partial s_{bm}} = -(2m)! \kappa_b^{2m}, \quad \frac{\partial q_{b,2m}}{\partial s_{1-b,m}} = 0. \quad (17.2)$$

Indeed, at the right contact write the boundary as $x = x(y)$. Stationarity of $(h(\theta) - y \sin \theta) / \cos \theta$ at its supporting normal gives, at fixed y , $\partial_s x = \partial_s h(\theta(y)) / \cos \theta(y)$. Since the radius is R_b , $\theta(y) = \kappa_b y + O(y^3)$. The contact graph is $\psi_b = 1 - x$, giving the negative coefficient in (17.2). At the left contact use the outward horizontal normal and the reflected longitudinal coordinate; the same calculation applies. Even powers make the transverse orientation immaterial.

A perturbation of order $2m$ cannot change graph jets of smaller order. Derivatives of the compensator also have order $2M + 2$. Hence the map from the $2(M - 1)$ variables s to the displayed graph jets has a block lower triangular Jacobian, with the invertible blocks (17.2). It is an analytic local diffeomorphism. Compose it with Theorem 11.1 to obtain the asserted statistical coordinates. Compactly restricting the coordinate neighborhood bounds both inverse derivatives. Finally, the quadratic endpoint Hessian and its normalized mixed derivative depend only on g, κ_0, κ_1 . The full-phase leading count coefficient also uses only A . These four data are fixed. The endpoint-metric leading coefficients and the selected count leading coefficients are therefore fixed at every order, by (5.1)–(5.2) and Proposition A.17.1. This assertion concerns this channel and its reverse, not the leading data of every other channel. $\square$

Corollary 17.2 (Finite-order physical fibres). *For $N > M \geq 2$, use the family of Theorem 17.1 with N in place of M . In a sufficiently small coordinate neighborhood, each nonempty fibre of the two-law jet map through order $M - 1$ is an analytic submanifold of dimension $2(N - M)$. The full two-law jet map through order $N - 1$ is locally injective. All these statements hold with the selected leading hierarchy fixed.*

Proof. In the statistical coordinates furnished by the theorem the first map is the coordinate projection onto its first $2(M - 1)$ coordinates. Its fibres have the stated dimension; the full map is the coordinate chart itself. These are fibres of specified finite physical families, not a classification of the fibres of arbitrary smooth asymmetric contact geometry. $\square$

17.2. A fixed physical experiment. Fix M and one compact coordinate neighborhood from the preceding theorem. Write $n = M - 1$ and $\vartheta = (f_{b,k})_{b=0,1; 1 \leq k \leq n} \in \mathbb{R}^{2n}$. The exact family, its labels, and g, A, κ_0, κ_1 are supplied in the following experiment. A preparation returns the indicator of one selected alternating event, at a specified orientation and time. It does not return an endpoint, a derivative, or a limiting law.

Theorem 17.3 (Finite-window observation and physical separation). *After shrinking the parameter neighborhood, there are $h_0 > 0$ and one even integer $j_0 \geq 2$ such that the $2n$ physical means*

$$\mathcal{P}(\vartheta) = \left(\Pr_{\vartheta} E_{j_0, b}(\ell h_0) \right)_{b=0,1; 1 \leq \ell \leq n} \quad (17.3)$$

form a bi-Lipschitz coordinate on a compact parameter ball Θ with nonempty interior. Here the physical observation time is $j_0 g + \ell h_0$. The constants and the windows depend on the fixed family and M , but not on the requested accuracy.

For $0 < \varepsilon < \varepsilon_0$ and $0 < \eta < 1/4$, a Borel estimator based on independent binary preparations in these windows satisfies

$$\sup_{\vartheta \in \Theta} \Pr_{\vartheta} \{ \|\hat{\vartheta} - \vartheta\| > \varepsilon \} \leq \eta, \quad N_{\text{tot}} \leq C \varepsilon^{-2} \log(C/\eta).$$

The count is deterministic and includes all failures. The same order holds for the corresponding finite graph-jet vector.

Moreover, for all sufficiently large N , the minimax squared risk in this fixed finite-window experiment satisfies

$$\frac{c}{N} \leq \inf_{\hat{\vartheta}, \mathcal{D}_N} \sup_{\vartheta \in \Theta} \mathbb{E}_{\vartheta} \|\hat{\vartheta} - \vartheta\|^2 \leq \frac{C}{N}. \quad (17.4)$$

The infimum allows adaptive and randomized choices of these windows and at most N fresh preparations. Its lower bound uses pairs of physical tables in the stated family, with exactly the same selected leading hierarchy.

Proof. Analyticity of the boundary law in the physical parameters and offset, on a common neighborhood, gives

$$\mathcal{F}_{bb}(d; \vartheta) = 1 + \sum_{k=1}^n \vartheta_{b,k} d^k + d^{n+1} R_b(d; \vartheta),$$

where R_b and its first parameter derivatives are uniformly bounded. At $d = \ell h$ the Jacobian with respect to ϑ is the block diagonal matrix with blocks $V_h = ((\ell h)^k)_{1 \leq \ell, k \leq n}$, plus a matrix of norm $O(h^{n+1})$. The matrix $V_h = V_1 \text{diag}(h, \dots, h^n)$ is invertible: its determinant is a nonzero Vandermonde determinant multiplied by $\prod_{\ell=1}^n \ell$. Also $\|V_h^{-1}\| \leq C h^{-n}$ for $h \leq 1$. Choose a fixed $h = h_0$ so small that $n h_0$ lies in the common collar and the transformed remainder norm is less than $1/4$.

The differentiated relative theorem bounds the difference between $F_{j,b}$ and $\mathcal{F}_{bb}$, including its first parameter derivatives, by $C\tau^j$. Choose a fixed even $j = j_0$ so large that $\|V_{h_0}^{-1}\|C\tau^{j_0} < 1/4$ (with the fixed block norms included in C). The Jacobian of the actual normalized finite law is then invertible. Passing to the physical means multiplies its rows by the positive constants $(\ell h_0)^2/(2A \sinh(j_0\gamma))$. These constants do not depend on ϑ , since all leading geometric data in this family are fixed. Thus the physical Jacobian is invertible as well.

Choose a small convex ball about an interior reference parameter. The Jacobian, after multiplication by its inverse at the center, is uniformly within $1/2$ of the identity on the ball. Integrating along a line segment shows that the physical mean map is bi-Lipschitz there. Take its compact closure as Θ . By further restriction of the offset collar if needed, each of its finitely many probabilities is strictly between zero and one; compactness gives common positive margins. These margins may be very small, but are fixed constants of this experiment.

Take the same number L of fresh preparations at each of the $2n$ windows, and let Y be the vector of sample means. For any $t > 0$, Hoeffding's elementary bounded-summand bound and a union bound give

$$\Pr\{\|Y - \mathcal{P}(\vartheta)\|_\infty > t\} \leq 4n \exp(-2Lt^2).$$

Choose a finite ordered net of Θ fine enough that its image approximates $\mathcal{P}(\Theta)$ within t in Euclidean norm. Minimize the Euclidean discrepancy from Y , resolving ties by the first index. This is a Borel estimator. On the displayed event its image error is at most C_nt ; the inverse Lipschitz bound gives parameter error at most Ct . Set $t = c\varepsilon$ and $L = \lceil Ct^{-2} \log(4n/\eta) \rceil$. This proves the deterministic charge and confidence statement. The graph-jet inverse is Lipschitz on this neighborhood, so the assertion transfers to that vector. For squared risk, use $L = \lfloor N/(2n) \rfloor$ and a net of mesh $N^{-1/2}$. The same deterministic discrepancy inequality and $\mathbb{E}\|Y - \mathcal{P}(\vartheta)\|^2 \leq C/L$ give the upper bound in (17.4).

For the lower bound choose an interior point ϑ_* , a unit vector v , and the two parameters $\vartheta_\pm = \vartheta_* \pm aN^{-1/2}v$. They represent actual analytic obstacles from Theorem 17.1. Every allowed window has a probability in a fixed interval $[p_*, 1 - p_*]$. The Bernoulli relative entropy satisfies

$$D(\text{Ber}(p) \parallel \text{Ber}(q)) \leq \frac{(p - q)^2}{q(1 - q)}.$$

The smooth mean map therefore bounds this entropy by Ca^2/N for each window. In an adaptive experiment, condition on the complete past and any auxiliary random seed. The rule for choosing the next window is the same under the two parameters; the conditional entropy is bounded by the same uniform bound. The chain rule, with padding by parameter-independent observations after an early stop, gives entropy at most Ca^2 for the entire record. Choose $a > 0$ fixed and small so that the total variation is at most $1/2$, using $\text{TV}^2 \leq D/2$.

Any estimator induces the test selecting the nearer of the two parameters. On a test error its squared parameter loss is at least a^2/N . The sum of the two testing error probabilities is at least $1 - \text{TV} \geq 1/2$. The maximum risk is therefore at least $a^2/(4N)$. This proves the lower bound, including adaptive window choices. All the alternatives keep g, A, κ_0, κ_1 and the selected leading hierarchy exactly fixed. $\square$

The fixed-window theorem resolves a physical finite-jet inverse problem, not just a formal coefficient problem: a fixed finite record length and finitely many strictly positive offsets already give local coordinates on independently varying contact jets. The windows and conditioning constants can deteriorate with M . The theorem neither passes this bound to an increasing-dimensional model nor substitutes its supplied exact family for the unknown smooth remainder experiment of Section 36.

18. TWO-FLIGHT DETERMINATION AND THE FINITE OBSERVATION DESIGN

The geometric inverse admits a finite-flight counterpart. This result separates recovery of a fixed contact jet from the relative limit that constructs the general boundary energy invariant. Throughout this section the two contact graphs are individually even, their labels are retained, and g, κ_0, κ_1 are supplied. Their higher jets need not agree. The finite-flight calculation and its observation consequence were pointed out in the independent memorandum [23]. We give a direct proof from the stationary action and the physical phase measure.

For $b \in \{0, 1\}$ define

$$G_b(d) = \frac{2A \sinh(2\gamma)}{d^2} \Pr(E_{2,b}(d)), \quad \xi_{b,k} = [d^k]G_b(d), \quad G_b(0) = 1. \quad (18.1)$$

The input to the exact germ inverse is the normalized germ G_b . If instead the full unnormalized probability germ $p_b(d)$ is supplied, its leading coefficient and its normalization are

$$\alpha_2 = \lim_{d \downarrow 0} \frac{p_b(d)}{d^2} = \frac{1}{2A \sinh(2\gamma)}, \quad G_b(d) = \frac{p_b(d)}{\alpha_2 d^2}.$$

This observation concerns exact germs, not a finite-sample estimate of a limit at zero. The finite-family experiment below supplies the fixed free area in its family specification; the general unknown-normalization experiment uses the separately charged smooth-class pilot. No derivative or leading coefficient is supplied as a free statistical observation.

Here two flights means three successive impacts, with itinerary $b, 1-b, b$; the probability also integrates the initial residual time. Put

$$z = c_0 c_1 - 1 = \sinh^2 \gamma > 0, \quad L_b = \frac{g}{2c_b z}, \quad k_m = \frac{4}{2^m(m+1)(m!)^2}, \quad (18.2)$$

$$U_m = (1+2z)^m, \quad V_m = 1+2mz.$$

The letters L_b in this section denote inverse-Hessian scales, not flight lengths. As before, $q_m = (q_{0,2m}, q_{1,2m})^t$, and write $\xi_{m-1} = (\xi_{0,m-1}, \xi_{1,m-1})^t$.

Theorem 18.1 (Two-flight inverse for independent even contacts). *For every $m \geq 2$, ξ_{m-1} depends only on the leading geometry and $q_2, \dots, q_m$, is affine in q_m , and has last-jet derivative*

$$D_{q_m} \xi_{m-1} = - \begin{pmatrix} U_m & V_m \\ V_m & U_m \end{pmatrix} \text{diag}(k_m L_0^m, k_m L_1^m). \quad (18.3)$$

The symmetric factor is positive definite. In particular,

$$U_m - V_m = \sum_{r=2}^m \binom{m}{r} (2z)^r > 0. \quad (18.4)$$

For each fixed $M \geq 2$, the map

$$(q_2, \dots, q_M) \mapsto (\xi_1, \dots, \xi_{M-1})$$

has a block-triangular analytic inverse on its image. The inverse is locally Lipschitz on compact finite-jet sets and compact positive leading-geometry boxes, including coincident curvatures. Equality of the two oriented normalized two-flight germs for real-analytic even contacts at the same labelled leading geometry determines both graph germs. For connected analytic closed boundaries it determines the two participating boundary images in their contact frames.

Proof. Fix b and put $o = 1-b$. Let u, v be the endpoints and let w be the intermediate impact coordinate. The quadratic part of the broken action, after subtracting $2g$, is

$$\frac{c_b u^2 + 2c_o w^2 + c_b v^2 - 2w(u+v)}{2g}.$$

Its stationary intermediate point is $w_0 = (u + v)/(2c_o)$. The finite implicit-function theorem, or Lemma 5.2, gives a unique smooth actual point $w = w_0 + O((|u| + |v|)^3)$. Evenness is used for this cubic order. Eliminating w_0 gives $E_0(y) = \frac{1}{2}y^t H_b y$, where $y = (u, v)^t$ and

$$H_b = \frac{1}{g} \begin{pmatrix} c_b - (2c_o)^{-1} & -(2c_o)^{-1} \\ -(2c_o)^{-1} & c_b - (2c_o)^{-1} \end{pmatrix}, \quad \det H_b = \frac{c_b z}{c_o g^2} = a_b^2. \quad (18.5)$$

For a linear form $\ell \cdot y$ set $\nu(\ell) = \ell^t H_b^{-1} \ell$. Inversion of this two-by-two matrix yields

$$\nu(u) = \nu(v) = L_b(1 + 2z), \quad \nu(w_0) = L_o. \quad (18.6)$$

The notation $\nu(u)$ denotes the value for the coordinate linear form $y \mapsto u$, and similarly for v, w_0 .

Write $E_2 = W_{2,b} - 2g$ and $b_2 = -W_{2,b,wv}/d_2^0$. The reference mixed derivative is $d_2^0 = (2gc_o)^{-1} = a_b/\sinh(2\gamma)$. The exact full-phase formula (6.1), including the residual-time integration, gives

$$G_b(d) = \frac{a_b}{\pi d^2} \int (d - E_2(y))_+ b_2(y) \, dy, \quad I_0(d) := \int (d - E_0(y))_+ \, dy = \frac{\pi d^2}{a_b}. \quad (18.7)$$

For a sufficiently small offset collar the whole active sublevel is inside the physical first-hit charts. Thus this identity has no unobserved endpoint restriction or substituted ensemble.

Keep all lower graph jets fixed. By stationarity, the first variation of the reduced action is obtained by varying the broken action at its stationary intermediate point. The first varying homogeneous length term is a pure endpoint power divided by $(2m)!$. The boundary sites are counted once and the interior site twice. Consequently

$$\begin{aligned} \partial_{q_{b,2m}} E_2 &= \frac{u^{2m} + v^{2m}}{(2m)!} + O(|y|^{2m+2}), \\ \partial_{q_{o,2m}} E_2 &= \frac{2w_0^{2m}}{(2m)!} + O(|y|^{2m+2}). \end{aligned} \quad (18.8)$$

All error estimates here may be differentiated to the required finite orders. Differentiate these identities in both endpoints and divide by $-d_2^0$. Since d_2^0 uses only the fixed leading geometry,

$$\begin{aligned} \partial_{q_{b,2m}} b_2 &= O(|y|^{2m}), \\ \partial_{q_{o,2m}} b_2 &= -\frac{g}{c_o} \frac{w_0^{2m-2}}{(2m-2)!} + O(|y|^{2m}). \end{aligned} \quad (18.9)$$

In particular, omitting the normalized twist variation would omit an essential contribution from the other contact.

We record both moment identities needed for the integration. The change $y = H_b^{-1/2} x$, followed by a rotation, sends $\ell \cdot y$ to $\sqrt{\nu(\ell)} x_1$. The angular average of $\cos^{2k} \theta$ is $\binom{2k}{k}/4^k$. Radial integration gives

$$\frac{1}{I_0(d)} \int_{E_0 < d} (\ell \cdot y)^{2m} \, dy = \frac{2(2m)! \nu(\ell)^m}{2^m (m!)^2 (m+1)} d^{m-1}, \quad (18.10)$$

$$\frac{1}{I_0(d)} \int (d - E_0)_+ (\ell \cdot y)^{2k} \, dy = \frac{2 \binom{2k}{k} \nu(\ell)^k}{2^k (k+1)(k+2)} d^k. \quad (18.11)$$

For example, the respective radial integrals before normalization are $(2d)^{m+1}/(2m+2)$ and $2^k d^{k+2}/((k+1)(k+2))$. The residual weight vanishes on the sublevel boundary, so the first action variation contributes the negative unweighted integral, with no additional moving-boundary term.

There is no replacement of a nonlinear domain by a quadratic domain at an uncontrolled order. Indeed, the smooth Morse construction of Lemma 6.1 fixes the integration domain. At scale $y = \sqrt{d} Y$, the first action variation has order d^{m-1} in the normalized law, and the first

amplitude variation has the same order. The unvaried nonlinear action and amplitude start two endpoint degrees above their quadratic and constant parts. Every product with either first variation therefore has at least one additional power of d . Taylor's formula in the common Morse coordinates justifies this degree calculation with smooth remainders at $d = 0$.

By (18.6) and (18.10), the own-contact action contribution is $-k_m L_b^m (1 + 2z)^m$, and the other-contact action contribution is $-k_m L_o^m$. There is no own-contact twist contribution at this order. Substituting $k = m - 1$ in (18.11), the other-contact twist contribution is

$$-\frac{g}{c_o} \frac{4L_o^{m-1}}{2^m(m+1)m!(m-1)!} = -k_m L_o^m (2mz),$$

since $g/(c_o L_o) = 2z$. These are precisely the first row of (18.3). Reversing the labels proves the other.

For completeness, finite-jet dependence does not require analyticity of the original graphs. Differentiate the finite stationary equation at zero. At each degree the highest derivative of w is multiplied by the same positive scalar $2c_o/g$; the remaining terms involve lower derivatives and only the graph jets through that degree. The normalized law coefficient in question uses the reduced action through degree $2m$ and its twist through degree $2m - 2$. Hence it uses no graph derivative above $2m$. The displayed last-jet derivative is independent of that jet and of all lower jets. This proves affinity, also away from the quadratic graph. The finite operations are analytic in their finite-jet arguments and in positive g, κ_0, κ_1 .

Let R_m be the value of ξ_{m-1} with $q_m = 0$ and the lower jets fixed. The explicit recursive inverse is

$$q_m = \text{diag}((k_m L_0^m)^{-1}, (k_m L_1^m)^{-1}) \frac{1}{U_m^2 - V_m^2} \begin{pmatrix} U_m & -V_m \\ -V_m & U_m \end{pmatrix} (R_m - \xi_{m-1}). \quad (18.12)$$

The binomial identity (18.4) and $U_m + V_m > 0$ make every denominator positive. A fixed finite collection has a positive minimum on compact positive leading-geometry boxes. The recursion proves the inverse and its local Lipschitz bounds without any factor $(\kappa_0 - \kappa_1)^{-1}$. Equality of all recovered jets gives equality of analytic even graph germs. For analytic connected closed boundaries, the arclength-curvature continuation argument in Corollary 11.2 applies to each contact frame and proves the last assertion. $\square$

At $g = 1$, $c_0 = c_1 = 2$, $m = 2$, the block is

$$D_{q_2} \xi_1 = -\frac{1}{1728} \begin{pmatrix} 49 & 13 \\ 13 & 49 \end{pmatrix}.$$

Its antisymmetric eigenvalue has magnitude $1/48$. This finite normalization check is consistent with, but does not substitute for, the all-order proof.

Corollary 18.2 (Finite-order change of invariant coordinates). *At fixed labelled leading geometry and fixed M , the two-flight coefficient vector $\Xi = (\xi_1, \dots, \xi_{M-1})$ and the limiting coefficient vector $f = (f_1, \dots, f_{M-1})$ are related on the even physical jet image by mutually inverse block-triangular analytic maps. On sufficiently small compact finite-jet neighborhoods these changes of coordinates are bi-Lipschitz, also at equal curvatures.*

Proof. Let $\Psi_M : q \mapsto \Xi$ be the map of Theorem 18.1 and let $\Phi_M : q \mapsto f$ be that of Theorem 11.1. Then $f = \Phi_M \circ \Psi_M^{-1}(\Xi)$ and $\Xi = \Psi_M \circ \Phi_M^{-1}(f)$. Each map is triangular and analytic. On a sufficiently small compact coordinate neighborhood, the maps and their inverses have bounded derivatives. Integration along segments in the relevant coordinate neighborhoods gives the Lipschitz bounds; neither inverse uses a curvature separation. $\square$

This is an equivalence of fixed finite-jet coordinates, not an equality of G_b and $\mathcal{F}_{bb}$ and not a comparison of noisy function-valued experiments. Its constants may depend on M . For smooth contacts, equality of all jets does not replace the full-profile uniqueness argument of Theorem 10.1.

Theorem 18.3 (Two-flight physical observation). *Fix $M \geq 2$ and a physical analytic family of Theorem 17.1. Put $n = M - 1$. There are $h_0 > 0$ and a compact parameter ball Θ with nonempty interior such that the $2n$ actual means*

$$\mathcal{P}_2(\vartheta) = (\Pr_{\vartheta}(E_{2,b}(\ell h_0)))_{b=0,1; 1 \leq \ell \leq n} \quad (18.13)$$

are bi-Lipschitz coordinates. The physical times are $2g + \ell h_0$. Thus the finite-window conclusion of Theorem 17.3 holds with $j_0 = 2$ for every fixed M .

Using independent binary preparations, a Borel estimator satisfies, for sufficiently small $\varepsilon > 0$ and $0 < \eta < 1/4$,

$$\sup_{\vartheta \in \Theta} \Pr_{\vartheta} \{ \|\hat{\vartheta} - \vartheta\| > \varepsilon \} \leq \eta, \quad N_{\text{tot}} \leq C\varepsilon^{-2} \log(C/\eta).$$

All failures are charged. For at most N fresh preparations, with adaptive or randomized choice among these windows, the minimax squared risk lies between c/N and C/N for large N . The same conclusions hold for the corresponding finite graph jets. Constants and offsets may depend on the fixed family and M ; the flight number does not. The family and its leading data are supplied.

Proof. The support realization and Theorem 18.1 give local analytic coordinates Ξ on the physical family. Analyticity of the finite stationary integral on a common collar yields

$$G_b(d; \Xi) = 1 + \sum_{k=1}^n \Xi_{b,k} d^k + d^{n+1} R_b(d; \Xi),$$

where R_b and its first parameter derivatives are bounded after compact restriction. The nodal Jacobian at $d = \ell h$ is block diagonal with blocks $V_h = ((\ell h)^k)_{\ell,k=1}^n$, plus $O(h^{n+1})$. Since $V_h = V_1 \text{diag}(h, \dots, h^n)$ and V_1 is invertible, $\|V_h^{-1}\| \leq Ch^{-n}$. Choose a fixed $h = h_0 > 0$ small enough that all offsets lie in the collar and the transformed error has norm less than $1/2$. There is no finite-to-infinite bridge error in this argument.

Passing to physical probabilities multiplies rows by the positive constants $(\ell h_0)^2 / (2A \sinh(2\gamma))$. They are fixed on the family. Corollary 18.2 transfers invertibility to the original limiting-jet parameter ϑ . On a sufficiently small convex parameter ball, the physical Jacobian, multiplied by its inverse at the center, stays within $1/2$ of the identity. Integrating along line segments gives both Lipschitz bounds. Use the compact closure as Θ . Decrease the collar if necessary so that every probability is below one. Positivity and compactness then place the finitely many means in $[p_*, 1 - p_*]$ for some fixed $p_* > 0$.

Here are the statistical details in this direct two-flight design. Take L independent preparations in each window and let Y be the vector of sample means. The bounded-summand exponential estimate and a union bound give

$$\Pr \{ \|Y - \mathcal{P}_2(\vartheta)\|_{\infty} > t \} \leq 4ne^{-2Lt^2}.$$

Choose a finite ordered net in Θ whose image has mesh at most t , and minimize its Euclidean discrepancy from Y , breaking ties by the first index. This is Borel measurable. On the event that all nodal errors are at most t , its image error is at most $C_n t$ and its parameter error at most Ct . Taking $t = c\varepsilon$ and $L = \lceil Ct^{-2} \log(4n/\eta) \rceil$ proves the confidence bound and deterministic charge. With $L = \lfloor N/(2n) \rfloor$ and a net of mesh $N^{-1/2}$, the deterministic discrepancy inequality and $\mathbb{E} \|Y - \mathcal{P}_2(\vartheta)\|^2 \leq C/L$ give risk at most C/N .

For the lower bound choose an interior ϑ_* and a unit vector v , and use the physical alternatives $\vartheta_{\pm} = \vartheta_* \pm aN^{-1/2}v$. They have exactly the same selected leading hierarchy. Smoothness of the mean map and its probability margins give, uniformly over the allowed windows,

$$D(\text{Ber}(p_+) \| \text{Ber}(p_-)) \leq \frac{(p_+ - p_-)^2}{p_-(1 - p_-)} \leq \frac{Ca^2}{N}.$$

Conditioning on the past and on the procedure's independent random seed, the rule for choosing the next window is the same under both parameters. The entropy chain rule therefore bounds the entropy of the full record by Ca^2 , including adaptive choices. An early stop can be padded by parameter-independent observations. Fix $a > 0$ small so that total variation is at most $1/2$. The test choosing the nearer parameter from any estimator has summed error at least $1/2$; on a test error the squared estimation loss is at least a^2/N . Hence the maximum risk is at least $a^2/(4N)$. Finally the two parameter systems and the graph-jet vector are locally bi-Lipschitz, so the same orders hold in all of them. $\square$

The preceding theorem supplies the direct finite-flight design. The long-bridge construction in Theorem 17.3 is retained as an alternative construction using limiting coordinates. Neither construction requires j_0 to grow with requested accuracy on its fixed physical family. The present theorem does not assert that two flights are minimal for every restricted family. In particular, the identical-contact one-flight comparison in Section A.15 remains applicable in its stated class. The general, nonsymmetric energy-profile experiment of the next part has neither this supplied finite-dimensional family nor its parametric risk conclusion.

19. THE SIGNED ONE-FLIGHT BENCHMARK AND THE RESTRICTION TO LONG BRIDGES

The signed support observation is richer than an integrated probability germ. It therefore has a short-flight inverse different from the two-flight integrated benchmark in the preceding section. Recording this distinction specifies the contribution of the long-bridge results.

Proposition 19.1 (One-flight signed support inverse). *Suppose mixed-type one-flight endpoint support is observed as a function of physical time in calibrated signed transverse coordinates, with its onset retained. Then the two labelled facing graph germs are determined without symmetry, equal contacts or supplied curvatures.*

Proof. On a sufficiently small contact box the one-flight length is

$$\ell(u, v) = \sqrt{(g + \psi_0(u) + \psi_1(v))^2 + (v - u)^2}.$$

The physical flux factor is positive there. Hence the threshold in physical time at which an endpoint pair enters the support is $\ell(u, v)$: the entrance residual time can be any sufficiently small positive number, and no record is possible before the flight length. Equivalently, in excess time the support threshold is $\ell - g$; the onset supplies g . Since $\psi_0(0) = \psi_1(0) = 0$, the coordinate slices give

$$\psi_0(u) = \sqrt{\ell(u, 0)^2 - u^2} - g, \quad \psi_1(v) = \sqrt{\ell(0, v)^2 - v^2} - g. \quad (19.1)$$

The positive square-root branch is fixed by $g + \psi_b > 0$. Thus the graph functions themselves, and not just finitely many of their jets, are recovered. For analytic boundaries the same continuation and intrinsic gluing questions can subsequently be considered. $\square$

This proposition does not identify its observations with the same-type limiting laws. It assumes access to a mixed-type one-flight support function $\ell(u, v)$. The signed long-bridge theorem instead recovers two half-line actions from the limiting 00 and 11 endpoint laws, and then inverts their interaction with both contact graphs through the order-by-order determinant-one blocks. The

two-flight benchmark concerns integrated probability germs and is a third observation family. No Markov ordering between these three families is asserted without a common acquisition model and an actual kernel.

An unrestricted finite-flight consistency theorem would not, by itself, establish a need for long bridges: the identity (19.1) already gives a direct short-flight route when its support data are available. The physical global theorem here has a more specific conclusion. For any prescribed lower bound on the flight number, its pilot and all inference batches use only even same-type records above that bound. Along the consistent reconstruction sequence that lower bound tends to infinity. Its pilot uses shrinking near-onset windows at those same long even flight numbers, and not a hidden one-flight geometric scan. Thus its contact inverse is genuinely implemented in the restricted long-bridge observation family.

The one-flight identity neither yields a relative law uniform on a fixed offset collar as the flight number grows nor describes the information lost by deleting residual time or waiting counts. Those are separate assertions of the relative boundary law, the explicit Poisson equivalence, and the Gaussian coarsening theorems. In particular, the necessity of the long-bridge analysis in this article is tied to these observation and limit restrictions, not to a claim that a billiard contact cannot be recovered from any shorter experiment.

20. DIRECT GEOMETRIC SAMPLING IN THE SAME LONG-EVEN EXPERIMENT

The restriction to long even bridges does not by itself make the half-line inverse necessary when exact planar endpoint positions are observed. We give a comparison within the physical record space (38.1), using its charged pilot and no short or mixed-type flight. The comparison preserves the global reconstruction problem; it identifies the additional content of the single-offset *transverse-law* inverse.

Proposition 20.1 (Charged direct acquisition of contact jets). *Fix an integer $M \geq 2$ and a contact class with uniform C^{M+1} graph bounds and the uniform localization and flux bounds of Lemma 38.1. In particular, the compact analytic class $\mathfrak{K}$ has these properties. Fix a target error probability $\beta \in (0, 1)$, and any prescribed even flight number $J \geq 2$. For every sufficiently small $a > 0$ and pilot-frame tolerance $\varepsilon > 0$ satisfying $\varepsilon \leq c_M a$, there is a finite capped policy, every preparation of which uses J flights, that produces estimates of the two graph jets with, uniformly on the class,*

$$\max_{e,b} \left| \widehat{\psi}_{e,b}^{(m)}(0) - \psi_{e,b}^{(m)}(0) \right| \leq C_M \{a^{M+1-m} + \varepsilon a^{-m}\}, \quad 0 \leq m \leq M, \quad (20.1)$$

except on an event of probability at most β . All failed preparations and discarded off-bin successes are charged. No half-line action is estimated or inverted.

Proof. Choose $M + 1$ distinct fixed numbers ξ_i in a bounded interval and pairwise disjoint closed bins I_i about them. The bins have positive length and a positive minimum separation. The observed transverse bins will be aI_i . Constants below may depend on these fixed choices and on M .

Run Theorem 38.2 with $j = J$, error probability $\beta/2$, and a resolution h such that

$$C\sqrt{h} \leq \varepsilon, \quad h \leq c_M a^2.$$

On its good event the estimated contact frames have error at most ε , and $J|\widehat{g}_e - g_e| \leq h$. The pilot is charged and its flight number is never increased afterwards.

Choose a fixed sufficiently large D_M and program $t_e = J\widehat{g}_e + D_M a^2$. For small a these windows lie in the common positive-offset collar. The finite bridge action obeys $E_J(u, v) \leq C(u^2 + v^2)$ on a common contact box. In each target bin restrict the true first endpoint to a smaller interior

sub-bin of length comparable to a , the second endpoint to an interval of length comparable to a , and the residual time to an interior interval of length comparable to a^2 . Increasing D_M first ensures $r + E_J < t_e - Jg_e$ throughout these sets; taking a small afterwards keeps them in the collar. The frame error, after reducing c_M , ensures that each smaller true-coordinate bin lies in the corresponding observed bin. Positivity of the exact flux and its uniform lower bound give a one-preparation probability at least

$$p_M(J, a) = c_M a^4 e^{-J\gamma_+} > 0 \quad (20.2)$$

for every target bin, conditional on any good pilot history. Alternatively, for a fixed J, a, M , compactness and strict positivity alone give a positive lower bound; no effective uniform-in- J estimate is needed for the existential comparison.

For each (e, b, i) , take at most

$$N_M = \left\lceil p_M(J, a)^{-1} \log \frac{4|E|(M+1)}{\beta} \right\rceil$$

fresh preparations at the stated window, stopping when the first endpoint lies in its observed bin. The chance of exhausting any one cap is at most $e^{-N_M p_M}$. A union bound over $2|E|(M+1)$ bins bounds their total failure probability by $\beta/2$. The post-pilot preparation cap is $2|E|(M+1)N_M$; success outside the target bin does not evade this count. Every decision is a Borel function of the physical records and the pilot.

For a retained point X_i , form its measured transverse coordinate $\hat{y}_i = \hat{t}_e \cdot (X_i - \hat{p}_{e,b})$ and signed normal coordinate $\hat{z}_i$ in the estimated frame, with the prescribed facing-side sign. The true graph representation and the frame bounds give

$$|\hat{z}_i - \psi_{e,b}(\hat{y}_i)| \leq C\varepsilon.$$

Indeed the transverse-coordinate error is $O(\varepsilon)$, the normal error is $O(\varepsilon)$, and the graph has a uniform local Lipschitz bound. The normalized observed nodes $\hat{y}_i/a$ belong to the fixed separated bins I_i .

Let P_M interpolate the $M+1$ measured pairs. The scaled Vandermonde matrix has a uniformly bounded inverse: its entries vary continuously on the compact product of the bins, and its determinant is a product of nonzero node differences with a uniform lower bound. Taylor's formula on the common C^{M+1} graph patch shows that the discrepancy between the observations and the Taylor polynomial of degree M is at most $C_M(a^{M+1} + \varepsilon)$ at every node. Inverting the scaled matrix and converting its degree- m coefficient into a derivative gives

$$|P_M^{(m)}(0) - \psi_{e,b}^{(m)}(0)| \leq C_M a^{-m} (a^{M+1} + \varepsilon).$$

This is (20.1). Default estimates on flagged failures make the rule defined on every transcript. Adding pilot and bin failures completes the proof. $\square$

Taking $\varepsilon \leq a^{M+1}$ gives error $O_M(a)$ simultaneously through order M . The pilot can achieve this precision at finite cost by reducing h ; its cost is included rather than supplied externally. Thus, at stage s , one may first choose any even $J_s \geq s$, then choose the geometric resolution and its pilot, all at that same J_s . The contact graph jets and gaps determine the action jets by the finite smooth forward map. Alternatively their exact infinite collection is injective on the same signature-rigid analytic class. The compact inverse modulus, finite-template matching, and increasing-order diagonal argument therefore give global consistency from these direct graph samples as well. No claim of an effective analytic-continuation rate is needed for this conclusion.

Theorem 13.4 proves a different assertion. Its successful observations retain (u, v) , not $(u, \psi_b(u), v, \psi_b(v))$. The four-density identity reconstructs the unsymmetrized half-line actions from the distribution of those two scalar coordinates, and the determinant-one recursion recovers

the contact geometry encoded by repeated alternating returns. Graph interpolation has no graph ordinates to use in that data map. The two results are compared under their actual observations; a diverging flight number is not offered as an argument excluding direct interpolation in the richer position experiment.

Part 2. Boundary information and physical local experiments

21. RELATIVE BOUNDARY LAWS FOR PHYSICAL EXPERIMENTS

The two-boundary law also approximates finite experiments, rather than only their limiting offset coefficients. It is important to retain the probability of obtaining a selected record in this comparison. We use $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$ for probability measures. For finite signed measures, $|\mu|(\Omega)$ denotes the total variation mass. Constants below are uniform on the compact geometric families of Theorem 3.1. Neither symmetry nor finite horizon is added in this section.

Fix a channel, suppress its index, and put

$$t_j = -W_{j,uv}(0,0) > 0, \quad p = j \bmod 2.$$

$$E_\infty(u, v) = S_0(u) + S_p(v), \quad b_\infty(u, v) = B_0(u)B_p(v).$$

The physical time from preparation to the first impact is denoted by r . On a selected event it lies between zero and the unused excess time. At $d > 0$ use the common coordinates

$$(z_0, z_1, s) = (u/\sqrt{d}, v/\sqrt{d}, r/d), \quad z = (z_0, z_1).$$

The finite and boundary subprobability measures on this space are

$$d\mu_{j,d} = \frac{d^2 t_j}{2\pi A} b_j(\sqrt{d}z) \mathbf{1}_{\{s \geq 0, s+E_j(\sqrt{d}z)/d < 1\}} dz ds, \quad (21.1)$$

$$d\mu_{j,d}^\partial = \frac{d^2 t_j}{2\pi A} b_\infty(\sqrt{d}z) \mathbf{1}_{\{s \geq 0, s+E_\infty(\sqrt{d}z)/d < 1\}} dz ds. \quad (21.2)$$

Their densities are zero outside the common contact box. After decreasing the offset collar, all their nonzero supports lie in a fixed bounded z -ball and in $0 \leq s \leq 1$. The first measure is exactly the full-phase selected-record measure, by the physical flux formula. Write $p_{j,d} = \mu_{j,d}(\mathbb{R}^3)$ and $p_{j,d}^\partial = \mu_{j,d}^\partial(\mathbb{R}^3)$, and normalize them to $\Pi_{j,d}$ and $\Pi_{j,d}^\partial$. Finally set

$$P_{j,d} = (1 - p_{j,d})\delta_\partial + \mu_{j,d}, \quad P_{j,d}^\partial = (1 - p_{j,d}^\partial)\delta_\partial + \mu_{j,d}^\partial.$$

Here ∂ records that the preparation did not produce the selected event. Thus $P_{j,d}$ is a coarsening of a raw physical record, not an experiment in which unsuccessful preparations have been discarded for free.

Theorem 21.1 (Uniform transfer to the boundary experiment). *There are $d_* > 0$, $C < \infty$ and $0 < \tau < 1$ such that, for all $j \geq 1$ and $0 < d \leq d_*$, the displayed measures are well defined and*

$$p_{j,d} \asymp p_{j,d}^\partial \asymp d^2 e^{-j\gamma}, \quad (21.3)$$

$$|\mu_{j,d} - \mu_{j,d}^\partial|(\mathbb{R}^3) \leq C p_{j,d} \tau^j, \quad \|\Pi_{j,d} - \Pi_{j,d}^\partial\|_{\text{TV}} \leq C \tau^j, \quad (21.4)$$

$$\|P_{j,d} - P_{j,d}^\partial\|_{\text{TV}} \leq C p_{j,d} \tau^j. \quad (21.5)$$

For n independent preparations, respectively k independently prepared successful records, these imply

$$\|P_{j,d}^{\otimes n} - (P_{j,d}^\partial)^{\otimes n}\|_{\text{TV}} \leq \min\{1, C n p_{j,d} \tau^j\}, \quad (21.6)$$

$$\|\Pi_{j,d}^{\otimes k} - (\Pi_{j,d}^\partial)^{\otimes k}\|_{\text{TV}} \leq \min\{1, C k \tau^j\}. \quad (21.7)$$

The same bounds hold after applying any common measurable observation map, including retention of the endpoint positions alone. At $d = 0$, the conditional statement holds for the corresponding quadratic right limits in the scaled coordinates.

Proof. The positive endpoint Hessians and the uniform nonlinear bounds give, on a sufficiently small common box,

$$c|Y|^2 \leq E_j(Y), E_\infty(Y) \leq C|Y|^2, \quad c \leq b_j(Y), b_\infty(Y) \leq C.$$

These inequalities give the asserted common bounded scaled support and upper and lower bounds on the scaled integrals. They also imply that both masses are less than $1/2$ after decreasing d_* , since t_j is uniformly bounded. The Jacobi formula gives $t_j \asymp e^{-j\gamma}$, which proves (21.3).

The action difference has zero value and gradient at $Y = 0$. Taylor's integral formula and the C^2 factorization estimate imply

$$|E_j(Y) - E_\infty(Y)| \leq C\tau^j|Y|^2, \quad |b_j(Y) - b_\infty(Y)| \leq C\tau^j.$$

Consequently, on the common scaled ball,

$$\left| \frac{E_j(\sqrt{d}z) - E_\infty(\sqrt{d}z)}{d} \right| \leq C\tau^j|z|^2.$$

For each fixed z , the two allowed s -fibres have lengths $(1 - E_j(\sqrt{d}z)/d)_+$ and $(1 - E_\infty(\sqrt{d}z)/d)_+$. They start at the same point zero. Since the positive-part map is 1-Lipschitz, the length of their symmetric difference is at most $C\tau^j|z|^2$. Integrating the bounded amplitudes on this difference, and the amplitude difference on the common part, proves

$$|\mu_{j,d} - \mu_{j,d}^\partial|(\mathbb{R}^3) \leq Cd^2t_j\tau^j/A.$$

This proves the first part of (21.4). In particular $|p_{j,d} - p_{j,d}^\partial|$ obeys the same bound. Dividing by the positive masses and adding the error of their normalizers gives the conditional estimate. Adding the missing-event atom instead gives (21.5). This argument estimates a moving domain by its fibres; it does not differentiate its indicator function.

For probability measures P_i, Q_i , telescope their product measures, or couple each pair maximally, to obtain

$$\left\| \bigotimes_{i=1}^n P_i - \bigotimes_{i=1}^n Q_i \right\|_{\text{TV}} \leq \sum_{i=1}^n \|P_i - Q_i\|_{\text{TV}}.$$

This proves the two product bounds. A measurable pushforward cannot increase total variation. Lastly $E_j(\sqrt{d}z)/d$ and its boundary analogue converge to their positive quadratic forms as $d \downarrow 0$, and the amplitudes converge to one. The same fibre argument proves L^1 convergence of these bounded densities and of their normalizations. Passing to the limit proves the final assertion. $\square$

Corollary 21.2 (Transfer of bounded risks). *For any estimator or test based on the stated coarsened records, and any measurable loss taking values in $[0, 1]$, replacing the finite experiment by its boundary model changes its risk by at most the right-hand side of (21.6) or (21.7), respectively. The estimate is uniform in the parameter family. More generally, a fixed design of independent preparations at (j_i, d_i) has risk error at most $C \sum_i p_{j_i, d_i} \tau^{j_i}$.*

Proof. Integrate the bounded loss against the difference of the laws, using its layer-cake representation. Apply the product estimate, with possibly different factors, and then take the supremum over parameters. $\square$

The comparison also applies in unscaled physical endpoint-position and residual-time coordinates, by using the same boundary embeddings for both laws at each parameter. For predetermined physical windows t_i , set $d_i(\xi) = t_i - j_i g_e(\xi)$ and restrict the parameter family so

that $0 < d_i(\xi) \leq d_*$; the bounds hold uniformly at these offsets. Scaling is then only a device in the proof, not an extra observation of an unknown gap. In a design programmed directly at fixed excess d , the onset must instead be supplied or calibrated. No onset-calibration cost is removed by the total-variation comparison.

The effective sample size in (21.6) is the expected number $np_{j,d}$ of selected records. In particular, an experiment with $np_{j,d} = O(k_j)$ is approximated uniformly whenever $k_j \tau^j \rightarrow 0$. This is a simultaneous large-record and large-sample conclusion that separate fixed-flight Taylor expansions do not supply. It concerns the specified endpoints and residual time. The full growing collision array lies on an embedded surface; no total-variation comparison of distinct such surfaces is asserted. Nor does this theorem supply a parameter-free sampling algorithm for an unknown table. It compares the physical statistical families in common observation coordinates and preserves the cost of obtaining a selected event.

22. ADAPTIVE OBSERVATION AND STOPPED EXPERIMENTS

The preceding comparison concerns a fixed list of preparations. In an observation procedure the next window may instead depend on the records already obtained, and the procedure may stop after enough successful records. We prove a comparison for the entire stopped transcript. In particular, the failure symbols and the stopping time are not removed from the experiment.

We use the unscaled physical endpoint and residual-time coordinates at the end of Section 21. A design a specifies a labelled channel, a flight number $j(a)$ and a physical observation time $t(a)$. At geometric parameter ξ its excess is $d_\xi(a) = t(a) - j(a)g_\xi$. Write $P_{\xi,a}$ for the raw coarsened physical law and $Q_{\xi,a}$ for the corresponding boundary law. The success probability is $p_\xi(a) = P_{\xi,a}(\text{success})$. The chosen designs must lie in the common physical collar: $0 < d_\xi(a) \leq d_*$. An unknown excess is used only in proving the comparison; it is not supplied to the observation rule.

The design and record spaces are standard Borel spaces. A policy consists of measurable rules for choosing a design or stopping, using the observed past and auxiliary randomness independent of ξ . The same rules are used in the two experiments. Conditional on the past and the chosen design, a fresh preparation has law $P_{\xi,a}$ or $Q_{\xi,a}$, respectively. The policy stops after $T \leq N$ preparations, where N is a prescribed deterministic cap. The transcript includes every design, every success or failure record, and T . These conventions include randomized pilot procedures followed by data-dependent designs.

Lemma 22.1 (A stopped comparison for observation kernels). *Let $P_{\xi,a}$ and $Q_{\xi,a}$ be jointly measurable probability kernels on a common record space, and suppose that their total variation distance is at most a nonnegative measurable function $e_\xi(a)$. Assume that a measurable maximal coupling is available for each pair of kernels. For any policy as above, let P_ξ^π and Q_ξ^π be the laws of its stopped transcript. Then*

$$\|P_\xi^\pi - Q_\xi^\pi\|_{\text{TV}} \leq \min \left\{ 1, \mathbb{E}_{P_\xi^\pi} \sum_{i=1}^T e_\xi(a_i) \right\}. \quad (22.1)$$

One may interchange P and Q in the expectation. The conclusion is unchanged by recording the independent random seed or by applying a common randomized observation map to the transcript.

Proof. Use the same auxiliary randomness for both policies while their pasts agree. They then choose the same design, or both stop. At a common active past, couple the next two records maximally. After their first disagreement continue each marginal according to its own policy and observation kernel. This constructs a coupling of the required transcript laws. Let D_i be the event that their first disagreement is at preparation i . Given a common past and its next design

a_i , its conditional probability is at most $e_\xi(a_i)$. Therefore

$$\Pr(D_i) \leq \mathbb{E}[\mathbf{1}_{\{\text{pasts agree before } i, T_P \geq i\}} e_\xi(a_i^P)] \leq \mathbb{E}_{\mathbf{P}_\xi^\pi}[\mathbf{1}_{\{T \geq i\}} e_\xi(a_i)].$$

Here T_P and a_i^P refer to the physical marginal; no independence of the random designs is used. The events D_i are disjoint. If none occurs, the two policies also stop together and their complete transcripts agree. For any measurable set B , a coupling satisfies $|\Pr(X \in B) - \Pr(Y \in B)| \leq \Pr(X \neq Y)$. Summing the last display proves (22.1). Interchanging the marginals gives the other expectation. Coupling the randomized output conditionally in the same way proves the last assertion. $\square$

The coupling inequality is classical; see, for example, [28, Proposition 4.7]. For the physical kernels, a common scalar dominating measure suffices; their planar pushforwards need not have ambient densities.

Lemma 22.2 (Measurable coupling in physical observation coordinates). *Let K be a fixed bounded scalar endpoint-time box, and equip $K_\partial = K \sqcup \{\partial\}$ with the measure $\nu = \text{Leb}_K + \delta_\partial$. Suppose $f_{\xi,a}, g_{\xi,a}$ are jointly measurable probability densities with respect to ν . Let Y be a standard Borel physical record space and let $\iota_{\xi,a} : K_\partial \rightarrow Y$ be jointly measurable and, for each (ξ, a) , a Borel isomorphism onto a Borel subset of Y . Assume that it sends ∂ to the common failure symbol. Then the probability kernels*

$$P_{\xi,a} = (\iota_{\xi,a})_*(f_{\xi,a}\nu), \quad Q_{\xi,a} = (\iota_{\xi,a})_*(g_{\xi,a}\nu)$$

admit a jointly measurable maximal coupling. In particular,

$$\|P_{\xi,a} - Q_{\xi,a}\|_{\text{TV}} = 1 - \int_{K_\partial} \min(f_{\xi,a}, g_{\xi,a}) d\nu. \quad (22.2)$$

No domination of $P_{\xi,a}$ and $Q_{\xi,a}$ by ambient Lebesgue measure on Y , or domination across different geometric parameters, is required.

Proof. Suppress (ξ, a) and put $h = \min(f, g)$ and $m = \int h d\nu$. For a measurable set $D \subset Y \times Y$ define

$$\begin{aligned} C(D) &= \int_{K_\partial} \mathbf{1}_D(\iota(x), \iota(x)) h(x) d\nu(x) \\ &\quad + \frac{1}{1-m} \int_{K_\partial^2} \mathbf{1}_D(\iota(x), \iota(y)) (f(x) - h(x))(g(y) - h(y)) d\nu(x) d\nu(y), \end{aligned}$$

where the second term is defined to be zero when $m = 1$. Both residual measures have mass $1 - m$, so C is a probability measure with marginals P and Q . Its dependence on (ξ, a) is measurable by integration of jointly measurable functions; on $\{m = 1\}$ the specified zero branch is measurable as well.

The residual measures are supported, up to null sets, on the disjoint sets $\{f > g\}$ and $\{g > f\}$. Injectivity of ι makes their images disjoint. Thus the second term puts no mass on the diagonal and the coupling disagrees with probability $1 - m$. Conversely, for fixed (ξ, a) the assumed Borel inverse makes the image of $\{f > g\}$ measurable, and the difference of its P and Q probabilities is

$$\int_{\{f > g\}} (f - g) d\nu = 1 - m.$$

The coupling bound and this measurable set prove (22.2) and maximality. $\square$

For the billiard kernels, K is the common unscaled scalar contact box for (u, v, r) . The finite and boundary densities, including the failure atom, are given by the physical flux formula and its two-boundary limit. The maps $\iota_{\xi,a}$ insert the two boundary positions in the plane and retain the

residual time and the chosen labels. The contact graphs defining the forward model make these maps jointly smooth on each channel component. Their inverses on the physical images are the transverse projections in the corresponding contact frames, together with the unchanged time and labels. Thus they are Borel isomorphisms onto their Borel images. A finite or countable disjoint union handles the flight numbers and channel labels. The preceding lemma therefore supplies the measurable maximal coupling used in Lemma 22.1 for the actual physical records.

When only scalar endpoint coordinates and residual time are observed, $\iota_{\xi,a}$ is the identity and the same construction applies directly. For planar endpoint positions the successful law is instead supported on an embedded boundary surface; it is not an ambient Lebesgue density. The parameter-dependent embedding is used only to prove a comparison at one fixed table. It is neither an extra observation nor a parameter-free simulator for an unknown table, and does not compare densities on the different boundary surfaces of two distinct tables.

Theorem 22.3 (Adaptive physical transfer with a success-weighted budget). *For the compact physical families of Theorem 21.1, let π be a policy whose designs are in the common collar on every reachable history. There are constants $C < \infty$ and $0 < \tau < 1$, independent of π and N , such that*

$$\|P_\xi^\pi - Q_\xi^\pi\|_{\text{TV}} \leq \min\{1, C\mathcal{B}_\xi(\pi)\}, \quad (22.3)$$

$$\mathcal{B}_\xi(\pi) := \mathbb{E}_{P_\xi^\pi} \sum_{i=1}^T p_\xi(a_i) \tau^{j(a_i)} = \mathbb{E}_{P_\xi^\pi} \sum_{i=1}^T Y_i \tau^{j(a_i)}, \quad (22.4)$$

where Y_i is the indicator of a successful preparation. Hence replacing the physical experiment by the boundary experiment changes any risk for a loss in $[0, 1]$ by at most $\min\{1, C\mathcal{B}_\xi(\pi)\}$. The bounds are uniform in ξ and survive any common coarsening.

In particular, if $j(a_i) \geq J$ and the policy stops no later than its k th success, then

$$\|P_\xi^\pi - Q_\xi^\pi\|_{\text{TV}} \leq \min\{1, Ck\tau^J\}. \quad (22.5)$$

The physical cost remains the number T of raw preparations, including all failures; it is not replaced by the number of successes.

Proof. Theorem 21.1, transported to the common unscaled physical coordinates, supplies $e_\xi(a) = Cp_\xi(a)\tau^{j(a)}$. Apply Lemma 22.1. The event $\{T \geq i\}$ and the design a_i are known before the i th fresh observation. Its conditional success probability is $p_\xi(a_i)$. Conditional expectation therefore gives

$$\mathbb{E}[\mathbf{1}_{\{T \geq i\}} Y_i \tau^{j(a_i)}] = \mathbb{E}[\mathbf{1}_{\{T \geq i\}} p_\xi(a_i) \tau^{j(a_i)}].$$

Summing the finite list proves (22.4). Under the stopping assumption the realized sum is at most $k\tau^J$. For a loss $0 \leq L \leq 1$, write $L = \int_0^1 \mathbf{1}_{\{L > s\}} ds$ and apply the total variation bound. This proves both the risk assertion and the stated uniformities. $\square$

Corollary 22.4 (Pilot observations and growing adaptive experiments). *The same bound applies when a charged pilot is kept exact in both experiments and only the subsequent preparations are replaced by their boundary laws. In that case the sum in (22.4) runs over the replaced preparations only, whereas the total cost includes the pilot. If the pilot certifies the common collar on an event of probability at least $1 - \eta_0$, the bound becomes*

$$\|P_\xi^\pi - Q_\xi^\pi\|_{\text{TV}} \leq \eta_0 + C\mathbb{E}_{P_\xi^\pi} \left[\mathbf{1}_{\{\text{good pilot}\}} \sum_{i \in I_{\text{post}}} Y_i \tau^{j(a_i)} \right]. \quad (22.6)$$

Here, on bad pilot histories, the comparison experiment is completed by any specified probability kernels on the same transcript space; no boundary formula outside its collar is assumed. For

a sequence of policies, if $\eta_0 \rightarrow 0$ and the displayed success-weighted expectations tend to zero uniformly on the physical family, every bounded transcript risk has a vanishing comparison error.

Proof. Couple the exact pilot identically, including its stopping decision. It has the same law in both experiments. Condition on the entire pilot transcript. On a good transcript apply the preceding proof to the post-pilot policy, and on a bad one bound disagreement by one. Integration over this common pilot law proves (22.6). The conditional-expectation identity for successes holds with the good-pilot indicator, which is measurable before every post-pilot observation. The final assertion follows by taking a uniform bound. $\square$

For example, any sequence of policies capped after at most k_n post-pilot successes and using flight numbers at least J_n has vanishing error if $k_n \tau^{J_n} \rightarrow 0$ and $\eta_{0,n} \rightarrow 0$. This includes adaptive selection of positive physical offsets and early stopping. Neither this criterion nor the exact pilot removes a preparation charge. It is a consequence of the relative physical theorem for complete experiments; it does not replace the separate full-profile regularization argument, or assert an optimal rate for choosing J_n .

23. BOUNDARY INFORMATION AFTER RESIDUAL-TIME INTEGRATION

Residual-time integration preserves the smooth relative law of Theorem 2.1, but it changes the singularity of the finite-versus-boundary experiment. Before integration the tangent success density is positive up to a moving boundary. After integration it vanishes there to first order. The resulting logarithmic information is a boundary quantity. The theorem below isolates that quantity before the billiard specialization.

The one-dimensional change of rate caused by parameter-dependent support is classical. Akahira [30, 31] and Ibragimov–Has’minskii [32] belong to the general nonregular background, and Smith [33] explicitly treats the linearly vanishing endpoint case. Limits of experiments for models with parameter-dependent support are also developed by Hirano–Porter [34]. We do not use these references as a black-box proof: the form needed here has a moving regular hypersurface, an intrinsic coefficient, and an independent failure atom, and we prove it directly.

23.1. A linearly vanishing density. Let $U \subset \mathbb{R}^m$ be bounded and open. For $|t| < t_0$ let a_t and w_t be real functions which are C^3 jointly on a neighborhood of $[-t_0, t_0] \times \bar{U}$. Assume that a_t is bounded above and bounded away from zero, that w_t is uniformly negative near ∂U , and that

$$D = \{x \in U : w_0(x) > 0\} \neq \emptyset, \quad \Sigma = \{x \in U : w_0(x) = 0\}, \quad |\nabla w_0| > 0 \quad \text{on } \Sigma.$$

Thus $\bar{D} \subset U$ and $\Sigma = \partial D$ is a compact regular hypersurface. Decreasing t_0 if necessary, suppose

$$f_t(x) = a_t(x)(w_t(x))_+, \quad \int_U f_t(x) dx = 1. \quad (23.1)$$

Normalizing an initially unnormalized density is allowed: a common tubular chart for the moving boundary makes its positive normalizing integral C^3 . Put

$$v(x) = \partial_t w_t(x)|_{t=0}, \quad \mathcal{I}_\Sigma = \int_\Sigma \frac{a_0(x)v(x)^2}{|\nabla w_0(x)|} d\sigma(x). \quad (23.2)$$

We use

$$H^2(P, Q) = \int (\sqrt{dP/d\nu} - \sqrt{dQ/d\nu})^2 d\nu, \quad \|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|,$$

so $0 \leq H^2 \leq 2$, and write Φ for the standard normal distribution function.

Theorem 23.1 (Logarithmic boundary information). *Under (23.1),*

$$H^2(f_t dx, f_0 dx) = \frac{\mathcal{I}_\Sigma}{4} t^2 \log \frac{1}{|t|} + O(t^2), \quad t \rightarrow 0. \quad (23.3)$$

Assume $\mathcal{I}_\Sigma > 0$. Let $t_n \rightarrow 0$, $t_n \neq 0$, let $0 < p_n \leq 1$, and consider n independent observations from

$$P_{n,0} = (1 - p_n)\delta_{\dagger} + p_n f_0 dx, \quad P_{n,1} = (1 - p_n)\delta_{\dagger} + p_n f_{t_n} dx, \quad (23.4)$$

where the failure symbol $\dagger$ is retained. If

$$np_n t_n^2 \log(1/|t_n|) \rightarrow b \in [0, \infty], \quad (23.5)$$

then

$$\|P_{n,1}^{\otimes n} - P_{n,0}^{\otimes n}\|_{\text{TV}} \rightarrow 2\Phi(\sqrt{\mathcal{I}_\Sigma b}/2) - 1, \quad (23.6)$$

with the right-hand side interpreted as one when $b = \infty$.

If $0 < b < \infty$, the log likelihood of the absolutely continuous part of $P_{n,1}^{\otimes n}$ relative to $P_{n,0}^{\otimes n}$ converges under the latter law to

$$\mathcal{N}(-\mathcal{I}_\Sigma b/2, \mathcal{I}_\Sigma b), \quad (23.7)$$

and the minimum equal-prior testing error tends to $\Phi(-\sqrt{\mathcal{I}_\Sigma b}/2)$.

If $b < \infty$ (including $b = 0$), the probability that at least one of the n observations falls in a support-exclusive region tends to zero under either hypothesis. No support-exclusivity assertion is made under the sole assumption $b = \infty$; the conclusion (23.6) in that regime follows from Hellinger affinity and remains valid even when $np_n t_n^2$ does not tend to zero.

The last paragraph is a scope distinction, not a weakening of the supercritical theorem. At finite information scale, $np_n t_n^2 = (np_n t_n^2 \log(1/|t_n|))/\log(1/|t_n|) \rightarrow 0$, so exclusive support is negligible. At supercritical scale that implication is false in general, but it is unnecessary for total-variation separation.

23.2. Coarea estimates. For $x \in D$ define

$$S(x) = \frac{\dot{a}_0(x)}{a_0(x)} + \frac{v(x)}{w_0(x)}.$$

It is integrable under $f_0 dx$ but, when $\mathcal{I}_\Sigma > 0$, has infinite second moment. For $h > 0$ put $D_h = \{w_0 \geq h\}$ and $S_h = S \mathbf{1}_{D_h}$.

Lemma 23.2 (Truncated score moments). *As $h \downarrow 0$,*

$$\Pr_0\{0 < w_0 < h\} = O(h^2), \quad \mathbb{E}_0 S_h = O(h), \quad (23.8)$$

$$\mathbb{E}_0 S_h^2 = \mathcal{I}_\Sigma \log(1/h) + O(1), \quad (23.9)$$

$$\mathbb{E}_0 |S_h|^3 = O(h^{-1}), \quad \mathbb{E}_0 |S_h|^4 = O(h^{-2}). \quad (23.10)$$

For $|t|$ sufficiently small, the f_t -mass of its support outside D and the f_0 -mass of the region where $f_t = 0$ are both $O(t^2)$. The estimates are uniform on compact families with the stated strict positivity and regularity bounds.

Proof. Choose $h_0 > 0$ so that $|\nabla w_0|$ is bounded away from zero on $|w_0| \leq 2h_0$. The flow of $\nabla w_0/|\nabla w_0|^2$ identifies the level sets $w_0 = z$ for $|z| \leq h_0$ with Σ , with uniformly bounded C^2 Jacobians. By coarea,

$$J(z) = \int_{\{w_0=z\}} \frac{a_0 v^2}{|\nabla w_0|} d\sigma = \mathcal{I}_\Sigma + O(|z|).$$

The collar contribution of v^2/w_0^2 to the second moment is therefore $\int_h^{h_0} J(z) dz/z$. The remaining terms in S^2 are integrable, which proves (23.9). Since $f_0 = O(w_0)$ in the collar,

$$\int_{\{0 < w_0 < h\}} f_0 dx = O(h^2), \quad \int_{D_h} |S|^r f_0 dx \leq C \left(1 + \int_h^{h_0} z^{1-r} dz \right), \quad r = 3, 4.$$

Differentiating $\int f_t = 1$ at zero gives $\int_D (\dot{a}_0 w_0 + a_0 v) dx = 0$, hence $\mathbb{E}_0 S = 0$. The omitted collar integral of $S f_0$ is $O(h)$, proving (23.8). Finally $|w_t - w_0| \leq C|t|$, so both support-exclusive regions lie in $|w_0| \leq C|t|$; there the relevant density is $O(|t|)$ and the collar volume is $O(|t|)$. Their masses are $O(t^2)$. $\square$

23.3. Proof of the information theorem.

Proof of Theorem 23.1. Write $t \neq 0$ and $\ell = \log(1/|t|)$. On D_h , whenever $h \geq C_0|t|$ for a sufficiently large fixed C_0 , Taylor expansion of $a_t w_t$ before division by $a_0 w_0$ gives

$$\frac{f_t}{f_0} = 1 + tS + t^2 R_t, \quad |R_t(x)| \leq C(1 + w_0(x)^{-1}). \quad (23.11)$$

This expansion never differentiates a support indicator.

Take first $h = C_0|t|$. The integral of $f_0 + f_t$ outside the common part D_h is $O(t^2)$. On D_h set $z = tS + t^2 R_t$. With C_0 large and then t small, $|z| \leq 1/2$, and $(\sqrt{1+z} - 1)^2 = z^2/4 + O(|z|^3)$. The coarea bounds give

$$\int_{D_h} z^2 f_0 dx = t^2 \mathbb{E}_0 S_h^2 + O(|t|^3 \log(1/|t|)), \quad \int_{D_h} |z|^3 f_0 dx = O(t^2).$$

Together with (23.9) this proves (23.3).

Assume now $0 < b < \infty$. In the n th row write

$$t = t_n, \quad p = p_n, \quad \ell = \log(1/|t|), \quad L = \ell^{1/4}, \quad h = |t|L.$$

Represent a null observation by a Bernoulli success indicator Y_i of mean p and, on success, an independent X_i of density f_0 . With probability tending to one all successful X_i lie in D_h , because

$$Cnph^2 = C(npt^2\ell)L^2/\ell = o(1).$$

The same is true under the alternative. In particular the total support-exclusive probability is $O(npt^2) = o(1)$ in this finite critical regime.

On the high-probability common-support event,

$$\log \frac{dP_{n,1}^{\otimes n}}{dP_{n,0}^{\otimes n}} = t \sum_{i=1}^n Y_i S_h(X_i) - \frac{t^2}{2} \sum_{i=1}^n Y_i S_h(X_i)^2 + o_{P_{n,0}^{\otimes n}}(1). \quad (23.12)$$

Indeed the summed $t^2 R_t$ contribution is $O(npt^2) = o(1)$, the cubic logarithmic remainder has expectation $O(np|t|^3/h) = O((npt^2\ell)/(L\ell)) = o(1)$, and replacing the quadratic term by $t^2 S_h^2$ produces $o(1)$ by the same coarea estimates.

The mean of the first sum in (23.12) is $O(np|t|h) = o(1)$ and its variance tends to $\mathcal{I}_{\Sigma} b$. Each centered summand after multiplication by t is $O(1/L)$, so Lindeberg is automatic. The quadratic sum has mean $\mathcal{I}_{\Sigma} b + o(1)$ and variance bounded by

$$Cnpt^4 \mathbb{E}_0 |S_h|^4 \leq Cnpt^4 h^{-2} = C(npt^2\ell)/(L^2\ell) = o(1).$$

This proves (23.7).

Let Z_n be the log likelihood, with $Z_n = -\infty$ on the null-exclusive set. Even in the presence of an alternative singular part,

$$1 - \|P_{n,1}^{\otimes n} - P_{n,0}^{\otimes n}\|_{\text{TV}} = \mathbb{E}_{P_{n,0}^{\otimes n}} \min\{1, e^{Z_n}\}.$$

The integrand is bounded and continuous on the extended real line. If $v = \mathcal{I}_\Sigma b$ and $G \sim \mathcal{N}(0, 1)$, then

$$\mathbb{E} \min\{1, e^{\sqrt{v}G - v/2}\} = 2\Phi(-\sqrt{v}/2),$$

which proves the finite positive case of (23.6) and the testing-risk assertion.

For $b = 0$ and $b = \infty$ use affinity instead. The one-observation affinity of the thinned pair is exactly $1 - pH^2(f_t, f_0)/2$. Affinities multiply, and if $\mathcal{A}(P, Q)$ is the affinity then

$$1 - \mathcal{A}(P, Q) \leq \|P - Q\|_{\text{TV}} \leq \sqrt{1 - \mathcal{A}(P, Q)^2}.$$

Equation (23.3) therefore gives $\text{TV} \rightarrow 0$ when $b = 0$ and $\text{TV} \rightarrow 1$ when $b = \infty$, without any assumption on npt^2 . If merely $b < \infty$, the support-exclusive probability is $O(npt^2) = o(1)$ by the same division by ℓ used above. This proves all remaining statements. $\square$

Remark 23.3 (Intrinsic coefficient). Replace w_t by $\tilde{w}_t = c_t w_t$, $c_t > 0$ smooth, and a_t by a_t/c_t . On Σ , $\partial_t \tilde{w}_t|_0 = c_0 v$ and $|\nabla \tilde{w}_0| = c_0 |\nabla w_0|$, so (23.2) is unchanged. Equivalently, if $V_\nu = -v/|\nabla w_0|$ is the normal velocity of the support boundary,

$$\mathcal{I}_\Sigma = \int_\Sigma a_0 |\nabla w_0| V_\nu^2 d\sigma.$$

If this integral vanishes, the theorem supplies only $H^2 = O(t^2)$; no nontrivial logarithmic critical scale is asserted. When it is positive, the smooth interior Fisher contribution is lower order.

24. VECTOR BOUNDARY INFORMATION FOR A MOVING-SUPPORT LOCAL FAMILY

We formulate the moving-support experiment without assuming absolute continuity with respect to its reference member. The original family is jointly dominated by Lebesgue measure and its failure atom, but a likelihood ratio relative to the reference law need not exist on the support-exclusive part. We separate that mass and apply parameter-independent common-collared censoring. The censored family is dominated by its censored reference member and satisfies a uniform LAN expansion. The original family is uniformly asymptotically equivalent to that representative on every compact local set. We distinguish this equivalence statement from finite-subexperiment convergence to the Gaussian shift and formulate singular information on its identifiable quotient. The further compact-parameter conclusion, proved by a uniform Hellinger modulus in Theorem 27.2, is a separate step. Proposition 30.1 records the common measures explicitly.

Throughout this section we use the Hellinger convention

$$H^2(P, Q) = \int (\sqrt{p} - \sqrt{q})^2 d\mu = 2 - 2 \int \sqrt{pq} d\mu, \quad (24.1)$$

for any common dominating measure μ . With this convention $H^2(P^{\otimes k}, Q^{\otimes k}) \leq kH^2(P, Q)$.

Let $\Theta \subset \mathbb{R}^r$ be a neighborhood of the origin. On a fixed bounded open set $U \subset \mathbb{R}^m$ suppose

$$f_\theta(x) = a_\theta(x)(w_\theta(x))_+, \quad \int_U f_\theta(x) dx = 1. \quad (24.2)$$

Assume that a_θ, w_θ are C^3 jointly in (θ, x) on a common compact neighborhood of the relevant supports, uniformly for small θ , that a_θ is bounded above and bounded away from zero there, and that

$$\Sigma = \{w_0 = 0\} \cap \text{supp } a_0$$

is a compact regular hypersurface with $|\nabla w_0| \geq c_* > 0$. After shrinking Θ , all moving boundaries lie in one fixed tubular collar. Put

$$V(x) = D_\theta w_\theta(x)|_{\theta=0} \in \mathbb{R}^r, \quad \mathcal{J}_\Sigma = \int_\Sigma \frac{a_0(x)V(x)V(x)^t}{|\nabla w_0(x)|} d\sigma(x). \quad (24.3)$$

Multiplying w_θ by a positive smooth factor and dividing a_θ by the same factor leaves $\mathcal{J}_\Sigma$ unchanged.

Let $\delta_n \downarrow 0$, $0 < p_n \leq 1$, and assume

$$np_n\delta_n^2 \log(1/\delta_n) \longrightarrow 1. \quad (24.4)$$

For $h \in \mathbb{R}^r$ define on $\Omega = U \sqcup \{\dagger\}$

$$P_{n,h} = (1 - p_n)\delta_\dagger + p_n f_{\delta_n h}(x) dx. \quad (24.5)$$

No absolute continuity of $P_{n,h}$ with respect to $P_{n,0}$ is assumed.

24.1. Collar estimates and the singular part.

Lemma 24.1 (Uniform vector collar estimates). *Let*

$$S(x) = D_\theta \log a_\theta(x)|_0 + \frac{V(x)}{w_0(x)}, \quad D = \{w_0 > 0\}.$$

There are $q_0, C > 0$ such that, for $0 < q < q_0$ and unit vectors $u, v \in \mathbb{R}^r$,

$$\mathbb{E}_0[(u^t S)(v^t S) \mathbf{1}_{\{w_0 \geq q\}}] = u^t \mathcal{J}_\Sigma v \log(1/q) + O(1), \quad (24.6)$$

$$\mathbb{E}_0[|u^t S|^3 \mathbf{1}_{\{w_0 \geq q\}}] \leq Cq^{-1}, \quad (24.7)$$

$$\mathbb{E}_0[|u^t S|^4 \mathbf{1}_{\{w_0 \geq q\}}] \leq Cq^{-2}, \quad (24.8)$$

$$\|\mathbb{E}_0[S \mathbf{1}_{\{w_0 \geq q\}}]\| \leq Cq, \quad \mathbb{P}_0(0 < w_0 < q) \leq Cq^2. \quad (24.9)$$

The constants are uniform for the indicated unit directions and on compact families satisfying the displayed regularity bounds.

Proof. Use normal coordinates (y, s) with $s = w_0$ in a fixed tubular neighborhood of Σ . The Jacobian is $|\nabla w_0|^{-1} + O(s)$ and the null density is $a_0 s$. The only nonintegrable contribution to the second score moment is $a_0(u^t V)(v^t V)/s$, whose normal integral gives (24.6). The k th singular moment has normal integrand $O(s^{1-k})$, proving (24.7) and (24.8). The same coordinates give the quadratic collar mass. Differentiating $\int f_\theta = 1$ at zero gives $\mathbb{E}_0 S = 0$ in the improper integrable sense; the omitted collar has first score moment $O(q)$. $\square$

Lemma 24.2 (Support-exclusive mass). *For every compact $K \subset \mathbb{R}^r$ there is C_K such that, uniformly for $h \in K$ and all large n ,*

$$P_{n,h}^{\perp 0}(\Omega) \leq C_K p_n \delta_n^2, \quad (24.10)$$

$$P_{n,0}\{f_{\delta_n h} = 0 < f_0\} \leq C_K p_n \delta_n^2. \quad (24.11)$$

Consequently, under either product experiment, the probability of seeing at least one support-exclusive observation is at most $C_K np_n \delta_n^2 = o(1)$.

Proof. Joint C^1 regularity gives $|w_{\delta_n h} - w_0| \leq C_K \delta_n$ on the common collar. A point belonging to exactly one support therefore lies in $|w_0| \leq C_K \delta_n$. On that strip the relevant density is $O(\delta_n)$ and the strip has volume $O(\delta_n)$. Multiplication by p_n gives the one-observation estimates. A union bound and (24.4) give the product estimate. $\square$

24.2. A reference-dominated common-collar representative. Put

$$L_n = (\log(1/\delta_n))^{1/4}, \quad q_n = \delta_n L_n, \quad C_n = \{w_0 \geq q_n\}. \quad (24.12)$$

Introduce a second cemetery symbol ∂ and let $T_n : \Omega \rightarrow C_n \sqcup \{\dagger, \partial\}$ keep $\dagger$ and every $x \in C_n$, while sending every successful $x \notin C_n$ to ∂ . Write

$$\bar{P}_{n,h} = T_n P_{n,h} = (1 - p_n)\delta_\dagger + p_n f_{\delta_n h} \mathbf{1}_{C_n} dx + p_n r_{n,h} \delta_\partial, \quad r_{n,h} = \int_{C_n^c} f_{\delta_n h} dx. \quad (24.13)$$

Lemma 24.3 (Common-collar asymptotic equivalence). *For every compact $K \subset \mathbb{R}^r$,*

$$\sup_{h \in K} r_{n,h} \leq C_K q_n^2, \quad \sup_{h \in K} np_n r_{n,h} = o(1). \quad (24.14)$$

For all large n , $\{\bar{P}_{n,h} : h \in K\}$ is dominated by $\bar{P}_{n,0}$, and

$$\Delta\left(\{P_{n,h}^{\otimes n} : h \in K\}, \{\bar{P}_{n,h}^{\otimes n} : h \in K\}\right) \leq C_K np_n q_n^2 = O((\log(1/\delta_n))^{-1/2}) = o(1). \quad (24.15)$$

Both comparison kernels can be chosen independently of h .

Proof. For $h \in K$, the moving support lies in $\{w_0 > -C_K \delta_n\}$. Integrating $f_{\delta_n h} = O(w_0 + C_K \delta_n)$ over $-C_K \delta_n < w_0 < q_n$ gives $r_{n,h} = O(q_n^2)$. Equation (24.4) gives the second bound.

On C_n , $w_{\delta_n h} \geq q_n - C_K \delta_n \geq q_n/2$ for all large n , uniformly on K , so all successful densities are mutually positive there. Moreover $r_{n,0} > 0$: the null collar $0 < w_0 < q_n$ has positive mass. Hence the ∂ atom of the null censored law has positive mass, and the whole censored family is dominated by $\bar{P}_{n,0}$.

The forward kernel is the parameter-independent censoring map T_n . For a reverse kernel, keep C_n and $\dagger$ fixed and send ∂ to one fixed point of the original successful sample space. This reverse kernel is also parameter independent. Couple it with the original observation before censoring; disagreement is possible only if censoring occurred. A union bound gives probability at most $np_n r_{n,h}$, uniformly on K , proving (24.15). $\square$

24.3. Uniform LAN on the representative.

Lemma 24.4 (Uniform LAN). *For h in a fixed compact set define under $\bar{P}_{n,0}^{\otimes n}$*

$$\Delta_n = \delta_n \sum_{i=1}^n Y_i S(X_i) \mathbf{1}_{C_n}(X_i), \quad (24.16)$$

$$Q_n = \delta_n^2 \sum_{i=1}^n Y_i S(X_i) S(X_i)^t \mathbf{1}_{C_n}(X_i), \quad (24.17)$$

where Y_i is the success indicator and, conditionally on success under the null, X_i has density f_0 . Then

$$\log \frac{d\bar{P}_{n,h}^{\otimes n}}{d\bar{P}_{n,0}^{\otimes n}} = h^t \Delta_n - \frac{1}{2} h^t Q_n h + o_{\bar{P}_{n,0}}(1) \quad (24.18)$$

uniformly for h in compact subsets of $\mathbb{R}^r$, and

$$\Delta_n \Rightarrow N_r(0, \mathcal{I}_\Sigma), \quad Q_n \longrightarrow \mathcal{I}_\Sigma \quad \text{in probability.} \quad (24.19)$$

The probability that the atom ∂ occurs in the product experiment is $o(1)$ uniformly on compact local parameter sets.

Proof. The last assertion is Lemma 24.3. On its complement the likelihood ratio uses only the common support C_n and the unchanged failure atom. Taylor expansion before division by w_0 gives, uniformly for bounded h and $x \in C_n$,

$$\log \frac{f_{\delta_n h}(x)}{f_0(x)} = \delta_n h^t S(x) - \frac{\delta_n^2}{2} (h^t S(x))^2 + R_{n,h}(x).$$

The regular parameter remainder has expectation $O(\delta_n^2)$ per successful observation, while the cubic logarithmic remainder is bounded in expectation by $C_K \delta_n^3 \mathbb{E}_0(1 + \|S\|^3) \mathbf{1}_{C_n}$. Lemma 24.1 and (24.4) give

$$np_n \delta_n^2 = o(1), \quad np_n \delta_n^3 q_n^{-1} = o(1),$$

which proves (24.18).

The truncated-score mean has norm at most Cq_n , so $\|\mathbb{E}\Delta_n\| \leq Cnp_n\delta_n q_n = o(1)$. Its covariance tends to $\mathcal{J}_\Sigma$ because $\log(1/q_n)/\log(1/\delta_n) \rightarrow 1$. On C_n every scalar score contribution multiplied by δ_n is $O(L_n^{-1})$; Lindeberg therefore holds uniformly on the unit sphere and Cramér–Wold gives the first limit in (24.19). Finally

$$np_n\delta_n^4 \mathbb{E}_0[\|\mathbf{S}\|^4 \mathbf{1}_{C_n}] \leq Cnp_n\delta_n^4 q_n^{-2} = o(1),$$

while the mean of Q_n tends to $\mathcal{J}_\Sigma$. Hence $Q_n \rightarrow \mathcal{J}_\Sigma$ in probability. $\square$

Lemma 24.5 (Bounded-sequence contiguity). *If h_n is any bounded sequence, then $\overline{P}_{n,h_n}^{\otimes n}$ and $\overline{P}_{n,0}^{\otimes n}$ are mutually contiguous. The same is true for $P_{n,h_n}^{\otimes n}$ and $P_{n,0}^{\otimes n}$.*

Proof. Along every subsequence, take a further subsequence on which $h_n \rightarrow h$. Lemma 24.4 gives joint convergence of the log likelihood ratio under the null to $h^t Z - \frac{1}{2} h^t \mathcal{J}_\Sigma h$ with $Z \sim N(0, \mathcal{J}_\Sigma)$. The limiting likelihood is strictly positive and has mean one. The normalized-likelihood argument in Proposition 24.12 therefore applies: truncation gives uniform integrability and forward contiguity, while positivity of the null-law likelihood limit gives reverse contiguity. This argument uses only the null LAN expansion and unit likelihood means; it does not assume a central-sequence limit under the alternative. Since every subsequence admits such a further subsequence, the whole bounded sequence is mutually contiguous. The common-space censoring coupling in that proposition, with the uniform error from Lemma 24.3, transfers the assertion to the original experiments. $\square$

24.4. Finite likelihood experiments and normalization. We record the finite-experiment passage used below, including its normalization hypothesis and both comparison kernels. It is a standard statistical fact; the proof is included to specify exactly what is used. The statement by itself makes no assertion on an infinite parameter set.

Lemma 24.6 (Finite likelihood experiments). *For each n , let $(P_{n,i})_{i=0}^m$ be probability measures on a standard Borel space, with $P_{n,i} \ll P_{n,0}$, and put $L_{n,i} = dP_{n,i}/dP_{n,0}$. Suppose that, under $P_{n,0}$,*

$$(L_{n,1}, \dots, L_{n,m}) \Rightarrow (L_1, \dots, L_m) \quad \text{under } Q_0, \quad L_i \geq 0, \quad \mathbb{E}_{Q_0} L_i = 1.$$

Here Q_0 is a probability measure on a standard Borel space, the L_i are measurable, and $Q_i = L_i Q_0$. Then

$$\Delta(\{P_{n,i} : 0 \leq i \leq m\}, \{Q_i : 0 \leq i \leq m\}) \rightarrow 0. \quad (24.20)$$

The kernels in both directions are independent of the unknown index i . No uniform integrability assumption beyond the stated normalization is needed: it follows from the likelihood convergence and the unit means.

Proof. Write μ_n and μ for the respective laws of the likelihood vectors on $[0, \infty)^m$. Each coordinate has integral one under both measures. For every $A > 0$, weak convergence and bounded continuity of $x \mapsto x_i \wedge A$ give

$$\int (x_i - A)_+ d\mu_n = 1 - \int (x_i \wedge A) d\mu_n \rightarrow 1 - \int (x_i \wedge A) d\mu = \int (x_i - A)_+ d\mu.$$

Together with $x_i \mathbf{1}_{\{x_i > 2A\}} \leq 2(x_i - A)_+$, this proves uniform integrability of every coordinate (the finitely many initial indices are harmless), hence of the ℓ^1 norm.

There are couplings π_n of μ_n, μ for which

$$\int \|x - y\|_1 d\pi_n(x, y) \rightarrow 0. \quad (24.21)$$

For completeness, choose a large cube in $[0, \infty)^m$ whose outer faces are μ -null, and partition it into finitely many rectangles of ℓ^1 -diameter at most ε , with boundaries that are μ -null relative

to $[0, \infty)^m$. Weak convergence makes all rectangle masses converge. Couple the minimum of the two masses inside each matching rectangle; couple the remaining masses arbitrarily. The matched part costs at most ε ; the unmatched part in the fixed cube tends to zero. Contributions involving its complement tend uniformly to zero as the cube grows, by the just-proved uniform integrability. Taking the sample size limit, then enlarging the cube and sending ε to zero, and choosing a diagonal sequence, proves (24.21).

Put $x_0 = y_0 = 1$. The likelihood-vector statistic identifies the first experiment exactly with $(x_i \mu_n)_{i=0}^m$, and the second with $(y_i \mu)_{i=0}^m$. Indeed, the forward kernel is the statistic; a reverse kernel is a regular conditional distribution under the reference measure given that statistic. Multiplication by a likelihood coordinate recovers the corresponding alternative. Standard Borelness ensures the required conditional distributions exist.

Disintegrate π_n in the two directions. For its kernel from x to y , the difference between the simulated alternative and $y_i \mu$ is the y -marginal of $(x_i - y_i) \pi_n$. Since this signed measure has total mass zero,

$$\|K_n(x_i \mu_n) - y_i \mu\|_{\text{TV}} \leq \frac{1}{2} \int |x_i - y_i| d\pi_n.$$

The reversed disintegration gives the identical bound in the other direction. Composing with the preceding statistic and reconstruction kernels proves (24.20). The same coupling and the same disintegrations serve every index, including $i = 0$; no kernel receives the unknown index as input. $\square$

Remark 24.7 (Gaussian likelihoods and the scope of the lemma). If $Z \sim N(0, J)$ with J positive semidefinite, then, for each fixed h ,

$$L_h = \exp\{h^t Z - \frac{1}{2} h^t J h\} > 0, \quad \mathbb{E} L_h = 1.$$

Thus joint LAN limits on any finite parameter set satisfy the normalization required in Lemma 24.6, also when J is singular or zero. One may adjoin the reference index to the finite set and then restrict the resulting kernels. For a moving-support family the lemma is applied to its reference-dominated common-collar representative, not to nonexistent reference likelihoods on support-exclusive observations. Compact-parameter convergence additionally uses the uniform modulus and finite-net argument in Theorem 27.2; it does not follow from the finite lemma alone.

24.5. The Gaussian limit on the identifiable quotient. Let

$$\mathbf{H} = \text{Ran}(\mathcal{J}_\Sigma), \quad \Pi_{\mathbf{H}} : \mathbb{R}^r \rightarrow \mathbf{H} \tag{24.22}$$

be the orthogonal projection. The restriction $\mathcal{J}_{\mathbf{H}} = \mathcal{J}_\Sigma|_{\mathbf{H}}$ is positive definite. The Gaussian limit experiment is most cleanly written in central-sequence coordinates as

$$\Delta \sim N_{\mathbf{H}}(\mathcal{J}_{\mathbf{H}} \Pi_{\mathbf{H}} h, \mathcal{J}_{\mathbf{H}}). \tag{24.23}$$

Equivalently, $Z = \Pi_{\mathbf{H}} h + \mathcal{J}_{\mathbf{H}}^{-1/2} G$ with standard Gaussian G on $\mathbf{H}$. Directions in $\ker \mathcal{J}_\Sigma$ are not identifiable at the critical logarithmic scale.

Theorem 24.8 (Moving-support vector boundary Gaussian limit). *Under (24.2)–(24.4):*

- (1) *On every compact local parameter set, the original experiment is at $o(1)$ Le Cam distance from the reference-dominated common-collar representative, by (24.15). The representative satisfies the compact- uniform LAN expansion (24.18) and the central limits (24.19).*
- (2) *For every finite collection $h_1, \dots, h_m$, the corresponding finite local subexperiment converges in Le Cam distance to the restriction of the Gaussian experiment (24.23) to those parameters. More generally, every bounded local sequence satisfies the contiguity and third-lemma conclusions implied by the LAN expansion. We do not identify this finite-subexperiment statement with a separate assertion of global deficiency convergence for an*

uncountable parameter set. The stronger conclusion on every fixed compact set is proved in Theorem 27.2.

(3) For every fixed h ,

$$\|P_{n,h}^{\otimes n} - P_{n,0}^{\otimes n}\|_{\text{TV}} \rightarrow 2\Phi\left(\frac{\sqrt{h^t \mathcal{J}_\Sigma h}}{2}\right) - 1. \quad (24.24)$$

If $\mathcal{J}_\Sigma$ is positive definite, the reference-model statistic

$$\hat{h}_n = \mathcal{J}_\Sigma^{-1} \Delta_n \quad (24.25)$$

(with observations outside C_n contributing zero) satisfies $\hat{h}_n \Rightarrow N_r(h, \mathcal{J}_\Sigma^{-1})$ uniformly on compact local sets. If $\mathcal{J}_\Sigma$ is singular, the corresponding estimator of the identifiable coordinate is $\hat{h}_n^{\text{id}} = \mathcal{J}_\Sigma^+ \Delta_n$, with limit centered at $\Pi_{\mathbf{H}} h$ and covariance $\mathcal{J}_\Sigma^+$.

(4) Let W be positive semidefinite on $\mathbf{H}$ and $\ell_M(z) = \min\{z^t W z, M\}$. The standard local asymptotic minimax theorem applied to the LAN experiment on $\mathbf{H}$ gives

$$\lim_{R \rightarrow \infty} \liminf_{n \rightarrow \infty} \inf_{T_n} \sup_{\substack{h \in \mathbf{H} \\ \|h\| \leq R}} \mathbb{E}_{n,h} \ell_M(T_n - h) \geq \mathbb{E} \ell_M(Z_0), \quad Z_0 \sim N_{\mathbf{H}}(0, \mathcal{J}_\Sigma^{-1}). \quad (24.26)$$

Letting $M \rightarrow \infty$ gives $\text{tr}(W \mathcal{J}_\Sigma^{-1})$. When $\mathcal{J}_\Sigma$ is positive definite, this is the full-parameter bound.

Proof. Item 1 is Lemmas 24.3 and 24.4. For any finite parameter set, the joint likelihood-ratio vector is a continuous function of Δ_n, Q_n plus a uniform $o_P(1)$ term. Equations (24.19) therefore give joint weak convergence of the likelihood-ratio vector to the Gaussian likelihood vector in (24.23). Each prelimit likelihood has mean one under the censored reference law; each limiting coordinate has mean one by Remark 24.7. Adjoin the reference parameter to the finite set, apply Lemma 24.6, and then restrict. Lemma 24.3 transfers the resulting two-sided comparison to the original family. Lemma 24.5 gives the bounded-sequence statement.

Le Cam's third lemma translates the limiting central sequence by $\mathcal{J}_\Sigma h$. The equal-covariance Gaussian total-variation formula gives (24.24). The estimator statements follow by linear transformation of the central sequence; in the singular case the Moore–Penrose inverse acts only on $\mathbf{H}$.

We verify the moment assertion under the original local alternatives. Fix a compact set K and write $s = w_0(x)$. On C_n , Taylor expansion of $a_\theta w_\theta$ before division by $a_0 s$ gives

$$\frac{f_{\delta_n h}(x)}{f_0(x)} = 1 + \delta_n h^t \mathbf{S}(x) + O_K\left(\delta_n^2(1 + s^{-1})\right), \quad h \in K. \quad (24.27)$$

The bound is uniform because $s \geq q_n$ and $w_{\delta_n h} \geq q_n/2$ there. In particular, the ratio for one successful observation tends uniformly to one on C_n ; no such assertion is made for the product likelihood. In the normal coordinate s , $\|\mathbf{S}\| = O(1 + s^{-1})$ and $f_0 dx = O(s) ds d\sigma$. Thus integration of the remainder in (24.27) against $\mathbf{S} f_0$ yields

$$\begin{aligned} \mathbb{E}_{n,h} \Delta_n &= np_n \delta_n \mathbb{E}_0[\mathbf{S} \mathbf{1}_{C_n}] + np_n \delta_n^2 \mathbb{E}_0[\mathbf{S} \mathbf{S}^t \mathbf{1}_{C_n}] h \\ &\quad + O_K\left(np_n \delta_n^3 \log(1/q_n)\right) = \mathcal{J}_\Sigma h + o_K(1). \end{aligned} \quad (24.28)$$

Here the first term is $O(np_n \delta_n q_n) = o(1)$ by (24.9); the second converges uniformly on K by (24.6) and $\log(1/q_n)/\log(1/\delta_n) \rightarrow 1$. The displayed remainder also tends to zero. This establishes, in particular, a uniformly bounded mean.

Put $\xi_{n,i} = \delta_n Y_i \mathbf{S}(X_i) \mathbf{1}_{C_n}(X_i)$, with value zero for a failure or a successful observation outside C_n . Under each fixed local alternative these vectors are independent. The uniformly bounded

one-observation density ratio and the collar moment estimates give

$$\sum_{i=1}^n \mathbb{E}_{n,h} \|\xi_{n,i}\|^2 \leq C_K n p_n \delta_n^2 \log(1/q_n), \quad \sum_{i=1}^n \mathbb{E}_{n,h} \|\xi_{n,i}\|^4 \leq C_K n p_n \delta_n^4 q_n^{-2}.$$

Expanding the fourth power of the centered sum coordinate by coordinate therefore gives

$$\mathbb{E}_{n,h} \|\Delta_n - \mathbb{E}_{n,h} \Delta_n\|^4 \leq C_K \left\{ [n p_n \delta_n^2 \log(1/q_n)]^2 + n p_n \delta_n^4 q_n^{-2} \right\} = O_K(1). \quad (24.29)$$

Indeed, for $\ell_n = \log(1/\delta_n)$ and $B_n = n p_n \delta_n^2 \ell_n \rightarrow 1$, the relevant quantities are

$$n p_n \delta_n q_n = B_n \ell_n^{-3/4}, \quad n p_n \delta_n^2 \log(1/q_n) = B_n \left(1 - \frac{\log \ell_n}{4 \ell_n} \right), \quad n p_n \delta_n^4 q_n^{-2} = B_n \ell_n^{-3/2}.$$

Restoring the bounded mean in (24.28) proves $\sup_{n \geq n_0(K)} \sup_{h \in K} \mathbb{E}_{n,h} \|\Delta_n\|^4 < \infty$. Each excluded observation contributes zero; the other summands are retained even when censoring occurs elsewhere in the sample. This proves uniform integrability for compact-local quadratic risk, also after applying the fixed linear map $\mathcal{J}_\Sigma^+$ on $\mathbf{H}$. Finally apply the classical finite-dimensional LAN local asymptotic minimax theorem on the identifiable space $\mathbf{H}$; see [32, 34]. $\square$

Remark 24.9 (Reference-model nature of the central sequence). The statistics (24.16) and (24.25) use the score and information at the reference geometry. They are local-experiment statistics, not globally adaptive estimators for an unknown reference model. The physical theorem below uses them only after a reference channel and a finite local chart have been fixed.

24.6. Quantitative stability of the moving boundary.

Lemma 24.10 (Linearly vanishing support stability). *Let $f = A(w)_+$ and $\tilde{f} = \tilde{A}(\tilde{w})_+$ be normalized densities on the same bounded open set. Suppose $A, \tilde{A}$ are uniformly positive and C^1 , the zero sets of $w, \tilde{w}$ are uniformly regular in one fixed tubular collar, and*

$$\|w - \tilde{w}\|_{C^1} \leq \varepsilon < 1/2, \quad \|A - \tilde{A}\|_{C^0} \leq \eta.$$

Then, with the convention (24.1),

$$H^2(f \, dx, \tilde{f} \, dx) \leq C \left\{ \varepsilon^2 \log \frac{e}{\varepsilon} + \eta^2 \right\}. \quad (24.30)$$

The estimate is unchanged, up to the uniform constant, if a common positive smooth factor is moved between the amplitude and defining function.

Proof. Use the tubular coordinate $s = w$. On $|s| \leq C\varepsilon$ the total mass of both densities is $O(\varepsilon^2)$. On $s > C\varepsilon$ the supports agree and

$$|\sqrt{Aw} - \sqrt{\tilde{A}\tilde{w}}|^2 \leq C \left(\eta^2 s + \frac{\varepsilon^2}{s} \right).$$

The first term is $O(\eta^2)$ after integration and the second is $O(\varepsilon^2 \log(e/\varepsilon))$. The same estimate holds using $\tilde{w}$ as the normal coordinate on the other side of the collar; away from the collar all denominators are uniformly bounded. Multiplication of $w, \tilde{w}$ by the same uniformly positive smooth factor merely changes the tubular Jacobian and amplitude bounds by uniform constants. $\square$

Remark 24.11 (Terminology). We reserve “LAN expansion” for the reference-dominated common-collar representative. For the original family we first prove asymptotic equivalence to that representative and the finite-subexperiment Gaussian limit, testing profile, contiguity, and minimax consequences. The uniform moving-boundary modulus then upgrades the Gaussian comparison to two-sided Le Cam convergence on every fixed compact local set in Theorem 27.2;

Corollary 27.3 supplies its physical endpoint-output counterpart. The original family already has a common dominating measure. Reference absolute continuity, local asymptotic normality, and convergence of experiments are distinct assertions.

24.7. Likelihood tilting and compact-local quadratic risk. We give the likelihood-tilting and moment arguments underlying Lemma 24.5 and the estimator assertions of Theorem 24.8. This also specifies how the unit-mean normalization in Lemma 24.6 is used under local alternatives. The argument concerns the moving-support model (24.5); physical acquisition is handled by the separate stopped-transfer results.

Three distinct estimates enter the conclusion. The density expansion (24.27) gives the alternative mean (24.28); centering under that same alternative then gives (24.29). These are direct moment estimates for the original product law. The normalized likelihoods and the LAN expansion instead give the weak shifted limit (24.31). Combining that weak limit with the original-law fourth-moment bound yields (24.33). Thus the argument for unbounded quadratic loss does not rely on total-variation comparison alone. The further passage from finite parameter sets to compact experiments uses the uniform modulus in Theorem 27.2.

Proposition 24.12 (Likelihood tilting and uniform quadratic risk). *Under the hypotheses of Theorem 24.8, let $h_n \rightarrow h$ be a bounded local sequence. Under the reference-dominated common-collar experiment, set*

$$L_n = \frac{d\bar{P}_{n,h_n}^{\otimes n}}{d\bar{P}_{n,0}^{\otimes n}}.$$

The likelihoods L_n are uniformly integrable under their reference measures. The reference and alternative sequences are mutually contiguous, and, for every bounded continuous φ ,

$$\mathbb{E}_{\bar{P}_{n,h_n}^{\otimes n}} \varphi(\Delta_n) \longrightarrow \mathbb{E} \left[e^{h^t Z - h^t \mathcal{J}_\Sigma h/2} \varphi(Z) \right], \quad Z \sim N(0, \mathcal{J}_\Sigma). \quad (24.31)$$

The limit is the expectation under $N(\mathcal{J}_\Sigma h, \mathcal{J}_\Sigma)$, including singular information. The same central-sequence limit holds in the original experiment, with observations outside C_n contributing zero.

For every compact $K \subset \mathbb{R}^r$, the original product experiment satisfies

$$\sup_{n \geq n_0(K)} \sup_{h \in K} \mathbb{E}_{P_{n,h}^{\otimes n}} \|\Delta_n\|^4 < \infty. \quad (24.32)$$

Write $J = \mathcal{J}_\Sigma$, $\mathbf{H} = \text{Ran } J$, and $\Pi = \Pi_{\mathbf{H}}$. If W is positive semidefinite on $\mathbf{H}$, then

$$\sup_{h \in K} \left| \mathbb{E}_{P_{n,h}^{\otimes n}} [(J^+ \Delta_n - \Pi h)^t W (J^+ \Delta_n - \Pi h)] - \text{tr}(W J^+) \right| \longrightarrow 0. \quad (24.33)$$

Here J^+ acts only on the identifiable space; no risk assertion for an unidentifiable coordinate is made.

Proof. The uniform LAN expansion gives joint convergence under the reference law,

$$(\Delta_n, L_n) \Rightarrow (Z, L), \quad L = e^{h^t Z - h^t J h/2} > 0.$$

Both $\mathbb{E} L_n$ and $\mathbb{E} L$ equal one. For fixed $A > 0$, convergence of the bounded continuous truncation gives

$$\mathbb{E}(L_n - A)_+ = 1 - \mathbb{E}(L_n \wedge A) \longrightarrow 1 - \mathbb{E}(L \wedge A) = \mathbb{E}(L - A)_+.$$

Since $L_n \mathbf{1}_{\{L_n > 2A\}} \leq 2(L_n - A)_+$, this proves uniform integrability, without any assumption on higher likelihood moments. Apply joint weak convergence to $(L_n \wedge A) \varphi(\Delta_n)$, then let $A \rightarrow \infty$. The tail bound just proved yields (24.31). Completing the Gaussian exponential, or computing its characteristic function on $\text{Ran } J$, identifies the displayed shifted normal law. This calculation does not invert J on its kernel.

For completeness, let P_n and $Q_n = L_n P_n$ denote the censored reference and alternative product measures. For every event A_n ,

$$Q_n(A_n) \leq A P_n(A_n) + \mathbb{E}_{P_n}[L_n \mathbf{1}_{\{L_n > A\}}].$$

Uniform integrability proves contiguity of Q_n with respect to P_n . In the reverse direction, for every $c > 0$,

$$P_n(A_n) \leq P_n\{L_n \leq c\} + c^{-1} Q_n(A_n).$$

First take a continuity value c for the law of L , and then send $c \downarrow 0$. Positivity of L proves reverse contiguity. Subsequence extraction proves the assertion for every bounded sequence, not only a convergent one. Embed the original and censored observations in their common space with the extra symbol ∂ . Their laws differ only on an event of probability at most $C_K n p_n q_n^2 = o(1)$, uniformly on K . This transfers the contiguity assertions to the original experiments. The central sequence is unchanged by censoring, since every censored observation already contributes zero to its definition.

We next prove the moment estimate directly under the original alternatives. On the common collar C_n , joint smoothness and positivity give the *one-successful-observation* density bound

$$\left| \frac{f_{\delta_n h}}{f_0} - 1 \right| \leq C_K \delta_n (1 + w_0^{-1}), \quad \sup_{h \in K} \left\| \frac{f_{\delta_n h}}{f_0} - 1 \right\|_{L^\infty(C_n)} = O_K(\delta_n / q_n) = o(1). \quad (24.34)$$

This is not a bound tending to one for the product likelihood. The collar estimates and (24.34) imply

$$\left\| \int_{C_n} S f_{\delta_n h} dx \right\| \leq C q_n + C_K \delta_n \log(1/q_n), \quad (24.35)$$

$$\int_{C_n} \|S\|^2 f_{\delta_n h} dx \leq C_K \log(1/q_n), \quad \int_{C_n} \|S\|^4 f_{\delta_n h} dx \leq C_K q_n^{-2}. \quad (24.36)$$

For the first inequality, the reference truncated mean is $O(q_n)$. In normal coordinates $s = w_0$, the remaining integrand is bounded by $C_K \delta_n (s + 2 + s^{-1})$; integration from q_n to a fixed collar width gives the logarithm. The complement of the collar is uniformly regular. The other inequalities follow from the uniformly bounded density ratio and Lemma 24.1.

Put $\xi_{n,i} = \delta_n Y_i S(X_i) \mathbf{1}_{C_n}(X_i)$. These vectors are independent under every fixed alternative. Their sum is Δ_n , and (24.35) gives

$$\sup_{h \in K} \|\mathbb{E}_{n,h} \Delta_n\| \leq C n p_n \delta_n q_n + C_K n p_n \delta_n^2 \log(1/q_n) = O_K(1).$$

For centered independent vectors in fixed dimension, expansion of the fourth power, coordinate by coordinate, yields

$$\begin{aligned} \mathbb{E}_{n,h} \|\Delta_n - \mathbb{E}_{n,h} \Delta_n\|^4 &\leq C_r \left\{ \left(\sum_{i=1}^n \mathbb{E}_{n,h} \|\xi_{n,i}\|^2 \right)^2 + \sum_{i=1}^n \mathbb{E}_{n,h} \|\xi_{n,i}\|^4 \right\} \\ &\leq C_K \left\{ [n p_n \delta_n^2 \log(1/q_n)]^2 + n p_n \delta_n^4 q_n^{-2} \right\} = O_K(1). \end{aligned}$$

Indeed, with $q_n = \delta_n [\log(1/\delta_n)]^{1/4}$, the first mean term $n p_n \delta_n q_n$ and the final fourth-moment term tend to zero, whereas $n p_n \delta_n^2 \log(1/q_n) \rightarrow 1$. This proves (24.32); each excluded observation, rather than the entire product sample on a censoring event, contributes zero.

The tilted central-sequence limit and (24.32) imply convergence of quadratic risks along every convergent local sequence. The Gaussian error $J^+ Z - \Pi h$, for $Z \sim N(Jh, J)$, has covariance J^+ and mean zero. Its W -risk is therefore $\text{tr}(W J^+)$. If (24.33) were not uniform on K , a violating sequence would have a convergent subsequence in K , contradicting the just-proved sequential limit. This proves the compact-uniform assertion. $\square$

25. PHYSICAL REALIZATION OF ANCHORED FINITE CONTACT-JET FAMILIES

The statistical experiment below is formulated in one fixed laboratory chart. We record that the resulting finite-dimensional contact-jet models are realized by actual nearby billiard tables. The contact patch used by the statistical theorem is chosen inside the region where every cutoff is exactly one, so the displayed polynomial jet family is literally the physical graph on the whole observation patch.

Proposition 25.1 (Anchored realization of finite labelled jets). *Let a smooth periodic dispersing billiard contain a selected labelled closest pair with positive gap g_0 , positive contact curvatures, and a strict clearance margin separating this pair from every competing collision segment outside fixed contact neighborhoods. Fix $M \geq 2$. Choose one Euclidean laboratory chart in which the reference contacts have common transverse coordinate $y = 0$ and common tangent direction.*

There exist numbers $0 < u_ < u_{**}$ and a smooth finite-dimensional family of actual periodic billiard tables through the reference table, with coordinates*

$$(g, q_{0,2}, \dots, q_{0,M}, q_{1,2}, \dots, q_{1,M}),$$

such that on the whole statistical patch $|y| < u_$ the two participating boundaries are exactly*

$$(-\psi_0(u), u), \quad (g + \psi_1(v), v), \quad (25.1)$$

with

$$\psi_b(0) = \psi'_b(0) = 0, \quad \psi_b^{(n)}(0) = q_{b,n} \quad (2 \leq n \leq M).$$

The labels, transverse origin, and tangent orientation are the same throughout the family. Strict convexity, embeddedness of every participating obstacle, disjointness of distinct obstacle closures and of their relevant lattice translates, and the selected-channel clearance margin persist uniformly after the parameter neighborhood is made sufficiently small.

The same statement holds if the two contacts lie on different labelled obstacles, on two lattice translates, or in two disjoint contact neighborhoods of one connected periodic obstacle, provided lattice-related perturbations are made equivariantly.

Proof. Choose disjoint arclength neighborhoods of the two reference contacts so small that the fixed laboratory graph representation is valid. Select smooth cutoffs χ_b supported in $|y| < u_{**}$ and identically one on a smaller interval $|y| < 2u_*$. We choose the statistical endpoint box only afterwards, inside $|y| < u_*$. Thus every graph used by the physical statistical theorem lies strictly inside the plateau of the cutoff.

For small parameters set

$$\delta\psi_b(y) = \chi_b(y) \sum_{n=2}^M \frac{\delta q_{b,n}}{n!} y^n. \quad (25.2)$$

On $|y| < u_*$ this is exactly the displayed polynomial perturbation and hence changes precisely the prescribed finite jet without moving the contact origin or tangent line. A gap variation is realized by a normal displacement of the second patch which is constant on $|y| < 2u_*$ and is cut off before the boundary leaves the chosen coordinate neighborhood. Its constant value on the statistical patch is absorbed into g in (25.1).

If the two contacts lie on one connected obstacle, take the two cutoff supports disjoint. The remaining part of the boundary is unchanged. A sufficiently small C^1 perturbation of an embedded compact curve remains embedded; a sufficiently small C^2 perturbation preserves strict positive curvature. The reference table has positive separation of distinct obstacle closures and a strict channel-isolation margin away from the chosen contact patches. On a compact fundamental cell these inequalities persist under small C^1 perturbations. For periodic translates, the unchanged lattice and a uniform obstacle diameter leave only finitely many nearby translates

to check. Shrinking the parameter ball therefore preserves global embeddedness, disjointness and channel isolation, including when both patches belong to the same connected obstacle.

Because each cutoff is constant on the statistical patch, the map from bump coefficients to the displayed contact jets has identity Jacobian. The inverse function theorem gives an open finite-dimensional coordinate family. $\square$

Remark 25.2 (Analytic versus smooth realization). The proposition serves the finite-dimensional statistical local experiment, which needs only finite smoothness on compact parameter sets. The analytic continuation conclusions of Theorems 12.1 and 14.3 are separate deterministic statements for analytic boundary germs; no compactly supported analytic bump is used there.

26. A PHYSICAL ENDPOINT EXPERIMENT ON FIXED LABORATORY WINDOWS

We now realize the boundary-information theorem on a parameter-independent physical observation space. Three points are fixed at the outset. First, all acquisition times are programmed from the reference channel and are unchanged under local alternatives. Second, the statistical theorem is formulated for the anchored registered family of Section 25, in one laboratory chart. Third, we distinguish the sigma-fields used at the successive transfer steps. The finite-to-boundary theorem does not compare the complete growing collision array.

26.1. The declared physical record hierarchy. A physical preparation has a full raw collision history, denoted $\mathcal{R}^{\text{full}}$, consisting of all collision points, obstacle labels, flight times and collision count observed during that preparation. We use three successively coarser records.

- (1) The *per-preparation endpoint-time record* is the measurable coarsening

$$C_{j,t}(\mathcal{R}^{\text{full}}) = \begin{cases} (y_{\text{first}}, y_{\text{last}}, r), & \text{if the prescribed bridge succeeds,} \\ \dagger, & \text{otherwise,} \end{cases} \quad (26.1)$$

where r is the residual preparation-to-first-impact time used in Section 21. This is the record for which the historical finite-to-boundary total-variation theorem is proved.

- (2) A policy produces the *stopped endpoint-time acquisition transcript*: every chosen design, every record (26.1), the success/failure sequence and the stopping time are retained. The adaptive coupling theorem of Section 22 compares the finite and boundary laws of this entire stopped transcript at the declared record level.
- (3) The *endpoint-output experiment* is a further common Markov coarsening which retains the selected endpoint pairs needed for inference and the cap cemetery symbol. The Gaussian information theorem below is a theorem for this endpoint-output sigma-field.

The complete raw collision history $\mathcal{R}^{\text{full}}$ may contain more information than item 1, and the stopped endpoint-time transcript may contain more information than item 3 through the waiting counts. No theorem below identifies these different sigma-fields.

26.2. An anchored registered channel in one laboratory chart. Fix once and for all a Euclidean chart (x, y) , two obstacle labels, and two facing patches. In this chart the admissible local family has the form

$$\Gamma_{0,\vartheta} = \{(-\psi_{0,\vartheta}(u), u) : |u| < u_*\}, \quad \Gamma_{1,g,\vartheta} = \{(g + \psi_{1,\vartheta}(v), v) : |v| < u_*\}, \quad (26.2)$$

where

$$\psi_{b,\vartheta}(0) = \psi'_{b,\vartheta}(0) = 0, \quad \psi''_{b,\vartheta}(0) > 0. \quad (26.3)$$

The labels, transverse origin and tangent direction are fixed throughout the family. Varying g translates the second graph in the normal x direction; it does not re-center the observed transverse coordinate. The local parameter $\vartheta \in \Theta \subset \mathbb{R}^r$ changes only the anchored graph geometry. For

the finite-dimensional theorem below, ϑ is a finite labelled contact-jet coordinate realized by Proposition 25.1.

The endpoint observation used after the residual-time coarsening is the same measurable map at every parameter:

$$\mathcal{O}_{bb}(\mathcal{R}^{\text{full}}) = \begin{cases} (y_{\text{first}}, y_{\text{last}}), & \text{if the prescribed } bb \text{ bridge succeeds,} \\ \dagger, & \text{otherwise.} \end{cases} \quad (26.4)$$

Thus endpoint positions are literal laboratory y coordinates rather than parameter-dependent contact coordinates.

Let $S_{b,g,\vartheta}$ and $B_{b,g,\vartheta}$ denote the half-line actions and positive amplitudes constructed in the same fixed transverse coordinate. At $(g_0, 0)$ write

$$S_b = S_{b,g_0,0}, \quad B_b = B_{b,g_0,0}, \quad \dot{S}_b(u)h = D_{\vartheta}S_{b,g_0,\vartheta}(u)|_0[h].$$

The anchoring gives $S_{b,g,\vartheta}(0) = S'_{b,g,\vartheta}(0) = 0$ throughout the family.

Fix finitely many same-type designs (b_ℓ, d_ℓ) with $b_\ell \in \{0, 1\}$ and $0 < d_\ell < d_*$, $1 \leq \ell \leq L$. A same-type design is always implemented with an *even* number of alternating flights. For even j its programmed acquisition time is

$$t_{j,\ell} = jg_0 + d_\ell. \quad (26.5)$$

The parity convention is part of the physical design: an odd flight number would terminate at the opposite contact and would instead involve the mixed action $S_b \oplus S_{1-b}$.

26.3. The multirate local parameter and fixed-time expansion. Let

$$j_n \in 2\mathbb{N}, \quad j_n \longrightarrow \infty, \quad (26.6)$$

and let $\delta_n \downarrow 0$. For bounded $z = (a, h) \in \mathbb{R} \times \mathbb{R}^r$ define

$$g_n(a) = g_0 + \frac{\delta_n}{j_n}a, \quad \vartheta_n(h) = \delta_n h. \quad (26.7)$$

The fixed laboratory time has exact excess

$$t_{j_n,\ell} - j_n g_n(a) = d_\ell - \delta_n a. \quad (26.8)$$

Thus a gap displacement of order δ_n/j_n and a shape displacement of order δ_n enter the moving endpoint support at the same first order.

For a same-type design (b, d) define

$$w_{b,d}(u, v) = d - S_b(u) - S_b(v). \quad (26.9)$$

Let $A_{b,d} > 0$ be the normalized smooth amplitude, so the base conditional endpoint density is $f_{b,d} = A_{b,d}(w_{b,d})_+$. Set

$$U_b(u, v)z = a + \dot{S}_b(u)h + \dot{S}_b(v)h \quad (26.10)$$

and

$$\mathcal{K}_{b,d} = \int_{\{S_b(u)+S_b(v)=d\}} \frac{A_{b,d}(u, v) U_b(u, v)^t U_b(u, v)}{|\nabla(S_b \oplus S_b)(u, v)|} d\sigma. \quad (26.11)$$

The derivatives $\dot{S}_b$ in this display are the historical half-line derivatives in the fixed transverse coordinate at fixed reference gap. The physical acquisition time itself is held fixed. Its gap variation is the explicit term $-\delta_n a$ in (26.8); the remaining smooth gap dependence of the half-line action is only $(\delta_n a/j_n)\partial_g S_b$ and belongs to the quantitative remainder below. Thus channel-centered action differentiation is not confused with differentiation of the laboratory time.

Proposition 26.1 (Quantitative fixed-window reduction). *For every compact local set $K \subset \mathbb{R}^{1+r}$, the exact limiting boundary endpoint density at (26.5) can be written*

$$f_{n,z}^{\text{ex}}(u, v) = A_{n,z}^{\text{ex}}(u, v)(w_{n,z}^{\text{ex}}(u, v))_+, \quad z \in K, \quad (26.12)$$

where, uniformly on the common endpoint box,

$$\|w_{n,z}^{\text{ex}} - (w_{b,d} - \delta_n U_b z)\|_{C^2} \leq C_K r_n, \quad r_n := \delta_n^2 + \frac{\delta_n}{j_n}, \quad (26.13)$$

$$\|A_{n,z}^{\text{ex}} - A_{b,d}\|_{C^1} \leq C_K \delta_n. \quad (26.14)$$

Let $\tilde{f}_{n,z}$ be the normalized ideal density

$$\tilde{f}_{n,z}(u, v) = \frac{A_{b,d}(u, v)(w_{b,d}(u, v) - \delta_n U_b(u, v)z)_+}{\int A_{b,d}(w_{b,d} - \delta_n U_b z)_+ du dv}. \quad (26.15)$$

Then, with the Hellinger convention (24.1),

$$\sup_{z \in K} H^2(f_{n,z}^{\text{ex}}, \tilde{f}_{n,z}) \leq C_K \left\{ r_n^2 \log \frac{e}{r_n} + \delta_n^2 \right\}. \quad (26.16)$$

If

$$k_n \delta_n^2 \log(1/\delta_n) \longrightarrow 1, \quad (26.17)$$

then the exact and ideal k_n -record endpoint experiments have Le Cam distance $o(1)$, uniformly on K .

Proof. Equation (26.8) supplies the exact fixed-time contribution $-\delta_n a$. Smooth dependence of the half-line action on (g, ϑ) gives, with two endpoint derivatives and uniformly on K ,

$$S_{b, g_n(a), \vartheta_n(h)} = S_b + \delta_n \dot{S}_b h + \frac{\delta_n a}{j_n} \partial_g S_b + O_K(\delta_n^2 + \delta_n^2/j_n^2).$$

Substitution at both endpoints proves (26.13); smooth dependence of the flux amplitude and its normalizing constant gives (26.14).

The ideal normalizing denominator is $1 + O_K(\delta_n)$. Lemma 24.10 with support error $C_K r_n$ and amplitude error $C_K \delta_n$ proves (26.16). The product error is negligible at (26.17): if $j_n^{-1} \leq \delta_n$, then $r_n \leq 2\delta_n^2$ and $k_n r_n^2 \log(e/r_n) = O(\delta_n^2)$; if $j_n^{-1} > \delta_n$, then $r_n \leq 2\delta_n/j_n$ and the same quantity is $O(j_n^{-2})$. The amplitude term is $k_n \delta_n^2 = O(1/\log(1/\delta_n))$. Product Hellinger subadditivity and $\|P - Q\|_{\text{TV}} \leq H(P, Q)$ up to the fixed convention give the conclusion. $\square$

26.4. Finite positive designs from the tangent inverse.

Lemma 26.2 (Finite positive-offset design). *Fix a finite labelled contact-jet order M . Let ϑ consist of the two labelled contact jets through order M at fixed gap in the anchored chart. There exist finitely many same-type designs (b_ℓ, d_ℓ) and positive weights α_ℓ , $\sum_\ell \alpha_\ell = 1$, such that*

$$\mathcal{K}_{\text{design}} = \sum_{\ell=1}^L \alpha_\ell \mathcal{K}_{b_\ell, d_\ell} \quad (26.18)$$

is positive definite on the full local coordinate (a, h) . The finite design may depend on M and on the reference geometry; no universal finite design for the infinite jet or for all reference tables is asserted.

Proof. Suppose $z = (a, h)$ lies in the kernel of $\mathcal{K}_{b,d}$ for both labels and all sufficiently small $d > 0$. Positivity of the weight in (26.11) and continuity imply

$$a + \dot{S}_b(u)h + \dot{S}_b(v)h = 0 \quad \text{whenever} \quad S_b(u) + S_b(v) = d.$$

Take $v = 0$ and $d = S_b(u)$. Anchoring gives $\dot{S}_b(0)h = 0$, so $a + \dot{S}_b(u)h = 0$. Letting $u \rightarrow 0$ gives $a = 0$; hence $\dot{S}_b(u)h = 0$ for both labels near zero. All action-jet variations through order M vanish, and Proposition 12.7 gives $h = 0$. Thus the common kernel is trivial. Every unit vector is detected by some positive design; compactness of the unit sphere selects finitely many. Any positive convex combination of these designs is positive definite. $\square$

26.5. A single deterministic cap sequence for all compact local sets. Let $p_{n,\ell}^\partial(z)$ be the one-preparation success probability in the boundary model at design ℓ , with the fixed even flight number j_n . We use one reference-based cap sequence rather than a cap chosen after a compact local set has been specified.

Lemma 26.3 (Reference-based cap). *Assume*

$$j_n \delta_n \rightarrow 0. \quad (26.19)$$

Then, for every fixed compact local set K and each design ℓ ,

$$\sup_{z \in K} \left| \frac{p_{n,\ell}^\partial(z)}{p_{n,\ell}^\partial(0)} - 1 \right| \rightarrow 0. \quad (26.20)$$

Choose, before quantifying over K ,

$$N_{n,\ell} = \left\lceil \frac{4k_{n,\ell}}{p_{n,\ell}^\partial(0)} \right\rceil. \quad (26.21)$$

Then for every fixed compact K there is n_K such that, for $n \geq n_K$,

$$\inf_{z \in K} N_{n,\ell} p_{n,\ell}^\partial(z) \geq 2k_{n,\ell}. \quad (26.22)$$

Consequently

$$\sup_{z \in K} \Pr_z \{ \text{cap reached before } k_{n,\ell} \text{ successes} \} \leq e^{-ck_{n,\ell}} \quad (26.23)$$

for all large n , with the same cap sequence for every compact K .

Proof. The boundary success law has the form, uniformly on compact local parameter sets,

$$p_{n,\ell}^\partial(z) = C_\ell(z, n) [d_\ell + O_K(\delta_n)]^2 \exp\{-j_n \gamma(z, n)\},$$

where $C_\ell(z, n)/C_\ell(0, n) = 1 + O_K(\delta_n)$ and $\gamma(z, n) - \gamma(0, n) = O_K(\delta_n)$. This is the uniform threshold law and smooth parameter dependence of its prefactor. Therefore

$$\log \frac{p_{n,\ell}^\partial(z)}{p_{n,\ell}^\partial(0)} = O_K(\delta_n) + O_K(j_n \delta_n) = o(1),$$

uniformly on K , proving (26.20). In particular the ratio is at least $1/2$ for all $z \in K$ and all $n \geq n_K$. Substituting (26.21) proves (26.22). A standard binomial Chernoff bound then gives (26.23). $\square$

26.6. The boundary endpoint-output Gaussian experiment. Choose a finite design from Lemma 26.2. Let $k_{n,\ell}$ be target success counts and put $k_n = \sum_\ell k_{n,\ell}$. Assume

$$\frac{k_{n,\ell}}{k_n} \rightarrow \alpha_\ell, \quad k_n \delta_n^2 \log(1/\delta_n) \rightarrow 1, \quad j_n \delta_n \rightarrow 0, \quad j_n \in 2\mathbb{N}. \quad (26.24)$$

Use the fixed cap sequence (26.21).

Theorem 26.4 (Fixed-window multirate endpoint Gaussian experiment). *The capped boundary endpoint-output experiment for $z = (a, h)$ is, uniformly on every fixed compact local parameter set, at $o(1)$ Le Cam distance from a dominated uniformly LAN representative with information matrix $\mathcal{K}_{\text{design}}$. In particular, its finite local subexperiments have the Gaussian limit with that positive-definite information matrix. Corollary 27.3 proves the stronger two-sided Le Cam convergence on every fixed compact local set, at this same endpoint-output observation level.*

After applying to each design the common-collar censoring of Section 24, the representative has central sequence $\Delta_n \Rightarrow N_{1+r}(0, \mathcal{K}_{\text{design}})$ and

$$\log \frac{d\bar{\mathbf{Q}}_{n,z}^{\text{end}}}{d\bar{\mathbf{Q}}_{n,0}^{\text{end}}} = z^t \Delta_n - \frac{1}{2} z^t \mathcal{K}_{\text{design}} z + o_P(1) \quad (26.25)$$

uniformly on compact z -sets. No finite-sample likelihood ratio is asserted for the original endpoint-output family when its support extends beyond the null support.

The gap is visible at rate δ_n/j_n and the anchored registered contact geometry at rate δ_n . The Gaussian testing profile, reference-model central-sequence estimator and local asymptotic minimax conclusions of Theorem 24.8 hold with $\mathcal{J}_\Sigma$ replaced by $\mathcal{K}_{\text{design}}$.

Proof. Before applying the cap, Proposition 26.1 puts each exact boundary endpoint experiment at $o(1)$ Le Cam distance from the ideal linearly moving-support experiment. Theorem 24.8 applies to the latter, with boundary velocity U_b and information matrix $\mathcal{K}_{b,d}$. Independent design batches add their central sequences and information matrices; the proportions $k_{n,\ell}/k_n \rightarrow \alpha_\ell$ give $\mathcal{K}_{\text{design}}$, which is positive definite by Lemma 26.2. Lemma 26.3 changes the experiment by $o(1)$ in total variation on every fixed compact set, using the same deterministic cap sequence independently of that set. $\square$

26.7. Transfer to actual long finite bridges at the declared record level.

Theorem 26.5 (Uniform stopped finite-to-boundary transfer). *Let $\mathbf{P}_{n,z}^{\text{fin,ET}}$ be the law of the stopped endpoint-time acquisition transcript of item 2 in Section 26.1, under the deterministic caps, fixed physical times and target success counts above. Thus it retains every design, every success or failure record (26.1), and the stopping time. Let $\mathbf{Q}_{n,z}^{\partial,ET}$ be the boundary transcript generated by the same policy on the same laboratory record space. If*

$$k_n \tau^{j_n} \rightarrow 0, \quad (26.26)$$

then, for every fixed compact local set K ,

$$\sup_{z \in K} \|\mathbf{P}_{n,z}^{\text{fin,ET}} - \mathbf{Q}_{n,z}^{\partial,ET}\|_{\text{TV}} \leq C_K k_n \tau^{j_n} = o(1). \quad (26.27)$$

The bound survives every common Markov coarsening. In particular the actual finite-bridge endpoint-output experiment inherits Theorem 26.4.

This theorem makes no total-variation comparison of the full raw collision histories $\mathcal{R}^{\text{full}}$. No failed preparation is free. If $T_{n,\ell}$ is the number of raw preparations in batch ℓ , then

$$\mathbb{E}_z T_{n,\ell} \leq \frac{k_{n,\ell}}{p_{n,\ell}(z)} \leq C_K k_{n,\ell} d_\ell^{-2} e^{j_n \gamma_0} \quad (26.28)$$

uniformly on K for all large n ; the deterministic cap gives an absolute upper bound.

Proof. For bounded local z , the excess (26.8) stays in a common compact subcollar and the anchored graphs remain in a compact smooth family. The policy satisfies the deterministic bounded-stopping hypotheses of Theorem 22.3. At each preparation, that theorem uses exactly the per-preparation endpoint-time coarsening of Section 21; it does not compare the full collision array. Its success-weighted coupling gives $C_K k_n \tau^{j_n}$ for the entire stopped transcript at this

declared record level. Total variation contracts under a common Markov kernel, proving the endpoint-output statement. The cost estimate follows from domination by the uncapped negative-binomial stopping time and the uniform success law. $\square$

Remark 26.6 (Endpoint information is not complete physical information). The matrix $\mathcal{K}_{\text{design}}$ characterizes the endpoint-output experiment of item 3. It is not an efficiency bound for every procedure based on the stopped endpoint-time transcript of item 2, and still less for the full raw collision history. The success probability contains the factor $\sinh(j\gamma)^{-1}$, so waiting counts can carry a different and possibly faster information channel in directions changing the hyperbolic exponent. Charging preparation cost is not the same as proving those counts ancillary.

26.8. A charged pilot for centered windows.

Corollary 26.7 (Pilot-centered endpoint-output equivalence). *Assume the coarse-bracket hypothesis of Theorem A.18.3. Let $r_n \downarrow 0$ and let an independent pilot produce $\hat{g}_n$ with*

$$j_n |\hat{g}_n - g| \leq Cr_n^{m+1}$$

with probability $1 - \eta_n$, where $\eta_n \rightarrow 0$. Suppose

$$k_n r_n^{2m+2} \log(1/r_n) \rightarrow 0. \quad (26.29)$$

After the pilot use the common data-dependent rule $t_{n,\ell} = j_n \hat{g}_n + d_\ell$ and the same reference-based capped success targets. On the post-pilot endpoint-output sigma-field alone, this experiment is at vanishing Le Cam distance from the oracle centered endpoint experiment using $t_{n,\ell} = j_n g + d_\ell$. Including the finite-bridge replacement adds at most

$$\eta_n + C_K k_n \tau^{j_n} + o(1). \quad (26.30)$$

The pilot transcript and programmed times are used operationally and their costs are charged, but no equivalence is claimed after those richer random coordinates are added to the statistical output.

Proof. On the good pilot event the post-pilot excess differs from its oracle value by at most Cr_n^{m+1} . Lemma 24.10 gives one-record squared Hellinger distance at most $Cr_n^{2m+2} \log(1/r_n)$, uniformly over the finite design. Product Hellinger bounds and (26.29) give $o(1)$ for endpoint output. Add η_n for the bad pilot event and use Theorem 26.5 for the finite-bridge replacement. $\square$

26.9. Generic nearby tables. The anchored model is a physical submodel, not a parameter-dependent coordinate convention. A generic nearby-table family can still be described on one laboratory record space, but translations and rotations of the contact frames become nuisance directions. If $\chi_{b,\eta}$ maps a fixed laboratory transverse coordinate y to the contact-centered coordinate for table parameter η , then

$$\mathcal{S}_{b,\eta}(y) = S_{b,\eta}(\chi_{b,\eta}(y))$$

and, at a reference with $\chi_{b,0}(y) = y$,

$$D_\eta \mathcal{S}_b(y)[h] = D_\eta S_b(y)[h] + S'_b(y) D_\eta \chi_b(y)[h]. \quad (26.31)$$

The abstract non-dominated theorem of Section 24 applies once these nuisance coordinates and their full laboratory support velocity are included. We do not identify the anchored information matrix with the generic-table one after silently removing those directions. A charged registration pilot with coordinate error ε_n is negligible for endpoint output whenever $k_n \varepsilon_n^2 \log(1/\varepsilon_n) \rightarrow 0$, by Lemma 24.10.

27. COMPACT LOCAL EXPERIMENTS

The finite-subexperiment conclusions above identify the Gaussian limit. We now establish the separate assertion of convergence in Le Cam distance on an arbitrary fixed compact local parameter set. The additional step is a uniform continuity estimate for the experiments, not a reinterpretation of finite-dimensional likelihood convergence. Throughout, total variation is $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$, and the Hellinger convention is (24.1).

Lemma 27.1 (Passage from finite sets to compact sets). *Let K be a compact metric space and let $\mathbf{E}_n = (P_{n,t} : t \in K)$ and $\mathbf{E} = (Q_t : t \in K)$ be experiments on standard Borel spaces. Suppose that their restrictions to each finite $F \subset K$ satisfy $\Delta(\mathbf{E}_n|_F, \mathbf{E}|_F) \rightarrow 0$. Put*

$$\omega_n(r) = \sup_{d(s,t) \leq r} \|P_{n,s} - P_{n,t}\|_{\text{TV}}, \quad \omega(r) = \sup_{d(s,t) \leq r} \|Q_s - Q_t\|_{\text{TV}}.$$

If

$$\lim_{r \downarrow 0} \limsup_{n \rightarrow \infty} \omega_n(r) = 0, \quad \lim_{r \downarrow 0} \omega(r) = 0, \quad (27.1)$$

then $\Delta(\mathbf{E}_n, \mathbf{E}) \rightarrow 0$. Neither experiment is required to be dominated by one of its own members.

Proof. Fix $r > 0$ and a finite r -net $F \subset K$. For every $\varepsilon > 0$, choose a Markov kernel R_n whose error on F is at most $\delta(\mathbf{E}_n|_F, \mathbf{E}|_F) + \varepsilon$. For $t \in K$, choose $s \in F$ with $d(s, t) \leq r$. Contraction of total variation gives

$$\begin{aligned} \|R_n P_{n,t} - Q_t\|_{\text{TV}} &\leq \|P_{n,t} - P_{n,s}\|_{\text{TV}} + \|R_n P_{n,s} - Q_s\|_{\text{TV}} + \|Q_s - Q_t\|_{\text{TV}} \\ &\leq \omega_n(r) + \delta(\mathbf{E}_n|_F, \mathbf{E}|_F) + \varepsilon + \omega(r). \end{aligned}$$

The same argument for a reverse kernel, followed by $\varepsilon \downarrow 0$, proves

$$\Delta(\mathbf{E}_n, \mathbf{E}) \leq \Delta(\mathbf{E}_n|_F, \mathbf{E}|_F) + \omega_n(r) + \omega(r). \quad (27.2)$$

First let $n \rightarrow \infty$ with this finite net fixed, and then let $r \downarrow 0$. The net point is used only to estimate the error of a single kernel; it is not an unknown-parameter input to that kernel. $\square$

Theorem 27.2 (Compact moving-support Gaussian approximation). *Under the hypotheses of Theorem 24.8, for every compact $K \subset \mathbb{R}^r$,*

$$\Delta\left(\{P_{n,h}^{\otimes n} : h \in K\}, \{N_{\mathbf{H}}(\mathcal{J}_{\mathbf{H}}\Pi_{\mathbf{H}}h, \mathcal{J}_{\mathbf{H}}) : h \in K\}\right) \rightarrow 0. \quad (27.3)$$

Here $\mathbf{H}$, $\Pi_{\mathbf{H}}$ and $\mathcal{J}_{\mathbf{H}}$ are as in (24.22)–(24.23). The assertion includes singular information, with the zero-dimensional Gaussian interpreted as the one-point experiment.

Proof. Fix K and a compact neighborhood containing every line segment between its points. Joint smoothness and the common regular collar imply, for $h, h' \in K$ and $s = \|h - h'\| \leq 1$,

$$\|w_{\delta_n h} - w_{\delta_n h'}\|_{C^1} + \|a_{\delta_n h} - a_{\delta_n h'}\|_{C^0} \leq C_K \delta_n s.$$

Lemma 24.10, with constants uniform in this neighborhood, therefore yields

$$H^2(f_{\delta_n h}, f_{\delta_n h'}) \leq C_K \delta_n^2 s^2 \left(1 + \log \frac{1}{\delta_n} + \log \frac{1}{s}\right), \quad 0 < s \leq 1. \quad (27.4)$$

The failure atoms have identical masses. Hence their one-record squared Hellinger distance is p_n times the successful-density distance. Product subadditivity and (24.4) give, for all large n ,

$$H^2(P_{n,h}^{\otimes n}, P_{n,h'}^{\otimes n}) \leq C_K s^2 (1 + \log(1/s)). \quad (27.5)$$

Since $\|P - Q\|_{\text{TV}} \leq H(P, Q)$ with our convention, this supplies the first modulus in (27.1).

For the displayed Gaussian laws, writing $v = h - h'$, direct Gaussian integration gives

$$H^2(Q_h, Q_{h'}) = 2 \left(1 - \exp\{-v^t \mathcal{J}_{\Sigma} v / 8\}\right).$$

This identity is evaluated on $\mathbf{H}$; it does not require an inverse of $\mathcal{J}_\Sigma$ on its kernel. It supplies the second modulus. The finite-set convergence is item 2 of Theorem 24.8. Apply Lemma 27.1.

In particular, the comparison here is for the original moving-support family. The common-collar representative is needed to identify its finite limits, but no likelihood ratio with respect to the original reference law has been asserted on support-exclusive observations. $\square$

Corollary 27.3 (Compact fixed-window endpoint experiments). *With the finite design, rate conditions and reference-based caps of Theorem 26.4, the capped boundary endpoint-output experiment converges in Le Cam distance, on every fixed compact set of $z = (a, h)$, to the Gaussian experiment with information $\mathcal{K}_{\text{design}}$. The same conclusion holds after restriction to any linear subspace of the local coordinate. Under (26.26), it holds also for the actual finite-bridge endpoint-output experiment, at the same declared observation level.*

Proof. Use the normalized ideal densities (26.15) in the independent design batches. Their defining functions and normalized amplitudes satisfy the uniform bounds used in (27.4); the normalization is positive and smooth on a common small parameter neighborhood. Sum the resulting one-record bounds over the $\sum_\ell k_{n,\ell} = k_n$ records. The critical success scale gives the product modulus (27.5) with np_n replaced by k_n . The finite limits have information $\sum_\ell \alpha_\ell \mathcal{K}_{b_\ell, d_\ell} = \mathcal{K}_{\text{design}}$, by Theorem 26.4. Lemma 27.1 and Gaussian continuity give compact convergence for the ideal experiment.

Proposition 26.1 transfers it to the exact boundary densities with a uniform $o(1)$ error. The cap contributes at most $\sum_\ell e^{-ck_{n,\ell}}$, by Lemma 26.3. Restriction cannot increase deficiency. Finally, Theorem 26.5 contributes $C_K k_n \tau^{j_n} = o(1)$ on the common endpoint-output space. $\square$

Remark 27.4. Compact local convergence is not an assertion of uniform equivalence over an unbounded local parameter space or over the complete analytic class of tables. The kernels may depend on the fixed compact set, the reference channel and the declared design, but never on the unknown local parameter.

28. THE COMPLETE ENDPOINT-TIME ACQUISITION EXPERIMENT AT ITS FASTEST LOCAL SCALE

The endpoint-output experiment of Section 26 is a strict coarsening of the record for which the finite-to-boundary theorem is proved. We now restore the residual preparation time and the complete waiting sequence. The resulting local experiment is not Gaussian at its fastest shape scale. Its fastest geometric information is an extreme boundary process. This gives a theorem for the entire stopped endpoint-time acquisition transcript of item 2 in Section 26.1, rather than only for its endpoint-output coarsening.

Fix a finite labelled contact-jet order M in the anchored realization of Section 25. Choose a finite positive design (b_ℓ, d_ℓ) , $1 \leq \ell \leq L$, as in Lemma 26.2. Write

$$w_\ell(u, v) = d_\ell - S_{b_\ell}(u) - S_{b_\ell}(v), \quad D_\ell = \{(u, v) : w_\ell(u, v) > 0\}.$$

The boundary selected-record measure of Section 21, after conditioning on success and returning to unscaled residual time, has density

$$q_{\ell,0}(u, v, r) = \rho_\ell(u, v) \mathbf{1}_{\{0 < r < w_\ell(u, v)\}}, \quad \int_{D_\ell} \rho_\ell w_\ell du dv = 1, \quad (28.1)$$

where ρ_ℓ is positive and smooth on the common endpoint box. Up to the normalizing constant it is the half-line amplitude $B_{b_\ell}(u)B_{b_\ell}(v)$. Thus the conditional density has a positive trace at the moving ceiling $r = w_\ell(u, v)$; integrating out r multiplies the endpoint density by w_ℓ and creates the linear vanishing studied in Section 24.

Let $k_{n,\ell} \rightarrow \infty$, put $k_n = \sum_{\ell} k_{n,\ell}$, and assume

$$\frac{k_{n,\ell}}{k_n} \rightarrow \alpha_{\ell} > 0, \quad \sum_{\ell=1}^L \alpha_{\ell} = 1. \quad (28.2)$$

Use an even flight number $j_n \rightarrow \infty$ for every same-type design and the fixed physical windows $t_{n,\ell} = j_n g_0 + d_{\ell}$. At the endpoint–time scale the local parameter is

$$g_n(a) = g_0 + \frac{a}{j_n k_n}, \quad \vartheta_n(h) = \frac{h}{k_n}, \quad z = (a, h) \in \mathbb{R}^{1+r_M}. \quad (28.3)$$

For design ℓ , use the support velocity of Section 26,

$$U_{\ell}(u, v)z = a + \dot{S}_{b_{\ell}}(u)h + \dot{S}_{b_{\ell}}(v)h. \quad (28.4)$$

Proposition 26.1, with $\delta_n = k_n^{-1}$, gives uniformly on compact z -sets

$$w_{n,\ell,z}^{\text{ET}}(u, v) = w_{\ell}(u, v) - \frac{1}{k_n} U_{\ell}(u, v)z + o_K(k_n^{-1}) \quad (28.5)$$

in C^1 , while the conditional density trace at the ceiling is $\rho_{\ell} + O_K(k_n^{-1})$. The residual gap-dependence of the half-line action is $O((j_n k_n)^{-1})$ and is absorbed by the remainder.

28.1. The boundary point-process limit. For a successful endpoint–time record (u, v, r) , define its null ceiling slack and scaled slack by

$$s_{\ell} = w_{\ell}(u, v) - r, \quad y = k_n s_{\ell}.$$

Under the reference model $y \geq 0$. Under a local alternative the same reference coordinate may be negative when the ceiling expands.

For a compact local set K , choose

$$R_K > 1 + \max_{1 \leq \ell \leq L} \sup_{(u,v) \in \overline{D}_{\ell}, z \in K} |U_{\ell}(u, v)z|. \quad (28.6)$$

Let $\mathcal{N}_{n,\ell}$ be the point measure formed from the $k_{n,\ell}$ successful records by retaining only points with $|y| \leq R_K$ and recording (u, v, y) .

Theorem 28.1 (Endpoint–time Poisson boundary experiment). *Assume (28.2) and let the local parameter be (28.3). Then, uniformly for z in every fixed compact set K , the conditional successful endpoint–time experiment is asymptotically equivalent to independent Poisson point processes $\mathcal{N}_1, \dots, \mathcal{N}_L$ on $D_{\ell} \times [-R_K, R_K]$ with intensities*

$$d\Lambda_{\ell,z}(u, v, y) = \alpha_{\ell} \rho_{\ell}(u, v) \mathbf{1}_{\{y > U_{\ell}(u, v)z\}} \mathbf{1}_{\{|y| \leq R_K\}} du dv dy. \quad (28.7)$$

The part of the successful sample outside the displayed boundary layer is asymptotically parameter-free after conditioning on the retained point process. Equivalently, the local experiment is the projective Poisson boundary-shift experiment obtained by increasing R_K .

The finite-strip family has a common dominating Poisson envelope, as shown in Proposition 30.1. It need not be absolutely continuous with respect to the zero-shift member: if $U_{\ell}z < 0$ on a set of positive $\rho_{\ell} du dv$ -measure, the alternative assigns positive probability to points in a region where the reference intensity vanishes. The resulting support-exclusive events prevent contiguity in such expanding directions at this scale, and hence prevent a finite-information Gaussian LAN representation there. The obstruction is reference absolute continuity, not common domination.

Proof. Fix one design and suppress ℓ . The level set $\{w = 0\}$ is smooth for $d > 0$. Hence the coarea formula gives, uniformly for fixed R ,

$$\text{Leb}\{(u, v) : 0 < w(u, v) \leq 2R/k_n\} = O(k_n^{-1}). \quad (28.8)$$

On this endpoint collar the residual interval has length at most $2R/k_n$, so its one-record probability is $O(k_n^{-2})$. Over $k_{n,\ell} = O(k_n)$ records the probability of seeing any point from this corner is therefore $O(k_n^{-1}) = o(1)$. This removes the codimension-two intersection between the moving ceiling and the fixed face $r = 0$ from the first-order experiment.

Away from that negligible collar, the condition $r > 0$ imposes no additional restriction on a fixed y -strip. By (28.5) and smoothness of the conditional trace, the one-record law in reference coordinates is, uniformly on K ,

$$\frac{1}{k_n}(\rho(u, v) + o_K(1))\mathbf{1}_{\{y > U(u, v)z + o_K(1)\}} du dv dy \quad (28.9)$$

on $|y| \leq R_K$. In particular the probability $p_{n,R}$ of the whole retained layer is $O(k_n^{-1})$. Conditional on entering the layer, the location law converges in total variation to the normalized version of (28.7). The standard binomial–Poisson coupling therefore gives

$$d_{\text{TV}}(\mathcal{L}(\mathcal{N}_{n,\ell}), \text{PPP}(\Lambda_{\ell,z})) \leq Ck_{n,\ell}p_{n,R}^2 + o_K(1) = o_K(1). \quad (28.10)$$

The same estimate for a finite measurable partition of the layer, followed by monotone refinement, gives the point-process statement; independent design batches give the product process.

It remains to verify that no random first-order information is hidden in the bulk. Remove a reference ceiling layer of thickness R_K/k_n . The choice (28.6) makes the remaining support common to every $z \in K$. There the density ratio is a smooth perturbation of the form

$$1 + k_n^{-1}a_z(u, v, r) + o_K(k_n^{-1})$$

with uniformly bounded a_z . Its centered one-record log-likelihood has variance $O_K(k_n^{-2})$, hence the centered sum over $k_{n,\ell} = O(k_n)$ records has variance $O_K(k_n^{-1})$ and converges to zero uniformly on K . The deterministic first-order normalizing term is forced, by total mass one, to equal the negative of the mass gained or lost in the ceiling layer. In the Poisson likelihood this is exactly the exponential compensator $\exp\{-\Lambda_{\ell,z} + \Lambda_{\ell,0}\}$. Thus, conditional on the boundary-layer point process, the bulk likelihood ratio tends to one in probability uniformly on K .

The extraction of $\mathcal{N}_{n,\ell}$ is a parameter-independent Markov kernel. Conversely, generate the asymptotically parameter-free bulk from the reference conditional law and attach it to the Poissonized boundary layer; (28.10) and the preceding bulk estimate make the reverse deficiency tend to zero. The explicit layer-count, bulk, and corner-safe reverse kernels, with the verified relative-density hypothesis, are given in Theorem 29.2. They prove the asserted two-sided asymptotic equivalence, including exceptional configurations. $\square$

Corollary 28.2 (Identification at the k^{-1} geometric scale). *Choose the positive finite design of Lemma 26.2. Then the map*

$$z \longmapsto (U_\ell(\cdot, \cdot)z)_{\ell=1}^L$$

is injective. Consequently the Poisson limit experiment of Theorem 28.1 identifies the full finite coordinate (a, h) : gap is visible at scale $(j_n k_n)^{-1}$, while the anchored labelled contact geometry through order M is visible at scale k_n^{-1} .

Proof. If $U_\ell z = 0$ on D_ℓ for every ℓ , it vanishes in particular on the endpoint boundary $\{S_{b_\ell}(u) + S_{b_\ell}(v) = d_\ell\}$. Hence $z^t \mathcal{K}_{\text{design}} z = 0$. Positive definiteness from Lemma 26.2 gives $z = 0$. $\square$

28.2. Waiting counts and transfer to finite bridges. For a fixed design, stop after $k_{n,\ell}$ successful preparations and let $T_{n,\ell}$ be the total number of preparations. Before the deterministic cap is applied, the boundary law factorizes exactly into

$$\mathcal{L}_z(T_{n,\ell}) \otimes (\Pi_{n,\ell,z}^{\partial, \text{ET}})^{\otimes k_{n,\ell}}, \quad (28.11)$$

where $T_{n,\ell}$ is negative binomial with success probability $p_{n,\ell}^\partial(z)$. Thus the successful marks are independent of the waiting count.

Theorem 28.3 (Complete stopped endpoint–time local experiment). *In addition to (28.2), assume*

$$j_n \in 2\mathbb{N}, \quad j_n \rightarrow \infty, \quad j_n = o(\sqrt{k_n}), \quad k_n \tau^{j_n} \rightarrow 0. \quad (28.12)$$

Use the reference-based deterministic caps of Lemma 26.3, with the target success counts $k_{n,\ell}$. Then, uniformly on every fixed compact local parameter set:

- (1) *the waiting-count component of the boundary stopped transcript is asymptotically ancillary at the scale (28.3);*
- (2) *the entire capped boundary endpoint–time acquisition transcript is at vanishing Le Cam distance from the Poisson boundary experiment of Theorem 28.1, up to a parameter-free factor;*
- (3) *the actual finite-bridge stopped endpoint–time transcript is at vanishing total-variation distance from that boundary transcript, and hence has the same Poisson local experiment.*

Thus the declared endpoint–time acquisition process has a complete local limit at its fastest geometric scale; this statement retains failures, waiting counts, designs, successful residual times and stopping times. It does not identify the still richer full growing collision array.

Proof. The selected-channel term in Theorem 3.1, (3.1), and its smooth fixed-order parameter dependence give

$$\log \frac{p_{n,\ell}^\partial(z)}{p_{n,\ell}^\partial(0)} = O_K(k_n^{-1}) + O_K(j_n k_n^{-1}) = O_K(j_n/k_n).$$

For a negative-binomial experiment stopped after $k_{n,\ell}$ successes, the Fisher information for $\log p$ is $k_{n,\ell}/(1-p)$. Hence the squared Hellinger distance between the waiting-count laws is $O_K(k_n(j_n/k_n)^2) = O_K(j_n^2/k_n) = o(1)$. This proves asymptotic ancillarity. The same success-ratio estimate is $1 + o(1)$, so the single reference-based cap remains adequate and its failure probability is exponentially small.

The exact factorization (28.11), cap negligibility and Theorem 28.1 prove the boundary statement. Finally Theorem 26.5 applies to the complete stopped endpoint–time transcript and gives total-variation error $O_K(k_n \tau^{j_n}) = o(1)$. Le Cam distance contracts under composition, which proves the finite-bridge conclusion. $\square$

Remark 28.4 (Why the endpoint Gaussian theorem is slower). At the scale k_n^{-1} , positive ceiling density makes the extreme residual-time layer carry order-one information. Integrating out residual time replaces the positive ceiling density by the linearly vanishing endpoint density $\rho_\ell w_\ell$; the k_n^{-1} Poisson layer disappears and the next surviving endpoint scale is $k_n \delta_n^2 \log(1/\delta_n) \asymp 1$. Thus the Gaussian endpoint experiment of Theorem 26.4 is the slower experiment created by residual-time coarsening.

29. TWO-SIDED DEFICIENCY FOR THE MOVING-CEILING EXPERIMENT

We give parameter-independent kernels in both directions for the local endpoint–time experiment. Besides convergence of the ceiling and its trace, the argument uses a uniform first-order relative density bound on the common bulk. We state this bound explicitly: convergence of a trace in C^0 alone would not imply the required product Hellinger estimate.

For one design write $k = k_n$, $w = w_\ell$, $\rho = \rho_\ell$, and $U = U_\ell$. Let K be a fixed compact local parameter set containing zero. On a fixed bounded endpoint box V , the exact successful density has the form

$$q_{n,z}(u, v, r) = \rho_{n,z}(u, v, r) \mathbf{1}_{\{0 < r < w_{n,z}(u,v)\}}.$$

All density factors extend to a common bounded residual-time box, with uniform upper and positive lower bounds. Put $D = \{w > 0\}$, whose boundary is regular in the endpoint box, and assume

$$\varepsilon_{n,K} := \sup_{z \in K} \left(k \|w_{n,z} - w + k^{-1} U z\|_{C^1} + \|\rho_{n,z} - \rho\|_{C^0} \right) \longrightarrow 0. \quad (29.1)$$

Here $\rho = \rho(u, v)$ is independent of r and $\int_D \rho w \, du \, dv = 1$. In addition suppose that on the common bulk specified below,

$$\sup_{z \in K} \left\| \frac{\rho_{n,z}}{\rho_{n,0}} - 1 \right\|_{\infty} \leq C_K / k. \quad (29.2)$$

A bounded expansion $1 + k^{-1} a_{n,z} + o_K(k^{-1})$ is sufficient, but not necessary, for this bound.

These hypotheses hold for the anchored fixed-window boundary model of Section 28. Its parameter displacement is $O_K(k^{-1})$, its normalized smooth density factor is independent of r , and its reference factor is bounded away from zero. Smooth dependence of that factor and its positive normalizer gives (29.2) uniformly on the box. Equation (28.5) gives the support estimate; the residual gap term is $O((j_n k_n)^{-1}) = o(k_n^{-1})$. Thus the additional displayed hypothesis is a verified dependency of the physical application, not a deduction from bare trace convergence.

Choose

$$R > 2 + \sup_{z \in K, (u,v) \in \overline{V}} |U(u, v) z|$$

and define the reference layer

$$L_{n,R} = \{|k(w(u, v) - r)| \leq R\}. \quad (29.3)$$

It is independent of z . For large n , the support outside this layer is the common set $0 < r < w - R/k$; the alternative ceilings cannot reach the other side $r > w + R/k$. This is the bulk in (29.2).

Let $p_{n,z} = Q_{n,z}(L_{n,R})$, let $A_{n,z}$ be its conditional layer law after $y = k(w - r)$, and let $B_{n,z}$ be the conditional bulk law. The target intensity is the finite measure on $D \times [-R, R]$ with density

$$d\Lambda_z^R = \rho(u, v) \mathbf{1}_{\{y > U(u, v) z\}} \mathbf{1}_{\{|y| \leq R\}} \, du \, dv \, dy. \quad (29.4)$$

The factor in this formula is the previously defined density ρ .

Lemma 29.1 (Uniform layer and bulk bounds). *Uniformly for $z \in K$,*

$$p_{n,z} = O(k^{-1}), \quad (29.5)$$

$$\left| k Q_{n,z} \circ (u, v, r \mapsto u, v, k(w - r))^{-1} \Big|_{L_{n,R}} - \Lambda_z^R \right| (V \times [-R, R]) \leq C_K (\varepsilon_{n,K} + k^{-1}), \quad (29.6)$$

$$H^2(B_{n,z}, B_{n,0}) \leq C_K k^{-2}. \quad (29.7)$$

The notation in the second line means that the layer submeasure is restricted before pushforward, and that the target intensity is extended by zero from D to V . The vertical bars denote total variation mass of a finite signed measure.

Proof. The layer has residual thickness $2R/k$ over a bounded endpoint set, which gives (29.5). Regularity of $\{w = 0\}$ implies $\text{Leb}\{|w| \leq C/k\} = O(k^{-1})$. On this endpoint strip any physical residual interval involved has length $O(k^{-1})$, so its one-record mass is $O(k^{-2})$. The part of the target intensity on $0 < w \leq C/k$ is $O(k^{-1})$ as well. These estimates handle both the intersection with the fixed face $r = 0$ and any alternative support created just outside D .

Off that strip, $r > 0$ imposes no additional restriction in the reference layer. The change of variable has Jacobian k^{-1} , and the ceiling condition becomes

$$y > U(u, v) z + e_{n,z}(u, v), \quad \|e_{n,z}\|_{\infty} \leq \varepsilon_{n,K}.$$

Fiber integration bounds the symmetric difference by the endpoint area times $\varepsilon_{n,K}$, while the trace-density difference is bounded by the second term of (29.1). Adding the corner error proves (29.6). In particular, the unscaled mass attributable to the support remainder is at most $C_K \varepsilon_{n,K}/k$.

On the common bulk write $\rho_{n,z}/\rho_{n,0} = 1 + e_{n,z}^B$, with $\|e_{n,z}^B\|_\infty \leq C_K/k$. Dividing by the bulk normalizers gives exactly

$$\frac{dB_{n,z}}{dB_{n,0}} = \frac{1 + e_{n,z}^B}{1 + B_{n,0}e_{n,z}^B} = 1 + O_K(k^{-1}).$$

The inequality $(\sqrt{1+x} - 1)^2 \leq Cx^2$ for small x proves (29.7). This proof identifies the relative-density hypothesis actually used in the bulk and controls the normalized conditional law, not merely a marginal fluctuation. $\square$

Let $\mathbf{E}_{n,K}^R = \{Q_{n,z}^{\otimes k} : z \in K\}$ and $\mathbf{P}_K^R = \{\text{PPP}(\Lambda_z^R) : z \in K\}$.

Theorem 29.2 (Explicit two-sided deficiency). *Under the displayed support, positivity and relative-density hypotheses,*

$$\Delta(\mathbf{E}_{n,K}^R, \mathbf{P}_K^R) \leq C_K(\varepsilon_{n,K} + k^{-1/2}). \quad (29.8)$$

Both deficiencies admit Markov kernels independent of z . For finitely many independent batches the bounds add. In particular this proves the two-sided equivalence claimed in Theorem 28.1; it is stronger than convergence of the extracted point process alone.

Proof. The forward kernel retains records in $L_{n,R}$, replaces r by $y = k(w - r)$, and discards points outside D . All operations use reference quantities only. Before the last discard, the retained sample is a binomial point process. Its distance from the Poisson process with the same intensity is at most $Ckp_{n,z}^2 = O(k^{-1})$. Poisson coupling and (29.6) bound the further intensity error by $C_K(\varepsilon_{n,K} + k^{-1})$. Discarding exterior points has already been accounted for in the corner estimate and cannot increase distance. This proves the forward deficiency bound.

For the reverse kernel, let m be the number of input Poisson points. If $m \leq k$, map each point to $(u, v, r = w - y/k)$, draw $k - m$ independent records from the reference bulk law $B_{n,0}$, and interleave the two collections by a uniformly random permutation. If a reconstructed point has $r \leq 0$, replace it by one fixed reference record. Such a point has $0 < w \leq R/k$, so the probability that this convention is used is $O(k^{-1})$. If $m > k$, output one fixed k -sample. The Poisson means are uniformly bounded, hence $\sup_{z \in K} \Pr_z\{m > k\} = o(k^{-A})$ for every fixed A . This defines a valid parameter-independent kernel on every configuration, including corner and count exceptions.

For the original iid sample, conditional on its layer count, the layer records and bulk records have respective iid laws $A_{n,z}$ and $B_{n,z}$, are independent of each other, and are uniformly interleaved. The layer count and locations can be coupled with the Poisson input with the forward error just proved. Replacing the conditional bulk by its reference law costs, uniformly in $m \leq k$,

$$H^2(B_{n,z}^{\otimes(k-m)}, B_{n,0}^{\otimes(k-m)}) \leq (k-m)H^2(B_{n,z}, B_{n,0}) \leq C_K/k.$$

Its total variation cost is therefore $O_K(k^{-1/2})$. The common permutation does not increase distance. Adding the corner and count tails gives (29.8). Independent-product coupling proves tensorization over finitely many batches. For the proportions in (28.2), scale by total k_n ; the intensity in a batch is multiplied by $k_{n,\ell}/k_n$, converging to α_ℓ . The resulting additional intensity error tends to zero. $\square$

Remark 29.3 (The projective Poisson experiment). On $D \times \mathbb{R}$, the intensity

$$\rho(u, v) \mathbf{1}_{\{y > U(u, v)z\}} du dv dy$$

is locally finite. Restriction from a larger finite strip to a smaller one is a common Markov kernel. Once R exceeds every displacement on the compact parameter set, the added positive tail has parameter-independent intensity and the negative tail is empty. It can thus be adjoined as an ancillary factor. This precise projective interpretation does not require an informal limit in the reverse reconstruction kernel.

30. COMMON DOMINATION AND THE INFORMATION IN EXACT POSITIONS

Three properties must be separated: domination of a statistical family by one sigma-finite measure, absolute continuity with respect to a designated reference member, and mutual absolute continuity of the members. Moving support need not prevent the first property. The distinction also separates the scalar-coordinate local experiments from the richer position protocol.

Proposition 30.1 (Common reference measures). *The family (24.5) is dominated, for every n , by $\mu = \text{Leb}|_U + \delta_{\dagger}$. Its censored family is dominated by $\text{Leb}|_U + \delta_{\dagger} + \delta_{\partial}$. The finite products have the corresponding product dominating measures. The original members need not be absolutely continuous with respect to $P_{n,0}$; the common-collar construction establishes this stronger property for the censored members and their censored reference.*

On the fixed finite strip $B = D \times [-R, R]$, let

$$d\Lambda_z = \rho(u, v) \mathbf{1}_{\{y > U(u, v)z\}} du dv dy, \quad d\Lambda_* = \rho(u, v) du dv dy.$$

Then the Poisson law $\mathbf{P}_* = \text{PPP}(\Lambda_*)$ is a common dominating probability measure for the entire finite-strip family, and

$$\frac{d\mathbf{P}_z}{d\mathbf{P}_*}(N) = \exp\{\Lambda_*(B) - \Lambda_z(B)\} \prod_{q \in N} \mathbf{1}_{\{y(q) > U(q)z\}}. \quad (30.1)$$

The empty product is one. Multiplication by a fixed batch proportion gives the same assertion for each batch of (28.7).

Proof. The first statements follow by writing the displayed Lebesgue densities and atomic masses. They do not involve division by the reference density. For example, on $[0, 2]$,

$$f_\theta(x) = \frac{2(1 + \theta - x)_+}{(1 + \theta)^2}, \quad 0 \leq \theta < 1,$$

is a family of probability densities, but $P_\theta((1, 1 + \theta)) = \theta^2/(1 + \theta)^2 > 0$ for $\theta > 0$, whereas $P_0((1, 1 + \theta)) = 0$.

For the Poisson assertion write $A_z = \{y > Uz\} \cap B$. Under $\mathbf{P}_*$, restrictions to A_z and its complement are independent Poisson processes. The probability of no point in the complement is $\exp\{-\Lambda_*(B \setminus A_z)\}$. Multiplying its indicator by the reciprocal of that probability has expectation one and leaves the independent process on A_z unchanged. The resulting law has intensity $\Lambda_*|_{A_z} = \Lambda_z$, proving (30.1). The argument includes the zero-intensity and empty-configuration cases. $\square$

A zero-shift Poisson member need not dominate the others. If $Uz < 0$ on a set of positive weighted endpoint measure, there is positive intensity in $\{Uz < y < 0\}$, where the reference has none. The event of at least one point there has positive alternative probability and zero reference probability. This is a failure of reference absolute continuity, not of common domination. It also excludes a Gaussian shift with mutually absolutely continuous members as the local limit in such a direction. The common-collar Gaussian limit and the moving-ceiling Poisson limit remain different experiments for the reasons supplied by their boundary rates and support events, not by an incorrect classification of their dominating measures.

For a fixed compact parameter set, choose R beyond all displacements. The positive tail outside the strip is parameter independent and the negative tail is empty. Adjoining the common positive-tail process preserves the common domination just proved, also for the projective experiment of Remark 29.3. This makes no uniform assertion about an unbounded parameter set without a common displacement bound.

30.1. Matched records and the observed contact type. We use the convention $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$, so a deficiency belongs to $[0, 1]$. For experiments $\mathbf{E} = (P_\lambda)_{\lambda \in \Theta}$ and $\mathbf{F} = (Q_\lambda)_{\lambda \in \Theta}$, write

$$\delta(\mathbf{E}, \mathbf{F}) = \inf_K \sup_{\lambda \in \Theta} \|KP_\lambda - Q_\lambda\|_{\text{TV}}, \quad \Delta(\mathbf{E}, \mathbf{F}) = \max\{\delta(\mathbf{E}, \mathbf{F}), \delta(\mathbf{F}, \mathbf{E})\},$$

where the kernels are independent of the unknown parameter.

Fix an even-flight design starting at contact type b . Both retained endpoints then have type b . In this subsection a successful position record is exactly $(X_{b,\lambda}(u), X_{b,\lambda}(v))$, and its scalar coarsening is exactly (u, v) . Residual time is integrated out in both records; intermediate collisions and waiting counts occur in neither. The fixed transverse projection is one parameter-independent kernel. All claims concern these matched records, conditional on success. They do not remove preparation costs from the unconditional acquisition theorems.

Lemma 30.2 (Dominated input and disjoint output supports). *Suppose Θ is uncountable, $P_\lambda \ll \nu$ for one probability measure ν , and there are pairwise disjoint measurable sets A_λ with $Q_\lambda(A_\lambda) = 1$. Then $\delta((P_\lambda), (Q_\lambda)) = 1$.*

Proof. For any kernel K , put $Q = K\nu$. If $Q(A) = 0$, nonnegativity gives $K(x, A) = 0$ for ν -almost every x , hence $KP_\lambda(A) = 0$ for every λ . Thus $KP_\lambda \ll Q$. A probability measure gives positive mass to at most countably many pairwise disjoint measurable sets. Choose λ outside that countable set for Q . Then $KP_\lambda(A_\lambda) = 0$, whereas $Q_\lambda(A_\lambda) = 1$. The worst-case error of K is therefore one. Infimizing over K proves the assertion. No integration over the uncountable parameter set is used. $\square$

Proposition 30.3 (Observed-type dichotomy for anchored homotheties). *Let Θ be a nondegenerate compact interval of positive homothety parameters. In a persistent registered analytic facing channel let the obstacle of type b_* vary as $K_\lambda = p + \lambda(K - p)$, with $p \in \partial K$ its fixed contact point and fixed tangent. Keep the facing obstacle fixed, with positive anchored gap. Choose one admissible even-flight design and common small contact windows. The successful endpoint law has positive smooth flux and a density with respect to the two boundary arclength coordinates. Let $\mathbf{E}_{\text{sc},b}^{(k)}$ and $\mathbf{E}_{\text{pos},b}^{(k)}$ denote $k \geq 1$ independent matched successful records of the preceding definition.*

If the design starts at $b = b_$, the distinct position laws are mutually singular and admit no common sigma-finite dominating measure. For every finite $k \geq 1$,*

$$\delta(\mathbf{E}_{\text{sc},b_*}^{(k)}, \mathbf{E}_{\text{pos},b_*}^{(k)}) = 1, \quad \delta(\mathbf{E}_{\text{pos},b_*}^{(k)}, \mathbf{E}_{\text{sc},b_*}^{(k)}) = 0. \quad (30.2)$$

If instead the design starts at the fixed obstacle, $b = 1 - b_$, then*

$$\Delta(\mathbf{E}_{\text{sc},b}^{(k)}, \mathbf{E}_{\text{pos},b}^{(k)}) = 0. \quad (30.3)$$

In this case the position family has a common dominating measure and a common positive-density region, and distinct members are not mutually singular. In either case the scalar laws are continuous in total variation as λ varies.

Proof. First suppose the observed type is b_* . Along each ray from p entering the strictly convex body K , there is exactly one other boundary point. A homothety multiplies its distance from p by λ . Thus two distinct homothetic boundaries meet only at p . Removing the anchor gives pairwise

disjoint Borel curves $\Gamma_\lambda = \partial K_\lambda \setminus \{p\}$. The first endpoint has an arclength density and hits p with probability zero. The event that the first recorded position lies in Γ_λ has probability one under that member, also for any finite product of successful records. These events are pairwise disjoint. They prove mutual singularity. A sigma-finite measure has at most countably many disjoint measurable sets of positive measure: intersect with a countable finite-measure cover, and then group by a positive lower bound on the intersection measure. A common dominating measure is therefore impossible.

In the fixed transverse coordinates all scalar laws have densities on one bounded box $U \subset \mathbb{R}^2$. They are dominated by normalized Lebesgue measure on U ; their k -fold products are dominated by its k -fold product. Apply Lemma 30.2 to the events just constructed. This gives the first equality in (30.2). The fixed transverse projection gives the reverse deficiency zero. In particular the Le Cam distance is one.

Now suppose $b = 1 - b_*$. The observed contact arc is a single fixed graph X_b , independent of λ . The maps

$$F_b(u, v) = (X_b(u), X_b(v)), \quad T_b(x, x') = (\text{transverse}(x), \text{transverse}(x'))$$

are measurable and mutually inverse on the supported contact boxes. Moreover $P_{\lambda, b}^{\text{pos}} = (F_b)_* P_{\lambda, b}^{\text{psc}}$. Extend the maps arbitrarily off those boxes. They are parameter-independent kernels and give (30.3), also on products. The inverse may use the fixed obstacle: it is a known constant of this specified family, not an unknown parameter-dependent boundary.

For either observed type the scalar density is

$$f_{\lambda, b}(u, v) = Z_{\lambda, b}^{-1} A_{\lambda, b}(u, v)(d - E_{J, \lambda, b}(u, v))_+, \quad A_{\lambda, b} > 0. \quad (30.4)$$

The action and flux vary smoothly on the common box, the normalizers are positive, and compactness bounds them away from zero. Uniform continuity and $|x_+ - y_+| \leq |x - y|$ prove total-variation continuity. Since $E_{J, \lambda, b}(0, 0) = 0$, a sufficiently small common square has $E_{J, \lambda, b} < d/4$ for every λ . On this square all scalar densities are positive. For the fixed observed obstacle, $(F_b)_* \text{Leb}|_U$ is a common dominating measure, with a common positive-density region. No two position members can be mutually singular. The same continuity argument applies to the limiting factorized endpoint law. $\square$

Remark 30.4 (Matched residual-time retention). The same dichotomy holds if residual time is retained in both records. Use (u, v, r) and $(X_{b, \lambda}(u), X_{b, \lambda}(v), r)$, respectively. Scalar densities are dominated on a fixed bounded three-dimensional box. For a fixed observed obstacle use $F_b \times \text{id}$; for a varying observed obstacle use the same first-position disjoint events. Total-variation continuity follows by integrating the difference of ceiling indicators, whose integral is bounded by the ceiling displacement. The common square above, together with $d/4 < r < d/2$, is a positive-density core. Comparing a time-retaining record with a time-deleted record is a different question and is not an instance of (30.3).

30.2. A fixed-obstacle design and nonsymmetric examples. The dependence on the observed type already occurs for two circular contacts. Set

$$K_0 = \{(x, y) : (x+1)^2 + y^2 \leq 1\}, \quad K_{1, \lambda} = \{(x, y) : (x-1-\lambda)^2 + y^2 \leq \lambda^2\}, \quad \lambda \in [1-\epsilon, 1+\epsilon].$$

Their closest points are $(0, 0)$ and $(1, 0)$, and their gap is one. The second disk is obtained by homothety about $(1, 0)$. For small $\epsilon > 0$, take $J = 2$, physical time $2 + d$, and the word $K_0 \rightarrow K_{1, \lambda} \rightarrow K_0$. Both observed positions belong to the fixed arc

$$X_0(u) = (-1 + \sqrt{1 - u^2}, u).$$

The finite stationarity equations have a nondegenerate solution at the normal orbit and hence a common smooth solution on a small box. Its excess vanishes at the origin. Formula (30.4)

therefore has the common positive core used in the proof, and F_0 gives exact endpoint-only equivalence. Unobserved intermediate visits to the varying disk do not turn these endpoint-position laws into singular laws. This construction explains why the type qualification in (30.2) cannot be omitted.

The varying-type conclusion does not require circular symmetry. A strictly convex analytic oval with support function

$$h(\theta) = 1 + \frac{1}{50} \cos(2\theta) + \frac{1}{100} \sin(3\theta)$$

has $h + h'' \geq 43/50 > 0$. Its nonzero harmonic orders two and three exclude a nonidentity rotational symmetry. Fix a boundary anchor and use nearby homothetic copies, with a facing obstacle kept at positive gap. A sufficiently small parameter interval preserves the local channel. Starting on the oval gives (30.2); starting on the fixed facing obstacle gives (30.3). These are local examples, without an assertion of the global cycle and spanning-tree hypotheses for an unspecified periodic table.

Exact injectivity of the transverse *law* map in Theorem 13.4 is compatible with maximal finite-sample deficiency on the varying-type family. The supremum over an uncountable parameter interval and the finite-sample record are essential. Distribution-level determination, recovery of a particular homothety parameter from an exact position, and reconstruction of an arbitrary analytic obstacle are distinct assertions. The global law inverse remains a statement using both contact types and the separately retained gap.

31. WAITING COUNTS AND ENDPOINT INFORMATION AFTER RESIDUAL-TIME COARSENING

Theorem 28.3 identifies the fastest local experiment when residual time is retained. We now remove residual time but retain the waiting counts. The positive-density ceiling is thereby integrated out. Two information mechanisms remain: the rare-event probability detects the hyperbolic exponent, while the endpoint mark detects motion of its linearly vanishing support. We prove that their product is a Gaussian experiment in *two-sided Le Cam distance on every fixed compact local parameter set*.

Fix one anchored labelled channel and a finite contact-jet model of order $M \geq 2$. Let $\vartheta \in \mathbb{R}^M$ be its shape coordinate at fixed gap, and let $\gamma(g, \vartheta)$ be the channel hyperbolic exponent. At the reference point $(g_0, 0)$, put

$$\lambda = D_{\vartheta} \gamma(g_0, 0), \quad H_{\gamma} = \ker \lambda.$$

The coordinate contains the two contact curvatures, on which γ depends nontrivially, so $\lambda \neq 0$. Choose $v_{\gamma} \in \mathbb{R}^M$ with $\lambda(v_{\gamma}) = 1$. This choice gives a fast hyperbolic direction and an iso-hyperbolic tangent space; changing the complement merely changes the local coordinates.

Choose the finite positive design of Lemma 26.2, with target counts $k_{n,\ell}$, total $k_n = \sum_{\ell} k_{n,\ell}$, and proportions $k_{n,\ell}/k_n \rightarrow \alpha_{\ell} > 0$. Let $\delta_n \downarrow 0$ satisfy

$$k_n \delta_n^2 \log(1/\delta_n) \rightarrow 1. \quad (31.1)$$

For bounded $(b, a, h) \in \mathbb{R} \times \mathbb{R} \times H_{\gamma}$, set

$$\eta_n = (j_n \sqrt{k_n})^{-1}, \quad g_n(a) = g_0 + \delta_n a / j_n, \quad \vartheta_n(b, h) = \eta_n b v_{\gamma} + \delta_n h. \quad (31.2)$$

All physical windows remain $t_{j_n, \ell} = j_n g_0 + d_{\ell}$, and the caps are the parameter-independent reference caps (26.21). In batch ℓ , retain the waiting count $T_{n, \ell}$ and the successful laboratory transverse pairs $(y_{\text{first}}, y_{\text{last}})$ of (26.4), with the cap cemetery convention. This is a common coarsening of the stopped endpoint-time transcript, not of an experiment in which preparation failures are free.

Lemma 31.1 (Negative-binomial hyperbolic score). *Suppose $j_n \rightarrow \infty$ and $j_n \delta_n \rightarrow 0$. Uniformly on compact (b, a, h) -sets,*

$$\log \frac{p_{n,\ell}^\partial(b, a, h)}{p_{n,\ell}^\partial(0, 0, 0)} = -b/\sqrt{k_n} + o(k_n^{-1/2}). \quad (31.3)$$

For the uncapped boundary waiting experiment, define under its reference law

$$S_{n,\ell} = \frac{k_{n,\ell} - p_{n,\ell}^\partial(0)T_{n,\ell}}{1 - p_{n,\ell}^\partial(0)}, \quad Z_n = -k_n^{-1/2} \sum_{\ell=1}^L S_{n,\ell}.$$

Then $Z_n \Rightarrow N(0, 1)$, and

$$\log \frac{dW_{n,b,a,h}}{dW_{n,0,0,0}} = bZ_n - \frac{1}{2}b^2 + o_P(1), \quad (31.4)$$

uniformly on compact sets. The slow coordinates contribute no limiting count score.

Proof. On the fixed positive offset collar the success probability has a smooth positive prefactor times $\exp\{-j_n \gamma(g, \vartheta)\}$; the logarithmic derivatives of the prefactor are uniformly bounded. Taylor expansion of γ , using $\lambda(h) = 0$, gives

$$\gamma(g_n(a), \vartheta_n(b, h)) - \gamma(g_0, 0) = \eta_n b + O_K(\delta_n/j_n) + O_K(\delta_n^2 + \delta_n \eta_n + \eta_n^2).$$

Consequently, the remainder in the success log ratio after the term $-b/\sqrt{k_n}$ is bounded by

$$C_K(\delta_n + \eta_n + j_n \delta_n^2 + j_n \delta_n \eta_n + j_n \eta_n^2).$$

After multiplication by $\sqrt{k_n}$, these terms tend to zero: $\sqrt{k_n} \delta_n \rightarrow 0$, $\sqrt{k_n} \eta_n = 1/j_n \rightarrow 0$, $\sqrt{k_n} j_n \delta_n^2 = (j_n \delta_n)(\sqrt{k_n} \delta_n) \rightarrow 0$, $\sqrt{k_n} j_n \delta_n \eta_n = \delta_n \rightarrow 0$, and $\sqrt{k_n} j_n \eta_n^2 = \eta_n \rightarrow 0$. This proves (31.3) without discarding the mixed fast-slow Taylor terms.

For a negative-binomial waiting time stopped at k successes, the score for $\phi = \log p$ is $(k - pT)/(1 - p)$, with variance $k/(1 - p)$. Represent each waiting time as a sum of independent geometric waiting times. For p small, the centered one-success log-probability scores have uniformly bounded third absolute moments: the variables pT have a uniform exponential tail. The Lyapunov bound for the normalized sum is $O(k_n^{-1/2})$. Since $p_{n,\ell}^\partial(0) \rightarrow 0$, this proves $Z_n \Rightarrow N(0, 1)$.

The second and third derivatives of the batch log likelihood are

$$-\frac{(T - k)p}{(1 - p)^2}, \quad -\frac{(T - k)p(1 + p)}{(1 - p)^3},$$

respectively. On the compact local neighborhoods, their orders are $O_P(k)$, and the normalized negative second derivative converges to its information. Taylor expansion with (31.3) gives (31.4). $\square$

Lemma 31.2 (Uniform comparison of waiting experiments). *For geometric laws G_p on $\{1, 2, \dots\}$, if $p, q \leq p_* < 1$,*

$$H^2(G_p, G_q) \leq \frac{(\log p - \log q)^2}{4(1 - p_*)}. \quad (31.5)$$

For the sum of k independent such waiting times, the bound is multiplied by at most k . Define ideal batch probabilities

$$p_{n,\ell}^*(b) = p_{n,\ell}^\partial(0) \exp\{-b/\sqrt{k_n}\}.$$

The uncapped joint waiting experiment is uniformly at $o(1)$ total variation distance, on compact (b, a, h) -sets, from the product of the negative-binomial laws with probabilities $p_{n,\ell}^(b)$. This ideal experiment converges in Le Cam distance on every compact b -set to $N(b, 1)$.*

Proof. As a curve in ℓ^2 , the square-root mass function of G_{e^ϕ} has derivative with squared norm $1/[4(1 - e^\phi)]$, one quarter of its log-probability information. Integrating this derivative on the segment between $\log p$ and $\log q$ proves (31.5). Product subadditivity and contraction under summation give the waiting-time bound. In particular, this proof uses the exact discrete waiting law, not a Poisson replacement.

By (31.3), the maximum difference between the true and ideal log probabilities is $o(k_n^{-1/2})$, uniformly on the compact set. All probabilities are at most $1/2$ for large n . Summing (31.5) over all k_n required successes gives squared Hellinger distance $o(1)$, hence the stated total variation comparison.

The count LAN calculation gives joint weak convergence of the ideal likelihood vector on each finite b -set to $(\exp\{bZ - b^2/2\})_b$, with $Z \sim N(0, 1)$. The discrete waiting laws have common support and each likelihood has reference mean one. Each Gaussian limiting coordinate also has mean one. Lemma 24.6, adjoining the reference value $b = 0$ before restriction when necessary, therefore proves the finite-subexperiment limit $N(b, 1)$. Its pairwise squared Hellinger distance is bounded by $C|b - b'|^2$, uniformly in n , because the log-probability difference is exactly $-(b - b')/\sqrt{k_n}$. The Gaussian family has a continuous Hellinger modulus as well. Lemma 27.1 proves compact Le Cam convergence. $\square$

Let $\mathcal{K}_H$ be the restriction of $\mathcal{K}_{\text{design}}$, defined in (26.18), to $\mathbb{R} \times H_\gamma$. It is positive definite on that space.

Theorem 31.3 (Two-speed count–endpoint local experiment). *Assume*

$$\begin{aligned} k_n \delta_n^2 \log(1/\delta_n) &\rightarrow 1, & j_n \delta_n &\rightarrow 0, \\ \log(j_n \sqrt{k_n})/j_n^2 &\rightarrow 0, & k_n \tau^{j_n} &\rightarrow 0, & j_n \in 2\mathbb{N}, & j_n \rightarrow \infty. \end{aligned} \quad (31.6)$$

Use (31.2), the fixed laboratory windows and the reference caps. For each fixed compact $K \subset \mathbb{R} \times \mathbb{R} \times H_\gamma$, let $\mathbf{E}_{n,K}^{\partial, CE}$ be the capped boundary experiment retaining waiting counts and the successful laboratory transverse pairs $(y_{\text{first}}, y_{\text{last}})$ defined by the common map (26.4). Normal endpoint coordinates, residual times and intermediate collisions are not retained in this output. Let $\mathbf{E}_{n,K}^{\text{fin}, CE}$ be the same coarsening of the actual finite-bridge transcript, with every preparation failure charged. Then

$$\Delta(\mathbf{E}_{n,K}^{\partial, CE}, \mathbf{G}_K) \rightarrow 0, \quad \Delta(\mathbf{E}_{n,K}^{\text{fin}, CE}, \mathbf{G}_K) \rightarrow 0, \quad (31.7)$$

where $\mathbf{G}_K$ observes, independently,

$$X \sim N(b, 1), \quad Y \sim N(\mathcal{K}_H(a, h), \mathcal{K}_H), \quad (b, a, h) \in K.$$

Thus the limiting information is $1 \oplus \mathcal{K}_H$. The hyperbolic shape direction is visible at scale $(j_n \sqrt{k_n})^{-1}$, the iso-hyperbolic shape directions at scale δ_n , and the gap at scale δ_n/j_n . Both comparison kernels are independent of the unknown (b, a, h) .

Proof. First remove the caps. In an iid Bernoulli-mark experiment the successful marks are independent of the waiting times and have their conditional success law. This factorization is exact for the uncapped experiment; the cap error will be added only after the product comparison.

Lemma 31.2 replaces the waiting factor by its ideal b -only experiment with a uniform $o(1)$ total variation error. For the endpoint factor, compare the parameter $(g_n(a), \eta_n b v_\gamma + \delta_n h)$ with $(g_n(a), \delta_n h)$ at the same physical time. Smooth dependence of the actions and amplitudes changes the defining function in C^1 and the amplitude in C^0 by at most $C_K \eta_n$. By Lemma 24.10, the total squared Hellinger error over the successful records is at most

$$C_K k_n \eta_n^2 \log(e/\eta_n) = C_K \frac{1 + \log(j_n \sqrt{k_n})}{j_n^2} = o(1). \quad (31.8)$$

The smooth amplitude term is bounded by $C_K k_n \eta_n^2$ and is included in this estimate. The remaining endpoint experiment depends only on (a, h) , and converges on compact sets to its Gaussian shift by Corollary 27.3, restricted to $\mathbb{R} \times H_\gamma$.

Take the compact projections of K onto its count and endpoint coordinates. The product of the two comparison kernels on those projection sets has error at most the sum of the component errors. Restriction from their Cartesian product to K cannot increase either deficiency. This proves compact two-sided convergence for the uncapped boundary experiment, with independent Gaussian factors; it is stronger than convergence of a fixed finite vector of likelihood ratios.

By (31.3), the true success probabilities are uniformly asymptotic to their reference values on K . The reference cap therefore has failure probability at most $\sum_\ell e^{-ck_{n,\ell}} = o(1)$, as in Lemma 26.3. Coupling to the uncapped acquisition sequence changes the experiment by at most this probability; a reverse kernel may assign any fixed completion on the cap event. This proves the first limit in (31.7).

The common count–endpoint coarsening of the stopped transcript preserves the total variation bound $C_K k_n \tau^{j_n}$ from Theorem 26.5. Its hypotheses hold here: the local graphs stay in the same compact collar, the fast perturbation tends to zero, and the policy and caps are parameter independent. The same success-weighted adaptive comparison consequently applies, without a free failure convention. The triangle inequality for Le Cam distance proves the second limit. $\square$

Remark 31.4 (Likelihoods and the retained record). Count likelihoods are ordinary negative-binomial likelihoods. For endpoint marks, likelihood expansions relative to a reference member are taken only after common-collar censoring. Formula (31.7) instead concerns the original capped experiments and therefore does not assert reference domination where the support moves. Nor does it compare complete raw collision histories.

Remark 31.5 (Three levels of information). Under the respective rate hypotheses, retaining residual time gives the non-Gaussian Poisson boundary experiment of Theorem 28.3, with shape scale k_n^{-1} and gap scale $(j_n k_n)^{-1}$. Removing residual time but keeping waiting counts gives Theorem 31.3. Removing both gives the endpoint-output Gaussian experiment of Theorem 26.4 and Corollary 27.3. These are different experiments related by common Markov coarsenings, not alternative names for one record.

32. INTRINSIC NORMAL–TANGENT GEOMETRY OF THE COUNT–ENDPOINT LIMIT

The splitting used in Section 31 is a convenient coordinate representative of a canonical quotient construction. We now state the product experiment invariantly and verify directly that the hyperbolic differential is nonzero.

Fix a finite anchored contact-jet model at a reference channel. Let $V = T_{\vartheta_0} \Theta_M$ be its shape tangent space at fixed gap and let

$$\lambda = d\gamma_{\vartheta_0} \in V^*. \quad (32.1)$$

The two contact curvatures are coordinates of this finite jet model. With the notation of (2.1),

$$c_b = 1 + g\kappa_b, \quad c_0 c_1 = \cosh^2 \gamma, \quad \gamma = \operatorname{arcosh} \sqrt{c_0 c_1}.$$

Consequently, at fixed $g > 0$,

$$\frac{\partial \gamma}{\partial \kappa_0} = \frac{gc_1}{2\sqrt{c_0 c_1} \sinh \gamma} > 0, \quad \frac{\partial \gamma}{\partial \kappa_1} = \frac{gc_0}{2\sqrt{c_0 c_1} \sinh \gamma} > 0. \quad (32.2)$$

In particular $\lambda \neq 0$ at every dispersing reference channel in the model; no generic non-cancellation hypothesis is needed.

Put

$$H = \ker \lambda, \quad N = V/H. \quad (32.3)$$

The covector λ induces the canonical isomorphism

$$\bar{\lambda} : N \longrightarrow \mathbb{R}, \quad [v] \longmapsto \lambda(v). \quad (32.4)$$

Thus the fast count coordinate is intrinsically an element of the one-dimensional normal quotient N , normalized by $\bar{\lambda}$, not a preferred vector in V .

Let $\mathcal{K}$ be the positive endpoint information quadratic form of the fixed positive design on $\mathbb{R}_{\text{gap}} \oplus V$. Its slow component is the restricted quadratic form

$$\mathcal{K}_{\text{tan}} = \mathcal{K}|_{\mathbb{R}_{\text{gap}} \oplus H}. \quad (32.5)$$

This restriction is positive definite by the full finite-design theorem.

Theorem 32.1 (Coordinate-free two-speed information theorem). *Under the rate assumptions of Theorem 31.3, the count-endpoint experiment has the intrinsic product limit*

$$\mathcal{G}_{\bar{b}}^{\text{count}} \otimes \mathcal{G}_{(a,h)}^{\text{end}}, \quad \bar{b} \in N, \quad a \in \mathbb{R}_{\text{gap}}, \quad h \in H, \quad (32.6)$$

where the count factor has quadratic information

$$I_{\text{count}}(\bar{b}) = |\bar{\lambda}(\bar{b})|^2 \quad (32.7)$$

and the endpoint factor has information tensor $\mathcal{K}_{\text{tan}}$. The factors are independent.

This formulation is invariant under every C^2 reparameterization of the finite jet model. In particular, choosing any splitting vector $v_\gamma \in V$ with $\lambda(v_\gamma) = 1$ identifies N with the coordinate called b in Section 31, but the limit experiment does not depend on that choice.

Proof. Let v_γ be any representative with $\lambda(v_\gamma) = 1$. Theorem 31.3 gives the product limit with count information one in the scalar coefficient of v_γ and endpoint information $\mathcal{K}_{\text{tan}}$ on the tangent coordinates. If v'_γ is another representative, then $v'_\gamma - v_\gamma \in H$. At the fast physical scale $\eta_n = (j_n \sqrt{k_n})^{-1}$, replacing one representative by the other adds an H -displacement of size $O(\eta_n)$. The endpoint experiment cannot see this displacement because the proof of Theorem 31.3 gives

$$k_n \eta_n^2 \log(1/\eta_n) \longrightarrow 0.$$

The count likelihood sees only the first differential of γ , hence only $\lambda(v)$, and therefore gives the same scalar shift for the two representatives. This proves that the limit descends to the quotient $N = V/H$ and that its quadratic form is (32.7).

Under a C^2 change of shape coordinates, $d\gamma$ pulls back as a covector, its kernel maps to the tangent space of the same level set, and $\mathcal{K}$ transforms as a quadratic tensor. Therefore both (32.7) and (32.5) are coordinate invariant. The exact waiting-count/mark factorization gives independence, as before. $\square$

Remark 32.2 (Geometry of the two rates). The faster Gaussian channel is the normal quotient to the level set of the hyperbolic exponent; the slower endpoint channel is tangent to that level set and includes the independently scaled gap direction. A choice of v_γ is analogous to choosing a connection for writing a quotient as a direct sum. It is useful for formulas but is not part of the experiment. Equation (32.2) also closes the previously implicit assertion that the normal quotient is genuinely one dimensional.

Remark 32.3 (One simultaneous admissible rate regime). The scaling assumptions used by the endpoint, count-endpoint and stopped transfer theorems have a common nonempty regime. For example, let

$$\delta_n = n^{-1}, \quad k_n = \left\lfloor \frac{n^2}{\log n} \right\rfloor, \quad j_n = 2 \lceil C \log n \rceil$$

with $C > |\log \tau|^{-1}$. Then

$$k_n \delta_n^2 \log(1/\delta_n) \rightarrow 1, \quad j_n \delta_n \rightarrow 0, \quad k_n \tau^{j_n} \rightarrow 0,$$

and also

$$j_n = o(\sqrt{k_n}), \quad \frac{\log(j_n \sqrt{k_n})}{j_n^2} \rightarrow 0.$$

Thus the stated physical and statistical asymptotics do not rely on an empty intersection of rate conditions.

Remark 32.4 (Exact source of the cap success-ratio estimate). The success-law input used in Lemma 26.3 is the selected channel summand in Theorem 3.1, equation (3.1), evaluated at the fixed physical window. That formula gives the factor $\sinh(j\gamma)^{-1}$, the quadratic offset factor and a smooth remainder with uniform fixed-order parameter derivatives. On a compact local parameter set, Taylor expansion of this exact formula gives the prefactor ratio $1 + O_K(\delta_n)$ and the exponent variation $j_n O_K(\delta_n)$ used in (26.20). Hence $j_n \delta_n \rightarrow 0$ yields the required uniform ratio $1 + o(1)$ for the single reference-based cap sequence.

33. THE ENDPOINT CRITICAL EXPERIMENT

We apply the preceding theorem to a fixed physical channel. The two hypotheses are its finite-flight law and its boundary model for the *same* table. The observation space, the failure symbol, and the physical preparation cost are those of Section 21. Write $P_{j,d}^{\text{end}}$ and $P_{\partial,j,d}^{\text{end}}$ for the pushforwards retaining the two endpoint positions on success and the failure symbol otherwise. Thus residual time is not reported. Their conditional versions are denoted by $\Pi_{j,d}^{\text{end}}$ and $\Pi_{\partial,j,d}^{\text{end}}$. Superscript 0 denotes the corresponding quadratic right-limit laws of Section A.7.

The common whitening in (A.7.1) is used only to prove the comparison for this fixed table. It does not give an observation rule knowledge of an unknown gap or curvature. Total variation and Hellinger distance are unchanged by this common invertible map.

33.1. Exact tangent density and its information. Set

$$q_j = e^{-j\gamma}, \quad \zeta_j = \log \coth(j\gamma/2) = \log \frac{1+q_j}{1-q_j}, \quad r_j = q_j^2 \log(1/q_j). \quad (33.1)$$

After whitening and a common orthogonal rotation, the endpoint density of the finite tangent law is

$$f_{\zeta_j}(x) = \frac{2}{\pi} (1 - e^{-\zeta_j} x_1^2 - e^{\zeta_j} x_2^2)_+, \quad x \in \mathbb{R}^2, \quad (33.2)$$

and the boundary density is $f_0(x) = 2(1 - |x|^2)_+/\pi$. The raw pair has the same mass

$$p_{j,d}^0 = \frac{d^2}{2A \sinh(j\gamma)} \quad (33.3)$$

on these successful densities, and the same failure mass $1 - p_{j,d}^0$.

Lemma 33.1 (The endpoint information coefficient). *The densities (33.2) are normalized. In the parameter ζ their boundary information coefficient is one, and*

$$H^2(f_{\zeta_j}, f_0) = r_j + O(q_j^2). \quad (33.4)$$

For every j, d , the raw tangent pair has identical success-indicator laws, even though its endpoint laws differ. These statements hold for unequal facing curvatures and both parities of j .

Proof. The eigenvalues of the whitened Hessian are $\lambda_j = \tanh(j\gamma/2) = e^{-\zeta_j}$ and $\lambda_j^{-1} = e^{\zeta_j}$. Integrating residual time in the uniform cap gives (33.2). The change $(y_1, y_2) = (e^{-\zeta/2}x_1, e^{\zeta/2}x_2)$ has determinant one, and $\int_{|y|<1} (1 - |y|^2) dy = \pi/2$; this proves normalization. The same determinant identity in (A.7.1) gives the common mass (33.3). In particular the two success bits are exactly Bernoulli with the same parameter, not merely equal to leading order in q_j .

Take $w_\zeta = 1 - e^{-\zeta}x_1^2 - e^{\zeta}x_2^2$ and $a_\zeta = 2/\pi$ on a fixed large disk. Then $v = x_1^2 - x_2^2$ and $|\nabla w_0| = 2$ on the unit circle. Consequently

$$\mathcal{I}_\Sigma = \frac{1}{\pi} \int_0^{2\pi} \cos^2(2\theta) d\theta = 1.$$

One can also see the logarithm directly. For $w = 1 - |x|^2$ the score is $S = (x_1^2 - x_2^2)/w$. Polar integration gives the exact identities

$$\Pr\{0 < w < h\} = h^2, \quad \mathbb{E}_0[S \mathbf{1}_{\{w \geq h\}}] = 0, \quad \mathbb{E}_0[S^2 \mathbf{1}_{\{w \geq h\}}] = \log(1/h) - \frac{3}{2} + 2h - \frac{h^2}{2}. \quad (33.5)$$

Indeed the last integral is $\int_h^1 (1-w)^2 dw/w$; the angular integral of $\cos^2(2\theta)$ is π . Theorem 23.1 now gives $H^2 = \zeta_j^2 \log(1/\zeta_j)/4 + O(\zeta_j^2)$. Since $\zeta_j = 2q_j + O(q_j^3)$, this is (33.4). All dependence on the individual curvatures was removed by a common invertible whitening, not by assuming that the curvatures agree. $\square$

Theorem 33.2 (Sharp endpoint testing profile). *Let $j \rightarrow \infty$. If $k_j r_j \rightarrow b \in [0, \infty]$, then*

$$\|(\Pi_j^{0,\text{end}})^{\otimes k_j} - (\Pi_{\partial,j}^{0,\text{end}})^{\otimes k_j}\|_{\text{TV}} \rightarrow 2\Phi(\sqrt{b}) - 1. \quad (33.6)$$

If $n_j p_{j,d_j}^0 r_j \rightarrow b \in [0, \infty]$, the same limit holds for the raw tangent endpoint laws with n_j preparations. At $0 < b < \infty$, the log likelihood under the boundary hypothesis converges to $\mathcal{N}(-2b, 4b)$. The minimum equal-prior testing error tends to $\Phi(-\sqrt{b})$. At $b = \infty$ the distance tends to one and the testing error to zero.

In either experiment, vanishing total variation is equivalent to the corresponding effective sample size times r_j tending to zero. In particular the sharp conditional endpoint scale is $e^{2j\gamma}/(j\gamma)$, rather than $e^{j\gamma}$.

Proof. Apply Theorem 23.1 with $t = \zeta_j$, first with $p = 1$ and then with $p = p_{j,d_j}^0$. The row size in that theorem may of course run through the integers k_j or n_j . The information scales satisfy

$$\zeta_j^2 \log(1/\zeta_j) = 4r_j(1 + o(1)).$$

Lemma 33.1 supplies coefficient one. This proves the limits and the likelihood normalization. For necessity in the last assertion, if the effective scale does not tend to zero, pass to a subsequence on which it is bounded below by a fixed positive number. A further subsequence either has a finite positive limit or diverges. The corresponding distance limit is strictly positive. Sufficiency is the case $b = 0$. $\square$

33.2. Positive-offset physical transfer. As in Lemma A.7.2, put

$$\omega(d) = \sqrt{d} \quad \text{for general smooth contacts,} \quad \omega(d) = d \quad \text{if both contact graphs are even.}$$

No assumption that the two even contacts coincide is made.

Theorem 33.3 (The physical endpoint transition). *Suppose $d_j > 0$ belongs to the common physical collar and*

$$\frac{\omega(d_j)}{r_j} \rightarrow 0. \quad (33.7)$$

If $k_j r_j \rightarrow b \in [0, \infty]$, then the actual conditional endpoint laws satisfy

$$\|(\Pi_{j,d_j}^{\text{end}})^{\otimes k_j} - (\Pi_{\partial,j,d_j}^{\text{end}})^{\otimes k_j}\|_{\text{TV}} \rightarrow 2\Phi(\sqrt{b}) - 1. \quad (33.8)$$

If $n_j p_{j,d_j}^0 r_j \rightarrow b \in [0, \infty]$, the same limit holds for the actual raw endpoint laws with n_j independent preparations. The minimum equal-prior errors have the same limits as in Theorem 33.2. Under (33.7), vanishing physical total variation is equivalent to the corresponding effective scale tending to zero. The estimates and conclusions are uniform on the compact geometric families under consideration, including variable geometry and either parity, provided the displayed scaling assumptions hold uniformly.

Proof. Projection onto the endpoint observation cannot increase total variation. Lemma A.7.2, telescoping independent products, and the reverse triangle inequality therefore give

$$|\text{TV}_{j,k,d}^{\text{end}} - \text{TV}_{j,k}^{0,\text{end}}| \leq Ck\omega(d), \quad (33.9)$$

$$|\text{TV}_{j,n,d}^{\text{end,raw}} - \text{TV}_{j,n,d}^{0,\text{end,raw}}| \leq Cnp_{j,d}^0\omega(d). \quad (33.10)$$

The symbols in these two inequalities denote the distances of the corresponding simple pairs of product laws. If $b < \infty$, their right sides tend to zero by (33.7). Theorem 33.2 proves the assertions in this case.

If $b = \infty$, fix $B > 0$. The conditional experiment eventually contains its first $\lfloor B/r_j \rfloor$ observations. Their accumulated tangent error is at most $CB\omega(d_j)/r_j = o(1)$, and their limiting distance is $2\Phi(\sqrt{B}) - 1$. Projection onto this subsample gives the same lower bound for the lower limit of the full distance. Let $B \rightarrow \infty$. For raw preparations use instead $\lfloor B/(p_{j,d_j}^0 r_j) \rfloor$ observations. Both rounding errors in the effective scales tend to zero. This proves the supercritical assertion without requiring the tangent error accumulated over the entire supercritical sample to vanish.

The same fixed positive subsample proves necessity of a vanishing effective scale, as in Theorem A.7.7. The uniform version uses the uniform tangent estimates and the fact that γ is bounded below by a positive number; the whitened family (33.2) itself is independent of the remaining geometry. Finally use the exact equal-prior testing identity. $\square$

The positive-offset conditions are nonempty. For any $\epsilon > 0$, $d_j = r_j^{2+\epsilon}$ in the general smooth case and $d_j = r_j^{1+\epsilon}$ in the even case satisfy (33.7). The preparation counts are finite for every j . Since $p_{j,d}^0 \asymp d^2 e^{-j\gamma}$, the raw endpoint transition occurs at order

$$d_j^{-2} e^{3j\gamma} / (j\gamma).$$

Failures have not been conditioned away in this statement.

33.3. A strict hierarchy of observations.

Corollary 33.4 (Three distinct comparison experiments). *Under (33.7), suppose $n_j p_{j,d_j}^0 r_j \rightarrow b \in (0, \infty)$. For the same two actual physical models and the same n_j preparations, the respective total variation distances have the limits*

retained observation on success	limiting distance	(33.11)
both endpoints and residual time	1	
both endpoints, without residual time	$2\Phi(\sqrt{b}) - 1$	
success indicator only	0.	

Every experiment retains the same failed preparations. In the tangent models the last distance is identically zero at every sample size. Conversely, at the full-record critical scale $n_j p_{j,d_j}^0 q_j \rightarrow c \in (0, \infty)$, assuming only $\omega(d_j)/q_j \rightarrow 0$, the full-record distance tends to $1 - e^{-2c/\pi}$ while the

endpoint-only distance tends to zero. The analogous assertions hold for a prescribed number of independently prepared successes, with the success-only conditional observation constant.

Proof. At the endpoint scale,

$$n_j p_{j,d_j}^0 q_j = \frac{n_j p_{j,d_j}^0 r_j}{q_j \log(1/q_j)} \longrightarrow \infty, \quad \frac{\omega(d_j)}{q_j} = \frac{\omega(d_j)}{r_j} q_j \log(1/q_j) \longrightarrow 0.$$

Theorem A.7.7 gives the first limit in (33.11), and Theorem 33.3 gives the second. For the last, both tangent success-indicator laws are exactly the same. Their actual counterparts differ from these by at most $C n_j p_{j,d_j}^0 \omega(d_j) = o(1)$ in total variation, by coarsening Lemma A.7.2. This proves the last limit without postulating any rate for a count experiment outside this tangent regime.

At the full-record critical scale the known support calculation gives $1 - e^{-2c/\pi}$, whereas $n_j p_{j,d_j}^0 r_j = (n_j p_{j,d_j}^0 q_j) q_j \log(1/q_j) \rightarrow 0$. The tangent endpoint distance tends to zero by Theorem 33.2; its physical approximation error is $O(n_j p_{j,d_j}^0 \omega(d_j)) = o(1)$ under the stated weaker collar condition. The conditional proof replaces $n_j p_{j,d_j}^0$ by k_j . $\square$

The full-record limit is an erasure limit because its tangent laws have equal densities on their common support (Theorem A.7.1). The endpoint limit is instead Gaussian: its supports are still different, but their exclusive mass is only $O(q_j^2)$, whereas its Hellinger information is $q_j^2 \log(1/q_j)(1 + o(1))$. The success bit loses this tangent shape information entirely through the exact determinant cancellation. Thus neither a one-record weak limit nor a coarsening of the full-record coupling bound identifies the sharp endpoint experiment.

These are intrinsic scales involving the exact hyperbolic multiplier, not a supplied upper convergence certificate τ . They establish a sharp observation-dependent validity law in an explicitly nonempty physical tangent regime. They do not identify two unknown tables, assert equivalence of arbitrary noisy smooth-profile experiments, or turn the sufficient full-profile acquisition exponent into a minimax one. The nonlinear relative law on a fixed collar and the complete boundary-profile inverse retain their separate statements and proofs.

34. THE ABEL GEOMETRY OF THE BOUNDARY INVARIANT

The scalar boundary law is obtained by two residual-time integrations of an autoconvolution. Its natural inverse topology is therefore better expressed on the integrated flux than on the law divided by d^2 . For two profiles define

$$H(V, W)(d) = d^2 T(V, W)(d), \quad H_V = H(V, V),$$

where T is the normalized positive integral (10.2). This notation agrees with (A.4.3) below. For $H \in C^3([0, D])$ set

$$\begin{aligned} (\mathcal{A}H)(x) &= \frac{\pi}{2} \left\{ H''(0) + \sqrt{x} \int_0^x \frac{H'''(s)}{\sqrt{x-s}} ds \right\}, \quad x > 0, \\ (\mathcal{A}H)(0) &= \frac{\pi}{2} H''(0), \quad |H|_{\mathcal{A}} = \|\mathcal{A}H\|_{C^0([0, D])}. \end{aligned} \tag{34.1}$$

The operator is linear. Equivalently, with $k_*(x) = x^{-1/2}$ and $Q = (\pi/2)H''$, it is $\sqrt{x}(k_* * Q)'$. Formula (34.1), not a derivative of a singular expression at zero, defines its endpoint value. The transform is unchanged by adding an affine function to H .

Theorem 34.1 (A nonlinear inverse in the Abel flux norm). *Let $V, W \in C^3([0, D])$, $V(0) = W(0) = 1$, and suppose $\|V'\|_\infty, \|W'\|_\infty \leq B$. Then*

$$\begin{aligned} |H_V - H_W|_{\mathcal{A}} &\leq 2\pi(1 + BD)\|V - W\|_\infty, \\ \|V - W\|_\infty &\leq \frac{1 + BDe^{BD/2}}{2\pi}|H_V - H_W|_{\mathcal{A}}. \end{aligned} \quad (34.2)$$

In particular, on every bounded normalized positive profile class of Section A.4, the physical same-type integrated law is a bi-Lipschitz coordinate for the full energy profile in this specified data norm. No smallness of $V - 1$ is assumed.

The nullspace of $|\cdot|_{\mathcal{A}}$ on C^3 consists exactly of affine functions. Consequently it is a norm on $\{H \in C^3 : H(0) = H'(0) = 0\}$, which contains all profile-law differences. For arbitrary $H \in C^3$ one also has

$$|H|_{\mathcal{A}} \leq \frac{\pi}{2}|H''(0)| + 2\pi\sqrt{D}\sqrt{\|H'' - H''(0)\|_\infty\|H'''\|_\infty}. \quad (34.3)$$

The same statements apply to $C^{m-1,1}$ profiles, $m \geq 4$.

Proof. Put $u = V - W$, $f = x^{-1/2}u$ and $a = x^{-1/2}(V + W)$. The normalization and the first derivative bounds make f continuous, with $f(0) = 0$. The convolution identities in Section 10 give, for $Q = (\pi/2)(H_V - H_W)''$,

$$Q = a * f, \quad k_* * a = 2\pi + R, \quad R(0) = 0.$$

Indeed,

$$R(x) = \int_0^1 \frac{V(xt) + W(xt) - 2}{\sqrt{t(1-t)}} dt, \quad \|R'\|_\infty \leq \pi B.$$

The constant follows from $\int_0^1 t\{t(1-t)\}^{-1/2} dt = \pi/2$. Differentiating $k_* * Q = (2\pi + R) * f$ gives

$$\mathcal{A}(H_V - H_W)(x) = 2\pi u(x) + \sqrt{x} \int_0^x R'(x-s) \frac{u(s)}{\sqrt{s}} ds. \quad (34.4)$$

All kernels are locally integrable; the differentiation follows first by writing the constant convolution as $2\pi \int_0^x f$ and then using $R(0) = 0$. The displayed identity extends to zero by continuity. Its integral term has absolute value at most $2\pi Bx\|u\|_\infty$, proving the first estimate.

Let $E = |H_V - H_W|_{\mathcal{A}}/(2\pi)$ and $v(x) = \int_0^x |u(s)|s^{-1/2} ds$. The same identity implies

$$|u(x)| \leq E + \frac{B}{2}\sqrt{x}v(x), \quad v'(x) \leq Ex^{-1/2} + \frac{B}{2}v(x)$$

almost everywhere. Multiplication by $e^{-Bx/2}$ and integration, with $v(0) = 0$, yield

$$v(x) \leq E \int_0^x e^{B(x-s)/2} s^{-1/2} ds \leq 2E\sqrt{x}e^{Bx/2}.$$

Substitution proves the inverse estimate, including its stated constant. This argument uses the weighted forcing directly; it does not estimate f by dividing a uniform forcing bound by $\sqrt{x}$.

For the nullspace assertion, $\mathcal{A}H = 0$ first gives $H''(0) = 0$. For $x > 0$ it then gives $k_* * H''' = 0$. Convolution once more with k_* , whose square is the constant π , gives $\pi(H'' - H''(0)) = 0$. Thus H is affine. Conversely every affine function is annihilated by the definition.

To prove (34.3), write $q = H'' - H''(0)$, $M_0 = \|q\|_\infty$ and $M_1 = \|q'\|_\infty$. If either is zero the assertion follows directly. Otherwise $q(0) = 0$ implies $\ell = M_0/M_1 \leq D$. For $x \leq \ell$, $|(k_* * q')(x)| \leq 2M_1\sqrt{x}$. For $x > \ell$, split the integral at $x - \ell$. Integration by parts on $[0, x - \ell]$, using $q(0) = 0$, bounds that portion by $2M_0/\sqrt{\ell}$; direct integration on the remaining interval bounds it by $2M_1\sqrt{\ell}$. Hence

$$\|k_* * q'\|_\infty \leq 4\sqrt{M_0M_1}.$$

Insert this into (34.1). The entire proof uses only the displayed derivatives and locally integrable convolutions, so also applies in the stated $C^{m-1,1}$ class. $\square$

The product on the right of (34.3) is an estimate, not the definition of a discrepancy norm. In particular, our regularization will minimize the seminorm of the linear transform $\mathcal{A}$, which satisfies the triangle inequality. Treating a geometric mean of two norms as a norm would not give that argument.

34.1. An exact half-order benchmark. The following calculation locates the loss between the two neighboring integer derivative norms. It does not identify an optimal statistical experiment or a sharp endpoint Hölder space.

Proposition 34.2 (Linearization and fixed-class instability). *Let $F(V) = T(V, V)$. Its linearization at the constant profile is*

$$DF_1[w](d) = \frac{16}{3\pi d^2} \int_0^d x^{-1/2} (d-x)^{3/2} w(x) \, dx. \quad (34.5)$$

For $w(x) = x^n$, $n \geq 0$, this equals $\lambda_n d^n$, where

$$\lambda_n = \frac{4(1/2)_n}{(n+2)!} = \frac{4 \binom{2n}{n}}{4^n (n+1)(n+2)} \sim \frac{4}{\sqrt{\pi}} n^{-5/2}. \quad (34.6)$$

At every such monomial the integrated transform is exactly inverted:

$$\mathcal{A}(\lambda_n d^{n+2}) = 2\pi d^n. \quad (34.7)$$

Fix $D = 1$, $m \geq 4$, $B > 1$ and $0 < \beta < 1$. On the single class $\mathcal{K}_m(B, \beta; 1)$ defined below there is no uniform Lipschitz inverse from the C^2 norm of $F(V) - F(W)$, or of $H_V - H_W$, to $\|V - W\|_\infty$.

Proof. Bilinearity gives $DF_1[w] = 2T(1, w)$. Integrating out the unperturbed variable in the simplex, with $\int_0^{d-x} (d-x-y)y^{-1/2} dy = \frac{4}{3}(d-x)^{3/2}$, gives (34.5). The beta integral, or (10.9), gives the first expression for λ_n . The central binomial identity and Stirling's formula give the other two. For $n \geq 1$,

$$\sqrt{x} (k_* * (x^n)')(x) = n B(n, 1/2) x^n = \frac{n! \sqrt{\pi}}{\Gamma(n+1/2)} x^n.$$

After two derivatives of $\lambda_n x^{n+2}$, all factors in (34.1) cancel to 2π . For $n = 0$, its endpoint term gives the same identity directly.

Choose $a > 0$ with $(m+1)a < B-1$ and $a < 1-\beta$ and, for $n \geq m$, put $V_n^\pm(x) = 1 \pm an^{-m}x^n$. Every derivative of the perturbation through order m is bounded by an^{r-m} . The last such bound controls the Lipschitz constant of the preceding derivative. Thus both signs belong to this same sum-norm class. Bilinearity cancels the quadratic perturbation terms exactly:

$$F(V_n^+) - F(V_n^-) = 2an^{-m}\lambda_n d^n.$$

For the maximum convention on C^2 , the ratio of profile error to law error is $\{\lambda_n n(n-1)\}^{-1}$; for integrated-law error it is $\{\lambda_n(n+1)(n+2)\}^{-1}$. Both tend to infinity like a positive multiple of $\sqrt{n}$. A sum convention changes these ratios by bounded factors. This proves the assertion on a fixed nontrivial class, rather than on classes with growing smoothness bounds. $\square$

The monomial and mixed-norm benchmarks were given in the independent memorandum [21]. The two-sided flux norm and its use in the sampling and calibration theorems below make a further observation statement. The alternatives in Proposition 34.2 are abstract profile alternatives; no billiard realization or preparation lower bound is deduced from them. The genuinely physical input remains the relative flux theorem and the geometric meaning of the profiles in (10.1).

35. THE INVERSE AUTOCONVOLUTION PROBLEM

After the physical relative law has been proved, the inverse in Section 10 belongs to the theory of nonlinear Volterra equations. The passage from a billiard record to that equation, and the topology of its inversion, must be distinguished from the convolution machinery itself.

Dai and Lamm [18] develop local regularization for inverse autoconvolution, preserving the causal structure and proving convergence for continuous and L^2 data. Their work supplies the relevant regularization context. We do not claim that causal inversion or a regularization principle is new, or apply an unverified theorem from that setting directly to the singular density here.

Hofmann, Werner and Deng [19] consider real L^2 functions supported in a cube, with full or limited convolution data. Their limited-data uniqueness result uses nonnegativity and support reaching the origin; it belongs to a convolution-support framework. Our normalized kernel $k(x) = x^{-1/2}V(x)$ has support at the origin and is locally integrable, but is not in L^2 there: $V(0) = 1$ implies $\int_0^a k(x)^2 dx = \infty$. Thus neither an L^2 uniqueness theorem nor L^2 ill-posedness is silently substituted for the statements proved here.

There are four specific features of the present inverse. First, the observed scalar law is $F = T(V, V)$, with two residual integrations between F and $k * k = (\pi/2)(d^2 F)''$. Second, normalization fixes the leading singularity of k , so the difference of two candidate kernels is continuous at zero and Abel reduction gives a forced second-kind equation. Third, all convolutions are restricted to a finite collar: reconstruction at energy d uses no values beyond d , no full-line Fourier inversion and no analytic continuation. Fourth, the deterministic stability statement uses C^3 law error and uniform profile loss. It is a full-function statement for smooth profiles, rather than a determination of their Taylor series.

These features identify the exact functional setting of Theorem 10.2; they are not a claim of a new general deautoconvolution method. Its geometric content is the physical realization of the singular weighted energy measure and the relative estimate that makes it observable. Its statistical use requires another step. Section A.4 regularizes finite scalar data on a positive-node grid, retaining the preparation cost of the rare selected event. This uses a bounded smoothness class; it does not replace C^3 by C^0 in the unregularized theorem. The exponent obtained there is sufficient and is not asserted to be the optimal derivative loss or minimax exponent. Nor is an arbitrary regularization candidate asserted to be a realizable contact graph.

The Abel norm in Section 34 is a norm of integrated flux differences with zero initial value and slope; on arbitrary flux functions it has an affine nullspace. Its two-sided estimate is on the image of the normalized positive profile map. It does not assert that every scalar law is an autoconvolution square. The refinement in the observation theorem uses this linear discrepancy together with a globally smooth reconstruction and a different variance allocation. The subsequent calibration argument uses the physical identity that an onset error translates the integrated flux. These are specified consequences for this billiard invariant, not claims that Abel inversion, local polynomial regularization or Bernoulli concentration are new.

36. REGULARIZED OBSERVATION OF THE COMPLETE INVARIANT

We return to general contacts, without evenness, congruence or equal curvature assumptions. The observations in this section recover the full symmetrized energy profiles. The physical class is not replaced by the finite-dimensional family of the preceding section. The integrated-flux identity supplies two additional derivatives on exactly the profile class used below. This gain and its quantitative consequences were pointed out in the author-requested independent memorandum [22]. We include the proof and the complete modified observation argument; the earlier, weaker constructions remain in Sections A.4–A.6.

Fix $m \geq 4$, $D > 0$, $B > 0$ and $\beta > 0$. Write

$$\|V\|_{r,1} = \sum_{a=0}^r \|V^{(a)}\|_{\infty} + \text{Lip}(V^{(r)}),$$

$$\mathcal{K}_m = \{V \in C^{m-1,1}([0, D]) : V(0) = 1, V \geq \beta, \|V\|_{m-1,1} \leq B\}.$$

The sum convention is part of this definition. The two actual profiles belong to this class. For two such functions define

$$H(V, W)(d) = d^2 T(V, W)(d) = \frac{2}{\pi} \int_{x+y < d} (d-x-y) \frac{V(x)W(y)}{\sqrt{xy}} dx dy, \quad H_V = H(V, V),$$

where the integral is over positive x, y . The target laws are $T(\mathcal{V}_b, \mathcal{V}_c)$. Constants below may depend on m, D, B, β and the specified physical bounds, but not on the accuracy, mesh or flight number.

36.1. The regularity of integrated flux.

Lemma 36.1 (Two-derivative gain). *For $V, W \in \mathcal{K}_m$ one has*

$$\|H(V, W)\|_{m+1,1} \leq CB^2, \quad H(V, W)(0) = H(V, W)'(0) = 0. \quad (36.1)$$

In particular, the candidate fluxes belong to a bounded $C^{m+1,1}$ class even though the profiles are only bounded in $C^{m-1,1}$.

Proof. Fubini's theorem and two integrations of a continuous convolution give

$$Q(d) := H(V, W)''(d) = \frac{2}{\pi} \int_0^1 \frac{V(ds)W(d(1-s))}{\sqrt{s(1-s)}} ds. \quad (36.2)$$

For $0 \leq r \leq m-1$, dominated differentiation gives

$$Q^{(r)}(d) = \frac{2}{\pi} \sum_{a=0}^r \binom{r}{a} \int_0^1 \frac{s^a (1-s)^{r-a} V^{(a)}(ds) W^{(r-a)}(d(1-s))}{\sqrt{s(1-s)}} ds.$$

The weight is integrable, including at both endpoints, and every profile derivative displayed is bounded by B . Thus $\|Q^{(r)}\|_{\infty} \leq 2^{r+1} B^2$. Every derivative of either profile through order $m-1$ is Lipschitz with constant at most B : for a lower derivative use its next derivative, and for the last use the norm assumption. The difference of each product in the formula at d, e is consequently bounded by $CB^2|d-e|$. It follows that $Q^{(m-1)}$ is uniformly Lipschitz. Finally

$$H'(d) = \int_0^d Q(s) ds, \quad H(d) = \int_0^d (d-s)Q(s) ds.$$

These identities control the two lower derivatives and prove (36.1). No extension of the profiles across zero, additional derivative observation, or higher profile norm is used. $\square$

36.2. Positive-node reconstruction. Put $q = m+2$, let $L \geq q$, and write $h = D/L \leq 1$. The reconstruction $J = J_h^{[q]}$ uses the positive nodes $h, 2h, \dots, Lh$ only. To specify it, set

$$\omega(u) = \begin{cases} e^{-1/(1-u^2)}, & |u| < 1, \\ 0, & |u| \geq 1, \end{cases} \quad \theta_{\ell}(x) = \frac{\omega(x/h - \ell)}{\sum_{r=0}^L \omega(x/h - r)}.$$

For $0 \leq \ell \leq L$, interpolate the q nodal values starting at $s_{\ell} = \max\{1, \min\{\ell - \lfloor (q-1)/2 \rfloor, L-q+1\}\}$ by a polynomial $P_{\ell}z$ of degree at most $q-1$, and define $Jz = \sum_{\ell=0}^L \theta_{\ell} P_{\ell}z$. This is a globally smooth function on the closed interval. Its first stencil is $h, \dots, qh$; the value at zero is extrapolated and is not inserted as an additional observation.

Lemma 36.2 (Three reconstruction estimates). *Uniformly for the indicated meshes,*

$$|JH_V - H_V|_{\mathcal{A}} \leq Ch^{m-1/2}, \quad V \in \mathcal{K}_m, \quad (36.3)$$

$$|Jf|_{\mathcal{A}} \leq C\|f\|_{C^3}, \quad f \in C^3([0, D]), \quad (36.4)$$

$$|Jz|_{\mathcal{A}} \leq Ch^{-5/2} \max_k |z_k|. \quad (36.5)$$

The seminorm is the linear Abel seminorm of Theorem 34.1, including its endpoint term.

Proof. At each point at most two weights are positive. A nearest grid index lies within rescaled distance $1/2$, so the denominator is bounded below uniformly. Derivatives of the weights of order r are bounded by $C_r h^{-r}$. Every active stencil is within distance $(q+1)h$. The rescaled Lagrange polynomials and their fixed-order derivatives are bounded on these compact intervals. Since q is fixed, the product rule gives $\|Jz\|_{C^r} \leq Ch^{-r} \max_k |z_k|$ for $r \leq 3$. The reconstruction reproduces every polynomial of degree at most $q-1 = m+1$. Subtracting the Taylor polynomial of that degree at the evaluation point, using a Lipschitz last derivative, gives

$$\|(JH_V - H_V)''\|_{\infty} \leq Ch^m, \quad \|(JH_V - H_V)'''\|_{\infty} \leq Ch^{m-1}.$$

Lemma 36.1 bounds the remainders uniformly. These estimates are valid at zero and D , since the rescaled stencils remain bounded there. For $f \in C^3$, subtract instead its quadratic Taylor polynomial. Its nodal remainder is $O(h^3)\|f\|_{C^3}$, so $\|Jf\|_{C^3} \leq C\|f\|_{C^3}$.

Apply the mixed Abel estimate

$$|G|_{\mathcal{A}} \leq \frac{\pi}{2} |G''(0)| + 2\pi\sqrt{D} \sqrt{\|G'' - G''(0)\|_{\infty} \|G'''\|_{\infty}}.$$

The approximation product has square root $h^{m-1/2}$, and its endpoint term is $O(h^m)$. The nodal-error product has square root $h^{-5/2}$, dominating its h^{-2} endpoint term. The C^3 bound treats the smooth input with no mesh amplification. This proves all three claims. In particular, the gain is used only for limiting candidate fluxes, not to assume two extra derivatives of the physical finite-bridge error. $\square$

36.3. Calibrated binary observations. For the moment suppose g, A, γ are supplied, as well as the selected patch labels, a common collar, and a relative convergence certificate. For even $j \geq 2$ assume the physical selected means have

$$c_j = 2A \sinh(j\gamma), \quad H_{j,b} = c_j p_{j,b}, \quad 0 \leq H_{j,b}(d) \leq M_0 d^2, \quad \|H_{j,b} - H_{\mathcal{V}_b}\|_{C^3([0,D])} \leq M_1 \tau^j. \quad (36.6)$$

These are consequences of the relative physical law on the supplied compact class. The values M_0, M_1 and $\tau \in (0, 1)$, rather than the unknown profiles or finite-flight remainders, are supplied.

Theorem 36.3 (Complete-profile preparation bound). *For $L \geq m+2$, $h = D/L \leq 1$, even $j \geq 2$ and $0 < \delta \leq 1$, use*

$$N_{b,k} = \left\lceil 2c_j(M_0 d_k^2 + t/3)t^{-2} \log \frac{4L}{\eta} \right\rceil, \quad d_k = kh, \quad (36.7)$$

fresh binary preparations in orientation b at time $jg + d_k$. There is a Borel estimator with values in $\mathcal{K}_m^2$ such that, with probability at least $1 - \eta$,

$$\max_b \|\widehat{V}_b - \mathcal{V}_b\|_{\infty} \leq C\{h^{m-1/2} + \tau^j + th^{-5/2} + \delta\}. \quad (36.8)$$

The predicted odd law has at most $2B$ times this error. The total number of preparations is deterministically bounded by

$$N_{\text{tot}} \leq 2L + CA \sinh(j\gamma) h^{-1} t^{-2} \log \frac{4L}{\eta}. \quad (36.9)$$

In particular, uniform accuracy ε for both profiles and the odd-law prediction, at confidence $1 - \eta$, is attained using

$$N_{\text{tot}} \leq C\varepsilon^{-(2+\frac{6}{m-1/2}+\frac{\gamma}{|\log \tau|})} \log \frac{C}{\eta\varepsilon}. \quad (36.10)$$

Every failure is charged. No value at zero or derivative is observed. This is a sufficient preparation bound, not a minimax or computation-time bound.

Proof. The class $\mathcal{K}_m$ is compact in C^{m-1} by Arzelà–Ascoli and the closed sum-norm bound. For fixed h , the map $V \mapsto \mathcal{A}JH_V$ is continuous through its finite vector of nodal values, so its image in C^0 is compact. Choose a finite ordered dictionary $\mathcal{D}_{h,\delta} \subset \mathcal{K}_m$ whose transformed nodal images form a δ -net. For a data vector z , take the first minimizer over the dictionary of $|JH_W - Jz|_{\mathcal{A}}$. Each discrepancy is continuous in the finite data vector. The first-index rule therefore defines a Borel map, including ties. No physical realization of dictionary elements is required for this upper bound.

Let $Z_{b,k} = c_j \bar{Y}_{b,k}$. Its mean is $H_{j,b}(d_k)$ and

$$\text{Var}(Z_{b,k}) = \frac{c_j H_{j,b}(d_k)(1 - p_{j,b}(d_k))}{N_{b,k}} \leq \frac{c_j M_0 d_k^2}{N_{b,k}}.$$

A centered summand is bounded by c_j . The Bernoulli exponential moment bound consequently gives

$$\Pr\{|Z_{b,k} - H_{j,b}(d_k)| > t\} \leq 2 \exp\left\{-\frac{N_{b,k} t^2}{2c_j(M_0 d_k^2 + t/3)}\right\}.$$

The allocations and a union bound make every nodal error at most t with probability $1 - \eta$. On this event, (36.4) treats the smooth bridge error and (36.5) treats only the centered scalar errors, giving

$$|JZ_b - JH_{\mathcal{V}_b}|_{\mathcal{A}} \leq C(th^{-5/2} + \tau^j).$$

The minimizing property and the dictionary net bound the discrepancy of the reconstructed candidate images by twice this quantity plus δ . Apply (36.3) to the true and fitted candidate fluxes. The triangle inequality for the linear seminorm gives their unregularized Abel discrepancy at most $C(h^{m-1/2} + th^{-5/2} + \tau^j + \delta)$. The two-sided inverse of Theorem 34.1 proves (36.8). The positive mass-one kernel of T gives the $2B$ bound for the odd-law prediction.

Summing (36.7) uses $\sum d_k^2 \leq D^2 L$ and $L = D/h$, and includes each ceiling. This proves (36.9). Put $\nu = m - 1/2$. Choose an integer mesh with $h \asymp \varepsilon^{1/\nu}$, $t = c\varepsilon h^{5/2}$, and $\delta = c\varepsilon$. Take the least even $j \geq 2$ with $\tau^j \leq c\varepsilon$. Then

$$h^{-1}t^{-2} = c^{-2}\varepsilon^{-2}h^{-6}, \quad e^{j\gamma} \leq C\varepsilon^{-\gamma/|\log \tau|}.$$

Integer rounding changes only fixed constants. Substituting in (36.9), and absorbing the smaller ceiling term, proves (36.10). $\square$

Corollary 36.4 (A uniform-data conditional modulus). *For $V, W \in \mathcal{K}_m$, put $e = \|H_V - H_W\|_{\infty}$. Then*

$$\|V - W\|_{\infty} \leq C e^{(m-1/2)/(m+2)}. \quad (36.11)$$

This is a conditional estimate on a fixed regularity class of forward images. No optimality of its exponent is asserted.

Proof. Use the inverse theorem, insert JH_V, JH_W , and apply (36.3) and (36.5). This gives $\|V - W\|_{\infty} \leq C(h^{m-1/2} + eh^{-5/2})$. For sufficiently small $e > 0$, choose an integer mesh with $h \asymp e^{1/(m+2)}$. If $e = 0$, let $h \downarrow 0$; for larger e , the bound $\|V - W\|_{\infty} \leq 2B$ permits an enlargement of C . These cover all cases. $\square$

36.4. Unknown gap, area and multiplier. We next remove the actual values of g, A, γ , not the supplied labels or class certificates. Assume precisely the pilot model (A.6.5): a supplied positive coarse gap bracket of width at most d_c , positive compact area and multiplier boxes, and one- and two-flight physical probabilities with exact leading coefficients $[2A \sinh(r\gamma)]^{-1}$ and unknown $C^{m-1,1}$ remainders F_r , normalized at zero and bounded below. Their common norm and positivity constants are supplied. In addition, assume the relative C^3 estimates on $[0, D_+]$, a uniform $C^{3,1}$ bound for $H_{j,b}$ there, and a fixed $D < D_+$. These are the same assumptions as in Theorem A.6.3; the profile class is not replaced by $\mathcal{K}_{m+2}$.

Theorem 36.5 (Self-calibrated complete-profile observation). *Under these supplied smooth physical bounds, for sufficiently small $\varepsilon > 0$ and $0 < \eta < 1/4$, a Borel procedure recovers both complete energy profiles on $[0, D]$ and their predicted odd law to uniform error ε , with probability at least $1 - \eta$. It uses a separately charged one-/two-flight pilot and fresh positive-offset bits at one subsequent even flight number. Its deterministic total preparation count is at most*

$$C\varepsilon^{-(2+\frac{6}{m-1/2}+\frac{\gamma_+}{\lceil \log \tau \rceil})} \log \frac{C}{\eta\varepsilon}. \quad (36.12)$$

The estimate includes the energy origin and does not infer separate contact curvatures or unsymmetrized contact branches.

Proof. We give the plug-in argument because an error in the programmed time must be handled before regularization. Suppose a pilot yields, on an event of probability $1 - \eta_0$,

$$0 \leq \hat{g} - g \leq \rho, \quad |\hat{A}/A - 1| \leq \alpha, \quad |\hat{\gamma} - \gamma| \leq \zeta, \quad j\zeta \leq 1, \quad j\rho \leq D_+ - D.$$

All estimates are projected onto their supplied compact boxes even on bad histories. Program time $j\hat{g} + d$ and scale a new bit by $\hat{c}_j = 2\hat{A} \sinh(j\hat{\gamma})$. Conditionally on the pilot its mean is exactly

$$RH_{j,b}(d + \Delta), \quad \Delta = j(\hat{g} - g), \quad R = \frac{\hat{A} \sinh(j\hat{\gamma})}{A \sinh(j\gamma)}.$$

Since $\gamma \geq \gamma_- > 0$, differentiation of $\log \sinh(j\gamma)$ gives $|R - 1| \leq C(\alpha + j\zeta)$. The one-sided gap estimate keeps every translated argument in $[0, D_+]$. The $C^{3,1}$ flux bound gives a C^3 translation error $O(\Delta)$, including at zero. Hence

$$\|RH_{j,b}(\cdot + \Delta) - H_{j,b}\|_{C^3([0,D])} \leq C(\tau^j + \alpha + j\zeta + j\rho). \quad (36.13)$$

No factor Δ/d occurs, since the bits have not been divided by d^2 .

On the good pilot event the mean is bounded by a supplied constant M_H . Allocate at each node

$$\left\lceil 2\hat{c}_j(M_H + t/3)t^{-2} \log(4L/\eta_1) \right\rceil$$

fresh preparations. Conditional Bernoulli concentration gives nodal error at most t with probability $1 - \eta_1$. Applying (36.4) to (36.13), and using the same dictionary argument, yields error at most

$$C\{h^{m-1/2} + th^{-5/2} + \delta + \tau^j + \alpha + j\zeta + j\rho\}.$$

The smooth calibration term has no inverse power of h . The charge on every history is bounded by $2L + Ce^{j\gamma}h^{-1}t^{-2} \log(4L/\eta_1)$, since the pilot estimates are always in their compact boxes.

Apply Lemma A.6.2 with $\xi = c\varepsilon/j$ and $\eta_0 = \eta/2$. Its proof, retained in full below, brackets the onset using charged one-flight trials and extrapolates the two unknown smooth remainders from positive nodes; it does not assume an exact-family probability oracle. It supplies $\alpha + \zeta \leq C\xi$ and $\rho \leq C\xi^{1+1/m}$ with pilot charge

$$C(j/\varepsilon)^{2+2/m} \log \frac{C \log(Cj/\varepsilon)}{\eta}.$$

Choose the least even j with $\tau^j \leq c\varepsilon$, $\eta_1 = \eta/2$, and the mesh, t and δ from Theorem 36.3. For small enough c and ε , all the displayed pilot constraints hold and the profile error is at most ε . Conditioning and the two failure allocations give total failure probability at most η . Finally

$$2 + \frac{6}{m-1/2} + \frac{\gamma_+}{|\log \tau|} > 2 + \frac{2}{m}.$$

This strict power gap absorbs the pilot cost and its fixed powers of $\log(1/\varepsilon)$ into (36.12). The allocation is bounded independently of pilot success, proving the unconditional deterministic preparation claim. $\square$

For $m = 4$, the profile-stage accuracy power, excluding the common bridge factor, is $26/7$ rather than the power 6 of the earlier m -node Abel construction. Both are sufficient bounds for specified procedures. A larger admissible value of τ weakens the displayed bridge factor without changing the billiard. Neither this factor nor the complete exponent is asserted to be an intrinsic information law. The physical finite-dimensional lower bound of Theorem 17.3 and the separate nuisance lower bounds below do not imply a lower bound for this full-profile experiment. The two even laws identify the invariant; separately measured odd records can validate, rather than further identify, its predicted compatibility law.

Part 3. Observable calibration and global physical reconstruction

37. FROM NOISY FINITE JETS TO GLOBAL ANALYTIC TABLE LOSS

The exact analytic rigidity theorem uses all contact jets, whereas every finite-sample information theorem is first proved in a fixed finite jet model. We now give a precise bridge between those two levels. The result is a compactness/sieve theorem: it does not postulate a uniform-in-order condition number, and it does not perform unstable numerical analytic continuation. Instead it uses injectivity of the intrinsic lifted-channel data map on a compact analytic class to show that sufficiently many accurately recovered finite coordinates resolve the global table uniformly.

Fix the marked abstract Euclidean lattice, obstacle-orbit labels and a connected measured lifted-channel graph. Let $\mathfrak{K}$ be a compact family of labelled periodic billiard tables, modulo one simultaneous common element of $\text{SE}(2)$, with the following properties.

- (1) Every obstacle boundary is connected, real analytic and strictly convex, and the family has uniform positive curvature, separation and channel-collar bounds.
- (2) The chosen lifted channels persist throughout $\mathfrak{K}$ with the same labels, deck displacements and orientations.
- (3) The admissible periodic intrinsic gluing space of Theorem 14.3 is a singleton throughout $\mathfrak{K}$. A sufficient intrinsic model is a fixed signature-rigid lattice-anchoring cycle together with a signature-rigid spanning tree rooted on an obstacle orbit met by that cycle. If an oriented Euclidean realization of the lattice and one root channel frame relative to that lattice are supplied as ambient calibration, the anchoring cycle may instead be omitted.
- (4) The topology of $\mathfrak{K}$ is strong enough that analytic boundary images and the corresponding half-line actions vary continuously with all fixed finite derivatives. A uniformly bounded holomorphic extension to a fixed complex collar is a sufficient model.

For every integer $q \geq 0$, let d_q be any metric inducing the labelled C^q -topology on this quotient class.

For a table $T \in \mathfrak{K}$ and an integer $M \geq 2$, define its finite intrinsic action data

$$\mathcal{D}_M(T) = \left(g_e, S_{e,0}^{(m)}(0), S_{e,1}^{(m)}(0) : e \in E, 2 \leq m \leq M \right). \quad (37.1)$$

The fixed lifted-channel labels (a_e, b_e, ℓ_e) are part of the experimental design and are suppressed from this numerical vector. The full sequence $\mathcal{D}_\infty$, together with those labels, is determined by the signed endpoint support germs, and Theorem 14.3 makes it injective on $\mathfrak{K}$.

Theorem 37.1 (Uniform finite-coordinate resolution of a compact analytic class). *For every integer $q \geq 0$ and every $\varepsilon > 0$, there exist an integer $M = M(q, \varepsilon)$ and a number $\eta = \eta(q, \varepsilon) > 0$ such that for all $T, T' \in \mathfrak{K}$,*

$$\|\mathcal{D}_M(T) - \mathcal{D}_M(T')\| < \eta \implies d_q(T, T') < \varepsilon. \quad (37.2)$$

Thus global analytic table loss can be controlled by finitely many intrinsic channel action jets, with the required order allowed to depend on the desired global accuracy.

Proof. Suppose the assertion fails for some q and $\varepsilon > 0$. For each integer $m \geq 2$ there then exist $T_m, T'_m \in \mathfrak{K}$ such that

$$d_q(T_m, T'_m) \geq \varepsilon, \quad \|\mathcal{D}_m(T_m) - \mathcal{D}_m(T'_m)\| < m^{-1}.$$

By compactness, after passing to a subsequence,

$$T_m \rightarrow T, \quad T'_m \rightarrow T'$$

in $\mathfrak{K}$. Fix any finite order M . For all sufficiently large $m \geq M$, continuity of $\mathcal{D}_M$ and the preceding estimate give

$$\mathcal{D}_M(T) = \mathcal{D}_M(T').$$

Since M is arbitrary, $\mathcal{D}_\infty(T) = \mathcal{D}_\infty(T')$. The all-order inverse and the singleton admissible periodic gluing hypothesis imply that $T = T'$ in the $\text{SE}(2)$ -quotient. Continuity of d_q then gives $d_q(T_m, T'_m) \rightarrow 0$, contradicting the fixed lower bound ε . $\square$

This theorem avoids treating analytic continuation of noisy Taylor series as an algorithm. On a fixed compact analytic class there are no globally separated tables whose intrinsic action data remain arbitrarily close through arbitrarily high finite order.

Theorem 37.2 (Diagonal growing-order reconstruction principle). *Let $\mathbf{E}_n = \{P_{n,T} : T \in \mathfrak{K}\}$ be any sequence of physical or boundary experiments. Assume that for every fixed M there exists an estimator $\hat{\mathcal{D}}_{n,M}$ of the finite data $\mathcal{D}_M(T)$ such that, for every $\eta > 0$,*

$$\sup_{T \in \mathfrak{K}} P_{n,T}(\|\hat{\mathcal{D}}_{n,M} - \mathcal{D}_M(T)\| > \eta) \longrightarrow 0. \quad (37.3)$$

Then there exist a single deterministic nondecreasing sequence $M_n \rightarrow \infty$ and estimators $\hat{T}_n \in \mathfrak{K}$ such that for every fixed integer $q \geq 0$ and every $\varepsilon > 0$,

$$\sup_{T \in \mathfrak{K}} P_{n,T}\{d_q(\hat{T}_n, T) > \varepsilon\} \longrightarrow 0. \quad (37.4)$$

The same sequence M_n works simultaneously for all fixed q . It may be chosen to increase arbitrarily slowly. No uniform-in- M conditioning of the determinant-one recursion is required.

Proof. For every integer $m \geq 1$ and every $0 \leq q \leq m$, apply Theorem 37.1 with accuracy m^{-1} to obtain $M(m, q)$ and $\eta(m, q)$. Put

$$M_m^* = \max_{0 \leq q \leq m} M(m, q), \quad \eta_m^* = \frac{1}{3} \min_{0 \leq q \leq m} \eta(m, q). \quad (37.5)$$

Increase M_m^* if necessary so that it is strictly increasing. Since closeness of $\mathcal{D}_{M_m^*}$ implies the same closeness for every lower truncation, an $3\eta_m^*$ -accurate M_m^* -vector resolves every metric d_q , $q \leq m$, to accuracy m^{-1} .

By (37.3), choose a strictly increasing sequence n_m so large that for every $n \geq n_m$,

$$\sup_{T \in \mathfrak{K}} P_{n,T} \left(\|\widehat{\mathcal{D}}_{n,M_m^*} - \mathcal{D}_{M_m^*}(T)\| > \eta_m^* \right) < 2^{-m}. \quad (37.6)$$

For $n_m \leq n < n_{m+1}$, set $M_n = M_m^*$. Choose $\widehat{T}_n$ to be a measurable η_m^* -approximate minimizer of

$$T' \mapsto \|\mathcal{D}_{M_m^*}(T') - \widehat{\mathcal{D}}_{n,M_m^*}\|.$$

For example, fix a countable dense subset of the compact metric space $\mathfrak{K}$ and choose the first element whose criterion is within η_m^* of the infimum; this gives a measurable selection.

On the event in which the finite-data estimator is within η_m^* of the truth, the minimizing property and the triangle inequality give

$$\|\mathcal{D}_{M_m^*}(\widehat{T}_n) - \mathcal{D}_{M_m^*}(T)\| < 3\eta_m^*.$$

Hence, simultaneously for every $q \leq m$,

$$d_q(\widehat{T}_n, T) < m^{-1}.$$

The exceptional probability is at most 2^{-m} . Given a fixed q and $\varepsilon > 0$, take $m \geq q$ with $m^{-1} < \varepsilon$ and let $n \rightarrow \infty$. This proves (37.4) with one common sequence M_n . $\square$

Corollary 37.3 (How the physical finite-order theorems feed the global bridge). *Whenever a compact analytic class $\mathfrak{K}$ admits a common finite acquisition atlas for which every fixed finite action-jet vector $\mathcal{D}_M$ is uniformly consistently estimable, the actual finite-bridge experiments of this paper admit a growing-order global table reconstruction in every fixed C^q loss by Theorem 37.2.*

The fixed-order ingredients used to construct such local estimators are the signed-jet tangent isomorphism, finite positive physical designs, the endpoint-time Poisson experiment (or its endpoint Gaussian coarsening), and the stopped finite-to-boundary transfer. The present theorem supplies the missing $M_n \rightarrow \infty$ and global-loss passage. Sharp rates in M_n are not claimed.

Remark 37.4 (What has and has not been proved). The bridge is stronger than simply placing an exact analytic theorem beside a fixed-dimensional statistical theorem: it gives a genuine growing-order sieve statement and global table loss. It is deliberately weaker than a sharp analytic minimax theorem. Obtaining an explicit optimal relation among analytic radius, jet order, physical preparation cost and global C^q risk would require uniform quantitative control of the finite-order inverse as $M \rightarrow \infty$. No such bound is inserted by assumption here.

38. COMMON PHYSICAL OBSERVATIONS AND CHARGED CHANNEL CALIBRATION

A channel coordinate is not an observation until its origin and direction have been supplied or estimated. We construct the latter from the physical records themselves. The construction uses only even, same-type alternating bridges; their flight numbers may be required to exceed any prescribed integer. In particular, no mixed-type one-flight experiment is used.

38.1. The acquisition space. Fix the marked channel labels $e = (a_e, b_e, \ell_e)$ and their transverse sign conventions. A channel detector reports positions in one oriented Euclidean sensor frame for that channel. The same frame is used for both endpoint types of the channel, but the rigid transformations relating *different* channel frames are neither supplied nor observed. A sensor reports both components of a collision position in physical length units. This is the endpoint-position observation, before taking its transverse projection. Unit speed and the physical clock fix the length unit. The marked obstacle and deck labels are acquisition labels, not supplied deck vectors. As in the physical preparation model, each preparation is an independent draw from normalized billiard phase volume. Its distribution may depend on the table; the instructions for taking and recording a preparation do not.

The design and one-preparation record spaces are the standard Borel spaces

$$\mathcal{A} = \{(e, b, j, t) : e \in E, b \in \{0, 1\}, j \geq 2 \text{ even}, t > 0\}, \quad \mathcal{X} = \{\partial\} \sqcup (\mathbb{R}^2 \times \mathbb{R}^2 \times [0, \infty)). \quad (38.1)$$

For a design $a = (e, b, j, t)$, monitor the marked collision word up to the programmed time t . If it is the specified alternating word of $j+1$ impacts beginning at type b , record (X, Y, r) , where X, Y are the first and last positions in that channel's sensor frame and r is the time from preparation to the first impact. Otherwise record ∂ . This is a single measurable map of the marked timed trajectory, defined by labels, comparisons with t , and coordinate readings. It contains no contact point, tangent, gap or unknown boundary graph. All failures, designs and stopping decisions are retained in the transcript. Extra trajectory information used by the apparatus to select the word is then discarded.

The selected labels refer to the facing lifted obstacle pair, not to a parameter-dependent spatial gate. If detector gates are used, they are fixed before observation and contain the persistent contact neighborhoods for the declared class. Near onset, the localization below makes the two implementations agree. Discovery of unlabelled channels is a different observation problem and is not assumed as part of this record map.

Let $\mathfrak{K}$ be the compact marked analytic class of Section 16. In this section only its uniform positive curvature, gap, area, separation and persistent-channel bounds are needed. Write $p_{e,0}, p_{e,1}$ for the two contact positions in the channel sensor frame, and put

$$n_e = (p_{e,1} - p_{e,0})/g_e, \quad t_e = R_{\pi/2} n_e.$$

The prescribed sign convention fixes the possible simultaneous sign change of the transverse axes. These quantities are used in the analysis, not in the acquisition instructions. Bounds below are invariant under arbitrary independent rigid motions of the channel sensor frames.

Lemma 38.1 (Uniform physical localization and success mass). *There are $h_0, c_*, C_*, \gamma_+ > 0$, uniform on $\mathfrak{K}$, such that a selected j -flight record is impossible for $t \leq jg_e$. If $0 < t - jg_e \leq 2h \leq 2h_0$, each recorded endpoint of type b satisfies*

$$|X - p_{e,b}| + |Y - p_{e,b}| \leq C_* \sqrt{h}. \quad (38.2)$$

For every even $j \geq 2$ and $h \leq t - jg_e \leq 2h \leq 2h_0$, the selected-record probability is at least

$$p_*(j, h) := c_* h^2 e^{-j\gamma_+} > 0. \quad (38.3)$$

The constants can be decreased so that $p_(j, h) < 1$.*

Proof. Every flight between the specified two lifted obstacles has length at least g_e . Their unique closest pair is an interior nondegenerate minimum of the two-endpoint distance. In contact arclength coordinates its Hessian is positive definite, uniformly on the compact class. Taylor's formula therefore gives

$$\ell_e(s, s') - g_e \geq c(s^2 + (s')^2)$$

on a common neighborhood. On its complement a positive uniform distance margin follows from uniqueness, channel persistence and compactness. Thus if the sum of the j flight lengths is less than $jg_e + 2h$, for h sufficiently small every flight is in that neighborhood. Applying the displayed bound to the first and last flights proves (38.2); the constant does not grow with j . Nonnegative entrance and exit times prove the assertion before onset.

For the lower mass use the exact flux density in (21.1), in unscaled coordinates. Uniformly on the contact box, $E_j(u, v) \leq C(u^2 + v^2)$, $b_j \geq c > 0$, and $-W_{j,uv}(0, 0) \geq ce^{-j\gamma_+}$. Restrict (u, v) to a ball of radius $c\sqrt{h}$ and r to an interval of length ch , so that $r + E_j < h$. Integrating gives $ch^2 e^{-j\gamma_+}/A$. The uniform upper bound on the free area gives (38.3). These bounds follow from the finite bridge and relative-law estimates with strict uniform curvature and hyperbolicity margins; they require no finite-dimensional parametrization of $\mathfrak{K}$. $\square$

38.2. A finite capped scan. Choose known bounds $0 < g_- \leq g_e \leq g_+$ for the class. Fix an even flight number j , a resolution $h < h_0$, and an error probability $\beta \in (0, 1)$. All of these are chosen before the pilot. Set

$$t_l = jg_- + lh, \quad 0 \leq l \leq L, \quad L = \lceil j(g_+ - g_-)/h \rceil + 2, \quad N = \left\lceil p_*(j, h)^{-1} \log(2|E|/\beta) \right\rceil. \quad (38.4)$$

For each (e, b) , scan these times in increasing order, taking at most N independent preparations at each time, and stop at the first successful record. Denote its time by $T_{e,b}$ and its first endpoint by $\hat{p}_{e,b}$. If a scan exhausts the grid, report pilot failure. The total preparation cap is

$$B_{\text{pil}} = 2|E|(L + 1)N. \quad (38.5)$$

No intermediate failed preparation is omitted from this count.

Theorem 38.2 (Observable contact-frame and onset calibration). *The preceding policy has, with probability at least $1 - \beta$ uniformly on $\mathfrak{R}$, a successful outcome in every scan and*

$$jg_e < T_{e,b} \leq jg_e + 2h.$$

On this event define

$$\hat{g}_e = \frac{\min_b T_{e,b} - h}{j}, \quad \hat{n}_e = \frac{\hat{p}_{e,1} - \hat{p}_{e,0}}{|\hat{p}_{e,1} - \hat{p}_{e,0}|}, \quad \hat{t}_e = R_{\pi/2} \hat{n}_e. \quad (38.6)$$

For all sufficiently small h , the denominator is at least $g_-/2$ and

$$\max_e j|\hat{g}_e - g_e| \leq h, \quad \max_{e,b} |\hat{p}_{e,b} - p_{e,b}| + \max_e |\hat{t}_e - t_e| \leq C\sqrt{h}. \quad (38.7)$$

The policy and estimates are measurable functions of the transcript on (38.1). They require no supplied contact center, tangent, curvature, gap or inter-channel placement.

Proof. For each (e, b) there is a grid point whose true excess is in $[h, 2h]$. At that point the probability of N failures is at most $(1 - p_*)^N \leq \exp(-Np_*) \leq \beta/(2|E|)$. If the scan has stopped earlier, its first success cannot precede onset and its time is even smaller. A union bound proves the claimed simultaneous event. In particular, this argument does not require a lower success bound at times far beyond the near-onset collar, nor a decision that a finite run of failures proves zero probability.

The gap bound follows by subtracting the midpoint h from a number in $(0, 2h]$. Lemma 38.1 gives the two position bounds. For vectors v, w with $|v| \geq g_-$ and $|v - w| \leq g_-/2$,

$$\left| \frac{v}{|v|} - \frac{w}{|w|} \right| \leq 4|v - w|/g_-.$$

Apply this with the true and estimated contact displacements. Each operation in (38.6) is Borel on the good event. On a failed scan or a small denominator assign a fixed default value and a failure flag. This defines the policy and estimators on every transcript. Rigidly moving a channel sensor moves both estimated contact positions by the same rigid motion and rotates the estimated tangent. Hence the bounds are uniform in these unsupplied sensor placements. $\square$

The pilot flight number may be any prescribed even j , however large. Later we set it equal to the already chosen post-pilot flight number J . The pilot therefore controls $J|\hat{g}_e - g_e|$, not a different quantity at an earlier, smaller flight number. Both types are measured in the same channel sensor frame; this does not register different channels.

38.3. Observable normalization and continuity of the tested laws. For a fresh successful same-type record set

$$\widehat{C}_{e,b}(X, Y, r) = (\widehat{t}_e \cdot (X - \widehat{p}_{e,b}), \widehat{t}_e \cdot (Y - \widehat{p}_{e,b}), r). \quad (38.8)$$

Its ideal counterpart $C_{T,e,b}$ replaces the estimates by the true contact frame. On the pilot event and a common positive-offset collar,

$$|\widehat{C}_{e,b}(X, Y, r) - C_{T,e,b}(X, Y, r)| \leq C\sqrt{h}. \quad (38.9)$$

Indeed the projected displacement error is bounded by the position error plus the tangent error times $|X - p_{e,b}|$, which is uniformly bounded on the selected collar. No bound on the absolute sensor origin is needed.

Let $Q_{T,e,b,d}^\infty$ be the centered successful endpoint-time law, with density, extended by zero to a fixed box,

$$q_{T,e,b,d}(u, v, r) = Z_{T,e,b,d}^{-1} B_{e,b}(u) B_{e,b}(v) \mathbf{1}_{\{0 < r < d - S_{e,b}(u) - S_{e,b}(v)\}}. \quad (38.10)$$

For any compact positive-offset interval inside the common collar, Z is bounded below by a positive constant. If S, B, d and $\tilde{S}, \tilde{B}, \tilde{d}$ are two such descriptions, fiber integration and $|x_+ - y_+| \leq |x - y|$ give

$$\|Q - \tilde{Q}\|_{\text{TV}} \leq C(\|S - \tilde{S}\|_\infty + \|B - \tilde{B}\|_\infty + |d - \tilde{d}|). \quad (38.11)$$

To verify this estimate, integrate the amplitude difference on the common fibers and the uniformly bounded amplitudes on their symmetric difference; the latter length is bounded by the action and offset differences. The same bound applies to the normalizers, after which division by their positive lower bound proves the assertion. The half-line contraction and relative operator estimates imply continuous dependence of S, B on the table in this fixed collar. Thus the centered laws are continuous in total variation. This is not a claim of total-variation continuity for ambient position laws supported on different boundary curves.

Proposition 38.3 (Uniform implementation of a finite family of tests). *Let $(e_r, b_r, d_r, \phi_r)_{r=1}^R$ be a fixed finite family with positive offsets in the interior of the collar and bounded Lipschitz functions $|\phi_r| \leq 1$. Program the post-pilot time $J\widehat{g}_{e_r} + d_r$, where J is chosen before a pilot with $j = J$. Conditional on any good pilot history, a fresh successful record satisfies*

$$\left| \mathbb{E}_T \phi_r(\widehat{C}_{e_r, b_r}(X, Y, r)) - Q_{T, e_r, b_r, d_r}^\infty \phi_r \right| \leq C_{\mathcal{F}}(\sqrt{h} + h + \tau^J). \quad (38.12)$$

The actual statistic and the programmed design are parameter independent functions of the transcript.

Proof. On the good history the true offset differs from d_r by at most h . Choose h below half the smallest offset and below the upper collar margin. The conditional physical-to-boundary comparison (21.4) costs at most $2C\tau^J$ for a test bounded by one. The same boundary-position embedding is used for both measures at the fixed true table. Applying the *observable* map (38.8) cannot increase that distance. Replacing this map by its ideal counterpart costs at most the Lipschitz constant times $C\sqrt{h}$. Finally (38.11) bounds the offset error by Ch . Summing proves (38.12). The ideal map appears only in this error calculation. It is not a kernel that the experimenter must apply. $\square$

39. SINGLE-OFFSET STATISTICAL SEPARATION AND GLOBAL PHYSICAL RECONSTRUCTION

We construct the estimator on the common observation space of Section 38. The identifiable datum consists of the onsets and the centered conditional laws. The onsets enter the estimation criterion, not merely the programmed times. The single-offset inverse permits every post-pilot test to ignore residual time and both longitudinal coordinates. Only the two endpoint types

at one fixed offset per channel are needed. No finite-dimensional model for the entire compact analytic class is imposed.

Fix one $d_0 > 0$ inside the common collar. For $M \geq 2$, write

$$\mathcal{D}_M(T) = (g_e, S_{e,0}^{(m)}(0), S_{e,1}^{(m)}(0) : e \in E, 2 \leq m \leq M).$$

Use the normalized finite-coordinate norm of Theorem 16.4, with a fixed positive gap scale g_* . The notation Q_{T,e,b,d_0}^∞ denotes the centered successful endpoint-time law of (38.10); all tests below are independent of its residual-time coordinate.

Lemma 39.1 (Single-offset mixed onset-law separators). *For every $M \geq 2$ and $\eta > 0$, there are finitely many labelled bounded Lipschitz tests $(e_r, b_r, \varphi_r)_{r=1}^R$, with $|\varphi_r| \leq 1$, and $c > 0$ such that*

$$\|\mathcal{D}_M(T) - \mathcal{D}_M(T')\|_M \geq \eta \implies \|F(T) - F(T')\|_\infty \geq 8c, \quad (39.1)$$

where the continuous moment map is

$$F(T) = \left((g_e(T)/g_*)_e, \left(Q_{T,e_r,b_r,d_0}^\infty \varphi_r(u, v) \right)_{r=1}^R \right). \quad (39.2)$$

All tests use the same fixed offset d_0 . The list and the separation constant depend only on the compact class, M , and η .

Proof. The set of pairs on the left of (39.1) is compact. For any such pair, either a gap differs, or all gaps agree. In the latter case, equality of both endpoint marginals at d_0 for every channel would, by Theorem 13.1, imply equality of every half-line action germ. It would therefore imply equality of $\mathcal{D}_M$, a contradiction. Hence one endpoint law separates the pair. Bounded Lipschitz functions separate finite Borel probability measures on the fixed compact endpoint box: continuous functions do so, and Lipschitz functions are uniformly dense in the continuous functions. Normalize a separating function to absolute value at most one; no uniform bound on its Lipschitz constant is required before the finite list is chosen.

Gap continuity and (38.11), followed by marginalization, make each strict separation persist on a neighborhood of the pair. A finite subcover gives finitely many tests and a positive minimum margin $8c$. Include every gap coordinate whether or not it was needed in a particular neighborhood. Add constant tests for any missing (e, b) , so all $D = 2|E|$ design groups are represented and $R \geq D$. If the compact pair set is empty, any positive c and these constant tests suffice. This proves the lemma. $\square$

A conditional law alone is not here asserted to recover the gap. Already at quadratic order, fixing $a_0 = a_1 = a$ permits

$$\kappa_0(g) = \kappa_1(g) = \frac{\sqrt{1 + g^2 a^2} - 1}{g}.$$

This explains the separate onset coordinate without asserting equality of the full nonlinear laws along this quadratic example.

39.1. A finite minimum-distance rule. Choose fixed templates $T_1, \dots, T_H \in \mathfrak{K}$ whose images form a c -net of $F(\mathfrak{K})$. For each (e, b) , collect k fresh successful records after the pilot. Apply the observable transverse projection in (38.8), and evaluate all tests belonging to that group on the same sample. Put

$$\hat{F} = \left((\hat{g}_e/g_*)_e, \left(k^{-1} \sum_{i=1}^k \varphi_r(\hat{u}_{e_r,b_r,i}, \hat{v}_{e_r,b_r,i}) \right)_{r=1}^R \right).$$

Select

$$\hat{T} = T_{\hat{h}}, \quad \hat{h} = \min_{1 \leq h \leq H} \argmin \|F(T_h) - \hat{F}\|_\infty, \quad \hat{\mathcal{D}}_M = \mathcal{D}_M(\hat{T}). \quad (39.3)$$

On a flagged pilot or batch failure use T_1 . This finite rule is Borel with a fixed tie convention. It does not hide a selection theorem or an unbounded optimization over unknown analytic functions. Compactness gives the library, not an effective rate for generating it from collar constants.

If $\|\hat{F} - F(T)\|_\infty \leq c$, a template within c of $F(T)$ has criterion at most $2c$. Consequently

$$\|F(\hat{T}) - F(T)\|_\infty \leq 3c < 8c, \quad \|\mathcal{D}_M(\hat{T}) - \mathcal{D}_M(T)\|_M < \eta. \quad (39.4)$$

The first norm includes the gap coordinates.

Theorem 39.2 (Uniform finite-order physical estimation at one offset). *For every $M \geq 2$, $\eta > 0$, $\delta \in (0, 1)$, and prescribed minimum flight number J_0 , there is one deterministically capped policy on $\mathcal{A}, \mathcal{X}$ such that every preparation uses the same chosen even $J \geq \max(2, J_0)$, and*

$$\sup_{T \in \mathcal{R}} \Pr_T \{ \|\hat{\mathcal{D}}_M - \mathcal{D}_M(T)\|_M \geq \eta \} < \delta. \quad (39.5)$$

No gap, contact frame, inter-channel registration or lattice metric is supplied. After the pilot there are exactly $2|E|$ design groups, all programmed at $J\hat{g}_e + d_0$; the estimator uses only their transverse endpoint pairs and the pilot summaries. Every preparation is charged, and all design and stopping decisions are measurable.

Proof. Choose the separators, $c \leq 1$, and the template library first. Take

$$k \geq 8c^{-2} \log(8R/\delta). \quad (39.6)$$

Choose the final even $J \geq \max(2, J_0)$ so that the finite-flight bias in Proposition 38.3 is at most $c/4$ and

$$CDk\tau^J < \delta/8. \quad (39.7)$$

Only then choose $h > 0$, below both pilot and offset-collar margins, with

$$C_{\mathcal{F}}(\sqrt{h} + h) \leq c/4, \quad h/(Jg_*) \leq c. \quad (39.8)$$

Run the calibration theorem with $j = J$, $\beta = \delta/8$, its finite grid and its deterministic cap B_{pil} . Thus the pilot controls $J|\hat{g} - g|$ for the final flight number, which is not changed later.

On a good pilot history program each group at $(e, b, J, J\hat{g}_e + d_0)$, take fresh preparations, and stop at its k -th success. The true offsets remain in a fixed positive subcollar. Uniform flux gives a success lower bound $p_0 > 0$, depending on the fixed design but not on the table or the good history. Impose the per-group cap

$$B = \left\lceil \frac{2k + 8 \log(8D/\delta)}{p_0} \right\rceil. \quad (39.9)$$

Since $Bp_0 \geq 2k$, the binomial lower-tail bound gives probability at most $\exp(-Bp_0/8) \leq \delta/(8D)$ of not obtaining k successes. The union of batch-cap failures has probability at most $\delta/8$.

Conditional on the pilot, successive marks in an uncapped fixed-design success sequence are iid with its successful physical law. The capped and uncapped constructions agree whenever the cap is not hit. Apply concentration to the uncapped sequence and add the cap probability; conditioning on completion of the cap is not used to assert iid sampling. Proposition 38.3, applied to $\phi_r(u, v, r') = \varphi_r(u, v)$, gives bias at most $c/2$. Hoeffding's inequality on $[-1, 1]$ and a union bound give

$$\Pr_T \left\{ \max_r |\hat{\mu}_r - \mathbb{E}_T(\hat{\mu}_r \mid \text{pilot})| > c/2 \mid \text{good pilot} \right\} \leq 2R \exp(-kc^2/8) \leq \delta/4.$$

Tests sharing a sample need not be independent for this union bound. On the complementary event the test and gap coordinates are all within c of $F(T)$. Apply (39.4). The sum of pilot, empirical and cap probabilities is at most $\delta/2$, hence strictly less than δ . The full cap is $B_{\text{pil}} + DB$.

The complete-transcript comparison is retained as well. Keep the pilot exact. Conditional on each good history the post-pilot designs are fixed and observable. The stopped comparison of Theorem 22.3 costs at most $CDk\tau^J$, since each group stops no later than its k -th success. Corollary 22.4 adds the pilot exceptional probability when histories are integrated. At each fixed table the finite and boundary position laws use the same physical embedding, and estimated projection is a common transcript map. This proves a distance bound without identifying an unknown exact centering map with an observation. The comparison includes all failures, counts, residual times and stopping decisions, even though the estimator discards some of them. The direct concentration proof already applies to the actual finite physical experiment. $\square$

39.2. Increasing order and a prescribed preparation budget.

Theorem 39.3 (Uniform global reconstruction from long same-type records). *On the compact marked analytic class of Section 16, there are finite capped policies π_s and table estimators $\hat{T}_s$ such that every flight number used by π_s is even and at least s , and, for every fixed $q \geq 0$ and $\epsilon > 0$,*

$$\sup_{T \in \mathfrak{R}} \Pr^{\pi_s}_T \{d_q(\hat{T}_s, T) > \epsilon\} \longrightarrow 0. \quad (39.10)$$

The recovered table includes its Euclidean period lattice. Only one simultaneous global element of $SE(2)$ is quotiented out. After calibration each policy uses the same fixed offset d_0 and only the two endpoint types in each channel.

Proof. Use Theorem 16.4 to choose increasing $M_s \geq s$ and decreasing $\eta_s > 0$ with

$$\max_{0 \leq q \leq s} \omega_q(C\eta_s + C2^{-M_s}) < 1/s.$$

Each of the finitely many moduli tends to zero. Apply the preceding theorem with $(M_s, \eta_s, 2^{-s}, J_0 = s)$ and keep its table-valued template estimator. Except on an event of probability at most 2^{-s} , the finite-data modulus gives $d_q(\hat{T}_s, T) < 1/s$ for every $q \leq s$. This proves the uniform assertion. The exact inverse used by the modulus includes the marked deck holonomies and $L = VM^{-1}$, so no lattice Gram form is inserted into the statistical conclusion.

For a prescribed total budget N , let b_s be the deterministic preparation cap of π_s , and set

$$C_s = \max\{s, b_1, \dots, b_s\}, \quad s(N) = \max\{s : C_s \leq N\}.$$

For $N < C_1$ use a fixed default estimate with no preparations. Otherwise run $\pi_{s(N)}$ afresh and spend no remaining budget. This is a budget-indexed family, not a cumulative execution of all preceding stages. It uses at most N preparations, $s(N) \rightarrow \infty$, and every flight number in its entire acquired transcript is at least $s(N)$. This proves the same consistency statement with the claimed flight restriction for the prescribed-budget implementation. A cumulative historical transcript would still contain the earlier flight numbers; no diverging minimum is asserted for that different implementation. $\square$

The library, separation constants and inverse moduli may be non-effective functions of the compact class. The theorem gives finite measurable acquisition once these constants are fixed, not polynomial-time design generation or a sharp analytic minimax rate. Its caps expose the rare-event cost. The direct-position benchmark also respects a diverging even flight number. The extra intrinsic conclusion is therefore not the exclusion of that benchmark, but the law inverse and its implementation using scalar post-pilot tests at one fixed offset.

40. COMPATIBLE ANALYTIC VARIATIONS AND UNIFORM FINITE POSITIVE DESIGNS

The compact analytic class used for global consistency need not be a finite-dimensional manifold. Nor does retaining finitely many target jets make a positive-offset statistical law depend on finitely many parameters. We give a precise finite-rank variation construction for the separate local information assertion. The global estimator of Theorem 39.3 does not depend on this construction.

Fix a compact family $\mathfrak{K}_0$ with a fixed pattern of coincident and distinct marked contacts on each obstacle. Coincident contacts are identified before specifying variations. Assume the normal angles of distinct contacts on the same obstacle have a positive uniform separation. This is a compact contact stratum; no such stratification is needed in the global separator theorem. Choose an oriented ambient angular coordinate for each obstacle in a fixed root-frame gauge. Its analytic support function is $h_i(\theta)$, with radius of curvature $R_i = h_i + h_i'' > 0$. At a contact angle θ_a , its point is

$$X_i(\theta_a) = h_i(\theta_a)n(\theta_a) + h_i'(\theta_a)t(\theta_a).$$

Consequently a support perturbation with zero value and first derivative at θ_a preserves that point and its oriented tangent exactly. It also preserves the gaps of the anchored channel network when this is imposed at every marked incidence. We vary the obstacle shapes in an ambient analytic family of dispersing tables; an arbitrary closed set $\mathfrak{K}_0$ is not asserted to contain these variations.

Lemma 40.1 (Uniform analytic Hermite right inverse). *Fix an integer $M \geq 2$. For l distinct contact angles there is a finite-dimensional space $\mathcal{T}_D$ of real trigonometric polynomials such that the jet evaluation map*

$$E_T : \mathcal{T}_D \longrightarrow \mathbb{R}^{l(M+1)}, \quad E_T f = (f^{(r)}(\theta_a))_{a, 0 \leq r \leq M}$$

is onto. One may take $D = (l-1)(M+1) + M$. In a fixed real Fourier basis,

$$R_T = E_T^t (E_T E_T^t)^{-1} \tag{40.1}$$

is a continuous right inverse, uniformly bounded on the compact contact stratum. It depends smoothly on the distinct contact angles.

Proof. For each contact a , put

$$F_a(\theta) = \prod_{b \neq a} (1 - \cos(\theta - \theta_b))^{M+1}.$$

This polynomial is nonzero at θ_a and vanishes to order $2(M+1)$ at all other contact angles. The $M+1$ functions $F_a(\theta) \sin^r(\theta - \theta_a)$, $0 \leq r \leq M$, have zero jets through order M at other contacts. At θ_a their jet matrix is triangular, with diagonal $r!F_a(\theta_a) \neq 0$. Their degrees are at most D . They therefore interpolate arbitrary jets at this contact, independently of the other contacts. Summing over a proves surjectivity. The Gram matrix in (40.1) is positive definite and varies smoothly with the contact angles. Its least eigenvalue has a positive minimum on the compact separated-angle set. This proves the remaining assertions. $\square$

Proposition 40.2 (Realization of compatible finite contact variations). *Over $\mathfrak{K}_0$, assign one labelled oriented graph jet $(q_{a,2}, \dots, q_{a,M})$ to each distinct physical contact. Repeated incidences use the same jet, transformed to their prescribed oriented frames. These coordinates define a finite-rank vector bundle $\mathcal{V}_M$ with a continuous norm. Its vectors have analytic realizations through their base tables which preserve every marked contact point, tangent and gap. The realizations vary continuously with the base and smoothly with the fiber variable; on a uniform fiber neighborhood they remain strictly convex separated dispersing tables with the same persistent channels.*

Proof. Take the direct sum of the right inverses of Lemma 40.1 over the obstacles. Require the zeroth and first support-jet increments to vanish. The remaining support jets through order M are local coordinates for graph jets through order M at a fixed point and tangent. Here is the triangular reason. Since $dX/d\theta = Rt$, at the contact the transverse graph coordinate satisfies $dy/d\theta = R > 0$. The graph curvature is $q_2 = R^{-1}$. Repeated differentiation with $d/dy = R^{-1}d/d\theta$ at the contact shows that, after lower support jets have been fixed, the coefficient of a variation of $h^{(m)}$ in q_m is nonzero (in the convex graph convention it is $-R^{-m}$). Thus the differential from support jets $h^{(2)}, \dots, h^{(M)}$ to graph jets $q_2, \dots, q_M$ is triangular and invertible. Changing the oriented transverse sign multiplies the corresponding odd-order coordinates by minus one and does not affect invertibility.

The finite-dimensional inverse function theorem now assigns support-jet increments to each sufficiently small desired graph-jet increment. Extend these increments by R_T , with the first two jets zero. The result is an actual real-analytic support function, not independent formal contact series. The interpolation imposes all incidences on each obstacle simultaneously, so compatibility is automatic. Uniform curvature and separation margins, compactness and the uniform bounds on R_T give a common small fiber neighborhood on which the realized tables are admissible. The periodic lattice and obstacle reference placements are kept fixed.

In each choice of labelled frames the fibers are the direct sum of the finite jet-coordinate spaces just described. On overlaps, frame changes and relabellings induce the differentials of the corresponding jet-coordinate changes. They are continuous invertible matrices and satisfy the cocycle identity because the underlying coordinate changes do. This specifies the bundle and its transition maps. A finite frame atlas and a partition of unity give a continuous fiber norm. Its unit sphere bundle over the compact base is compact. This construction does not give a tangent bundle to an arbitrary closed analytic class; it gives a specified bundle of ambient, compatible analytic variations through it. $\square$

For $v \in \mathcal{V}_M(T)$, let $\dot{S}_{e,b}[v]$ be the action variation along this realization. Define the positive-offset information form by (26.11) with boundary velocity

$$U_{e,b}(u, w)[v] = \dot{S}_{e,b}(u)[v] + \dot{S}_{e,b}(w)[v].$$

The gap is fixed in this bundle. A gap direction in a separately specified anchored one-channel local family is handled by the additional constant velocity in Lemma 26.2; no independent gap directions are assigned to an arbitrary globally compatible network here.

Lemma 40.3 (Uniform finite positive design on the specified bundle). *For each fixed M , there are finitely many labelled same-type positive offsets (e_r, b_r, d_r) , positive weights α_r summing to one, and $c_M > 0$, independent of the base table in $\mathfrak{R}_0$, such that*

$$\sum_r \alpha_r \mathcal{K}_{T, e_r, b_r, d_r}[v, v] \geq c_M \|v\|^2 \quad (v \in \mathcal{V}_M(T)). \quad (40.2)$$

This statement concerns the explicitly realized finite-rank bundle, not all variations of the infinite-dimensional analytic class.

Proof. If a vector lies in the kernels for both types of every channel and every small positive offset, positivity of the boundary weight implies

$$\dot{S}_{e,b}(u)[v] + \dot{S}_{e,b}(w)[v] = 0 \quad \text{when } S_{e,b}(u) + S_{e,b}(w) = d.$$

Set $w = 0$ and $d = S_{e,b}(u) > 0$. The anchored variation satisfies $\dot{S}_{e,b}(0)[v] = 0$, hence every $\dot{S}_{e,b}$ vanishes near the contact. The fixed-gap finite action-jet inverse in Theorem 12.1 forces the variations of all marked contact jets through order M to vanish. These jets are the fiber coordinates, so $v = 0$.

The forms are continuous in the base, direction and each fixed positive offset. Indeed the realized support families and their finite derivatives are continuous; the half-line inverse transports this continuity to the actions and amplitudes; and each positive-offset level has a regular boundary with a common local coordinate description. The coarea formula then gives continuity of the information integral. Every unit vector in the compact sphere bundle is detected by a design, and this remains true on a neighborhood. A finite subcover supplies finitely many designs. Their positive weighted sum has a positive continuous minimum on that sphere bundle, proving (40.2). $\square$

The hypotheses about the contact stratum and the compatible analytic realizations belong to this local-design lemma alone. In particular, a finite-rank information argument is not needed to pass from the compact analytic class to the global physical consistency theorem. That passage uses the augmented observable separator lemma and its finite capped policy.

Acknowledgments. The nuisance-splicing construction used in Section A.14 was supplied in the author-requested memorandum [17]; the author acknowledges this contribution. The author-requested referee-style memoranda were not commissioned journal reports. Revision records and source provenance are archived separately from the mathematical article.

Part 4. Complete auxiliary proofs and applications

APPENDIX A.1. INFORMATION BEYOND THE LEADING HIERARCHY

The nonlinear law of the preceding section is not determined by the contact Hessian. We prove this by an analytic deformation with fixed area, gap and contact curvatures. Its higher coefficient can be computed from the half-line action and determinant, without extrapolating an uncontrolled finite-length error.

A.1.1. The quartic variation of a boundary law. Assume first that the two facing graphs are identical and even, and write

$$\psi(y) = \frac{\kappa}{2}y^2 + \frac{q}{24}y^4 + O(y^6), \quad c = 1 + g\kappa, \quad \gamma = \operatorname{arcosh} c, \quad a = \frac{\sinh \gamma}{g}, \quad \lambda = e^{-\gamma}. \quad (\text{A.1.1})$$

Here q is a fourth graph derivative, not a source variable. Fix g and κ and vary q in a smooth family. Higher graph jets may vary as well. The two boundary actions and amplitudes now agree; write them as S and B . Both limiting parities give one function $\mathcal{F}$.

Proposition A.1.1 (A nonlinear coefficient and its quartic derivative). *Let $\mathcal{R}_\infty(0) = \partial_d \mathcal{F}(0)$. At fixed g, κ , its derivative with respect to q is*

$$\partial_q \mathcal{R}_\infty(0) = -\frac{\cosh(2\gamma) + 2}{12a^2 \sinh(2\gamma)}. \quad (\text{A.1.2})$$

This derivative does not depend on the graph derivatives of order six or higher. More precisely, if $S(u) = au^2/2 + tu^4/24 + O(u^6)$ and $B(u) = 1 + \beta u^2 + O(u^4)$, then

$$\partial_q t = \coth(2\gamma), \quad \partial_q \beta = -\frac{1}{4a \sinh(2\gamma)}, \quad \mathcal{R}_\infty(0) = \frac{2\beta}{3a} - \frac{t}{12a^2}. \quad (\text{A.1.3})$$

Proof. The linear half-line segment is $x_i = \lambda^i u$. Because the quadratic action is stationary with its first endpoint fixed, its change upon substituting the cubic bridge correction begins at degree six. The quartic part of the stationary action is therefore the quartic part of the broken action

evaluated on that linear segment. In one flight, the q -dependent quartic term of the length is $q(x_i^4 + x_{i+1}^4)/24$. The endpoint is counted once and every interior site twice. It follows that

$$\partial_q t = 1 + 2 \sum_{i \geq 1} \lambda^{4i} = \frac{1 + \lambda^4}{1 - \lambda^4} = \coth(2\gamma).$$

This use of stationarity may equivalently be obtained by differentiating finite stationary actions and passing to the convergent half-line sum.

To degree two in u , the q variation of an edge's mixed derivative vanishes: the quartic term just displayed is a sum of two pure fourth powers. Thus the product term U in (7.6) has no q -dependent quadratic coefficient. The interior Hessian variation at that degree is diagonal, with entries $q\lambda^{2i}u^2$. Changes in the cubic bridge do not enter degree two, and products of two Hessian perturbations enter only degree four. Consequently the quadratic coefficient of the logarithmic determinant has derivative $\sum_{i \geq 1} G_{ii}\lambda^{2i}$. At equal curvatures, (7.1) gives $G_{ii} = (1 - \lambda^{2i})/(2a)$. Hence

$$\partial_q \beta = -\frac{1}{2a} \left(\frac{\lambda^2}{1 - \lambda^2} - \frac{\lambda^4}{1 - \lambda^4} \right) = -\frac{1}{4a \sinh(2\gamma)}.$$

The quadratic coefficient of B equals that of $\log B$ because both have vanishing linear coefficient by evenness.

Finally use (7.12). The density correction is $\beta(u^2 + v^2)$, whose expectation under the quadratic residual-time weight is $2\beta d/(3a)$. For the quartic action correction, first variation of the positive-part integral gives the negative integral of $t(u^4 + v^4)/24$ over $a(u^2 + v^2)/2 < d$; there is no boundary term because the residual weight vanishes there. The two elementary integrals are

$$\int (d - ar^2/2)_+ \, du \, dv = \frac{\pi d^2}{a}, \quad \int_{r^2 < 2d/a} (u^4 + v^4) \, du \, dv = \frac{2\pi d^3}{a^3}.$$

Taylor's formula in the uniform Morse coordinates justifies the remaining $O(d^2)$ relative error. This gives the last identity in (A.1.3), and substitution proves (A.1.2). Terms of degree six or higher cannot contribute to this first offset coefficient. $\square$

For an additional finite-length identity, put $L_i = (\sinh((j-i)\gamma), \sinh(i\gamma))/\sinh(j\gamma)$, $M_j = H_j^{-1}$ and $v_i = L_i M_j L_i^t$ in the equal-curvature case. Thus $L_0 = (1, 0)$ and $L_j = (0, 1)$. The same calculation before taking the limit yields

$$\partial_q \mathcal{R}_j(0) = -\frac{1}{3} \sum_{i=1}^{j-1} (G_j)_{ii} v_i - \frac{1}{24} \left(v_0^2 + v_j^2 + 2 \sum_{i=1}^{j-1} v_i^2 \right). \quad (\text{A.1.4})$$

Indeed, the logarithmic amplitude variation is $-\sum (G_j)_{ii} (L_i Y)^2$, and its quadratic conditional expectation is $-d \sum (G_j)_{ii} v_i/3$. The unweighted fourth moment in the quadratic sublevel, divided by its residual-time integral, is dv_i^2 . These observations prove (A.1.4), including $j = 1$ with an empty interior sum. Its limit is (A.1.2). This identity provides a finite check of the normalization but is not needed to define the infinite boundary determinant.

A.1.2. An analytic family invisible to all leading data.

Theorem A.1.2 (Identical leading metrics and distinct nonlinear laws). *On the lattice $3\mathbb{Z} \times 4\mathbb{Z}$ there is a real-analytic family of strictly convex scatterers D_s , $|s| < s_*$, with D_0 the unit disk, for which*

$$g = 1, \quad A = 12 - \pi, \quad \kappa_0 = \kappa_1 = 1$$

at the two horizontal contacts. The horizontal orientations exhaust the ground-onset event. All leading count amplitudes and all leading endpoint covariance matrices, at every flight number, are independent of s . Nevertheless their nonlinear boundary coefficient satisfies

$$\left. \frac{d}{ds} \mathcal{R}_\infty^{(s)}(0) \right|_{s=0} = \frac{\sqrt{3}}{2}. \quad (\text{A.1.5})$$

It therefore determines s locally with a bounded inverse derivative. For every sufficiently small fixed positive d , the limiting normalized probability $\mathcal{F}^{(s)}(d)$ also varies with s .

Proof. Consider the support function

$$h_{s,z}(\theta) = 1 + s \sin^4 \theta + z \sin^6 \theta. \quad (\text{A.1.6})$$

Its obstacle area is $\mathcal{A}(h) = \frac{1}{2} \int_0^{2\pi} (h^2 - (h')^2) d\theta$. At $(s, z) = (0, 0)$ its z derivative is $\int_0^{2\pi} \sin^6 \theta d\theta = 5\pi/8 > 0$. The analytic implicit-function theorem gives $z = z(s)$, $z(0) = 0$, with $\mathcal{A}(h_{s,z(s)}) = \pi$ exactly. Since the s derivative is $3\pi/4$, one also has $z'(0) = -6/5$.

At $\theta = 0, \pi$, both perturbations and their first two derivatives vanish. Hence $h = 1$, $h' = 0$ and the curvature radius $h + h'' = 1$. The support points remain $(1, 0)$ and $(-1, 0)$, their curvatures remain one, and the supporting lines of horizontal neighbors have separation one. These points attain that separation. At the disk the vertical gap is two; all other nonhorizontal gaps are larger. A small analytic parameter interval preserves this strict gap, disjointness, positive curvature, and clearance. Thus only the horizontal pair contributes to the ground onset. Reflection and central symmetry of (A.1.6) make the two facing graphs identical and even. Area and the two contact Hessians are fixed exactly, not merely to first order. Equations (5.2) and (A.16.1) give the assertion about every leading datum.

We compute the first higher graph jet. Let $r(\theta) = h(\theta) + h''(\theta)$ be the curvature radius. Near the right contact the boundary parametrization has derivatives $x'(\theta) = -r(\theta) \sin \theta$ and $y'(\theta) = r(\theta) \cos \theta$. Since $r'(0) = 0$, integrating these expansions and eliminating θ gives for the facing graph

$$\psi(y) = \frac{y^2}{2r(0)} + \frac{3r(0) - r''(0)}{24r(0)^4} y^4 + O(y^6).$$

In our family $r(0) = 1$ and $r''(0) = h''(0) + h^{(4)}(0) = 24s$; $\sin^6 \theta$ has zero fourth jet and does not affect this identity. Thus $q_s = 3 - 24s$. Here $\gamma = \text{arcosh } 2$ and $a = \sqrt{3}$, so $\cosh(2\gamma) = 7$, $\sinh(2\gamma) = 4\sqrt{3}$. Multiplying (A.1.2) by $q'_s = -24$ yields (A.1.5). The inverse-function theorem proves the local scalar reconstruction. Finally the joint analytic expansion $\mathcal{F}^{(s)}(d) = 1 + d\mathcal{R}_\infty^{(s)}(0) + O(d^2)$ has uniform differentiated remainder. Its s derivative at zero is $d\sqrt{3}/2 + O(d^2)$, nonzero for all sufficiently small positive d . $\square$

The family in Theorem A.16.3 varies the two contact curvatures while preserving only their scalar product. The family in Theorem A.1.2 instead keeps both contact curvatures fixed, so even the entire leading conditional metric is unchanged. It is separated by the nonlinear law, not by a relabelling of the leading inverse. Neither construction claims equality of finite-offset probabilities or of marked length spectra. Higher-jet information from other billiard observations has its own inverse theory; the assertion here is its explicit realization in a full-phase statistical threshold law with a collision-order-uniform limiting construction.

APPENDIX A.2. STATISTICAL DETERMINATION OF AN ANALYTIC CONTACT

The scalar fourth-jet calculation extends to a triangular inverse at every finite order. In this section the two facing graphs are identical and even. This hypothesis is additional to those of

the general threshold theorem; it is satisfied by the analytic obstacles constructed below. Write

$$\psi(y) = \frac{\kappa}{2}y^2 + \sum_{m \geq 2} \frac{q_{2m}}{(2m)!}y^{2m}, \quad a = \frac{\sinh \gamma}{g}, \quad \gamma = \operatorname{arcosh}(1 + g\kappa), \quad \lambda = e^{-\gamma}. \quad (\text{A.2.1})$$

For smooth graphs this notation denotes their Taylor coefficients, not an assertion that the series converges. Both limiting parities coincide. Put

$$f_k = \frac{1}{k!} \partial_d^k \mathcal{F}(0), \quad f_{j,k} = \frac{1}{k!} \partial_d^k F_j(0).$$

The functions F_j and $\mathcal{F}$ are the normalized physical probabilities of Theorem 7.3.

Theorem A.2.1 (Triangular contact-jet recovery). *Fix $g, \kappa > 0$. For every $m \geq 2$, the coefficient f_{m-1} depends only on $g, \kappa, q_4, \dots, q_{2m}$, is affine in its last argument, and has derivative*

$$\frac{\partial f_{m-1}}{\partial q_{2m}} = -K_m \{\coth(m\gamma) + 2mD_m\} < 0, \quad (\text{A.2.2})$$

where

$$K_m = \frac{4}{2^m(m+1)(m!)^2 a^m}, \quad D_m = \frac{1}{e^{2(m-1)\gamma} - 1} - \frac{1}{e^{2m\gamma} - 1} > 0. \quad (\text{A.2.3})$$

Consequently, for every fixed M , the map

$$(q_4, \dots, q_{2M}) \mapsto (f_1, \dots, f_{M-1})$$

has a triangular smooth inverse, locally Lipschitz on compact finite-jet sets. It is analytic in finite-jet coordinates. There is no separation assumption on the higher jets. If two real-analytic even facing graphs have the same g, κ and the same germ of $\mathcal{F}$, they coincide as graph germs.

For each fixed M , inserting the finite-bridge coefficients $(f_{j,1}, \dots, f_{j,M-1})$ in the limiting inverse recovers the actual jet vector with error at most $C_M \tau^j$, uniformly on a sufficiently small compact parameter family. The constants may depend on M and on higher boundary norms; no inverse bound uniform in M is asserted.

Proof. We first justify the finite-jet dependence. Assign degree one to each endpoint coordinate. The coefficient of degree n of a stationary segment is obtained by differentiating its equation n times at the zero segment. The highest segment derivative is multiplied by the same quadratic interior Hessian at every step. Its inverse is the Green operator in (7.1); all other terms involve already determined lower derivatives and derivatives of the local length. The weighted bounds in Lemma 7.1 justify this induction also on the half-line. The action and the logarithmic edge and determinant sums can be differentiated term by term, using their summable entrywise bounds. Thus their coefficients through any given degree depend on only finitely many graph derivatives. In particular, the normalized coefficient of d^{m-1} uses the action through degree $2m$ and the amplitude through degree $2m-2$. This last assertion follows also directly from the smooth Morse construction: after the change $(u, v) = \sqrt{d}(U, V)$, an action term of degree n carries degree $d^{n/2-1}$ and an amplitude term degree $d^{n/2}$. Odd terms vanish under simultaneous sign reversal. Taylor remainders in the common Morse coordinates justify this bookkeeping with differentiated bounds.

Keep the jets below $2m$ fixed and vary q_{2m} . In the local length, the first varying homogeneous term is

$$\frac{u^{2m} + v^{2m}}{(2m)!}.$$

It has no mixed derivative. The linear half-line segment is $x_i = \lambda^i u$. By the envelope identity for a stationary action, the varying homogeneous term of its stationary value is obtained by

substituting this linear segment. This can first be done on finite segments; the weighted derivative estimates pass its absolutely convergent sum to the half-line. If $s_{2m} = S^{(2m)}(0)$, then

$$\partial_{q_{2m}} s_{2m} = 1 + 2 \sum_{i \geq 1} \lambda^{2mi} = \coth(m\gamma). \quad (\text{A.2.4})$$

The boundary site is counted once, each interior site twice. Lower nonlinear corrections to x_i have degree at least three, and cannot alter the displayed degree- $2m$ term.

Write $\beta_{2m-2} = [u^{2m-2}]B(u)$, with ordinary rather than factorial coefficients. At degree $2m-2$ the logarithmic edge product has zero variation. The varying interior Hessian is diagonal at this degree, with entries

$$\frac{2\lambda^{(2m-2)i} u^{2m-2}}{(2m-2)!}, \quad i \geq 1.$$

The change of the stationary segment itself has degree at least $2m-1$. Since the Hessian perturbation for even graphs starts in degree two, that change cannot enter degree $2m-2$. For the same reason, only the linear term of the logarithmic determinant contributes at this degree; every product with an unvaried Hessian perturbation has at least two additional degrees. Using $G_{ii} = (1 - \lambda^{2i})/(2a)$ gives

$$\partial_{q_{2m}} \beta_{2m-2} = -\frac{1}{a(2m-2)!} \sum_{i \geq 1} (1 - \lambda^{2i}) \lambda^{(2m-2)i} = -\frac{D_m}{a(2m-2)!}. \quad (\text{A.2.5})$$

Passing from $\log B$ to B does not change this first varying coefficient. All products with a lower positive-degree term again lie above the degree under consideration.

It remains to compute how these two variations enter the probability. Set $k = m-1$. Under the quadratic residual-time density proportional to $(d - a(u^2 + v^2)/2)_+$, polar integration gives

$$\mathbb{E}(u^{2k}) = \frac{2 \binom{2k}{k}}{2^k (k+1)(k+2) a^k} d^k, \quad \int (d - a(u^2 + v^2)/2)_+ du dv = \frac{\pi d^2}{a}. \quad (\text{A.2.6})$$

Indeed, the angular average of $\cos^{2k} \theta$ is $\binom{2k}{k}/4^k$, while the ratio of the radial integrals is $2(2d/a)^k / ((k+1)(k+2))$. First variation with respect to the action gives the negative integral of $(u^{2m} + v^{2m})/(2m)!$ over the quadratic sublevel. There is no boundary contribution because the residual-time weight vanishes there. Dividing by $\pi d^2/a$, its coefficient of d^{m-1} is $-K_m$. The amplitude variation contributes

$$\frac{4 \binom{2m-2}{m-1}}{2^{m-1} m(m+1) a^{m-1}} \partial_{q_{2m}} \beta_{2m-2} = -2m K_m D_m.$$

Lower-order terms in the action and amplitude multiply either variation only in higher degrees, so the calculation at the quadratic part is valid for arbitrary fixed lower jets. Equations (A.2.4)–(A.2.5) prove (A.2.2). They also show its independence of all higher jets and of the value of q_{2m} itself. The finite-jet dependence then proves affinity in that argument.

Inductively write $f_{m-1} = d_m q_{2m} + P_m(g, \kappa, q_4, \dots, q_{2m-2})$, with d_m given by (A.2.2), and solve this identity for q_{2m} . Each d_m is nonzero; for a fixed finite set of orders it is bounded away from zero on compact positive gap–curvature boxes. The recursion and its derivatives give the asserted local Lipschitz bounds. The finite-jet operations are analytic: they consist of the local analytic length expansion, inversion of its positive quadratic operator, and absolutely convergent geometric sums. Equality of every recovered jet implies equality of analytic graph germs by convergence of their Taylor series.

Finally Theorem 7.3 gives $|f_{j,k} - f_k| \leq C_M \tau^j$ for $k \leq M-1$. Apply the finite-dimensional local Lipschitz inverse just constructed. For all sufficiently large j the argument is in its neighborhood. The recursion defines an inverse for arbitrary finite coefficient vectors at fixed g, κ ; the finitely

many remaining j are covered by enlarging the constant on the compact parameter set. This proves the final assertion. $\square$

Corollary A.2.2 (Analytic obstacle determination from a selected onset). *Within the identical-even contact class, the gap, the leading selected probabilities, and the germ of their normalized limiting law determine the analytic contact graph. If the participating obstacles have connected real-analytic closed boundaries, this determines each such boundary in its contact frame. It does not determine the lattice or other obstacles which do not participate in this channel.*

Proof. Let $L_j = \lim_{d \downarrow 0} d^{-2} \Pr(E_{j,e}(d))$. The ratio $L_1/(2L_2) = \cosh \gamma$ determines $\kappa = (\cosh \gamma - 1)/g$. Also $F_j(d) = \Pr(E_{j,e}(d))/(d^2 L_j)$, so the normalized datum does not require a supplied area. The theorem determines the contact germ. For the global assertion, parametrize two candidate boundaries by oriented arclength from the common contact. Their signed curvature functions are real analytic on the real line, by continuation around each closed regular curve. They agree on an interval, hence everywhere by the real-analytic identity theorem. Integrating the planar Frenet equations with their common initial frame makes the two boundary images identical. This is exact analytic continuation, not a stability estimate for continuation from noisy local observations. $\square$

A.2.1. Finite-order geometric realization.

Theorem A.2.3 (Independent statistical jet coordinates at fixed leading data). *For every fixed $M \geq 2$ there is an $(M - 1)$ -parameter real-analytic family near the unit disk on $3\mathbb{Z} \times 4\mathbb{Z}$ which preserves $g = 1$, $A = 12 - \pi$, both horizontal contact curvatures, and all collision-order leading count and endpoint-metric coefficients of the selected horizontal ground channel, with its reverse orientation, exactly. Its coordinates can be taken to be $(q_4, \dots, q_{2M})$, or equivalently $(f_1, \dots, f_{M-1})$. Both coordinate changes have bounded inverse derivatives on a sufficiently small compact parameter neighborhood.*

Proof. Use the support functions

$$h_{s,z}(\theta) = 1 + \sum_{m=2}^M s_m \sin^{2m} \theta + z \sin^{2M+2} \theta.$$

At the disk the derivative of the obstacle area with respect to z is $\int_0^{2\pi} \sin^{2M+2} \theta d\theta > 0$. The analytic implicit-function theorem supplies $z = z(s)$ keeping that area equal to π . All the perturbations vanish to order at least four at $0, \pi$. Thus the two horizontal support points and their curvature radii remain fixed, and so do the gap and free area. Positive curvature, clearance, and the strict gap to nonhorizontal translates persist on a small neighborhood, as in Theorem A.1.2.

The compensator vanishes to order $2M + 2$ and cannot affect graph jets through order $2M$. More generally q_{2m} depends only on $s_2, \dots, s_m$, and

$$\partial_{s_m} q_{2m} = -(2m)!. \quad (\text{A.2.7})$$

To see the sign and coefficient without fixing the normal parameter, write the right boundary as $x = x(y)$. At its supporting normal $\theta(y)$, stationarity of $(h(\theta) - y \sin \theta)/\cos \theta$ gives, at fixed y , $\partial_{s_m} x = \sin^{2m} \theta(y)/\cos \theta(y)$. The contact radius is one, so $\theta(y) = y + O(y^3)$. Since $\psi(y) = 1 - x(y)$, the first varying coefficient is $-y^{2m}$, proving (A.2.7). Lower graph coefficients cannot vary because the perturbation starts in that order. The compensator term has still higher order even after its implicit differentiation. The triangular Jacobian is invertible. Combine it with Theorem A.2.1. The constancy of this channel's full leading hierarchy follows from Proposition A.17.1 and the unchanged contact Hessians, including both parities. $\square$

A.2.2. The one-flight comparison. The preceding inverse information is nonlinear, but is not exclusive to long records. This distinction can be made at every order, not just at the fourth jet.

Proposition A.2.4 (One-flight triangular diagonal). *Let $\nu_1 = gc/(c^2 - 1)$, where $c = 1 + g\kappa$. For $m \geq 2$,*

$$\partial_{q_{2m}} f_{1,m-1} = -\frac{4\nu_1^m}{2^m(m+1)(m!)^2} < 0. \quad (\text{A.2.8})$$

Thus the normalized one-flight nonlinear germ also determines the analytic even contact graph at known g, κ . For the family of Theorem A.1.2,

$$\partial_s f_{1,1}^{(s)} \Big|_0 = \frac{8}{9}, \quad \partial_s f_1^{(s)} \Big|_0 = \frac{\sqrt{3}}{2}.$$

Proof. There are no interior variables or interior determinant at one flight. The inverse quadratic endpoint Hessian has both diagonal entries ν_1 . The first varying action term is $(u^{2m} + v^{2m})/(2m)!$ and its mixed derivative is zero. Hence the amplitude does not vary in degree $2m - 2$. Transform the quadratic ellipse to a disk. The even moment of either endpoint linear functional is the disk moment with its squared length ν_1 in place of $1/a$. The unweighted sublevel integral used in the proof of (A.2.2) gives exactly (A.2.8). The same finite-jet recursion proves germ determination. For $m = 2$, $g = 1, c = 2$, this derivative is $-1/27$; multiplication by $q'_s = -24$ gives $8/9$. The other identity is (A.1.5). $\square$

The uniform long-bridge theorem establishes a limiting physical law and relative two-boundary factorization, with uniform convergence of every fixed statistical jet. It is not an assertion that increasing flight number creates contact parameters absent from one-flight nonlinear data. The distinction from the entire leading hierarchy is genuine at arbitrarily high finite jet order, as the realized families show.

APPENDIX A.3. FINITE-FLIGHT CONTACT RECOVERY AND COEFFICIENT SENSITIVITY

We continue in the identical-even contact class of Section A.2. The half-line inverse has a finite-flight counterpart with constants uniform in the flight number at each fixed jet order. Write $Y = (u, v)^\top$ and

$$H_j = a \begin{pmatrix} \coth(j\gamma) & -\operatorname{csch}(j\gamma) \\ -\operatorname{csch}(j\gamma) & \coth(j\gamma) \end{pmatrix}, \quad L_i = \left(\frac{\sinh((j-i)\gamma)}{\sinh(j\gamma)}, \frac{\sinh(i\gamma)}{\sinh(j\gamma)} \right). \quad (\text{A.3.1})$$

For $0 \leq i \leq j$ put $\nu_i = L_i H_j^{-1} L_i^\top$. For interior sites set

$$G_{ii} = \frac{\sinh(i\gamma) \sinh((j-i)\gamma)}{a \sinh(j\gamma)}.$$

Let $w_0 = w_j = 1$ and $w_i = 2$ for $1 \leq i < j$, and set $\alpha_m = 2/[2^m(m+1)(m!)^2]$.

Theorem A.3.1 (Uniform finite-flight triangular inverse). *At each $j \geq 1$ and $m \geq 2$, the normalized coefficient $f_{j,m-1}$ depends only on $g, \kappa, q_4, \dots, q_{2m}$, is affine in its last argument, and satisfies*

$$-\partial_{q_{2m}} f_{j,m-1} = \alpha_m \left\{ \sum_{i=0}^j w_i \nu_i^m + 4m \sum_{i=1}^{j-1} G_{ii} \nu_i^{m-1} \right\} \geq \frac{2\alpha_m}{a^m} > 0. \quad (\text{A.3.2})$$

Fix M and a compact finite-jet set with g, κ in a compact positive box. The triangular maps $J_{j,M} : q \mapsto (f_{j,1}, \dots, f_{j,M-1})$ have locally Lipschitz inverses on neighborhoods of their images whose radii and Lipschitz constants can be chosen independently of j . In particular, for coefficient errors at most δ in such a neighborhood,

$$\|J_{j,M}^{-1}(\hat{f}) - q\| \leq C_M \delta. \quad (\text{A.3.3})$$

Using the limiting inverse instead gives $C_M(\delta + \tau^j)$. Every fixed finite-flight nonlinear germ also determines the analytic even contact at known g, κ . No constant uniform in M is claimed.

Proof. The finite-degree argument in the proof of Theorem A.2.1 applies to every finite bridge. To calculate its diagonal, vary only q_{2m} while fixing lower jets. The first varying term in one flight is $(u^{2m} + v^{2m})/(2m)!$. The linear stationary segment is $x_i = L_i Y$, so the envelope identity gives

$$\partial_{q_{2m}} W_j^{[2m]}(Y) = \frac{1}{(2m)!} \sum_{i=0}^j w_i (L_i Y)^{2m}.$$

The interior sites occur in two lengths and the endpoints in one. The first varying local length term has zero mixed derivative. Therefore the edge product contributes nothing at degree $2m - 2$. The first varying diagonal of the interior Hessian is $2(L_i Y)^{2m-2}/(2m - 2)!$. Differentiating its relative determinant gives

$$\partial_{q_{2m}} b_j^{[2m-2]}(Y) = -\frac{2}{(2m - 2)!} \sum_{i=1}^{j-1} G_{ii} (L_i Y)^{2m-2}.$$

The change of the stationary segment starts at degree $2m - 1$. The unvaried Hessian perturbation starts at degree two, since the graphs are even. Thus that change, products of Hessian perturbations, and products with lower nonconstant amplitude terms cannot enter the displayed degree. The same observation justifies using the quadratic sublevel to compute the coefficient under consideration.

The quadratic residual integral equals $\pi d^2 / \sqrt{\det H_j}$. Changing variables from the ellipse to a disk, a linear functional $L_i Y$ has squared metric length ν_i . The unweighted sublevel moment from action variation contributes $-\alpha_m w_i \nu_i^m$ to the normalized coefficient. The residual-weighted moment from amplitude variation contributes $-4m \alpha_m G_{ii} \nu_i^{m-1}$. These constants follow from the angular average $\binom{2k}{k}/4^k$ and radial integration, exactly as in (A.2.6). The residual weight is zero on the sublevel boundary, so its first variation has no boundary term. Addition proves the identity in (A.3.2).

All its summands are positive. Moreover $\nu_0 = \nu_j = a^{-1} \coth(j\gamma) \geq a^{-1}$, proving the lower bound. This also proves that the coefficient is affine in the last jet: finite-degree dependence excludes higher jets, and the displayed derivative is independent of every higher and lower nonquadratic jet. The coefficient maps and their fixed derivatives are uniformly bounded on compact finite-jet sets by the differentiated threshold estimates. One may apply those estimates to the local even polynomial graphs with the specified finite jets on a sufficiently small common contact box; no unprescribed higher jet can enter these coefficients.

Solve successively the equations $f_{j,m-1} = d_{j,m} q_{2m} + P_{j,m}(q_4, \dots, q_{2m-2})$. The denominators $d_{j,m}$ are bounded away from zero uniformly in j for $m \leq M$. The derivatives of the numerators are uniformly bounded on an enlarged compact finite-jet box. Induction gives bounded inverse derivatives, and a common admissible coefficient perturbation radius: at each step choose that radius small enough to keep the recovered lower jets in the enlarged box. There are only $M - 1$ steps. This proves (A.3.3), without differentiating an infinite vector of jets. The limiting bound follows from Theorem A.2.1 and the triangle inequality. Equality of all recovered Taylor coefficients implies equality of analytic germs. It does not imply equality of merely smooth graphs with identical Taylor jets. $\square$

For $j = 1$ the interior sum is empty and $\nu_0 = \nu_1 = g \cosh \gamma / \sinh^2 \gamma$, so (A.3.2) reduces to (A.2.8). At each fixed m , the two separated endpoint layers give

$$\sum_i w_i \nu_i^m \longrightarrow 2a^{-m} \coth(m\gamma), \quad \sum_{i=1}^{j-1} G_{ii} \nu_i^{m-1} \longrightarrow a^{-m} D_m.$$

Indeed the left layer has $\nu_i \rightarrow a^{-1} e^{-2i\gamma}$ and $G_{ii} \rightarrow (1 - e^{-2i\gamma})/(2a)$, and the right layer has its reversed copy. The common geometric majorants supplied by endpoint localization justify both sums. This recovers (A.2.2).

Proposition A.3.2 (Limiting versus one-flight diagonal sensitivity). *Let $D_m^\infty = |\partial_{q_{2m}} f_{m-1}|$ and $D_m^1 = |\partial_{q_{2m}} f_{1,m-1}|$. Then*

$$\frac{D_m^\infty}{D_m^1} = (\tanh \gamma)^m \{\coth(m\gamma) + 2m D_m\} = (\tanh \gamma)^m \{1 + O_\gamma(m e^{-2(m-1)\gamma})\}. \quad (\text{A.3.4})$$

Proof. Divide (A.2.2) by (A.2.8) and use $a\nu_1 = \coth \gamma$. Expanding $(e^{2k\gamma} - 1)^{-1} = e^{-2k\gamma}/(1 - e^{-2k\gamma})$ gives the stated remainder at fixed $\gamma > 0$. $\square$

At fixed lower jets and equal absolute error in the corresponding coefficient, the inverse diagonal amplification is the reciprocal of (A.3.4). This comparison is unchanged by using ordinary rather than factorial graph coefficients. It does not compare full triangular condition numbers, the cost of extracting coefficients from noisy probability values, or minimax risks of different experiments. The fixed- M uniformity in Theorem A.3.1 is uniformity in j , not in the number of recovered jets. Exact analytic continuation in Corollary A.2.2 remains a uniqueness statement, not a noise-stable continuation procedure.

A.3.1. The channel index in geometric realization. The unchanged leading data in Theorem A.2.3 are all collision-order leading count and endpoint-metric coefficients of the selected horizontal ground channel, with its reverse orientation. The global free area is also fixed. Other selected channels need not have the same leading data. This distinction can be checked within its $M = 2$ realization. Write $h_s = 1 + s \sin^4 \theta + z(s) \sin^6 \theta$. The area constraint gives

$$z'(0) = -\frac{\int_0^{2\pi} \sin^4 \theta \, d\theta}{\int_0^{2\pi} \sin^6 \theta \, d\theta} = -\frac{6}{5}.$$

For the clear vertical channel on $3\mathbb{Z} \times 4\mathbb{Z}$, put $g_v = 4 - 2h_s(\pi/2)$ and $r_v = (h_s + h_s'')(\pi/2)$. Then

$$(g_v(0), g_v'(0)) = (2, 2/5), \quad (r_v(0), r_v'(0)) = (1, 3),$$

$$(c_v(0), c_v'(0)) = (3, -28/5), \quad c_v = 1 + g_v/r_v.$$

These equalities follow by differentiating the displayed trigonometric polynomials. Positive curvature and clearance persist by openness. Thus the oriented one-flight coefficient at its own onset satisfies

$$L_{1,v} = \frac{1}{2(12 - \pi)\sqrt{c_v^2 - 1}}, \quad \left. \frac{d}{ds} \log L_{1,v} \right|_{s=0} = \frac{21}{10}.$$

The horizontal realization is therefore a fibre of the complete leading hierarchy of a specified channel, not a fibre of all channel-resolved observations of the table. The general forward theorem and the full ground-onset count law retain their original scope.

APPENDIX A.4. RECONSTRUCTION OF BOUNDARY PROFILES FROM PHYSICAL COUNTS

The stability theorem of Section 10 takes a function and three of its derivatives as data. Here the observations are instead finitely many binary reports from independent physical preparations. We recover both full profiles on a closed energy collar, including its origin. No derivatives of empirical proportions are supplied as observations.

Fix $0 < D \leq d_*$ and an even flight number j . Write $e_0 = e$, $e_1 = \bar{e}$ and

$$p_{j,b}(d) = \Pr(E_{j,e_b}(d)), \quad a_j(d) = \frac{2A \sinh(j\gamma)}{d^2}, \quad F_{j,b}(d) = a_j(d)p_{j,b}(d). \quad (\text{A.4.1})$$

The gap g , free area A , multiplier parameter γ , and the selected facing patches are supplied. Thus the window $jg + d$ and its normalization are known before sampling. Each preparation reports only whether the selected event occurred; a failure is recorded and charged. This is a labelled binary experiment, not the unlabelled experiment of Section A.11.

A.4.1. Uniform classes and the statement. Let $m \geq 4$, $0 < \beta \leq 1$ and $B \geq 1$. Set

$$\|V\|_{m-1,1} = \sum_{r=0}^{m-1} \|V^{(r)}\|_{\infty} + \text{Lip}(V^{(m-1)}),$$

and let $\mathcal{K} = \mathcal{K}_m(B, \beta; D)$ be the class of real $C^{m-1,1}([0, D])$ functions with $V(0) = 1$, $V \geq \beta$ and $\|V\|_{m-1,1} \leq B$. The true profiles belong to $\mathcal{K}$. Assume supplied uniform constants C_0, M and $0 < \tau < 1$ such that, for both orientations and every even $j \geq 2$,

$$\|F_{j,b} - \mathcal{F}_{bb}\|_{C^3([0,D])} \leq C_0 \tau^j, \quad 0 \leq F_{j,b} \leq M. \quad (\text{A.4.2})$$

These are conclusions of Theorem 2.1 on each fixed compact smooth family, with B, β chosen for that family. The theorem below applies to any collection of physical experiments with these common bounds. It does not postulate that every member of $\mathcal{K}$ is realizable by a billiard. The larger class is used only for regularization. Constants in this section may depend on $m, D, B, \beta, C_0, M, A, \gamma, \tau$, not on the accuracy or confidence.

For $V, W \in \mathcal{K}$ define the explicit bilinear map

$$T(V, W)(d) = \frac{2}{\pi} \int_{s,t>0, s+t<1} \frac{1-s-t}{\sqrt{st}} V(ds) W(dt) \, ds \, dt, \quad T(V, W)(0) = 1. \quad (\text{A.4.3})$$

Thus $\mathcal{F}_{bc} = T(\mathcal{V}_b, \mathcal{V}_c)$.

Theorem A.4.1 (Finite-preparation recovery of full profiles). *Under (A.4.2), choose an integer $L \geq m$, $h = D/L \leq 1$, an even $j \geq 2$, and $0 < t, \delta \leq 1$. At each positive node $d_k = kh$, $1 \leq k \leq L$, and for each $b = 0, 1$, make*

$$N_{b,k} = \left\lceil 2a_j(d_k)(M + t/3)t^{-2} \log \frac{4L}{\eta} \right\rceil \quad (\text{A.4.4})$$

independent preparations, where $0 < \eta < 1/2$. There is a measurable finite-dictionary estimator $(\hat{V}_0, \hat{V}_1) \in \mathcal{K}^2$, defined below, for which with probability at least $1 - \eta$,

$$\max_{b=0,1} \|\hat{V}_b - \mathcal{V}_b\|_{\infty} \leq C(h^{m-3} + \tau^j + th^{-3} + \delta). \quad (\text{A.4.5})$$

Its prediction $T(\hat{V}_0, \hat{V}_1)$ of the odd limiting law has at most $2B$ times the error on the left. Its deterministic total preparation count satisfies

$$N_{\text{tot}} \leq 2L + CA \sinh(j\gamma) h^{-2} t^{-2} \log \frac{4L}{\eta}. \quad (\text{A.4.6})$$

Only positive-offset, even-flight observations are used. Dictionary cardinality and computational running time are not bounded by this preparation estimate.

Corollary A.4.2 (An explicit accuracy budget). *For sufficiently small $\varepsilon > 0$, the two profiles and the odd limiting law can be recovered in uniform norm to accuracy ε with confidence $1 - \eta$, using at most*

$$C\varepsilon^{-(2+\frac{8}{m-3}+\frac{\gamma}{|\log \tau|})} \log \frac{C}{\eta\varepsilon} \quad (\text{A.4.7})$$

independent preparations. The windows remain in the fixed collar, and the even flight number is $O(\log(1/\varepsilon))$. This is a sufficient cost in the stated labelled experiment; no optimality over designs or billiard observations is asserted.

A.4.2. Positive-node differentiation and regularization. For each cell $J_\ell = [\ell h, (\ell + 1)h]$, $0 \leq \ell < L$, put

$$s_\ell = \max\{1, \min\{\ell, L - m + 1\}\}.$$

Let $I_h z$ on J_ℓ be the polynomial of degree at most $m - 1$ interpolating z_k at the m nodes $d_{s_\ell}, \dots, d_{s_\ell+m-1}$. We also write $I_h f$ for interpolation of the sampled values of a function f . Polynomials on adjacent cells need not have matching derivatives. Their norm is

$$\|P\|_{3,h} = \max_{0 \leq \ell < L} \max_{0 \leq r \leq 3} \|P^{(r)}\|_{C^0(J_\ell)}.$$

The stencil on the first cell uses $h, \dots, mh$. There is neither a preparation at zero nor an assumed exact value of $F_{j,b}(0)$. The limited extrapolation to zero uses the same fixed stencil bounds.

Lemma A.4.3 (Three interpolation estimates). *With constants depending only on m, D , one has*

$$\|I_h f - f\|_{3,h} \leq Ch^{m-3} \|f\|_{m-1,1}, \quad f \in C^{m-1,1}([0, D]), \quad (\text{A.4.8})$$

$$\|I_h f\|_{3,h} \leq C \|f\|_{C^3([0, D])}, \quad f \in C^3([0, D]), \quad (\text{A.4.9})$$

$$\|I_h z\|_{3,h} \leq Ch^{-3} \max_k |z_k|. \quad (\text{A.4.10})$$

In particular a smooth C^3 model error is not amplified by h^{-3} , whereas an arbitrary scalar error can be.

Proof. Rescale a cell by $d = \ell h + hu$. Its interpolation nodes are distinct integers lying within distance m of $[0, 1]$. There are only finitely many possible relative stencils. Their Lagrange basis polynomials and derivatives through order three are therefore bounded uniformly on $[0, 1]$. Scaling back gives $\|(I_h z)^{(r)}\|_{C^0(J_\ell)} \leq Ch^{-r} \max_k |z_k|$; since $h \leq 1$, this proves (A.4.10).

For the first estimate, subtract the degree- $(m - 1)$ Taylor polynomial of f at ℓh . The Lipschitz bound on its last derivative gives remainder $C\|f\|_{m-1,1} h^{m-r}$ after $r \leq 3$ derivatives on the cell, and remainder $C\|f\|_{m-1,1} h^m$ at every stencil node. The interpolant reproduces the Taylor polynomial. Applying the scaled basis bound to the nodal remainder and adding the differentiated function remainder proves (A.4.8).

For (A.4.9), instead subtract the quadratic Taylor polynomial. Its interpolant is itself, since $m \geq 4$; its derivatives through order three are bounded by $C\|f\|_{C^3}$. Its nodal remainder is bounded by $Ch^3\|f\|_{C^3}$, including in the first and last cells. Scaling the remainder gives $Ch^{3-r}\|f\|_{C^3}$ for $r \leq 3$, proving the assertion. $\square$

Lemma A.4.4 (Compact regularization class). *The class $\mathcal{K}$ is compact in $C^{m-1}([0, D])$. The functions $T(V, V)$, $V \in \mathcal{K}$, have uniformly bounded $C^{m-1,1}$ norms. For each h and $\delta > 0$, there is a finite ordered set $\mathcal{D}_{h,\delta} \subset \mathcal{K}$ such that*

$$\inf_{W \in \mathcal{D}_{h,\delta}} \|I_h T(V, V) - I_h T(W, W)\|_{3,h} \leq \delta \quad (V \in \mathcal{K}). \quad (\text{A.4.11})$$

Moreover for $V, W \in \mathcal{K}$,

$$\|V - W\|_\infty \leq C \|T(V, V) - T(W, W)\|_{C^3([0, D])}. \quad (\text{A.4.12})$$

Proof. The derivatives through order $m-1$ are uniformly bounded and equicontinuous. Successive use of the fundamental theorem of calculus identifies the limits of these derivatives under an Arzelà–Ascoli subsequence. The norm bound, positivity, normalization and Lipschitz bound pass to the limit. This proves compactness.

The weight in (A.4.3) is integrable and has total mass one. Differentiating under the integral through order $m-1$ uses bounded products of derivatives of V , with factors $s, t \leq 1$. The same product estimate bounds the Lipschitz seminorm of the last derivative. It gives a uniform $C^{m-1,1}$ bound. At a fixed grid, $V \mapsto I_h T(V, V)$ is continuous even for the uniform topology: the forward nodal values are continuous, and interpolation is a linear map between finite-dimensional spaces. Its image is compact, so it admits a finite δ -net with representatives in $\mathcal{K}$. Fix any order of those representatives in advance.

Finally the proof of Theorem 10.2 uses only normalization, the first-derivative bound and the three derivatives of the forward law. It applies to the present $C^{m-1,1}$ functions without a smoothness extension. More explicitly, put $f = x^{-1/2}(V - W)$ and $a = x^{-1/2}(V + W)$. Their Abel convolution has the form $x^{-1/2} * a = 2\pi + R$, with $R(0) = 0$ and $\|R'\|_\infty \leq \pi B$. If $Q = (\pi/2)(d^2[T(V, V) - T(W, W)])''$, then $2\pi f + R' * f = x^{-1/2} * Q'$. The normalization makes f continuous at zero. The Abel bound and Gronwall give $\|V - W\|_\infty \leq (D/\pi)e^{BD/2}\|Q'\|_\infty$. The identity

$$Q' = \frac{\pi}{2} (6\Delta F' + 6d\Delta F'' + d^2\Delta F'''), \quad \Delta F = T(V, V) - T(W, W),$$

proves (A.4.12) in the stated regularity class. $\square$

Given nodal observations $z = (z_k)_{k=1}^L$, define

$$\widehat{V}(z) = \text{the first minimizer over } W \in \mathcal{D}_{h,\delta} \text{ of } \|I_h T(W, W) - I_h z\|_{3,h}. \quad (\text{A.4.13})$$

Each criterion is continuous in the finite data vector. The prescribed tie rule makes the estimator Borel measurable. Neither the dictionary nor its order depends on the unknown profile or the observed counts. This construction specifies a statistical estimator; it does not claim a tractable enumeration of the dictionary.

A.4.3. Proof of the acquisition theorem.

Proof of Theorem A.4.1. For orientation b and node k , let $\overline{X}_{b,k}$ be the empirical proportion of successes and put $Z_{b,k} = a_j(d_k)\overline{X}_{b,k}$. Its expectation is exactly $F_{j,b}(d_k)$, and its variance is

$$\text{Var}(Z_{b,k}) = \frac{a_j(d_k)F_{j,b}(d_k)(1 - p_{j,b}(d_k))}{N_{b,k}}. \quad (\text{A.4.14})$$

For a Bernoulli variable X of mean p , $\log \mathbb{E}e^{s(X-p)} \leq p(e^s - 1 - s)$. The exponential series gives $e^s - 1 - s \leq s^2/[2(1 - s/3)]$ for $0 < s < 3$; for the negative exponent use $e^{-s} - 1 + s \leq s^2/2$. Independence, exponential Markov bounds, and the parameter $s = t/(M + t/3)$ for $X - p$ give

$$\Pr\{|Z_{b,k} - F_{j,b}(d_k)| > t\} \leq 2 \exp \left\{ -\frac{N_{b,k}t^2}{2a_j(d_k)(M + t/3)} \right\}. \quad (\text{A.4.15})$$

The allocation (A.4.4) and a union bound over $2L$ observations yield an event of probability at least $1 - \eta$ on which all scalar errors are at most t .

Work on this event and fix b . Abbreviate $\mathcal{F} = \mathcal{F}_{bb}$ and $\hat{F} = T(\hat{V}_b, \hat{V}_b)$, where $\hat{V}_b = \hat{V}((Z_{b,k})_k)$. The last two bounds of Lemma A.4.3 give

$$\|I_h Z_b - I_h \mathcal{F}\|_{3,h} \leq C(th^{-3} + C_0 \tau^j). \quad (\text{A.4.16})$$

In particular the deterministic relative error in (A.4.2) has not been multiplied by h^{-3} . Choose a dictionary representative for the true profile using (A.4.11). The minimizing property and the triangle inequality imply

$$\|I_h \hat{F} - I_h \mathcal{F}\|_{3,h} \leq \delta + 2\|I_h Z_b - I_h \mathcal{F}\|_{3,h}.$$

The first interpolation bound, applied separately to $\hat{F}$ and $\mathcal{F}$, now gives

$$\|\hat{F} - \mathcal{F}\|_{C^3([0,D])} \leq C(h^{m-3} + th^{-3} + \tau^j + \delta).$$

Both functions are globally C^3 ; the broken norm of their difference therefore bounds each of its derivatives on the whole closed collar. Apply (A.4.12) to prove (A.4.5). The mass-one positive weight in (A.4.3) gives

$$\|T(V_0, V_1) - T(W_0, W_1)\|_\infty \leq B(\|V_0 - W_0\|_\infty + \|V_1 - W_1\|_\infty),$$

which proves the odd-law prediction bound.

To count preparations, the ceilings contribute at most $2L$. Since $\sum_{k=1}^L d_k^{-2} \leq h^{-2} \sum_{k \geq 1} k^{-2} \leq 2h^{-2}$, summing (A.4.4) gives (A.4.6). All preparations, including failures, have been counted. This completes the proof of Theorem A.4.1. $\square$

Proof of Corollary A.4.2. Choose $h = D/L$ comparable to a sufficiently small constant times $\varepsilon^{1/(m-3)}$, with $L \geq m$, and put $t = c\varepsilon h^3$, $\delta = c\varepsilon$. Choose the least even $j \geq 2$ with $\tau^j \leq c\varepsilon$. Writing $\lambda = |\log \tau|$, rounding changes j by less than two once the lower bound $j \geq 2$ is inactive. Consequently

$$e^{j\gamma} \leq C\varepsilon^{-\gamma/\lambda}, \quad h^{-2}t^{-2} = c^{-2}\varepsilon^{-2}h^{-8}.$$

A small enough fixed c , and the corresponding constant in the mesh, make both profile errors and the odd-law prediction error at most ε . The logarithm of $4L/\eta$ is bounded by $C \log(C/(\eta\varepsilon))$. Substitute these estimates into (A.4.6); its $2L$ term is of smaller order. This proves (A.4.7) with a deterministic design. $\square$

A.4.4. What the budget measures. The three accuracy-dependent contributions to the power in (A.4.7) have distinct origins. Scalar accuracy costs ε^{-2} . Resolving three derivatives while sampling near a quadratic onset costs h^{-8} . Suppressing finite-bridge bias costs $e^{j\gamma}$, with j chosen from the relative decay τ^j . No equality between τ and $e^{-\gamma}$ is assumed; a slower certified relative rate gives a larger sufficient budget.

The same experiment was implicit in the direct-variance calculation of the referee memorandum [20]. Formula (A.4.14) retains that cost rather than treating the limiting law as a freely observed function. Regularization is needed because the inverse differentiates the data. Its finite smoothness bound is an additional quantitative assumption for acquisition, not a new restriction on the general smooth forward theorem. The recovered objects remain the weighted, symmetrized energy profiles, not arbitrary asymmetric boundary graphs. The two oriented even sequences suffice for their reconstruction and for prediction of the odd law. Separate odd records provide an independent compatibility test.

APPENDIX A.5. ACQUISITION FROM INTEGRATED PHYSICAL FLUX

The preceding construction controls three integer derivatives after normalizing every observation by d^2 . We now retain the integrated flux and use the linear Abel discrepancy of Section 34. This changes both the approximation loss and the allocation of preparations. The observation is still exactly the selected-event bit; no endpoint coordinate or derivative is added.

Keep $\mathcal{K}_m(B, \beta; D)$, the supplied g, A, γ , the labels and (A.4.2). Write

$$c_j = 2A \sinh(j\gamma), \quad H_{j,b}(d) = c_j p_{j,b}(d) = d^2 F_{j,b}(d), \quad H_b = H_{\mathcal{V}_b}. \quad (\text{A.5.1})$$

The relative law gives $\|H_{j,b} - H_b\|_{C^3} \leq C\tau^j$. Multiplication by d^2 has not altered that estimate. In contrast, the variance of a scaled empirical integrated flux decreases quadratically at the onset.

Theorem A.5.1 (Abel-discrepancy acquisition). *Let $L \geq m$, $h = D/L \leq 1$, $j \geq 2$ be even, and $0 < t, \delta \leq 1$. At every positive node $d_k = kh$ and in each orientation $b = 0, 1$, make*

$$N_{b,k} = \left\lceil 2c_j(Md_k^2 + t/3)t^{-2} \log \frac{4L}{\eta} \right\rceil \quad (\text{A.5.2})$$

independent preparations. There is a Borel finite-dictionary estimator $(\widehat{V}_0, \widehat{V}_1) \in \mathcal{K}^2$ for which, with probability at least $1 - \eta$,

$$\max_b \|\widehat{V}_b - \mathcal{V}_b\|_\infty \leq C\{h^{m-5/2} + \tau^j + th^{-5/2} + \delta\}. \quad (\text{A.5.3})$$

The prediction of the odd limiting law has at most $2B$ times this error. Every preparation, including every failure, is included in the deterministic bound

$$N_{\text{tot}} \leq 2L + CA \sinh(j\gamma)h^{-1}t^{-2} \log \frac{4L}{\eta}. \quad (\text{A.5.4})$$

No value or derivative at zero is observed. Computational complexity of the finite dictionary is not included in this preparation bound.

Corollary A.5.2 (A refined preparation bound). *For sufficiently small $\varepsilon > 0$, both complete profiles and their predicted odd law are recovered to uniform accuracy ε , with confidence $1 - \eta$, using at most*

$$C\varepsilon^{-(2 + \frac{6}{m-5/2} + \frac{\gamma}{|\log \tau|})} \log \frac{C}{\eta\varepsilon} \quad (\text{A.5.5})$$

preparations. The flight number is even and of order $\log(1/\varepsilon)$. This is a sufficient bound for this design, not a minimax assertion. The factor involving τ depends on the supplied relative convergence certificate; it is not asserted to be an invariant of the billiard.

A.5.1. A globally smooth positive-node reconstruction. The Abel discrepancy is applied to globally smooth functions, not to distributional derivatives across interpolation cells. We give a specific reconstruction with this property. Let

$$\omega(u) = \begin{cases} \exp\{-1/(1-u^2)\}, & |u| < 1, \\ 0, & |u| \geq 1, \end{cases} \quad \theta_\ell(x) = \frac{\omega(x/h - \ell)}{\sum_{r=0}^L \omega(x/h - r)}, \quad 0 \leq \ell \leq L.$$

The denominator is positive on $[0, D]$. For each ℓ , let $P_\ell z$ interpolate the m positive nodal values starting at

$$s_\ell = \max\{1, \min\{\ell - \lfloor (m-1)/2 \rfloor, L - m + 1\}\}.$$

Thus the nodes are $s_\ell h, \dots, (s_\ell + m - 1)h$. Define

$$J_h z = \sum_{\ell=0}^L \theta_\ell P_\ell z. \quad (\text{A.5.6})$$

Every polynomial of degree at most $m - 1$ is reproduced. At a positive grid point, only its central weight is nonzero and its polynomial contains that node, so the nodal values are interpolated. At zero, the first stencil uses $h, \dots, mh$; it extrapolates rather than inserting a known datum. The output is C^∞ on the closed interval in the usual extension sense.

Lemma A.5.3 (Smooth reconstruction in the Abel seminorm). *With constants depending only on m, D , one has*

$$|J_h f - f|_{\mathcal{A}} \leq Ch^{m-5/2} \|f\|_{m-1,1}, \quad f \in C^{m-1,1}, \quad (\text{A.5.7})$$

$$|J_h f|_{\mathcal{A}} \leq C \|f\|_{C^3}, \quad f \in C^3, \quad (\text{A.5.8})$$

$$|J_h z|_{\mathcal{A}} \leq Ch^{-5/2} \max_k |z_k|. \quad (\text{A.5.9})$$

In particular, a smooth bridge error and an arbitrary nodal error have different amplification factors.

Proof. At each x at most two weights are positive. A nearest integer to x/h is at distance at most $1/2$, so the denominator of the weights has a fixed positive lower bound. Its derivatives of order r are bounded by $C_r h^{-r}$, as are those of θ_ℓ . Each active stencil lies within distance $(m+1)h$ of x . After rescaling, there are only finitely many Lagrange basis configurations; their derivatives of every order at most $m-1$ on the relevant compact intervals are uniformly bounded. The product rule therefore gives

$$\|J_h z\|_{C^r} \leq Ch^{-r} \max_k |z_k|, \quad 0 \leq r \leq 3.$$

For $f \in C^{m-1,1}$, subtract its degree- $(m-1)$ Taylor polynomial at the point at which a derivative is evaluated. The interpolator reproduces that polynomial; its nodal remainders are bounded by $Ch^m \|f\|_{m-1,1}$. The differentiated remainder estimate and the product rule give

$$\|J_h f - f\|_{C^r} \leq Ch^{m-r} \|f\|_{m-1,1}, \quad r = 2, 3. \quad (\text{A.5.10})$$

This argument is valid at both endpoints because all active stencils remain a bounded rescaled distance away. Subtracting instead the quadratic Taylor polynomial for $f \in C^3$ gives $\|J_h f\|_{C^3} \leq C \|f\|_{C^3}$.

Apply (34.3). In (A.5.10) the product of the second- and third-derivative errors is of order h^{2m-5} ; its square root is $h^{m-5/2}$. The endpoint second-derivative error is $O(h^{m-2})$, a smaller bound when $h \leq 1$. For nodal noise the two derivative scales are h^{-2} and h^{-3} , whose geometric mean is $h^{-5/2}$. Finally the C^3 stability just proved bounds the smooth-bias term without any negative power of h . $\square$

A.5.2. The estimator and its full preparation charge. The maps $V \mapsto H_V$ take $\mathcal{K}$ into a bounded $C^{m-1,1}$ set. This follows from Lemma A.4.4 and multiplication by d^2 . For fixed h , the map $V \mapsto \mathcal{A} J_h H_V$ has compact image in C^0 : it is a continuous map of the compact class $\mathcal{K}$ through a finite-dimensional vector of nodal values. Choose, independently of the unknown profile, a finite ordered set $\mathcal{D}_{h,\delta}^{\mathcal{A}} \subset \mathcal{K}$ such that

$$\inf_{W \in \mathcal{D}_{h,\delta}^{\mathcal{A}}} |J_h H_V - J_h H_W|_{\mathcal{A}} \leq \delta \quad (V \in \mathcal{K}). \quad (\text{A.5.11})$$

For a data vector z , let $\widehat{V}(z)$ be the first minimizer of $|J_h H_W - J_h z|_{\mathcal{A}}$ over this ordered set. Each criterion is continuous in z ; the fixed tie rule makes the estimator Borel. The affine nullspace of $\mathcal{A}$ causes no problem for this finite minimization or for the norm on actual profile differences.

Proof of Theorem A.5.1. Let $Z_{b,k} = c_j \bar{Y}_{b,k}$, with $\bar{Y}$ the empirical success proportion. Its mean is $H_{j,b}(d_k)$, and

$$\text{Var}(Z_{b,k}) = \frac{c_j H_{j,b}(d_k)(1 - p_{j,b}(d_k))}{N_{b,k}} \leq \frac{c_j M d_k^2}{N_{b,k}}.$$

The centered summands are bounded by c_j . The elementary Bernoulli exponential-moment bound used in Section A.4 therefore gives

$$\Pr\{|Z_{b,k} - H_{j,b}(d_k)| > t\} \leq 2 \exp \left\{ -\frac{N_{b,k} t^2}{2c_j(M d_k^2 + t/3)} \right\}.$$

The allocation and a union bound over $2L$ means give simultaneous nodal error at most t with probability $1 - \eta$.

On this event, Lemma A.5.3 and the relative C^3 bound give

$$|J_h Z_b - J_h H_b|_{\mathcal{A}} \leq C(th^{-5/2} + \tau^j).$$

The minimizing property and (A.5.11) bound $|J_h H_{\hat{V}_b} - J_h H_b|_{\mathcal{A}}$ by twice this quantity plus δ . The interpolation error for the two candidate fluxes then gives

$$|H_{\hat{V}_b} - H_b|_{\mathcal{A}} \leq C(h^{m-5/2} + th^{-5/2} + \tau^j + \delta).$$

Here the triangle inequality is that of the linear Abel seminorm. The inverse estimate (34.2) proves (A.5.3). The positive mass-one bilinear kernel of T gives the stated odd-law error.

Finally $\sum_{k=1}^L d_k^2 \leq D^2 L$ and $L = D/h$. Summing (A.5.2), including each ceiling, yields (A.5.4). This sums the allocations themselves, not the number of successes or a waiting time conditional on success. $\square$

Proof of Corollary A.5.2. Put $\nu = m - 5/2 > 0$. Choose L so that h is comparable to a sufficiently small multiple of $\varepsilon^{1/\nu}$, take $t = c\varepsilon h^{5/2}$ and $\delta = c\varepsilon$, and choose the least even $j \geq 2$ with $\tau^j \leq c\varepsilon$. All constants can be chosen to control also the odd-law error. Rounding j changes it by less than two once the lower endpoint is inactive, so $\sinh(j\gamma) \leq C\varepsilon^{-\gamma/|\log \tau|}$. Moreover

$$h^{-1}t^{-2} = c^{-2}\varepsilon^{-2}h^{-6}.$$

The logarithmic factor is bounded by $C \log\{C/(\eta\varepsilon)\}$, and the ceiling contribution $2L$ is smaller than the displayed bound. This proves the claim. $\square$

There are two separate improvements over Corollary A.4.2: the approximation/noise loss is $5/2$ rather than three, and the sum of integrated-flux variances has order h^{-1} rather than the h^{-2} sum produced by dividing first by d_k^2 . Neither improvement follows by merely substituting a mixed-norm inequality in the earlier discrepancy criterion. In particular, when $m = 4$ the accuracy exponent becomes $6 + \gamma/|\log \tau|$ rather than $10 + \gamma/|\log \tau|$. Both remain sufficient exponents with a supplied convergence certificate.

APPENDIX A.6. CALIBRATION WITHIN THE SMOOTH PHYSICAL EXPERIMENT

The preceding two acquisition theorems supply the gap, area and multiplier. These quantities cannot be imported from an exact-family inverse when the unknown experiment has arbitrary smooth remainders. We instead construct a calibration procedure inside the same selected physical experiment. Patch labels, a coarse parameter box and common smoothness bounds remain supplied. The target continues to be the symmetrized energy invariant, not an arbitrary asymmetric contact graph.

A.6.1. Why the integrated flux is stable under calibration. Let $D < D_+$ be fixed collars. In this subsection assume the relative C^3 estimates on $[0, D_+]$ and a supplied uniform $C^{3,1}$ bound on $H_{j,b}$ there. These follow on a fixed compact smooth physical family from the relative theorem with four derivatives. Suppose a pilot experiment supplies positive values $\hat{A}, \hat{\gamma}$ and an upper gap estimate $\hat{g}$. For a programmed window $j\hat{g} + d$, put

$$\Delta = j(\hat{g} - g), \quad R = \frac{\hat{A} \sinh(j\hat{\gamma})}{\hat{A} \sinh(j\gamma)}.$$

The mean of the scaled bit $2\hat{A} \sinh(j\hat{\gamma})Y$ is exactly

$$RH_{j,b}(d + \Delta). \quad (\text{A.6.1})$$

This is a translation of an integrated flux followed by a constant rescaling. It has no factor d^{-2} . A small calibration error can therefore be treated as a smooth model error in the Abel discrepancy.

Theorem A.6.1 (Stability of the plug-in experiment). *Assume $A, \hat{A}$ and $\gamma, \hat{\gamma}$ lie in fixed positive compact intervals, with the multiplier interval bounded away from zero. On a pilot event suppose*

$$0 \leq \hat{g} - g \leq \rho, \quad |\hat{A}/A - 1| \leq \alpha, \quad |\hat{\gamma} - \gamma| \leq \zeta, \quad j\zeta \leq 1, \quad j\rho \leq D_+ - D.$$

There is a constant M_H , determined by the supplied bounds, which bounds the means in (A.6.1) on this event. Conditionally on any pilot outcome in the event, take fresh independent bits at the positive grid windows in both orientations. Set $\hat{c}_j = 2\hat{A} \sinh(j\hat{\gamma})$ and allocate

$$N_{b,k} = \left\lceil 2\hat{c}_j(M_H + t/3)t^{-2} \log \frac{4L}{\eta_1} \right\rceil. \quad (\text{A.6.2})$$

The Abel-dictionary estimator, applied to $Z_{b,k} = \hat{c}_j \bar{Y}_{b,k}$, then satisfies, with conditional probability at least $1 - \eta_1$,

$$\max_b \|\hat{V}_b - \mathcal{V}_b\|_\infty \leq C\{h^{m-5/2} + \tau^j + th^{-5/2} + \delta + \alpha + j\zeta + j\rho\}. \quad (\text{A.6.3})$$

The allocation has the deterministic bound $2L + C\hat{c}_j h^{-1} t^{-2} \log(4L/\eta_1)$. Thus calibration errors in this procedure are not multiplied by an inverse power of the grid spacing. If the pilot event has probability at least $1 - \eta_0$, the unconditional success probability is at least $1 - \eta_0 - \eta_1$.

Proof. The true offset of the programmed window is $d + \Delta$. Applying (A.5.1) at this offset proves (A.6.1). On the fixed multiplier interval,

$$\left| \log \frac{\sinh(j\hat{\gamma})}{\sinh(j\gamma)} \right| \leq j \coth(j\gamma_-) \zeta \leq Cj\zeta.$$

The compact area bounds and $j\zeta \leq 1$ then give $|R - 1| \leq C(\alpha + j\zeta)$ and a fixed upper bound for R . The $C^{3,1}$ bound and the fundamental theorem of calculus through order two, and the Lipschitz bound at order three, give

$$\|H_{j,b}(\cdot + \Delta) - H_{j,b}\|_{C^3([0,D])} \leq C\Delta.$$

The upper gap estimate ensures that only nonnegative offsets in the supplied collar occur, even when the translated function is evaluated at zero in this analysis. No observation there is made. Together with the relative law this proves

$$\|RH_{j,b}(\cdot + \Delta) - H_b\|_{C^3([0,D])} \leq C(\tau^j + \alpha + j\zeta + j\rho). \quad (\text{A.6.4})$$

It also supplies M_H . Conditionally on the pilot, the variance of a scaled mean is at most $\hat{c}_j M_H / N_{b,k}$ and a summand is bounded by $\hat{c}_j$. The same Bernoulli bound and union argument as before therefore give nodal error at most t with conditional probability $1 - \eta_1$.

Apply (A.5.8) to (A.6.4), and (A.5.9) only to the centered scalar observations. The dictionary and inverse arguments of Theorem A.5.1 give (A.6.3). Summing the ceilings gives the budget; conditioning and a union bound give the final probability assertion. $\square$

For comparison, normalizing first by d^2 would give the mean $R(1 + \Delta/d)^2 F_{j,b}(d + \Delta)$, the sensitivity formula in [21]. The two calculations concern different processing of the same bits and are consistent. The present proof avoids the singular factor before regularization. In the quadratic benchmark $H(d) = d^2$, translation adds only $2\Delta d + \Delta^2$, which $\mathcal{A}$ annihilates exactly. For a general smooth physical flux, (A.6.4) provides the required bound instead of asserting such an exact cancellation.

A.6.2. A charged calibration procedure with smooth remainders. We now specify sufficient uniform hypotheses for obtaining the pilot. Let the unknown gap lie in a supplied interval $[g_-, g_+]$ of width $w_0 > 0$, with $g_- > 0$ and $w_0 \leq d_c$. The area and multiplier lie in supplied boxes

$$0 < A_- \leq A \leq A_+ < \infty, \quad 0 < \gamma_- \leq \gamma \leq \gamma_+ < \infty.$$

For one selected orientation let $P_r(t)$ denote the success probability of the r -flight event in the physical window t . For $r = 1, 2$ assume

$$\begin{aligned} P_r(t) &= 0 \quad (t \leq rg), \\ P_r(rg + d) &= c_r d^2 F_r(d), \quad c_r = \{2A \sinh(r\gamma)\}^{-1}, \quad 0 \leq d \leq d_c, \\ F_r(0) &= 1, \quad F_r \geq \beta_c > 0, \quad \|F_r\|_{m-1,1} \leq B_c. \end{aligned} \tag{A.6.5}$$

All constants in these conditions are supplied, not the functions F_r . The exact physical leading coefficient and smooth finite-flight law give these hypotheses after choosing a common local collar on a compact smooth family. No equality of the two remainders and no analytic finite-dimensional parameterization are assumed.

Lemma A.6.2 (Finite-preparation calibration). *Under (A.6.5), for sufficiently small $\xi > 0$ and $0 < \eta_0 < 1/2$ there is a measurable procedure using only selected-event bits at flight numbers one and two which outputs $\hat{g} \in [g_-, g_+]$, $\hat{A} \in [A_-, A_+]$ and $\hat{\gamma} \in [\gamma_-, \gamma_+]$. With probability at least $1 - \eta_0$,*

$$0 \leq \hat{g} - g \leq C\xi^{1+1/m}, \quad |\hat{A}/A - 1| + |\hat{\gamma} - \gamma| \leq C\xi. \tag{A.6.6}$$

Its total number of preparations, including failures, is deterministically at most

$$C\xi^{-(2+2/m)} \log \frac{C \log(C/\xi)}{\eta_0}. \tag{A.6.7}$$

The constants are uniform over the specified smooth physical class.

Proof. We first obtain a one-sided gap bracket. The bounds imply a supplied $c_* > 0$ such that $P_1(g + d) \geq c_* d^2$ on $[0, d_c]$. Fix a target bracket width $\rho < w_0$. Starting from $[l, u]$ of width w , sample at $q = l + 3w/4$. Use

$$N(w) = \left\lceil \frac{16}{c_* w^2} \log \frac{2I}{\eta_0} \right\rceil, \quad I = 1 + \left\lceil \frac{\log(w_0/\rho)}{\log(4/3)} \right\rceil.$$

If any success occurs, replace the interval by $[l, q]$; otherwise replace it by $[l + w/2, u]$. A success certifies $g < q$, because a success at or below the onset has probability zero. On a correct incoming bracket a false lower update can occur only if $g \leq l + w/2$. In that case $q - g \geq w/4$ and $q - g \leq w \leq d_c$, so its probability is at most $\exp\{-N(w)c_* w^2/16\} \leq \eta_0/(2I)$. The bracket width decreases by a factor either $3/4$ or $1/2$ at each step, irrespective of the observations. Stop at width at most ρ . There are at most I steps. A union bound over correct incoming histories gives a correct final bracket with probability $1 - \eta_0/2$. Its upper endpoint $\hat{g}$ then satisfies $0 \leq \hat{g} - g \leq \rho$.

The total of w^{-2} over the steps is at most $\rho^{-2} \sum_{r \geq 0} (3/4)^{2r}$: read the widths backward from the last pre-stopping width, which is larger than ρ . The ceilings add at most I . Consequently this stage costs at most $C\rho^{-2} \log\{C \log(C/\rho)/\eta_0\}$ preparations. This bound is deterministic even on the exceptional incorrect-bracket outcomes.

Choose $s = c_s \xi^{1/m}$ and $\rho = c_\rho \xi s$ with fixed positive constants small enough that $ms + 2\rho \leq d_c$ and $\rho \leq s/4$. At fresh preparations, for $r = 1, 2$ and $1 \leq k \leq m$, sample the window $r\hat{g} + ks$. Let $\bar{Y}_{r,k}$ be the corresponding proportion and put

$$\tilde{c}_r = \sum_{k=1}^m (-1)^{k-1} \binom{m}{k} \frac{\bar{Y}_{r,k}}{(ks)^2}. \quad (\text{A.6.8})$$

The weights reproduce the value at zero of every polynomial of degree at most $m-1$. Taylor's formula with a Lipschitz highest derivative therefore bounds the extrapolation error for $c_r F_r$ by Cs^m . If the gap bracket is correct, writing $e = \hat{g} - g$ gives the exact mean at the k th normalized node as

$$c_r \left(1 + \frac{re}{ks}\right)^2 F_r(ks + re).$$

Its difference from $c_r F_r(ks)$ is at most $C\rho/s$; the finite sum of absolute weights depends only on m .

For completeness, at these windows $P_r \leq Cs^2$ uniformly in k , since $k \leq m$ and $0 \leq re \leq 2\rho$. A summand of $\bar{Y}_{r,k}/(ks)^2$ is bounded by s^{-2} and its variance is at most Cs^{-2} . Thus

$$N_{r,k} = \left\lceil Cs^{-2} \xi^{-2} \log \frac{8m}{\eta_0} \right\rceil$$

with a sufficiently large supplied constant gives simultaneous normalized nodal error at most a fixed small multiple of ξ with conditional probability $1 - \eta_0/2$. This follows from the same Bernoulli exponential-moment inequality; its denominator is bounded by $Cs^{-2}(1 + \xi)$. Hence, on the joint good event,

$$|\tilde{c}_r - c_r| \leq C(s^m + \rho/s + \xi) \leq C\xi.$$

Project each $\tilde{c}_r$ onto the known positive interval of possible c_r values and call the result $\hat{c}_r$. Projection does not increase its distance from the true c_r . Since $c_1/(2c_2) = \cosh \gamma$, project $\hat{c}_1/(2\hat{c}_2)$ onto $[\cosh \gamma_-, \cosh \gamma_+]$ and apply arcosh to obtain $\hat{\gamma}$. Finally project $\{2\hat{c}_1 \sinh \hat{\gamma}\}^{-1}$ onto $[A_-, A_+]$ to obtain $\hat{A}$. All these maps are Lipschitz on the specified boxes, since $\gamma_- > 0$. They give (A.6.6). The threshold decisions, finite sums, and projections are measurable.

The second stage costs $Cs^{-2} \xi^{-2} \log(8m/\eta_0)$. With the selected s, ρ , both $s^{-2} \xi^{-2}$ and ρ^{-2} have order $\xi^{-(2+2/m)}$. Adding the two stages proves (A.6.7), including all ceilings and unsuccessful preparations. $\square$

A.6.3. The full physical experiment with unknown normalization.

Theorem A.6.3 (Self-calibrated recovery of boundary profiles). *Consider a class of physical channels satisfying the general relative law, the supplied parameter boxes and calibration conditions (A.6.5), common profile bounds $\mathcal{K}_m(B, \beta; D_+)$, and common relative C^3 and finite $C^{3,1}$ flux bounds on $[0, D_+]$. Fix $0 < D < D_+$. Only the selected patch labels, these common bounds and the coarse gap bracket are known; the actual g, A, γ are unknown.*

For sufficiently small $\varepsilon > 0$ and $0 < \eta < 1/2$, a measurable procedure based exclusively on finitely many selected-event bits recovers both full profiles on $[0, D]$ and predicts the odd law with uniform error at most ε , with probability at least $1 - \eta$. Its deterministic total preparation count is at most

$$C\varepsilon^{-(2 + \frac{6}{m-5/2} + \frac{\gamma_+}{|\log \tau|})} \log \frac{C}{\eta\varepsilon}. \quad (\text{A.6.9})$$

The pilot uses flight numbers one and two; the profile stage uses one even flight number of order $\log(1/\varepsilon)$. No exact-law oracle or finite-dimensional remainder model is used. The conclusion is a uniform sufficient bound, not a minimax lower bound.

Proof. Choose j, h, t, δ as in Corollary A.5.2, with constants small enough to accommodate the plug-in error. In the pilot lemma take $\xi = c\varepsilon/j$ and $\eta_0 = \eta/2$. On its good event, $\alpha \leq C\xi$, $\zeta \leq C\xi$ and $\rho \leq C\xi^{1+1/m}$. Thus

$$\alpha + j\zeta + j\rho \leq Cc\varepsilon$$

for sufficiently small ε . The same choice ensures $j\zeta \leq 1$ and $j\rho \leq D_+ - D$. The latter fixed margin is why the theorem is stated on a prescribed subcollar. It never loses the energy origin. Apply Theorem A.6.1 with fresh observations, $\eta_1 = \eta/2$, and a finite dictionary accurate enough also for the $2B$ odd-law factor. The union bound and (A.6.3) give the error assertion.

The pilot estimates are projected into the supplied boxes even on bad outcomes. Consequently $\hat{c}_j \leq 2A_+ \sinh(j\gamma_+)$ always, and the profile allocation has the deterministic bound in (A.6.9). Before absorption, the pilot adds at most

$$C(j/\varepsilon)^{2+2/m} \log \frac{C \log(Cj/\varepsilon)}{\eta}.$$

Since $j \leq C \log(C/\varepsilon)$ and

$$2 + \frac{6}{m-5/2} + \frac{\gamma_+}{|\log \tau|} > 2 + \frac{2}{m},$$

this additional cost is absorbed into (A.6.9), uniformly for small ε and $0 < \eta < 1/2$. All observations are independent fresh preparations conditionally on the chosen windows; their success or failure is immaterial to the preparation charge. $\square$

The unknown-normalization theorem is a new observation statement, not an interpretation of the earlier exact-family four-window theorem. Its smooth-remainder calibration is proved above and its pilot cost is included. Relative to a calibrated design controlled by the same supplied parameter box, it does not increase the sufficient accuracy exponent. This does not replace γ_+ by an unknown intrinsic complexity, remove the convergence certificate, or remove patch labels. Nor does symmetrized energy recovery become recovery of the two asymmetric branches of a contact graph. Separately observed odd records test the compatibility predicted by the two recovered even laws; they are not a third identifying datum for this reconstruction.

APPENDIX A.7. CRITICAL BOUNDARY EXPERIMENTS AND SUPPORT SEPARATION

The relative comparison has a statistical scale which is not captured by weak convergence of a single record. We determine it first for the quadratic experiments, then transfer it to actual positive-offset laws in an explicit joint limit. The calculation allows unequal facing curvatures and either parity of the flight number. Throughout this section the channel and table are fixed; constants are uniform on the compact geometric families specified in Section 2. The two hypotheses compared here are the finite law and its boundary model for that same table, not two unknown billiard tables.

Write $\iota = j \bmod 2$, $a_0 = S_0''(0)$, $a_\iota = S_\iota''(0)$, and $D = \text{diag}(\sqrt{a_0}, \sqrt{a_\iota})$. Lemma 5.1 gives

$$H_j = DM_j D, \quad H_{\partial, \iota} = D^2, \quad M_j = \begin{pmatrix} \coth(j\gamma) & -\text{csch}(j\gamma) \\ -\text{csch}(j\gamma) & \coth(j\gamma) \end{pmatrix}. \quad (\text{A.7.1})$$

Both determinants equal $a_0 a_\iota$. On the common endpoint-residual coordinate space define

$$\mathcal{D}(H) = \{(z, s) \in \mathbb{R}^2 \times \mathbb{R} : s \geq 0, \ s + z^\top H z / 2 < 1\}.$$

Its volume is $\pi/\sqrt{\det H}$, as follows by first integrating s and then making a linear change to a disk. Let Π_j^0 and $\Pi_{\partial,j}^0$ have the common density $\sqrt{a_0 a_i}/\pi$ on $\mathcal{D}(H_j)$ and $\mathcal{D}(H_{\partial,i})$, respectively. The second subscript in $\Pi_{\partial,j}^0$ retains parity. The actual conditional laws $\Pi_{j,d}$, $\Pi_{\partial,j,d}$ and the raw laws $P_{j,d}$, $P_{\partial,j,d}$ are those of Theorem 21.1; the boundary raw law retains the finite normalizing twist. Put

$$p_{j,d}^0 = \frac{d^2}{2A \sinh(j\gamma)}, \quad P_{j,d}^0 = (1 - p_{j,d}^0)\delta_{\dagger} + p_{j,d}^0 \Pi_j^0, \quad P_{\partial,j,d}^0 = (1 - p_{j,d}^0)\delta_{\dagger} + p_{j,d}^0 \Pi_{\partial,j}^0. \quad (\text{A.7.2})$$

Shrink the common collar so these are probability measures. We use $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$.

Theorem A.7.1 (Exact support separation). *For every $j \geq 1$, independently of the two facing curvatures except through γ ,*

$$\delta_j := \|\Pi_j^0 - \Pi_{\partial,j}^0\|_{\text{TV}} = \frac{2}{\pi} \arcsin(e^{-j\gamma}). \quad (\text{A.7.3})$$

For independent observations at any finite list $(j_i, d_i)_{i=1}^n$,

$$\left\| \bigotimes_i \Pi_{j_i}^0 - \bigotimes_i \Pi_{\partial,j_i}^0 \right\|_{\text{TV}} = 1 - \prod_i (1 - \delta_{j_i}), \quad (\text{A.7.4})$$

$$\left\| \bigotimes_i P_{j_i, d_i}^0 - \bigotimes_i P_{\partial, j_i, d_i}^0 \right\|_{\text{TV}} = 1 - \prod_i (1 - p_{j_i, d_i}^0 \delta_{j_i}). \quad (\text{A.7.5})$$

As a simple binary experiment, each displayed pair of product laws is exactly equivalent, by measurable randomizations independent of the binary hypothesis, to an erasure experiment: with probability equal to the corresponding product on the right, the observation is an erasure, and otherwise it reveals the hypothesis. Consequently its minimum equal-prior testing error is half that product.

Proof. The common transformation $x = Dz/\sqrt{2}$ and a subsequent orthogonal rotation reduce every residual section to a disk and an equal-area ellipse. The eigenvalues of M_j are

$$\lambda = \tanh(j\gamma/2), \quad \lambda^{-1}, \quad 0 < \lambda < 1.$$

For the unit section the ellipse is $\lambda x^2 + \lambda^{-1} y^2 < 1$. Its boundary crosses the unit circle in the first quadrant at $\theta_0 = \arctan \sqrt{\lambda}$. Polar integration gives the intersection area

$$2 \left\{ \theta_0 + \int_{\theta_0}^{\pi/2} \frac{d\theta}{\lambda \cos^2 \theta + \lambda^{-1} \sin^2 \theta} \right\} = 4\theta_0.$$

For the equality, set $t = \tan \theta$ in the integral; it becomes $[\arctan(t/\lambda)]_{\sqrt{\lambda}}^{\infty} = \theta_0$. Every residual section is obtained by the same dilation for both supports, so integration in s leaves the overlap fraction $4\theta_0/\pi$. Equal densities on the intersection imply $\delta_j = 1 - 4\theta_0/\pi$. If $q = e^{-j\gamma}$, then $\lambda = (1 - q)/(1 + q)$ and $\pi/2 - 2 \arctan \sqrt{\lambda} = \arcsin q$; this proves (A.7.3). The whitening in (A.7.1) was common to both laws and loses no information. In particular the computation does not require equal curvatures or an even flight number.

For a pair with equal densities on its overlap, the measure with density the minimum of the two densities is exactly their common restriction. In the product the same property holds, and its mass is the product of the individual common masses. This proves (A.7.4). In the raw observation space the common restriction also includes the failure atom; its mass in observation i is $1 - p_{j_i, d_i}^0 \delta_{j_i}$. This proves (A.7.5).

Here is the asserted equivalence, including the reverse randomization. Send the common support to an erasure symbol, the support exclusive to the first law to 0, and the support exclusive

to the second to 1. The common restricted measures of the two laws coincide. Given an erasure, draw from their normalized common restriction; given 0 or 1, draw from the corresponding normalized exclusive restriction. These kernels recover the two original laws and do not depend on which of the two hypotheses is true. Zero-mass restrictions may be assigned an arbitrary fixed probability. Under equal prior probabilities there is no error on an exclusive observation, and the posterior probabilities on an erasure remain equal. The minimum error is therefore half the common mass. This construction is for the fixed simple pair specified above; its kernel may depend on the known table and design. $\square$

The next lemma controls the approximation error in the same topology before allowing the sample size to grow. Define

$$\omega(d) = \sqrt{d}, \quad \text{or } \omega(d) = d \text{ when both contact graphs are even.}$$

Evenness here does not require the two graphs to be identical.

Lemma A.7.2 (Uniform tangent approximation). *On a common positive collar, uniformly in j and in the specified compact geometric family,*

$$\|\Pi_{j,d} - \Pi_j^0\|_{\text{TV}} + \|\Pi_{\partial,j,d} - \Pi_{\partial,j}^0\|_{\text{TV}} \leq C\omega(d), \quad (\text{A.7.6})$$

$$\|P_{j,d} - P_{j,d}^0\|_{\text{TV}} + \|P_{\partial,j,d} - P_{\partial,j,d}^0\|_{\text{TV}} \leq Cp_{j,d}^0\omega(d). \quad (\text{A.7.7})$$

Proof. Write $E_j = W_j - jg$ and $b_j = -W_{j,uv}/[-W_{j,uv}(0)]$. The uniform Taylor estimates in Lemma 5.2 give, on bounded scaled endpoint sets,

$$\frac{E_j(\sqrt{d}z)}{d} = \frac{1}{2}z^\top H_j z + O(\sqrt{d}|z|^3), \quad b_j(\sqrt{d}z) = 1 + O(\sqrt{d}|z|).$$

All sublevels under discussion lie in a common bounded set by uniform coercivity. At each z , the actual and quadratic residual intervals start at zero and have lengths $(1 - E_j(\sqrt{d}z)/d)_+$ and $(1 - z^\top H_j z/2)_+$. Their symmetric-difference length is at most $C\sqrt{d}|z|^3$. Integrating it and the amplitude error over that bounded set bounds the total variation norm of the unnormalized densities by $C\sqrt{d}$. Their normalizing integrals are bounded away from zero; dividing therefore proves the first finite-law estimate.

The same argument applies to $S_0(u) + S_\ell(v)$ and $B_0(u)B_\ell(v)$. Their uniform derivatives follow from Lemma 7.1. If both graphs are even, uniqueness of the stationary segment makes E_j and b_j invariant under simultaneous endpoint sign reversal. Their remainders improve to $O(d|z|^4)$ and $O(d|z|^2)$, respectively; the half-line expressions have the same improvement. This proves (A.7.6) with either choice of ω .

The successful subprobabilities have the common scalar prefactor $d^2[-W_{j,uv}(0)]/(2\pi A)$. The quadratic integral is $\pi/\sqrt{a_0 a_\ell}$, and the Hessian identity gives $[-W_{j,uv}(0)]/\sqrt{a_0 a_\ell} = 1/\sinh(j\gamma)$. Multiplying the unnormalized estimate by this prefactor bounds the successful-measure difference by $Cp_{j,d}^0\omega(d)$. The difference of the failure masses is no larger than that bound. Restoring these atoms proves (A.7.7). $\square$

Theorem A.7.3 (Joint physical experiment limit). *For any finite independent schedule in the common collar, the actual positive-offset laws satisfy*

$$\left\| \bigotimes_i \Pi_{j_i, d_i} - \bigotimes_i \Pi_{\partial, j_i, d_i} \right\|_{\text{TV}} - \left(1 - \prod_i (1 - \delta_{j_i}) \right) \leq C \sum_i \omega(d_i), \quad (\text{A.7.8})$$

$$\left\| \bigotimes_i P_{j_i, d_i} - \bigotimes_i P_{\partial, j_i, d_i} \right\|_{\text{TV}} - \left(1 - \prod_i (1 - p_{j_i, d_i}^0 \delta_{j_i}) \right) \leq C \sum_i p_{j_i, d_i}^0 \omega(d_i). \quad (\text{A.7.9})$$

In particular, let $j \rightarrow \infty$, $d_j > 0$, and set $q_j = e^{-j\gamma}$. For k_j conditional successes, if

$$k_j q_j \rightarrow b \in [0, \infty], \quad k_j \omega(d_j) \rightarrow 0, \quad (\text{A.7.10})$$

then their total variation distance from the boundary experiment tends to $1 - e^{-2b/\pi}$. For n_j raw preparations the same limit holds if

$$n_j p_{j,d_j}^0 q_j \rightarrow b \in [0, \infty], \quad n_j p_{j,d_j}^0 \omega(d_j) \rightarrow 0. \quad (\text{A.7.11})$$

At $b = \infty$ the limit is interpreted as one. The minimum equal-prior testing errors tend to $e^{-2b/\pi}/2$. These limits hold for unequal facing curvatures and across both parity subsequences.

Proof. Telescoping product measures gives $\|\otimes_i P_i - \otimes_i Q_i\|_{\text{TV}} \leq \sum_i \|P_i - Q_i\|_{\text{TV}}$. Apply this separately on both sides of each simple binary comparison, use the reverse triangle inequality for their distances, and invoke Lemma A.7.2 and Theorem A.7.1. This proves the finite inequalities.

Now $\delta_j = (2/\pi)q_j + O(q_j^3)$. If $k_j q_j$ has a finite limit, $k_j \delta_j^2 \rightarrow 0$ and $k_j \log(1 - \delta_j) \rightarrow -2b/\pi$; if it diverges, use $(1 - \delta_j)^{k_j} \leq e^{-k_j \delta_j}$. This proves the conditional limit. The raw calculation is identical with $p_{j,d_j}^0 \delta_j$ in place of δ_j ; this quantity tends to zero, and the required first-order sum is $(2/\pi)n_j p_{j,d_j}^0 q_j (1 + O(q_j^2))$. The error terms vanish by the second condition in each displayed scaling assumption. For any simple pair P, Q , integration of the smaller of their two densities gives minimum equal-prior error $(1 - \|P - Q\|_{\text{TV}})/2$. This proves the final assertion. $\square$

The conditions are nonempty. In the general smooth case take $d_j = q_j^{2+\epsilon}$, with any fixed $\epsilon > 0$, and $k_j \sim b/q_j$, or $n_j \sim b/(p_{j,d_j}^0 q_j)$, for $0 < b < \infty$. In the even case $d_j = q_j^{1+\epsilon}$ suffices. Thus the critical number of successes is of order $e^{j\gamma}$, while the critical number of raw preparations is of order $d_j^{-2} e^{2j\gamma}$. These are distinct resources. In the tangent experiment the probability that every observation lies in the common support is the product in Theorem A.7.1; its limiting exponential is the source of the testing profile, rather than a Gaussian fluctuation argument.

More generally, in a triangular schedule, if the largest δ_{j_i} tends to zero, $\sum_i \delta_{j_i} \rightarrow \lambda$, and $\sum_i \omega(d_i) \rightarrow 0$, the conditional distance tends to $1 - e^{-\lambda}$. Replace δ_{j_i} by $p_{j_i,d_i}^0 \delta_{j_i}$ and the last sum by $\sum_i p_{j_i,d_i}^0 \omega(d_i)$ for raw preparations. The logarithm argument in the proof applies without change. The identity maps to the tangent laws and the kernels in Theorem A.7.1 also give two-way randomizations whose errors tend to zero under these same assumptions. This last statement concerns the specified simple binary experiments only.

The uniform relative nonlinear theorem is still needed outside this joint tangent regime: at fixed positive d it compares the actual nonlinear boundary law, whereas replacing it by its quadratic law would retain an $O(\omega(d))$ error in each success. The rate τ in Theorem 21.1 is not asserted to be optimal. Nor do the preceding identities extend total variation to full embedded growing collision arrays. They show, in an actual physical joint limit, why exponentially small support differences and the number of successful preparations must be controlled together.

A.7.1. Adaptive critical experiments. The exact overlap calculation also has a stopped, adaptive form. The relevant overlap mass is generally not a product of unconditional one-step overlaps, because later windows depend on earlier records. We give its exact expression and a physical approximation bound. Throughout this subsection the comparison is the same simple binary pair as in Section A.7: a fixed table's finite law versus its boundary model. The policy is independent of the binary hypothesis, but may use the specified table and the past records.

For a design $a = (j, d)$ write $P_a^0 = P_{j,d}^0$ and $Q_a^0 = P_{\partial,j,d}^0$. Define

$$h(a) = p_{j,d}^0 \delta_j, \quad \delta_j = \frac{2}{\pi} \arcsin(e^{-j\gamma}), \quad p_{j,d}^0 = \frac{d^2}{2A \sinh(j\gamma)}. \quad (\text{A.7.12})$$

Their densities agree on the common support, including the failure atom. Let C_a be their common restriction and put $R_a = C_a/(1 - h(a))$. The chosen collar ensures $h(a) < 1$. For a policy π with cap N , let R^π be the law obtained by following the same policy with record kernel R_a . This auxiliary law is supported on histories that have not revealed the binary hypothesis. Its stopping rule is still the specified rule of π .

Theorem A.7.4 (Exact stopped overlap and physical transfer). *The tangent stopped transcript laws satisfy*

$$\|P^{0,\pi} - Q^{0,\pi}\|_{TV} = 1 - m_\pi, \quad m_\pi = \mathbb{E}_{R^\pi} \prod_{i=1}^T (1 - h(a_i)). \quad (\text{A.7.13})$$

As a simple binary experiment, this pair is exactly equivalent by hypothesis-independent randomizations to an erasure experiment with erasure probability m_π . Its minimum equal-prior testing error is $m_\pi/2$.

Let P^π, Q^π instead be the actual positive-offset finite and boundary transcript laws. All reachable designs are assumed to lie in the common collar. Put

$$\varepsilon_\pi = \mathbb{E}_{P^\pi} \sum_{i=1}^T p_{a_i}^0 \omega(d_i) + \mathbb{E}_{Q^\pi} \sum_{i=1}^T p_{a_i}^0 \omega(d_i), \quad (\text{A.7.14})$$

where $\omega(d) = \sqrt{d}$, or $\omega(d) = d$ when both contact graphs are even. Uniformly on the compact geometric families under consideration,

$$|\|P^\pi - Q^\pi\|_{TV} - (1 - m_\pi)| \leq C\varepsilon_\pi. \quad (\text{A.7.15})$$

The randomizations in the first assertion concern the specified known simple pair and policy, not an unknown family of billiard tables.

Proof. At each design, the measures P_a^0, Q_a^0 have the same restriction C_a to their overlap; their other restrictions are supported on disjoint sets. A transcript compatible with both hypotheses can therefore contain only common-support records at every step. Conversely, along any such history every observation density agrees under the two hypotheses. The design kernels and stopping probabilities agree as well, since they are functions of the same observed past. Thus the two full transcript densities agree on their overlap. A first exclusive record makes the entire transcript exclusive, regardless of all later decisions.

The common transcript restriction is obtained by using the subprobability kernels C_a at active steps and the original policy kernels for choosing designs and stopping. Since $C_a = (1 - h(a))R_a$, its density relative to R^π is

$$Z = \prod_{i=1}^T (1 - h(a_i)).$$

This identity follows by induction over the at most N active steps, keeping the stopping branch at each step; it also covers $T = 0$ by the empty-product convention. Its total mass is m_π . Equal densities on the overlap give total variation $1 - m_\pi$.

Map the overlap to an erasure symbol and each exclusive part to its hypothesis label. For the reverse randomization, given an erasure draw from the normalized common transcript restriction; given a label draw from that hypothesis's normalized exclusive restriction. The common restriction is $Z R^\pi$, not in general R^π itself. Assign any fixed distribution to a zero-mass branch. These kernels recover both laws and do not depend on the true binary label. With equal priors the posterior remains equal on an erasure and the hypothesis is known otherwise, proving the testing assertion.

By Lemma A.7.2, the actual raw kernel and its own tangent kernel have total variation at most $Cp_a^0\omega(d)$, for either hypothesis. Apply Lemma 22.1 separately to these two approximations,

under the same policy. The sum of the resulting bounds is $C\varepsilon_\pi$. The reverse triangle inequality for total variation then proves (A.7.15). $\square$

Corollary A.7.5 (An adaptive critical limit). *Consider a sequence of the preceding experiments and capped policies. Under the corresponding auxiliary laws $\mathbf{R}^{\pi_n}$, suppose*

$$\sum_{i=1}^{T_n} h(a_i) \longrightarrow b \in [0, \infty], \quad \max_{i \leq T_n} h(a_i) \longrightarrow 0 \quad \text{in probability}, \quad (\text{A.7.16})$$

where the maximum of an empty list is zero. If $\varepsilon_{\pi_n} \rightarrow 0$, the actual stopped transcript distances converge to $1 - e^{-b}$ and their minimum equal-prior testing errors converge to $e^{-b}/2$. For $b = \infty$, convergence of the sum to infinity and $\varepsilon_{\pi_n} \rightarrow 0$ suffice without the maximum condition.

Proof. For $0 \leq x < 1$, the series of $-\log(1 - x)$ gives

$$0 \leq -\log(1 - x) - x \leq \frac{x^2}{2(1 - x)}.$$

For finite b , the sum of squared hazards is at most their maximum times their sum, hence tends to zero in probability. Applying the displayed inequality on the event that the maximum is at most $1/2$ shows that the product in (A.7.13) converges in probability to e^{-b} . Products lie in $[0, 1]$, so their expectations converge as well: for each $\epsilon > 0$, the expectation of their absolute error is at most ϵ plus the probability that the error exceeds ϵ . If $b = \infty$, bound the product by the exponential of minus the hazard sum and use the same boundedness argument. Finally apply (A.7.15) and the binary testing identity. $\square$

For predetermined windows and a deterministic stopping index, (A.7.13) reduces to (A.7.5). For adaptive windows it retains the observed common-history distribution instead of incorrectly averaging each hazard in isolation. The conditions concern the specified binary comparison; they do not assert optimal design or a minimax lower bound for geometric reconstruction. All transcript laws in this subsection retain the failure atom and the actual number of preparations.

A.7.2. Random accumulated hazard. The deterministic limit in Corollary A.7.5 has the following distributional form. This clarification is recorded in the independent memorandum [35, Section 4.4].

Corollary A.7.6 (Random-hazard mixture). *Use the same capped policies, common-history laws and physical approximation budgets as in Theorem A.7.4. Suppose, under $\mathbf{R}^{\pi_n}$,*

$$H_n := \sum_{i \leq T_n} h(a_i) \implies B, \quad M_n := \max_{i \leq T_n} h(a_i) \longrightarrow 0 \quad \text{in probability},$$

where B is a finite nonnegative random variable. If $\varepsilon_{\pi_n} \rightarrow 0$, then

$$\|\mathbf{P}^{\pi_n} - \mathbf{Q}^{\pi_n}\|_{\text{TV}} \longrightarrow 1 - \mathbb{E}e^{-B}, \quad \inf\{\text{equal-prior testing error}\} \longrightarrow \frac{1}{2}\mathbb{E}e^{-B}.$$

Proof. The convergence in distribution makes H_n tight. On $M_n < 1$, the logarithmic inequality in the proof of Corollary A.7.5 gives

$$0 \leq -\log \prod_{i \leq T_n} (1 - h(a_i)) - H_n \leq \frac{M_n H_n}{2(1 - M_n)}.$$

The right side tends to zero in probability. Consequently the product converges in distribution to e^{-B} . It takes values in $[0, 1]$, so its expectation tends to $\mathbb{E}e^{-B}$ by bounded continuous convergence. Apply (A.7.13), (A.7.15) and the exact binary testing identity. No averaging under the physical history law is substituted for averaging under $\mathbf{R}^{\pi_n}$. $\square$

In particular the limit is not determined by the mean of B . If $\mathbb{E}B < \infty$, strict convexity gives $\mathbb{E}e^{-B} \geq e^{-\mathbb{E}B}$, with equality only when B is almost surely constant. This observation does not assert optimality of an adaptive policy or make failed preparations free.

A.7.3. Completion of the critical transition. The accumulated remainder conditions in Theorem A.7.3 are useful for heterogeneous finite schedules. For a homogeneous sequence a condition on the error of one observation suffices throughout the entire transition, even when the full accumulated error diverges.

Theorem A.7.7 (Zero, finite, and infinite effective sample limits). *Let $j \rightarrow \infty$, $d_j > 0$, $q_j = e^{-j\gamma}$, and let ω have the meaning in Lemma A.7.2. The geometry may vary with j in a fixed compact family; γ, A and the laws then take their corresponding parameter values. Suppose*

$$\omega(d_j)/q_j \rightarrow 0. \quad (\text{A.7.17})$$

If $k_j q_j \rightarrow b \in [0, \infty]$, then

$$\|\Pi_{j,d_j}^{\otimes k_j} - \Pi_{\partial,j,d_j}^{\otimes k_j}\|_{\text{TV}} \rightarrow 1 - e^{-2b/\pi}.$$

If $n_j p_{j,d_j}^0 q_j \rightarrow b \in [0, \infty]$, the same limit holds for $\|P_{j,d_j}^{\otimes n_j} - P_{\partial,j,d_j}^{\otimes n_j}\|_{\text{TV}}$. At $b = \infty$ the right side is one. The minimum equal-prior errors converge to $e^{-2b/\pi}/2$. Both statements permit unequal facing curvatures and either parity. Under (A.7.17), vanishing total variation is equivalent to $k_j q_j \rightarrow 0$ in the conditional experiment, and to $n_j p_{j,d_j}^0 q_j \rightarrow 0$ in the raw experiment.

Proof. If $b < \infty$, then $k_j \omega(d_j) = (k_j q_j) \omega(d_j)/q_j \rightarrow 0$ in the conditional case, and $n_j p_{j,d_j}^0 \omega(d_j) = (n_j p_{j,d_j}^0 q_j) \omega(d_j)/q_j \rightarrow 0$ in the raw case. Theorem A.7.3 applies, including $b = 0$.

Suppose $b = \infty$ and fix $B > 0$. For all sufficiently large j the conditional sample contains its first $\lfloor B/q_j \rfloor$ observations. Projection onto them cannot increase total variation. Their effective sample size tends to B and their accumulated tangent error tends to zero by (A.7.17). Hence the lower limit of the full-sample variation distance is at least $1 - e^{-2B/\pi}$. Let $B \rightarrow \infty$ and use the upper bound one. For raw preparations project instead onto the first $\lfloor B/(p_{j,d_j}^0 q_j) \rfloor$ observations. Here $p_{j,d_j}^0 q_j \rightarrow 0$, so rounding changes their effective sample size by a quantity tending to zero. The identical argument proves the raw assertion. All estimates used above are uniform on the compact geometric family, and its positive lower bound for γ ensures $q_j \rightarrow 0$ even when the geometry varies. If an effective sample size fails to tend to zero, a subsequence is bounded below by a positive constant. Project on that subsequence onto a fixed positive critical subsample smaller than this constant. Its limiting distance is positive by the finite case, proving necessity in the final equivalences. Sufficiency is the case $b = 0$. The testing formula follows from $\inf(\text{equal-prior error}) = (1 - \|P - Q\|_{\text{TV}})/2$. $\square$

For a general smooth contact, (A.7.17) is $d_j = o(q_j^2)$; when both graphs are even it is $d_j = o(q_j)$. The finite critical subsamples therefore approach their quadratic experiments in total variation. This is a sharp physical testing profile in an asymptotically tangent regime, not a replacement for the nonlinear fixed-offset theorem or a proof that its rate τ is sharp. The supercritical conclusion no longer requires the tangent error of all retained observations to vanish. The subsample argument applies to the known simple pair of laws, and does not supply a simulator for an unknown table.

APPENDIX A.8. CURVATURES INVISIBLE TO ALL THRESHOLD LOCATIONS

The following construction separates geometric uncertainty from its statistical measurement. Set $\theta_r = r\pi/3$, $r = 0, 1, 2$, and let

$$h_{R,\alpha,\beta,\zeta}(\theta) = R + (1 - \cos 6\theta)b(\theta), \quad b(\theta) = \alpha + \beta \cos 2\theta + \zeta \sin 2\theta. \quad (\text{A.8.1})$$

Here R ranges over a compact interval in $(0, 1/2)$, and the three other parameters are sufficiently small. These are smooth centrally symmetric strictly convex scatterers, generally noncircular. At each θ_r and its opposite, $h = R$, $h' = 0$, and

$$\kappa_r^{-1} = h(\theta_r) + h''(\theta_r) = R + 36b(\theta_r). \quad (\text{A.8.2})$$

The map from (α, β, ζ) to the three $b(\theta_r)$ is invertible: its rows are $(1, 1, 0)$, $(1, -1/2, \sqrt{3}/2)$ and $(1, -1/2, -\sqrt{3}/2)$.

Theorem A.8.1 (Equal gaps and identifiable curvature amplitudes). *In (A.8.1) all six shortest gaps are exactly $g = 1 - 2R$, independently of (α, β, ζ) . The entire set of maximal-count threshold locations is therefore the same. Nevertheless, write*

$$C_j = \lim_{d \downarrow 0} d^{-2} \Pr(N_{jg+d} = j + 1), \quad \gamma_r = \operatorname{arcosh}(1 + g\kappa_r).$$

Together with the observed spacing g , the unlabelled sequence $(C_j)_{j \geq 1}$ determines the unordered triple $(\kappa_0, \kappa_1, \kappa_2)$, including multiplicities, and the free area A , without prior knowledge of A . It obeys

$$C_j = \frac{1}{A} \sum_{r=0}^2 \operatorname{csch}(j\gamma_r), \quad F_j := \sum_{\substack{k \geq 1 \\ k \text{ odd}}} \mu(k) C_{kj} = \frac{2}{A} \sum_{r=0}^2 e^{-j\gamma_r}, \quad (\text{A.8.3})$$

where $\mu(k)$ is the arithmetic Möbius function. Both series involved in the inversion are absolutely convergent.

In the subfamily $\beta = \zeta = 0$, two coefficients already determine its shape parameter:

$$\alpha = \frac{1}{36} \left\{ \frac{g}{C_1/(2C_2) - 1} - R \right\}. \quad (\text{A.8.4})$$

Threshold locations alone cannot determine this parameter.

Proof. For small parameters $h + h''$ remains positive, which proves smooth strict convexity and (A.8.2). For each nearest lattice vector e at angle θ_r , the two supporting lines have separation $1 - h(\theta_r) - h(\theta_r + \pi) = g$. Every displacement from the first body to its translate has projection at least g in direction e , and hence length at least g . Since $h' = 0$ at the two normals, the supporting points themselves lie on the axis and attain that distance. More distant translates have distance at least $\sqrt{3} - 2 \sup h > g$ after a uniform decrease of the parameter neighborhood. Thus the global shortest length is exactly g and all six channels attain it. The three curvatures can vary independently by the displayed invertible matrix.

Central symmetry makes both curvatures of the r th pair equal. Theorem 3.1 gives the first formula in (A.8.3), with two orientations per pair. Put $x_r = e^{-\gamma_r} \in (0, 1)$. Expanding $\operatorname{csch}(j\gamma_r) = 2 \sum_{m \geq 1, m \text{ odd}} x_r^{mj}$ and interchanging the absolutely convergent sums gives

$$F_j = \frac{2}{A} \sum_r \sum_{\substack{n \geq 1 \\ n \text{ odd}}} x_r^{nj} \sum_{k|n} \mu(k) = \frac{2}{A} \sum_r x_r^j.$$

Absolute convergence follows, for example, by bounding the double sum by $\sum_{n \geq 1} n x_{\max}^{nj} < \infty$; the number of divisors is at most n . The divisor identity leaves only $n = 1$.

After the odd-index transform, recovery is a finite-exponential-sum problem of Prony type [10]. We retain the complete Hankel–Vandermonde argument, rather than replace it by attribution. Let $x_1, \dots, x_d$ be its distinct nodes, $d \leq 3$, with weights $w_l = 2m_l/A > 0$, where m_l are their multiplicities. Then $F_j = \sum_l w_l x_l^j$. The Hankel matrices

$$H^{(0)} = (F_{p+q+1})_{0 \leq p, q < d}, \quad H^{(1)} = (F_{p+q+2})_{0 \leq p, q < d}$$

satisfy $H^{(0)} = V \operatorname{diag}(w_l x_l) V^t$ and $H^{(1)} = V \operatorname{diag}(w_l x_l^2) V^t$, where $V_{pl} = x_l^p$. The rank of the full Hankel sequence is d , and $H^{(0)}$ is positive definite. Thus the generalized eigenvalues of this pair are exactly x_l . A Vandermonde system using $F_1, \dots, F_d$ recovers the w_l . Since $\sum_l m_l = 3$, we recover $A = 6/(\sum_l w_l)$ and then $m_l = Aw_l/2$. Finally $\kappa_l = (\cosh(-\log x_l) - 1)/g$ proves the curvature claim. No label ordering or global determination of the boundary is asserted.

If $b \equiv \alpha$, the three curvatures coincide and $C_1/C_2 = 2 \cosh \gamma = 2(1 + g/(R + 36\alpha))$. Solving gives (A.8.4). All threshold locations are independent of α on a nontrivial open interval, while this ratio varies strictly with it. This proves the claimed nonredundancy. $\square$

The infinite original sequence enters the Möbius sums; this is an exact identification theorem, not a finite-noisy-data reconstruction guarantee. The special two-amplitude formula needs no such infinite transform. In a circle, by contrast, $R = (1 - g)/2$ and $1/C_j$ already satisfies a second-order recurrence. Corollary A.32.2 remains correct; Theorem A.8.1 supplies the additional geometric uncertainty which is absent from that one-parameter model.

Corollary A.8.2 (Instability selection at equal metric thresholds). *In (A.8.1), conditional on the maximal-count event in a common small collar, the mass of a channel with $\gamma_r > \min_s \gamma_s$ is at most $C \exp[-j(\gamma_r - \min_s \gamma_s)]$, uniformly in the offset. Thus channels with the same exact threshold length can have exponentially different probabilities. The limiting leading-amplitude weights among minimizing channels are equal.*

Proof. The relative remainders in (3.1) are bounded above and below by positive constants. Divide a channel's coefficient by one with minimal γ and use the exponential expression for csch . At offset zero the equal minimal coefficients give equal leading-amplitude weights. This last assertion concerns amplitudes; it does not identify the fixed-positive-offset limiting corrections. $\square$

A.8.1. Area dependence and coalescence of unlabelled amplitudes. The normalizer recovered in Theorem A.8.1 is not an additional independent shape parameter in its chosen family. The following identity also identifies a loss of sensitivity which is absent from the channel-resolved observation of Theorem 3.2.

Proposition A.8.3 (The normalizer and the circular differential). *For (A.8.1),*

$$|D| = \pi \left[R^2 + 2R\alpha - \frac{33}{2}\alpha^2 - \frac{45}{4}(\beta^2 + \zeta^2) \right]. \quad (\text{A.8.5})$$

The free area $A = \sqrt{3}/2 - |D|$ is a symmetric function of the gap and the unordered contact-curvature triple. In particular it is not a fourth independent inverse invariant in this three-parameter family.

Fix R , and restrict to $\alpha = \zeta = 0$, $\beta = s$. The curvature radii are $(R + 36s, R - 18s, R - 18s)$ and $A(s) = A(0) + 45\pi s^2/4$. For every fixed finite J ,

$$\max_{1 \leq j \leq J} |C_j(s) - C_j(0)| = O_J(s^2), \quad \operatorname{dist}_{\text{match}}(\kappa(s), \kappa(0)) \asymp |s|, \quad (\text{A.8.6})$$

where $\operatorname{dist}_{\text{match}}$ is the minimum maximum coordinate distance over permutations of the two triples. In contrast, conditional endpoint variances on the three fixed channel patches recover their contact curvatures with a common Lipschitz constant, including at $s = 0$.

Proof. For a positively curved support function, the boundary parametrization $q = hn + h't$ has derivative $q' = (h + h'')t$. Green's area identity and integration by parts give

$$|D| = \frac{1}{2} \int_0^{2\pi} (h^2 - (h')^2) d\theta.$$

Write the nonconstant part of h as

$$h = R + \alpha - \alpha \cos 6\theta + \beta \cos 2\theta - \frac{\beta}{2} \cos 4\theta - \frac{\beta}{2} \cos 8\theta \\ + \zeta \sin 2\theta + \frac{\zeta}{2} \sin 4\theta - \frac{\zeta}{2} \sin 8\theta.$$

Orthogonality assigns to a sine or cosine coefficient of order k its square times $\pi(1-k^2)/2$. Thus the α contribution is $\pi[(R+\alpha)^2 - 35\alpha^2/2]$, and the β, ζ contribution is $\pi[-3 - 15/4 - 63/4](\beta^2 + \zeta^2)/2$. This is (A.8.5). Set $b_r = (\kappa_r^{-1} - R)/36$. The three values in (A.8.2) give

$$\alpha = \frac{1}{3} \sum_r b_r, \quad \beta^2 + \zeta^2 = \frac{2}{3} \sum_r (b_r - \alpha)^2.$$

Both expressions are symmetric; $R = (1 - g)/2$ is known. This proves the normalizer assertion without changing its recovery formula.

Let $\Phi_j(r) = \text{csch}(j \text{arcosh}(1 + g/r))$. Along the specified one-parameter family,

$$C_j(s) = \frac{\Phi_j(R + 36s) + 2\Phi_j(R - 18s)}{A(s)}.$$

The numerator derivative at zero is $(36 - 18 - 18)\Phi'_j(R) = 0$, and $A'(0) = 0$. Taylor's theorem on a compact interval proves the first bound in (A.8.6), simultaneously for a fixed finite J . The largest curvature displacement is comparable to $|s|$ because $r \mapsto r^{-1}$ has derivative bounded away from zero there. This proves the matching-distance assertion. The last statement is Theorem 3.2 applied to each fixed channel; opposite curvatures agree by central symmetry, but that equality is not needed by its inverse. $\square$

Thus exact recovery of a multiset from the infinite amplitude sequence does not imply Lipschitz recovery from any fixed finite amplitude vector at coalescence. The observation underlying the stable theorem is different: it retains physical endpoint positions in a selected channel, rather than only an unlabelled count. Its improvement is therefore not attributed to an algebraic regularization of the same datum. The cost of acquiring the conditional records is included in Proposition A.16.2. The restricted amplitude theorem, including its exact multiplicity recovery and the removal of an initially unknown normalizer, remains a separate conclusion.

APPENDIX A.9. A SHARP PAIRWISE INVERSE THROUGH A TRIPLE COLLISION

We now compare two arbitrary nearby curvature triples, rather than a triple with the circular reference alone. In the ambient model with an independent area normalizer, four amplitudes give an inverse without separating repeated nodes. The constrained physical family admits a three-amplitude refinement near $R = 1/4$, proved in Section A.10.

Fix $g, R, A_0 > 0$ and consider the observation model

$$C_j(A, r) = \frac{1}{A} \sum_{i=1}^3 \Phi_j(r_i), \quad \Phi_j(r) = \text{csch}(j \text{arcosh}(1 + g/r)). \quad (\text{A.9.1})$$

Here A is near A_0 and r_i are near R . The equal-gap geometric family of Theorem A.8.1 is a subfamily of this model, with $r_i = \kappa_i^{-1}$ and its area given by (A.8.5). Allowing A to vary independently in the observation model proves a stronger nuisance-parameter statement; it does not make area an independent shape coordinate in that geometric family. When r_i^{-1} is interpreted as an actual contact curvature, each channel is assumed to have equal facing curvatures. Without that equality, the same scalar model describes the effective channel curvatures (A.10.3), not both endpoint curvatures.

Theorem A.9.1 (Four-amplitude pairwise stability). *There is a neighborhood of (A_0, R, R, R) on which, for every two members of (A.9.1),*

$$|A - \tilde{A}| \leq C\delta_4, \quad \text{dist}_{\text{match}}(\kappa, \tilde{\kappa}) \leq C\delta_4^{1/3}, \quad \delta_4 = \max_{1 \leq j \leq 4} |C_j - \tilde{C}_j|. \quad (\text{A.9.2})$$

The inverse determines multiplicities and requires neither channel labels nor a lower bound on the distance between distinct curvatures. In particular the first four amplitudes determine the area and unordered triple locally, even at complete coalescence. With deterministic error at most ϵ in these four amplitudes, a reconstruction constrained to a sufficiently small compact neighborhood has area error $C\epsilon$ and matching curvature error $C\epsilon^{1/3}$.

Proof. Put $x_i = r_i - R$ and let e_1, e_2, e_3 be their elementary symmetric polynomials. Define

$$p_e(z) = z^3 - e_1 z^2 + e_2 z - e_3.$$

For a small fixed circle $|z| = b$ around zero, $\Phi_j(R + z)$ is analytic inside it, with the positive-real branch at R . If e is sufficiently small, all roots of p_e lie strictly inside that circle. The symmetric sum in (A.9.1) has the analytic extension

$$M_j(e) = \frac{1}{2\pi i} \int_{|z|=b} \Phi_j(R + z) \frac{p'_e(z)}{p_e(z)} dz. \quad (\text{A.9.3})$$

This formula counts roots with multiplicity and defines an analytic function on a full coefficient neighborhood, not merely on its real-rooted part. It is real for real e . Expanding the Newton sums at $e = 0$ gives

$$M_j(0) = 3\Phi_j(R), \quad \partial_{e_1} M_j = \Phi'_j(R), \quad \partial_{e_2} M_j = -\Phi''_j(R), \quad \partial_{e_3} M_j = \frac{1}{2}\Phi'''_j(R) \quad (e = 0). \quad (\text{A.9.4})$$

For clarity, the first three Newton sums are e_1 , $e_1^2 - 2e_2$, and $e_1^3 - 3e_1e_2 + 3e_3$; all sums of degree at least four have zero first differential at $e = 0$. This also follows by differentiating the contour integral.

Consider the map $\Psi : (A, e_1, e_2, e_3) \mapsto (A^{-1}M_j(e))_{j=1}^4$. Its Jacobian determinant at $(A_0, 0, 0, 0)$ is

$$\det D\Psi = \frac{3}{2A_0^5} \det[\Phi_j^{(k)}(R)]_{1 \leq j \leq 4, 0 \leq k \leq 3} > 0. \quad (\text{A.9.5})$$

We verify the strict sign explicitly. Put $c = 1 + g/R > 1$ and write $\Phi_j(r) = w(c(r))f_j(c(r))$, where $c(r) = 1 + g/r$, $w(c) = (c^2 - 1)^{-1/2}$, and

$$f_1 = 1, \quad f_2 = \frac{1}{2c}, \quad f_3 = \frac{1}{4c^2 - 1}, \quad f_4 = \frac{1}{4c(2c^2 - 1)}.$$

Column elimination in the product and chain rules gives

$$\det[\Phi_j^{(k)}(R)] = \left(\frac{g}{R^2}\right)^6 (c^2 - 1)^{-2} \det[f_j^{(k)}(c)].$$

The last determinant reduces, using $f_1 = 1$, to the three by three matrix of the first, second, and third derivatives of f_2, f_3, f_4 . Differentiating these displayed rational functions and putting the entries over their common denominator yields

$$\det[f_j^{(k)}(c)] = \frac{12(64c^8 + 32c^6 + 116c^4 + 4c^2 + 1)}{c^4(4c^2 - 1)^4(2c^2 - 1)^4}. \quad (\text{A.9.6})$$

Every factor is strictly positive for $c > 1$. The column factors in (A.9.4) are $-3/A_0^2$, $1/A_0$, $-1/A_0$ and $1/(2A_0)$, which proves the factor $3/(2A_0^5)$ in (A.9.5). Thus the analytic inverse-function theorem applies in the full four-dimensional coefficient space. Choose a convex ball in its data

image and then a smaller parameter neighborhood whose image lies in that ball. A bounded derivative of the inverse on the ball gives

$$|A - \tilde{A}| + \sum_{k=1}^3 |e_k - \tilde{e}_k| \leq C\delta_4 \quad (\text{A.9.7})$$

for any two members of this smaller neighborhood, including its real-rooted boundary strata. This is the step for which using the symmetric coordinates, rather than differentiating individually labelled roots, is essential.

We give the needed root estimate with multiplicities. For two monic cubics p, q whose roots lie in a fixed bounded set and whose coefficient distance is ϵ , one has $|p(z) - q(z)| \leq C_0\epsilon$ on a common bounded disk containing the small neighborhoods below. Set $\rho = (2C_0\epsilon)^{1/3}$. On the boundary of the union of the open radius- ρ disks about the three roots of p , every root distance is at least ρ . Hence $|p(z)| \geq \rho^3 > |p(z) - q(z)|$ there. Rouché's theorem on each connected component shows that the two polynomials have the same number of roots in that component, counted with multiplicities. If two boundary arcs are tangent, perturb the radius slightly and pass to the limit; the strict inequality persists. Every component consists of at most three intersecting disks and has diameter at most 6ρ . Match the roots inside each component. This proves a matching distance at most $C\epsilon^{1/3}$. At $\epsilon = 0$ the polynomials coincide. The assertion for bounded, not necessarily small, ϵ follows by enlarging the constant.

Apply this estimate to (A.9.7). The radii stay in a compact interval bounded away from zero, so $r \mapsto r^{-1}$ is Lipschitz there. This proves (A.9.2). For noisy data, minimize the maximum discrepancy over a small compact admissible parameter set containing the true member. A minimum exists by continuity. Its exact amplitude vector differs from the true one by at most 2ϵ , since the true member is itself feasible. Apply the pairwise estimate. Uniqueness of this minimizer is not required for the error bound, nor is an efficient numerical optimization algorithm being assumed in its proof. $\square$

A.9.1. Sharpness in the physical geometric family. For $a \geq 0$ use the same absolute-amplitude topology as before, $\|D\|_a = \sup_{j \geq 1} e^{aj} |D_j|$. Choose a small fixed- R neighborhood in (A.8.1) on which all exponents exceed some $\gamma_- > a$.

Theorem A.9.2 (Optimal pairwise exponent). *On a sufficiently small neighborhood in that physical family,*

$$\text{dist}_{\text{match}}(\kappa, \tilde{\kappa}) \leq C_a \|C - \tilde{C}\|_a^{1/3}.$$

The exponent 1/3 cannot be increased, even if the entire sequence is observed. On the two-sided path $\alpha = \zeta = 0$, $\beta = s$,

$$\|C(s) - C(-s)\|_a \asymp |s|^3, \quad \text{dist}_{\text{match}}(\kappa(s), \kappa(-s)) \asymp |s|. \quad (\text{A.9.8})$$

The circular-reference exponent 1/2 of Theorem A.19.2 remains valid and sharp for its different, reference-point quantifiers.

Proof. The first assertion follows from Theorem A.9.1, since the first four errors are bounded by the sequence norm. For sharpness use exactly the realized path of Proposition A.8.3:

$$r(s) = (R + 36s, R - 18s, R - 18s), \quad A(s) = A_0 + 45\pi s^2/4.$$

Here A_0 is the actual circular free area. Differentiation of $2e^{-j\gamma}/(1 - e^{-2j\gamma})$, together with the bounded first three derivatives of $\gamma(r)$ and $A(s)^{-1}$, gives

$$\sup_{|s| \leq s_*} |C_j'''(s)| \leq B(1 + j)^3 e^{-j\gamma_-}.$$

The first derivative at zero vanishes. Subtract the Taylor expansions at s and $-s$: their constant and quadratic terms cancel, as does the zero linear term. The strict margin $a < \gamma_-$ absorbs the polynomial in j , proving $\|C(s) - C(-s)\|_a \leq B_a |s|^3$.

The first amplitude supplies a nonzero lower bound. Direct differentiation gives

$$\Phi_1(r) = \frac{r}{\sqrt{g(g+2r)}}, \quad \Phi_1'''(r) = \frac{3(3g+r)}{\sqrt{g}(g+2r)^{7/2}} > 0.$$

Since $36^3 + 2(-18)^3 = 34992$ and $A(s)$ is even,

$$C_1(s) - C_1(-s) = \frac{11664\Phi_1'''(R)}{A_0} s^3 + O(s^5). \quad (\text{A.9.9})$$

This proves the other sequence bound. In one real dimension sorted matching minimizes the maximum coordinate distance: an inverted pair can be uncrossed without increasing either extreme distance. Sorting the two curvature triples, their middle pair is the largest difference for all sufficiently small $s > 0$, and gives

$$\text{dist}_{\text{match}}(\kappa(s), \kappa(-s)) = \frac{36s}{R^2 - 324s^2}.$$

The negative case follows by symmetry. If a pairwise exponent $\theta > 1/3$ held uniformly, these estimates would imply $b|s| \leq C|s|^{3\theta}$ for all small nonzero s , a contradiction. This proves sharpness without replacing the infinite sequence by a finite numerical prefix. $\square$

The reference and pairwise results describe different singularities of one observation map. Distance from the circular point first detects quadratic dispersion. Comparing two noncircular triples on opposite sides of it can cancel that dispersion as well, leaving a cubic term. The four-amplitude inverse proves that the latter cancellation is the worst local pairwise scale in the stated three-channel model. This positive conclusion does not use relative accuracy at arbitrarily small amplitudes or a separation-dependent Prony bound.

APPENDIX A.10. THREE AMPLITUDES IN THE CONSTRAINED PHYSICAL FAMILY

The four-amplitude theorem permits an independent area normalizer. The physical family (A.8.1) has a constraint on that normalizer, and this constraint can reduce the required number of amplitudes. We prove the reduction near $R = 1/4$. The four-amplitude result remains available on its full positive parameter range and with an independent area.

Put $x_i = r_i - R$ for the three contact radii and let e_k be their elementary symmetric functions. The three values in (A.8.2) give

$$\alpha = \frac{e_1}{108}, \quad \beta^2 + \zeta^2 = \frac{e_1^2}{2916} - \frac{e_2}{972}. \quad (\text{A.10.1})$$

Substituting in (A.8.5) yields the exact identity

$$A(R, e) = A_0(R) - \frac{\pi R}{54} e_1 + \frac{41\pi}{7776} e_1^2 - \frac{5\pi}{432} e_2, \quad A_0(R) = \frac{\sqrt{3}}{2} - \pi R^2. \quad (\text{A.10.2})$$

Indeed, the sum of the three x_i is 108α , while their squared sum is $3888\alpha^2 + 1944(\beta^2 + \zeta^2)$. The identity $2e_2 = e_1^2 - \sum_i x_i^2$ proves (A.10.1), and collecting its terms in the area formula proves (A.10.2).

The contact radii in this physical family are actual radii at both ends of each channel: central symmetry makes the two facing curvatures equal. This is additional structure not required by

the general forward law. Without equality of the facing curvatures, the scalar count coefficient retains only

$$\kappa_e^{\text{eff}} = \frac{\sqrt{(1 + g\kappa_{e,0})(1 + g\kappa_{e,1})} - 1}{g}. \quad (\text{A.10.3})$$

The ambient scalar inverse then recovers these effective parameters, not the two actual endpoint curvatures. The conditional variances in Theorem 3.2 recover the latter without this symmetry.

Theorem A.10.1 (Physical three-amplitude reconstruction). *There is a neighborhood of $R = 1/4$, $\alpha = \beta = \zeta = 0$ in (A.8.1) such that the spacing $g = 1 - 2R$ and the first three unlabelled leading amplitudes determine the unordered contact-curvature triple, including multiplicities, and the area. For two members of a sufficiently small such neighborhood, put*

$$\Delta = |g - \tilde{g}| + \max_{1 \leq j \leq 3} |C_j - \tilde{C}_j|.$$

Then

$$|R - \tilde{R}| + |A - \tilde{A}| + \sum_{k=1}^3 |e_k - \tilde{e}_k| \leq C\Delta, \quad \text{dist}_{\text{match}}(\kappa, \tilde{\kappa}) \leq C\Delta^{1/3}. \quad (\text{A.10.4})$$

The pairwise exponent $1/3$ is sharp even at fixed $R = 1/4$ and known g . Thus no fourth amplitude is required in this constrained family near that reference point. The assertion does not impose a minimum separation between distinct curvatures.

Proof. For fixed R , set $g = 1 - 2R$, and use the analytic contour sums $M_j(R, e)$ of (A.9.3). They depend analytically also on R in a small real neighborhood of $1/4$, with a common complex contour and branches. On a full coefficient neighborhood, define

$$\Psi_3(R, e) = \left(\frac{M_j(R, e)}{A(R, e)} \right)_{j=1}^3.$$

The area stays positive there. At $e = 0$, the derivative with respect to the three coefficient coordinates has rows

$$\frac{1}{A_0} \left(\Phi'_j(R) + \frac{\pi R}{18A_0} \Phi_j(R), \quad -\Phi''_j(R) + \frac{5\pi}{144A_0} \Phi_j(R), \quad \frac{1}{2} \Phi'''_j(R) \right). \quad (\text{A.10.5})$$

Derivatives on Φ_j here keep g fixed. The area terms follow from $M_j(R, 0) = 3\Phi_j(R)$ and $A_{e_1} = -\pi R/54$, $A_{e_2} = -5\pi/432$, $A_{e_3} = 0$.

We verify nondegeneracy without assuming it numerically. At $R = 1/4$ and $g = 1/2$, the entries $\Phi_j^{(k)}(R)/\sqrt{2}$ are

j	$k = 0$	$k = 1$	$k = 2$	$k = 3$
1	1/4	3/4	-5/4	21/4
2	1/24	17/72	35/216	-491/216
3	1/140	297/4900	36243/171500	-4458537/6002500

These values follow by differentiating $\Phi_1(r) = r/\sqrt{g(g+2r)}$, $\Phi_2 = \Phi_1/[2(1+g/r)]$ and $\Phi_3 = \Phi_1/[4(1+g/r)^2 - 1]$. Substitution into (A.10.5) gives

$$\det D_e \Psi_3(1/4, 0) = -\frac{2\sqrt{2}(15804720A_0 + 64253\pi)}{72930375A_0^4} \neq 0, \quad A_0 = \frac{\sqrt{3}}{2} - \frac{\pi}{16} > 0. \quad (\text{A.10.6})$$

To include changes in the spacing, apply the analytic inverse-function theorem to

$$(R, e) \mapsto (1 - 2R, \Psi_3(R, e)).$$

Its first row is $(-2, 0, 0, 0)$, and the determinant in the other three columns is (A.10.6). Hence it is locally invertible in a full four-dimensional coefficient neighborhood, before restriction to real roots. Choose a convex data ball on which the inverse has bounded derivative and then a smaller parameter neighborhood whose image lies in that ball. This gives the coefficient and R estimates in (A.10.4). Analyticity of (A.10.2) gives the area estimate.

The root-matching proof in Theorem A.9.1 gives $\text{dist}_{\text{match}}(x, \tilde{x}) \leq C\Delta^{1/3}$. Adding the change of R and applying the Lipschitz map $r \mapsto r^{-1}$ on a compact positive interval proves the curvature estimate, after decreasing the neighborhood so $\Delta \leq 1$. This retains all repeated-root strata. On the physical path $\alpha = \zeta = 0$, $\beta = s$, compared with $-s$, the gap stays fixed and Theorem A.9.2 gives amplitude discrepancy $O(|s|^3)$ and matching curvature distance comparable to $|s|$. The nonzero first-amplitude cubic coefficient gives the reverse amplitude bound for the first three amplitudes as well. An exponent larger than $1/3$ would contradict this path. $\square$

With deterministic error at most δ in the observed spacing and these three amplitudes, choose an admissible member minimizing their maximum discrepancy on a sufficiently small compact physical parameter set containing the truth. Its exact data differ from the true data by at most 2δ . The theorem therefore gives area error $C\delta$ and matching curvature error $C\delta^{1/3}$. This existence argument neither requires a unique minimizer nor asserts a particular efficient optimization algorithm. More importantly, it concerns errors in amplitudes, not yet errors in individual collision counts. We next account for the experiment that estimates those amplitudes.

APPENDIX A.11. EXACT INVERSION FROM FOUR FIXED PHYSICAL WINDOWS

We now keep the offsets fixed while the requested accuracy improves. This experiment uses the exact analytic support family (A.8.1), not an unspecified smooth remainder in its onset expansion. The true gap is unknown. The reference circle has $R_c = 1/4$ and $g_c = 1/2$. Put $x_i = r_i - R$, and let $e = (e_1, e_2, e_3)$ be the elementary symmetric functions of the three radius increments. Thus $R = (1 - g)/2$, the unordered curvatures are $(R + x_i)^{-1}$, and $A = A(R, e)$ is the exact function in (A.10.2). Throughout this section the physical parameter set is a sufficiently small fixed compact neighborhood of the reference member. In particular it contains coalescing and repeated-root configurations. It is not a set of labelled root parameters with a separation condition.

A.11.1. Analytic descent through coalescence. For $j = 1, 2, 3$, define at positive offset

$$F_j(g, x, d) = d^{-2} \Pr_{g,x} \{N_{jg+d} \geq j + 1\}.$$

These are normalized exact finite-offset probabilities. The analytic extension needed for inversion at a triple root is stronger than continuity in the labelled radii.

Lemma A.11.1 (Symmetric analytic finite-offset coordinates). *The functions F_j have jointly analytic extensions to a neighborhood of $(g_c, 0, 0)$ in the variables (g, e, d) , before restriction to the real-rooted physical locus. On a smaller such neighborhood,*

$$F_j(g, e, d) = C_j(g, e) + dB_j(g, e, d), \tag{A.11.1}$$

where B_j is analytic and C_j is the physical leading-amplitude map in Theorem A.10.1. Every fixed derivative is bounded on a smaller closed neighborhood.

Proof. The linear map from (α, β, ζ) to (x_1, x_2, x_3) is invertible by (A.8.2). The support family is analytic in these variables and g . For each of the finitely many channels and flight numbers under consideration, Lemma 5.2 gives the analytic stationary action and its twist on a common complex neighborhood. The positive real endpoint Hessian has an analytic square-root branch

on a smaller complex neighborhood. The constructive Morse map in Lemma 6.1 and its inverse are consequently analytic there, by the analytic inverse-function theorem.

After physical residual-time integration and the Morse change of variables, the probability divided by d^2 is an integral over a fixed disk of an analytic amplitude at $\sqrt{2d}z$, with weight $1 - |z|^2$. Expand the amplitude in its convergent endpoint power series on a strictly smaller polydisk. The series is normally convergent, uniformly in the remaining complex parameters. Integration annihilates all odd total degrees and leaves a normally convergent series in integer powers of d . This constructs an analytic extension across $d = 0$; it does not assign a nonzero physical probability below onset. The six channels all have exactly the same gap in the present family, so adding them introduces no moving activation interface.

A rotation of the triangular lattice by $\pi/3$ cycles the three radius increments. Reflection in the horizontal axis fixes the first and interchanges the other two. These transformations generate their full permutation group. They preserve phase volume and the unlabelled count event, hence $F_j(g, x, d)$ is symmetric in x on the real physical open set. The analytic identity principle gives the same symmetry on a permutation-invariant complex polydisk.

We spell out descent at the discriminant. Choose a smaller coefficient polydisk on which all roots of

$$z^3 - e_1 z^2 + e_2 z - e_3$$

stay strictly inside the original root polydisk; this is possible by continuity of the root multiset at $e = 0$. Away from the discriminant, local holomorphic root branches define F_j as a holomorphic function of (g, e, d) . Symmetry makes this definition independent of the branches and their permutations. It is locally bounded uniformly up to the discriminant, since all roots stay in the smaller root polydisk and the original function is bounded on its closure. The removable-singularity theorem for locally bounded holomorphic functions across an analytic hypersurface therefore extends it jointly across the discriminant. One can apply the theorem in the product domain with g, d as additional variables, so joint analyticity is retained.

At $d = 0$, the physical onset formula identifies the extension with $C_j(g, e)$ on distinct real-rooted configurations. Such configurations contain real open subsets of the coefficient domain, so uniqueness of analytic extension identifies the two analytic functions everywhere near $(g_c, 0)$. Dividing their difference by d , or using its convergent Taylor series, gives (A.11.1). Cauchy estimates on smaller polydisks give the final boundedness assertion. $\square$

A.11.2. The unknown gap as a fourth coordinate. Fix a positive number a , to be chosen sufficiently small once and then kept independent of accuracy and confidence. Use the four windows

$$(j, t) = (1, g_c + a), \quad (1, g_c + 2a), \quad (2, 2g_c + a), \quad (3, 3g_c + a). \quad (\text{A.11.2})$$

Their normalizations use the known programmed excesses, not the unknown actual ones. If $v = g - g_c$, put

$$Q_a(g, e) = \begin{pmatrix} (a - v)^2 F_1(g, e, a - v)/a^2 \\ (2a - v)^2 F_1(g, e, 2a - v)/(4a^2) \\ (a - 2v)^2 F_2(g, e, a - 2v)/a^2 \\ (a - 3v)^2 F_3(g, e, a - 3v)/a^2 \end{pmatrix}. \quad (\text{A.11.3})$$

On $|v| < a/6$ these are the normalized physical means of the four binary reports; all actual offsets are positive. Decrease a and the neighborhood so they lie in the common physical collar.

Theorem A.11.2 (Four-window physical inverse). *For some fixed $a > 0$, the exact map Q_a is analytically invertible on a full neighborhood of $(g_c, 0)$ in coefficient coordinates. There is a*

compact physical neighborhood K of the circular member such that, for $\theta, \tilde{\theta} \in K$,

$$|g - \tilde{g}| + \|e - \tilde{e}\|_\infty + |A - \tilde{A}| \leq C \|Q_a(\theta) - Q_a(\tilde{\theta})\|_\infty, \quad (\text{A.11.4})$$

$$\text{dist}_{\text{match}}(\kappa, \tilde{\kappa}) \leq C \|Q_a(\theta) - Q_a(\tilde{\theta})\|_\infty^{1/3}. \quad (\text{A.11.5})$$

The neighborhood and constants do not depend on a root separation. Only the known family formula, reference window constants, and the fixed neighborhood are supplied. The true gap, area, channel labels, and impact positions are not supplied.

Proof. Let $D = D_e(C_1, C_2, C_3)(g_c, 0)$. By (A.10.6),

$$\det D = -\frac{2\sqrt{2}(15804720A_c + 64253\pi)}{72930375A_c^4}, \quad A_c = \frac{\sqrt{3}}{2} - \frac{\pi}{16} > 0. \quad (\text{A.11.6})$$

In particular $\det D \neq 0$. The derivation of this exact determinant, including its twelve amplitude derivatives and the physical area constraint, is given in Theorem A.10.1.

For a window of order j with programmed excess la at the reference gap, differentiation at $(g_c, 0)$ gives

$$\begin{aligned} \partial_g \left[\frac{(la - jv)^2}{(la)^2} F_j(g, e, la - jv) \right] &= -\frac{2jC_j}{la} + O(1), \\ D_e \left[\frac{(la - jv)^2}{(la)^2} F_j(g, e, la - jv) \right] &= D_e C_j + O(a). \end{aligned}$$

Here Lemma A.11.1 bounds derivatives in the full coefficient neighborhood; estimates only in the labelled roots would not justify the second formula at coalescence.

In DQ_a , subtract row one from row two, place this difference first, and multiply the gap column by a . The resulting matrix converges as $a \downarrow 0$ to

$$\begin{pmatrix} C_1 & 0 & 0 & 0 \\ -2C_1 & D_{11} & D_{12} & D_{13} \\ -4C_2 & D_{21} & D_{22} & D_{23} \\ -6C_3 & D_{31} & D_{32} & D_{33} \end{pmatrix}_{(g_c, 0)}. \quad (\text{A.11.7})$$

Its determinant is $C_1 \det D \neq 0$. These row and column operations are invertible for every $a > 0$. Fix a small enough that DQ_a is nonsingular. The analytic inverse-function theorem now applies to the exact finite-offset map, not to its leading approximation.

Choose a convex data ball with closure inside the image of an inverse chart, and a smaller compact physical neighborhood K whose image lies in that ball. The inverse has bounded derivative on the ball, giving the gap and coefficient bounds in (A.11.4). The analytic area identity gives its area bound. The root-matching argument of Theorem A.9.1 gives $\text{dist}_{\text{match}}(x, \tilde{x}) \leq C \|e - \tilde{e}\|_\infty^{1/3}$. Adding the change in $R = (1 - g)/2$ and using the bounded derivative of $r \mapsto r^{-1}$ on the common positive radius interval proves (A.11.5). Both the analytic inverse and the root estimate were established before any exclusion of a repeated root, so no such exclusion is needed. $\square$

Corollary A.11.3 (Deterministic-budget fixed-window acquisition). *There are constants $c, C > 0$ such that for $0 < \varepsilon, \delta < c$ and $0 < \eta < 1/4$, four-window binary observations give a Borel estimator satisfying, uniformly on K ,*

$$\Pr\{\text{dist}_{\text{match}}(\hat{\kappa}, \kappa) \leq \varepsilon, \quad |\hat{g} - g| \leq \delta\} \geq 1 - \eta, \quad (\text{A.11.8})$$

using a deterministic number of independent preparations bounded by

$$C(\varepsilon^{-6} + \delta^{-2}) \log(C/\eta). \quad (\text{A.11.9})$$

The same estimator has coefficient and area errors at most $C \min\{\varepsilon^3, \delta\}$ on this event. In particular, curvature-only accuracy ε costs at most $C\varepsilon^{-6} \log(C/\eta)$ and also gives gap and area errors $O(\varepsilon^3)$.

Proof. Take n fresh preparations in each window. The centered Bernoulli moment-generating function is at most $\exp(t^2/8)$: the second derivative of its log is the variance of a tilted Bernoulli law and is at most $1/4$, while its value and first derivative at zero vanish. Exponential Markov and optimization therefore give $\Pr\{|\hat{p} - p| > u\} \leq 2e^{-2nu^2}$. Dividing the four fractions by their fixed design normalizers and taking a union bound gives

$$\Pr\{\|\hat{Q} - Q_a(\theta)\|_\infty > C_a \sqrt{\log(8/\eta)/n}\} \leq \eta.$$

The dependence on a is a constant because a is fixed.

Minimize $\|Q_a(\vartheta) - \hat{Q}\|_\infty$ on K , using the Borel selection of Lemma A.20.1. This is defined on all outcomes, including data outside the inverse chart. Since the truth lies in K , its fitted exact data differ from the true data by at most $2\|\hat{Q} - Q_a(\theta)\|_\infty$. Apply Theorem A.11.2. Taking

$$n = \left\lceil C \min\{\varepsilon^3, \delta\}^{-2} \log(C/\eta) \right\rceil$$

with a sufficiently large constant proves all stated error bounds. Finally $\min\{\varepsilon^3, \delta\}^{-2} = \max\{\varepsilon^{-6}, \delta^{-2}\} \leq \varepsilon^{-6} + \delta^{-2}$, and there are exactly $4n$ preparations. This specifies a statistical estimator, not an efficient algorithm for exact evaluation of the nonlinear probability map. $\square$

The fixed-offset analytic descent and the four-window design were suggested in the referee comparison supplied with the two independent reports of September 9, 2026. The preceding proofs incorporate that comparison. The joint lower bound below identifies the additional cost of demanding a finer gap estimate.

APPENDIX A.12. JOINT CURVATURE AND ONSET ACCURACY

We specify the class of experiments before giving the lower bound. Fix an integer $J \geq 3$, a sufficiently small common collar $d_* > 0$ for flight numbers at most J , and the compact physical neighborhood K of Theorem A.11.2, decreased if necessary. A query chooses a physical time $t \geq 0$ and $1 \leq j \leq J$ and returns only $\mathbf{1}_{\{N_t \geq j+1\}}$ from a fresh independent full-phase preparation. The policy may use previous returned bits and independent randomization, and may stop at a stopping time T . Its rules are the same for every unknown parameter. On every possible history, every selected query satisfies

$$t - jg(\theta) \leq d_* \quad \text{for every } \theta \in K. \quad (\text{A.12.1})$$

Below-onset queries are allowed and are deterministic failures. We keep J, d_*, K fixed as the requested errors decrease. The four windows of (A.11.2) belong to this class after the preceding choices are made small enough. A fixed known neighborhood is thus permitted; the true gap is not revealed.

Let $\mathcal{N}_K(\varepsilon, \delta, \eta)$ be the infimum of $\sup_{\theta \in K} \mathbb{E}_\theta T$ over these policies and Borel estimators satisfying (A.11.8). All information retained from each trajectory is its one binary report. In particular an unrecorded lower collision count is not an extra observation.

A.12.1. Testing the unknown onset. The curvature splitting alternatives of Lemma A.22.1 have the same gap. To measure the price of timing, we need a second pair of physical alternatives.

Lemma A.12.1 (One-sided entropy for a displacement of the gap). *In the circular subfamily $e = 0$, consider gaps g_c and $g_c + \Delta$ with $0 < \Delta$ small. For any common query satisfying (A.12.1), denote its probabilities by p_0, p_Δ . Then*

$$D_{\text{KL}}(\text{Ber}(p_\Delta) \parallel \text{Ber}(p_0)) \leq C\Delta^2. \quad (\text{A.12.2})$$

For a stopped policy with $\mathbb{E}_\Delta T < \infty$, its complete transcript laws satisfy

$$D_{\text{KL}}(\mathbf{P}_\Delta \parallel \mathbf{P}_0) \leq C\Delta^2 \mathbb{E}_\Delta T. \quad (\text{A.12.3})$$

The constant is independent of how near to onset the queries are.

Proof. Put $d = t - jg_c$ and $z = j\Delta$. For the circular subfamily the uniform physical expansion, at these finitely many flight numbers, can be written

$$p_0 = d_+^2 H_j(g_c, d_+), \quad p_\Delta = (d - z)_+^2 H_j(g_c + \Delta, (d - z)_+).$$

Here H_j is positive, bounded above and below, and has bounded first derivatives on the common closed collar. This follows from Theorem 3.1; the area and all curvatures vary smoothly with $R = (1 - g)/2$. Decrease d_* so all nonzero probabilities are less than $1/2$.

If $d \leq 0$, both reports are deterministic failures. If $0 < d \leq z$, then $p_\Delta = 0$ and $p_0 \leq Cd^2 \leq CJd\Delta$. If $d > z$, then

$$|(d - z)^2 - d^2| \leq 2dz \leq 2Jd\Delta,$$

and the mean-value bound for H_j gives an additional difference at most $Cd^2(\Delta + z) \leq Cd\Delta$, since $d \leq d_*$. Thus in both positive- d cases

$$|p_\Delta - p_0| \leq Cd\Delta, \quad p_0 \geq cd^2.$$

The Bernoulli entropy inequality used in (A.22.4), including its continuous extension when the first probability is zero, yields

$$D_{\text{KL}}(\text{Ber}(p_\Delta) \parallel \text{Ber}(p_0)) \leq \frac{(p_\Delta - p_0)^2}{p_0(1 - p_0)} \leq C\Delta^2.$$

The direction of entropy is important: the higher-gap law has no successes where the lower-gap law is identically zero, so no unsupported absolute continuity assertion across the onset is used.

Conditioning on a common finite history and the parameter-independent random seed selects the same query under the two alternatives. The chain rule for the first M positions of the transcript, padded with stop symbols, gives $C\Delta^2 \mathbb{E}_\Delta(T \wedge M)$. Relative entropy on the increasing transcript sigma-fields converges to the entropy of the full transcript, as in the truncation proof of (A.22.4). Monotone convergence proves (A.12.3). A Borel statistic of the transcript adds no relative entropy. $\square$

Theorem A.12.2 (Joint local binary minimax order). *For fixed J, d_*, K as above there are $c, C, \varepsilon_0, \delta_0 > 0$ such that, for $0 < \varepsilon < \varepsilon_0$, $0 < \delta < \delta_0$, and $0 < \eta < 1/4$,*

$$c(\varepsilon^{-6} + \delta^{-2}) \log(1/\eta) \leq \mathcal{N}_K(\varepsilon, \delta, \eta) \leq C(\varepsilon^{-6} + \delta^{-2}) \log(1/\eta). \quad (\text{A.12.4})$$

The upper bound uses four fixed physical windows and a deterministic preparation count. The lower bound permits adaptive queries and random stopping. It is a statement for this exact physical family and this bounded-flight, local binary experiment.

Proof. The upper bound is Corollary A.11.3; for $\eta < 1/4$ its logarithm is comparable to $\log(1/\eta)$.

We first record the confidence version of the testing step. If a test chooses its first hypothesis on an event B with $P(B) \geq 1 - \eta$ and $Q(B) \leq \eta$, the log-sum inequality and monotonicity of binary relative entropy on this rectangle give

$$D_{\text{KL}}(P \parallel Q) \geq (1 - 2\eta) \log \frac{1 - \eta}{\eta} \geq \frac{1}{4} \log(1/\eta), \quad 0 < \eta < 1/4. \quad (\text{A.12.5})$$

For the last inequality use $1 - 2\eta \geq 1/2$ and $1 - \eta \geq \sqrt{\eta}$. The rectangle monotonicity follows by differentiating the binary relative entropy in its two arguments when the first exceeds the second.

For the curvature lower bound, take the physical tables $R = 1/4$, $\alpha = \zeta = 0$, $\beta = \pm s$ inside K . Their matching curvature distance is at least $c_0|s|$, and Lemma A.22.1 together with the stopped entropy argument gives

$$D_{\text{KL}}(\mathbf{P}_s \parallel \mathbf{P}_{-s}) \leq Cs^6 \mathbb{E}_s \sum_{i \leq T} d_{i,+}^2 \leq Cd_*^2 s^6 \mathbb{E}_s T.$$

Common below-onset queries contribute zero. Choose $s = L\varepsilon$ with L fixed large enough that the curvature separation exceeds 2ε . The nearest-alternative test based on the estimated multiset has both errors at most η by (A.11.8) and the triangle inequality for matching distance. Applying (A.12.5) yields

$$\sup_{\theta \in K} \mathbb{E}_\theta T \geq c\varepsilon^{-6} \log(1/\eta).$$

Small enough ε_0 keeps both alternatives in K .

For the gap lower bound use the circular tables of Lemma A.12.1 with $\Delta = 3\delta$. They remain in K for small δ_0 . Choosing the nearer gap to $\hat{g}$ gives a test with both errors at most η . Apply (A.12.5) in the higher-gap to lower-gap direction and then (A.12.3). It follows that

$$\sup_{\theta \in K} \mathbb{E}_\theta T \geq c\delta^{-2} \log(1/\eta).$$

A policy with infinite worst-case expected cost already satisfies the lower bound; otherwise both stopped entropy arguments apply. Combining the two bounds and using $\max\{u, v\} \geq (u + v)/2$ proves the lower half of (A.12.4). $\square$

Corollary A.12.3 (Curvature loss, finer timing, and shrinking designs). *In the fixed-collar class, curvature-only recovery has optimal order $\varepsilon^{-6} \log(1/\eta)$. For each fixed $m \geq 1$, the joint loss with curvature tolerance ε and gap tolerance $\delta = \varepsilon^{3+3/m}$ instead has optimal order*

$$\varepsilon^{-(6+6/m)} \log(1/\eta). \quad (\text{A.12.6})$$

Separately, any all-history design whose positive offsets are at most Hh and whose flight numbers are at most J , even with the exact gap supplied, needs worst-case expected cost at least

$$ch^{-2}\varepsilon^{-6} \log(1/\eta) \quad (\text{A.12.7})$$

for curvature accuracy ε on the fixed-gap splitting subfamily. Consequently the supplied-bracket acquisition theorem has matching upper and lower orders at $h \asymp \varepsilon^{3/m}$, including the confidence logarithm.

Proof. The first assertion is the curvature part of the preceding proof and Corollary A.11.3. Substitute the displayed δ in (A.12.4) to obtain (A.12.6). For the shrinking class use instead $\mathbb{E}_s \sum d_{i,+}^2 \leq H^2 h^2 \mathbb{E}_s T$ in the curvature entropy argument; the alternatives have the same gap, so supplying it has no effect. This proves (A.12.7).

In Theorem A.21.2 the supplied bracket is $|g - g_0| \leq h/4$. Its pilot offsets are in $[3h/4, (m + 5/4)h]$, and all its second-stage offsets are in $[(L - 3J/4)h, (Lm + 3J/4)h]$ with $L = J + 1$, even on an unsuccessful pilot history. They fit such a shrinking collar with a fixed H_m . Its expected upper bound and (A.12.7) agree after substituting $h \asymp \varepsilon^{3/m}$. $\square$

Thus the fixed-offset comparison improves the curvature-only benchmark, without invalidating the finer joint timing result. Theorem A.12.2 also identifies a timing obstruction in the larger fixed-collar class; its proof does not use the shrinking-design restriction. At the joint cost in (A.12.6), the exact four-window procedure may recover curvature more accurately than requested, because it estimates all symmetric coordinates to the finer gap tolerance. These statements make

no assertion about the computational cost of evaluating exact finite-offset probabilities or about a uniform limit $m \rightarrow \infty$.

APPENDIX A.13. RECONSTRUCTION WITH UNKNOWN SMOOTH REMAINDER FUNCTIONS

The exact inverse in Section A.11 uses the known nonlinear family at positive offset. To separate that assumption from the acquisition of leading invariants, we now enlarge the observation model explicitly. The underlying parameter is still $(g, e) \in K$ and its leading amplitudes are still the physical $C_j(g, e)$. The nuisance functions below need not come from that exact analytic support family.

Fix $m \geq 1$, constants $0 < c_* < C_* < \infty$, and a sufficiently small $d_* > 0$. For each $j = 1, 2, 3$, let the unknown function H_j satisfy

$$\begin{aligned} H_j &\in C^m([0, d_*]), \quad c_* \leq H_j \leq C_*, \quad \|H_j\|_{C^m} \leq C_*, \quad H_j(0) = C_j(g, e), \\ p_j(t) &= (t - jg)_+^2 H_j((t - jg)_+), \quad t - jg \leq d_*. \end{aligned} \quad (\text{A.13.1})$$

Choose $C_* d_*^2 < 1$. Each query returns an independent Bernoulli variable with this probability. The fixed neighborhood K , the common bounds, and a fixed interval $[L_0, U_0]$ containing the possible gaps are known. No value or derivative of H_j away from zero is supplied. The class can be chosen to contain the exact physical finite-offset probabilities on a smaller compact neighborhood. It is an observation envelope: we do not assert that every arbitrary nuisance triple in it is realized by a single billiard table. The target area and curvatures are defined from (g, e) by the same physical leading inverse.

Theorem A.13.1 (Uniform count recovery over a smooth envelope). *Let $0 < U_0 - L_0 = W < d_*/12$. For fixed m and the preceding bounds, there are $h_m, C_m > 0$ such that for $0 < h < h_m$ and $0 < \eta < 1/4$ a Borel count-only procedure satisfies, uniformly over (A.13.1),*

$$\Pr \left\{ \begin{array}{l} |\hat{g} - g| \leq C_m h^{m+1}, \\ \|\hat{e} - e\|_\infty + |\hat{A} - A| \leq C_m h^m, \\ \text{dist}_{\text{match}}(\hat{\kappa}, \kappa) \leq C_m h^{m/3} \end{array} \right\} \geq 1 - \eta. \quad (\text{A.13.2})$$

Its number of preparations is bounded on every history by

$$C_m h^{-(2m+2)} \log(C_m/\eta). \quad (\text{A.13.3})$$

It uses only flight numbers 1, 2, 3 and programmed times. It supplies its own accuracy-dependent bracket from the fixed interval, and requires no exact finite-offset formula, contact positions, labels, or independently measured area.

Proof. We give a deterministic-budget form of the calibration and extrapolation construction to specify its dependence on the nuisance functions.

First use the bracket search of Lemma A.23.1 with failure allowance $\eta/3$. Its proof needs only zero probability below onset and the lower bound $p_1(g + d) \geq c_* d^2$, both part of (A.13.1). It returns a midpoint $g_0 \in [L_0, U_0]$ and, on its good event, $|g - g_0| \leq h/4$. Its all-history cost is $Ch^{-2} \log((2 + \log(W/h))/\eta)$.

Next take a predetermined

$$n = \left\lceil C_m h^{-(2m+2)} \log(C_m/\eta) \right\rceil$$

fresh preparations at each pilot time $g_0 + lh$, $1 \leq l \leq m + 1$. On bracket success all actual offsets lie between $3h/4$ and $(m + 5/4)h$, and the probabilities are between positive multiples of h^2 . The Bernoulli concentration bound proved in Lemma A.21.1, divided by those probabilities, gives relative errors at most $c_m h^m$ simultaneously, with conditional probability at least $1 - \eta/3$, after increasing C_m .

Interpolate the square roots of the empirical fractions by a polynomial of degree at most m in the variable $z = t - g_0$ at nodes lh . Use its unique-zero-or-zero convention on $[-h/2, h/2]$ and call the result $\hat{\tau}$. This rule is defined and Borel also when a fraction is zero. Put $\hat{g} = g_0 + \hat{\tau}$. On every history $|\hat{g} - g_0| \leq h/2$. On the good pilot event, write

$$d\sqrt{H_1(d)} = \sqrt{C_1} d\sqrt{H_1(d)/C_1}.$$

The second square-root factor has value one at zero and bounded C^m norm, uniformly over the envelope. Multiplying its degree- $(m-1)$ Taylor polynomial by d gives a polynomial of degree m with a simple zero at onset. The fixed-node Lagrange bounds and the root argument of Lemma A.18.2 therefore give $|\hat{g} - g| \leq C_m h^{m+1}$. Only right derivatives through order m were used; no probability was evaluated below onset for this Taylor argument.

For the last stage put $L = 4$ and take n fresh preparations at $j\hat{g} + Llh$ for $j = 1, 2, 3$ and $l = 1, \dots, m$. Use

$$\omega_l = (-1)^{l-1} \binom{m}{l}, \quad \hat{C}_j = \sum_{l=1}^m \omega_l \frac{\hat{p}_{j,l}}{(Llh)^2}.$$

On bracket success every actual offset of this stage lies in a fixed positive interval of multiples of h , even if the pilot estimates are inaccurate. On the two good events, its displacement from Llh is at most $3C_m h^{m+1}$. The derivative of $d^2 H_j(d)$ is $O_m(h)$ on these intervals, so the resulting normalized timing error is $O_m(h^m)$. The identities

$$\sum_l \omega_l = 1, \quad \sum_l \omega_l l^r = 0 \quad (1 \leq r < m)$$

and the bounded m th derivative of H_j give extrapolation bias $O_m(h^m)$. Conditional Bernoulli concentration on these fresh preparations gives the same stochastic order, with additional failure probability at most $\eta/3$. Hence $\max_j |\hat{C}_j - C_j| \leq C_m h^m$ on their intersection.

Fit $(\hat{g}, \hat{C}_1, \hat{C}_2, \hat{C}_3)$ on the fixed compact physical leading-data set using Lemma A.20.1. The jointly Lipschitz coefficient inverse of Theorem A.10.1 gives coefficient and area errors $O_m(h^m)$, and its cubic root bound gives curvature error $O_m(h^{m/3})$. Retain the pilot $\hat{g}$ as the gap estimate; the fitted physical parameter and the pilot need not have identical gaps. This distinction preserves the finer asserted timing error.

All queries are legitimate even after a failed search. The returned g_0 remains in $[L_0, U_0]$ and $|\hat{g} - g_0| \leq h/2$. Thus pilot positive offsets are at most $W + (m+1)h$, and final positive offsets at most $3(W + h/2) + 4mh$. The fixed restriction on W and small enough h_m place both in $[0, d_*]$; negative offsets are deterministic failures. No waiting time depends on seeing a success. The pilot and final stage have exactly $(m+1+3m)n$ preparations on every history. As in Theorem A.23.2, their bound absorbs the bracket search cost because $h^{2m} \log(2 + \log(W/h))$ is bounded. The three conditional failure allowances sum to at most η , proving the result. $\square$

Taking $h \asymp \varepsilon^{3/m}$ yields curvature error $O(\varepsilon)$ and gap error $O(\varepsilon^{3+3/m})$ with the preparation power $6 + 6/m$. This result is uniform over unknown smooth higher-order remainders. The theorem gives an upper bound; the matching envelope lower bound, under the additional strict interior condition, is proved in Theorem A.14.1. In the exact analytic subexperiment, the same joint accuracy has the lower bound of Theorem A.12.2; for curvature loss alone its fixed-window benchmark has power six instead. The two experiments therefore differ in the information supplied to the estimator, even when their stated joint guarantees have the same preparation order.

APPENDIX A.14. THE SHARP COST OF UNKNOWN SMOOTH REMAINDERS

We prove a matching lower bound for the smooth-remainder upper bound. The experiment retains three binary reports with the physical leading-amplitude map. Its nuisance functions range over the stated smooth envelope; they are not asserted to be exact finite-offset probabilities of billiard tables. The construction uses a fixed interior norm margin, physical splitting paths for the leading parameters, and nuisance splices that agree outside a shrinking collar. The stopped-transcript argument permits adaptive queries and random stopping throughout.

Use the sum convention $\|H\|_{C^m} = \sum_{r=0}^m \|H^{(r)}\|_\infty$ in (A.13.1). Fix all its constants, a sufficiently small compact physical neighborhood K of $\theta_c = (g_c, e) = (1/2, 0)$, and the interval $[L_0, U_0]$ of Theorem A.13.1. The set K is to contain the two-sided physical splitting paths and the circular gap variations used below. In addition suppose that some $\rho > 0$ obeys

$$c_* + 4\rho < C_j(\theta_c) < C_* - 4\rho, \quad j = 1, 2, 3. \quad (\text{A.14.1})$$

This condition gives strict positivity and a norm budget around the constant reference nuisances. It can be arranged when the envelope is chosen, but is not a consequence of its non-strict bounds alone.

An admissible policy uses parameter-independent randomization and, after each history, either stops or chooses $j \in \{1, 2, 3\}$ and a time $t \geq 0$. Require $t - jg(\theta) \leq d_*$ for every $\theta \in K$ on every history, including histories at which a previous calibration failed. Each fresh preparation returns the bit with probability (A.13.1). The observation is the stopped transcript of these queries and bits, together with the recorded common randomization. For $0 < \eta < 1/4$, let $\mathcal{N}_m^{\text{env}}(\varepsilon, \delta, \eta)$ be the infimum of the worst-case expected number of preparations among such policies and Borel estimators satisfying

$$\Pr_{\theta, H} \{ \text{dist}_{\text{match}}(\hat{\kappa}, \kappa(\theta)) \leq \varepsilon, \quad |\hat{g} - g| \leq \delta \} \geq 1 - \eta \quad (\text{A.14.2})$$

uniformly over the envelope. The two outputs are target estimates; they are not required to be coordinates of one exactly fitted table.

Theorem A.14.1 (Joint minimax preparation cost over the smooth envelope). *Under these fixed choices and (A.14.1), there are $c, C, \varepsilon_0, \delta_0 > 0$ such that*

$$\begin{aligned} c(\varepsilon^{-(6+6/m)} + \delta^{-2}) \log(1/\eta) &\leq \mathcal{N}_m^{\text{env}}(\varepsilon, \delta, \eta) \\ &\leq C(\varepsilon^{-(6+6/m)} + \delta^{-2}) \log(1/\eta) \end{aligned} \quad (\text{A.14.3})$$

for $0 < \varepsilon < \varepsilon_0$, $0 < \delta < \delta_0$ and $0 < \eta < 1/4$. The upper bound has a deterministic, all-history preparation budget. The lower bound permits adaptive queries and random stopping. For curvature loss alone the order is $\varepsilon^{-(6+6/m)} \log(1/\eta)$, even when the common gap in the lower-bound construction is supplied.

Proof. We first construct the curvature alternatives within one fixed envelope, then charge the stopped experiment, and finally treat the timing loss and the upper bound.

Physical leading parameters. In the support family of Section A.8, set $R = 1/4$, $\alpha = \zeta = 0$ and $\beta = \pm s$. The two radius multisets are

$$(R + 36s, R - 18s, R - 18s), \quad (R - 36s, R + 18s, R + 18s).$$

Their gap is $g_c = 1/2$ and their elementary coefficient triples are

$$e^\pm = (0, -972s^2, \pm 11664s^3). \quad (\text{A.14.4})$$

For small $s > 0$, sorting the positive reciprocals gives

$$\text{dist}_{\text{match}}(\kappa^+, \kappa^-) = \frac{36s}{R^2 - 324s^2} \asymp s. \quad (\text{A.14.5})$$

For example the middle sorted coordinates differ by the displayed quantity; the two outer differences are smaller when s is small. The leading amplitudes are analytic in a full coefficient neighborhood by the leading inverse proved above. The two points in (A.14.4) differ by $O(s^3)$ there, so, with constants independent of s ,

$$|C_j^+(s) - C_j^-(s)| \leq Cs^3, \quad j = 1, 2, 3. \quad (\text{A.14.6})$$

This uses physical leading parameters, but not a free variation of area or a separated-root inverse.

Splicing the nuisance functions. Set $\psi(u) = (1-u)_+^{m+1}$ on $[0, \infty)$. It is C^m , equals one at zero, and it and its first m derivatives vanish at one. Write $\bar{C}_j = (C_j^+ + C_j^-)/2$ and $D_j = (C_j^+ - C_j^-)/2$. Choose a fixed sufficiently large L and put

$$h = Ls^{3/m}, \quad H_j^\pm(d) = \bar{C}_j \pm D_j\psi(d/h). \quad (\text{A.14.7})$$

These functions have exactly the prescribed leading values at zero. For $0 \leq r \leq m$,

$$\|(H_j^\pm - \bar{C}_j)^{(r)}\|_\infty \leq C_m L^{-r} s^{3(1-r/m)}. \quad (\text{A.14.8})$$

Choose L first so that the order- m bound is smaller than a specified part of ρ . Then make s small so that the sum of all lower-order bounds, and $|\bar{C}_j - C_j(\theta_c)|$, use less than the remaining margin. Condition (A.14.1) proves all the positivity and sum-norm bounds in the same fixed envelope. Also $h < d_*$ for sufficiently small s . Only the alternatives depend on accuracy; the parameter set and the nuisance norm ball do not.

Information at every allowed query. A common query has the same offset $d = t - jg_c$ under the two alternatives. At $d \leq 0$ both probabilities are zero, and at $d \geq h$ they agree exactly. For $0 < d < h$,

$$|p_j^+ - p_j^-| \leq Cd^2s^3, \quad p_j^- \geq c_*d^2, \quad 1 - p_j^- \geq 1 - C_*d_*^2 > 0.$$

The inequality $\log x \leq x - 1$ gives the Bernoulli entropy bound $D_{\text{KL}}(\text{Ber}(p) \parallel \text{Ber}(q)) \leq (p - q)^2/[q(1 - q)]$ for $0 < q < 1$. Consequently every query has

$$D_{\text{KL}}(\text{Ber}(p_j^+) \parallel \text{Ber}(p_j^-)) \leq Cd^2s^6 \mathbf{1}_{\{0 < d < h\}} \leq Cs^{6+6/m}. \quad (\text{A.14.9})$$

Queries are not restricted to shrinking offsets. It is the nuisance pair itself that makes larger offsets uninformative.

Stopping and confidence. Condition on the common randomization and on the same realized finite history. The policy's next query or stop decision is then the same under the two hypotheses. Pad the transcript by a fixed stop symbol after stopping. The entropy chain rule for its first M positions and (A.14.9) give

$$D_{\text{KL}}(\mathbf{P}_+^{[M]} \parallel \mathbf{P}_-^{[M]}) \leq Cs^{6+6/m} \mathbb{E}_+(T \wedge M).$$

Entropy on the increasing finite-transcript sigma-fields converges to the entropy on the full transcript: one may use its variational formula and conditional approximations to each bounded test function. Monotone convergence on the right therefore yields

$$D_{\text{KL}}(\mathbf{P}_+ \parallel \mathbf{P}_-) \leq Cs^{6+6/m} \mathbb{E}_+ T. \quad (\text{A.14.10})$$

An infinite worst-case expected cost already satisfies the lower bound. No deterministic stopping assumption has been imposed.

Choose s a fixed sufficiently large multiple of ε so that (A.14.5) exceeds 2ε . Classifying the curvature estimate by its nearer alternative gives a test with both errors at most η . Binary data processing implies

$$D_{\text{KL}}(\mathbf{P}_+ \parallel \mathbf{P}_-) \geq (1 - 2\eta) \log \frac{1 - \eta}{\eta} \geq \frac{1}{4} \log(1/\eta), \quad 0 < \eta < 1/4. \quad (\text{A.14.11})$$

The first inequality follows by minimizing the binary entropy over the two error constraints. For the second, $1 - 2\eta \geq 1/2$ and $(1 - \eta)/\eta \geq \eta^{-1/2}$ on this interval. Together with (A.14.10), this proves the curvature lower bound, also for curvature-only loss with supplied gap.

Timing. Use the circular leading parameters $(g_c, 0)$ and $(g_c + \Delta, 0)$, and constant nuisance functions $H_{j,0} = C_j(g_c, 0)$ and $H_{j,\Delta} = C_j(g_c + \Delta, 0)$. They are in the same envelope by strict slack and continuity. At a common query, put $d = t - jg_c$ and $z = j\Delta$. Then

$$p_0 = d_+^2 C_j(g_c, 0), \quad p_\Delta = (d - z)_+^2 C_j(g_c + \Delta, 0).$$

Both vanish for $d \leq 0$. For $d > 0$, splitting the cases $d \leq z$ and $d > z$, expanding the squares and using the bounded derivative of $C_j(g, 0)$ gives $|p_\Delta - p_0| \leq Cd\Delta$. Since $p_0 \geq cd^2$ and $1 - p_0$ is bounded away from zero, the entropy in the support-compatible direction satisfies

$$D_{\text{KL}}(\text{Ber}(p_\Delta) \parallel \text{Ber}(p_0)) \leq C\Delta^2.$$

The argument includes $p_\Delta = 0$ and does not reverse this entropy direction. Set $\Delta = 3\delta$ and repeat the stopped chain rule and the classification argument using the gap estimate. It gives the lower cost $c\delta^{-2} \log(1/\eta)$. The curvature and timing pairs need not be the same pair. Their two worst-case bounds combine by $\max(a, b) \geq (a + b)/2$, proving the lower half of (A.14.3).

The upper bound. In Theorem A.13.1, choose

$$h = c_m \min\{\varepsilon^{3/m}, \delta^{1/(m+1)}\}$$

with fixed $c_m > 0$ sufficiently small. Both target errors then meet (A.14.2). The all-history preparation bound is at most a constant times

$$\max\{\varepsilon^{-(6+6/m)}, \delta^{-2}\} \log(1/\eta),$$

which is bounded by the required sum. The known fixed gap interval and all-history admissibility were part of that theorem. Its pilot gap estimate is retained separately from the fitted coefficients, as required for the finer timing loss. Omitting the timing loss and using $h = c_m \varepsilon^{3/m}$ proves the curvature-only upper bound and completes the proof. $\square$

Corollary A.14.2 (Information in a finite positive-window design). *For any prescribed finite set of query windows whose offsets at g_c are all strictly positive, there are two different curvature targets in the fixed smooth envelope which give exactly the same law for every stopped experiment using only those windows, including adaptive choices and arbitrary repetitions among them.*

Proof. Let $d_{\min} > 0$ be the smallest such offset. Choose $s > 0$ in (A.14.7) so small that $h < d_{\min}$. The two nuisance functions, and hence their probabilities, agree at every permitted query. Induction on transcript length proves equality on every finite history, and thus on the stopped observation. The curvature targets differ by (A.14.5). $\square$

This corollary identifies the extra information used in the exact four-window inverse; it does not contradict that inverse. Under the exact analytic physical family, Theorem A.12.2 gives order $(\varepsilon^{-6} + \delta^{-2}) \log(1/\eta)$. For unknown C^m remainders with strict slack, Theorem A.14.1 gives order $(\varepsilon^{-(6+6/m)} + \delta^{-2}) \log(1/\eta)$. At the finer timing accuracy $\delta = \varepsilon^{3+3/m}$ these orders coincide. The envelope alternatives establish an information comparison, not a new geometrically realized lower bound for exact billiard probabilities or a restriction on the general forward law.

APPENDIX A.15. RELATION TO ACTION, SPECTRAL, AND STATISTICAL INVERSE PROBLEMS

The analysis starts from a discrete variational problem but ends with a normalized physical probability. Bolotin and Treschev [2] develop Hill-type identities connecting action Hessians and monodromy in discrete and continuous Lagrangian systems. Such determinant identities explain the linear structure used here. The further step in Theorem 7.2 is a relative nonlinear determinant

limit with two separated boundary perturbations, followed by full-phase residual-time integration. Propositions 8.1–8.3 separate the exact finite identity, trace-class transport and normalized integration. The additional physical consequence is Theorems 10.1 and 10.2: the two same-type scalar laws determine the complete energy profiles stably in a specified norm, and predict the mixed-type law. The finite-record compatibility defect in Corollary 10.5 is controlled by bridge length. It does not follow from just the value of the linear multiplier.

Zelditch [14] relates successive localized wave-trace invariants to analytic boundary geometry. His construction uses multiple oscillatory boundary integrals and a stationary-phase expansion; the inverse spectral conclusions concern the specified symmetric classes of simply connected analytic plane domains, including a symmetry reversing a bouncing-ball orbit. This is an important predecessor for the coefficient-to-boundary strategy, not the same observation theorem. Our coefficients arise from a positive full-phase residual-time integral, not a Dirichlet or Neumann wave trace. The highest-jet diagonal in (A.3.2) separates the stationary-action contribution from the relative interior determinant contribution. Exact analytic germ recovery and finite-jet coefficient conditioning are kept distinct from noise-stable global analytic continuation.

Marked-length observations provide a different route to geometry. Bálint, De Simoi, Kaloshin and Leguil [1] recover period-two curvatures and periodic Lyapunov exponents from the marked length spectrum of open dispersing billiards. Osterman [13] studies analytic rigidity by homoclinic asymptotics and normal forms, including recovery with two scatterers supplied. De Simoi, Kaloshin and Leguil [5] prove marked-length determination of analytic chaotic billiards with axial symmetries, under symmetry and genericity hypotheses in a no-eclipse open configuration. Their theorem concerns the geometry of the table, not merely a specified contact jet.

The data in Corollary A.2.2 are instead a selected onset probability and its offset coefficients under normalized phase volume. The identical-even contact hypothesis makes its inverse triangular. Analytic continuation then determines each participating boundary in its contact frame, but supplies neither a lattice nor the other obstacles. There is no implication here between equality of these statistical data and equality of marked length spectra. In particular, the corollary is not a replacement for the global table determination in [5]. Its coefficient formula and its physically realized finite-jet coordinates identify the precise inverse observation being used. Proposition A.2.4 also shows explicitly that this information is not exclusive to long records.

For periodic finite-horizon Sinai billiards, Finamore and Leguil [12] introduce an enriched marked length spectrum and prove isometry from equality of that datum, using geodesic approximations and CAT(0) geometry. Their enriched observation and finite-horizon assumptions should not be identified with the selected count probabilities considered here. Conversely our general boundary-law construction does not require finite horizon.

The exact exponential-sum step in Theorem A.8.1 belongs to the theory of Prony reconstruction. Batenkov and Yomdin [10] analyze local accuracy for confluent Prony systems. The quantitative result here uses a specific feature of the hyperbolic-sine observation: its first four functions have the nonzero Wronskian (A.9.6). They give regular symmetric coefficient coordinates even when all three radii coincide. The classical coefficient-to-root estimate then gives the pairwise exponent $1/3$; the realized physical path proves its sharpness. No new general root perturbation theorem or universally improved Prony algorithm is needed.

The source and onset series use a different physical window at each coefficient. Accordingly their first poles and convergence domains are consequences of the relative boundary law, not assertions about a pressure function for all long itineraries. Return-time and compound Poisson laws such as [6, 9], and the anisotropic statistical frameworks in [3, 4], address other observations and asymptotic regimes. The fixed-window companion [7] retains its separate transversality and moving-level argument. The historical Fourier-inversion analysis [8] remains available with its

own explicit hypotheses; those hypotheses are not substituted for the finite-threshold proofs in this article.

Finally, polynomial extrapolation and binomial concentration are classical ingredients of the acquisition statements. Their uniform use here depends on the differentiated physical onset expansion. The self-calibration theorem adds a concrete statistical use of its quadratic vanishing: square-root probabilities have a simple onset zero, independent of the unknown normalizer. Its proof accounts for both the calibration preparations and the subsequent conditional records, and preserves a positive sampling window on every pilot outcome. The sufficient cost is not represented as a minimax rate.

The relative boundary law has an experiment-level consequence beyond coefficient identification. Theorem 21.1 controls total variation for the endpoints and residual time, uniformly down to zero offset, and retains the rare-event mass when unsuccessful preparations are restored. It permits bounded-risk replacement for growing independent samples whenever their expected number of successes times τ^j tends to zero. Its proof uses the quadratic vanishing of the action error together with the normalized determinant estimate; an absolute approximation to a vanishing flux would not give this conclusion.

The finite-flight inverse of Theorem A.3.1 and the sensitivity formula (A.3.4) concern fixed jet order and varying collision length. The area constraint (A.10.2) explains why three leading amplitudes suffice in the constrained family while four allow an independent normalizer. Section A.11 uses a different fourth datum: a second first-flight window separates an unknown gap from the three symmetric shape coordinates in the exact physical model.

The minimax statement in Theorem A.12.2 concerns a specified binary experiment, not every observation available from a billiard. Its curvature and timing obstructions are realized by two different physical pairs. The smooth envelope in Theorem A.13.1 removes the supplied exact finite-offset formula while preserving a positive, fully charged reconstruction procedure. The earlier supplied-bracket and fixed-bracket procedures remain valid with their complete proofs. They are neither counterexamples to the fixed-window benchmark nor general lower bounds for all data.

Finally, the support calculation in Theorem A.7.1 is an elementary disk–ellipse comparison after a common whitening. Its role is to identify the constant in a physical testing profile, not to replace the nonlinear relative law. The projection argument in Theorem A.7.7 completes that profile when the full sample’s tangent approximation error no longer tends to zero. At fixed positive offset, the genuinely nonlinear comparison is still supplied by Theorems 2.1 and 21.1.

The smooth-envelope optimum in Theorem A.14.1 uses the additional strict-slack condition and allows free nuisance functions. It is not a lower bound obtained from a pair of exact nonlinear billiard laws. Its attribution to the referee memorandum is separate from the boundary compatibility and profile inverse developed here. Neither the count theorem nor its envelope comparison is used as a substitute for the long-bridge consequence.

The whole-profile count theorem, Theorem A.4.1, occupies a different place from these finite-dimensional benchmarks. It uses the same labelled scalar events as the relative boundary law, not their endpoint positions, and supplies no exact finite-offset probability formula to the estimator. The known normalization and uniform relative estimates allow its finite-bridge bias to be separated from scalar sampling noise. The positive-node regularization makes recovery possible on the full collar without an observed derivative oracle. Its preparation bound records both the cost of differentiating noisy scalar data and the exponentially decreasing event mass. This is the operational interpretation of the boundary energy invariant; it neither subsumes nor borrows the lower bounds of the other experiments.

The self-calibrated profile theorem in Section A.6 is distinct from the calibrated construction just described. It permits unknown g, A, γ because it supplies and charges a new one-/two-flight

pilot on a uniform smooth physical class. Patch labels, regularity certificates and coarse boxes remain supplied. Its proof neither applies the exact four-window inverse to a free remainder nor imports the envelope minimax lower bound.

APPENDIX A.16. CONDITIONAL ENDPOINT METRICS AND CURVATURE OBSERVABILITY

The scalar coefficient in (3.1) contains only the ratio in (5.2). We now retain the metric which appears before that ratio is formed. All expectations in this section are for the actual full-phase preparation, conditioned on $E_{j,e}(d)$.

Proposition A.16.1 (Quadratic endpoint statistics). *In the common transverse endpoint coordinates $Y = (u, v)$,*

$$\mathbb{E}Y = d m_{j,e}(\xi, d), \quad \frac{1}{d} \text{Cov}(Y) = \frac{1}{3} \mathbf{H}_{j,e}^{-1} + d B_{j,e}(\xi, d). \quad (\text{A.16.1})$$

The vector and matrix remainders are smooth for $d \geq 0$ with uniform fixed-order derivative bounds. For physical positions the scalar variance in (3.2) has the corresponding leading diagonal entry of $\mathbf{H}_{j,e}^{-1}/3$, with the same $O(d)$ remainder. In particular the leading metric is determined without a phase-space normalizer or a chosen transverse initial law.

Proof. Integrating the first residual time in (6.2) gives the conditional endpoint density, in $w = \sqrt{2d}z$ coordinates,

$$\frac{(1 - |z|^2) a_{j,e}(\xi, \sqrt{2d}z)}{\int_{|z| < 1} (1 - |z|^2) a_{j,e}(\xi, \sqrt{2d}z) dz} dz, \quad |z| < 1.$$

Its denominator is uniformly positive. The area and the exponentially small flux coefficient cancel in this conditional expression, but were included in the event probability. At $d = 0$ the density is $2(1 - |z|^2)/\pi$. Direct radial integration gives

$$\mathbb{E}_0 z = 0, \quad \mathbb{E}_0(z z^t) = I_2/6. \quad (\text{A.16.2})$$

For example $\mathbb{E}_0 |z|^2 = 1/3$ and rotational symmetry divides this between the two coordinates.

The endpoint map is $Y = \Phi_{j,e}(\xi, w)$, with linear part $\mathbf{H}_{j,e}^{-1/2} w$. For any smooth amplitude, integration on this radial domain annihilates its odd Taylor terms. Applied to each component of $Y a$ and $Y Y^t a$, the smooth radial lemma gives

$$\mathbb{E}Y = d m_{j,e}, \quad \mathbb{E}(Y Y^t) = \frac{d}{3} \mathbf{H}_{j,e}^{-1} + d^2 \tilde{B}_{j,e}.$$

All statements include smooth right extensions and uniform derivatives: the endpoint maps and amplitudes have the bounds in Lemma 6.1, and each division by d uses the vanishing constant term of a smooth even integral. Subtracting $(\mathbb{E}Y)(\mathbb{E}Y)^t$ proves (A.16.1).

The physical endpoint graph has the form $Q_b = q_b + \mathbf{t} y_b \pm \mathbf{n} \psi_b(y_b)$, with $|\mathbf{t}| = |\mathbf{n}| = 1$ and $\psi_b = O(y_b^2)$. Its mean is $q_b + d \tilde{m}_b$. The leading term in $|Q_b - q_b|^2$ is y_b^2 , since the two frame vectors are orthogonal. The remaining squared term starts at degree four. Apply the same radial argument and subtract the squared mean displacement. This proves the physical scalar variance formula. In particular no unknown q_b occurs in the observable itself: it is eliminated by centering the measured positions. $\square$

Proof of Theorem 3.2. The determinant-one middle matrix in (5.1) has inverse obtained by changing the signs of its off-diagonal entries. Thus

$$\mathbf{H}_{j,e}^{-1} = \frac{g}{c \sinh \gamma} D_j \begin{pmatrix} \coth(j\gamma) & \text{csch}(j\gamma) \\ \text{csch}(j\gamma) & \coth(j\gamma) \end{pmatrix} D_j. \quad (\text{A.16.3})$$

For odd j , $\sigma_0^2 = c_1$ and $\sigma_j^2 = c_0$. Proposition A.16.1 gives (3.3), with its differentiated error estimate. Taking the geometric mean and ratio gives (3.4). The function

$$F_j(\gamma) = \frac{\coth(j\gamma)}{\sinh \gamma}$$

is strictly decreasing from infinity to zero on $(0, \infty)$, since

$$-F'_j(\gamma) = \frac{j \operatorname{csch}^2(j\gamma)}{\sinh \gamma} + \frac{\coth(j\gamma) \cosh \gamma}{\sinh^2 \gamma} > 0. \quad (\text{A.16.4})$$

This proves unique recovery of γ , then of the two curvatures. At $j = 1$, the inversion is even explicit: $F_1(\gamma) = c/(c^2 - 1)$, so if $P = 3\sqrt{v_0 v_1}/g$, then $c = (1 + \sqrt{1 + 4P^2})/(2P)$.

We verify uniform conditioning, not just nonvanishing of a fixed-length Jacobian. On a compact positive gap–curvature box, $0 < \gamma_- \leq \gamma \leq \gamma_+ < \infty$. The second term of (A.16.4) gives a lower bound for $-F'_j$ independent of j . Its upper bound is also uniform because $j \operatorname{csch}^2(j\gamma_-) \leq C$ and $\coth(j\gamma_-) \leq \coth \gamma_-$. Both variances are bounded above and bounded away from zero. Taking their ratio and geometric mean is consequently Lipschitz; monotone inversion of F_j , then the last line of (3.4), is uniformly Lipschitz. Forward differentiation of (3.3) gives the reverse bound with the same type of constants. More explicitly, for data from two points of this box,

$$C^{-1} \|(g, \kappa_0, \kappa_1) - (\tilde{g}, \tilde{\kappa}_0, \tilde{\kappa}_1)\|_\infty \leq \|(g, v_0, v_1) - (\tilde{g}, \tilde{v}_0, \tilde{v}_1)\|_\infty \leq C \|(g, \kappa_0, \kappa_1) - (\tilde{g}, \tilde{\kappa}_0, \tilde{\kappa}_1)\|_\infty. \quad (\text{A.16.5})$$

No factor $(\kappa_0 - \kappa_1)^{-1}$ appears. Nor do we divide by the exponentially small off-diagonal entry in (A.16.3).

For data outside the exact image, minimize its distance to that image on the compact curvature box at the known g ; a minimizer exists. The true pair is an admissible comparison, so the selected image point is within twice the data error of the true image. The error is bounded by $\delta + Cd$ by Proposition A.16.1. Apply the inverse bound in (A.16.5). This proves the last assertion and also specifies a reconstruction for noisy or finite-offset data. A small error in the supplied g adds its size to the same bound, provided the conditional variances are still those of the stated physical window; we do not replace an onset-location error by a cost-free timing change. $\square$

The distinction between odd and even lengths is informational. At an even length the first and last impacts are on the same obstacle, and the two diagonal leading variances coincide. At an odd length they sample the two endpoints. The theorem therefore uses the two diagonal entries at an odd length rather than an exponentially small covariance or an assumed equality of endpoint curvatures.

Proposition A.16.2 (Independent-preparation estimation). *Suppose that $2K$ independent full-phase preparations are retained upon success of $E_{j,e}(d)$, for a fixed odd j . The two variances have estimators $\hat{V}_0, \hat{V}_1$, using only their endpoint positions and the known offset d , such that*

$$\Pr \left(\max_b |\hat{V}_b - V_{j,b}(d)| > \delta \right) \leq 4 \exp(-cK\delta^2).$$

A reconstruction as in Theorem 3.2 consequently has error at most $C[d + \sqrt{\log(4/\eta)/K}]$ with probability at least $1 - \eta$. All constants are independent of odd j . The expected number of independent unconditioned preparations needed for these $2K$ successes is comparable to

$$Kd^{-2}e^{j\gamma_e}. \quad (\text{A.16.6})$$

No independence of successive impacts in one trajectory is assumed.

Proof. Pair the independent successful records and use, at each endpoint,

$$\widehat{V}_b = \frac{1}{K} \sum_{k=1}^K \frac{|Q_b^{(2k)} - Q_b^{(2k-1)}|^2}{2d}.$$

Independence gives its expectation as the variance in (3.2). Coercivity confines each endpoint to a set of diameter $C\sqrt{d}$, independently of j . The summands are therefore in one bounded interval $[0, C_0]$. For completeness, a random variable in an interval of length C_0 satisfies $\mathbb{E}e^{t(X-\mathbb{E}X)} \leq e^{t^2 C_0^2/8}$: the second derivative of the log moment generating function is a variance under a tilted law, bounded by $C_0^2/4$, and its value and first derivative vanish at zero. Multiplication over the independent pairs, the exponential Markov inequality, and optimization over t yield $2e^{-2K\delta^2/C_0^2}$ at each endpoint. A union bound proves the displayed estimate. The deterministic inverse estimate gives the curvature bound. Rejection from independent preparations gives independent conditional successes, and the expected waiting time is $2K/\Pr(E_{j,e}(d))$. Theorem 3.1 and $\sinh(j\gamma) \asymp e^{j\gamma}$ prove (A.16.6). Uniform conditional conditioning is therefore not a uniform unconditioned sampling cost. $\square$

A.16.1. Realized scalar fibers and their separation by records. At known g, A , all leading selected-channel count coefficients depend on c_0, c_1 only through $c_0 c_1$. Conversely, their first two values recover this product, since their ratio is $2 \cosh \gamma$. The following construction shows that this scalar fiber occurs for actual periodic billiards, not just for algebraic local jets.

Theorem A.16.3 (A geometric fiber of the full leading count hierarchy). *On the rectangular lattice $3\mathbb{Z} \times 4\mathbb{Z}$ there is a real-analytic family of strictly convex scatterers D_s , $|s| < s_0$, with D_0 the unit disk, for which the only shortest unoriented channel is horizontal,*

$$g(s) = 1, \quad A(s) = 12 - \pi, \quad c_0(s) = 2e^s, \quad c_1(s) = 2e^{-s}.$$

All maximal-count threshold locations and every leading coefficient $\lim_{d \downarrow 0} d^{-2} \Pr_s(N_{j+d} = j+1)$ are independent of s . The ordered contact curvatures vary. The two endpoint variances at any odd order distinguish s , with constants uniform in that order.

Proof. Use the support function of the disk, $h_0 \equiv 1$. Put

$$\chi_0 = \frac{1}{4}(1 + \cos \theta) \sin^2 \theta, \quad \chi_\pi = \frac{1}{4}(1 - \cos \theta) \sin^2 \theta, \quad \chi_* = \sin^4 \theta.$$

At $0, \pi$ all three functions and their first derivatives vanish. The second derivatives of χ_0 at these two points are $(1, 0)$, those of χ_π are $(0, 1)$, and those of χ_* are $(0, 0)$. These explicit trigonometric polynomials permit independent contact jets and a compensating change of area. Set

$$r_0(s) = (2e^s - 1)^{-1}, \quad r_1(s) = (2e^{-s} - 1)^{-1},$$

and

$$h_{s,a} = 1 + (r_0(s) - 1)\chi_0 + (r_1(s) - 1)\chi_\pi + a\chi_*.$$

For small s, a , $h + h'' > 0$, so this is a smooth strictly convex body. At normals $0, \pi$, its support values stay one and its first support derivatives stay zero. Its two curvature radii are exactly $r_0(s), r_1(s)$, since the compensator has zero second jet at both normals. The supporting lines between neighboring horizontal translates therefore stay a distance one apart, and their support points attain that distance. The closest horizontal segment is unchanged.

The obstacle area is $\mathcal{A}(h) = \frac{1}{2} \int_0^{2\pi} [h^2 - (h')^2] d\theta$. Its derivative in a at $(s, a) = (0, 0)$ is $\int \chi_* d\theta > 0$. Since $\int \chi_* d\theta = 3\pi/4$, the analytic implicit function theorem supplies a real-analytic $a = a(s)$ with $a(0) = 0$ and $\mathcal{A}(h_{s,a(s)}) = \pi$ exactly. This variation has not altered either contact jet. At the disk, the vertical gap is two and every other nonhorizontal pair has still larger gap. The positive gap to the horizontal minimum, disjointness of translates, and clearance persist for

small s . Hence the two horizontal orientations are the entire ground-onset event. Their sum in (3.1) has leading coefficient

$$\frac{1}{(12 - \pi) \sinh(j \operatorname{arcosh} 2)},$$

independent of s . This assertion concerns leading coefficients, not equality of the finite-offset remainders or all other billiard data. Finally (3.3) gives $v_{j,1}/v_{j,0} = e^{2s}$ at every odd j . It separates the family; uniform conditioning follows from Theorem 3.2. $\square$

The fixed contact patches can be chosen once for this entire family. Even when several minimizing pairs compete, a recorded first-impact patch and its outgoing direction select the oriented channel. On a nonminimal channel one must also use its own window and the itinerary selection specified in Section 2. Counts in the ground collar alone eventually suppress a fixed gap difference $\Delta > 0$, since activation requires $j\Delta < \varepsilon_0$. The marked theorem neither changes that condition nor implicitly provides unobserved channel labels.

APPENDIX A.17. LEADING-DATA FACTORIZATION AND QUANTITATIVE ACQUISITION

A.17.1. What increasing collision order does not add at leading order. For one selected channel write $A_j = \lim_{d \downarrow 0} d^{-2} \Pr(E_{j,e}(d))$; this uppercase subscripted symbol is distinct from the free area A .

Proposition A.17.1 (One-flight factorization of the leading data). *At known gap g , the entire leading selected-channel probability and endpoint-metric hierarchy factors through $(A_1, v_{1,0}, v_{1,1})$. More explicitly, the two one-flight variances determine γ, c_0, c_1 , and*

$$A_j = A_1 \frac{\sinh \gamma}{\sinh(j\gamma)}, \quad v_{j,b} = v_{1,b} \frac{\coth(j\gamma)}{\coth \gamma} \quad (j \text{ odd}, b = 0, 1). \quad (\text{A.17.1})$$

The full endpoint matrix at either parity is then given by (A.16.3). Also $A = [2A_1 \sinh \gamma]^{-1}$, when the normalizer is unknown.

Proof. The one-flight case of Theorem 3.2 recovers both curvatures. Divide (3.1) at zero offset by its one-flight version and do the same in (3.3). These are precisely (A.17.1). The displayed Hessian formula covers the even orders as well, including their off-diagonal entries. Solving the first amplitude for A proves the final assertion. $\square$

Thus access to long odd records preserves access to the same leading contact parameters; it does not produce new leading inverse coordinates. The collision-order-uniform nonlinear calculus supplies something else: the relative boundary factorization, fixed-offset limiting law and higher-jet separation of Sections 7 and A.1. Its all-order smooth remainder also permits the following controlled bias cancellation. The latter uses classical polynomial extrapolation, rather than a new interpolation algorithm.

A.17.2. Finite-order bias cancellation. Fix an integer $m \geq 1$. All constants with subscript m below may depend on this fixed order; no uniform assertion as $m \rightarrow \infty$ is made. Put

$$\omega_l = (-1)^{l-1} \binom{m}{l}, \quad d_l = lh, \quad 1 \leq l \leq m, \quad (\text{A.17.2})$$

where $mh < \varepsilon_0$. Each observation uses the same selected channel at the different physical window $jg + d_l$, with j odd. The true gap and the indicated offsets are assumed known unless otherwise stated. At each window use $2K$ successful records from independent full-phase preparations, independently across the windows, and form the paired-position estimators $\hat{V}_{l,b}$ of Proposition A.16.2.

Theorem A.17.2 (Extrapolated recovery and preparation cost). *Set $\hat{v}_b^{[m]} = \sum_{l=1}^m \omega_l \hat{V}_{l,b}$. There is a reconstruction on the compact gap-curvature boxes of Theorem 3.2 such that, with probability at least $1 - \eta$,*

$$\max_b |\hat{\kappa}_b - \kappa_b| \leq C_m \left(h^m + \sqrt{\frac{\log(8m/\eta)}{K}} \right). \quad (\text{A.17.3})$$

The expected total number T of independent unconditioned preparations is comparable, with constants depending on m , to $Kh^{-2}e^{j\gamma}$. In addition, the reconstruction and the bound

$$T \leq C_m e^{j\gamma} h^{-2} [K + \log(2m/\eta)] \quad (\text{A.17.4})$$

hold together with probability at least $1 - \eta$, after adjusting constants in (A.17.3). For each sufficiently small accuracy $\epsilon > 0$, this yields the sufficient expected and high-probability acquisition order

$$C_m e^{j\gamma} \epsilon^{-(2+2/m)} \log(C_m/\eta) \quad (\text{A.17.5})$$

for curvature error at most ϵ . The experiment counts independent preparations, not independent successive impacts.

Proof. The m th forward difference annihilates every polynomial of degree less than m . Apply it at the integers $0, 1, \dots, m$ to obtain

$$\sum_{l=1}^m \omega_l = 1, \quad \sum_{l=1}^m \omega_l l^r = 0 \quad (1 \leq r < m).$$

The smooth right extension of $V_{j,b}(d)$ in Proposition A.16.1 has uniformly bounded derivatives through order m . Taylor's integral remainder at zero consequently proves

$$\left| \sum_{l=1}^m \omega_l V_{j,b}(lh) - v_{j,b} \right| \leq C_m h^m. \quad (\text{A.17.6})$$

This step uses the uniform differentiated nonlinear theorem. A pointwise $O(d)$ statement with a length-dependent constant would not suffice for the stated uniform assertion.

Each paired summand at offset lh lies in a common bounded interval, by endpoint coercivity and its normalization by $2lh$. Proposition A.16.2 and a union bound over the m windows give simultaneous errors $C\sqrt{\log(8m/\eta)/K}$ at both endpoints with failure probability at most $\eta/2$. Multiplication by $\sum_l |\omega_l| = 2^m - 1$, together with (A.17.6), gives the same form for the leading variance estimate. Projection onto the compact image of the leading observation map, followed by its uniform Lipschitz inverse, proves (A.17.3). It also defines the estimator when the extrapolated values do not belong to the exact image.

For the cost, put $p_l = \Pr(E_{j,e}(lh))$. The uniform threshold law and a smaller collar give

$$c(lh)^2 e^{-j\gamma} \leq p_l \leq C(lh)^2 e^{-j\gamma}.$$

Waiting for $2K$ independent successes has mean $2K/p_l$. Summing over the fixed number of windows proves the expected-cost assertion. For a tail estimate let T_l be that waiting time. Then $\Pr(T_l > N) = \Pr(\text{Bin}(N, p_l) < 2K)$. If $Np_l \geq 4K$, the exponential Markov inequality for the binomial law gives

$$\Pr(\text{Bin}(N, p_l) < Np_l/2) \leq e^{-Np_l/8}.$$

For example this follows by multiplying the Bernoulli moment generating functions and optimizing a negative parameter; the bound $\Pr(X \leq (1-t)\mathbb{E}X) \leq \exp(-t^2\mathbb{E}X/2)$ for $0 < t < 1$ follows from $1+x \leq e^x$ and $t + (1-t)\log(1-t) \geq t^2/2$. Choose N with $Np_l \geq \max(4K, 8\log(2m/\eta))$. The preceding lower bound on p_l and a union bound show that (A.17.4) fails with probability at most

$\eta/2$. Combining with the estimation event proves the simultaneous assertion; no independence between those two events is needed.

Finally take $h = c_m \epsilon^{1/m}$, with a sufficiently small fixed c_m , and K at least $C_m \epsilon^{-2} \log(C_m/\eta)$. Equation (A.17.3) is then at most ϵ , and both cost bounds have order (A.17.5). $\square$

At $m = 1$ the sufficient cost is $e^{j\gamma} \epsilon^{-4} \log(C/\eta)$, the explicit accuracy–cost interpretation of the original paired estimator. At $m = 2$ it is $e^{j\gamma} \epsilon^{-3} \log(C_2/\eta)$. These are guarantees for the specified procedures, not minimax lower bounds or a statistical advantage of high collision order. The exponential preparation factor persists, and the extrapolation constants and required boundary differentiability grow with m . There is no independent sensor-noise model hidden in the estimates.

A.17.3. Gap calibration and the physical window.

Proposition A.17.3 (Propagation of a timing error). *Suppose the programmed windows are $j\hat{g} + lh$, whereas the true gap is $g = \hat{g} + \Delta g$. Assume $|j\Delta g| \leq h/2$ and $mh + |j\Delta g| < \epsilon_0$. Use the programmed offset lh in each paired-position denominator. The reconstruction based on the supplied gap $\hat{g}$ then has error, with the same independent-preparation confidence convention,*

$$C_m \left[h^m + \sqrt{\frac{\log(8m/\eta)}{K}} + \frac{j|\Delta g|}{h} + |\Delta g| \right]. \quad (\text{A.17.7})$$

A sufficient calibration for preserving an $O(h^m)$ deterministic bias is $|\Delta g| \leq c_m h^{m+1}/j$. In particular the original $m = 1$ procedure requires an $O(h^2/j)$ gap accuracy by this direct normalization rule.

Proof. The actual offset is $d_l = lh - j\Delta g > 0$. The expectation of the paired statistic divided by $2lh$ is

$$\frac{d_l}{lh} V_{j,b}(d_l).$$

Uniform boundedness and a bounded offset derivative imply that its difference from $V_{j,b}(lh)$ is at most $C[j|\Delta g|/(lh) + j|\Delta g|]$. Since h is bounded above on the chosen collar, the first term absorbs the second after changing C_m . Sum with the fixed weights in (A.17.2) and use (A.17.6). The random summands remain uniformly bounded because $d_l/(lh)$ lies between $1/2$ and $3/2$. The concentration proof is unchanged. The joint gap–variance inverse bound (A.16.5) adds $C|\Delta g|$ for using the supplied gap. This proves (A.17.7) and its stated sufficient calibration. Thus a gap error is not treated as a cost-free shift of the physical observation window. $\square$

APPENDIX A.18. LOCAL CALIBRATION FROM THE ONSET PROBABILITIES

The known-gap experiment of Theorem A.17.2 and an experiment which must calibrate that gap are different. We first compute the sensitivity of the former statistic, and then give a count-based calibration procedure whose preparation cost can be included explicitly. Throughout this section m is fixed and j is odd. Constants may depend on m and on the compact geometric family, but not on j .

Proposition A.18.1 (The harmonic timing coefficient). *Let $V(d)$ denote either of the normalized physical endpoint variances at the chosen flight number, and let $v = V(0)$. For the programmed-offset normalization in Proposition A.17.3, put $\tau = j\Delta g$. The extrapolated population statistic is*

$$W_m(\tau) = \sum_{l=1}^m \omega_l \left(1 - \frac{\tau}{lh} \right) V(lh - \tau).$$

Its first timing derivative satisfies

$$W'_m(0) = -\frac{vH_m}{h} + O_m(1), \quad H_m = \sum_{l=1}^m \frac{1}{l}. \quad (\text{A.18.1})$$

The remainder is uniform in j . In particular extrapolation does not cancel the leading normalization error.

Proof. Differentiate the displayed statistic to obtain $-\sum_l \omega_l [V(lh)/(lh) + V'(lh)]$. The uniform estimates $V(d) = v + O(d)$ and $V'(d) = O(1)$ reduce the leading term to $-v \sum_l \omega_l / lh$. The binomial identity

$$\sum_{l=1}^m \frac{(-1)^{l-1} \binom{m}{l}}{l} = \int_0^1 \frac{1 - (1-x)^m}{x} dx = \sum_{l=1}^m \frac{1}{l}$$

proves (A.18.1). The last equality follows on setting $y = 1 - x$ and summing $1 + y + \dots + y^{m-1}$. $\square$

A.18.1. A population calibration lemma. Suppose that a coarse location g_0 is supplied with

$$|j(g - g_0)| \leq h/4. \quad (\text{A.18.2})$$

This is a local bracketing assumption, not a fine calibration at scale h^{m+1}/j . The cost of locating this coarse bracket is not included in the experiment below. Write $\tau = j(g - g_0)$ and observe the selected channel at the $m + 1$ programmed times $jg_0 + lh$, $1 \leq l \leq m + 1$. Let p_l be the respective probabilities. By the uniform threshold law,

$$p_l = K_j(lh - \tau)^2 F_j(lh - \tau), \quad K_j = \frac{1}{2A \sinh(j\gamma)}. \quad (\text{A.18.3})$$

All actual offsets are positive. The common factor K_j is unknown and will not have to be estimated.

Given positive approximations $\hat{p}_l$, interpolate their square roots at $z = lh$ by a polynomial $\hat{P}$ of degree at most m . If it has a unique real zero in $[-h/2, h/2]$, define $\hat{\tau}$ to be that zero. Otherwise set $\hat{\tau} = 0$. This convention always keeps $\hat{\tau}$ in the indicated interval.

Lemma A.18.2 (Stable extrapolation of the onset zero). *There are $h_m, \epsilon_m > 0$ such that, if $0 < h \leq h_m$ and $\max_l |\hat{p}_l/p_l - 1| \leq \epsilon \leq \epsilon_m$, then $\hat{P}$ has exactly one zero in $[-h/2, h/2]$ and*

$$|\hat{\tau} - \tau| \leq C_m(h^{m+1} + h\epsilon). \quad (\text{A.18.4})$$

After decreasing h_m, ϵ_m , the right side is at most $h/8$. The assertion holds uniformly in j without knowing K_j .

Proof. Put $H_j(d) = \sqrt{F_j(d)}$. Positivity and the uniform differentiated threshold law give $H_j(0) = 1$ and bounded right derivatives through order m . At the positive nodal arguments $d_l = lh - \tau$,

$$K_j^{-1/2} \sqrt{p_l} = d_l H_j(d_l).$$

Let $Q(z)$ be the degree- m Taylor polynomial obtained by multiplying $z - \tau$ by the degree- $(m - 1)$ right Taylor polynomial of H_j at zero. Then $Q(\tau) = 0$, and $Q'(z) = 1 + O_m(h)$ for $|z| \leq h/2$. Taylor's integral remainder at the positive d_l shows that the exact nodal values differ from $Q(lh)$ by at most $C_m h^{m+1}$.

For the fixed nodes $1, \dots, m + 1$, let L_l be their Lagrange basis. On $[-1/2, 1/2]$ both $\sum |L_l|$ and $\sum |L'_l|$ are bounded by constants depending only on m . Therefore interpolation of the nodal Taylor remainders changes the value by at most $C_m h^{m+1}$ and the z derivative by at most $C_m h^m$ throughout $[-h/2, h/2]$. This argument uses Taylor coefficients and polynomial evaluation only; it does not evaluate a physical probability at a negative offset.

If the probability errors are at most ϵ relatively, the square-root errors are at most $C\epsilon$ relatively. The normalized nodal square roots have size $O_m(h)$. The same Lagrange bounds thus add at most $C_m h \epsilon$ to the value error and $C_m \epsilon$ to the derivative error. Hence $K_j^{-1/2} \hat{P}$ has derivative at least $1/2$ on the interval for sufficiently small h, ϵ , and its value at τ is at most $C_m(h^{m+1} + h\epsilon)$ in absolute value. Since $|\tau| \leq h/4$, its values at the two interval endpoints have opposite signs after decreasing these constants' allowed neighborhood. Monotonicity proves existence and uniqueness of the root and the mean-value theorem proves (A.18.4). The common unknown positive factor does not change any zero. $\square$

A.18.2. Calibration and conditional recovery in one cost bound.

Theorem A.18.3 (Locally self-calibrated curvature acquisition). *Assume the coarse bracket (A.18.2), a fixed selected channel, and independent full-phase preparations. For $0 < h \leq h_m$ and $0 < \eta < 1/2$, there is an experiment with the following properties. It estimates the gap by $\hat{g} = g_0 + \hat{\tau}/j$ and the two contact curvatures by $\hat{\kappa}_b$, and, with probability at least $1 - \eta$,*

$$|\hat{g} - g| \leq C_m h^{m+1}/j, \quad \max_b |\hat{\kappa}_b - \kappa_b| \leq C_m h^m. \quad (\text{A.18.5})$$

Its total number T of unconditioned preparations, including calibration and conditional-position acquisition, satisfies

$$\mathbb{E}T \leq C_m e^{j\gamma} h^{-(2m+2)} \log(C_m/\eta), \quad T \leq C_m e^{j\gamma} h^{-(2m+2)} \log(C_m/\eta) \quad (\text{A.18.6})$$

with the latter bound holding together with (A.18.5) with probability at least $1 - \eta$. Thus for curvature accuracy ϵ a sufficient preparation order remains

$$C_m e^{j\gamma} \epsilon^{-(2+2/m)} \log(C_m/\eta).$$

Unlike Theorem A.17.2, this statement includes the fine-gap calibration experiment. It still presupposes the local coarse bracket, fixed channel patches, and noiseless collision recording.

Proof. For each of the $m+1$ calibration windows, start independent preparations until r selected events have occurred. Let T_l be this stopping time and put $\hat{p}_l = r/T_l$. We record the waiting-time estimate used here. For independent Bernoulli trials of probability p ,

$$\Pr\{|(r/T_r)/p - 1| > \epsilon\} \leq 2e^{-cr\epsilon^2}, \quad \mathbb{E}T_r = r/p, \quad (\text{A.18.7})$$

for $0 < \epsilon < 1/4$ and $r \geq C\epsilon^{-2}$, with absolute constants. Indeed $T_r > n$ is equivalent to fewer than r successes in n trials, and $T_r \leq n$ is equivalent to at least r successes. Apply the binomial exponential Markov bound at the two integers nearest $r/(p(1-\epsilon))$ and $r/(p(1+\epsilon))$. Rounding errors are absorbed by $r\epsilon \geq C$. The relative deviations of the binomial means from r are comparable to ϵ , which gives the exponential bound. The expectation is the sum of r independent geometric waiting times of mean $1/p$.

Choose $\epsilon = c_m h^m$ with a sufficiently small fixed c_m and

$$r = \lceil C_m h^{-2m} \log(C_m/\eta) \rceil.$$

Use (A.18.7) and a union bound over the calibration windows. Except on an event of probability at most $\eta/4$, their relative errors are at most $c_m h^m$. Lemma A.18.2 then gives $|\hat{\tau} - \tau| \leq C_m h^{m+1}$ and the first estimate in (A.18.5).

Use fresh independent preparations for the conditional-position stage, at the m programmed windows $j\hat{g} + lh$. At each window collect $2K$ selected records and form the paired variance estimator of Proposition A.16.2, normalized by the programmed lh . Take $K = \lceil C_m h^{-2m} \log(C_m/\eta) \rceil$ and use the extrapolation weights (A.17.2). Conditional on the entire calibration experiment,

these new preparations are independent, and all the variance concentration bounds are uniform in the chosen windows. On the calibration success event,

$$\frac{j|g - \hat{g}|}{h} = \frac{|\tau - \hat{\tau}|}{h} \leq C_m h^m.$$

For small h this is in the timing collar of Proposition A.17.3. That proposition and the conditional concentration bound give the second estimate of (A.18.5), with a further failure probability at most $\eta/4$. Use a fixed compact reconstruction box enlarged slightly to include all the supplied nearby gap values.

We check cost also outside the successful calibration event. The root rule always gives $|\hat{\tau}| \leq h/2$, whereas $|\tau| \leq h/4$. Consequently every actual endpoint-stage offset is

$$lh + \hat{\tau} - \tau \in [h/4, (m + 3/4)h].$$

The calibration offsets are in $[3h/4, (m + 5/4)h]$. Decrease h_m so both ranges lie in the common threshold collar. In both stages every success probability is bounded above and below by fixed multiples of $h^2 e^{-j\gamma}$, uniformly over all calibration outcomes. In particular a failed polynomial fit cannot produce a nonpositive offset or an infinite waiting-time expectation. The expectation identity in (A.18.7), conditioning on the pilot in the second stage, gives

$$\mathbb{E}T \leq C_m e^{j\gamma} h^{-2} (r + K).$$

For the high-probability cost, the equivalence with binomial counts and $\Pr\{\text{Bin}(n, p) < np/2\} \leq e^{-np/8}$ show that waiting for r or $2K$ successes costs at most $Cp^{-1}(r + K + \log(C_m/\eta))$, simultaneously over the fixed number of windows. These bounds also hold conditionally on the pilot, with the same constants. A union bound, allocating the remaining $\eta/2$ to the waiting-time tails, proves the simultaneous assertion in (A.18.6). Substituting r, K proves its order, and $h = c_m \varepsilon^{1/m}$ gives the final accuracy statement. $\square$

The calibration gain comes from estimating the zero of the square root of an onset probability, not from cancelling the nonzero harmonic coefficient in (A.18.1). The unknown area and the rare-event prefactor cancel in that zero. The factor $e^{j\gamma}$ nevertheless persists in the number of preparations needed to locate it. No minimax claim, independence of successive impacts, or uncharged fine timing measurement is used.

APPENDIX A.19. A SHARP STABILITY SCALE AT THE CIRCULAR CONFIGURATION

The exact infinite-sequence identification in Theorem A.8.1 does not resolve conditioning at a repeated node. We now give both a sequence-level estimate for the coalescing path and a matching positive estimate relative to the circular configuration. The latter is explicitly a reference-point estimate; it is not asserted for every pair of noncircular triples.

A.19.1. An observable dispersion statistic. Use the equal-gap family (A.8.1), with R fixed and $g = 1 - 2R$. Define, from its first three positive amplitudes,

$$m = \frac{2C_2}{C_1}, \quad J = \frac{C_3}{C_1} - f(m), \quad f(x) = \frac{x^2}{4 - x^2}. \quad (\text{A.19.1})$$

For the circular member let $\kappa_* = 1/R$ and $x_* = (1 + g\kappa_*)^{-1}$.

Theorem A.19.1 (Three-amplitude circular-reference stability). *On a sufficiently small compact positive-curvature neighborhood in the fixed- R family, $J \geq 0$ and there are constants $c, C > 0$ such that*

$$c(|m - x_*| + \sqrt{J}) \leq \max_{0 \leq r \leq 2} |\kappa_r - \kappa_*| \leq C(|m - x_*| + \sqrt{J}). \quad (\text{A.19.2})$$

In particular, if C_j^* are the circular amplitudes and $\delta = \max_{1 \leq j \leq 3} |C_j - C_j^*| \leq 1$, then

$$\text{dist}_{\text{match}}(\kappa, \kappa_*) \leq C\sqrt{\delta}. \quad (\text{A.19.3})$$

The statistic J vanishes exactly when the three contact curvatures coincide. It measures their dispersion without a supplied area normalizer or channel labels.

Proof. Set $x_r = \text{sech } \gamma_r = (1 + g\kappa_r)^{-1}$ and

$$p_r = \frac{\text{csch } \gamma_r}{\sum_s \text{csch } \gamma_s}.$$

The x_r lie in a fixed compact interval inside $(0, 1)$, while $p_r \geq p_* > 0$ and $\sum p_r = 1$. The identities $\sinh(2\gamma) = 2 \sinh \gamma \cosh \gamma$ and $\sinh(3\gamma) = \sinh \gamma (4 \cosh^2 \gamma - 1)$ give

$$m = \sum_r p_r x_r, \quad J = \sum_r p_r f(x_r) - f(m). \quad (\text{A.19.4})$$

On the common interval,

$$f''(x) = \frac{8(4 + 3x^2)}{(4 - x^2)^3}$$

has positive lower and finite upper bounds. Taylor's integral formula at m , followed by $\sum p_r(x_r - m) = 0$, therefore gives

$$\frac{c_f}{2} \sum_r p_r (x_r - m)^2 \leq J \leq \frac{C_f}{2} \sum_r p_r (x_r - m)^2. \quad (\text{A.19.5})$$

This proves nonnegativity and the equality criterion. Since all weights are bounded below, the square root in (A.19.5) is comparable to $\max_r |x_r - m|$. Also

$$|m - x_*| \leq \max_r |x_r - x_*|, \quad \max_r |x_r - x_*| \leq |m - x_*| + \max_r |x_r - m|.$$

The reverse comparison follows from $\max_r |x_r - m| \leq 2 \max_r |x_r - x_*|$. Finally the scalar change $\kappa \mapsto (1 + g\kappa)^{-1}$ is bi-Lipschitz on the compact interval. This proves (A.19.2).

All amplitude denominators C_1 are bounded away from zero. At the circle $m = x_*$ and $J = 0$. The two ratios in (A.19.1) are Lipschitz in the first three amplitudes on this set, and f has bounded derivative. Hence $|m - x_*| \leq C\delta$ and $0 \leq J \leq C\delta$. Substitution gives (A.19.3). Because the reference triple has all entries equal, its matching distance is exactly the maximum coordinate distance, independent of ordering. $\square$

The theorem quantifies distance to a specified circular table. It does not recover an arbitrary unordered triple from only three numbers, nor does it claim a one-half-Hölder inverse between every two nearby noncircular triples. The exact infinite-sequence theorem and its multiplicity recovery remain separate statements.

A.19.2. The infinite absolute-amplitude topology. For a fixed $a \geq 0$ write

$$\|D\|_a = \sup_{j \geq 1} e^{aj} |D_j|. \quad (\text{A.19.6})$$

Restrict the coalescing path of Proposition A.8.3 to a small closed interval on which all exponents satisfy $\gamma_r(s) \geq \gamma_- > a$.

Theorem A.19.2 (Sharp circular-reference exponent for the full sequence). *Along $\alpha = \zeta = 0$, $\beta = s$ one has*

$$\|C(s) - C(0)\|_a \leq B_a s^2, \quad \text{dist}_{\text{match}}(\kappa(s), \kappa(0)) \asymp |s|. \quad (\text{A.19.7})$$

For every table in a sufficiently small fixed- R neighborhood of the circle, the positive estimate

$$\text{dist}_{\text{match}}(\kappa, \kappa_*) \leq C_a \|C - C^*\|_a^{1/2} \quad (\text{A.19.8})$$

holds when the right side is small. No exponent larger than one half can replace it throughout that neighborhood, even when the entire sequence is observed. This sharpness concerns the absolute-error norm (A.19.6), with a strict exponential margin.

Proof. On the indicated path the three radii are $R + 36s$, $R - 18s$, $R - 18s$, and $A(s) = A(0) + 45\pi s^2/4$. With $\Phi_j(r) = \text{csch}(j \text{arcosh}(1 + g/r))$ the amplitude is

$$C_j(s) = \frac{\Phi_j(R + 36s) + 2\Phi_j(R - 18s)}{A(s)}.$$

Both numerator and denominator have zero first derivative at zero. On the compact interval the derivatives through order two of the radius-to-exponent map and of A^{-1} are bounded. Differentiate

$$\text{csch}(j\gamma) = \frac{2e^{-j\gamma}}{1 - e^{-2j\gamma}}$$

twice. The denominator is bounded below by $1 - e^{-2\gamma_-}$; each differentiation contributes at most a factor $C(1 + j)$. Consequently

$$\sup_{|s| \leq s_*} |C_j''(s)| \leq B(1 + j)^2 e^{-j\gamma_-}. \quad (\text{A.19.9})$$

Taylor's integral remainder and $C_j'(0) = 0$ imply the first part of (A.19.7), since $\sup_j (1 + j)^2 e^{-(\gamma_- - a)j} < \infty$. The curvature displacement is comparable to $|s|$ because reciprocal radius has a derivative bounded above and away from zero there.

For a general nearby table, the first three absolute amplitude errors are bounded by (A.19.6). Thus Theorem A.19.1 gives (A.19.8); no unstable truncation of a Möbius inversion is used. If an exponent $\theta > 1/2$ were valid in its place, the specified path would give $b|s| \leq LB_a^\theta |s|^{2\theta}$ for arbitrarily small nonzero s , which is impossible. This proves sharpness. In particular, (A.19.8) also supplies a quadratic lower bound for the sequence discrepancy along that path; the order in (A.19.7) is not merely a one-sided diagnosis. $\square$

Exact identification and stability in an absolute-amplitude norm are different questions. The two theorems give a positive sharp scale for one specified singular reference, while the channel-resolved endpoint observation remains Lipschitz there because it retains different data. None of these assertions treats relative error at arbitrarily small amplitudes as equivalent to (A.19.6).

APPENDIX A.20. MEASURABLE RECONSTRUCTION ON COMPACT PARAMETER SETS

All compact discrepancy minimizations in the sequel use the same measurable convention. This removes a possible ambiguity at repeated roots or at data having several best fits; uniqueness of a minimizing parameter is unnecessary.

Lemma A.20.1 (A specified Borel minimizing selection). *Let K be a nonempty compact metric space, let $\mathcal{Y}$ be a standard Borel data space, and suppose $G : \mathcal{Y} \times K \rightarrow \mathbb{R}$ is Borel measurable and continuous in its second argument. There is a Borel map $\hat{\theta} : \mathcal{Y} \rightarrow K$ such that*

$$G(y, \hat{\theta}(y)) = \min_{\theta \in K} G(y, \theta).$$

A single choice of finite closed-ball covers of K specifies such a map for every data value.

Proof. Fix, for each integer $r \geq 1$, a finite cover of K by closed balls $B_{r,1}, \dots, B_{r,N_r}$ of radius 2^{-r} , with centers in K , and fix their indicated order. The minimum $v(y) = \min_K G(y, \cdot)$ is Borel: it equals the infimum of $G(y, \cdot)$ over any fixed countable dense subset of K . For any nonempty fixed compact subset $F \subset K$, the same argument shows that $v_F(y) = \min_F G(y, \cdot)$ is Borel. For $F = \emptyset$, set $v_F = +\infty$.

At the first stage choose the least index i_1 for which $v_{K \cap B_{1,i_1}}(y) = v(y)$. At stage r , after the earlier choices, choose the least i_r for which

$$v_{K \cap B_{1,i_1} \cap \dots \cap B_{r,i_r}}(y) = v(y).$$

Such a choice always exists because the current nonempty compact set of minimizers is covered by the balls at the next stage. For each fixed finite prefix of indices the compact set in this expression is fixed, so its equality event is Borel. Induction and the finite number of indices at each stage show that every selected index is Borel.

Intersect the minimizer set with the selected balls. These are nested nonempty compact sets with diameter at most 2^{1-r} at stage r . Their intersection consists of a single point $\hat{\theta}(y)$. The centers of the selected balls converge to it, since their distances from this point are at most 2^{-r} . The centers form Borel maps into K ; their pointwise limit is Borel and is a minimizer by continuity. $\square$

Apply the lemma with G equal to the maximum discrepancy between the measured finite vector and the continuous physical observation map. The physical compact set, its covers, and their order are fixed before observations. Every resulting estimator in Theorems A.9.1, A.10.1, and A.21.2 is then a Borel statistic. The proof is an existence and specification result, not an assertion of computational efficiency. It applies equally to compact constrained variance fits.

The polynomial root-or-zero convention in Lemma A.18.2 is Borel as well. One direct verification is to minimize the squared polynomial on a fixed compact subinterval; the minimum is a continuous function of its coefficients. Existence of a zero on any closed rational interval is therefore a closed condition. Existence of two distinct zeros in the prescribed interval is the countable union of conditions asserting zeros in two disjoint closed intervals with rational endpoints, also allowing the two fixed end points. Existence of exactly one zero is consequently Borel. On that set its location is recovered measurably from these interval tests; on its complement the declared value is zero. This includes the zero polynomial, for which uniqueness fails. Thus the pilot and the final compact fit both define statistics on all data outcomes.

APPENDIX A.21. LOCALLY CALIBRATED RECONSTRUCTION FROM UNLABELLED COUNTS

This section uses counts only, with neither channel labels nor impact positions. There are three unoriented equal-gap ground channels. For an actual-curvature conclusion we assume equal facing curvatures within each channel. The asymmetry allowed in the general forward theorem is not silently removed there: without the additional equality, the same scalar reconstruction concerns (A.10.3). All preparations are independent and have normalized full phase volume. Collision counts and programmed physical times are recorded without instrument noise. Statistical noise arises from the random preparation.

We consider either a sufficiently small compact subfamily of the physical family in Theorem A.10.1, with $J = 3$, or a compact geometric family whose leading data lie in the four-amplitude neighborhood of Theorem A.9.1, with $J = 4$. In the latter case the normalizer is treated as an independent nuisance parameter in inversion; the observed probabilities still come from a physical configuration. For $1 \leq j \leq J$, the ground-onset law reads

$$p_j(d) := \Pr\{N_{jg+d} \geq j+1\} = d^2\{C_j + dB_j(d)\}, \quad 0 \leq d \leq d_*. \quad (\text{A.21.1})$$

The functions in braces are positive, bounded away from zero, and have bounded derivatives at each fixed order, uniformly on the chosen family. At these offsets the event is equivalently $N_{jg+d} = j+1$.

Lemma A.21.1 (Amplitude error from finitely many Bernoulli experiments). *Suppose g is known. Fix $m \geq 1$, and at each of the Jm times $jg + lh$, $1 \leq j \leq J$, $1 \leq l \leq m$, perform n independent preparations. Write $\hat{p}_{j,l}$ for the observed success fraction and put*

$$\omega_l = (-1)^{l-1} \binom{m}{l}, \quad \hat{C}_j = \sum_{l=1}^m \omega_l \frac{\hat{p}_{j,l}}{(lh)^2}.$$

For small h and $0 < \eta < 1/2$, with probability at least $1 - \eta$,

$$\max_{j \leq J} |\hat{C}_j - C_j| \leq C_m \left(h^m + \sqrt{\frac{\log(C_m/\eta)}{nh^2}} + \frac{\log(C_m/\eta)}{nh^2} \right). \quad (\text{A.21.2})$$

Proof. The weights satisfy $\sum_l \omega_l = 1$ and $\sum_l \omega_l l^k = 0$ for $1 \leq k < m$. Taylor's formula for $p_j(d)/d^2$ in (A.21.1), with its uniformly bounded m th derivative, gives a population bias at most $C_m h^m$. For a Bernoulli variable X of mean p , exponential Markov bounds give

$$\Pr\{|\hat{p} - p| > \sqrt{2px/n} + 2x/(3n)\} \leq 2e^{-x}.$$

For completeness, for $0 < t < 3$ the power series of the exponential and $\mathbb{E}|X - p|^k \leq \mathbb{E}(X - p)^2 \leq p$ for $k \geq 2$ give $\log \mathbb{E}e^{t(X-p)} \leq pt^2/[2(1-t/3)]$; the same bound holds for $-t$. Optimization for the sum of n independent variables gives the stated two-sided bound with these harmless enlarged constants. Here $p_j(lh) \leq C_m h^2$. Dividing by $(lh)^2$, summing the fixed weights, and taking $x = \log(2Jm/\eta)$ proves the stochastic terms by a union bound. Independence is used between preparations, not between collisions of one trajectory. $\square$

We now include fine timing instead of assuming it. A coarse bracket $|g - g_0| \leq h/4$ is supplied. Its acquisition is not charged below. A pilot at order $j = 1$ suffices for all the finitely many subsequent orders. The safety margin in the second stage will be enlarged to cover every possible pilot outcome.

Theorem A.21.2 (Count-only acquisition with charged fine calibration). *Fix $m \geq 1$ and one of the preceding compact geometric families. There are $h_m > 0$ and $C_m < \infty$ such that, for $0 < h \leq h_m$, $0 < \eta < 1/2$ and a supplied bracket $|g - g_0| \leq h/4$, an experiment using only unlabelled counts produces estimates $\hat{g}, \hat{A}, \hat{\kappa}$ with*

$$|\hat{g} - g| \leq C_m h^{m+1}, \quad |\hat{A} - A| \leq C_m h^m, \quad \text{dist}_{\text{match}}(\hat{\kappa}, \kappa) \leq C_m h^{m/3} \quad (\text{A.21.3})$$

with probability at least $1 - \eta$. Its total number T of raw preparations, including the pilot, satisfies

$$\mathbb{E}T \leq C_m h^{-(2m+2)} \log(C_m/\eta), \quad T \leq C_m h^{-(2m+2)} \log(C_m/\eta) \quad (\text{A.21.4})$$

with the second estimate holding jointly with (A.21.3) with probability at least $1 - \eta$. Consequently a sufficient raw-preparation order for matching curvature accuracy ε is

$$C_m \varepsilon^{-(6+6/m)} \log(C_m/\eta). \quad (\text{A.21.5})$$

No channel labels, selected positions, or supplied fine-gap estimate enter this experiment. The result is local and sufficient, not a minimax theorem or a global procedure for locating the initial bracket.

Proof. Put $\tau = g - g_0$. At each of the pilot times $g_0 + lh$, $1 \leq l \leq m + 1$, count successes of $N_t \geq 2$ until r successes have occurred, using fresh preparations at every window. Let T_l be the waiting time and let $\hat{p}_l = r/T_l$. Formula (A.21.1) at $j = 1$ has the form

$$p_1(d) = C_1 d^2 F(d), \quad F(0) = 1,$$

where F is positive and has the required bounded right derivatives. The unknown $C_1 > 0$ replaces the unknown factor K_j in Lemma A.18.2; its proof uses no channel label. Interpolate the square roots of the pilot fractions at $z = lh$ and use precisely that lemma's root-or-zero rule. Thus $\hat{g} = g_0 + \hat{\tau}$ with $|\hat{\tau}| \leq h/2$ on every outcome. Choose

$$r = \left\lceil C_m h^{-2m} \log(C_m/\eta) \right\rceil.$$

The negative-binomial bound (A.18.7) gives relative errors at most $c_m h^m$, simultaneously, outside an event of probability $\eta/4$. The root estimate then gives $|\hat{g} - g| \leq C_m h^{m+1}$. Every pilot offset lies in $[3h/4, (m+5/4)h]$, so its success probability is bounded below by $c_m h^2$. The expected pilot cost is at most $C_m h^{-2}(m+1)r$. The same binomial waiting-time tail used in Theorem A.18.3 bounds the pilot cost by $C_m h^{-(2m+2)} \log(C_m/\eta)$ except on a further event of probability at most $\eta/4$.

For the second stage fix $L = J + 1$ and perform n fresh independent preparations at each time $j\hat{g} + Ll h$, with $j \leq J$, $l \leq m$. This stage observes only $\mathbf{1}_{\{N_i \geq j+1\}}$ and uses the estimator

$$\hat{C}_j = \sum_{l=1}^m \omega_l \frac{\hat{p}_{j,l}}{(Llh)^2}, \quad n = \left\lceil C_m h^{-(2m+2)} \log(C_m/\eta) \right\rceil.$$

The total second-stage cost Jmn is deterministic. Even if the pilot fails, $|\hat{g} - g| \leq 3h/4$. Hence every actual second-stage offset is between $(L - 3J/4)h$ and $(Lm + 3J/4)h$, and in particular exceeds h . Decreasing h_m places this entire range in the common collar. Thus unsuccessful pilots cannot produce a nonpositive offset or an undefined normalization.

Condition on the pilot. The fresh fractions have Bernoulli means $p_j(Llh + j(\hat{g} - g))$, each bounded by $C_m h^2$ uniformly over all pilot outcomes. The concentration proof of Lemma A.21.1, with nodes Ll , therefore still applies. On the successful pilot event, differentiation of (A.21.1) gives $|p'_j(d)| \leq C_m h$ throughout the relevant offset intervals. The mean-value theorem consequently gives

$$\left| \frac{p_j(Llh + j(\hat{g} - g)) - p_j(Llh)}{(Llh)^2} \right| \leq C_m \frac{|\hat{g} - g|}{h} \leq C_m h^m.$$

This explicitly accounts for the uncanceled denominator/timing error. Taylor extrapolation at the unshifted nodes Llh has bias $C_m h^m$, and the chosen n makes both concentration terms at most $C_m h^m$. A conditional union bound therefore gives $\max_{j \leq J} |\hat{C}_j - C_j| \leq C_m h^m$, with an additional failure probability at most $\eta/2$.

In the three-amplitude physical case, fit the observations $(\hat{g}, \hat{C}_1, \hat{C}_2, \hat{C}_3)$ over a sufficiently small compact admissible family containing the true member and apply Theorem A.10.1. Its true spacing discrepancy is $O_m(h^{m+1})$ and its amplitude discrepancy is $O_m(h^m)$, so (A.21.3) follows, retaining the pilot as the stated gap estimate. In the four-amplitude case, augment the coefficient map in Theorem A.9.1 by the coordinate g . Its derivative has the nonsingular four-by-four block already proved there, and the identity in the g coordinate. Thus its coefficient inverse is jointly locally Lipschitz in g and the four amplitudes. The same compact minimum-discrepancy argument and the cubic root bound give the asserted area and curvature estimates. On the failure set arbitrary compact fallback estimates may be used; no success assertion depends on them. Adding the pilot and deterministic second-stage costs proves (A.21.4). Finally choose $h = c_m \varepsilon^{3/m}$, with c_m sufficiently small, to obtain (A.21.5). $\square$

The exponent in (A.21.5) has two different sources: amplitude acquisition costs $h^{-(2m+2)}$, whereas coalescing unlabelled curvatures have matching error $h^{m/3}$. The selected-position experiment of Theorem A.18.3 instead has a Lipschitz curvature inverse and sufficient cost $e^{j\gamma} \varepsilon^{-(2+2/m)}$. These are different observations and different experiments. The sharp deterministic

exponent $1/3$ does not by itself prove optimality of a statistical sampling rate. All constants here may depend on fixed m , on $J \leq 4$, and on the compact family; the statement makes no uniform extrapolation-order or large- j count-only cost claim.

APPENDIX A.22. PHYSICAL INDISTINGUISHABILITY IN LOCAL COUNT DESIGNS

The pairwise cubic inverse has a statistical counterpart in the actual finite-offset count laws. We prove it within the specified Bernoulli query experiment, including adaptive stopping. This result concerns fixed bounded flight numbers; it is logically distinct from the long-bridge experiment of Section A.7.

Fix $R = 1/4$ in the physical support family (A.8.1), put $\alpha = 0$, and let $\theta = (\beta, \zeta)$ vary near zero. The common gap is $g = 1/2$. A query at (j, d) consists of one fresh full-phase preparation and the binary report $\mathbf{1}_{\{N_{jg+d} \geq j+1\}}$. We do not give the experiment the lower values of N_{jg+d} or a collision position. Write its probability as $p_{j,d}(\theta)$.

Lemma A.22.1 (Cubic physical indistinguishability). *For each fixed $J < \infty$ there are $d_*, s_*, c, C > 0$ such that, for $1 \leq j \leq J$, $0 < d \leq d_*$, and $|s| \leq s_*$,*

$$|p_{j,d}(s, 0) - p_{j,d}(-s, 0)| \leq Cd^2|s|^3, \quad cd^2 \leq p_{j,d}(\pm s, 0) \leq Cd^2 < \frac{1}{2}. \quad (\text{A.22.1})$$

The matching distance between their unordered curvature triples is bounded above and below by positive multiples of $|s|$ for $s \neq 0$.

Proof. A rotation of the triangular lattice and the entire table by $\pi/3$ preserves the preparation measure and the unlabelled count. On (β, ζ) it acts as a rotation through $2\pi/3$, up to reversal of its sign. This follows directly by replacing ϑ by $\vartheta - \pi/3$ in $(1 - \cos 6\vartheta)(\beta \cos 2\vartheta + \zeta \sin 2\vartheta)$. No nonzero linear functional is invariant under that rotation. The gradient of $p_{j,d}(\theta)$ at zero must therefore vanish.

Theorem 3.1 and the equal-gap property give a smooth extension of $d^{-2}p_{j,d}(\theta)$ to the closed collar, with bounded third parameter derivatives uniformly for $j \leq J$. Taylor expansion along $(s, 0)$ and $(-s, 0)$ cancels the constant and quadratic terms; the linear term is zero. This proves the first estimate for the full physical probabilities, not merely their leading amplitudes. Their positive continuous leading coefficients and a smaller common collar prove the second estimates. The two radii multisets are

$$(R + 36s, R - 18s, R - 18s), \quad (R - 36s, R + 18s, R + 18s).$$

For $s > 0$ sufficiently small, ordered matching of their reciprocals has distance $36s/(R^2 - 324s^2)$, as in Theorem A.9.2. Replacing s by $|s|$ proves the last assertion. $\square$

For a stopped adaptive design, a policy chooses the next query and whether to stop using only the previous binary reports and independent randomization. Its rules are the same under the two possible tables. Suppose all its queries have $j \leq J$ and $0 \leq d \leq d_*$ on every history. Define its weighted exposure by

$$\mathcal{B} = \max_{\theta \in \{(s,0), (-s,0)\}} \mathbb{E}_\theta \sum_{i=1}^T d_i^2.$$

Queries at $d = 0$ are deterministic failures and contribute zero. Bounds uniform over a fixed neighborhood mean the same exposure bound holds in particular for each pair used below.

Theorem A.22.2 (A lower bound for count queries). *There is $c > 0$ such that any predetermined or stopped adaptive design as above with weighted exposure at most B uniformly on the local physical family has worst-case expected matching curvature error at least $c(1 + B)^{-1/6}$. In*

particular, if every queried offset satisfies $d_i \leq Lh$ on every history and $\sup_\theta \mathbb{E}_\theta T \leq N$, the lower bound is

$$c(1 + L^2 h^2 N)^{-1/6}. \quad (\text{A.22.2})$$

For each fixed confidence level $1 - \eta_0 > 1/2$, accuracy ε throughout this local family consequently requires

$$N \geq c_{L, \eta_0} h^{-2} \varepsilon^{-6} \quad (\text{A.22.3})$$

for sufficiently small ε . The conclusion remains valid when the exact common gap is supplied to the procedure.

Proof. For $0 < p, q < 1$, the inequality $\log x \leq x - 1$ gives

$$D_{\text{KL}}(\text{Ber}(p) \parallel \text{Ber}(q)) \leq \frac{(p - q)^2}{q(1 - q)}.$$

Thus Lemma A.22.1 bounds the divergence of a single query under the alternatives s and $-s$ by $Cd^2 s^6$. For a predetermined schedule, the log likelihood is a sum and its expectation is at most $Cs^6 \sum_i d_i^2$.

For an adaptive policy, condition on each common observed history and on its random seed. The next query, or stop decision, is then the same in the two experiments. The conditional divergence is again at most $Cd_i^2 s^6$; a stop symbol contributes zero. Applying the chain rule to the first M steps, with the stopped process extended by stop symbols, gives

$$D_{\text{KL}}(\mathbf{P}_s^{[M]} \parallel \mathbf{P}_{-s}^{[M]}) \leq Cs^6 \mathbb{E}_s \sum_{i \leq \min(T, M)} d_i^2.$$

The bound passes to the complete stopped transcript. Indeed relative entropy is the supremum of its finite measurable-partition values; conditioning these partitions on increasing finite histories yields that supremum, while monotone convergence bounds the right side by $Cs^6 B$. Equivalently, this is the chain rule for a stopped sequence obtained by truncation. A finite expected number of queries ensures that the stop time is finite almost surely; if only weighted exposure is bounded, the same finite-history argument applies to any measurable estimator based on the infinite transcript. Hence

$$D_{\text{KL}}(\mathbf{P}_s \parallel \mathbf{P}_{-s}) \leq Cs^6 B. \quad (\text{A.22.4})$$

We recall the testing step to fix constants and topology. Coarsening to any event cannot increase relative entropy, by the log-sum inequality. For Bernoulli laws the second derivative in the first probability of relative entropy is $1/[x(1 - x)] \geq 4$, so Taylor's formula at equality yields $D_{\text{KL}}(\text{Ber}(x) \parallel \text{Ber}(y)) \geq 2(x - y)^2$. Taking the supremum over events therefore gives $\|P - Q\|_{\text{TV}} \leq \sqrt{D_{\text{KL}}(P \parallel Q)/2}$. Any test has sum of its two error probabilities at least $1 - \|P - Q\|_{\text{TV}}$, directly from the definition of total variation.

The curvature separation Δ_s is at least $c_0|s|$. Given an estimator, classify it by its nearer alternative, with fixed tie breaking. On a wrong classification its matching error is at least $\Delta_s/2$, by the triangle inequality. Its maximum expected error is consequently at least $\Delta_s(1 - \|\mathbf{P}_s - \mathbf{P}_{-s}\|_{\text{TV}})/4$. Choose $s = c_1(1 + B)^{-1/6}$, with $c_1 > 0$ small enough to stay inside the family and make the variation distance at most $1/2$ by (A.22.4). This proves the expected-risk bound. The bound $B \leq L^2 h^2 N$ proves (A.22.2).

For confidence accuracy choose $s = c_2 \varepsilon$ so that $\Delta_s > 2\varepsilon$. Success with probability at least $1 - \eta_0$ under both alternatives would give a test whose error sum is at most $2\eta_0$. Hence its variation distance is at least $1 - 2\eta_0 > 0$. The entropy bound forces $B \geq c_{\eta_0} \varepsilon^{-6}$, proving (A.22.3). The two alternatives have the same known $g = 1/2$, so supplying the gap does not change this test. $\square$

For the fixed-order design of Theorem A.21.2, all actual offsets are at most $L_m h$, including all pilot outcomes, and $h \asymp \varepsilon^{3/m}$. Thus the power $\varepsilon^{-(6+6/m)}$ in its preparation bound has a matching obstruction within this shrinking-offset, bounded-flight-number binary experiment, apart from confidence logarithms. This does not identify a minimax exponent for unrestricted collision observations: nonshrinking offsets, an unbounded choice of flight numbers, lower count values, and position data are different experiments. All constants here are at fixed m, J, L_m ; sending m to infinity is not justified.

APPENDIX A.23. COUNT-ONLY RECONSTRUCTION FROM A FIXED LOCAL BRACKET

The preceding acquisition theorem presupposes a bracket whose width shrinks with the desired accuracy. We now acquire that bracket using counts, starting from a fixed local interval. A deterministic cap on all later waiting times prevents an unsuccessful search from producing an infinite preparation cost. The original locally supplied-bracket theorem remains useful, with its sharper description of each stage.

Fix a sufficiently small compact physical neighborhood in Theorem A.10.1. The constants of the forward law on this neighborhood, including a positive lower bound, are regarded as design information. This is local reconstruction in a specified family, not a procedure requiring no prior geometric information. For $j = 1$ there are known constants $c_* > 0$, $d_* > 0$ such that

$$\Pr(N_{g+d} \geq 2) \geq c_* d^2 \quad (0 < d \leq d_*), \quad \Pr(N_t \geq 2) = 0 \quad (t \leq g). \quad (\text{A.23.1})$$

These follow from Theorem 3.1 and compactness.

Lemma A.23.1 (Count acquisition of a coarse bracket). *Suppose $g \in [L_0, U_0]$, with fixed width $0 < W = U_0 - L_0 \leq d_*$. For $0 < h < W$ and $0 < \delta < 1/2$, a binary count procedure returns g_0 satisfying $|g - g_0| \leq h/4$ with probability at least $1 - \delta$. On every history it uses at most*

$$Ch^{-2} \log \left(\frac{2 + \log(W/h)}{\delta} \right) \quad (\text{A.23.2})$$

independent preparations. Its queries use only the indicator of two or more collisions at programmed times inside $[L_0, U_0]$.

Proof. Maintain an interval $[L, U]$, initially $[L_0, U_0]$, and stop when its width w is at most $h/2$. At a nonterminal stage query $t = (L + U)/2$ on

$$n(w) = \left\lceil \frac{16}{c_* w^2} \log \frac{M+1}{\delta} \right\rceil$$

fresh preparations, where $M = \lceil \log(2W/h) / \log(4/3) \rceil$. If at least one query succeeds, replace U by t ; otherwise replace L by $t - w/4$. In either case the interval is nested in the previous one and its width is either $w/2$ or $3w/4$. There are at most M stages, regardless of the observations.

Conditional on an interval containing g , a success certifies $g < t$ by (A.23.1). A failure branch can discard g only if $g < t - w/4$. In this case $t - g > w/4$ and $t - g \leq W \leq d_*$, so the probability of seeing no success is at most $\exp(-n(w)c_* w^2/16) \leq \delta/(M+1)$. A union bound over the first possible incorrect discard proves that all retained intervals contain g with probability at least $1 - \delta$. The midpoint of the final interval then has error at most $h/4$.

For cost, each nonterminal width exceeds $h/2$ and contracts by at least a factor $3/4$. Summing reciprocal squared widths backwards from the last such stage gives $\sum w^{-2} \leq Ch^{-2}$. The ceiling contributions add at most M , also bounded by Ch^{-2} on the stated range, with a constant depending on fixed W . This proves (A.23.2) and its all-history form. There is no waiting for an event of unknown positive probability in this search. $\square$

Theorem A.23.2 (Fully charged local count reconstruction). *Fix $m \geq 1$, a compact physical neighborhood as above, and a fixed interval $[L_0, U_0]$ containing its possible gaps and satisfying the hypotheses of Lemma A.23.1. There exist $h_m > 0$ and $C_m < \infty$ such that, for $0 < h \leq h_m$ and $0 < \eta < 1/2$, a count-only procedure estimates the gap, area, and unordered curvature triple with*

$$|\hat{g} - g| \leq C_m h^{m+1}, \quad |\hat{A} - A| \leq C_m h^m, \quad \text{dist}_{\text{match}}(\hat{\kappa}, \kappa) \leq C_m h^{m/3} \quad (\text{A.23.3})$$

with probability at least $1 - \eta$. Its total preparation count is bounded on every history by

$$C_m h^{-(2m+2)} \log(C_m/\eta). \quad (\text{A.23.4})$$

The procedure uses only binary collision-count reports with flight numbers 1, 2, 3, independent full-phase preparations, and noiseless elapsed-time recording. In particular curvature accuracy ε has sufficient total cost $C_m \varepsilon^{-(6+6/m)} \log(C_m/\eta)$ from a bracket independent of ε . The supplied information includes the fixed local family and its uniform bounds, but no accuracy-dependent gap bracket, contact positions, or separately measured area.

Proof. Use Lemma A.23.1 with failure budget $\eta/4$ to acquire g_0 . Conditional on its success, the hypothesis $|g - g_0| \leq h/4$ of Theorem A.21.2 holds. All subsequent preparations are fresh. Run that theorem's pilot and amplitude stages with $J = 3$, and allocate failure probability $\eta/2$ to their estimation bounds. Their constants are uniform over all admissible g_0 , so conditioning on the entire first stage is legitimate. On these good events their estimates satisfy (A.23.3).

There is an additional issue on a failed first stage: a programmed pilot window could lie below onset. We therefore modify each pilot waiting time as follows. If its target $r = \lceil C_m h^{-2m} \log(C_m/\eta) \rceil$ successes have not occurred within

$$B_m = \lceil C_m h^{-(2m+2)} \log(C_m/\eta) \rceil$$

preparations, stop the entire procedure and return a fixed admissible parameter as a fallback. On first-stage success, every pilot probability is at least $c_m h^2$, uniformly over its programmed nodes. Choose the constant in B_m so that $B_m c_m h^2 \geq 2r$ and $B_m c_m h^2 \geq 8 \log(4(m+1)/\eta)$. The exponential Markov bound for a binomial variable gives

$$\Pr\{T_r > B_m\} = \Pr\{\text{Bin}(B_m, p) < r\} \leq e^{-B_m p/8} \leq \eta/[4(m+1)].$$

A union bound over the $m+1$ pilots charges at most $\eta/4$ for a cap on a good first stage. Before a cap the procedure agrees exactly with the original one, so this modification does not condition or bias the claimed estimation guarantee. The final amplitude stage has a predetermined number of preparations, whether or not its estimates are good. Its compact minimizer is selected by Lemma A.20.1.

Every history now has a deterministic preparation bound: the first stage has (A.23.2), each of the fixed number of pilots has cap B_m , and the amplitude stage has the deterministic bound in the proof of Theorem A.21.2. For fixed $m \geq 1$, the function $h^{2m} \log(2 + \log(W/h))$ is bounded for $0 < h \leq h_m < W$. Thus the first-stage cost is absorbed into (A.23.4), uniformly in $\eta < 1/2$. The three allocated failure probabilities sum to at most η . Finally choose $h = c_m \varepsilon^{3/m}$, with fixed c_m small enough to make the curvature bound at most ε . $\square$

The same initial search and cap construction applies to the independent-normalizer four-amplitude experiment of Theorem A.21.2 on any specified compact physical family satisfying its hypotheses; then $J = 4$. The search uses only its positive one-flight onset bound, and no area constraint is needed for that stage.

The first-stage windows are allowed to have offsets larger than h . Accordingly Theorem A.23.2 is an upper bound for a larger design class than the all-history shrinking-collar class in (A.22.3). We do not infer global minimax optimality for this enlarged class from that lower bound. The weighted-exposure inequality (A.22.4) remains applicable whenever its own query hypotheses

hold. The selected-position acquisition theorem has different data and is unchanged by this count-only construction.

APPENDIX A.24. MARKED ONSETS AND SOURCE DOMAINS

Let $x_{e,0}, x_{e,1}$ be the normal outgoing states and f_ξ a smooth collision mark near them. Put $a_{e,b} = f_\xi(x_{e,b})$, $\bar{f}_e = (a_{e,0} + a_{e,1})/2$, and $w_{j,e} = \sum_{i=0}^j a_{e,i \bmod 2}$.

Theorem A.24.1 (Marked response and the exact axial source domain). *At the selected channel's own threshold,*

$$Z_{j,e}(\xi, q, d) = \frac{d^2 e^{q w_{j,e}(\xi)}}{2A(\xi) \sinh(j\gamma_e(\xi))} [1 + d\mathcal{R}_{j,e}^f(\xi, q, d)]. \quad (\text{A.24.1})$$

Every fixed mixed derivative of the remainder is bounded uniformly in j on compact parameter and source sets. The complete marked event in the ground collar is the sum at its positive channel offsets. For fixed ξ , real q , and sufficiently small fixed $d \geq 0$, the series $\sum_{j,e} Z_{j,e}(\xi, q, d)/d^2$ converges exactly when

$$\max_{e \in \mathcal{E}} (q\bar{f}_e - \gamma_e) < 0. \quad (\text{A.24.2})$$

Right limits are used at $d = 0$. Every fixed mixed derivative is summable on compact subsets of this domain. Finitely many real sources are allowed, with the scalar product in place of $q\bar{f}_e$.

This series ranges over different onset windows, not over all itineraries in one unrestricted long window. The source formula, interface jumps and record response are consequences of the same uniform construction, not separate claims of a general limit theory. The exact equal-gap inverse theorem is retained in Section A.8. It uses central symmetry to remove the two-endpoint ambiguity and equal gaps to retain all channels at every collision order. The recovered area is a symmetric function of the three recovered curvatures in that chosen family, not a fourth independent unknown. Section A.8.1 proves this identity and the loss of first-order sensitivity at coalescence.

APPENDIX A.25. RECORD TOPOLOGY AND DIFFERENTIATED MOVING TESTS

A record limit is not, by itself, a response theorem. We specify both the metric and the differentiable test class. This also fixes the meaning of the bounded Lipschitz tests at the end of Theorem A.31.1. The unaltered circular estimates in that theorem continue to apply.

In a fixed oriented channel put $s = \sqrt{2d}$ and use the exact Morse coordinates $w = sz$, together with first residual time $r = d\eta$. The latent domain is the same $\mathcal{U}$ as in (6.5). Its unweighted conditional density relative to $\mathfrak{m}$ is

$$f_{j,e,\xi,s}(z, \eta) = \frac{a_{j,e}(\xi, sz)}{\int_{\mathcal{U}} a_{j,e}(\xi, sz) \, d\mathfrak{m}}. \quad (\text{A.25.1})$$

This is a density on common latent coordinates, not a total-variation comparison between physical measures on distinct embedded record surfaces. In general $f = 1 + O(s)$; for the circular unweighted density its evenness gives the stronger $O(s^2)$ already proved in Theorem A.31.1.

Let $\mathcal{R}_{j,e}(\xi, z, \eta, s)$ be the array consisting of y_i/s , $(t_i - ig_e)/d$, and both transverse velocity traces divided by s , for $0 \leq i \leq j$. The transverse frame is that of the closest chord; the axial values have zero transverse velocity. At $s = 0$ use the continuous limits. Equip each such array with the maximum norm,

$$d_{\infty,j}(\mathcal{R}, \tilde{\mathcal{R}}) = \max_{0 \leq i \leq j} \max\{|Y_i - \tilde{Y}_i|, |T_i - \tilde{T}_i|, |V_i^- - \tilde{V}_i^-|, |V_i^+ - \tilde{V}_i^+|\}. \quad (\text{A.25.2})$$

For tests involving only positions and times omit the other entries. An unnormalized Euclidean norm of the entire growing array is not used. The limiting positions are the linear bridge (5.3) applied to $\mathbf{H}_{j,e}^{-1/2}z$. Its limiting excess flight times are

$$Q_i = \kappa_{i \bmod 2} Y_i^2 + \kappa_{1-(i \bmod 2)} Y_{i+1}^2 + (Y_{i+1} - Y_i)^2 / g_e, \quad \sum_{i < j} Q_i = |z|^2.$$

Thus the limiting time array is $\eta + \sum_{h < i} Q_h$.

Proposition A.25.1 (Differentiated smooth record tests). *Fix any derivative order m . Let $\varphi_{j,\xi}$ be smooth tests of these scaled arrays with bounded derivatives through order $m+1$, in operator norm for the maximum-norm array space, and with the corresponding mixed parameter derivatives bounded independently of j . Then the conditional expectations $I_j(\xi, s)$ are smooth in (ξ, s) at $s = 0$ on the right, and*

$$\sup_{\xi, e, j} |\partial_\xi^\alpha (I_j(\xi, s) - I_j(\xi, 0))| \leq C_m s, \quad |\alpha| \leq m. \quad (\text{A.25.3})$$

A sufficient test class consists of a fixed finite number of selected record entries and smooth functions with uniformly bounded derivatives. For uniformly bounded Lipschitz tests without differentiability, the same statement without ∂_ξ^α follows from (A.25.2). The circular unweighted convergence in Theorem A.31.1 retains its stronger $O(d)$ rate for its positions and times.

Proof. The bridge, Morse inverse, chord directions and reflections are smooth in ξ, w . They vanish to the required order at $w = 0$: positions and transverse velocities vanish to first order, and every excess flight length to second order. Taylor's integral formula therefore extends the divisions by s and $s^2/2 = d$ smoothly to $s = 0$. For times, the estimates are summed along the bridge. The sum of the endpoint-decaying bounds from Lemma 5.2 is independent of j . Hence $\mathcal{R}_j$, all its fixed derivatives, and $s^{-1}(\mathcal{R}_j(s) - \mathcal{R}_j(0))$ have uniform maximum-norm bounds. The same claims hold for the conditional density (A.25.1); its denominator is bounded below.

Now write the expectation exactly on the fixed domain:

$$I_j(\xi, s) = \int_{\mathcal{U}} f_{j,e,\xi,s}(z, \eta) \varphi_{j,\xi}(\mathcal{R}_{j,e}(\xi, z, \eta, s)) \, d\mathbf{m}.$$

The chain rule in the maximum norm has no dimension factor by the hypothesis on the test derivatives. Dominated differentiation and the fundamental theorem of calculus in s prove (A.25.3). This differentiates the actual parametrization and density; it does not differentiate a total-variation inequality. For Lipschitz tests, instead bound the difference by the pathwise maximum-norm error plus the density error. In the circular positions–times specialization these errors are $O(d)$ as already proved. The entire statement also holds under bounded smooth additive sources after replacing the density by its positive normalized factor $e^{qV_{j,e}}$, with the same general $O(s)$ rate. $\square$

For a physical sampling cut, write the exact excess time of impact k as $T_{j,k}(\xi, z, s) = \sum_{i < k} (\ell_i - g_e)/d$. It has all fixed derivative bounds uniformly in j, k . A sample at $t^s = kg_e + da(\xi)$ precedes that impact precisely when

$$\eta > \psi_{j,k}(\xi, z, s) := a(\xi) - T_{j,k}(\xi, z, s).$$

Here a is smooth with bounded derivatives; the physical sample moves with the geometry, and is not an extra random coordinate.

Proposition A.25.2 (A differentiated transverse sampling interface). *Let a smooth test be supported in a fixed compact subset of $|z| < 1$, and suppose that on this support, for all parameters under consideration,*

$$\nu \leq \psi_{j,k}(\xi, z, s) \leq 1 - |z|^2 - \nu \quad (\text{A.25.4})$$

for one fixed $\nu > 0$. For a smooth test as in Proposition A.25.1, its conditional expectation restricted to the before-impact event is smooth in ξ, s , with uniform fixed-order bounds. If $F_j(\xi, z, \eta, s)$ denotes the conditional density times the test, relative to $dz d\eta$, then its first derivative in any scalar parameter direction is exactly

$$\partial_\xi I_j = \int \left\{ \int_{\psi_{j,k}}^{1-|z|^2} \partial_\xi F_j d\eta - F_j(\xi, z, \psi_{j,k}, s) \partial_\xi \psi_{j,k} \right\} dz. \quad (\text{A.25.5})$$

Repeated differentiation gives every fixed higher derivative and the analogue of (A.25.3). The boundary term is part of the formula.

Proof. The derivative of the cut function with respect to η is one. Condition (A.25.4) separates it uniformly from the two other boundaries of the latent domain on the support of the test. Its integral is therefore exactly $\int \int_{\psi_{j,k}}^{1-|z|^2} F_j d\eta dz$. Leibniz' rule gives (A.25.5), with no other boundary piece. Smooth bounded density and test derivatives and the uniform bounds for $T_{j,k}$ justify repeated differentiation under the outer integral. The fundamental theorem in s gives its differentiated limit estimate. This proves response of the selected moving physical interface, rather than response inferred from its small mismatch strip. $\square$

The margin is nonempty uniformly even for interior impacts. Take $a = 1/2$ and support the test on $|z| < 1/2$. Every flight excess is nonnegative, and the exact Morse identity gives $0 \leq T_{j,k} \leq |z|^2$ for every k . Both margins are therefore at least $1/4$. A finite collection of cuts with a fixed strict order and the same margins is handled by the same interval integration. At a fixed physical sample time t^s , instead, $a = (t^s - kg_e)/d$ and its derivatives contain factors k/d . Formula (A.25.5) still holds where the margins hold, but its order- m bounds can grow as $C_m(1 + k/d)^m$. No uniform onset response for all fixed schedules is asserted. When margins fail, or for an arbitrary bounded decoder, the convergence theorem alone gives no derivative estimate. The original finite-cut convergence and the smooth additive response remain separate valid statements.

APPENDIX A.26. THE CIRCULAR COLLISION HIERARCHY

A time window which admits two collisions need not test dependence between successive complete free flights. Under stationary preparation, its count law can depend only on the marginal law of one flight. The onset of higher collision tails is different: it requires an actual orbit segment, and its probability measures how a single two-dimensional collision state shadows a prescribed sequence of reflections. We determine this hierarchy for a periodic Lorentz gas, including its dependence on the length of the segment.

Let $\Lambda = \mathbb{Z}(1, 0) + \mathbb{Z}(1/2, \sqrt{3}/2)$ and remove the closed radius- R disk from $\mathbb{R}^2/\Lambda$. The particle has unit speed and reflects specularly. Throughout,

$$0 < R_- \leq R \leq R_+ < \frac{1}{2}, \quad K = [R_-, R_+], \quad g_R = 1 - 2R, \quad A_R = \frac{\sqrt{3}}{2} - \pi R^2, \quad \lambda_R = \frac{2R}{A_R}. \quad (\text{A.26.1})$$

Preparation is normalized phase volume $dq d\vartheta/(2\pi A_R)$. Write N_t for the number of collisions in $(0, t]$. The outgoing collision section has angular coordinates $x = (\alpha, \phi)$, with $|\phi| < \pi/2$, and invariant probability

$$d\nu = \frac{\cos \phi}{4\pi} d\alpha d\phi. \quad (\text{A.26.2})$$

Its collision map is B_R , its physical roof is τ_R , and $S_j \tau_R = \sum_{i=0}^{j-1} \tau_R \circ B_R^i$. These are the actual mechanical map and roof. In particular, the summands are not independent. Finite horizon is not required: the proof uses the positive lower bound $\tau_R \geq g_R$, not an upper bound.

Put

$$c_R = \frac{1-R}{R}, \quad \chi_R = \operatorname{arcosh} c_R, \quad s_R = \sinh \chi_R = \frac{\sqrt{g_R}}{R}, \quad \chi_* = \min_{R \in K} \chi_R > 0. \quad (\text{A.26.3})$$

There are six unit lattice vectors e . Let x_e denote the outgoing normal state facing e ; then $B_R x_e = x_{-e}$. In angular coordinates these states do not depend on R .

A.26.1. Unweighted threshold law.

Theorem A.26.1 (Uniform onset at every collision order). *There are $\varepsilon_0 > 0$ and $C < \infty$, depending on K but not on j , such that $\varepsilon_0 < \min_K g_R/4$ and, for all $j \geq 1$, $R \in K$ and $0 < \varepsilon < \varepsilon_0$,*

$$\Pr_R\{N_{jg_R+\varepsilon} \geq j+1\} = \Pr_R\{N_{jg_R+\varepsilon} = j+1\} = C_j(R)\varepsilon^2(1+\varepsilon h_j(R, \varepsilon)), \quad C_j(R) = \frac{3\lambda_R}{2R \sinh(j\chi_R)}. \quad (\text{A.26.4})$$

The functions h_j are smooth up to $\varepsilon = 0$. Every fixed mixed derivative of h_j in R and ε is bounded uniformly in j and $R \in K$, after decreasing ε_0 if necessary. For $-\varepsilon_0 < \varepsilon \leq 0$ the probability in (A.26.4) is zero.

The theorem concerns the entire equilibrium event, not a prescribed transverse initial subset. Its threshold collar has width independent of the collision order. The observation times $jg_R + \varepsilon$ grow with j ; the events remain the maximal possible collision counts in those windows. This is a uniform extremal regime, not a statement about typical long-time fluctuations.

A.26.2. Sources and summable response. Let f_R be a smooth real-valued function of the outgoing collision state. It is enough that it be smooth on fixed neighborhoods of the six normal states, uniformly on a neighborhood of K ; choose any bounded extension elsewhere. Let $F_* \geq 1$ bound $|f_R|$ on those neighborhoods. On the event in (A.26.4), write $X_0, \dots, X_j$ for the outgoing states at its successive impacts and set

$$Z_j(R, q, \varepsilon) = \mathbb{E}_R \left[\mathbf{1}_{\{N_{jg_R+\varepsilon}=j+1\}} \exp \left(q \sum_{i=0}^j f_R(X_i) \right) \right], \quad w_{j,e}(R) = \sum_{i=0}^j f_R(x_{(-1)^i e}).$$

The source is attached to a correlated impact record, not independently to a list of sampled roofs.

Theorem A.26.2 (Marked threshold expansion). *For every fixed $q_0 < \infty$, there are smooth functions $h_{j,e}(R, q, \varepsilon)$, uniformly bounded with every fixed mixed derivative for $R \in K$, $|q| \leq q_0$ and $0 \leq \varepsilon < \varepsilon_0$, such that*

$$Z_j(R, q, \varepsilon) = \frac{\lambda_R \varepsilon^2}{4R \sinh(j\chi_R)} \sum_{|e|=1} e^{qw_{j,e}(R)} (1 + \varepsilon h_{j,e}(R, q, \varepsilon)). \quad (\text{A.26.5})$$

All constants are independent of j . The offset ε is held fixed in radius derivatives in this formula. If $q_0 F_ < \chi_*$, then for each fixed $a, b, c \geq 0$,*

$$\sup_{\substack{R \in K, |q| \leq q_0 \\ 0 \leq \varepsilon < \varepsilon_0}} \left| \partial_R^a \partial_q^b \partial_\varepsilon^c \frac{Z_j(R, q, \varepsilon)}{\varepsilon^2} \right| \leq C_{a,b,c} (1+j)^{a+b} e^{-(\chi_* - q_0 F_*)j}. \quad (\text{A.26.6})$$

The quotient is understood by its smooth right extension at zero. Consequently its series over j and every fixed mixed derivative converge uniformly on the displayed set. The same conclusions hold for finitely many sources, with qf replaced by a scalar product and $q_0 F_$ by a uniform bound for its absolute value per impact.*

Uniformity in j is the main issue. An initial-value differentiation of B_R^j would amplify errors exponentially. Instead we parametrize an orbit segment by the transverse positions of its first and last impacts. Its effective action remains uniformly nondegenerate, while the flux density has size $\sinh(j\chi_R)^{-1}$. A relative determinant estimate, rather than an absolute error estimate, preserves this scale.

A.26.3. Consequences and relation to the fixed-window theorem. Theorem A.26.2 gives the finite-order response at all maximal-count onsets, including the derivative jumps across them. Section A.31 retains collision times, positions, both velocities and sampling-time cuts. Section A.32 compares the hierarchy with independent roofs having exactly the same marginal and recovers the normal periodic-orbit multiplier from the coefficients C_j .

The accompanying article [7] retains the full proof of the $T = 11/100$, $R \in [9/20, 47/100]$ three-point count law and its regular-level response. That is a complementary window: its second-collision probability is already positive throughout the radius interval. Here the count changes support at $t = jg_R$, and its second response is singular there. The one-flight onset coefficient suggested by that calculation is the case $j = 1$ of (A.26.4); the uniform multi-flight determinant and the full marked record follow from the uniform boundary-value argument and the explicit specialization below.

APPENDIX A.27. STATIONARY FLUX AND THE COMPLETE MARKED EVENT

For an outgoing boundary state write $ds = R d\alpha$ and $p = \sin \phi$. Its unnormalized flux is $ds dp$, of total mass $4\pi R$. A flight tube has phase volume $ds dp dr$. Except for a null set of directions, a line on the lattice torus meets the open disk in both time directions: irrational lines are dense, and the remaining rational directions are countable. Tangencies and their finite regular preimages have zero flux. The positive gap prevents finite-time accumulation. Thus the flight tubes cover phase volume up to a null set, also when corridors give infinite horizon. It follows that

$$\bar{\tau}_R = \int \tau_R d\nu = \frac{A_R}{2R}, \quad d\mu_R = \lambda_R d\nu(x) ds, \quad 0 \leq s < \tau_R(x). \quad (\text{A.27.1})$$

Reflection preserves flux, and B_R is invertible and ν -preserving on the full-measure regular section.

For completeness, integration of the allowable suspension positions gives

$$\Pr_R(N_t \geq k) = \lambda_R \int [(t - S_{k-1}\tau_R)_+ - (t - S_k\tau_R)_+] d\nu. \quad (\text{A.27.2})$$

Indeed, from (x, s) the k th hit occurs at $S_k\tau_R(x) - s$. The length of the allowed part of $[0, \tau_R(x))$ is $(t - S_{k-1}\tau_R(B_R x))_+ - (t - S_k\tau_R(x))_+$; invariance proves (A.27.2). This is a stationary suspension identity, not a renewal theorem. Since $S_k\tau_R \geq kg_R$, only finitely many tails are nonzero at each fixed time. Their sum telescopes to $\mathbb{E}_R N_t = \lambda_R t$.

A marked version must not shift the first term by invariance while leaving its mark unchanged. We use the following direct form instead.

Proposition A.27.1 (First-impact coordinates). *Let $t = jg_R + \varepsilon$, where $j \geq 1$ and $0 < \varepsilon < g_R$. Let y be the outgoing state at the first future impact and let r be its time. For any bounded Borel function G of the complete impact record,*

$$\mathbb{E}_R[\mathbf{1}_{\{N_t=j+1\}} G] = \lambda_R \int_{S_j\tau_R(y) < t} \int_0^{t-S_j\tau_R(y)} G(\mathcal{P}_{j,R}(y, r; t)) dr d\nu(y). \quad (\text{A.27.3})$$

Here $\mathcal{P}_{j,R}$ is the actual path record, with impact times $r + S_i\tau_R(y)$, $0 \leq i \leq j$, and collision states $B_R^i y$. The formula includes the initial and terminal free-flight portions.

Proof. Replace the last collision state x in (A.27.1) by $y = B_R x$ and the suspension height by the first residual time $r = \tau_R(x) - s$. The measure is $\lambda_R d\nu(y) dr$ with $0 < r < \tau_R(B_R^{-1}y)$. The event of at least $j + 1$ hits is $r + S_j \tau_R(y) \leq t$. Its upper bound on r is at most $\varepsilon < g_R \leq \tau_R(B_R^{-1}y)$, so no additional truncation occurs. A $(j + 2)$ nd hit would require $t \geq (j + 1)g_R$, which is impossible. This proves the integration limits and the event identity.

Write q_i, v_i^+, v_i^- for the impact position and the outgoing and incoming velocities. Starting at $q_0 - rv_0^-$, the initial segment lasts r . Subsequent straight segments and specular reflections are prescribed by $B_R^i y$. After the last impact, the path continues for $t - r - S_j \tau_R(y) \geq 0$. Both end portions are shorter than g_R and cannot contain an additional impact. This gives the claimed complete mechanical record, including both velocity traces at each hit. $\square$

Taking $G = 1$ recovers

$$P_j(R, \varepsilon) := \Pr_R(N_{jg_R + \varepsilon} = j + 1) = \lambda_R \int (jg_R + \varepsilon - S_j \tau_R)_+ d\nu. \quad (\text{A.27.4})$$

For the source in Theorem A.26.2, G is independent of r , so the same formula holds with the correlated weight $\exp(q \sum_{i=0}^j f_R(B_R^i y))$ inside the integral. For tests involving collision times or physical samples, the inner r integral in (A.27.3) must be retained.

APPENDIX A.28. LOCALIZATION TO ACTUAL ALTERNATING ITINERARIES

The smallest distance between distinct disks is g_R , attained only by the six normal chords between nearest neighbors. The next center distance is $\sqrt{3}$. An outgoing ray cannot return to the same convex disk without another reflection.

Lemma A.28.1 (Uniform localization). *There are a transverse coordinate radius $a_0 > 0$ and an excess $\varepsilon_0 > 0$, independent of $j \geq 1$ and $R \in K$, with the following property. Every regular j -flight segment satisfying $S_j \tau_R < jg_R + \varepsilon_0$ alternates between one nearest-neighbor pair. Its impacts lie in the facing arcs $|y_i| < a_0$ about the normal chord. Every segment in these arcs which satisfies the local specular stationarity equations is an actual, unobstructed billiard segment.*

Proof. Every flight has length at least g_R . If their sum has excess less than ε_0 , each individual length is at most $g_R + \varepsilon_0$. Its target center e satisfies $|e| \leq 2R + g_R + \varepsilon_0 = 1 + \varepsilon_0$; taking $\varepsilon_0 < \sqrt{3} - 1$ forces $|e| = 1$.

For nearest disks, the minimum of the distance between their boundary points is unique. By compactness in $R \in K$, chords whose length is within ε_0 of g_R have both endpoints and direction uniformly close to that normal chord. This compactness statement can also be read from the endpoint cosine bound

$$n_0 \cdot v, -n_1 \cdot v \geq \frac{1 - 2R - \tau^2}{2R\tau},$$

whose right side tends to one as $\tau \downarrow g_R$, uniformly on K . Reflection at the target therefore sends the direction close to $-e$. Distinct nearest-neighbor directions are separated by $\pi/3$. Taking the neighborhoods sufficiently small forces the next short flight to go back to the preceding disk. This argument uses the same neighborhood at every step and hence does not deteriorate with j .

The closed normal chord between the pair has positive distance from all other disks, uniformly on K . For the pair $(0, (1, 0))$, the other nearest centers have vertical coordinate $\sqrt{3}/2$, while $R \leq R_+ < 1/2$; more distant centers are farther away. Small facing arcs preserve this separation. Their connecting chords are outgoing and incoming with normal cosines bounded away from zero. Convexity prevents a second crossing of the originating disk before the target. Thus these are first physical hits, and the stationarity equations, which express the equality of tangential velocity components, are exactly the specular reflection law. There is no grazing cutoff on a preselected preparation: the localization follows from the complete event's excess constraint. $\square$

Fix the pair $(0, (1, 0))$ and use a global transverse coordinate y . The facing boundary points are $(\sqrt{R^2 - y^2}, y)$ on the first disk and $(1 - \sqrt{R^2 - y^2}, y)$ on the second. The same symmetric length function applies in both directions:

$$\ell_R(u, v) = \sqrt{(1 - \sqrt{R^2 - u^2} - \sqrt{R^2 - v^2})^2 + (v - u)^2}. \quad (\text{A.28.1})$$

For endpoints $u = y_0, v = y_j$, define the local broken-length action

$$\mathcal{L}_{j,R}(y_0, \dots, y_j) = \sum_{i=0}^{j-1} \ell_R(y_i, y_{i+1}). \quad (\text{A.28.2})$$

The interior variables satisfy $\partial_{y_i} \mathcal{L}_{j,R} = 0$. We write $W_{j,R}(u, v)$ for the stationary value, whose existence, uniqueness and uniform estimates are proved next.

The length has the even analytic expansion

$$\ell_R(u, v) = g_R + \frac{c_R(u^2 + v^2) - 2uv}{2g_R} + O((u^2 + v^2)^2). \quad (\text{A.28.3})$$

All derivatives of the remainder have uniform analytic bounds in a fixed small neighborhood for $R \in K$. In particular, $-\partial_{uv} \ell_R(0, 0) = 1/g_R$. There is also the elementary bound

$$\ell_R(u, v) - g_R \geq R - \sqrt{R^2 - u^2} + R - \sqrt{R^2 - v^2} \geq \frac{u^2 + v^2}{2R}. \quad (\text{A.28.4})$$

Consequently any localized orbit of total excess ε satisfies $y_0^2 + y_j^2 + 2 \sum_{i=1}^{j-1} y_i^2 \leq 2R\varepsilon$. This bounds all interior variables at once, without a factor j .

APPENDIX A.29. A BOUNDARY-VALUE CALCULUS UNIFORM IN THE ITINERARY LENGTH

The circular formulas follow from the general boundary-value lemma. We record their explicit expressions, including the even analytic improvements, to keep this specialization self-contained at the level of its calculations. The uniform contraction and determinant argument are those already proved in Lemma 5.2.

Lemma A.29.1 (Uniform stationary segment). *For endpoints $|u|, |v| < a_1$, with $a_1 > 0$ independent of j , the interior equations of (A.28.2) have a unique small solution. The solution and $W_{j,R}$ are real analytic in R, u, v . There are $C < \infty$ and $0 < \rho < 1$, independent of j , such that, writing $a = |u| + |v|$,*

$$y_i = \frac{\sinh((j-i)\chi_R)}{\sinh(j\chi_R)} u + \frac{\sinh(i\chi_R)}{\sinh(j\chi_R)} v + O(a^3(\rho^i + \rho^{j-i})), \quad (\text{A.29.1})$$

$$|y_i| \leq Ca(\rho^i + \rho^{j-i}), \quad 0 \leq i \leq j.$$

The analytic bounds for the bridge and for $W_{j,R} - jg_R$, and every fixed parameter derivative of these functions, are uniform in j . Moreover

$$W_{j,R}(u, v) = jg_R + \frac{1}{2}(u, v)H_{j,R}(u, v)^t + O(a^4), \quad (\text{A.29.2})$$

with uniform analytic remainder bounds.

Proof. Here $\psi_0 = \psi_1 = R - \sqrt{R^2 - y^2}$ and $c_0 = c_1 = c_R$. The endpoint solution of the constant Jacobi equation is the displayed linear term. Formula (5.5) becomes

$$G_{ik} = g_R \frac{\sinh(i\chi_R) \sinh((j-k)\chi_R)}{\sinh \chi_R \sinh(j\chi_R)}, \quad G_{ki} = G_{ik} \quad (1 \leq i \leq k < j). \quad (\text{A.29.3})$$

For any fixed $e^{-\chi_*} < \rho < 1$ and $w_i = \rho^i + \rho^{j-i}$,

$$\sum_k e^{-\chi_*|i-k|} w_k \leq C_\rho w_i. \quad (\text{A.29.4})$$

Splitting at $k = i$ proves this by the geometric ratios $e^{-\chi_*}/\rho$ and $e^{-\chi_*}\rho$. The graphs are even and analytic on a common complex neighborhood for $R \in K$. The even analytic case of Lemma 5.2 therefore gives the cubic bridge correction, quartic action remainder, uniqueness throughout the fixed small box and all uniform analytic bounds asserted here. Its proof applies with interior row margin $2/R$ and endpoint margin $1/R$, so none of these constants depends on j . For $j = 1$ there are no interior variables and (A.28.1) gives the same assertions directly. $\square$

Lemma A.29.2 (Effective Hessian and twist). *The endpoint Hessian and mixed derivative at the normal segment are*

$$H_{j,R} = \frac{s_R}{g_R} \begin{pmatrix} \coth(j\chi_R) & -\operatorname{csch}(j\chi_R) \\ -\operatorname{csch}(j\chi_R) & \coth(j\chi_R) \end{pmatrix}, \quad d_{j,R}^0 := -W_{j,R,uv}(0,0) = \frac{s_R}{g_R \sinh(j\chi_R)}. \quad (\text{A.29.5})$$

In particular

$$\sqrt{\det H_{j,R}} = \frac{s_R}{g_R} = \frac{1}{R\sqrt{g_R}}, \quad \frac{d_{j,R}^0}{\sqrt{\det H_{j,R}}} = \frac{1}{\sinh(j\chi_R)}. \quad (\text{A.29.6})$$

Both eigenvalues of $H_{j,R}$ are bounded above and below by positive constants independent of j and $R \in K$.

Proof. For the quadratic action the derivative at its first endpoint is $(c_R u - y_1^{\text{lin}})/g_R$. Insert the bridge formula and use $c_R \sinh(j\chi_R) - \sinh((j-1)\chi_R) = s_R \cosh(j\chi_R)$. Reversal gives the other endpoint derivative, proving (A.29.5). The identity $\coth^2 x - \operatorname{csch}^2 x = 1$ proves (A.29.6). The eigenvalues are $(s_R/g_R) \tanh(j\chi_R/2)$ and $(s_R/g_R) \coth(j\chi_R/2)$. Their asserted bounds follow from $j \geq 1$, $\chi_R \geq \chi_* > 0$ and compactness of K . $\square$

An absolute bound $W_{uv} = -d_{j,R}^0 + O(a^2)$ would be useless for large j . The next estimate is the relevant replacement.

Lemma A.29.3 (Relative flux determinant). *On a fixed endpoint neighborhood,*

$$-W_{j,R,uv}(u,v) = d_{j,R}^0 b_{j,R}(u,v), \quad b_{j,R} = 1 + O(u^2 + v^2) > 0. \quad (\text{A.29.7})$$

The functions $b_{j,R}$ are even under $(u,v) \mapsto (-u,-v)$ and have uniform analytic bounds, including every fixed radius and endpoint derivative, independently of j .

Proof. The even analytic case of Lemma 5.2 proves the relative estimate and every fixed derivative bound. To identify its exact circular normalization, put $b_i = -\ell_{R,uv}(y_i, y_{i+1})$ along the stationary segment. Its corner-cofactor identity is

$$-W_{j,R,uv} = \frac{\prod_{i=0}^{j-1} b_i}{\det H_{\text{int}}}, \quad (\text{A.29.8})$$

with empty determinant one when $j = 1$. Evenness gives $\sum_i |\log(g_R b_i)| \leq C a^2$. If $E = H_{\text{int}} - H_{\text{int}}^0$, the same even improvement gives $\|E\|_1 \leq C a^2$ and the exact convergent expansion

$$\log \frac{\det H_{\text{int}}}{\det H_{\text{int}}^0} = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} \operatorname{tr}[(H_{\text{int}}^0)^{-1} E]^m = O(a^2). \quad (\text{A.29.9})$$

The trace-norm and differentiated estimates for this expansion were proved in the general lemma; no factor equal to the matrix dimension is introduced. Comparing (A.29.8) with its value at zero and using (A.29.5) proves (A.29.7). Uniqueness and simultaneous reversal of the coordinates prove evenness. Positivity follows from the positive twists and interior Hessian. $\square$

Proposition A.29.4 (Physical flux in endpoint coordinates). *For each of the six oriented nearest-neighbor segments, the small endpoint coordinates (u, v) parametrize its outgoing initial collision state one-to-one, and*

$$d\nu = \frac{-W_{j,R,uv}(u, v)}{4\pi R} du dv. \quad (\text{A.29.10})$$

Moreover $S_j \tau_R = W_{j,R}(u, v)$ on this chart.

Proof. The first variation of length gives $\partial_u W = -v_0^+ \cdot q_0'(u)$. If s_0 is boundary arclength, the outgoing tangential momentum is $p_0 = -W_u / (ds_0/du)$. Thus $|ds_0 dp_0| = |W_{uv}| du dv$. The mixed derivative is nonzero by Lemma A.29.3, so this is a local coordinate change. At fixed u , p_0 is strictly monotone in v throughout the small box, proving injectivity. The stationary path is physical by Lemma A.28.1; uniqueness identifies its successive states with the iterates of B_R . Normalization of $ds_0 dp_0$ by its total mass $4\pi R$ proves the measure formula. The length identity follows by adding the actual flight lengths. $\square$

APPENDIX A.30. UNIFORM THRESHOLD INTEGRATION AND ITS RESPONSE

We now integrate the actual flux on the whole onset event. The dimension of this integration is two, regardless of the number of flights.

Lemma A.30.1 (Uniform even Morse coordinates). *There is an odd analytic change of variables $z = \Phi_{j,R}(w)$ on a fixed neighborhood of zero in $\mathbb{R}^2$, with uniform inverse and derivative bounds, such that*

$$W_{j,R}(\Phi_{j,R}(w)) - jg_R = \frac{1}{2}|w|^2, \quad D\Phi_{j,R}(0) = H_{j,R}^{-1/2}. \quad (\text{A.30.1})$$

In these variables the endpoint flux is

$$d\nu = \frac{1}{4\pi R \sinh(j\chi_R)} a_{j,R}(w) dw, \quad a_{j,R}(w) = 1 + O(|w|^2), \quad (\text{A.30.2})$$

where $a_{j,R}$ is even, positive and uniformly analytic, including all fixed parameter derivatives.

Proof. Put $E(z) = W_{j,R}(z) - jg_R$. Taylor's integral identity gives

$$E(z) = \frac{1}{2} z^t M(z) z, \quad M(z) = 2 \int_0^1 (1-t) D^2 E(tz) dt.$$

The matrix M is symmetric, even and analytic, with $M(0) = H_{j,R}$ and $M(z) - H_{j,R} = O(|z|^2)$ uniformly. The positive definite square root is analytic on a fixed neighborhood of the uniformly positive matrices $H_{j,R}$. Thus $w = M(z)^{1/2} z$ is an odd analytic map with uniformly invertible differential at zero. Its nonlinear differential is a uniformly small perturbation on a fixed ball. Contraction applied to the equation $z = H_{j,R}^{-1/2} w + O(|z|^3)$ gives an analytic inverse on a common smaller ball, with uniform derivative estimates. This proves (A.30.1) without a dimension-dependent Morse neighborhood.

Insert (A.29.7) into (A.29.10) and use (A.29.6). The factor $b_{j,R}(\Phi(w)) \det D\Phi(w) \sqrt{\det H_{j,R}}$ is positive, even, uniformly analytic and equals $1 + O(|w|^2)$. It is the required $a_{j,R}$. Orientation may be chosen so the local Jacobian is positive. $\square$

The energy bound (A.28.4) shows that, for a common smaller ε_0 , the complete sublevel set in each of the six charts lies inside these coordinates. Conversely all of the small ball $|w|^2 < 2\varepsilon$ gives physical segments. Thus no extraneous cutoff or boundary of a chart enters the following integral.

Proof of Theorem A.26.1. By (A.27.4) and (A.30.2), the contribution of one orientation is

$$\frac{\lambda_R}{4\pi R \sinh(j\chi_R)} \int_{|w|^2 < 2\varepsilon} (\varepsilon - |w|^2/2) a_{j,R}(w) dw.$$

The unweighted quadratic integral equals $\pi\varepsilon^2$. Indeed, rescale $w = \sqrt{2\varepsilon}z$ and use $\int_{|z|<1} (1 - |z|^2) dz = \pi/2$. All six orientations have the same flux by lattice symmetry. The analytic even amplitude has a convergent expansion in homogeneous even polynomials, with uniform bounds. Integration on the unit disk therefore gives $1 + \varepsilon h_j(R, \varepsilon)$ after division by the leading term; its analytic bounds are uniform in j . This proves (A.26.4) and every stated derivative bound. The roof lower bound proves vanishing on the other side of the onset. $\square$

Lemma A.30.2 (Additive marks on the endpoint bridge). *In the e chart define*

$$V_{j,e}(R, u, v) = \sum_{i=0}^j (f_R(X_i(R, u, v)) - f_R(x_{(-1)^i e})).$$

Then $V_{j,e}(R, 0, 0) = 0$, and $V_{j,e}$ and every fixed derivative in R, u, v have bounds independent of j . In particular $|V_{j,e}| \leq C(|u| + |v|)$.

Proof. Each impact position depends on y_i . The direction of its adjacent flight is the chord divided by (A.28.1); its denominator is at least $\min_K g_R$. Interior and final outgoing velocities are obtained by the specular formula $v^+ = v^- - 2(v^- \cdot n)n$. These maps are analytic on the fixed arcs. The last velocity uses (y_{j-1}, y_j) ; no additional endpoint is needed.

Consequently the deviation of X_i from its normal state is bounded by $Ca(w_{i-1} + w_i + w_{i+1})$, omitting unavailable indices. The same weighted bounds hold for every fixed derivative of these deviations: use the complex estimates in Lemma A.29.1, or differentiate the local maps and the bridge. The normal states in angular coordinates are independent of R . Applying the finite chain rule to f_R and subtracting its value at the normal state gives these weighted bounds for the summands as well. Since $\sum_i w_i \leq 2/(1 - \rho)$, their sum has uniform derivative bounds. The value at zero and the linear bound then follow from the mean value formula. $\square$

Proof of Theorem A.26.2. In one orientation, factor the source into $e^{qw_{j,e}(R)} e^{qV_{j,e}(R, u, v)}$. Lemmas A.30.1 and A.30.2 reduce its normalized threshold integral to

$$F_{j,e}(R, q, \varepsilon) = \frac{2}{\pi} \int_{|z|<1} (1 - |z|^2) a_{j,R}(\sqrt{2\varepsilon}z) \exp(qV_{j,e}(R, \Phi_{j,R}(\sqrt{2\varepsilon}z))) dz. \quad (\text{A.30.3})$$

It equals one at $\varepsilon = 0$. Replace the smooth amplitude in this integral by its even part in z . The integral is unchanged, and every odd Taylor polynomial disappears. Taylor's formula to any prescribed finite order, including its differentiated remainder, shows that $F_{j,e} = 1 + \varepsilon h_{j,e}$ with all fixed mixed derivative bounds uniform in j . More explicitly, to control c derivatives in ε use the even Taylor expansion to degree $2c + 4$ on a slightly larger ball; its integral remainder after the substitution $w = \sqrt{2\varepsilon}z$ is $O(\varepsilon^{c+2})$ with the corresponding differentiated bounds. Radius and source derivatives obey the same argument by Lemma A.30.2. This is finite-order smooth calculus and makes no analyticity assumption on f_R . Multiplying back the constants in (A.30.2) proves (A.26.5).

For every fixed a , direct differentiation of $\text{csch}(j\chi_R) = 2e^{-j\chi_R}/(1 - e^{-2j\chi_R})$ gives

$$|\partial_R^a \text{csch}(j\chi_R)| \leq C_a(1 + j)^a e^{-j\chi_*}.$$

Also $|\partial_R^a w_{j,e}| \leq C_a(j + 1)$ and $|e^{qw_{j,e}}| \leq e^{q_0 F^*(j+1)}$. The finite product and chain rules, together with the uniform bounds for $h_{j,e}$, prove (A.26.6). Polynomially weighted geometric series converge, proving termwise differentiation and the source-vector version. $\square$

A.30.1. The singularity across the onset surface. Set

$$C_j(R, q) = \frac{\lambda_R}{4R \sinh(j\chi_R)} \sum_{|e|=1} e^{qw_{j,e}(R)}.$$

This is positive for real sources. Formula (A.26.5), with zero on the negative side, is a full expansion of the form $\varepsilon_+^2[C_j(R, q) + O(\varepsilon)]$ with smooth right-hand coefficients. Thus it is C^1 , and its second t derivative has jump $2C_j(R, q)$. It is not C^2 across any of these onsets. At fixed physical t and a radius R_c with $t = jg_{R_c}$, the jump of the second radius derivative is

$$8j^2 C_j(R_c, q). \quad (\text{A.30.4})$$

Indeed $\varepsilon = t - jg_R = 2j(R - R_c)$; all other second-derivative terms have a factor ε and vanish at the surface. For higher distributional derivatives, differentiation of ε_+^2 supplies the corresponding delta terms; for example its third t derivative contributes $2C_j(R, q)\delta(t - jg_R)$. At fixed t , radius differentiation is $\partial_R + 2j\partial_\varepsilon$ in onset coordinates. Therefore all fixed-order coefficient and one-sided derivative estimates remain polynomially weighted, exponentially summable in j in the source range of (A.26.6). This statement concerns the onset collars, not singularities of other itinerary families at other times.

APPENDIX A.31. THE JOINT IMPACT RECORD AND PHYSICAL SAMPLING CUTS

The endpoint parametrization gives more than a count or a marginal roof. It reconstructs all positions, directions and collision times in the window. We describe its onset law explicitly.

Let

$$\mathcal{U} = \{(z, \eta) \in \mathbb{R}^2 \times \mathbb{R} : |z| < 1, 0 < \eta < 1 - |z|^2\}, \quad \text{dm} = \frac{2}{\pi} \mathbf{1}_{\mathcal{U}} \text{d}z \text{d}\eta. \quad (\text{A.31.1})$$

This is a probability measure. For $z \in \mathbb{R}^2$, set $(U, V)^t = H_{j,R}^{-1/2} z$ and define

$$\begin{aligned} Y_i(z) &= \frac{\sinh((j-i)\chi_R)}{\sinh(j\chi_R)} U + \frac{\sinh(i\chi_R)}{\sinh(j\chi_R)} V, \quad 0 \leq i \leq j, \\ Q_i(z) &= \frac{Y_i^2 + Y_{i+1}^2}{R} + \frac{(Y_{i+1} - Y_i)^2}{g_R}, \quad 0 \leq i < j. \end{aligned} \quad (\text{A.31.2})$$

These quadratic functions satisfy $Q_i \geq 0$ and $\sum_{i=0}^{j-1} Q_i = |z|^2$.

Theorem A.31.1 (Marked onset record). *Condition on $N_{jg_R+\varepsilon} = j+1$. The orientation e is exactly uniform on the six nearest lattice vectors. In that orientation, use the exact Morse coordinates w of the endpoints and the first-impact time r , and set $z = w/\sqrt{2\varepsilon}$, $\eta = r/\varepsilon$. Their law has total variation distance at most $C\varepsilon$ from $\mathfrak{m}$, uniformly in $j \geq 1$ and $R \in K$.*

On these same coordinates, the transverse impact positions and impact times t_i satisfy

$$\frac{y_i}{\sqrt{2\varepsilon}} = Y_i(z) + O(\varepsilon), \quad \frac{t_i - ig_R}{\varepsilon} = \eta + \sum_{h=0}^{i-1} Q_h(z) + O(\varepsilon), \quad 0 \leq i \leq j, \quad (\text{A.31.3})$$

with errors uniform in the index, itinerary length, radius and $(z, \eta) \in \mathcal{U}$. Both velocity traces are obtained from the same bridge by the chord and reflection formulas.

Consider any fixed finite number of samples at $t^s = kg_R + \kappa\varepsilon \in [0, jg_R + \varepsilon]$, with $0 \leq k \leq j$ and bounded fixed κ . In the limiting record, the sample precedes impact k exactly on

$$\eta + \sum_{h < k} Q_h(z) > \kappa. \quad (\text{A.31.4})$$

The joint probabilities of the resulting before/after cuts converge with error $O(\varepsilon)$, uniformly in j and R . Thus the sampling-time interfaces are retained, not suppressed by pointwise differentiation away from event times.

Proof. Keep the r integral in (A.27.3). The changes $w = \sqrt{2\varepsilon}z$, $r = \varepsilon\eta$ give exactly the domain $\mathcal{U}$ and, up to its normalizing constant, density $a_{j,R}(\sqrt{2\varepsilon}z)$. By (A.30.2), this density is $1 + O(\varepsilon)$ uniformly. Since $\text{vol}\mathcal{U} = \pi/2$, normalization proves the total variation assertion. Rotation through multiples of $\pi/3$ preserves phase volume and the event, proving the exact orientation weights.

The odd Morse inverse has expansion $\Phi_{j,R}(w) = H_{j,R}^{-1/2}w + O(|w|^3)$. Insert it into (A.29.1). This proves the first formula in (A.31.3), with the stronger error $O(\varepsilon(\rho^i + \rho^{j-i}))$. Expanding each chord with (A.28.3) gives $\ell_R(y_i, y_{i+1}) = g_R + \varepsilon Q_i(z) + O(\varepsilon^2(\rho^i + \rho^{j-i})^2)$. The sum of these error bounds is uniform, proving the time formula. The identity for $\sum Q_i$ is the quadratic endpoint action evaluated at $H_{j,R}^{-1/2}z$.

For clarity, in the pair's fixed global transverse direction the leading outgoing transverse velocity at impacts $i < j$, divided by $\sqrt{2\varepsilon}$, is $(Y_{i+1} - Y_i)/g_R$. At the final impact it is $(Y_j - Y_{j-1})/g_R + 2Y_j/R$; the initial incoming value is $(Y_1 - Y_0)/g_R - 2Y_0/R$. The longitudinal components and the unscaled velocities are given exactly by the unit chord and specular formulas. Uniform analyticity on the fixed arcs gives their corresponding Taylor bounds. Together with the initial and terminal pieces in Proposition A.27.1, this reconstructs the entire physical path.

Consecutive impacts are separated by at least g_R , so a sample within $O(\varepsilon)$ of kg_R can cross only the k th impact for small ε . Formula (A.31.3) gives (A.31.4). The cut has derivative one in η . For each fixed z , the set where its left side lies within δ of κ has η -length at most 2δ . Its m -mass is therefore at most 4δ . Taking $\delta = C\varepsilon$ and adding the $O(\varepsilon)$ density error bounds the mismatch probability. A union bound proves the result for any fixed finite set of cuts. Bounded Lipschitz tests of the scaled positions and times can be added using their uniform pathwise errors. No differentiability of a discontinuous sampling decoder is assumed. $\square$

For the source-weighted conditional law, the orientation weights tend to $e^{qw_{j,e}}/\sum_{e'} e^{qw_{j,e'}}$. Given an orientation, the limiting law is still (A.31.1); its total variation error is $O(\sqrt{\varepsilon})$ uniformly for bounded sources, by Lemma A.30.2. The integrated source normalization has the stronger $O(\varepsilon)$ relative error of Theorem A.26.2 because its odd part cancels. The distinction between these two errors is important for a full marked law.

APPENDIX A.32. MECHANICAL CONSEQUENCES OF THE THRESHOLD COEFFICIENTS

A.32.1. A multi-roof effect not determined by the one-roof law.

Corollary A.32.1 (The nonrenewal onset exponent). *Fix $R \in (0, 1/2)$. Let an independent-roof stationary suspension have the same section roof marginal as the Lorentz gas. Its intensity is therefore the same λ_R . For each fixed $j \geq 1$, its maximal count onset satisfies*

$$\Pr^{\text{ind}}(N_{jg_R+\varepsilon} = j+1) = \frac{\lambda_R}{(j+1)!} \left(\frac{3}{\sqrt{g_R}} \right)^j \varepsilon^{j+1} (1 + O_j(\varepsilon)). \quad (\text{A.32.1})$$

For the actual Lorentz gas the exponent is two for every j , with coefficient (A.26.4). In particular the two laws have different leading powers for every $j \geq 2$, despite their identical one-roof marginal and intensity.

Proof. The case $j = 1$ of the action calculation gives $\int (g_R + \varepsilon - \tau_R)_+ d\nu = 3\varepsilon^2/(2\sqrt{g_R}) + O(\varepsilon^3)$. The smooth threshold density proved above permits differentiation. Thus the excess roof $u = \tau_R - g_R$ has density $3/\sqrt{g_R} + O(u)$ near $u = 0$. For j independent excess roofs, integrate their

product density on $u_i \geq 0$, $\sum_i u_i < \varepsilon$ with weight $\varepsilon - \sum_i u_i$. The leading integral is $\varepsilon^{j+1}/(j+1)!$, as follows by integrating successively on this simplex. The error is $O_j(\varepsilon)$ relative to it. The stationary identity (A.27.2), with $\varepsilon < g_R$, supplies the factor λ_R and excludes the next count. This proves (A.32.1). The actual law is Theorem A.26.1. $\square$

For example the first genuinely multi-roof count has

$$\Pr_R(N_{2g_R+\varepsilon} = 3) = \frac{3\lambda_R R}{4(1-R)\sqrt{g_R}} \varepsilon^2 (1 + O(\varepsilon)). \quad (\text{A.32.2})$$

The independent-roof substitute instead has coefficient $3\lambda_R/(2g_R)$ multiplying ε^3 . The stationary identity is the same in both systems; the difference comes from the specular boundary-value geometry of $S_{2\tau_R}$.

A.32.2. Recovery of periodic-orbit instability.

Corollary A.32.2 (Instability in count probabilities). *The unstable multiplier of the normal period-two return map is $e^{2\chi_R}$. The measured threshold coefficients satisfy*

$$\lim_{j \rightarrow \infty} \frac{C_{j+1}(R)}{C_j(R)} = e^{-\chi_R}, \quad \frac{C_1(R)}{C_2(R)} = 2 \cosh \chi_R. \quad (\text{A.32.3})$$

Thus they determine that multiplier. The unweighted coefficient series has the exact representation

$$\sum_{j \geq 1} C_j(R) z^j = \frac{3\lambda_R}{R} \sum_{m \geq 0} \frac{z e^{-(2m+1)\chi_R}}{1 - z e^{-(2m+1)\chi_R}}, \quad |z| < e^{\chi_R}. \quad (\text{A.32.4})$$

It extends meromorphically to the complex plane with simple poles at $z = e^{(2m+1)\chi_R}$, $m \geq 0$.

Proof. The Jacobi recurrence in the proof of Lemma A.29.1 has transfer matrix $\begin{pmatrix} 2c_R & -1 \\ 1 & 0 \end{pmatrix}$ and eigenvalues $e^{\chi_R}, e^{-\chi_R}$. The local change between successive transverse positions and a collision position-momentum pair is invertible since the one-flight twist is $1/g_R$. After two flights the coordinate frame returns to the original orientation, so the physical period-two derivative is conjugate to the square of this transfer matrix. This proves the multiplier claim. The ratios follow from the explicit C_j .

Expand $\text{csch}(j\chi) = 2 \sum_{m \geq 0} e^{-(2m+1)j\chi}$ and sum the absolutely convergent geometric series in j . On every compact set avoiding the displayed poles, the tail in m converges uniformly, since its terms are $O(|z|e^{-(2m+1)\chi})$. This gives the meromorphic continuation. At each indicated point only one summand has a pole and its residue is nonzero, so there is no cancellation. $\square$

Series (A.32.4) organizes threshold coefficients belonging to different observation windows. It is not the ordinary pressure or the generating function of counts in one fixed window. The recovery of instability here uses equilibrium tail amplitudes; it does not claim that the relation between periodic action Hessians and monodromy is new.

A.32.3. The retained regular-window response. The complete companion [7] proves its fixed-window theorem, including the all-order moving-level formula, without using the present threshold asymptotic. We record the terminal-level witness requested for that calculation. At $T = 11/100$, $R \in [9/20, 47/100]$, choose $\alpha = 0$ facing $(1, 0)$ and

$$\cos \vartheta = \frac{1 - 2R + T^2}{2T(1 - R)}. \quad (\text{A.32.5})$$

The right side lies strictly between zero and one. The point $(R, 0) + T(\cos \vartheta, \sin \vartheta)$ lies on the target circle, since substitution gives squared distance R^2 from its center. It is an incoming hit because

$$(1 - R) \cos \vartheta - T = \frac{1 - 2R - T^2}{2T} > 0.$$

The outgoing sign and the no-obstruction estimate in the companion make it a genuine first hit. The terminal level is therefore nonempty. Its uniform gradient bound and $\partial_{RT} < 0$ show directly that the level integral in the companion's second-derivative formula is strictly positive. This is compatible with the onset singularity: the fixed window remains strictly above g_R on its specified interval, whereas (A.26.4) crosses the critical level itself.

REFERENCES

- [1] P. Bálint, J. De Simoi, V. Kaloshin and M. Leguil, Marked length spectrum, homoclinic orbits and the geometry of open dispersing billiards, *Comm. Math. Phys.* **374** (2020), no. 3, 1531–1575. doi:10.1007/s00220-019-03448-x.
- [2] S. V. Bolotin and D. V. Treschev, Hill's formula, *Russian Math. Surveys* **65** (2010), no. 2, 191–257. doi:10.1070/RM2010v065n02ABEH004671.
- [3] M. Demers, I. Melbourne and M. Nicol, Martingale approximations and anisotropic Banach spaces with an application to the time-one map of a Lorentz gas, *Nonlinearity* **33** (2020), 4095–4113. doi:10.1088/1361-6544/ab7d22.
- [4] M. F. Demers and H.-K. Zhang, A functional analytic approach to perturbations of the Lorentz gas, *Comm. Math. Phys.* **324** (2013), no. 3, 767–830. doi:10.1007/s00220-013-1820-0.
- [5] J. De Simoi, V. Kaloshin and M. Leguil, Marked length spectral determination of analytic chaotic billiards with axial symmetries, *Invent. Math.* **233** (2023), 829–901. doi:10.1007/s00222-023-01191-8.
- [6] J. Marklof, Entry and return times for semi-flows, *Nonlinearity* **30** (2017), no. 2, 810–824. doi:10.1088/1361-6544/aa518b.
- [7] Q. Qi, *Two-collision counting response in a periodic Lorentz gas*, September 8, 2026, source-pinned companion to this article. The complete original is retained as `two_collision.tex`.
- [8] Q. Qi, *A2: nonempty arithmetic and a checkable Fourier-inversion budget*, Round 33 source reconstruction, September 3, 2026. The original chapter and source identity are retained in the submission history, with the hypotheses printed in that chapter.
- [9] M. Carney, M. Nicol and H.-K. Zhang, Compound Poisson law for hitting times to periodic orbits in two-dimensional hyperbolic systems, *J. Stat. Phys.* **169** (2017), 804–823. doi:10.1007/s10955-017-1893-9.
- [10] D. Batenkov and Y. Yomdin, On the accuracy of solving confluent Prony systems, *SIAM J. Appl. Math.* **73** (2013), no. 1, 134–154. doi:10.1137/110836584.
- [11] A. Florio and M. Leguil, *Smooth conjugacy classes of 3D Axiom A flows*, arXiv:2010.04120v5, June 3, 2021. <https://arxiv.org/abs/2010.04120v5>.
- [12] D. Finamore and M. Leguil, *A CAT(0)-approach to the marked length spectral rigidity of Sinai billiards*, arXiv:2510.18983v1, October 21, 2025. <https://arxiv.org/abs/2510.18983v1>.
- [13] O. V. Osterman, On length spectrum rigidity of dispersing billiard systems, *J. Mod. Dyn.* **19** (2023), 847–878. doi:10.3934/jmd.2023025.
- [14] S. Zelditch, Inverse spectral problem for analytic domains, II: $\mathbb{Z}_2$ -symmetric domains, *Ann. of Math. (2)* **170** (2009), no. 1, 205–269. doi:10.4007/annals.2009.170.205.
- [15] G. Meurant, A review on the inverse of symmetric tridiagonal and block tridiagonal matrices, *SIAM J. Matrix Anal. Appl.* **13** (1992), no. 3, 707–728. doi:10.1137/0613045.
- [16] R. Nabben, Decay rates of the inverse of nonsymmetric tridiagonal and band matrices, *SIAM J. Matrix Anal. Appl.* **20** (1999), no. 3, 820–837. doi:10.1137/S0895479897317259.
- [17] Independent AI referee-style memorandum, *A2 v8: relative boundary laws*, September 9, 2026, and its companion *The smooth-envelope minimax experiment*, repository `TrillionniumFoundation/theta-theory`, review commit `79ff2f96fc2e41899666065355238915f21273b0`.
- [18] Z. Dai and P. K. Lamm, Local regularization for the nonlinear inverse autoconvolution problem, *SIAM J. Numer. Anal.* **46** (2008), no. 2, 832–868. doi:10.1137/070679247.
- [19] B. Hofmann, F. Werner and Y. Deng, *On uniqueness and ill-posedness for the deautoconvolution problem in the multi-dimensional case*, arXiv:2212.06534v1, December 13, 2022. <https://arxiv.org/abs/2212.06534v1>. Related journal publication: doi:10.1088/1361-6420/acd07b.

- [20] Independent AI referee-style memorandum, *Completed A2 v9 nonlinear-boundary revision*, September 10, 2026, repository `TrillionniumFoundation/theta-theory`, review commit `b756f4851669e34c7074ba61b5bcf48689756406`.
- [21] Independent AI referee-style memorandum, *A2 v10: observable boundary profiles*, September 10, 2026, repository `TrillionniumFoundation/theta-theory`, review commit `2b893b931a894dfc9e7730b853f575124ffb5c55`. The report is an author-requested independent review, not a journal editorial decision; its source is retained in the revision history.
- [22] Independent AI referee-style memorandum, *A2 v11: Abel stability and integrated-flux acquisition*, September 10, 2026, repository `TrillionniumFoundation/theta-theory`, review commit `bcde40b514e4dc00a3252f2d0710002a2caeac7b`. Section 5 contains the two-derivative gain, the improved sufficient preparation bound and a conditional modulus. This is an author-requested memorandum, not a commissioned journal report or editorial decision.
- [23] Independent AI referee-style memorandum, *A2 v12: independent harsh review and two-flight comparison*, September 10, 2026, repository `TrillionniumFoundation/theta-theory`, review commit `2ae2751f61224b66f314915fd5fc22f6321b606f`. Section 4 gives the all-order two-flight block and the direct fixed-window consequence. An author-requested independent AI memorandum, not a commissioned journal report or an editorial decision.
- [24] Independent AI-assisted referee-style memorandum, *Independent referee report on A2 v13 and analytic normal-form benchmark*, author-requested review, September 10, 2026, `TrillionniumFoundation/theta-theory`, review commit `b3f0ad5843782c221651c9189fc156d20865cab1`.
- [25] S. Sternberg, Local C^n transformations of the real line, *Duke Math. J.* **24** (1957), 97–102.
- [26] H. Eynard-Bontemps and A. Navas, *On the failure of linearization for germs of C^1 hyperbolic vector fields in dimension one*, arXiv:2212.13646v2, November 5, 2023. <https://arxiv.org/abs/2212.13646v2>. The introduction recalls the classical one-dimensional smooth theorem and distinguishes it from the low-regularity failures studied there.
- [27] Independent AI-assisted referee-style memorandum, *A2 v14: independent review of the intrinsic boundary density*, with the companion *Scalar linearization and width benchmark*, author-requested review, September 11, 2026, `TrillionniumFoundation/theta-theory`, review commit `0c523413ef7ffa2dedd1ab471ce58b3275896a93`. This is an independent AI memorandum, not a journal-commissioned report or an editorial decision.
- [28] D. A. Levin and Y. Peres, with contributions by E. L. Wilmer, *Markov Chains and Mixing Times*, second edition, American Mathematical Society, 2017. <https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf>.
- [29] Independent AI-assisted referee-style memorandum, *A2 v15: independent harsh review of scalar transport*, author-requested review, September 11, 2026, `TrillionniumFoundation/theta-theory`, review commit `a108adf17c4b9360e340a2d5708b20a375d69283`. The report records the exact strict-margin mixed-parameter benchmark. This is an independent memorandum, not a journal-commissioned report or an editorial decision.
- [30] M. Akahira, Asymptotic theory for estimation of location in non-regular cases, I: Order of convergence of consistent estimators, *Rep. Stat. Appl. Res., JUSE* **22** (1975), 8–26.
- [31] M. Akahira, Asymptotic theory for estimation of location in non-regular cases, II: Bounds of asymptotic distributions of consistent estimators, *Rep. Stat. Appl. Res., JUSE* **22** (1975), 99–115.
- [32] I. A. Ibragimov and R. Z. Has'minskii, *Statistical Estimation: Asymptotic Theory*, Springer, New York, 1981; English reprint 2013. doi:10.1007/978-1-4899-0027-2.
- [33] R. L. Smith, Maximum likelihood estimation in a class of nonregular cases, *Biometrika* **72** (1985), no. 1, 67–90. doi:10.1093/biomet/72.1.67.
- [34] K. Hirano and J. R. Porter, Asymptotic efficiency in parametric structural models with parameter-dependent support, *Econometrica* **71** (2003), no. 5, 1307–1338. doi:10.1111/1468-0262.00451.
- [35] Independent AI-assisted referee-style memorandum, *A2 v16: independent harsh review of adaptive physical experiments*, author-requested review, September 11, 2026, `TrillionniumFoundation/theta-theory`, review commit `c5fed5340df3147f64cb3479cf1837357f73e7a8`. Section 4.4 records the random-hazard mixture clarification. This is an independent memorandum, not a journal-commissioned report or an editorial decision.
- [36] A. Meister and M. Reiss, *Asymptotic equivalence for nonparametric regression with non-regular errors*, arXiv:1101.5248v1, January 27, 2011. <https://arxiv.org/abs/1101.5248v1>.